FRM Part II 2026

Operational Risk and Resilience

Gold-source question bank

746 questions, sorted by chapter and learning objective

Scope and LO wording: SchweserNotes 2026 (final authority). Every question solved and independently verified.

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Reading 42. Introduction to Operational Risk and Resilience

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 1.

LO	Learning objective	Qs
42.a	describe an operational risk management framework and assess the types of risks that can fall within the scope of such a framework.	4
42.b	describe the seven Basel II event risk categories and identify examples of operational risk events in each category.	10
42.c	explain characteristics of operational risk exposures and operational loss events, and challenges that can arise in managing operational risk due to these characteristics.	3
42.d	describe operational resilience, identify the elements of an operational resilience framework, and summarize regulatory expectations for operational resilience.	10

LO 42.a

Describe an operational risk management framework and assess the types of risks that can fall within the scope of such a framework.

Q1 AP Mock 1 Q39

Which of the four main tasks that make up the risk management cycle involves evaluating the likelihood of a risk occurring and its severity (impact) if it does occur?

A. Risk identification.
B. Risk assessment.
C. Risk mitigation.
D. Risk monitoring.

Answer: B

Step by step

1. The cycle has four activities: identification, assessment, mitigation, monitoring.
2. Evaluating likelihood and severity of previously identified risks, to prioritize the response, is the definition of risk assessment.
3. Answer: B.

Why B is correct: Risk assessment rates each identified risk on likelihood and severity so that those too likely or too severe can be prioritized for mitigation.

Why the others are wrong

- **A:** Identification only defines, discovers and categorizes risks; it does not rate them. Confuses 'finding' risks with 'rating' them.
- **C:** Mitigation is the response (controls, insurance, exposure reduction) applied after assessment; it does not evaluate likelihood and impact.
- **D:** Monitoring checks that the process works as intended (KRIs, incident reports) and feeds lessons back into the cycle; it is not the initial rating of risks.

Q2 AnalystPrep Q.5044

Which of the following risks falls within the scope of an operational risk management (ORM) framework?

A. Credit risk associated with a company's investments in the stock market.
B. Political risk resulting from changes in government policies.
C. Legal risk arising from breach of contracts, laws and legislation.
D. Interest rate risk associated with fluctuations in interest rates.

Answer: C

Step by step

1. The BCBS definition of operational risk is the risk of loss from inadequate or failed internal processes, people, systems and external events.
2. This definition explicitly includes legal risk (enforceability or breach of contracts, laws, regulations) but excludes strategic and reputational risk.
3. Credit, interest rate and political risks are financial or other risks managed under separate frameworks.

4. Only legal risk (C) is in scope.

Why C is correct: The Basel operational risk definition includes legal risk, so contract and legal breaches are managed within the ORM framework.

Why the others are wrong

- **A:** Investment-related credit risk is a financial (credit/market) risk managed under its own framework, not ORM.
- **B:** Political risk from policy change is a strategic/external exposure outside the four operational causes; the mistake is treating any external factor as an operational 'external event'.
- **D:** Interest rate risk is a market risk with its own framework; the mistake is treating every loss source as operational.

Q3 AnalystPrep Q.5045

REBUILT — Option B ('umbrella of all other actions') was loosely worded and arguably also questionable; replaced it with a statement taken directly from the chapter so that only D is incorrect.

A risk analyst presents on operational risk management (ORM) frameworks. Which of the following statements she makes is INCORRECT?

A. Firms should develop a holistic picture of their risk management practices that clarifies the relationships between actions, tools and techniques.

B. Regulated financial services companies are required to define their risk appetite and tolerance, which drives the priorities of risk exposure and mitigation.

C. Risk governance relates to assigning roles and responsibilities in the organization to execute and report on risk management.

D. Risk monitoring focuses on the definition, discovery, selection and categorization of the risks faced by a business.

Answer: D

Step by step

1. Check each statement against the ORM framework description.
2. A: a holistic view linking actions, tools and techniques is recommended - correct.
3. B: risk appetite and tolerance is a requirement for regulated financial firms and drives exposure and mitigation priorities - correct.
4. C: risk governance is the assignment of roles and responsibilities - correct.
5. D: definition, discovery, selection and categorization describe risk identification; monitoring checks that everything works as intended and tracks the evolution of risks - incorrect.

Why D is correct: D describes risk identification, not monitoring, so it is the incorrect statement.

Why the others are wrong

- **A:** Accurate; picking it would mean misreading an 'INCORRECT' question as 'correct'.
- **B:** Accurate; risk appetite is a regulatory requirement and a main ORMF component.
- **C:** Accurate definition of risk governance; not the error.

Q4 Schweser Module Quiz 42.1 Q1

During which step of the risk management process would scenario analysis and stress testing most likely be used?

A. Risk mitigation.

B. Risk monitoring.

C. Risk assessment.

D. Risk identification.

Answer: C

Step by step

1. Assessment determines likelihood and severity of identified risks for prioritization.
2. Scenario analysis and stress testing estimate how severe losses could be under adverse conditions, so they are assessment tools.
3. Answer: C.

Why C is correct: Scenario analysis and stress testing quantify probability and impact of identified risks, which is risk assessment.

Why the others are wrong

- **A:** Mitigation uses controls, insurance and exposure reduction, not scenario modeling.

- **B:** Monitoring uses incident reports and key risk indicators.
- **D:** Identification uses brainstorming, interviews and logs to list risks; it does not size them.

LO 42.b

Describe the seven Basel II event risk categories and identify examples of operational risk events in each category.

Q5 AnalystPrep Q.5048

BCBS divides operational risk into seven "Basel level 1" event types, further subdivided into level 2 regulatory types and level 3 examples. Which of the following statements about these categories is correct?

A. Internal fraud and external fraud belong to the same event risk category.
B. Acts that violate laws designed to protect the health, safety and well-being of employees and customers fall under clients, products, and business practices.
C. Data entry errors and incomplete legal documentation fall under execution, delivery, and process management.
D. Losses from theft and hacking are examples of the damage to physical assets category.

Answer: C

Step by step

1. Recall the seven Basel level 1 types: IF, EF, EPWS, CPBP, DPA, BDSF, EDPM.
2. EDPM examples: processing errors, missing documentation, vendor disputes. Data entry errors and incomplete legal documents are processing/documentation failures, so C is correct.
3. A is false: internal fraud (employees) and external fraud (outsiders) are separate types.
4. B is false: employment, health and safety law breaches belong to EPWS.
5. D is false: theft and hacking are external fraud; DPA is about destruction of equipment and natural disasters.

Why C is correct: Processing errors and missing or incomplete documentation are the standard examples of execution, delivery, and process management (EDPM).

Why the others are wrong

- **A:** Merges two distinct categories; the split is by perpetrator (employee vs outsider).
- **B:** Confuses CPBP (client/product/regulatory issues) with EPWS, which covers employment, health and safety.
- **D:** Theft and hacking are external fraud; DPA covers physical destruction such as natural disasters or equipment destruction.

Q6 AnalystPrep Q.5049

Which of the following is most likely an example of execution, delivery, and process management (EDPM) among the Basel level 1 event types?

A. Destruction of equipment
B. Employment contract termination
C. Vendor disputes
D. Unauthorized activities by employees

Answer: C

Step by step

1. EDPM examples in the Basel taxonomy: processing errors, missing documentation, vendor disputes.
2. Vendor disputes therefore match C.
3. Destruction of equipment is damage to physical assets; contract termination is EPWS; unauthorized employee activity is internal fraud.

Why C is correct: Disputes with vendors and trade counterparties arise from failed process management, which is EDPM.

Why the others are wrong

- **A:** This is the damage to physical assets (DPA) example, not a process failure.
- **B:** Contract termination issues are an employment practices and workplace safety (EPWS) example.
- **D:** Unauthorized activities by employees are internal fraud (IF).

Q7 AnalystPrep Q.5074

Which Basel level 1 event type covers losses from acts inconsistent with employment, health or safety laws, and from diversity and discrimination events?

A. Clients, products, and business practices
B. Employment practices and workplace safety
C. Damage to physical assets
D. Business disruption and system failures

Answer: B

Step by step

1. Breaches of employment, health or safety laws, and discrimination events concern the workforce and workplace.
2. This is the definition of EPWS (examples: contract termination, discrimination, employer's liability).
3. CPBP relates to clients/products, DPA to physical destruction, BDSF to IT outages.

Why B is correct: EPWS captures losses from employment-related law breaches, workplace safety failures and discrimination.

Why the others are wrong

- **A:** CPBP covers client misinformation, complaints, product errors and regulatory fines, not employee matters.
- **C:** DPA covers destruction of equipment and natural disasters.
- **D:** BDSF covers IT breakdowns and outages.

Q8 BT answers P2.T7.23.1.1 · BT notes P2.T7.23.1.1

REBUILT — Merged two near-identical versions and replaced the awkward 'true EXCEPT which is false' wording with a direct 'which is FALSE' question.

An operational risk analyst maps her bank's historical losses to the Basel level 1 categories (for example, an employee embezzling customer funds is classified as internal fraud, and a network outage blocking online access is classified as business disruption and system failures). Assume her frequency and severity profile resembles the industry (ORX) findings summarized by GARP. Which of the following statements is FALSE?

A. External fraud is high-frequency but heterogeneous, containing both low-severity and high-severity events.
B. In terms of severity, the largest losses accrue to the clients, products, and business practices (CPBP) category.
C. She excluded legal-risk losses because they cannot be cleanly mapped to any single level 1 category.
D. Damage to physical assets (DPA) tends to produce among the least severe losses and has among the lowest frequencies.

Answer: C

Step by step

1. A: EF is about 30% of frequency but 9% of severity; many small card frauds coexist with severe hacking events. True.
2. B: CPBP is about 52% of severity (from 22% of frequency), driven by regulatory fines. True.
3. D: DPA is about 1% of both frequency and severity. True.
4. C: Operational risk (Basel) includes legal risk, so legal losses are not excluded from the mapping. False.

Why C is correct: The Basel operational risk definition includes legal risk, so excluding legal losses is wrong; the other three statements match the ORX frequency/severity profile.

Why the others are wrong

- **A:** True statement (EF 30% of frequency, 9% of severity, heterogeneous), so not the answer to a 'FALSE' question.
- **B:** True statement (CPBP 52% of severity).
- **D:** True statement (DPA about 1% frequency and 1% severity).

Q9 BT answers P2.T7.600.3 · BT notes P2.T7.600.3

REBUILT — Merged duplicates and replaced the giveaway 'none of the above' distractor with 'damage to physical assets'.

Goodholdings Bank released a major upgrade of its custom-built enterprise resource planning (ERP) software platform, which exposed problems in the existing codebase (technical debt, the development effort needed to fix structural software problems caused by sub-optimal programming). Consultants were hired to re-factor the code, and management treats the consulting cost as an operational loss. To which Basel level 1 event type should the loss be assigned?

A. Internal fraud

B. Business disruption and system failures
C. Employment practices and workplace safety
D. Damage to physical assets

Answer: B
Step by step
1. The loss stems from defective software, a systems problem rather than a people, fraud or physical event.
2. Software and IT failures are the core of BDSF (examples: IT breakdown, outages).
3. No fraud (A), employment issue (C) or physical damage (D) is involved.

Why B is correct: Losses arising from software and IT system failures belong in business disruption and system failures.
Why the others are wrong
- **A:** No employee misconduct or intentional act is described.
- **C:** Nothing relates to employment, health or safety.
- **D:** The assets damaged are software code, not physical assets such as buildings or equipment.

Q10
BT notes P2.T7.603.1

REBUILT — Original relied on external business-line loss data (Cruz et al., BIS LDCE) outside the reading; rebuilt to test the GARP/ORX frequency versus severity profile by Basel event type.

A bank's loss data (2014-2019, consistent with industry ORX data) show that one Basel level 1 event type accounts for roughly 30% of all loss events but only about 9% of total loss severity. Most events are small (for example, thousands of card fraud incidents), though the category also includes rare, severe hacking events. Which event type is this?

A. External fraud
B. Clients, products, and business practices
C. Employment practices and workplace safety
D. Execution, delivery, and process management

Answer: A
Step by step
1. Compare frequency/severity shares (ORX): EF 30%/9%; EPWS 15%/5%; CPBP 22%/52%; EDPM 28%/27%.
2. Only EF has about 30% of frequency with about 9% of severity.
3. Hacking is part of EF, which explains the heterogeneity (many small card frauds plus rare severe hacks).

Why A is correct: External fraud is over-represented in frequency relative to severity because most incidents (card fraud) are small.
Why the others are wrong
- **B:** CPBP is the reverse: about 22% of frequency but 52% of severity because of large regulatory fines.
- **C:** EPWS has about 15% of frequency and 5% of severity, and no hacking component.
- **D:** EDPM has about 28% of frequency but 27% of severity, so it is proportional, not skewed to small losses.

Q11
Schweser Module Quiz 42.1 Q2

Which of the following Basel II event risk categories most likely results in the greatest loss severity for a financial institution?

A. External fraud (EF).
B. Client, products, and business practices (CPBP).
C. Employment practices and workplace safety (EPWS).
D. Execution, delivery, and process management (EDPM).

Answer: B
Step by step
1. ORX loss data (2014-2019) severity shares: CPBP 52%, EDPM 27%, EF 9%, EPWS 5%.
2. CPBP is largest, driven by regulatory fines that can reach billions of dollars.

Why B is correct: CPBP produces over half of total loss severity because of large fines and client compensation.
Why the others are wrong
- **A:** EF is high frequency but only about 9% of severity.

- **C:** EPWS is about 5% of severity.
- **D:** EDPM is second at about 27% of severity, not the greatest.

Q12 AP Mock 1 Q45

The head of risk management at a large bank must identify the event category of a loss suffered last year. Which event category covers losses from failed transaction processing and from relations with trade counterparties and vendors?

A. Internal fraud
B. Execution, delivery, and process management
C. Employment practices and workplace safety
D. Business disruption and system failures

Answer: B

Step by step

1. Failed transaction processing and vendor/counterparty relationships are process-management failures.
2. This matches EDPM (processing errors, missing documentation, vendor disputes).
3. IF is employee fraud, EPWS employment matters, BDSF system outages.

Why B is correct: EDPM covers failures to process transactions and manage processes, including vendor and counterparty disputes.

Why the others are wrong

- **A:** Internal fraud involves intentional misconduct by employees.
- **C:** EPWS covers termination, discrimination and workplace safety.
- **D:** BDSF covers IT outages where the system itself fails; here the process, not the system, failed.

Q13 BT answers P2.T7.600.1 · BT notes P2.T7.600.1

Trader Ben Smith enters a foreign exchange (FX) spot transaction, buying euros for dollars. He then quickly enters another transaction selling euros for dollars. He makes a quick profit from these two transactions; however, due to confusion in the back office, settlement to the counterparty is delayed to four days later. The counterparty demands compensation for the delayed settlement, and the claim includes lost (forgone) interest. The cost of the mistake exceeds the profit on the trade, resulting in an operational loss. To which Basel Level 1 event risk category should this loss be assigned?

A. Improper Business or Market Practices
B. Business Disruption and System Failures
C. Clients, Products, and Business Practices
D. Execution, Delivery, and Process Management

Answer: D

Step by step

1. The trade itself was valid and profitable; the loss arose because back-office settlement was delayed (a processing/transaction-management failure).
2. Basel Level 1 event types: internal fraud, external fraud, employment practices and workplace safety, clients/products/business practices, damage to physical assets, business disruption and system failures, execution/delivery/process management (EDPM).
3. Processing errors and failed settlement with a trade counterparty are classic EDPM examples, so the loss maps to EDPM.

Why D is correct: The loss stems from a back-office process failure (late settlement) toward a trade counterparty, which is Execution, Delivery, and Process Management.

Why the others are wrong

- **A:** 'Improper business or market practices' is a Level 2 sub-category (under CPBP), not a Level 1 category, and involves misconduct such as market manipulation, not a settlement error.
- **B:** No IT outage or system failure is described; the cause was 'confusion in the back office', a process error.
- **C:** CPBP covers failures toward clients (suitability, disclosure, product flaws); this is a settlement process error with a trading counterparty and no breach of a client obligation.

Q14 BT answers P2.T7.600.2 · BT notes P2.T7.600.2

REBUILT — Option A was a Level 2 sub-category and 'None of the above' was a giveaway; replaced with Level 1 categories and asked explicitly for the Level 1 type.

An Investment Advisor at a division of an international bank was sued by an employer-based retirement plan sponsor for fiduciary breach. The client claimed that the Advisor's recommendation put the Advisor's intention to earn a commission and the firm's profit motive above the client's best interest. The lawsuit was settled. To which Basel Level 1 event risk category should this loss be assigned?

A. Internal Fraud
B. Business Disruption and System Failures
C. Clients, Products, and Business Practices
D. Execution, Delivery, and Process Management

Answer: C

Step by step

1. The loss arises from a lawsuit by a client alleging the advisor failed a fiduciary/suitability obligation.
2. Failure to meet a professional obligation to specific clients (fiduciary, suitability, disclosure) falls under Clients, Products, and Business Practices (CPBP).
3. No theft by an employee, system outage, or processing error is involved, so the other Level 1 categories do not fit.

Why C is correct: A fiduciary-breach claim by a client is a failure to meet a professional obligation to a client, which is CPBP.

Why the others are wrong

- **A:** Internal fraud covers fraud and unauthorized activities by employees; the claim here alleges a breach of duty to a client, and the firm settled a civil suit.
- **B:** No system or IT failure is described.
- **D:** EDPM covers processing and transaction-management errors (settlement, documentation, vendor disputes), not breach of duty to a client.

LO 42.c

Explain characteristics of operational risk exposures and operational loss events, and challenges that can arise in managing operational risk due to these characteristics.

Q15 AnalystPrep Q.5051

An investment firm has contracted a risk professional to discuss the characteristics of operational loss events and the challenges they create for managing operational risk. Which of the following characteristics correctly matches its description?

A. Heterogeneous - operational risk is highly varied because it encompasses diverse risks, such as fraud in retail transactions and damage to physical assets
B. Idiosyncratic and diffuse - the distribution of operational losses is highly skewed, with most of the density concentrated in the lowest part of the distribution
C. Interconnected - operational risk arises from every person and process in the firm, so everyone has to take part in managing it
D. Heavy-tailed - operational risk evolves with the activities and development of the industry

Answer: A

Step by step

1. Recall the five characteristics: heterogeneous, idiosyncratic (and diffuse), heavy-tailed, interconnected, dynamic.
2. Test each pairing against its definition.
3. A: diverse risks (retail fraud to physical asset damage) with different causes, consequences and loss distributions = heterogeneous. Correct.
4. B: a skewed distribution with most density at small losses describes heavy-tailed.
5. C: risk arising from every person and process, managed by everyone, describes idiosyncratic and diffuse; interconnected means correlated risks and boundary events.
6. D: evolving with the industry describes dynamic.

Why A is correct: Heterogeneity means operational risk is an eclectic set of risks with different causes, consequences and loss distributions, which is exactly what A describes.

Why the others are wrong

- **B:** Gives the heavy-tailed description (skewed, many small losses, few huge) but labels it idiosyncratic and diffuse.
- **C:** Gives the idiosyncratic and diffuse description (managed by everyone, cannot be centralized) but labels it interconnected.
- **D:** Gives the dynamic description (evolving with the industry) but labels it heavy-tailed.

Q16

BT answers P2.T7.23.1.2 · BT notes P2.T7.23.1.2 · BT Mock Q28

Andrew is presenting a proposed operational risk management (ORM) framework to a board that is more familiar with market and credit risk than with operational risk. His presentation reviews the nature of operational risk, including its dynamic evolution and its skewed, heavy-tailed character. In addition to these features, which of the following statements about operational risk is TRUE?

A. The easiest way for a high-beta firm to lower its beta is to centralize the ORM function.

B. Operational risk is highly homogeneous because boundary events define clusters of loss events with shared features.

C. ORM is valuable because it reduces regulatory capital, and regulatory capital is a good indicator of the quality of a firm's risk management.

D. Most causes of operational risk trace to control weaknesses, human biases and behavioral failings, and/or changes in the operating environment.

Answer: D

Step by step

1. Check each statement against the characteristics of operational risk.
2. A: beta measures systematic (market) risk; operational risk is idiosyncratic and diffuse, so centralizing ORM does not change beta (and ORM cannot be centralized in the way market or credit risk management is).
3. B: operational risk is heterogeneous, not homogeneous; boundary events are events that begin in one risk category (e.g. a trade-booking error) and materialize in another (market loss), which reflects interconnectedness.
4. C: ORM's value is not defined by lower regulatory capital, and regulatory capital is not a reliable indicator of risk management quality.
5. D: control weaknesses, human behavioral failings and operating-environment changes are the common root causes of operational risk events. TRUE.

Why D is correct: Operational risk events mostly stem from weak controls, human biases and failings, and changes in the operating environment, which are also common root causes across risk types.

Why the others are wrong

- **A:** Confuses operational risk with systematic risk; beta captures market co-movement, and operational risk is idiosyncratic and not centralizable.
- **B:** States the opposite of the correct characteristic (heterogeneous) and misuses boundary events, which show interconnectedness, not homogeneity.
- **C:** Treats lower regulatory capital as the purpose of ORM and as a quality signal; neither is supported.

Q17

Schweser Module Quiz 42.2 Q1

Which of the following characteristics of operational risk best identifies the concept that operational risk cannot be fully eliminated through traditional methods, such as hedging?

A. Dynamic.

B. Idiosyncratic.

C. Heterogeneous.

D. Interconnected.

Answer: B

Step by step

1. Match the concept 'cannot be fully eliminated by avoidance, hedging or insurance' to a characteristic.
2. Operational risk is idiosyncratic and diffuse: it is driven by a firm's own people, processes and systems and cannot be centralized or hedged away.
3. Therefore some residual operational risk always remains: B.

Why B is correct: The idiosyncratic nature means traditional avoidance, hedging or insurance can reduce but never fully eliminate operational risk, leaving residual exposure.

Why the others are wrong

- **A:** Dynamic means exposures evolve with business practices and make modeling hard; it is not about irreducibility.
- **C:** Heterogeneous means diverse risks with different causes and loss distributions, which complicates classification.
- **D:** Interconnected means correlated risks and boundary events with market and credit risk, not residual risk after hedging.

LO 42.d

Describe operational resilience, identify the elements of an operational resilience framework, and summarize regulatory expectations for operational resilience.

Q18

AnalystPrep Q.5052

REBUILT — Stem attributed the five components to BCBS although GARP attributes them to the UK regulators (FCA, PRA, BoE); reworded the stem and kept the options.

Bank ABC wishes to strengthen its operational resilience. A consultant explains that UK regulators (FCA, PRA and Bank of England) do not define resilience in a single sentence but instead describe it through several components. Which of the following components is correctly described?

A. Continuity of business services: To contribute to the stability of the system, firms should respond to disruptions, maintain trust among key stakeholders, and provide clarity of communication during a crisis

B. Important business services: From a process-based view of continuity, the regulator moved to a service-based view to ensure continuity

C. Management of disruption: It protects vital business services from disruption

D. Lessons learned: Firms should learn from past events and cover predictable shocks only

Answer: B

Step by step

1. The UK components are: continuity of business services, important business services, impact tolerance levels, management of disruption, and lessons learned.
2. Important business services is the regulatory innovation: the focus moved from a process-based view of continuity to a service-based view, identifying services whose disruption would cause intolerable harm to consumers or market integrity.
3. Check each other option against its component: A and C are mismatched or misdescribed, and D is too narrow.

Why B is correct: Important business services marks the shift from a process-based to a service-based view of continuity, which is exactly what B states.

Why the others are wrong

- **A:** This describes management of disruption (response, stakeholder trust, clear communication), not continuity of business services, which is closest to classic business continuity planning and prevention.
- **C:** Management of disruption is about responding to disruption, maintaining trust and communicating clearly; protecting vital services from disruption is the continuity-of-services idea.
- **D:** Lessons learned requires firms to learn from internal and external events and to improve resilience to unanticipated extreme shocks, not only predictable ones.

Q19

AnalystPrep Q.5053

REBUILT — Stem was garbled and referred to BCBS although the expectations come from the Fed's 2020 sound practices; rewrote it as a clean 'which is NOT' question.

The Federal Reserve's 2020 Sound Practices to Strengthen Operational Resilience treat operational resilience as a key outcome of an effective operational risk management framework (ORMF). Which of the following is NOT a regulatory expectation for operational resilience?

A. Effective coordination of ORM relies on a solid foundation of governance and assigning roles and responsibilities to each party.

B. Firms should monitor and report on the coordination and maintenance of business continuity management

(BCM) and IT systems resilience.

C. A strong ORM framework is necessary in order to achieve operational resilience.

D. Firms are required to reduce their reliance on third parties.

Answer: D

Step by step

1. Recall the Fed building blocks: governance as the basis; ORM as the central component; third-party risk management and scenario analysis as supports; BCM and IT systems resilience as pillars; monitoring and reporting across all.
2. A is the governance/roles-and-responsibilities foundation; B is monitoring and reporting of BCM and IT resilience; C is the central role of ORM.
3. Third-party risk management means controlling and managing dependencies, not eliminating them, so D is not an expectation.

Why D is correct: The guidance asks firms to manage third-party risk (to protect supply-chain resilience), not to cut third-party reliance.

Why the others are wrong

- **A:** This is an actual expectation: governance and clear roles are the required starting point.
- **B:** This is an actual expectation: coordination and maintenance of BCM and IT resilience must be monitored and reported.
- **C:** This is an actual expectation: resilience is not possible without proper ORM, which is the central component.

Q20

BT answers P2.T7.20.16.1

REBUILT — Original relied on an outside (Oliver Wyman) BC/DR comparison not in the GARP/Schweser reading; rebuilt around the GARP contrast between process-based continuity and the service-based resilience view.

Which of the following statements best distinguishes the operational resilience approach of UK regulators from traditional business continuity planning?

A. Resilience concentrates solely on preventing disruptions, since severe disruptions are assumed to be avoidable with sufficient controls.

B. Resilience is built around the continuity of each individual internal business process, with impact tolerances set process by process.

C. Resilience is built around important business services, with impact tolerances tied to critical services rather than to processes per se.

D. Resilience learns only from the firm's own internal incidents, since events at other organizations are not comparable.

Answer: C

Step by step

1. GARP: UK regulators moved from a process-based view of continuity to a service-based view centred on important business services.
2. Impact tolerance is similar to a recovery time objective, but the nuance is that it relates to critical business services rather than to processes per se.
3. Firms and regulators also moved from a purely preventive approach to a balanced prevent, detect and correct approach, and lessons learned include events made public at other organizations.
4. Only C matches these points.

Why C is correct: Resilience differs from classic continuity planning by focusing on important services and service-level impact tolerances rather than on individual processes.

Why the others are wrong

- **A:** Wrong: the industry accepted that severe disruptions will occur and moved from pure prevention to preventing, detecting and correcting incidents.
- **B:** Wrong: this is the old process-based view that the regulators moved beyond.
- **D:** Wrong: lessons learned come from internal accidents and from events made public in other organizations.

Q21 BT answers P2.T7.23.1.3 · BT notes P2.T7.23.1.3

According to the 2020 U.S. Federal Reserve guidance, operational resilience is a key outcome of an effective operational risk management framework (ORMF), and governance, including clear roles and responsibilities, is the foundational building block. In addition, which of the following statements is TRUE about operational resilience?

A. Incident management involves establishing procedures for preventing all possible disruptions before they occur.

B. Among the necessary building blocks of operational resilience, the firm should include third-party risk management and scenario analysis

C. Because it is a BCBS concept, operational resilience is only relevant for financial institutions and not for other types of organizations.

D. The best operational resilience framework is a significant one-time effort that does not require ongoing attention or maintenance once it has been established

Answer: B

Key correction: Source explanations printed that A, C and D are each 'TRUE'; this is a typo for FALSE. The key B is correct.

Step by step

1. Building blocks (Fed/Chapelle): governance as the basis, ORM at the centre, supported by third-party risk management and scenario analysis, with BCM and IT systems resilience and ongoing monitoring.
2. B lists two of those building blocks, so it is true.
3. A confuses incident management (response and recovery) with prevention; C and D contradict the sources and the need for continuous monitoring.

Why B is correct: Third-party risk management (supply-chain resilience) and scenario analysis (preparedness for tail events) are explicit supporting pillars of ORM in the resilience framework.

Why the others are wrong

- **A:** Incident management (BCBS Principle 6) is about response and recovery plans; firms accept that disruptions will occur and cannot prevent them all.
- **C:** Unsupported: the stem attributes the concept to the BCBS only, but the UK, U.S. and BCBS regulators have all issued guidance, and nothing in the reading limits resilience to financial institutions.
- **D:** Wrong: coordination and maintenance of the activities must be monitored and reported, and firms must keep learning and improving; it is not a one-time effort.

Q22 GARP PE1 Q3

A UK-based retail brokerage firm has recently experienced rapid growth through a series of acquisitions and plans to improve its operational resilience in order to comply with new requirements issued by national regulators. The CRO asks an operational risk manager to assess best practices in this area and to suggest potential actions that the firm should take to meet this objective. Which of the following actions would be most appropriate for the manager to recommend in order to comply with the regulatory guidelines concerning operational resilience for important business services?

A. Increase the existing performance incentives for the firm's brokers and sales representatives to further enhance their productivity.

B. Reduce costs by consolidating the human resources and payroll functions of the acquired firms into a single firm-wide process.

C. Reserve additional economic capital for operational risk to provide an incremental capital cushion against potentially extreme operational losses.

D. Increase the bandwidth capacity for the firm's equity trading platform and procure remote backup capabilities so the platform can continue to function during an outage.

Answer: D

Step by step

1. UK guidance centres on identifying important business services and ensuring their continuity within tolerance levels.
2. For a brokerage, the equity trading platform is an important business service; added capacity and remote backup directly support continuity during an outage.
3. The other options address productivity, cost or loss absorption, not service continuity.

Why D is correct: Resilience means keeping important services running through disruptions, which extra capacity and remote backup for the trading platform directly support.

Why the others are wrong

- **A:** Incentives raise productivity but do nothing for continuity of important services.
- **B:** Consolidating internal HR/payroll is a cost action, not an important business service for clients or markets, and can create a single point of failure.
- **C:** Capital absorbs losses after the fact but does not keep services running within impact tolerance; resilience is about continuity, not capital.

Q23 Schweser Module Quiz 42.2 Q2

To date, which of the following entities is least likely to be considered a key regulator to have issued official guidance for operational resilience?

A. Bank of England.
B. U.S. Federal Reserve.
C. European Central Bank.
D. Basel Committee on Banking Supervision.

Answer: C

Step by step

1. As of May 2022, the key regulators with official guidance are the United Kingdom (BoE with FCA and PRA), the United States (Fed) and the BCBS.
2. The ECB only published a draft regulation (the Digital Operational Resilience Act, DORA) in 2020, so it is the least likely to be counted as a key issuer of official guidance.

Why C is correct: The UK, US and BCBS are the three main regulators with official guidance; the ECB's DORA was only a proposal.

Why the others are wrong

- **A:** The Bank of England, with the FCA and PRA, issued the first resilience guidance (2018 discussion paper).
- **B:** The Federal Reserve issued Sound Practices to Strengthen Operational Resilience in 2020.
- **D:** The BCBS issued its seven Principles for Operational Resilience in March 2021.

Q24 Schweser Module Quiz 42.2 Q3

Which of the following pairs of resilience principles directly addresses the issue of providing critical services with minimal or no disruption?

A. Third-party dependency management; incident management.
B. Mapping interconnections and interdependencies; incident management.
C. Business continuity planning and testing; third-party dependency management.
D. Business continuity planning and testing; mapping interconnections and interdependencies.

Answer: B

Step by step

1. BCBS principles: 1 governance, 2 ORM, 3 BCP and testing, 4 mapping interconnections and interdependencies, 5 third-party dependency management, 6 incident management, 7 ICT.
2. Principles 4 and 6 mirror the UK approach: know the dependencies needed to deliver critical services and have response and recovery plans, so critical services continue without unacceptable disruption.
3. Principles 3, 5 and 7 mirror the U.S. approach (BCP, third parties, ICT).

Why B is correct: Principles 4 and 6 are the ones tied to continuous delivery of critical operations, in the same spirit as the UK regulators' approach.

Why the others are wrong

- **A:** Principle 5 (third parties) is a U.S.-style principle; only incident management in this pair is tied to critical-service delivery.
- **C:** Both Principle 3 and Principle 5 are the U.S.-style elements, not the pair tied to critical services.
- **D:** Principle 4 fits, but Principle 3 (BCP and testing) is a U.S.-style element; the pair needs incident management (Principle 6).

Q25 BionicTurtle P2.T7.24.4.3

Zlich Finance, a fast-growing UK-based FinTech consumer lending firm, offers various services such as digital wallets, savings accounts, investment portfolios, debit cards, lending and borrowing, and FX remittances that allow customers to make payments in different currencies and meet their day-to-day consumer needs. A senior operations manager at the firm is developing a plan to comply with new UK regulatory requirements for operational resilience. Which of the following steps should the manager recommend for the firm to best comply with regulatory expectations in this area?

A. Develop an impact tolerance for each of the important business services provided by the firm.
B. Self-assessment by the board every 5 years to ensure longer-term coverage.
C. Use 1-year 99.9% VaR to calculate the operational risk for each of the firm's business divisions and use this result to reserve additional capital for each division.
D. Immediately establish lines of communication after a breach incident has occurred.

Answer: A

Step by step

1. UK guidance: identify important business services and quantify the amount of disruption that can be tolerated (impact tolerance) for each.
2. A does this. B is infrequent and inconsistent with continuous learning and monitoring; C is a capital approach by division, not by service; D leaves communication until after the incident.

Why A is correct: UK regulators ask firms to identify important business services and set impact tolerances for each.

Why the others are wrong

- **B:** A five-year board review is too infrequent for a dynamic risk and is not what the guidance asks; resilience needs ongoing monitoring and learning.
- **C:** Divisional VaR-based capital is a loss-absorption tool and ignores the service-based view; resilience is about continuity within tolerance.
- **D:** Communication must be planned in advance as part of management of disruption; setting it up only after a breach is too late.

Q26 Mock 1 Q52

The risk management department of a large bank needs to determine whether it has sufficient resilience to recover from a major operational event that causes loss of profits and reputation. In that regard, their goal is to enhance the bank's resilience by identifying, assessing, mitigating, and monitoring risks. When developing their operational resilience framework, the bank should consider the:

A. continuity of important business services in nonextreme situations only.
B. continuity of important business services in extreme and nonextreme situations.
C. continuity of each business process and recovery on a timely basis in nonextreme situations only.
D. continuity of each business process and recovery on a timely basis in extreme and nonextreme situations.

Answer: B

Step by step

1. Unit of analysis: UK regulators moved from a process-based to a service-based view, so the framework is built on important business services, not each process.
2. Breadth: firms and regulators accept that extreme disruptions will occur, so the framework must cover extreme as well as nonextreme situations.
3. Only B has both the right unit (important services) and the right breadth (extreme and nonextreme).

Why B is correct: Resilience is organised around important business services and must hold up in severe as well as ordinary disruptions.

Why the others are wrong

- **A:** Right unit, wrong breadth: ignores extreme events, which regulators expect to occur (infrequently).
- **C:** Wrong on both counts: process-based view and nonextreme scenarios only.
- **D:** Right breadth, wrong unit: the old process-based view rather than the service-based view.

Q27 BT answers P2.T7.20.16.2

REBUILT — Original relied on Oliver Wyman's single-incident-response-regime point (Exhibit 3 missing, not in GARP/Schweser); rebuilt as an EXCEPT question on the resilience elements and regulatory stance in the reading.

A bank's board is reviewing its operational resilience framework against the 2018 UK and 2020 U.S. regulatory approaches. Which of the following statements is INCONSISTENT with the elements of operational resilience and with regulatory expectations?

A. The bank should identify the key business services whose disruption would cause intolerable harm to consumers or market integrity, and ensure their continuity within tolerance levels.

B. The bank should set impact tolerance levels by quantifying the amount of disruption that could be tolerated, which resembles a recovery time objective but is tied to critical services rather than individual processes.

C. The bank should aim to prevent all severe operational disruptions, because regulators expect that with strong enough preventive controls such disruptions will not occur, so planning for response and recovery is secondary.

D. The bank should learn from its own incidents and from events made public at other firms and continuously improve its ability to withstand unanticipated extreme shocks.

Answer: C

Key correction: Source key (D) referred to Oliver Wyman's single-incident-response-regime question; that item was rebuilt, so the key no longer applies.

Step by step

1. Recall the elements of resilience: business continuity, key (critical) services, impact tolerance levels, management of disruption (response, stakeholder trust, communication), and lessons learned/feedback.
2. A restates the key-services element; B restates impact tolerance (similar to a recovery time objective but service-based); D restates the lessons-learned element.
3. Regulators and banks now accept that severe disruptions will occasionally occur, so the focus moved from purely preventing incidents to preventing, detecting and correcting them to minimize net impact.
4. C says prevention is sufficient and response/recovery secondary, which contradicts this, so C is the inconsistent statement.

Why C is correct: Operational resilience assumes extreme disruptions are inevitable at times, so frameworks balance prevention with detection, response and recovery rather than relying on prevention alone.

Why the others are wrong

- **A:** Consistent: this is the key-services element; choosing it means misreading the EXCEPT-type wording.
- **B:** Consistent: impact tolerance is similar to a recovery time objective but applies to critical business services, not processes.
- **D:** Consistent: lessons learned (feedback) from internal and external events is what differentiates resilience from traditional business continuity.

Reading 43. Risk Governance

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 2.

LO	Learning objective	Qs
43.a	explain the Basel regulatory expectations for the governance of an operational risk management framework.	6
43.b	describe and compare the roles of different committees and the board of directors in operational risk governance.	4
43.c	describe the "three lines of defense" model for operational risk governance and compare roles and responsibilities for each line of defense.	13
43.d	explain best practices and regulatory expectations for the development of a risk appetite for operational risk and for a strong risk culture.	6

LO 43.a

Explain the Basel regulatory expectations for the governance of an operational risk management framework.

Q28

AP Mock 1 Q38

REBUILT — Original relied on a Principle 3 item (following evolution of best practices) not in the Schweser/GARP text and had a vague correct option; rewrote options around the board duties listed in the reading.

Tom Mathews has been appointed a director of a multinational bank. With respect to Principle 3 (board review and approval of the operational risk management framework) of the Basel Committee's Principles for the Sound Management of Operational Risk, which of the following is the board LEAST likely to be responsible for?

A. Validating (approving) the operational risk management framework (ORMF).

B. Ensuring the ORMF is revised periodically.

C. Obtaining assurance that senior management implements the ORMF policies, processes, and systems effectively at all levels of the bank.

D. Designing and maintaining the day-to-day operational risk procedures and control systems used by the business lines.

Answer: D

Step by step

1. Principle 3 assigns the board an oversight role: validate/approve the ORMF, ensure periodic revision, and get assurance that senior management implements it.
2. Options A, B and C are exactly these three duties.
3. Option D describes designing and running procedures and controls, which is implementation work done by senior management, the first line and the CORF (Principle 5), not by the board.
4. Therefore D is the least likely board responsibility.

Why D is correct: The board approves, reviews and oversees; it does not build or run the procedures and controls, which is senior management's implementation role.

Why the others are wrong

- **A:** This is a core board duty under Principle 3; selecting it confuses approval with implementation.
- **B:** Periodic revision of the ORMF is a stated board responsibility; picking it overlooks the 'ensure periodic revision' duty.
- **C:** Obtaining assurance that senior management implements the framework is a board oversight duty; it is not delegated down.

Q29

AP Mock 3 Q35

REBUILT — Original distractors were giveaways (paper trail, oversee everything); replaced with plausible misconceptions and kept the same tested idea.

Mr Rihan is presenting to the board on Basel regulatory expectations for the governance of an operational risk management framework. Which of the following best describes the purpose of risk governance within the ORM framework of a bank?

A. To replace Pillar 1 regulatory capital with qualitative controls over operational risk.

B. To develop robust governance policies and processes and manage material risks in line with the firm's risk appetite.

C. To shift responsibility for operational risk decisions from the bank to its supervisors.

D. To restrict operational risk management to the second line of defense so that the business lines carry no risk

responsibility.

Answer: B

Step by step

1. The board is responsible for setting the risk appetite and for making sure the firm operates within its limits; governance provides the policies, processes and committee structure to do so.
2. ORM and Pillar 1 capital are complements: ORM cannot exist purely within capital calculations, but governance does not replace capital (rules out A).
3. Responsibility stays with the bank's board and management; supervisors review under Pillar 2 but do not take over decisions (rules out C).
4. Risk is owned by the first line (risk owners) and overseen by the second, so D is wrong. B is correct.

Why B is correct: Governance exists to put in place policies and processes that manage material operational risks within the board-approved risk appetite.

Why the others are wrong

- **A:** Confuses governance with Pillar 1: ORM must be balanced with, not substituted for, regulatory capital.
- **C:** Supervisors review the framework (Pillar 2) but accountability remains with the bank's board and management.
- **D:** Reverses the three-lines model: the first line (risk owners) owns risk, the second line only oversees and challenges.

Q30 AnalystPrep Q.5055

In an FRM presentation on Basel regulatory expectations for the governance of an operational risk management framework, a candidate wants to know how one can examine whether the ORM framework is being implemented at a firm. Which of the following questions should NOT be used for this purpose?

A. Is there evidence that all material events are captured in event reports?
B. Does the value of each risk indicator come from an independent source?
C. Are the presented data sufficient for decision-making?
D. Does the information pertain to the senior management?

Answer: D

Step by step

1. Regulators assess an ORMF through operational risk reports: incident reports (all significant incidents captured?), risk indicators (objective, unbiased?), and overall adequacy of information.
2. A tests completeness of incident reporting; B tests objectivity of risk indicators; C asks whether there is enough information to make proper decisions: all valid.
3. The proper question is whether the information is useful at the level of management it is intended for, not whether it pertains to senior management only.
4. So D is the question that should not be used.

Why D is correct: Reports must be useful at the relevant management level; restricting the test to senior management is not a valid implementation check.

Why the others are wrong

- **A:** A valid check: incident reports should account for all significant incidents.
- **B:** A valid check: risk indicators should be computed objectively and without bias.
- **C:** A valid check: regulators ask whether there is enough information to make proper decisions.

Q31 AnalystPrep Q.5057

Principle 3 of the Basel Principles for the Sound Management of Operational Risk outlines the role of the board of directors in operational risk governance. Which of the following board roles is in line with Principle 3?

A. Identify the types and levels of operational risk the bank is willing to assume, and approve risk appetite and risk tolerance statements.
B. Regularly review the appropriateness of the bank's risk appetite and tolerance statements.
C. Ensure the ORM framework is subject to independent review by sufficiently skilled personnel.
D. Ensure that all risks are considered when approving risk appetite and tolerance statements that detail risk limits and thresholds.

Answer: C

Step by step

1. Principle 3 is 'board review and approval of the ORMF': the board validates the ORMF, ensures periodic revision, obtains assurance on implementation, and ensures the framework is subject to independent review by skilled staff.
2. Principle 4 is a separate principle: risk appetite and tolerance statements are approved and periodically reviewed by the board.
3. A, B and D all concern appetite and tolerance statements, so they belong to Principle 4.
4. Only C concerns review of the ORMF itself, so C is Principle 3.

Why C is correct: Principle 3 concerns the board's review and validation of the ORMF itself, whereas appetite and tolerance statements fall under Principle 4.

Why the others are wrong

- **A:** Setting and approving appetite and tolerance is Principle 4, not Principle 3.
- **B:** Reviewing the appropriateness of appetite and tolerance statements is also Principle 4.
- **D:** Considering all risks when approving limits and thresholds in the statements is part of the appetite and tolerance principle (4).

Q32 BT answers P2.T7.800.2 · BT notes P2.T7.800.2

In order for a bank to achieve sound management of its operational risk, necessary features include a strong risk management culture, an integrated operational risk management framework (ORMF), an engaged board of directors, senior managers who assume responsibility for the framework, and an independent corporate operational risk management function (CORF; the second line of defense). In regard to the fundamental principles of operational risk management, each of the following is true EXCEPT which is false?

A. To maintain independence, the ORMF prohibits any reporting relationship (even dotted-line) from the CORF to either the Chief Risk Officer (CRO) or the board's risk committee.

B. The ORMF should provide for a common taxonomy of operational risk terms to ensure consistency of risk identification (e.g., operational loss event types), exposure rating, and risk management objectives.

C. Compensation plans should be aligned with the bank's risk appetite and risk tolerance; such plans may include incentive compensation explicitly linked to risk-adjusted measures, deferral mechanisms, or claw-backs.

D. The board of directors approves a risk appetite statement (a forward-looking, high-level view of risk acceptance that incorporates return expectations) and a risk tolerance statement (a more specific determination of the level of variation the bank is willing to accept around business objectives) for operational risk.

Answer: A

Step by step

1. Independence of the CORF means a reporting structure independent of the risk-generating business units, not isolation from risk governance.
2. The CORF normally reports to a chief-level function such as the CRO and has access to the board risk committee, so a ban on any such line is false.
3. B (common taxonomy), C (compensation aligned with appetite, with deferral/claw-backs) and D (board approves appetite and tolerance statements) are consistent with the reading.
4. A is the false statement.

Why A is correct: Independence is from the business units that generate risk, so the CORF can and does report to the CRO and the board risk committee.

Why the others are wrong

- **B:** True: a common taxonomy supports consistent identification and rating of operational risk; not the exception.
- **C:** True: pay is expected to align with appetite and tolerance, using tools such as deferral and claw-backs.
- **D:** True: the board approves appetite (high level, forward looking) and tolerance (more specific variation allowed) statements.

Q33 GARP PE1 Q4 · Mock 3 Q76

The board of directors of a midsize bank has recommended that the bank improve its processes for managing operational risk. The CEO asks an enterprise risk manager to review the bank's tools and processes and to suggest improvements consistent with Basel II and Basel III guidelines for operational risk governance. The bank also plans to adopt the new Basel III standardized approach (SA) to determine its regulatory capital for operational risk. Which of the following actions should the manager recommend?

A. Use third-party outsourcing agreements to replace most internal controls performed by senior managers and business line managers.
B. Develop an internal approach to model the distribution of operational risk losses and use it to determine the bank's regulatory capital.
C. Require an independent review of the bank's operational risk management framework.
D. Designate the risk management function as the primary owner of risk exposures within each business line.

Answer: C
Step by step
1. Basel governance expectations: the board validates the ORMF and ensures it is periodically revised, supported by independent review.
2. A: outsourcing cannot replace internal controls; accountability stays with the bank.
3. B: under the SA (effective January 2023) capital is ORC = BIC x ILM, a formula-based measure; internal loss-distribution modelling is not used for Pillar 1.
4. D: risk owners are the first line (business units); the risk function is the second line and oversees and challenges.
5. C is the consistent recommendation.

Why C is correct: Independent review of the ORMF is a core governance expectation, while the other actions conflict with the SA or the three-lines model.
Why the others are wrong
- **A:** Outsourcing may complement controls but cannot replace them or transfer accountability.
- **B:** Internal modelling approaches were replaced by the single formula-based SA, ORC = BIC x ILM.
- **D:** Confuses lines of defense: risk is owned by the business (first line), not the second-line risk function.

LO 43.b

Describe and compare the roles of different committees and the board of directors in operational risk governance.

Q34 BT answers P2.T7.800.3 · BT notes P2.T7.800.3

REBUILT — Removed the incorrect 'eleven principles' count (the 2021 Basel revision has twelve) and replaced the awkward double-EXCEPT wording with a direct question.

Under the Basel principles for the sound management of operational risk, the board of directors and senior management have distinct responsibilities for the operational risk management framework (ORMF). Which of the following is a responsibility of the board of directors rather than of senior management?

A. Establishing and maintaining robust challenge mechanisms and effective issue-resolution processes.
B. Translating the ORMF into specific policies and procedures that can be implemented and verified within each business unit.
C. Recruiting experienced technical staff and ensuring a sufficient level of operational risk training to implement the ORMF.
D. Validating the ORMF and ensuring it is reviewed periodically, so that the bank's operational risks are properly identified and managed.

Answer: D
Step by step
1. The board is ultimately responsible for the firm, including risk management: it validates (approves) the ORMF and ensures periodic revision of it.
2. The board also obtains assurance that senior management implements the policies, processes and systems of the ORMF effectively at all levels.
3. Implementation tasks sit with senior management: building challenge and issue-resolution mechanisms, turning the ORMF into business-unit policies and procedures, and staffing/training.
4. Only D describes approval and periodic review of the framework, so D is the board's responsibility.

Why D is correct: Approving/validating the ORMF and ensuring its periodic revision is a core board duty; the other three are implementation duties delegated to senior management.
Why the others are wrong
- **A:** Senior management responsibility: it builds the challenge mechanisms and issue-resolution processes that support the ORMF.
- **B:** Senior management responsibility: translating the framework into business-unit policies and procedures is

implementation, not board work.
- **C:** Senior management responsibility: hiring and training staff to implement the ORMF is an executive task; the board oversees management's implementation and ensures adequate resources.

Q35 Schweser Module Quiz 43.1 Q3

Within a bank, who is ultimately responsible for operational risk management and resilience?

A. Employees.
B. Chief risk officer.
C. Board of directors.
D. Senior management team.

Answer: C

Step by step

1. The board of directors is ultimately responsible for the general administration of the firm, including risk management and operational resilience.
2. It sets risk appetite/tolerance and oversees senior management's implementation; day-to-day tasks are delegated down.
3. Therefore the board, not management or staff, holds ultimate responsibility: C.

Why C is correct: Responsibility can be delegated but accountability cannot; the board owns operational risk and resilience and oversees management's execution.

Why the others are wrong

- **A:** Employees execute controls (first line) but do not hold ultimate responsibility.
- **B:** The CRO is part of senior management and runs the risk function; the role does not assume ultimate responsibility.
- **D:** Senior management implements the ORMF and reports to the board, but it is accountable to the board rather than ultimately responsible.

Q36 AP Mock 1 Q48

REBUILT — Original asked about an audit committee, whose tasks are not covered in the operational risk governance LO; rebuilt to test the board risk committee and the committee hierarchy.

A large bank has a pyramid of operational risk committees. Which of the following tasks is most likely performed by the board (enterprise-level) risk committee?

A. Making recommendations to the full board on risk exposures and risk-based decisions, and reviewing larger operational risk incidents escalated from lower committees.
B. Overseeing operational risk only within a single business line or country and passing information upward.
C. Setting the bank's risk appetite and approving the ORMF without any reference to the full board.
D. Preparing the day-to-day first-line risk and control self-assessments for each business unit.

Answer: A

Step by step

1. The lowest-level committees oversee a specific business type, function or geography and feed information upward (this is B).
2. Middle-level group/organization committees consolidate operational risk and report to the management committee and board risk committee.
3. The top-level board risk committee oversees all operational risk, makes recommendations to the full board on risk-based decisions and exposures, and reviews larger incidents.
4. The board itself, not a committee acting alone, sets risk appetite and validates the ORMF; first-line self-assessments are done by business units. So A.

Why A is correct: The board risk committee is a recommending and oversight body at the top of the pyramid: it advises the full board and reviews major, escalated incidents.

Why the others are wrong

- **B:** Describes a lowest-level business-line or country committee, not the board risk committee.
- **C:** The full board is responsible for setting risk appetite and validating the ORMF; the committee only makes recommendations to it.
- **D:** Risk and control self-assessments are first-line activities performed by business units, not a board committee task.

Q37 AnalystPrep Q.5569

REBUILT — Original was a case-specific statement about a named bank's CRO not covered by the Schweser/GARP text; rebuilt to test who holds ultimate responsibility for operational risk governance.

After a major operational loss, a review finds that a bank's chief risk officer (CRO) and senior management had given the full board only limited information on the bank's operational risks. Within the bank's operational risk governance, which party is ultimately responsible for operational risk management and resilience?

A. The chief risk officer.
B. The board of directors.
C. The senior management team.
D. The employees in the business lines.

Answer: B

Step by step

1. The board of directors is ultimately responsible for operational risk management, delegating tasks to senior management and employees.
2. The CRO is part of senior management, but that role does not assume ultimate responsibility.
3. Because the board is accountable, it depends on complete and adequate information from the CRO and management to discharge its oversight; the failure to supply it does not shift ultimate responsibility away from the board.
4. Answer: the board of directors.

Why B is correct: Ultimate responsibility rests with the board; management, the CRO and employees carry delegated responsibilities.

Why the others are wrong

- **A:** The CRO is part of senior management and has delegated, not ultimate, responsibility; blaming the person who failed to report is the tempting error.
- **C:** Senior management executes tasks delegated by the board, so it is accountable for them but not ultimately responsible.
- **D:** Business-line employees are first-line risk owners who manage day-to-day risk, but responsibility is delegated to them rather than ultimately held by them.

LO 43.c

Describe the "three lines of defense" model for operational risk governance and compare roles and responsibilities for each line of defense.

Q38 AP Mock 2 Q39

REBUILT — Options A and D were near-duplicates (both 'satisfactory') and B was a giveaway; rewrote options as distinct plausible errors anchored to CORF independence.

Prime Bank of India is a large lender to small and medium enterprises. It has established a risk management unit (RMU) to manage risk categories and ensure exposures are authorised and consistent with the bank's risk appetite. Under the reporting framework, the chief risk officer reports to the head of credit, who in turn reports to the board. Which of the following statements is most accurate?

A. The structure is appropriate because credit is the bank's dominant risk, so the RMU should be accountable to the head of credit.
B. Independence is satisfied because the RMU is a separate unit, whatever its reporting line.
C. The RMU lacks independence because it reports through a risk-generating business line; it should have a reporting structure independent of the risk-generating units.
D. Independence would be improved by moving the RMU inside the credit unit so it sits closer to the risk it monitors.

Answer: C

Step by step

1. The RMU plays the role of the corporate operational risk function (second line of defense).
2. GARP: in larger banks the CORF should have a reporting structure independent of the risk-generating business units.
3. Here the CRO reports to the head of credit, a risk-taking (first line) function, so independence and objective

challenge are compromised.
4. Therefore C is correct.

Why C is correct: The second line must be independent of the risk-generating units to challenge them objectively; reporting through the head of credit breaks that independence.

Why the others are wrong

- **A:** Confuses proximity to the risk with accountability; a CRO subordinate to a risk taker cannot challenge it.
- **B:** Treats a separate unit label as sufficient; independence depends on the reporting line, not the organisation box.
- **D:** Embedding risk management in a business unit removes independence entirely and creates a self-review threat.

Q39

MockExam 1 Q17

REBUILT — The 'fourth line' wording is not in the GARP chapter; reframed to the GARP content that external audit provides independent assurance outside the three internal lines.

A bank uses the three lines of defense model. Which of the following parties provides independent assurance to stakeholders such as regulators and investors from outside the bank's three internal lines (often informally called a 'fourth line')?

A. Business line staff
B. The bank's internal audit function
C. The credit risk management department
D. The independent external auditors

Answer: D

Step by step

1. First line: business units/risk owners (A).
2. Second line: corporate operational risk function and related risk oversight functions (C).
3. Third line: internal audit (B).
4. External auditors sit outside the firm, remain independent and give an opinion on the financial statements and financial-reporting controls, so D.

Why D is correct: External auditors are not part of the firm's three internal lines; they provide independent assurance from outside.

Why the others are wrong

- **A:** First line: risk owners who generate and manage risk.
- **B:** Internal audit is the third line, inside the bank.
- **C:** A risk management department is second-line oversight, inside the bank.

Q40

AP Mock 1 Q63

REBUILT — Stem was vague (firewalls could suggest IT security/second line); rewrote to the independent-assurance activity that identifies the third line.

Angela Myer is reviewing the bank's cyber-security controls, including firewalls, and independently assessing whether departments, including the risk function itself, comply with policies and procedures. She does not manage the risks and reports to a non-executive director. Ms Myer is most likely part of the bank's:

A. Internal audit team.
B. Senior management.
C. Risk management department.
D. Board of directors.

Answer: A

Step by step

1. Independent assessment of the control system and policy compliance, including that of the risk function, is the third line.
2. The third line is internal audit and reports in solid line to a non-executive director (e.g., audit committee chair).
3. So Ms Myer is in internal audit: A.

Why A is correct: Internal audit independently assesses controls and compliance across all departments, including risk management.

Why the others are wrong

- **B:** Management owns and runs controls (first and second line) and is not independent of them.

- **C:** The risk function is itself reviewed by internal audit and does not report to a non-executive director.
- **D:** The board provides oversight and accountability, not hands-on control testing.

Q41 AP Mock 3 Q36

Which of the following is most likely a role and responsibility of the second line of defense for the overall risk management of a firm under risk governance?

A. Making decisions for managing risks.
B. Establishing risk management methods and measurement methods.
C. Overseeing the risk management activities of the third line of defense.
D. Reporting independently to the board of directors.

Answer: B

Step by step

1. Second line develops and maintains ORM and measurement policies, standards and guidelines.
2. Risk decisions belong to the first line (risk owners).
3. Internal audit (third line) is independent of risk management, and is not overseen by it.
4. Hence B.

Why B is correct: The second line owns the methodology; the first line owns the risks.

Why the others are wrong

- **A:** Risk-taking decisions are first-line; the second line challenges rather than decides.
- **C:** The second line does not oversee internal audit; audit independently reviews the second line.
- **D:** Independent reporting to a non-executive director is the internal audit (third line) feature.

Q42 AnalystPrep Q.5058

REBUILT — Option B was truncated and C vague; D was a verbatim second-line item too close to A. Rewrote options as clear second/third-line roles.

A bank's operational risk specialist is explaining the three lines of defense. Which of the following is a role of the first line of defense?

A. Developing and maintaining operational risk measurement policies, standards and guidelines
B. Independently assessing the control system and compliance with policies across all departments, including the risk function
C. Monitoring and reporting the operational risk profile of the business unit and escalating loss events and control deficiencies to the second line
D. Challenging the consistency of the business units' use of operational risk tools and providing evidence of that challenge

Answer: C

Step by step

1. First line (risk owners) identify and assess risks, establish and assess controls, and monitor and report their own unit's operational risk profile.
2. They also escalate losses, control deficiencies and resource gaps to the CORF.
3. A and D are second-line duties; B is third-line (internal audit).
4. So C.

Why C is correct: Risk owners manage and report the risks they generate and escalate residual issues to the second line.

Why the others are wrong

- **A:** Policy and standard setting is a second-line CORF responsibility.
- **B:** Independent assurance over all departments is third line (internal audit).
- **D:** Effective challenge of business unit practice is the second line's role.

Q43 BionicTurtle P2.T7.23.2.2 · BT notes T7-23.2.2 · BT Mock Q18

The three lines of defense model (3LoD) has existed in some variation for over 30 years, although it achieved more visibility in 2013 when the Institute of Internal Auditors (IIA) published its initial position paper on the 3LoD, which it updated in 2021 (The IIA's Three Lines Model). In regard to the 3LoD model, each of the following is true EXCEPT which is false?

A. The first line defines the organization's risk appetite but should not attempt to directly influence its risk culture
B. The first line is responsible for implementing and maintaining effective controls to manage material operational risks
C. The second line is part of management (and owns the ORM methodology) but provides support to the first line in order to help the firm create and protect value
D. The third line has the clearest boundaries due to its independence

Answer: A

Step by step

1. B true: risk owners identify risks and establish and assess controls.
2. C true: the 2021 IIA model places the second line within management, providing expertise, support and challenge; the second line owns the methodology.
3. D true: internal audit is independent of risk management, the line with the clearest boundaries.
4. A false: risk appetite is a board/senior-management matter (not defined by the first line), and risk culture is shaped across the organization, including the first line.

Why A is correct: A misassigns risk appetite to the first line and denies the first line any influence on risk culture; B, C and D match the GARP description.

Why the others are wrong

- **B:** True statement of first-line responsibility.
- **C:** True; the IIA update stresses the second line's support role within management.
- **D:** True; independence gives internal audit the clearest boundaries.

Q44 BT answers P2.T7.800.1 · BT notes P2.T7.800.1

REBUILT — Merged two identical copies; replaced the vague 'verification and validation' and internal-vs-external audit wording (inconsistent with the third line being internal audit) with a cleaner statement set so exactly one option is true.

Risk manager Beth has drafted a re-design of her bank's operational risk management framework (ORMF) that assigns roles to the three lines of defense: (i) business line management, (ii) an independent corporate operational risk management function (CORF), and (iii) internal audit. Her draft includes the following responsibilities:

I. Business line management identifies and manages the risks inherent in the products, activities, processes, and systems for which it is accountable.

II. The CORF challenges the business lines' inputs to, and outputs from, the bank's operational risk management, measurement, and reporting systems.

III. Internal audit provides independent assurance and, to do so, designs and implements the ORMF.

Beth's draft looks reasonable, but it contains one mistake. Which of the following is a TRUE statement that explains her mistake?

A. Business line management is not a line of defense.
B. The CORF should not challenge the business lines' outputs because doing so creates a self-review threat.
C. Internal audit cannot design or implement the ORMF without compromising the independence of its assurance role.
D. Internal audit should report to the head of the CORF in order to remain independent.

Answer: C

Step by step

1. Test each responsibility against the roles of the three lines (Schweser 43.c / GARP Ch. 2).
2. I is correct: the first line consists of risk owners who identify, assess, and control the risks they generate.
3. II is correct: the second line (CORF) provides oversight and effective challenge of the first line's risk tools, measurement, and reporting.
4. III is wrong: the third line (internal audit) is completely separate from risk management and independently assesses the control system and the other lines; designing and implementing the ORMF is the CORF's job, and auditing one's own design is a self-review conflict.
5. Therefore the true statement is C.

Why C is correct: Internal audit's value is independent assurance over Lines 1 and 2; if it designs and implements the ORMF it would be reviewing its own work. Framework design belongs to the CORF (Line 2), with the first line owning the risks.

Why the others are wrong

- **A:** False: business line management (risk owners) is the first line of defense. Confuses 'business' with 'not a control function'.

- **B:** False: challenging the first line's inputs and outputs is exactly the second line's job. The self-review threat arises if the CORF takes part in first-line activities or coaches 'correct answers', not from challenging.
- **D:** False: internal audit must be independent of risk management and should report in solid line to a non-executive director (e.g., audit committee chair). Reporting to the CORF would tie it to a function it must assess.

Q45

BT answers P2.T7.603.3

REBUILT — Original EXCEPT item mixed taxonomy classification (outside 43.c) with governance statements from a non-curriculum text; rebuilt as an EXCEPT item focused on the three-lines roles per Schweser/GARP.

Regarding the three lines of defense model for operational risk governance, each of the following statements is correct EXCEPT which one?

A. Where a function such as legal or HR performs both first-line and second-line duties, the bank should document and distinguish those responsibilities and demonstrate the independence of the second line.

B. To preserve its independence, the second line should give the first line the correct answers to its risk and control self-assessments before the first line completes them.

C. Internal audit should not rely exclusively on the work of risk management, compliance, or finance, but should examine an appropriate sample of the activities itself.

D. What defines a line of defense is the role and responsibility performed by a function, not the name of its department.

Answer: B

Step by step

1. Check each statement against the three-lines description.
2. A is correct: hybrid functions (e.g., contract drafting vs. oversight of litigation in legal) must document the split and show second-line independence.
3. B is incorrect: the second line owns the methodology but must not pre-empt answers; it should run workshops, collect the first line's own answers, and only then challenge them. Coaching 'correct answers' creates a self-review threat.
4. C is correct: internal audit must test its own sample and may rely on others' work only after assessing its reliability.
5. D is correct: roles and duties, not names, delineate the lines.
6. The EXCEPT answer is B.

Why B is correct: The first line owns its risk assessments and controls; the second line challenges after the fact. Supplying answers in advance would destroy the second line's independence.

Why the others are wrong

- **A:** This is a correct statement (hybrid functions require documented, independent roles); picking it misses that the question asks for the incorrect one.
- **C:** This is correct IIA/GARP guidance on internal audit's independence; it is not the exception.
- **D:** This is correct: lines are defined by roles, which is why hybrid functions arise; it is not the exception.

Q46

GARP PE1 Q39 · Mock 3 Q41

REBUILT — Merged two near-identical items; combined the best distractors and removed the risk-culture/ERM twist and the risk-appetite-delegation option so the item tests only first/second/third line roles.

A community bank uses the three lines of defense approach for operational risk and the CRO decides to extend the approach to the bank's enterprise risk management (ERM) framework. Which of the following would be the most appropriate recommendation for allocating responsibilities and authority?

A. The risk management function should approve every transaction that creates a new risk exposure for the bank.

B. Business line managers, as the first line of defense, should have the authority to take on risk exposures within the bank's risk appetite limits.

C. The third line of defense should continuously monitor the implementation of the ERM framework to ensure its effectiveness.

D. As part of the second line of defense, the executive committee should perform an independent review of the risk management framework.

Answer: B

Step by step

1. Recall the roles: Line 1 = risk owners who generate, manage, and control risk; Line 2 = CORF/risk function that develops the framework and provides oversight and challenge; Line 3 = internal audit providing independent assurance.
2. B matches Line 1: risk is managed where it is generated, so business managers take risk within board-approved appetite and limits.
3. A would make the risk function a transaction-level decision maker, i.e., part of first-line risk taking, and destroy its independence; its role is oversight, challenge, and a veto over decisions breaching limits or regulation.
4. C describes monitoring that is a second-line activity; internal audit gives independent, periodic assurance rather than continuous monitoring.
5. D fails because the executive committee is management, not an independent reviewer; independent review is the third line.

Why B is correct: Under the three lines model, risk is owned by the business; business managers therefore hold the authority to take risk, but only within the limits set by the board's risk appetite.

Why the others are wrong

- **A:** Pulls the second line into first-line decision making; the second line challenges and may veto breaches, it does not approve every transaction.
- **C:** Assigns a second-line monitoring task to the third line; internal audit remains independent and provides periodic assurance.
- **D:** Misclassifies the executive committee as the second line and as independent; independent review belongs to internal audit (third line).

Q47 Schweser Module Quiz 43.2 Q1

Within the context of the three lines of defense model, risk champions (or risk specialists) are most likely to be included in which line(s)?

A. Line 1 only.
B. Lines 1 or 2.
C. Line 2 only.
D. Line 3 only.

Answer: A

Step by step

1. Risk champions/specialists sit within business groups as the key spokesperson for risk issues and gather information on incidents and losses.
2. They are sometimes called 'Line 1.5' because they act as a midway point toward the CORF, but they remain part of the first line.
3. Ownership of operational risk stays with the first line and is not transferred to the champion.
4. Therefore the answer is Line 1 only (A).

Why A is correct: Risk champions are embedded in the business and are a first-line resource; the 'Line 1.5' label describes their liaison role with Line 2, not membership of it.

Why the others are wrong

- **B:** Reads 'Line 1.5' as straddling both lines; they are classified in the first line only.
- **C:** Assumes interaction with the risk function makes them second line; they do not provide independent oversight and cannot be independent of the business.
- **D:** Internal audit is fully separate from risk management; champions are business staff, so they cannot be Line 3.

Q48 AP Mock 1 Q36

Which of the following tasks is performed under the second line of defense, an independent corporate operational risk management function?

A. Keeping track of risk events and losses through the collection and recording of data.
B. Following up on the implementation of control rules, risk management action plans, and audit tracking.
C. Identifying risks and controls in accordance with the group definitions.
D. Reviewing and taking part in the monitoring and reporting of the operational risk profile.

Answer: D

Step by step

1. Second-line duties: develop ORM policies and training, form an independent view of business activity, challenge the first line's tools, measurement, and reporting, and review and contribute to monitoring and reporting of the operational risk profile.
2. A, B, and C are the listed tasks of risk champions/specialists within the first line (collecting and recording events and losses, following up on control rules, action plans and audit tracking, and mapping risks and controls per group definitions).
3. Only D is a second-line task.

Why D is correct: Reviewing and contributing to the monitoring and reporting of the operational risk profile is an oversight task of the CORF, whereas A to C are hands-on first-line (risk champion) tasks.

Why the others are wrong

- **A:** Data collection of events and losses is a first-line/risk champion task.
- **B:** Follow-up on control rules, action plans, and audit tracking is carried out by risk champions in the first line.
- **C:** Mapping risks and controls per group definitions is a first-line/risk champion task; the second line only challenges it.

Q49

AnalystPrep Q.5152

REBUILT — Original option B was vaguely worded (second line 'works with' the first to assess all its activities) and the item tests the three-lines-of-defense LO; option B now clearly describes a self-review threat.

A bank applies the three lines of defense model, as used in operational risk management, to its model risk management (MRM) framework. Which of the following statements is correct?

A. Model developers and model owners, as risk owners, form the first line of defense

B. The second line of defense helps the first line build the models and co-signs their validation so that it can ensure consistency

C. The second line of defense owns the model risk it oversees

D. The first line of defense oversees and challenges the activities of the second line of defense

Answer: A

Step by step

1. First line = the business/risk owners who generate, measure and manage the risk; in MRM these are model developers and owners.
2. Second line = independent function that oversees and challenges the first line, without being involved in it (to avoid a self-review threat).
3. Third line = internal audit, which reviews both lines.
4. Only A matches this allocation.

Why A is correct: The first line owns and manages the risk it generates, so in MRM the developers and owners sit in the first line; the second line independently challenges them.

Why the others are wrong

- **B:** Second-line involvement in building or co-signing models creates a self-review threat and destroys independence.
- **C:** Confuses the lines: the first line, not the second, owns the risk.
- **D:** Reverses the direction of oversight: the second line oversees the first, not vice versa.

Q50

AnalystPrep Q.5568

REBUILT — Original depended on case-specific facts about a named bank not covered by the Schweser/GARP text; rebuilt to test the third-line (internal audit) role from the three lines of defense model.

A bank's board reviews a post-mortem of a large operational loss. It finds that the internal audit department relied entirely on the corporate operational risk function's (CORF) own assessments and never challenged management with its own testing. Which of the following best describes how internal audit, as the third line of defense, should have operated?

A. It should have owned and managed the operational risks generated by the business lines and designed the front-line controls.

B. It should have developed the operational risk policies and training programs and rewritten them where they were found inadequate.

C. It should have provided independent, objective assurance over the work of both the first and second lines,

performing its own sample testing and relying on others' work only after assessing its reliability.

D. It should have overturned business decisions that breached board-authorized risk limits and reported them as risk incidents.

Answer: C

Step by step

1. Third line = internal audit: an objective review of work performed in Lines 1 and 2, completely separate from risk management.
2. Internal audit evaluates the sufficiency and competence of the other functions and must not rely fully on their work; it performs its own independent analysis on a reasonable number of samples.
3. Reliance on others' risk assessment or testing is permitted only after internal audit has assessed its reliability.
4. The described failure (full reliance on the CORF, no challenge) therefore breaches the third-line role; option C states the correct role.

Why C is correct: Internal audit provides independent assurance over Lines 1 and 2 and must do its own testing rather than rely exclusively on the CORF's work.

Why the others are wrong

- **A:** Describes the first line (risk owners generate, measure and manage risk and set front-line controls); confuses audit with the business.
- **B:** Describes the second line/CORF (developing ORM policies and training); audit developing them would destroy its independence.
- **D:** Describes a second-line power (CORF overturning decisions that breach board risk limits); audit assesses rather than overrides decisions.

LO 43.d

Explain best practices and regulatory expectations for the development of a risk appetite for operational risk and for a strong risk culture.

Q51

AP Mock 2 Q40

REBUILT — Original relied on employment-law and 'all of the above' content not found in the reading; rebuilt around the no-blame culture and reinforcement guidance on risk culture.

A bank is considering publicizing the reasons for employee dismissals to show staff the conduct it expects and to signal discipline. Based on best practice for building a strong risk culture, what is the main danger of this approach if it is applied with unnecessary blame?

A. It may suppress the escalation of issues and incidents because staff fear being blamed for reporting them.

B. It will weaken the board's responsibility for the code of conduct.

C. It removes the need for risk training programs.

D. It makes the code of conduct inapplicable to board members.

Answer: A

Step by step

1. Reinforcement and signaling of discipline are effective ways to influence behavior, provided rules are taken seriously.
2. However, unnecessary blame must be avoided: a 'no-blame culture' is, with 'tone from the top', the most common recommendation for a good risk culture.
3. Blaming the person who reports or causes an incident suppresses escalation, so incidents stay hidden and risk culture deteriorates.
4. Hence A.

Why A is correct: Visible punishment helps only if it does not create fear of blame; blame discourages reporting and escalation, undermining risk culture.

Why the others are wrong

- **B:** The board remains responsible for the code of conduct and for leading risk culture; publicizing dismissals does not shift that.
- **C:** Training is a separate pillar of risk culture; discipline does not replace it.
- **D:** The code applies to both staff and board members regardless of how dismissals are communicated.

Q52 AnalystPrep Q.5061

According to the 4th principle of operational risk management, the board must identify the types and levels of operational risks the bank is willing to assume, as well as approve risk appetite and risk tolerance statements. Which of the following is not a correct feature of these statements?

A. Be easy to communicate and understand
B. Provide reasons for taking or avoiding certain operational risks
C. Be forward-looking and subject to scenario and stress testing
D. Perform scenario analysis retrospectively

Answer: D

Step by step

1. Appetite and tolerance statements should be easy to communicate and understand (A).
2. They should justify why the bank accepts, declines or limits specific risks, reflecting the risk-return trade-off (B).
3. They should be forward-looking and considered in scenario analysis and stress testing, including incidents that might push the bank beyond its limits (C).
4. Retrospective-only scenario analysis contradicts the forward-looking requirement, so D is the incorrect feature.

Why D is correct: Risk appetite must look forward and be stress tested; a retrospective approach is not a feature of a good statement.

Why the others are wrong

- **A:** Correct feature: statements must be straightforward to articulate and comprehend.
- **B:** Correct feature: BCBS expects the motivations for taking or avoiding risks to be stated.
- **C:** Correct feature: forward-looking and subject to scenario analysis and stress testing.

Q53 BT answers P2.T7.23.2.3 · BT questions T7-23.2.3

Steven is a new employee at a large publicly-traded financial services firm. He is learning about the firm's approach to operational risk, which is codified in an operational risk management framework. Before arriving, he had a rather straightforward view that risk is the possibility for bad things to happen. However, the firm's training material offers a more sophisticated perspective and includes specific vocabulary terms, including risk culture, risk appetite, risk tolerance, risk controls, and risk owners. In regards to these risk concepts (culture, appetite, tolerance, controls, and owners), which of the following statements is TRUE?

A. Risk tolerance is a broader, more general concept than risk appetite which is more specific
B. An organization's risk appetite should be sufficiently stable that it does not respond to external events or changes in market conditions
C. Because a firm's risk culture is intangible, subjective, and difficult to discern, it cannot be managed directly
D. It is effective, good practice to cascade down the firm-wide risk appetite into controls, limits, and monitoring metrics via risk, control, and metrics owners

Answer: D

Step by step

1. Good governance allocates a risk owner to each risk type in the appetite structure.
2. Appetite then cascades into limits, controls and monitoring metrics, with control owners and metrics owners designated.
3. This matches D. A reverses the usual relation; B and C contradict the idea that appetite is adjusted over time and that culture is actively led (tone from the top, code of conduct, training, compensation).

Why D is correct: Cascading firm-wide appetite into limits, controls and metrics through risk, control and metrics owners is the recommended governance structure.

Why the others are wrong

- **A:** Not supported: appetite is the broad, firm-wide expression of risk the bank is willing to take, while tolerance is the more specific threshold per risk or metric; the statement reverses this.
- **B:** Appetite should be forward-looking and reviewed, so it should respond to changing conditions rather than be frozen.
- **C:** Risk culture can be managed: the board leads it via code of conduct, tone from the top, reinforcement, compensation and training.

Q54 BT answers P2.T7.24.5.1 · BT questions T7-24.5.1

REBUILT — Original scenario was loosely worded (mixed bank names) and options B, C and D were giveaways; rewrote it to test cascading risk appetite into limits, controls and metrics with owners, with plausible distractors.

A bank launches a new real-time peer-to-peer payments feature with significant fraud and data-breach exposure. The CRO wants the feature governed in line with the board-approved operational risk appetite. Which action is most consistent with best practice?

A. Translate the appetite into exposure limits, key controls and monitoring metrics for the feature, and assign a risk owner, a controls owner and a metrics owner.

B. Set a single firm-wide statement of 'low tolerance for fraud' and leave the business to decide on controls after launch.

C. Have the board approve each individual control and metric directly, bypassing the risk committee.

D. Hand all monitoring of the feature to internal audit, which then owns the risk.

Answer: A

Step by step

1. Risk appetite sets exposure limits at business-unit level, minimum requirements for key controls, and limits on frequency and impact of acceptable incidents.
2. These cascade into limits, controls and monitoring metrics (KRIs).
3. Good governance assigns a risk owner (first line), controls owners (design and implement controls) and metrics owners (collect, report, monitor metrics).
4. Only A does this.

Why A is correct: Credible appetite must be actionable: translated into limits, controls and metrics, each with a named owner.

Why the others are wrong

- **B:** A bare statement is not credible: a stated low tolerance must be supported by documented controls and oversight, which must be in place, not deferred.
- **C:** The board sets appetite and limits, but the duty is usually delegated to the board risk committee with risk management support; the board does not approve every control.
- **D:** Internal audit is the third line and provides independent assurance; it does not own the risk, which sits with the first-line risk owner.

Q55 Mock 2 Q79

REBUILT — Original used an 'exposure owner' role absent from the reading; rebuilt around the risk owner, controls owner and metrics owner structure.

In a bank's risk appetite governance structure, a business line head is responsible for managing, maintaining and monitoring a risk within the stated limits of appetite and tolerance, and forms part of the first line of defense. This person is best described as which of the following?

A. Risk owner

B. Controls owner

C. Metrics owner

D. Internal audit

Answer: A

Step by step

1. Proper appetite governance allocates a risk owner to each risk type.
2. Risk owners are the managers responsible for managing, maintaining and monitoring risks within stated limits; they are the first line of defense.
3. Controls owners design/implement controls; metrics owners collect, report and monitor metrics; internal audit is third line.
4. The description fits the risk owner.

Why A is correct: Risk owners are first-line managers accountable for the risk within appetite and tolerance limits.

Why the others are wrong

- **B:** Controls owners are responsible for designing, implementing and the effectiveness of controls, not for managing the overall risk against limits.
- **C:** Metrics owners gather, report and monitor performance metrics against appetite; they do not own the risk itself.
- **D:** Internal audit is the third line providing independent assurance, not a first-line role.

Q56 Schweser Module Quiz 43.2 Q2

Which of the following items is least likely to appear in a bank's risk appetite statement?

A. Key controls.
B. Exposure limits.
C. Expected losses.
D. Tolerated incidents.

Answer: C

Step by step

1. Appetite sets exposure limits at the business-unit level (B).
2. It sets basic requirements for significant controls (A).
3. It sets limits on frequency and impact of acceptable incidents (D).
4. Expected losses are not a component, so C is least likely.

Why C is correct: Appetite statements are monitored through exposure limits, key controls and acceptable (tolerated) incidents, not expected losses.

Why the others are wrong

- **A:** Key controls are explicitly part of appetite statements.
- **B:** Exposure limits at business-unit level are a core appetite element.
- **D:** Tolerated incidents (acceptable frequency and impact) are a core appetite element.

Reading 44. Risk Identification

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 3.

LO	Learning objective	Qs
44.a	discuss different top-down and bottom-up approaches and tools for identifying operational risks.	12
44.b	describe best practices in extreme risk identification for operational risk.	6
44.c	describe and apply an operational risk taxonomy and give examples of different taxonomies of operational risks.	3
44.d	describe and apply the Level 1, 2, and 3 categories in the Basel operational risk taxonomy.	3

LO 44.a

Discuss different top-down and bottom-up approaches and tools for identifying operational risks.

Q57 AnalystPrep Q.5067

Which of the following risk identification processes begins at the executive level, then moves to the business units and finally to individual business processes?

A. Bottom-up risk identification
B. Event and loss data analysis
C. Top-down risk identification
D. Risk and control self-assessment

Answer: C

Step by step

1. Top-down identification starts with the board and senior management.
2. It then cascades to divisions or business units and finally to individual business processes.
3. This matches option C.

Why C is correct: Top-down identification is defined by its starting point (senior leadership) and its cascade downward through business units to processes.

Why the others are wrong

- **A:** Bottom-up starts with business units and staff and works upward; this reverses the sequence described.
- **B:** Event and loss data analysis is a bottom-up tool based on actual incidents, not a hierarchical process.
- **D:** RCSA is conducted at business-unit level (bottom-up), not initiated by executives.

Q58 AnalystPrep Q.5069

Which of the following is NOT a top-down risk identification tool?

A. Exposures and vulnerabilities
B. Risk wheel
C. Process mapping
D. Horizon scanning

Answer: C

Step by step

1. Top-down tools: exposures and vulnerabilities, risk wheel, emerging risks/horizon scanning (PESTLE).
2. Process mapping walks through the steps of a specific process to find what could go wrong, a bottom-up tool.
3. So C is the exception.

Why C is correct: Process mapping is a detailed, process-level (bottom-up) technique, whereas the other three are used in senior-level brainstorming.

Why the others are wrong

- **A:** A top-down tool: senior management lists significant exposures and weak spots.
- **B:** A top-down brainstorming tool showing causal links between risk types.
- **D:** A top-down tool for emerging risks, structured with PESTLE.

Q59 AnalystPrep Q.5070

Which of the following is NOT classified as an exposure under top-down risk identification tools?

A. Critical third parties
B. Key distribution channels
C. Systems overdue for updates
D. Main drivers of revenues

Answer: C

Step by step

1. Exposures are the activities, relationships and positions where a failure would cause large impact: critical third parties, key distribution channels, main revenue drivers, key persons, important clients.
2. Systems overdue for updates is a weak spot (a deficiency), i.e., a vulnerability.
3. Therefore C is not an exposure.

Why C is correct: Overdue maintenance or updates is a weakness in the firm (vulnerability), not an area of business dependence (exposure).

Why the others are wrong

- **A:** Critical third parties are a typical large exposure (outsourcing dependence).
- **B:** Distribution channels are listed exposures.
- **D:** Main drivers of revenue are listed exposures.

Q60 AnalystPrep Q.5071

Which of the following bottom-up risk identification tools relates to incidents that could have resulted in operational losses but did not, because of interventions outside the normal course of controls?

A. Internal losses
B. External losses
C. Near misses
D. Process mapping

Answer: C

Step by step

1. Identify the key feature: no loss actually occurred.
2. A near miss is an incident that could have caused a loss but was prevented by luck or intervention outside normal controls (e.g., misdirected payment reversed before withdrawal).
3. Answer C.

Why C is correct: Near misses are by definition incidents that did not produce a loss because of an intervention outside normal controls.

Why the others are wrong

- **A:** Internal losses are actual losses incurred by the firm.
- **B:** External losses are actual losses at other firms.
- **D:** Process mapping reviews process steps and risks; it is not a record of averted incidents.

Q61 BT answers P2.T7.23.3.2 · BT notes P2.T7.23.3.2

An operational risk management framework (ORMF) includes identification, assessment, mitigation, and monitoring. In regard to modern practice, which of the following statements about risk identification is TRUE?

A. Horizon scanning is a top-down approach that might be achieved in a structured way with a PESTLE analysis
B. Because bottom-up tools only identify risks that have already occurred (rather than potential future risks), their chief drawback is a tendency to overlook non-financial risks such as reputational or compliance
C. The bank should choose between either a top-down or bottom-up approach to risk identification in order to avoid confusing those in the middle of the organization
D. Although popular in the 80s and 90s, the risk wheel should be avoided because it is a novelty, and gamification tends to distract from the seriousness of the exercise

Answer: A

Step by step

1. Horizon scanning identifies emerging risks and is a top-down tool; PESTLE (political, economic, social, techno-

logical, legal, environmental) gives it structure: A is true.
2. Bottom-up tools such as RCSA also look forward and cover non-financial risks: B false.
3. Firms should combine top-down and bottom-up for a fuller picture: C false.
4. The risk wheel is a classic brainstorming tool that encourages creativity and shows cause-effect chains: D false.

Why A is correct: Horizon scanning is a top-down emerging-risk tool, and PESTLE is the structured method for performing it.

Why the others are wrong

- **B:** Bottom-up tools (e.g., RCSA) assess potential future risks and non-financial risks; they are not limited to past events.
- **C:** The two approaches are complementary and are combined, not chosen between.
- **D:** The risk wheel is a valued brainstorming tool, not a discredited one.

Q62

Mock 1 Q43

A risk manager at a bank is becoming familiar with top-down risk identification tools. The manager is presented with a list of issues. Which of the items is most likely to be considered an exposure?

A. Outdated processes.
B. Inadequate products.
C. Large company projects.
D. Systems overdue for maintenance.

Answer: C

Step by step

1. Exposures are significant areas of business dependence where a failure would cause large impact (large projects, critical third parties, key clients).
2. Outdated processes, inadequate products and systems overdue for maintenance are weaknesses, i.e., vulnerabilities.
3. Answer C.

Why C is correct: Large company projects are a typical large exposure; the other items are weaknesses in how the firm operates.

Why the others are wrong

- **A:** Outdated processes are a vulnerability.
- **B:** Inadequate or outdated products are a vulnerability.
- **D:** Systems overdue for maintenance and testing are a vulnerability.

Q63

Schweser Module Quiz 44.1 Q1

In the context of risk identification, which of the following items is most likely to be considered a vulnerability?

A. Principal regulator.
B. Critical third parties.
C. Stand-alone IT systems.
D. Main drivers of revenue.

Answer: C

Step by step

1. Vulnerabilities are the weakest links: stand-alone/unmaintained systems, unmonitored operations, overdue items, blind spots.
2. Principal regulator, critical third parties and main revenue drivers are dependencies, i.e., exposures.
3. Answer C.

Why C is correct: Stand-alone systems are a weak spot (often poorly monitored or updated), which makes them a vulnerability.

Why the others are wrong

- **A:** The principal regulator is an exposure (regulatory reporting and fines).
- **B:** Critical third parties are an exposure from outsourcing dependence.
- **D:** Main drivers of revenue are an exposure.

Q64 Schweser Module Quiz 44.1 Q2

The Chief Risk Officer (CRO) at a local bank favors assessing operational risk at a more local and detailed level. Which of the following risk identification tools is the CRO LEAST likely to recommend?

A. Risk wheel.
B. Process mapping.
C. Analysis of near misses.
D. Risk and control self-assessment.

Answer: A

Step by step

1. A local, detailed assessment is a bottom-up approach.
2. Process mapping, near-miss analysis and RCSA are bottom-up tools.
3. The risk wheel is a top-down brainstorming tool, so it is least likely to be recommended.

Why A is correct: The risk wheel is a senior-level top-down brainstorming tool, inconsistent with a local, detailed focus.

Why the others are wrong

- **B:** Process mapping is bottom-up and process-specific, so it fits.
- **C:** Near-miss analysis uses incident data at the working level, so it fits.
- **D:** RCSA is performed by business units, so it fits.

Q65 AnalystPrep Q.5065

REBUILT — Stem sharpened to match GARP wording (controls plus residual likelihood and impact) so that risk universe versus risk register is unambiguous; Schweser treats the terms as synonyms.

Which of the following is most likely a document that includes all operational risks of a firm, the controls applied to each risk, and an evaluation of the residual likelihood and impact of each risk?

A. Risk universe
B. Top-ten risks
C. Risk register
D. Shock scenarios

Answer: C

Step by step

1. The risk register is the central repository of operational risks, with controls and residual impact and likelihood for each.
2. A risk universe is only a list of all risks the firm believes it is exposed to, without assessment.
3. Answer C.

Why C is correct: The risk register combines the inventory of risks with controls and an assessment of residual likelihood and impact.

Why the others are wrong

- **A:** The risk universe lists risks but lacks the control and assessment elements.
- **B:** Top-ten risks is a prioritized subset, not all risks.
- **D:** Shock scenarios are separate stress analyses; scenarios are usually excluded from the register.

Q66 Mock 2 Q2

A newly established bank is creating its risk management framework prior to commencing operations. The framework consists of four main activities: risk assessment, risk identification, risk mitigation, and risk monitoring. Which of the following activities in the framework would use horizon scanning?

A. Risk assessment.
B. Risk identification.
C. Risk mitigation.
D. Risk monitoring.

Answer: B

Step by step

1. Horizon scanning (for example PESTLE analysis) is the technique used to find emerging risks that are less well

known and are only beginning to show up.
2. Finding risks that are not yet on the bank's risk register is the purpose of the risk identification step.
3. Assessment, mitigation and monitoring all act on risks that have already been identified, so they do not use horizon scanning.

Why B is correct: Horizon scanning is a top-down tool for spotting emerging risks, which is a risk identification activity.

Why the others are wrong

- **A:** Assessment measures the likelihood and impact of risks already identified; confusing 'looking ahead' with 'sizing known risks' leads here.
- **C:** Mitigation applies controls or transfers risk that is already known; it does not discover new risks.
- **D:** Monitoring tracks known risks and controls (for example via key risk indicators); it is ongoing oversight, not discovery of new emerging risks.

Q67 AnalystPrep Q.5072

Which of the following is not one of the six components of PESTLE that are used for scanning horizon risks?

A. Political component
B. Economic component
C. Labor market component
D. Environmental component

Answer: C

Step by step

1. PESTLE stands for Political, Economic, Social, Technological, Legal and Environmental.
2. Political (A), Economic (B) and Environmental (D) are all on the list.
3. Labor market is a cog in the risk wheel, not a PESTLE component, so C is the answer.

Why C is correct: PESTLE has exactly six components (Political, Economic, Social, Technological, Legal, Environmental), and labor market is not one of them.

Why the others are wrong

- **A:** Political is the 'P' in PESTLE, so it is a component.
- **B:** Economic is the 'E' in PESTLE, so it is a component.
- **D:** Environmental is the final 'E' in PESTLE, so it is a component.

Q68 AnalystPrep Q.5068

Which of the following is most likely the term for risks that a firm has identified as being on the horizon, relatively small but on the rise, with the potential for significant impact in the future?

A. Emerging risks
B. Risk universe
C. Taxonomies
D. Risk register

Answer: A

Step by step

1. The description (on the horizon, small but rising, potentially significant later, found by horizon scanning) defines emerging risks.
2. The risk universe is the comprehensive list of all risks the firm believes it is exposed to; a taxonomy is a classification scheme; a risk register records assessed risks and controls.
3. Therefore the answer is A.

Why A is correct: Emerging risks are nascent risks that are small today but increasing, and are found through horizon scanning rather than from existing risk lists.

Why the others are wrong

- **B:** The risk universe is the full inventory of all risks the firm believes it faces, not just those on the horizon; confuses breadth with trajectory.
- **C:** A taxonomy is a classification framework for describing risks, causes and impacts, not a set of rising risks.
- **D:** A risk register records assessed current risks and controls and usually excludes emerging risks, which are reported separately.

LO 44.b

Describe best practices in extreme risk identification for operational risk.

Q69

AnalystPrep Q.5073

REBUILT — Original distractors included non-biases (initiation of discussions, external loss data) and an arguable reversed-cause option; replaced with plausible bias-type distractors.

In a scenario analysis workshop, an external expert is invited to give an outside perspective. Which of the following is a bias that the external expert can most likely help mitigate?

A. Myopia, the undue emphasis on recent events
B. An excessive focus on scenarios driven by internal causes
C. Underestimation of the impact of recent events
D. Reliance on empirical evidence to support scenario assumptions

Answer: A

Step by step

1. GARP lists two biases an external expert helps mitigate: myopia (over-estimating recent events) and an excessive focus on scenarios driven by external causes.
2. Option A is myopia, which matches.
3. Option B reverses the second bias: the real bias is too much focus on external causes, since many large losses are internally caused.
4. Option C reverses myopia; D is a regulatory requirement, not a bias.

Why A is correct: Myopia overweights recent events (e.g., pandemics after 2020); an outsider is not anchored by the firm's recent experience and widens the scenario range.

Why the others are wrong

- **B:** Reversed: the documented bias is excessive focus on external causes, while internal causes (rogue trading, misselling, IT failures) are underweighted.
- **C:** Myopia is over-estimation, not under-estimation, of recent events.
- **D:** Basing assumptions on empirical evidence is required by regulators to reduce subjectivity; it is a safeguard, not a bias.

Q70

BT Mock Q63 · BT answers P2.T7.23.3.3 · BT notes P2.T7.23.3.3

Patricia is a risk analyst at a large international bank developing a new risk management framework. As part of extreme risk identification she will design and facilitate the firm's scenario analysis brainstorming session, for which she has prepared external loss data, internal loss data (including near misses), RCSA results, key risk indicator scores, audit issues, and concentrated exposures. Assuming she follows good practice, which of the following statements is TRUE?

A. Because most large operational losses result from external causes, her primary focus should be scenarios driven by external causes.
B. She should compare her firm's generated scenarios to industry-specific list(s) published by external sources, but only after her firm's own internal generation exercise.
C. The initial output of scenario identification should be limited to approximately 10 or 12 scenarios in order to avoid missing the forest for the trees.
D. Because the ultimate goal of scenario analysis is to predict the future with a sufficient degree of certainty, her process should acknowledge and elevate the role of human subjectivity.

Answer: B

Step by step

1. Good practice: compare own scenarios with an external industry list (e.g., ORX, ORIC) to find omissions, but only after generation so the external list does not bias it. B is true.
2. A: a substantial share of large losses is internally caused (rogue trading, misselling, IT failures), so over-focus on external causes is a bias.
3. C: initial output should contain at least 20 to 30 scenarios for a mid-size firm (about 50 not uncommon for international banks), narrowed to roughly 15 later.
4. D: scenario analysis is not prediction, and regulators demand that subjectivity and bias be minimized.

Why B is correct: Delaying the external comparison until after internal generation prevents anchoring on the

external list while still checking for gaps.

Why the others are wrong

- **A:** Reverses the evidence: large losses are often internal, so focusing mainly on external causes is a bias to be avoided.
- **C:** Understates the required breadth; the initial list should be at least 20-30 scenarios, with narrowing happening at the selection phase.
- **D:** Scenario analysis aims at plausible severe outcomes and repeatable results, and regulators want subjectivity minimized, not elevated.

Q71

BT answers P2.T7.602.3

REBUILT — Source drew on the older Cruz/Peters/Shevchenko reading (workshop formats, emerging-risk limitation) not in GARP/Schweser; rebuilt as an EXCEPT item on scenario selection per GARP.

Regarding the selection phase of an operational risk scenario workshop at a bank, each of the following statements is correct EXCEPT:

A. Scenarios with the same internal impact but different external causes, such as damage to physical assets from weather or from civil unrest, can be consolidated into one scenario.

B. Scenarios whose maximum likely loss is small enough to be absorbed by normal operating margin can be eliminated before assessment.

C. Large international banks generate fewer scenarios than small firms because their processes are more centralized.

D. The initial output of scenario identification should contain at least 20 to 30 scenarios for a mid-size firm, with roughly 15 remaining after selection.

Answer: C

Step by step

1. A is correct: GARP's example consolidates scenarios sharing an impact but with different causes.
2. B is correct: negligible scenarios are dropped to focus effort on larger ones.
3. D is correct: 20-30 initially for a mid-size firm, about 15 after selection.
4. C is false: small firms may produce fewer scenarios, while large international organizations generate more (about 50 is not uncommon). C is the exception.

Why C is correct: The number of scenarios rises with firm size and complexity, so larger banks have more, not fewer, scenarios.

Why the others are wrong

- **A:** True statement, so not the exception; consolidation by common impact is standard.
- **B:** True statement; scenarios with a moderate, absorbable maximum loss are removed at selection.
- **D:** True statement; this is the GARP guideline for the initial and post-selection list sizes.

Q72

GARP PE2 Q9 · Mock 4 Q10

A large global bank recently held an internal workshop to develop a set of scenarios involving extreme operational risk events. An internal auditor reviews the process used and some conclusions drawn by the participants. Which of the following would the auditor find to be most reasonable and aligned with best practices?

A. Participants began the workshop by reviewing a set of scenarios provided by an external industry source and used these as a guide to generate the scenarios discussed in the workshop.

B. Participants assigned a high probability to the occurrence of a severe new pandemic in the future given that one had occurred in the previous year.

C. Participants consolidated two scenarios in which different external events caused the same impact on the bank's infrastructure into a single scenario.

D. Participants emphasized scenarios arising from external causes, such as geopolitical crises and cyber-attacks, as losses from internal causes were considered to be unlikely.

Answer: C

Key correction: Mock 4 key said A; correct is C because an external industry list should be compared only after the firm's own scenario generation (Mock 4's option C was also garbled; restored from the GARP PE2 wording).

Step by step

1. Selection phase: scenarios with the same impact but different causes can be consolidated into one scenario

with various possible causes. C is best practice.
2. A: external lists should be used only after own generation, to avoid biasing it.
3. B: raising the probability because of a recent event is myopia.
4. D: over-focus on external causes ignores that many large losses are internal.

Why C is correct: Consolidating scenarios by common impact is explicitly endorsed in scenario selection; the other options each describe a recognized pitfall.

Why the others are wrong

- **A:** Using an external list as the starting guide anchors generation; it should be a post-generation completeness check.
- **B:** Overweighting a recent event is the myopia bias.
- **D:** Excessive focus on external causes is a known bias; internal causes (rogue trading, misselling, IT failures) produce many large losses.

Q73

Mock 2 Q26

REBUILT — Source was an authored replacement; option D wording ('anonymously') did not match GARP silent voting (written, silent), so it was reworded.

An operational risk manager is facilitating a scenario analysis workshop attended by senior business line heads and a few junior analysts. She is concerned that the most senior participants will dominate and that early contributions will anchor the group. Which technique is most appropriate during scenario identification?

A. Asking the most experienced participant to present an initial scenario for the group to refine.
B. Assigning probability estimates to each scenario before the group discusses its plausibility.
C. Restricting the workshop to participants of similar seniority to avoid hierarchical effects.
D. Having each participant write down their main concerns individually and in silence before any group discussion begins.

Answer: D

Step by step

1. The problem is dominance and anchoring by one voice.
2. Silent voting has everyone write down two or three concerns first, so all participants contribute and no individual steers the conversation.
3. A would deliberately anchor the group on a senior view.
4. B puts quantification before scenario discussion and does not address dominance; C removes the mix of business expertise and does not stop anchoring.

Why D is correct: Silent written contributions before discussion give every participant an equal voice and prevent a dominant individual from setting the agenda.

Why the others are wrong

- **A:** Maximizes both seniority dominance and anchoring.
- **B:** Estimating probabilities first skips plausibility discussion and does nothing about dominance.
- **C:** Participants should be senior managers across functions with relevant expertise; homogeneity does not remove anchoring.

Q74

Schweser Module Quiz 44.1 Q3

REBUILT — Original option A used 'can only be based on real-life data'; reworded to the GARP requirement of empirical evidence and documented rationale.

Which of the following statements regarding scenario analysis workshops and brainstorming sessions at a large bank is most accurate?

A. To meet regulators' requirements to minimize subjectivity and bias, scenario assumptions must be based on empirical evidence and their rationale documented.
B. The facilitators of the workshops should be drawn from the board of directors and senior management.
C. Participants should be drawn from a full range of seniority levels, including junior staff, in each business unit.
D. The most common procedure is to withhold the preparation pack from participants and distribute it after the generation phase.

Answer: A

Step by step

1. Regulators require repeatable results with minimal subjectivity: assumptions based on empirical evidence, explained rationale, documented process. A is accurate.

2. B: facilitators are typically ORM professionals.
3. C: participants are typically senior managers from the business and corporate functions.
4. D: the more common practice is to distribute documents before the first meetings; withholding is the less common alternative.

Why A is correct: Empirical grounding and documentation are what make scenario results repeatable and neutral, as regulators demand.

Why the others are wrong

- **B:** Facilitators are usually operational risk management professionals, not board members.
- **C:** Participants are normally senior managers with experience in their area, not a full seniority range.
- **D:** Reverses the norm: distributing before the first meetings is more common; withholding is occasional.

LO 44.c

Describe and apply an operational risk taxonomy and give examples of different taxonomies of operational risks.

Q75

BT answers P2.T7.23.3.1 · BT notes P2.T7.23.3.1

Peter is developing operational risk taxonomies for his financial services firm. His approach is influenced by the ORX taxonomy and their bow tie structure (Causes - Preventative controls - Risk events - Corrective controls - Impacts). Consequently, he is developing at least four taxonomies: causes, controls, risk events, and impacts. In regard to these operational risk taxonomies, each of the following statements is true EXCEPT which is false?

A. Causes include people, processes, systems, and external events (aka, PPSE)

B. Risk events include the seven Basel event types plus risk events that are elevated to highest-level (aka, Level 1), including third-party failure, statutory reporting & tax, data management, information/cyber security, and model risk

C. Controls are preventative, detective, corrective (i.e., to mitigate impact), or directive

D. Impacts include people, external fraud, internal fraud, physical security & safety, business continuity, transaction processing & execution, technology, conduct, legal, financial crime, and regulatory compliance

Answer: D

Step by step

1. Recall the ORX bow-tie structure: causes, preventive controls, risk events, corrective controls, impacts, each with its own taxonomy.
2. Causes: ORX uses the PPSE Level 1 categories (people, process, systems, external events), so A is true.
3. Risk events: ORX has 14 Level 1 risk types, i.e. the 7 Basel event types plus risks elevated to Level 1 (third-party failure, statutory reporting and tax, business continuity, data management, information/cyber security, model risk), so B is true.
4. Controls: four main categories (preventive, detective, corrective, directive), so C is true.
5. Impacts: ORX Level 1 impacts are direct financial impact, indirect financial impact and non-financial impact (with impact channels and stakeholders). The list in D is the list of Level 1 risk events, not impacts, so D is false.

Why D is correct: The list in D (people, external fraud, internal fraud, physical security, business continuity, technology, conduct, legal, financial crime, regulatory compliance) describes ORX Level 1 risk events, which map closely to the Basel event types; ORX impacts are direct financial, indirect financial and non-financial.

Why the others are wrong

- **A:** True statement: ORX causes follow PPSE (people, process, systems, external events); not the EXCEPT answer.
- **B:** True statement: ORX has 14 Level 1 risk types, the 7 Basel types plus elevated risks such as third-party failure, data management, cyber and model risk; the list is illustrative ('including'), not exhaustive.
- **C:** True statement: the four control categories are preventive, detective, corrective and directive; picking it means doubting the standard control taxonomy.

Q76

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Nadia, an operational risk analyst at a regional bank, is populating the bank's risk register. Following best practice, she wants each entry to describe a risk event (an adverse occurrence) and to record causes, impacts, and controls separately in their own taxonomies. Which of the following entries is most appropriately recorded as a risk event rather than as a cause or a constraint?

A. Errors in the valuation of funds that arise from manual processing of valuation inputs.
B. Manual processing of daily fund valuation inputs by the operations team.
C. Obligation to comply with newly issued operational resilience regulations.
D. Heavy reliance on technology across the bank's trading and payments platforms.

Answer: A

Step by step

1. Best practice separates causes, risk events, impacts, and controls; a risk event is a specific adverse occurrence ('what could go wrong?').
2. Technology is a resource, not a risk; the risks are incidents such as system interruptions or coding errors.
3. Manual processing is a cause (risk driver) that raises the likelihood of errors and omissions; it is not a risk by itself.
4. Compliance is a constraint from the operating environment; a compliance breach is the risk.
5. Errors in the valuation of funds are a specific adverse event, with manual processing as the cause, so A is the risk event.

Why A is correct: Errors in fund valuation are a concrete adverse occurrence. Manual processing is only the cause of that event, so the entry in A describes a risk event linked to its cause.

Why the others are wrong

- **B:** Manual processing is a cause or risk driver, not a risk event. It raises the likelihood of errors but is not itself an adverse occurrence. Confusing a cause with a risk event is the mistake.
- **C:** Compliance is a constraint from the operating environment, not a risk. The risk is a compliance breach. Confusing the obligation with the failure to meet it is the mistake.
- **D:** Technology is a resource that all firms rely on, not a risk. The risk would be a specific failure such as a system outage or coding error. Treating a resource as a risk is the mistake.

Q77 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Omar is the head of operational risk at a bank operating in a region heavily exposed to severe weather events and sociopolitical instability. He is deciding how to design the bank's operational risk taxonomy relative to the Basel taxonomy. Which of the following approaches is most consistent with how organizations tailor operational risk taxonomies?

A. Adopt the Basel Level 1, 2, and 3 categories without change because the regulator treats all three levels as mandatory reporting categories.
B. Include more physical and business disruption risk types in the taxonomy, particularly at the secondary level, to reflect the bank's exposures.
C. Expand the number of secondary-level categories as far as possible so that every specific incident type has its own category.
D. Classify cyber risk as external fraud, because all organizations place cyber risk in that category.

Answer: B

Step by step

1. Firms tailor taxonomies to the nature and size of their activities and their location.
2. Firms in regions heavily exposed to weather or sociopolitical instability include more physical and business disruption risk types, particularly at the secondary level, so B is correct.
3. The Basel Committee recognizes only Levels 1 and 2 as regulatory categories; Level 3 is illustrative.
4. Too much detail reduces information quality and wastes resources; detail belongs in Level 3 examples.
5. Cyber risk is categorized differently across firms (external fraud, IT risk, or a separate Level 1 risk).

Why B is correct: Taxonomies are tailored to a firm's exposures and location. A bank in a weather- and instability-prone region will add physical and business disruption risk types, mainly at the secondary level.

Why the others are wrong

- **A:** Only Levels 1 and 2 are regulatory categories; Level 3 gives examples. Firms also commonly use tailored taxonomies. Believing all three levels are mandatory is the mistake.
- **C:** Too much detail hurts information quality, is hard to review, and drains resources. Specific detail should be left to Level 3 examples. Equating more categories with a better taxonomy is the mistake.
- **D:** Organizations treat cyber risk differently: some as external fraud, some as IT risk, and others as a separate Level 1 risk. Assuming one universal treatment is the mistake.

LO 44.d

Describe and apply the Level 1, 2, and 3 categories in the Basel operational risk taxonomy.

Q78 AP Mock 1 Q44

A large bank operating in South America has been fined USD 20 million by the regulator for market manipulation and insider trading on the firm's own account. Under which of the Basel Level 1 event-type categories does this action most likely fall?

A. Business Disruption and System Failures.
B. Internal Fraud.
C. Employment Practices and Workplace Safety.
D. Clients, Products and Business Practices.

Answer: D

Step by step

1. Identify the activities: market manipulation and insider trading on the firm's account.
2. In the Basel taxonomy both are Level 3 examples under the Level 2 category Improper Business or Market Practices.
3. Improper Business or Market Practices sits under the Level 1 category Clients, Products and Business Practices.
4. Insider trading NOT on the firm's account would instead be Level 3 under Internal Fraud (Theft and Fraud), but that is not the case here.

Why D is correct: Market manipulation and insider trading on the firm's account are Level 3 examples of Improper Business or Market Practices, a Level 2 category under Clients, Products and Business Practices.

Why the others are wrong

- **A:** Business disruption covers system, hardware, telecom or utility failures; nothing here is a disruption.
- **B:** Internal Fraud includes insider trading only when it is not on the firm's account; this is the common trap.
- **C:** Employment Practices covers labor relations, workplace safety and discrimination; unrelated to trading misconduct.

Q79 GARP PE1 Q17 · Mock 3 Q63

An operational risk analyst at a large bank is preparing a year-end report of the bank's operational loss events. The analyst uses the Basel operational risk taxonomy to classify different events into Level 1, Level 2, and Level 3. As part of this process, the analyst considers the following four events that occurred during the year:

1. A sales representative in the lending department forged a senior manager's signature to approve an application for a large commercial loan that defaulted.
2. The bank failed to deliver sufficient collateral to support a swap position, which was later terminated at a loss.
3. Several managers in the bank's wealth management group were found to have misused confidential customer information by sharing this information with other employees and a third-party vendor representative.
4. Some personal customer information was stolen in a cyber-attack in which an external hacker broke into a database stored in the cloud.

In classifying the four events and considering the Basel taxonomy in general, which of the following actions would be most appropriate for the analyst to take?

A. The analyst should classify the forged signature under the Level 1 category of Clients, Products, and Business Practices.
B. The analyst should classify the failure to deliver sufficient collateral under the Level 3 category of Improper Business or Market Practices.
C. The analyst should classify the misuse of confidential information under the Level 2 category of Suitability, Disclosure, and Fiduciary.
D. The analyst should classify both the forged signature and the cyber-attack under the same Level 1 event-type category.

Answer: C

Step by step

1. Event 1 (forgery by an employee): Level 1 Internal Fraud, Level 2 Theft and Fraud, Level 3 Forgery.
2. Event 2 (collateral not delivered): Level 1 Execution, Delivery and Process Management, Level 2 Transaction Capture, Execution and Maintenance, Level 3 Collateral management failure.

3. Event 3 (misuse of confidential information): Level 1 Clients, Products and Business Practices, Level 2 Suitability, Disclosure and Fiduciary, Level 3 Misuse of confidential information.
4. Event 4 (external hacker): Level 1 External Fraud, Level 2 Systems Security, Level 3 Hacking damage/theft of information.
5. Only option C correctly names a category and its level.

Why C is correct: Misuse of confidential information is a Level 3 activity example whose Level 2 parent is Suitability, Disclosure and Fiduciary (under Clients, Products and Business Practices).

Why the others are wrong

- **A:** A forgery by an internal employee is Internal Fraud (Level 1), not Clients, Products and Business Practices, which covers unintentional or negligent failures toward clients.
- **B:** Improper Business or Market Practices is a Level 2 category (not Level 3) and the wrong branch; collateral failure is a Level 3 item under Execution, Delivery and Process Management.
- **D:** The forgery involves an internal party (Internal Fraud) whereas the hacker is a third party (External Fraud); they are different Level 1 categories.

Q80

Schweser Module Quiz 44.2 Q1

Regarding the Basel taxonomy of operational risks for banks and the Level 1 category of internal fraud, which of the following items would most appropriately be included at Level 3?

A. Insider trading.
B. Theft and fraud.
C. Unauthorized activity.
D. Losses due to acts of a type intended to defraud.

Answer: A

Step by step

1. Level 1 is the broad event type (Internal Fraud) with a general definition.
2. Level 2 divides it into categories: Unauthorized Activity and Theft and Fraud.
3. Level 3 lists specific activity examples, such as forgery, bribes/kickbacks and insider trading (not on the firm's account).
4. Insider trading is therefore a Level 3 item.

Why A is correct: Level 3 provides concrete activity examples, and insider trading is one such example under Internal Fraud.

Why the others are wrong

- **B:** Theft and Fraud is a Level 2 category of Internal Fraud, not a Level 3 example.
- **C:** Unauthorized Activity is the other Level 2 category of Internal Fraud.
- **D:** This is essentially the Level 1 definition of internal fraud (acts intended to defraud), not a specific Level 3 example.

Reading 45. Risk Measurement and Assessment

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 4.

LO	Learning objective	Qs
45.a	explain best practices for the collection of operational loss data and reporting of operational loss incidents, including regulatory expectations.	8
45.b	explain operational risk-assessment processes and tools, including risk control self-assessments (RCSAs), likelihood assessment scales, and heatmaps.	5
45.c	describe the differences among key risk indicators (KRIs), key performance indicators (KPIs), and key control indicators (KCIs).	5
45.d	describe the use of factor-based models that quantitatively assess operational risk, and explain the application of the Swiss cheese model and the bowtie tool.	6
45.e	estimate operational risk exposures based on the fault tree model given probability assumptions.	3
45.f	describe approaches used to determine the level of operational risk capital for economic capital purposes, including their application and limitations.	5
45.g	describe and explain the steps to ensure a strong level of operational resilience, and to test the operational resilience of important business services.	3

LO 45.a

Explain best practices for the collection of operational loss data and reporting of operational loss incidents, including regulatory expectations.

Q81

AP Mock 1 Q46 · AnalystPrep Q.5095

Why are near-miss occurrences included in the reporting of incidents in organizations with strong risk cultures?

A. To assess the cost of close calls.
B. To determine the importance of close calls.
C. To analyze the potential consequence that was unintentionally avoided.
D. To evaluate the frequency of close calls.

Answer: C

Step by step

1. A near miss follows the same causal chain as a loss, but the loss was avoided by chance rather than by a control working.
2. Recording it (impact type: near miss) lets the firm analyze what would have happened had luck not intervened.
3. This exposes control weaknesses and the severity narrowly missed, so the risk can be mitigated before a real loss occurs.
4. Hence the purpose is to analyze the potential consequence that was unintentionally avoided (C).

Why C is correct: Near misses are free lessons: they reveal control gaps and the size of the loss that was narrowly avoided, which supports prevention without suffering actual loss.

Why the others are wrong

- **A:** A near miss produces no actual loss, so there is no cost to assess; this confuses near misses with realized losses.
- **B:** Importance is not the reason for capture; it is a by-product of analysis. Choosing it treats reporting as ranking rather than learning.
- **D:** Frequency counts are a secondary statistic (e.g. for KRIs); the main value is learning from the avoided consequence.

Q82

BT answers P2.T7.23.4.1 · BT notes P2.T7.23.4.1

REBUILT — Fixed OCR-garbled currency symbol and reworded option C so it matches GARP (a non-zero threshold is acceptable if supportable); typo RSCA corrected to RCSA.

Paula is refactoring the operational incident (loss) database as part of an update to her firm's risk and control self-assessment (RCSA) procedure under a new operational risk management framework. One step is an audit of data quality. Regarding her database audit, each of the following statements is true EXCEPT which is false?

A. Each operational incident has (at least) four important dates: occurrence, discovery, reporting, and accounting (settlement).

B. The gap between the discovery date and the settlement date should rarely exceed three months and never exceed one year; otherwise, the IT reporting system itself probably requires an upgrade.
C. A non-zero minimum threshold for loss reporting (e.g., EUR 5,000) is acceptable, provided it is supportable and not set to reduce capital requirements.
D. Although the general ledger is a good source for benchmarking, it is not a perfect substitute for a dedicated operational event database.

Answer: B
Step by step
1. A: GARP lists four dates - occurrence, discovery, reporting, accounting (general ledger entry/settlement). True.
2. B: Long lags are normal. Events can go undiscovered for months or years, and large losses with legal/regulatory implications typically take about three to five years between reporting and settlement. A one-year cap is not a data-quality standard, and long gaps do not signal an IT failure. False.
3. C: The regulatory minimum collection threshold is EUR 20,000 and firms commonly use 1,000-10,000; a non-zero threshold is acceptable if justified. True.
4. D: The general ledger records only accounted financial effects, so it benchmarks but cannot replace the incident database. True.
5. The false statement is B.

Why B is correct: Settlement lags of several years are a normal feature of operational events (late discovery, legal processes), so a hard one-year cap is not a valid quality test.
Why the others are wrong
- **A:** True: these are the four dates GARP/Schweser define for every incident.
- **C:** True: non-zero thresholds are standard (typical 1,000-10,000; BCBS minimum EUR 20,000) if they are justified and not used to lower capital.
- **D:** True: the ledger shows accounted effects only and lacks cause, dates and near misses, so it benchmarks but does not replace the event database.

Q83 Schweser Module Quiz 45.1 Q1

REBUILT — Original options were nested (occurrence-to-reporting contains occurrence-to-discovery) and reporting-to-accounting could also be long; reframed to ask which interval drives the long lag, with non-overlapping options.

A bank employee has been manipulating suspense (transitory) accounts for a prolonged period before the fraud was detected, and the event was classified as a significant operational incident. Which interval between the incident dates is the main reason the settlement of such events typically occurs years after they began?

A. Date of reporting to date of accounting.
B. Date of discovery to date of reporting.
C. Date of occurrence to date of discovery.
D. All intervals contribute equally because each requires manual processing.

Answer: C
Step by step
1. Incident dates in order: occurrence, discovery, reporting, accounting.
2. Fraud in suspense accounts, salami frauds, rogue trading and data leaks can run undetected for months or years.
3. Time to discovery (occurrence to discovery) is therefore the main driver of the years-long gap before payment/settlement and it worsens impact.
4. Discovery to reporting is short for significant events (days); reporting to accounting is driven by legal/regulatory processes, not by the manipulation going unnoticed.

Why C is correct: Concealed manipulation is by nature slow to detect, so the occurrence-to-discovery interval reflects visibility of the problem and dominates the overall lag.
Why the others are wrong
- **A:** Reporting-to-accounting can be long for litigated events, but it is not what makes a concealed fraud slow; this mixes up settlement delay with detection delay.
- **B:** Significant incidents are reported within a few working days of discovery, so this interval is typically the shortest.
- **D:** No such equality exists; intervals differ and the detection interval is the dominant one for concealed fraud.

Q84 AnalystPrep Q.5157

REBUILT — Source key (C) conflicts with GARP: regulators require only direct financial losses (indirect ones are good practice), and option A had a garbled currency; rebuilt options so exactly one statement matches GARP.

Joel and Mark, FRM Part II candidates, are discussing regulatory expectations and best practice for reporting operational loss incidents. Which statement is most likely correct?

A. Setting the loss reporting threshold at zero is now the dominant best practice, especially in large institutions.

B. Regulators require firms to record non-financial impacts such as reputational damage and customer detriment as part of internal loss data.

C. Direct losses are usually poorly captured, whereas indirect losses are almost always monetized by every firm.

D. Different events that materialize from a single failure must be grouped as a single loss to reflect the common underlying cause.

Answer: D

Key correction: Source key (AnalystPrep) said C ('both direct and indirect losses must be reported'); per GARP regulators require only losses with direct financial effects, with indirect effects recorded as good management practice, so that statement is not correct.

Step by step

1. A: GARP says zero thresholds are a fading practice because of the burden of many small incidents. False.
2. B: Regulators require reporting only incidents with financial losses; non-financial impacts are recorded as good management practice. False.
3. C: Direct losses are usually well captured; only more mature firms monetize indirect losses. False.
4. D: Grouping losses with a common cause is a regulatory requirement, sound modelling practice and needed for correct management information. True.

Why D is correct: Losses arising from one underlying failure must be combined into a single grouped loss, which is a regulatory requirement.

Why the others are wrong

- **A:** Reverses GARP: zero thresholds are becoming rare; typical thresholds are 1,000-10,000 (BCBS minimum EUR 20,000).
- **B:** Overstates regulation; only direct financial losses are required, indirect effects are good practice.
- **C:** Inverts reality: direct losses are well captured, and only mature firms monetize indirect losses (often just a 1-4 rating).

Q85 BT answers P2.T7.601.1 · BT notes P2.T7.601.1

REBUILT — Original options relied on Basel 'de minimis' wording and a contestable 'primary motivation' claim not in GARP; rebuilt with GARP's threshold guidance (supportable, not used to cut capital, captures all material exposures).

Which of the following is TRUE about the selection of the threshold for reporting operational losses?

A. The threshold should be set so that only tail events that drive capital are collected.

B. Basel rules prohibit banks from applying any internal loss reporting threshold.

C. Thresholds may differ across firms, but the chosen threshold must be supportable, must not be set to reduce capital requirements, and must still capture all material activities and exposures.

D. A zero threshold is the dominant practice in large institutions because it avoids the effort of estimating losses.

Answer: C

Step by step

1. A: Internal loss data must be comprehensive, capturing all material activities and exposures, not only tail events.
2. B: Basel sets a minimum collection threshold of EUR 20,000; it does not prohibit thresholds. Many firms use lower ones (1,000-10,000).
3. C: GARP: the threshold should be supportable and not deliberately set to lower capital; data must be comprehensive. Correct.
4. D: Zero thresholds are fading in large institutions because of the burden of many small incidents.

Why C is correct: Firms have discretion over the threshold level, but it must be justified, not manipulated to reduce capital, and compatible with comprehensive capture of material exposures.

Why the others are wrong

- **A:** Confuses loss collection with capital modelling; restricting to tail events breaks the comprehensiveness requirement.
- **B:** Misreads the Basel minimum threshold (EUR 20,000) as a ban on thresholds.

- **D:** Zero thresholds still exist but are fading, especially in large institutions, so 'dominant' is wrong.

Q86

GARP PE1 Q10

REBUILT — Key (D, post-event upgrade costs excluded from gross loss) is a Basel-text point not supported by the GARP chapter; rebuilt around GARP's risk-based reporting timeline, grouping and threshold guidance.

A regional bank's CRO asks an operational risk manager to review the bank's process for reporting operational incidents against best practice. Which of the following statements would be most appropriate for the manager to make?

A. Material incidents should usually be reported within a few working days, while minor incidents can be included in periodic summary reporting.

B. Raising the loss reporting threshold increases the losses recorded and therefore increases operational risk capital.

C. Indirect effects such as reputational damage cannot be assessed in monetary terms because their consequences are non-financial.

D. Events caused by a single IT failure should be recorded as separate unrelated losses so that each stays below the reporting threshold.

Answer: A

Step by step

1. A: Best practice is risk-based reporting; material incidents are reported within a few working days and minor ones summarized periodically, so resources are not wasted on immaterial events. Correct.
2. B: A higher threshold captures fewer (small) losses, so recorded losses fall, not rise.
3. C: Indirect effects have real financial implications (remediation, attrition, regulatory costs) and mature firms monetize them.
4. D: Events from one failure must be grouped as a single loss; splitting them to stay under the threshold is non-compliant.

Why A is correct: GARP endorses a risk-based timeline for reporting incidents: quickly for material events, summary reporting for minor ones.

Why the others are wrong

- **B:** Gets the direction backwards: a higher threshold excludes small losses, lowering recorded losses.
- **C:** The word 'non-financial' is misleading; indirect consequences are costly and can be quantified.
- **D:** Violates the grouping requirement for losses with a common cause.

Q87

Mock 3 Q70

REBUILT — Original option A omitted that the recovery must still be recorded (separately), and the key's claim that lost revenue counts conflicts with the regulatory focus on direct losses; stem now uses concrete direct-loss figures and options are tightened to one defensible answer.

A severe storm damaged several branches of a regional bank. The bank paid USD 4.0 million in direct repair costs and later received a USD 2.5 million settlement from its property insurer. An operational risk analyst is preparing the loss event record for the internal loss database. Which treatment is most consistent with best practice and Basel loss data guidance?

A. Record a gross loss of USD 4.0 million and record the USD 2.5 million insurance settlement separately as a recovery, so that net loss (USD 1.5 million) can be derived.

B. Record only the net loss of USD 1.5 million, because the insurance payment has already offset the gross amount and a gross figure adds no information.

C. Delay recording the event until the insurer's final payment is received, then record a single loss of USD 1.5 million.

D. Do not record the event as an operational loss, because a natural disaster is an external cause outside the bank's control.

Answer: A

Step by step

1. Basel/GARP loss data standards require the gross loss amount for each event, the reference dates, and any recoveries (insurance and other), with causes recorded.
2. Gross loss = direct repair costs = USD 4.0 million.
3. Recovery = insurance settlement = USD 2.5 million, captured in its own field (recoveries often arrive later than the loss).

4. Net loss = gross loss - recovery = 4.0 - 2.5 = USD 1.5 million, derived from the two fields.
5. External events such as natural disasters (damage to physical assets) are a Basel event type and belong in the database.

Why A is correct: Loss databases keep gross loss and recoveries as separate fields, so exposure (gross) and mitigation effectiveness (recovery) are both visible and net loss is derived.

Why the others are wrong

- **B:** Storing only net loss hides the gross exposure and the effectiveness of insurance, understating the underlying risk; the standard requires gross loss plus recoveries.
- **C:** The event should be logged when it occurs/is discovered with the gross loss, and the recovery added when received; waiting delays reporting and loses the gross figure.
- **D:** External causes do not exempt an event: damage to physical assets from natural disasters is a standard Basel operational risk event type.

Q88

AP Mock 1 Q27

Banks are required to report not only the gross loss amount associated with each operational loss event but also the reference dates of the event. Which of the following is LEAST likely to be an important reference date for an operational loss incident?

A. Date of occurrence
B. Date of accounting
C. Date of reporting
D. Date of assessment

Answer: D

Step by step

1. The reference dates collected for an operational loss event are: occurrence (when the event first happened), discovery (when it was first identified), reporting (when it enters the reporting database) and accounting (when the financial impact enters the P&L).
2. Occurrence, reporting and accounting all appear on this list (options A, C, B).
3. 'Date of assessment' is not one of the reference dates, so it is the least likely to be important.

Why D is correct: The standard reference dates are occurrence, discovery, reporting and accounting; no 'assessment' date is part of the loss-event record.

Why the others are wrong

- **A:** Date of occurrence is a core reference date (when the event first happened); picking it confuses 'important' with 'least important'.
- **B:** Date of accounting is required: it is when the loss, reserve or provision hits the P&L and it drives the year to which the loss is allocated.
- **C:** Date of reporting (when the event enters the reporting database) is one of the four standard reference dates, so it is not the odd one out.

LO 45.b

Explain operational risk-assessment processes and tools, including risk control self-assessments (RCSAs), likelihood assessment scales, and heatmaps.

Q89

BT answers P2.T7.23.4.2 · BT notes P2.T7.23.4.2

REBUILT — Heatmap exhibit was garbled and option D said 'multiply likelihood by probability'; restated the scales in text and corrected D to 'likelihood by impact' so the false statement is unambiguous.

Ethan is designing the risk and control self-assessment (RCSA) process for his employer. His boss wants the process to generate reports for the board and eventually be automated. Ethan's initial design combines the two dimensions of likelihood and impact into a color-coded matrix (heatmap). Each dimension has a four-level scale numbered 1 to 4:

Likelihood	Remote = 1	Unlikely = 2	Possible = 3	Likely = 4
Impact	Low = 1	Moderate = 2	Major = 3	Extreme = 4

Regarding this RCSA process and heatmap, each of the following statements is true EXCEPT which is false?

A. A key function of the RCSA is to determine whether the control environment of an activity (or unit) is in line

with the firm's risk appetite.

B. At the end of the RCSA exercise, the firm should have a good understanding of its inherent risks, control effectiveness, and residual risks.

C. Impact types commonly include financial, regulatory, customer, and reputation (with service continuity a common addition), and likelihood scales are expressed either as percentages or as frequencies of occurrence.

D. To compare loss events on a single dimension, the algorithm should multiply likelihood by impact, giving a score of 4 to both a likely-but-low-impact event (4 x 1) and a remote-but-extreme-impact event (1 x 4).

Answer: D

Step by step

1. Read the question: three statements are true, one is false.
2. A is true: a key value of the RCSA is judging whether the control environment of a unit is in line with risk appetite.
3. B is true: an RCSA should leave the firm understanding inherent risk, control effectiveness and residual risk.
4. C is true: impact scales cover financial, regulatory, customer and reputation (plus service continuity); likelihood is expressed as a % or as 'once in x years'.
5. D is false: RCSA ratings are ordinal, qualitative, nonlinear scores. Multiplying them ($4 \times 1 = 1 \times 4 = 4$) treats a frequent minor event as equivalent to a remote extreme event, a still-widespread mistake that gives misleading results.

Why D is correct: Qualitative ratings are not numbers on a ratio scale, so arithmetic on them (likelihood × impact) is invalid; the heatmap colors, not products, express the risk level.

Why the others are wrong

- **A:** True: determining whether controls keep risk within appetite is a central RCSA purpose. Choosing it means missing that the question asks for the false statement.
- **B:** True: inherent risk, control effectiveness and residual risk are the standard RCSA outputs.
- **C:** True: this matches the standard impact types and the two ways of expressing likelihood; picking it confuses the listed variations with an error.

Q90 BT answers P2.T7.24.3.1

Sonic Asset Management ('the firm') is undergoing its annual risk control self-assessment (RCSA) to evaluate the operational risks within its IT department. The assessment involves both seasoned employees and newcomers who provide fresh perspectives on the IT processes. Participants use structured questionnaires to assess daily risks and the effectiveness of existing controls. Following the RCSA, the operational risk management team uses likelihood scales and heatmaps to prioritize and address the identified risks.

What is the primary purpose of employing RCSAs, likelihood scales, and heatmaps in the firm's operational risk management process?

A. To document control activities and compare operational efficiency across departments.

B. To uniformly classify and prioritize IT department risks based on impact and likelihood.

C. To evaluate only the residual risks post-control implementation, disregarding initial risk levels.

D. To assess the financial impact of control failures and integrate risk management with financial assessments.

Answer: B

Step by step

1. The main purpose of risk assessment is to prioritize risk management and mitigation responses.
2. Standardized likelihood and impact scales make ratings comparable across risks and units; the heatmap combines them into colors that rank risks.
3. Option B states exactly this: uniform classification and prioritization by impact and likelihood.

Why B is correct: Standardized scales and heatmaps give a consistent ordering of risks by likelihood and impact, which enables prioritization of mitigation actions.

Why the others are wrong

- **A:** RCSA is not an efficiency benchmarking or control-documentation tool; this confuses risk assessment with process/performance review.
- **C:** An RCSA covers both inherent and residual risk and control effectiveness; restricting it to residual risk is wrong.
- **D:** RCSA ratings are qualitative and do not give a precise financial measure; financial impact is only one impact type among several.

Q91 BT answers P2.T7.602.1 · BT notes P2.T7.602.1

Silverfind Financial International is planning to conduct a risk control self-assessment (RCSA) for each of its business units.

Which of the following is MOST LIKELY to be a feature or element of the firm's RCSA?

A. Staff within a business unit are excluded from providing input into their own business unit's RCSA scorecard in order to avoid conflicts of interest.

B. Any process with an inherent risk that is greater than its residual risk will be assigned a "red flag" because this is a situation that requires a process fix.

C. Managers will first identify and assess inherent risks by making no inferences about controls embedded in the process; i.e., controls are assumed to be absent.

D. In order to estimate realistic outcomes, risks are identified by assuming the mitigation impact of in-place controls; that is, the risk exposure sought is net of control and mitigation.

Answer: C

Step by step

1. An RCSA evaluates inherent risk (before the impact of controls) and residual risk (after evaluating existing controls).
2. Inherent risk is therefore assessed with controls assumed absent; this is option C.
3. The RCSA is a self-assessment run as workshops/questionnaires with the unit's own staff, so A is wrong.
4. Inherent risk exceeding residual risk is the normal result of effective controls, so B is wrong; D describes residual risk, not the first step.

Why C is correct: Inherent (gross) risk is by definition the exposure before controls are considered, so managers assess it assuming no controls.

Why the others are wrong

- **A:** RCSAs are self-assessments: the unit's staff provide the input (in workshops or questionnaires); excluding them defeats the purpose.
- **B:** Residual risk below inherent risk simply shows controls work; it is not a red-flag condition. A flag arises when residual risk exceeds appetite.
- **D:** A net-of-controls exposure is residual risk. Identifying risks net of controls skips the inherent assessment that the RCSA starts with.

Q92 Mock 2 Q13

REBUILT — Stem did not state the color convention, so red/amber were both arguable; added the standard red-amber-yellow-green meanings (without naming the cell's color).

A bank uses a risk and control self-assessment (RCSA) matrix (heatmap) in which risks are plotted against a likelihood axis (remote, unlikely, possible, likely) and an impact axis (low, moderate, major, extreme). Cells use the usual four colors: green (within risk appetite, regular monitoring only), yellow (within appetite but approaching the limit), amber (above the permissible limit; exposure must be lowered or escalated for approval) and red (far beyond appetite; immediate mitigation required). A risk is assessed as having a remote likelihood but an extreme impact. Which color is it most likely to be assigned?

A. Amber

B. Green

C. Red

D. Yellow

Answer: A

Step by step

1. Remote likelihood lowers the rating, but an extreme impact cannot be treated as acceptable: the risk is not green or merely yellow.
2. Red is reserved for exposures far beyond appetite that need urgent mitigation (e.g., likely with high impact, or unlikely with extreme impact).
3. A remote-but-extreme risk exceeds the permissible limit and must be lowered or escalated: amber.

Why A is correct: Very low likelihood offsets an extreme impact only partly: the exposure is outside appetite and needs reduction or escalation, which is the amber band.

Why the others are wrong

- **B:** Green is for exposures within appetite; ignoring the extreme impact because of low likelihood is the error.
- **C:** Red is for exposures far beyond appetite (e.g., unlikely with extreme impact); a remote likelihood moderates

the rating below red, so the risk needs reduction or escalation rather than urgent mitigation.
- **D:** Yellow is within appetite but near the limit (e.g., likely but low impact); it understates the extreme-impact tail.

Q93 Schweser Module Quiz 45.1 Q2

Which of the following statements regarding risk assessment is most accurate?

A. Risk and control self-assessments (RCSAs) are a good mixture of qualitative and quantitative aspects.
B. The main purpose of risk assessment is to prioritize risk management and risk mitigation responses.
C. When backtesting the results of risk assessment against past incident experience, there is a tendency to overestimate the likelihood of an operational risk event.
D. When backtesting the results of risk assessment against past incident experience, there is a tendency to underestimate the impact of an operational risk event.

Answer: B

Step by step

1. RCSAs are qualitative and largely judgment based, so A is wrong.
2. Backtesting against past incidents shows participants underestimate likelihood and overestimate impact, so C and D are reversed.
3. The main purpose of risk assessment is to prioritize risk management and mitigation responses: B.

Why B is correct: Comparable qualitative assessments allow risks to be ordered, so management can prioritize mitigation actions.

Why the others are wrong

- **A:** RCSAs are qualitative, judgment-based; calling them a quantitative mix is wrong.
- **C:** Reversed: backtests reveal underestimation of likelihood, not overestimation.
- **D:** Reversed: backtests reveal overestimation of impact, not underestimation.

LO 45.c

Describe the differences among key risk indicators (KRIs), key performance indicators (KPIs), and key control indicators (KCIs).

Q94 AP Mock 1 Q28

John, the CRO of Omega Bank, is implementing a new risk control system and introduces key risk indicators (KRIs) and key control indicators (KCIs). Some team members are confused about the difference. Which of the following best differentiates between KRIs and KCIs?

A. Both KRIs and KCIs measure the potential risks that may impact the achievement of strategic objectives.
B. KCIs identify potential risks, while KRIs measure the effectiveness of control mechanisms in mitigating those risks.
C. Both KRIs and KCIs monitor the effectiveness of risk control mechanisms.
D. KRIs identify potential risks that may hinder the achievement of an objective, while KCIs measure the effectiveness of mitigation mechanisms.

Answer: D

Step by step

1. KRIs indicate the bank's level of exposure to a risk at a point in time (a rise or fall in the likelihood or impact of the risk).
2. KCIs measure how effectively controls are operating, in design or performance.
3. Option D assigns exposure to KRIs and control effectiveness to KCIs, which is the correct split.

Why D is correct: A KRI points at the risk (its likelihood or impact), whereas a KCI points at the control and tests whether it works.

Why the others are wrong

- **A:** Treats both as risk measures; ignores that KCIs measure control effectiveness, not exposure.
- **B:** Swaps the definitions: KRIs signal risk exposure and KCIs measure control effectiveness, not the reverse.
- **C:** Treats both as control-effectiveness measures; KRIs measure exposure to risk.

Q95 BT notes P2.T7.23.4.3 · BT answers P2.T7.23.4.3

Sally is revising her firm's operational risk monitoring metrics. Her client attrition and retention metrics include likelihood key risk indicators (KRIs), such as drops in customer satisfaction, and impact KRIs, such as an increase in value generated by top-10 clients. She is also revising key performance indicators (KPIs) and key control indicators (KCIs). The net promoter score is an example of a potential KPI. Missed due diligence items are an example of a potential KCI.

In regard to these metrics (KRI, KPI, and KCI), each of the following is true EXCEPT which is false?

A. Some metrics can be both KPIs and KRIs, or even share elements of all three (KPI, KRI, and KCI).

B. KRI thresholds and governance vary according to the organization's risk appetite and risk tolerance levels.

C. Natural language processing (NLP) machine learning methods are a cost-effective substitute for risk and control self-assessment (RCSA) because they eliminate the need to select and design KRIs.

D. A KPI is more likely to be a lagging metric, while a KRI is more likely to be a leading metric.

Answer: C

Step by step

1. Read the stem as an EXCEPT question: find the false statement.
2. A is true: KPIs, KRIs and KCIs overlap, and a failed control is jointly a KPI, KRI and KCI.
3. B is true: thresholds express how strict management is, reflecting risk appetite and tolerance.
4. D is true: preventive KRIs are forward-looking (leading); KPIs report performance already achieved (lagging).
5. C is false: data analytics and machine learning (including NLP) improve the selection and design of preventive KRIs; they do not replace RCSA or remove the need for KRIs.

Why C is correct: Machine learning supports KRI selection and design but does not replace RCSA or eliminate KRIs, so C is the false statement.

Why the others are wrong

- **A:** True: the metrics overlap significantly; a control failure is jointly a KPI, KRI and KCI.
- **B:** True: KRI thresholds reflect the organization's risk appetite and tolerance.
- **D:** True: KRIs are preventive (leading) while KPIs typically report realized performance (lagging).

Q96 BT notes P2.T7.24.3.2 · BT answers P2.T7.24.3.2

REBUILT — Original stem was vague and its keyed option treated transaction error rates as a control metric, whereas the reading treats error rates on customer transactions as a KPI; rebuilt as a clear metric-classification item.

Wholesome Bank tracks several metrics for its retail operations: (i) error rates on customer transactions, (ii) the number of business continuity plans not reviewed before their due dates, (iii) the number of transactions processed per employee, and (iv) the sensitivity of data held on a server. The risk team wants to classify them as key risk indicators (KRIs), key performance indicators (KPIs), or key control indicators (KCIs). Which of the following classifications is most appropriate?

A. Error rates on customer transactions are a KCI.

B. An increase in transactions processed per employee is a KRI of likelihood (risk of error).

C. Business continuity plans not reviewed before due dates are a KPI.

D. An increase in the sensitivity of data held on a server is a KCI.

Answer: B

Step by step

1. KPIs measure how effectively the bank operates (e.g., error rates on customer transactions, IT downtime, complaints).
2. KCIs measure how effectively controls operate (e.g., business continuity plans not reviewed by due date).
3. KRIs measure exposure: more transactions per employee raises the likelihood of errors; more sensitive data on a server raises the impact of a leak.
4. Only B matches a metric to the correct category.

Why B is correct: A rising workload per employee is a cause of error, so it is a preventive KRI measuring likelihood.

Why the others are wrong

- **A:** Error rates on customer transactions are a standard KPI (how effectively the bank operates), not a control measure.
- **C:** Overdue plan reviews show a review control not operating effectively, so they are a KCI, not a KPI.
- **D:** Higher data sensitivity increases the impact if leakage occurs, so it is a KRI of impact; it says nothing about control effectiveness.

Q97 Schweser Module Quiz 45.2 Q1

One of the bank's compliance managers notices that there are two business continuity plans that have not been reviewed, and it is now past their review due date. The incomplete reviews of the business continuity plans are best described as a:

A. key assessment indicator.
B. key control indicator.
C. key performance indicator.
D. key risk indicator.

Answer: B

Step by step

1. Plans not reviewed or updated before their set due dates are a standard example of a KCI.
2. The review is a control; being overdue means the control is not operating effectively.
3. Therefore the best description is a key control indicator.

Why B is correct: KCIs measure how effectively the bank's controls are operating, and an overdue review is a control failure.

Why the others are wrong

- **A:** 'Key assessment indicator' is not one of the three metric types in the reading.
- **C:** KPIs measure how effectively the bank operates (e.g., complaints, downtime), not control operation.
- **D:** KRIs show exposure level (likelihood or impact drivers); the overdue review signals a control failure, not a change in exposure drivers.

Q98 BT answers P2.T7.602.2 · BT notes P2.T7.602.2

REBUILT — Original keyed answer (KRI collection automated at cost-center level) is not supported by the reading; rebuilt to test KRI thresholds, risk appetite and the leading/lagging nature of KRIs.

Streethigh Bank is designing its key risk indicator (KRI) program for the equity settlement process. Management wants the KRIs to be preventive and to reflect its risk appetite. Which of the following statements is most consistent with best practice?

A. KRI intervention thresholds should be set identically across all processes so that results are comparable, regardless of risk appetite.
B. KRI thresholds express how strict management wishes to be in controlling and mitigating each risk, and so reflect risk appetite and tolerance.
C. A KRI should only be used if it is also a KCI, since a KRI that does not measure control effectiveness gives no useful signal.
D. KRIs should be lagging measures of realized losses, since leading measures cannot signal a change in exposure.

Answer: B

Step by step

1. KRI thresholds and governance reflect the organization's risk appetite and risk tolerance.
2. The thresholds of intervention show how strict management is in controlling and mitigating those risks, so they differ by risk.
3. B states this; the other options contradict the reading.

Why B is correct: Chosen metrics and thresholds express the bank's priorities and tolerance, so they cannot be uniform or unrelated to risk appetite.

Why the others are wrong

- **A:** Uniform thresholds ignore that tolerance differs by risk; thresholds should reflect appetite.
- **C:** KRIs measure exposure; they may overlap with KCIs but need not be control measures.
- **D:** Preventive KRIs are leading indicators of rising cause intensity, which is their purpose; realized losses are lagging.

LO 45.d

Describe the use of factor-based models that quantitatively assess operational risk, and explain the application of the Swiss cheese model and the bowtie tool.

Q99 AP Mock 3 Q27

REBUILT — Original asked for the 'purpose' with a vague option set; reworded to the GARP/Schweser best-practice use of root-cause analysis and corrected option D (the second line supports or challenges the first line's analysis). Re-mapped from 45.a to 45.d (root-cause analysis sits with Swiss cheese/bowtie).

In an operational risk framework, root-cause analysis is performed after significant incidents and near misses. Which of the following describes the greatest value obtained from it?

A. Identifying only the immediate (proximate) trigger of the event.
B. Comparing the results of multiple investigations to find common causes and failures, revealing weaknesses that can be addressed across the organization.
C. Quantifying the financial impact of the event on operating performance.
D. Replacing the first line's analysis with an independent investigation run solely by the second line.

Answer: B

Step by step

1. Root-cause analysis (e.g. 5-whys, bowtie) goes beyond direct causes to indirect and root causes and the control failures behind them.
2. It is done by the first line; the second line adds to or queries it.
3. Its added value comes from comparing the current analysis with past ones to find common themes and patterns, so one action plan can address the causes of many events.
4. That matches B; A is the shallow approach it is meant to go beyond, C is impact assessment, D reverses the line-of-defense roles.

Why B is correct: Root-cause analysis digs below the proximate trigger, and comparing results across investigations exposes systemic weaknesses to fix at source.

Why the others are wrong

- **A:** The immediate cause is what root-cause analysis looks past; stopping there treats symptoms.
- **C:** Impact quantification belongs to loss measurement/impact assessment, not to causal analysis.
- **D:** The first line performs the analysis and the second line supports or challenges it; it is not replaced.

Q100 AnalystPrep Q.5159

What is the primary use of the Swiss cheese model?

A. A method to assess the impact of an attack on an asset.
B. A tool for generating scenarios with FAIR methodology.
C. A framework for analyzing and identifying security.
D. A concept that uses layers of defense to prevent hazards.

Answer: D

Step by step

1. The Swiss cheese (cumulative act) model, popularized by James Reason, views each defensive layer as a slice of cheese with holes.
2. The holes open, shut and shift over time; a hazard causes harm only when holes in many layers momentarily line up.
3. So the model is a conceptual framework for layered defenses (control layering) that prevents hazards from reaching their target.

Why D is correct: The model's purpose is to explain and design layered controls so that weaknesses in individual layers never align.

Why the others are wrong

- **A:** Impact assessment of an attack is not the purpose; the model describes layering of defenses, not loss severity.
- **B:** Scenario generation (asset, threat community, threat type, effect) belongs to FAIR, a different factor model.
- **C:** The model is not a security-analysis framework; it is a general layered-defense concept for any operational hazard.

Q101 AnalystPrep Q.5160

What is the purpose of Monte Carlo simulations in the FAIR model of managing operational risk?

A. To estimate the frequency and magnitude of a potential loss event.
B. To generate scenarios with an asset at risk, a threat community, a threat type and an effect.
C. To provide the distribution of simulated scenario losses as output.

D. To determine the best course of action to prevent a potential loss event.

Answer: C

Step by step

1. FAIR steps: identify factors and interrelationships, measure each factor, combine factors computationally.
2. Business experts estimate loss frequency and probable loss magnitude as distributions, not point values.
3. These distributions are inputs to Monte Carlo simulation, whose output is the distribution of simulated scenario losses.

Why C is correct: Monte Carlo is the computational combination step that turns factor distributions into a scenario loss distribution.

Why the others are wrong

- **A:** Frequency and magnitude are estimated by business experts; they are inputs to the simulation, not its purpose.
- **B:** Scenarios are built before simulation, as a separate step of the FAIR framework.
- **D:** Choosing a preventive course of action is a risk-management decision; the simulation only outputs a loss distribution.

Q102

BT Mock Q68 · BT answers P2.T7.23.5.1 · BT notes P2.T7.23.5.1

REBUILT — Source options paired each approach with a disadvantage (e.g., bowtie 'natively lacks a quantitative framework') that the GARP text does not support; the stem was shortened to a single-statement TRUE question with unambiguous options.

Sally is leading a causal analysis of operational risk and is choosing a factor model. Her candidates are fault tree analysis (FTA), root-cause analysis with the bowtie tool, the FAIR model and the Swiss cheese model. Which of the following statements is TRUE?

A. FTA is simple and does not rely on domain expertise, but it is unrealistic because it can only assume independent events.

B. The bowtie tool can be used proactively, because it assesses the likelihood and/or impact of risks before they materialize by examining controls and indirect causes.

C. The FAIR model is probabilistic only in a limited sense: it relies on subjective point estimates and does not produce a loss distribution.

D. The Swiss cheese model quantifies control weaknesses by retrieving clusters and their centroids through k-means clustering.

Answer: B

Step by step

1. Bowtie: the risk event sits at the centre; causes and preventive controls on the left, impacts and detective/corrective controls on the right.
2. GARP notes bowtie is useful for root-cause analysis of incidents and also for assessing likelihood and impact of risks before they materialize, looking at controls and indirect/root causes.
3. Check others: FTA needs expert knowledge and can use conditional probabilities for dependent events; FAIR uses distributions and Monte Carlo; Swiss cheese is conceptual.
4. Only B is true.

Why B is correct: Bowtie analysis is both a retrospective root-cause tool and a forward-looking assessment of likelihood and impact given the controls in place.

Why the others are wrong

- **A:** FTA relies on domain expertise to build the and/or logic and can incorporate dependence through conditional probabilities, so it is not limited to independent events.
- **C:** FAIR is explicitly probabilistic: factors are distributions fed into Monte Carlo to give a scenario loss distribution.
- **D:** Swiss cheese is a conceptual layered-control metaphor; k-means clustering has nothing to do with it.

Q103

BT answers P2.T7.24.3.3 · BT notes P2.T7.24.3.2

Omega Corp, a multinational corporation, is launching new trading software and assessing its operational risks. The risk management team uses factor-based models to quantify risks and has incorporated two specific models for visualizing and managing these risks. They focus on identifying overlapping security vulnerabilities and analyzing potential unauthorized transactions to map risk pathways and develop mitigation strategies effectively.
How is the Omega Corp team likely to structure their risk assessment?

A. The team is employing a hierarchical model with statistical tools for risk prediction and mitigation.
B. The team is focusing on a single-layer defense strategy to manage risks independently.
C. The team is using the Swiss Cheese Model to identify overlapping security vulnerabilities and the Bowtie Model to map the causes and impacts of transactions.
D. The team is adopting a linear model to trace risk sequences and use correlation analysis for loss forecasting.

Answer: C
Step by step
1. 'Overlapping vulnerabilities' across defenses is the Swiss cheese idea: holes in several layers aligning.
2. Mapping causes, preventive controls, impacts and corrective controls around an unauthorized-transaction event is the bowtie diagram.
3. Both are factor-based models, so C matches.

Why C is correct: Swiss cheese addresses alignment of weaknesses across control layers; bowtie maps causes and consequences around a central event.
Why the others are wrong
- **A:** A hierarchical statistical prediction approach is not what the two named visual factor models are.
- **B:** Swiss cheese is built on multiple layers; a single-layer defense contradicts it.
- **D:** Linear sequences with correlation-based forecasting describe neither the Swiss cheese nor the bowtie model.

Q104 GARP PE1 Q15 · Mock 3 Q65

An operational risk manager is giving a presentation on the advantages and disadvantages of different approaches firms use to manage operational risk. The manager provides an example of a financial institution that uses the Swiss cheese model as a control structure to manage the risk of operational losses due to incorrect transaction processing. Which of the following is correct about using this control structure?
A. The failure of a single key control would directly result in an operational loss.
B. Every layer of control would have to fail in order for an operational loss to occur.
C. The probability of each control failure depends on whether the previous control failed.
D. Extreme operational losses become less likely while small operational losses become more likely.

Answer: B
Step by step
1. In the Swiss cheese model each control is a slice with holes; a hole in one slice normally causes no harm.
2. A loss occurs only when holes in all layers line up, i.e., every layer fails.
3. Effective layering requires control failures to be uncorrelated: e.g., three independent controls each failing 10% of the time give 0.1^3 = 0.1% joint failure.
4. Hence B.

Why B is correct: Loss requires simultaneous failure of all layers, which is made unlikely by keeping layer failures independent and compensating.
Why the others are wrong
- **A:** A single weakness normally does not produce a loss; that is the point of layering.
- **C:** Dependence between failures is a flaw (common cause) that defeats layering; the model requires uncorrelated failure rates.
- **D:** The model lowers the probability of loss getting through generally; it does not shift loss severity from extreme to small events.

LO 45.e

Estimate operational risk exposures based on the fault tree model given probability assumptions.

Q105 BT answers P2.T7.24.4.1

FX Trade, a proprietary trading firm, is evaluating the adequacy of its capital amidst recent governance issues. A risk manager has used a fault tree model to estimate the expected loss from a rogue trader. The model identifies five conditions that must all occur for a loss to happen, with their respective probabilities:

Condition	Probability
Employee Unhappy (P1)	20%
Goes Rogue (P2)	1%
Back Office Failure (P3)	30%
Line 2 Misses (P4)	30%
Trades Lose Money (P5)	50%
Potential Loss per Event	$100,000,000

Assuming all conditions are independent, the firm has 1,000 traders, each with the same probabilities, and each rogue event results in a loss of $100,000,000. What is the expected annual loss (total exposure) for the firm?

A. $9,000
B. $9,000,000
C. $30,000
D. $50,000,000

Answer: B

Step by step

1. All five conditions are AND conditions, so multiply the independent probabilities: $0.20 \times 0.01 \times 0.30 \times 0.30 \times 0.50 = 0.00009 = 0.009\%$.
2. Expected loss per trader: $0.00009 \times \$100{,}000{,}000 = \$9{,}000$.
3. Scale to 1,000 traders: $\$9{,}000 \times 1{,}000 = \$9{,}000{,}000$.

Why B is correct: In a fault tree with AND gates the joint probability is the product of the independent probabilities; multiplying by the loss severity gives the per-trader expectation, and by the number of traders gives the firm exposure.

Why the others are wrong

- **A:** This is the exposure for a single trader; it omits scaling by the 1,000 traders.
- **C:** Not supported by the product of the five probabilities; the joint probability is 0.009% per trader, and $30,000 would require a different, incorrect joint probability.
- **D:** Applies only the 50% losing-trade probability to the $100,000,000 loss, ignoring the other four required conditions.

Q106

Schweser Module Quiz 45.2 Q2

In the context of a potential nonmalicious data leakage incident by a bank employee, the following information is provided:
There is a 99% chance that bank employees will receive a phishing email through their work email.
There is a 95% chance that the bank's firewalls will operate as intended.
There is a 90% chance that the employee will know immediately to delete the phishing email.
There is a 3% chance that the detective controls of suspicious network activity will fail.
There is a 1% chance that there will be a leak of information.
Using fault tree analysis (FTA) and assuming that the conditions just listed are fully independent, the likelihood of the risk of data leakage materializing for a given time period is closest to:

A. 0.0001485%.
B. 0.0004365%.
C. 0.0048015%.
D. 0.0253935%.

Answer: A

Step by step

1. Convert each control to its failure probability: firewall fails $1 - 0.95 = 0.05$; employee fails to delete $1 - 0.90 = 0.10$. The phishing email (0.99), detective-control failure (0.03) and leak (0.01) are already the event/failure probabilities.
2. AND gate with independence: $0.99 \times 0.05 \times 0.10 \times 0.03 \times 0.01$.
3. $0.99 \times 0.05 = 0.0495$; $\times 0.10 = 0.00495$; $\times 0.03 = 0.0001485$; $\times 0.01 = 0.000001485$.
4. $0.000001485 = 0.0001485\%$.

Why A is correct: The leakage requires every link in the chain to fail, so the joint probability is the product of the failure probabilities; controls stated as success rates must first be converted to failure rates.

Why the others are wrong

- **B:** Mixes success probabilities (0.90, 0.97) with a failure probability (0.05) and so is not the product of the correct

failure probabilities; it is about three times too large.
- **C:** Uses 0.97 (detective controls working) instead of the 3% failure probability, with the other control failure rates, inflating the product.
- **D:** Multiplies the control success probabilities as given ($0.99 \times 0.95 \times 0.90 \times 0.03 \times 0.01$) without converting them to failure probabilities.

Q107 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

An operational risk manager at a global bank is using fault tree analysis to assess a scenario in which a confidential client file is sent externally by mistake. The scenario requires four independent control failures to occur together: the email filter fails to flag the file (8%), the employee does not notice the warning prompt (5%), the supervisor's review is skipped (10%), and the data loss prevention tool fails to block the transmission (2.5%). The manager assumes that every one of the bank's 40,000 employees has the same failure probabilities. Which of the following is the best estimate of the likelihood that the scenario materializes for the bank as a whole, based on the stated assumptions?

A. 0.001%, because the joint probability of the four independent failures is the product of the individual probabilities.

B. 0.004%, because the joint probability of failure of the controls should be scaled by the 4 controls in the layered structure.

C. 40%, because the joint probability of the four independent failures should be multiplied by the number of employees exposed.

D. 25.5%, because the individual failure probabilities should be added since any one failure can start the scenario.

Answer: C

Step by step

1. All four conditions must occur together (AND conditions) and are independent, so the joint probability per employee is $0.08 \times 0.05 \times 0.10 \times 0.025 = 0.00001 = 0.001\%$.
2. The probability is per employee; assuming the same failure rate for all employees, scale by exposure: $0.001\% \times 40{,}000 = 0.40 = 40\%$.
3. Answer: C.

Why C is correct: Fault tree analysis multiplies the probabilities of AND conditions to get a per-employee likelihood of 0.001%, and the scenario assessment must then be multiplied by the unit of exposure (40,000 employees) to give about 40% for the bank as a whole.

Why the others are wrong

- **A:** This stops at the per-employee joint probability (0.001%) and ignores exposure; the reading stresses that a low failure rate across many employees can imply a high bank-wide likelihood.
- **B:** Scaling by the number of controls ($0.001\% \times 4$) has no basis; the correct scaling factor is the number of employees exposed, not the number of control layers.
- **D:** Adding probabilities (8% + 5% + 10% + 2.5% = 25.5%) treats the failures as OR conditions; the scenario requires all four to fail simultaneously, so probabilities are multiplied.

LO 45.f

Describe approaches used to determine the level of operational risk capital for economic capital purposes, including their application and limitations.

Q108 BT answers P2.T7.23.5.2 · BT notes P2.T7.23.5.2

In order to determine his firm's internal operational risk capital requirements, Robert is employing the loss distribution approach (LDA). He will take the following steps: collect and clean data; categorize data according to well-defined risk classes (aka, units of measure, UoM); estimate (aka, fit) the frequency distributions; fit the severity distributions; calculate expected loss (EL) and unexpected loss (UL) for each risk class; and aggregate.

In regard to this LDA, each of the following statements is true EXCEPT which is false?

A. Monte Carlo simulation convolutes frequency and severity distributions into an aggregated loss distribution

B. In practice, LDA frequency distributions tend to be either Poisson or negative binomial; severity distributions tend to be lognormal, Weibull, or Generalized Pareto Distribution (GPD)

C. Within each unit of measure (UoM; aka, risk class), losses should be correlated and heterogeneous; however, the many UoMs should be aggregated according to an independent and identically distributed (i.i.d.) assumption

D. Because the history of internal loss data is insufficient for the task of describing tails, the dataset likely needs to be completed with extreme value theory's (EVT's) tail distributions and/or external data must be added to the mix, where key methodological decisions include scaling, cut-off mix, and filtering

Answer: C

Step by step

1. LDA assumes losses within a UoM are independent and identically distributed (no correlation), so UoMs should be as homogeneous as practical.
2. Different UoMs are aggregated by modeling dependence between them (copulas, including tail dependence), not by assuming independence.
3. Statement C reverses both ideas, so it is the false statement.
4. A, B and D match the reading: Monte Carlo convolution, Poisson/negative binomial frequency with lognormal/Weibull/GPD severity, and supplementing thin tail data with EVT or external data (scaling, cut-off mix, filtering).

Why C is correct: C reverses the LDA assumptions: losses within a UoM are i.i.d. and uncorrelated, while dependence across UoMs is modeled (e.g., copulas).

Why the others are wrong

- **A:** True: Monte Carlo draws of frequency and severity are convoluted into the aggregate loss distribution; not the exception.
- **B:** True: Poisson/negative binomial for frequency and lognormal/Weibull/GPD for severity are standard; not the exception.
- **D:** True: internal data are thin in the tail, so EVT and/or external data (scaling, cut-off mix, filtering) are added; not the exception.

Q109 BT answers P2.T7.24.4.2

In the aftermath of a cyber-attack that cost FX Banco $500 million in losses, the bank is developing a model to calculate operational risk capital for economic capital purposes. The bank has comprehensive internal data on frequent operational losses, such as minor process failures, but it is also concerned with accurately modeling rare, extreme events like major fraud or cyber-attacks. The bank wants to use a method that allows it to address infrequent but large, catastrophic losses in a balanced way.

Which of the following approaches should FX Banco choose?

A. A method that estimates operational risk capital based on a percentage of the bank's gross income. While easy to implement, it does not account for specific loss events or their severity.

B. A method that separately models the frequency and severity of losses and combines them using statistical simulations. This approach is effective for regular losses but may not be as precise for extreme, catastrophic events.

C. A method that focuses on extreme events by using statistical techniques to model the tail of the loss distribution. This method is highly effective at estimating the impact of rare, large losses but may not adequately capture frequent, smaller losses.

D. A method that combines a general loss modeling approach with additional techniques specifically designed to improve the accuracy of extreme event estimation, effectively balancing both frequent, smaller losses and rare, catastrophic losses.

Answer: D

Step by step

1. Need both an accurate body (frequent losses, plenty of internal data) and an accurate tail (rare, severe events).
2. LDA handles the body but cannot cope with very high skewness; it must be supplemented by EVT, added large losses, or scenario analysis.
3. Hence a hybrid of LDA plus tail techniques (EVT/scenarios) is the balanced choice: D.

Why D is correct: LDA is the most common approach but needs supplementing by EVT or scenario analysis for the tail, so a hybrid balances frequent and extreme losses.

Why the others are wrong

- **A:** An income-based (indicator) approach ignores actual loss events and severity, so it cannot model the tail.
- **B:** LDA alone is good for the body but weak in the tail because of the high skewness of losses.

- **C:** EVT alone models only the tail and neglects the frequent small losses.

Q110 Schweser Module Quiz 45.3 Q1

A quantitative analyst is combining internal and external loss data for model calibration. The analyst has established specific criteria for the inclusion and exclusion of losses in the dataset. The method used by the analyst is best described as:

A. cut-off mix.
B. filtering.
C. scaling.
D. rule based.

Answer: B

Step by step

1. Three methods combine internal and external data: scaling, cut-off mix, filtering.
2. Filtering sets specific rules for including or excluding losses in the modeling dataset, which limits potential manipulation of results.
3. The analyst's inclusion/exclusion criteria match filtering.

Why B is correct: Filtering is the method defined by explicit inclusion/exclusion rules for losses in the modeling dataset.

Why the others are wrong

- **A:** Cut-off mix uses external losses only above a severity threshold where internal losses are sparse; it is not about general inclusion criteria.
- **C:** Scaling adjusts loss sizes to the bank's size (or for inflation); it does not decide which losses are included.
- **D:** 'Rule based' is not a recognized method; it is a plausible-sounding distractor for filtering.

Q111 AnalystPrep Q.5108

REBUILT — Original options were not mutually exclusive (Monte Carlo is part of LDA); reworded the stem to the LDA definition and made distractors clearly distinct.

A bank's operational risk manager asks an analyst to build a model that fits separate frequency and severity distributions for each unit of measure and then convolutes them into an aggregated annual loss distribution, whose 99.9th percentile gives stand-alone capital. Which approach is the analyst applying?

A. Reverse stress testing
B. Loss distribution approach
C. Fixed multiplier approach
D. Scenario aggregation using variance-covariance matrices

Answer: B

Step by step

1. Separately modeling frequency (e.g., Poisson) and severity (e.g., lognormal) per UoM, then convoluting by Monte Carlo, is the LDA.
2. The 99.9th percentile of the aggregate distribution gives the stand-alone capital.
3. Hence B.

Why B is correct: The LDA splits losses into frequency and severity, models each independently and convolutes them into an aggregate loss distribution.

Why the others are wrong

- **A:** Reverse stress testing starts from a failure outcome and works back to scenarios; it does not fit loss distributions.
- **C:** Fixed multipliers scale a business indicator; they involve no frequency/severity fitting.
- **D:** Scenario aggregation (variance-covariance/copulas) combines scenario estimates in the ICAAP; it does not fit frequency/severity distributions.

Q112 BT notes P2.T7.2.4

REBUILT — Source was a Crouhy-specific EXCEPT item with a vague false option; rebuilt around the reading's limitations of LDA and EVT.

Which of the following statements about the limitations of approaches used to determine operational risk capital is INCORRECT?

A. The LDA assumes frequency and severity are independent, which is questionable because empirical evidence shows many low-severity events and few high-severity events.
B. Segmenting losses into homogeneous units of measure reduces the data available in each UoM, making results less reliable.
C. LDA cannot properly handle the very high skewness of operational losses without adding large losses, applying EVT, or using scenario analysis.
D. EVT is well suited to operational risk because losses typically arise from one main underlying cause that will generate larger future losses.

Answer: D
Step by step
1. A, B and C are limitations stated for the LDA.
2. EVT works only when one main underlying cause generates all losses and future larger losses.
3. Operational losses are diverse, so a single main cause is unrealistic and this severely limits EVT's usefulness; D is incorrect.

Why D is correct: EVT's main limitation is that operational losses have diverse causes, so its single-underlying-cause requirement is unrealistic.
Why the others are wrong
- **A:** A correct statement of an LDA limitation; not the answer.
- **B:** Correct: more homogeneous UoMs mean less data and less reliable fits.
- **C:** Correct: LDA needs large losses, EVT or scenarios to capture extreme skewness.

LO 45.g

Describe and explain the steps to ensure a strong level of operational resilience, and to test the operational resilience of important business services.

Q113 GARP book PE2 Q10 · Mock 4 Q11

A fast-growing UK-based FinTech firm offers savings accounts, cryptocurrency accounts, domestic bill payment services, and services that allow customers to make payments in different currencies. A senior operations manager at the firm is developing a plan to comply with new UK regulatory requirements for operational resilience. Which of the following steps should the manager recommend that the firm take to best comply with the regulatory expectations in this area?

A. Develop an impact tolerance for each of the internal processes performed by the firm.
B. Identify important business services and map the dependencies between these services.
C. Calculate the 1-year 99.9% VaR for operational risk for each of the firm's business divisions and use this result to reserve additional capital for each division.
D. Create an operational resilience team that is led by the IT department, with its other members coming from the operational risk management, legal, and compliance functions.

Answer: B
Step by step
1. Operational resilience is built around important business services (IBSs), not individual processes: the first step is to identify the IBSs.
2. Next the firm maps the dependencies between the IBSs (and the resources needed to deliver them), then sets impact tolerances per IBS and tests them with severe but plausible scenarios.
3. Option B is the only option that is a genuine resilience step.

Why B is correct: Resilience starts with identifying important business services and mapping their dependencies and required resources; impact tolerances and scenario tests are then applied at the service level.
Why the others are wrong
- **A:** Impact tolerances are set for important business services, not for every internal process; a process-level focus is the business continuity (BCM) approach.
- **C:** Confuses resilience with capital measurement: a VaR-based capital reserve does not keep services running within impact tolerances.
- **D:** Resilience requires a multi-disciplinary team with operational risk management usually as coordinator, together with BCM, HR, third-party management, IT and operations; an IT-led team with only risk, legal and com-

pliance members is not the expected structure.

Q114 BT answers P2.T7.23.5.3 · BT notes P2.T7.23.5.3

Alice is developing her bank's operational resilience framework. Historically, her firm's language did not include the concept of resilience. According to the Bank for International Settlements (BIS), "operational resilience is an outcome that benefits from the effective management of operational risk."
Her workflow steps will include the following: identify important business services; set impact tolerances; map the important services end-to-end and identify required resources; design severe but plausible scenarios in order to test vulnerabilities; where tolerance(s) is exceeded, examine lessons learned and take action; ensure internal/external communications; get sign-off on an annual self-assessment document.
In regard to operational resilience, each of the following is true EXCEPT which is false?

A. Single points of failure (SPOF) are likely to be key risk indicators (KRI), and whenever possible, they should be mitigated or removed

B. The important business services (IBS) vary by the nature of the institution, and prioritization assumes that not every process is important to an institution

C. The key difference between resilience frameworks and business continuity management (BCM) is that resilience is articulated around important business services (IBS)

D. The framework should be qualitative (because it complements the factor models), non-complex (to achieve successful implementation), and resource-light in generation (because participants already working on the ORM framework should not bias the resilience framework with dual assignments)

Answer: D

Step by step

1. A: SPOFs are relevant resilience KRIs and should be mitigated or removed via backups and redundancies - true.
2. B: IBSs depend on the institution's nature, business and customers, and not everything is equally important - true.
3. C: BCM is articulated by business process, resilience around IBSs - true.
4. D: Mapping dependencies and critical paths is complex (complexity grows exponentially in large institutions) and needs large teams, resources and data and a multi-disciplinary approach (ORM as coordinator with BCM, HR, third-party management, IT, operations). So 'non-complex' and 'resource-light' is false.

Why D is correct: The resilience framework is complex, resource-intensive and multi-disciplinary, so describing it as non-complex and resource-light is the false statement.

Why the others are wrong

- **A:** True: SPOFs are standard resilience KRIs and should be removed through backups and redundancy where possible. Choosing it misreads the EXCEPT format.
- **B:** True: IBSs differ by institution and larger banks must prioritize because not all services are equally important.
- **C:** True: BCM is process-based while resilience is organized around important business services with impact tolerances.

Q115 BT answers P2.T7.20.16.3

REBUILT — Original tested an Oliver Wyman four-step program outside the GARP/Schweser text; rebuilt to test the seven-step resilience sequence in the reading.

A bank has just identified its important business services (IBSs) and established impact tolerances for each. Following the seven-step process for operational resilience, which of the following correctly gives the order of the next three steps?

A. Design severe but plausible scenarios; map the IBSs and required resources; ensure communication plans are ready

B. Map the IBSs and required resources; design severe but plausible scenarios; if impact tolerances are breached, review lessons learned and take action

C. Map the IBSs and required resources; ensure communication plans are ready; obtain board sign-off on the annual self-assessment

D. Set an impact tolerance for each internal process; map the IBSs; design severe but plausible scenarios

Answer: B

Step by step

1. The seven steps are: (1) identify IBSs, (2) set impact tolerances, (3) map IBSs and required resources, (4) design severe but plausible scenarios, (5) if tolerances are breached review lessons and act, (6) ensure communication

plans, (7) annual board-approved self-assessment.
2. Steps 1 and 2 are done, so the next three are steps 3, 4 and 5.
3. This matches option B.

Why B is correct: After identifying services and setting tolerances, the firm maps the services and resources, tests with severe but plausible scenarios, and then acts on lessons if tolerances are breached.

Why the others are wrong

- **A:** Reverses steps 3 and 4: scenarios cannot be designed to test delivery of a service before it is mapped end-to-end, and it skips step 5.
- **C:** Skips scenario testing and lessons learned (steps 4-5) and jumps to communication plans and the self-assessment (steps 6-7).
- **D:** Repeats step 2 at the wrong level: tolerances belong to important business services, not every internal process (the BCM view), and they were already set.

Reading 46. Risk Mitigation

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 5.

LO	Learning objective	Qs
46.a	explain and compare different ways firms address their operational risk exposures.	3
46.b	compare different types of internal controls and provide examples of each type of internal control.	9
46.c	describe control automation, internal control design, and control testing, including risks and challenges that arise in these processes and ways to make them more effective.	4
46.d	describe methods to improve the quality of an operational process and reduce the potential for human error.	6
46.e	explain how operational risk can arise with new products, new business initiatives, or mergers and acquisitions, and describe ways to mitigate these risks.	3
46.f	identify and describe approaches firms should use to mitigate the impact of operational risk events.	3
46.g	describe methods for the transfer of operational risks and the management of reputational risk, and assess their effectiveness in different situations.	4

LO 46.a

Explain and compare different ways firms address their operational risk exposures.

Q116 AnalystPrep Q.2507

A risk management consultant is developing a risk plan for a complex investment venture that launches an innovative product using cutting-edge technology in an emerging market. Numerous risks have been identified. The primary objective of the risk planning for this venture is to:

A. Eliminate all identified risks completely.
B. Transfer all risks to external parties.
C. Maximize the potential returns of the investment venture.
D. Reduce the impact of identified risks to an acceptable level.

Answer: D

Step by step

1. Risk mitigation aims to minimize operational risk to acceptable levels, consistent with the firm's risk appetite and the risk-return tradeoff.
2. Operational risk cannot be eliminated, only reduced (controls, insurance, reducing activities).
3. Transfer (insurance, outsourcing) is only one response and accountability stays with the firm.
4. Therefore the objective is to bring risk to an acceptable level: D.

Why D is correct: Risk mitigation seeks to bring risk down to a level consistent with risk appetite, because operational risk cannot be eliminated and some risk is the price of return.

Why the others are wrong

- **A:** Operational risk cannot be eliminated; complete elimination would require terminating the venture, which defeats its purpose.
- **B:** Transfer is only one of four responses, and accountability for a transferred risk remains with the original entity; not all risks can be transferred.
- **C:** Return maximization is a business objective; risk planning manages the risk-return tradeoff rather than maximizing return.

Q117 AnalystPrep Q.5075

According to ISO 31000, there are four ways to address risks (tolerate, treat, transfer, terminate). Which of the following statements is correct in this context?

A. All risks can be transferred to a third party.
B. Termination should be the first response action in case of an operational risk event.
C. Risk can be transferred through external insurance and outsourcing.
D. Tolerance involves all types of risk mitigation, especially internal controls aimed at reducing the probability.

Answer: C

Step by step

1. Tolerate = accept the risk and its consequences without mitigating it.
2. Treat = accept but mitigate (internal controls, automation, scenario planning).
3. Transfer = pass the risk to a third party, typically via insurance or outsourcing; accountability is retained.
4. Terminate = remove all exposure, a more extreme response when others will not work.
5. Only C matches these definitions.

Why C is correct: Insurance (paying a premium) and outsourcing are the standard ways to transfer operational risk to a willing third party.

Why the others are wrong

- **A:** Not all risks can be transferred, and even when risk is transferred, accountability stays with the original entity.
- **B:** Termination is the extreme response used when tolerating, treating and transferring will not work, not the first response.
- **D:** This describes treating risk; tolerating means accepting risk without any need to mitigate it.

Q118 Schweser Module Quiz 46.1 Q1

XYZ Bank just completed an internal reorganization in which it significantly improved its internal controls and management oversight of its businesses to address its operational risks. Which of the following risk responses best captures this scenario?

A. Treating risk.
B. Tolerating risk.
C. Transferring risk.
D. Terminating risk.

Answer: A

Step by step

1. The bank keeps its businesses (does not exit them) and does not pass the risk to a third party.
2. It strengthens internal controls and oversight, which are mitigants that reduce the risk's impact or likelihood.
3. Accepting a risk while mitigating it through controls is 'treating' risk: A.

Why A is correct: Treating risk means retaining the risk but mitigating it through internal controls, automation or scenario planning, which is what the bank did.

Why the others are wrong

- **B:** Tolerating means accepting risk without any mitigation; the bank actively improved controls.
- **C:** Transfer requires shifting risk to a third party (insurance, outsourcing); nothing was passed on.
- **D:** Terminating means removing the exposure by discontinuing products or services; the bank continued its businesses.

LO 46.b

Compare different types of internal controls and provide examples of each type of internal control.

Q119 AnalystPrep Q.2476

XYZ Bank has an advanced credit risk management system that is capable of pricing derivatives and supporting the generation of VaR reports. The bank has a proper VaR system in place with daily reporting to the Risk Management Committee (RMC). The bank has delegated joint responsibilities for front- and back-office functions to its few employees. In light of this information, which of the following is most likely true?

A. Since the bank has an advanced credit risk management system in place, its risk management practices are satisfactory.
B. Since the bank has the capability to generate VaR reports, its risk management practices are satisfactory.
C. The reporting frequency to the RMC must be increased.
D. The joint responsibility for front- and back-office functions makes the bank more vulnerable to trading frauds.

Answer: D

Step by step

1. Identify the control weakness: front-office (trade initiation) and back-office (confirmation and settlement) duties are combined in the same few employees.
2. Segregation of duties is perhaps the best-known preventive control for fraud: initiating, validating and settling a transaction should be performed by different parties.
3. Without it, one employee can book unauthorized trades and conceal them, so the bank is more exposed to trading fraud.
4. Sophisticated pricing, VaR and daily reporting do not compensate for a missing preventive control, so only D follows.

Why D is correct: Combining front- and back-office functions removes segregation of duties, the best-known preventive control for fraud, so one person can initiate and settle (and conceal) trades.

Why the others are wrong

- **A:** Wrong: a good credit risk system addresses credit risk only; it says nothing about operational control gaps such as missing segregation of duties.
- **B:** Wrong: the ability to produce VaR reports is a market risk measurement capability, not evidence of sound internal controls; it would not detect concealed unauthorized trades.
- **C:** Wrong: daily reporting is already frequent; the problem is the lack of a preventive control, which more frequent reporting does not fix.

Q120

Schweser Module Quiz 44.2 Q2

An investor records its investments on its internal systems and reconciles them with the investment listing on the brokerage statement each month. This reconciliation is best described as a:

A. corrective control.
B. detective control.
C. directive control.
D. preventative control.

Answer: B

Step by step

1. The reconciliation is performed monthly, after the investments have already been recorded.
2. It compares two records to find differences, i.e., it alerts the firm that an error has already occurred.
3. A control that alerts after an incident to speed resolution is a detective control.

Why B is correct: Because the reconciliation is done after the transactions are recorded, it uncovers errors that already exist rather than stopping them.

Why the others are wrong

- **A:** Corrective controls lessen the impact of an event (back-ups, continuity plans); a reconciliation finds errors but does not itself limit the loss.
- **C:** Directive controls are rules and guidance (policies, training); a reconciliation is an activity that tests records, not an instruction.
- **D:** Preventive controls act before the event; a reconciliation would be preventive only if it caught the error before the transaction executed, which is not the case with a monthly after-the-fact check.

Q121

AnalystPrep Q.5066

Which of the four main categories of internal controls is implemented to reduce the likelihood of risks materializing by addressing their possible causes?

A. Detective controls
B. Corrective controls
C. Preventative controls
D. Directive controls

Answer: C

Step by step

1. Likelihood reduction by addressing causes before the event describes preventive controls.
2. Examples: segregation of duties, access controls, levels of authorization.

Why C is correct: Preventive controls take place before potential risk events and address their causes, so they lower likelihood.

Why the others are wrong

- **A:** Detective controls alert after an incident occurs; they do not address causes in advance.
- **B:** Corrective controls lessen the impact of an event and do not influence likelihood.
- **D:** Directive controls set rules and guidance (policies, training) for executing processes; they are not defined as the control that targets causes to lower likelihood.

Q122 AnalystPrep Q.5076

Christian Grey, an FRM Part II candidate, is presenting on types of internal controls. Which of the following statements made by Grey is correct?

A. According to the Institute of Internal Auditors, controls can be of four types: preventive, detective, corrective, and directive controls.

B. Directive controls aim to alert the firm if an incident occurs to accelerate its resolution and limit the impact of the incident on the firm or its stakeholders.

C. Examples of preventive controls include smoke alarms and credit card notifications of potentially fraudulent transactions.

D. Directive controls are always part of control taxonomies.

Answer: A

Step by step

1. The IIA classification has four types: preventive, detective, corrective, directive.
2. B describes detective controls; C lists detective examples; D is contradicted by the fact that directive controls do not always appear in taxonomies.

Why A is correct: The IIA classification used in the reading consists of exactly the four types preventive, detective, corrective and directive.

Why the others are wrong

- **B:** Alerting the firm and limiting impact is the definition of a detective control, not a directive one.
- **C:** Smoke alarms and fraud notifications are detective controls, not preventive.
- **D:** Directive controls exist in every organization but do not always appear in control taxonomies, so 'always' is wrong.

Q123 AnalystPrep Q.5077

Controls can be classified in many ways. Which of the following is one of the control types in the classification of the Institute of Internal Auditors?

A. Preventive controls

B. Key controls

C. Manual controls

D. Automated controls

Answer: A

Step by step

1. IIA types: preventive, detective, corrective, directive.
2. Key vs non-key and manual vs automated are different classifications.

Why A is correct: Preventive is one of the four IIA types; the other options belong to other classification schemes.

Why the others are wrong

- **B:** Key (primary) controls are a classification by sufficiency to mitigate a risk alone; they span all four IIA types.
- **C:** Manual controls are a classification by execution method, not an IIA type.
- **D:** Automated controls are likewise classified by execution method, not an IIA type.

Q124 BT answers P2.T7.23.6.1 · BT notes P2.T7.23.6.1

Sarah is a senior risk manager at MetroBank designing a new risk mitigation system. Her plan includes: I. segregation of duties and authorization levels for high-value transactions to combat employee fraud; II. an intrusion detection system and real-time network monitoring for cyber threats; III. corrective controls such as IT system redundancies and crisis communication strategies; IV. new policies and procedures implemented through internal training; V. automation of as many controls as possible to increase reliability and free human resources. Which of the following statements about her plan is FALSE?

A. In regard to (I.), the segregation of duties is a preventative control.
B. In regard to (II.), the intrusion detection system is an example of a detective control.
C. In regard to (IV.), the policies and procedures are directive controls.
D. In regard to (V.), control automation transforms model risk into human error risk.

Answer: D
Step by step
1. A: segregation of duties and authorization levels are preventive controls: true.
2. B: intrusion detection systems alert to an incident: detective, true.
3. C: policies, procedures and training prescribe how to execute a process: directive, true.
4. D: automation removes human-error risk but creates technology and model risk, so the direction is reversed: false.

Why D is correct: Control automation transforms the risk of human error into technology and model risk, not the other way round.
Why the others are wrong
- **A:** True statement: segregation of duties is a preventive control.
- **B:** True statement: intrusion detection systems are detective controls.
- **C:** True statement: policies, procedures and training are directive controls.

Q125
GARP PE2 Q14 · Mock 4 Q15

An operational risk manager at a large retail bank reviews the framework for the bank's risk mitigation controls and classifies controls as preventive, detective, corrective, or directive. Which of the following should the manager classify as a directive control?
A. An employee training program that explains the policies and procedures for reviewing new account applications
B. A notification to a credit card customer about a potentially fraudulent transaction on that customer's account
C. An implementation of an antivirus software update across all of the bank's IT systems
D. A dual-factor authentication protocol that is used to control access to critical business systems

Answer: A
Step by step
1. Directive controls are prescriptions and rules for executing a process: policies, procedures, training, guidance.
2. A is training on policies and procedures: directive.
3. B alerts to an event: detective. C is patch/update management: corrective. D is an access control: preventive.

Why A is correct: Training that explains policies and procedures is guidance on how to execute a process, which is the defining feature of a directive control.
Why the others are wrong
- **B:** Fraud notifications alert after suspicious activity occurs, so they are detective controls.
- **C:** Software updates and patch management are listed as corrective controls, not instructions to staff.
- **D:** Dual-factor authentication is an access control that stops unauthorized entry, i.e., preventive.

Q126
Schweser Module Quiz 46.1 Q2

Business continuity planning (BCP) would most likely be described as a:
A. corrective control.
B. detective control.
C. directive control.
D. preventive control.

Answer: A
Step by step
1. BCP prepares the firm to keep operating after a disruption.
2. Continuity planning is listed as a corrective control: it lessens the impact of an event but does not change its likelihood.

Why A is correct: Corrective controls such as continuity plans, back-ups and redundancies reduce the impact once an adverse event occurs.

Why the others are wrong

- **B:** Detective controls alert to an incident (e.g., exception reports, intrusion detection); BCP does not detect anything.
- **C:** Directive controls are policies, training and guidance; although a BCP is a written plan, the reading classifies continuity planning as corrective.
- **D:** Preventive controls reduce likelihood; BCP cannot stop the disruption from happening.

Q127

AnalystPrep Q.2508

REBUILT — Original asked a generic risk-monitoring question not tied to the control-type LO; rebuilt as a classification question on monitoring-type controls with exactly one correct option.

Olivia Ross, Chief Risk Officer of Delta Energy Group, wants to show SolarTech's management that a broad set of controls is needed. She reviews four SolarTech controls: (1) a policy manual describing how purchase orders are processed, (2) an exception report that flags payments exceeding authorized limits after they are made, (3) a requirement that two managers approve any contract above a set threshold before signing, and (4) an offsite data back-up used to restore systems after a server failure. Which of the following correctly classifies control (2)?

A. Preventive control
B. Detective control
C. Corrective control
D. Directive control

Answer: B

Step by step

1. Control (2) flags payments after they are made, i.e., it alerts the firm that a limit breach has already occurred.
2. Exception reports are listed as detective controls.
3. For reference: (1) is directive, (3) is preventive (levels of authorization), (4) is corrective.

Why B is correct: An exception report alerts the firm to a breach after the fact so it can be resolved quickly, which defines a detective control.

Why the others are wrong

- **A:** Preventive controls act before the event; the two-manager pre-approval, control (3), is the preventive one, whereas the exception report operates after payment.
- **C:** Corrective controls lessen the impact of an event (back-ups, redundancies); the exception report only flags the breach.
- **D:** Directive controls are policies and guidance such as the manual in (1); the exception report does not instruct anyone how to execute a process.

LO 46.c

Describe control automation, internal control design, and control testing, including risks and challenges that arise in these processes and ways to make them more effective.

Q128

AnalystPrep Q.5078

David Hans, FRM, works as the risk manager at ABC Bank. In one of his presentations, David states that a firm's internal controls are its foundation for risk mitigation. He further goes ahead to state several issues concerning internal controls. Which of the following statements is incorrect in this regard?

A. A key control is a control that can sufficiently mitigate risk on its own.
B. Controls can either be manual or automated in nature.
C. Control automation can transform human error risk into technology and model risk.
D. Automated data back-up is an essential component of control testing.

Answer: D

Step by step

1. A key control (primary control) is sufficient, on its own, to prevent the risk: A is correct.
2. Controls may be manual or automated: B is correct.
3. Automation replaces the risk of a human failing to execute a control with the risk that the IT system fails or the

underlying model/code is wrong (technology and model risk): C is correct.
4. Automated data back-up is an example of control automation (and of a corrective control), and its failure, such as back-ups running on a full server, is an example of an automated-control breakdown. It is not a component of control testing, whose types are self-certification, examination, observation and reperformance: D is incorrect.

Why D is correct: Data back-up is a control (automation) example, not part of control testing, so the statement misclassifies it.

Why the others are wrong

- **A:** This is the standard definition of a key control, so it is a correct statement; not the answer.
- **B:** Controls can indeed be manual or automated; a correct statement.
- **C:** Automation turns the risk of human error into technology and model risk; a correct statement.

Q129 AnalystPrep Q.5079

An FRM candidate is preparing for May exam. In one of the open discussion forums, the candidate states that a control should be effectively designed so as to be applied effectively and hence be able to mitigate risk effectively. Ineffectively designed controls waste resources and may give unrealistic expectations resulting in vulnerabilities. He goes ahead to state the types of weakly designed controls. Which of the following is least likely a type of weakly designed control highlighted by the candidate?

A. "Optimistic control."
B. "More of the same."
C. "Collective controls."
D. "System-based data validation."

Answer: D

Step by step

1. GARP lists three types of weakly designed controls: optimistic controls, collective controls, and 'more of the same'.
2. Optimistic controls rely on exceptional ability or motivation of the controller (tick-box sign-offs); collective controls dilute accountability (four-eyes/maker-checker); 'more of the same' adds controls of the failed design.
3. System-based data validation (reasonableness checks) is listed as an example of control automation, not as a weak design.
4. Hence D is the least likely weak design type.

Why D is correct: System-based data validation is an automated control example, whereas the other three are the named weak-design types.

Why the others are wrong

- **A:** Optimistic controls (tick-box controls needing exceptional ability or motivation) are a named weak design.
- **B:** 'More of the same' (adding more controls of a design that already failed) is a named weak design.
- **C:** Collective controls (e.g., four-eyes checks that dilute accountability) are a named weak design.

Q130 AnalystPrep Q.5080

The Bank of India is in the process of implementing an effective control system. Its risk management unit has clarified that control designs should be assessed, and if satisfactory, they can be tested to check whether they are operationally effective. Which of the following is not a type of control testing?

A. Examination
B. Observation
C. Self-certification
D. Independence

Answer: D

Step by step

1. The four main types of control testing, in increasing scrutiny, are self-certification (inquiry), examination, observation and reperformance.
2. Examination, observation and self-certification are therefore all testing types (A, B, C).
3. Independence of the testing party is one of the factors that influence the effectiveness of testing (with frequency, scope and sample), not a testing type.

Why D is correct: Independence is an influence on testing effectiveness, not one of the four testing methods.

Why the others are wrong
- **A:** Examination (review of supporting documentation and evidence) is a testing type.
- **B:** Observation (real-time oversight of control execution) is a testing type.
- **C:** Self-certification or inquiry is the weakest testing type but is still a testing type.

Q131 AnalystPrep Q.5081

Joseph Bolts, FRM, is a risk manager at the Bank of Baroda. In his recent presentation to the board of directors, Joseph highlights that while the firm strives to establish effective control testing, we have several factors that determine the level of this effectiveness. Which of the following factors does not influence the effectiveness of control testing?

A. The independence of the testing party
B. The frequency of testing
C. Scope and sample
D. Reperformance

Answer: D

Step by step
1. GARP names the elements influencing testing effectiveness: independence of the testing party, frequency of testing, and scope and sample.
2. Reperformance (reproducing the control on a sample and comparing results) is the strongest of the four testing types, not an influencing factor.
3. So D is the answer.

Why D is correct: Reperformance is a method of control testing, not a factor that determines how effective testing is.

Why the others are wrong
- **A:** Independence avoids conflict of interest and bias; it is an influencing factor.
- **B:** Frequency should be proportionate to risk severity; it is an influencing factor.
- **C:** The scope and sample size directly affect the results; it is an influencing factor.

LO 46.d

Describe methods to improve the quality of an operational process and reduce the potential for human error.

Q132 Mock 2 Q7

REBUILT — Original wording (adjusting outcomes and metrics) could be read as Plan or Act and the topic belongs to Reading 46 (LO 46.d); rewrote the stem to match the Act-stage description and remapped LO.

A manager is improving an operational process using the plan-do-study-act (PDSA) cycle. The data from the initial implementation have been collected and analyzed against forecasts. The manager now makes the necessary adjustments to the process and sets up the process for the next cycle. Which stage of the PDSA cycle is the manager in?

A. Plan.
B. Do.
C. Study.
D. Act.

Answer: D

Step by step
1. Plan: define objective and expected outcome (what, when, where, who).
2. Do: execute the plan, collect data, document problems.
3. Study: analyze collected data versus forecasts and assess opportunities for improvement.
4. Act: make adjustments based on the analysis and create the process for the next cycle.
5. The described activity is Act (D).

Why D is correct: Adjusting the process after analysis and starting the next cycle is the Act stage.

Why the others are wrong

- **A:** Plan is the initial definition of objectives, before any data exist; confused because the next cycle begins with Plan.
- **B:** Do is executing the plan and collecting data, which has already been completed.
- **C:** Study is the analysis itself, which is already finished; the manager is now acting on its results.

Q133 AP Mock 2 Q49

REBUILT — Original said only 'in error', which leaves the error type ambiguous; added the distracted keying detail so that exactly one category fits.

An employee at ABC Bank, while distracted by a phone call, keyed a transposed digit in an account number and deposited a client's funds into the wrong account. The bank is now seeking to recover the money through litigation. Into which category of human error does this incident most likely fall?

A. Slip (skill-based error).
B. Rule-based mistake.
C. Knowledge-based mistake.
D. Violation.

Answer: A

Step by step

1. Identify the nature of the act: a transposed digit while distracted is an unintended, involuntary error in a routine task.
2. Involuntary errors caused by inattention or distraction are slips (skill-based errors).
3. Rule-based and knowledge-based mistakes involve voluntary but wrong choices; violations involve knowingly breaking a rule.
4. The later litigation to recover the money does not change the classification of the original error.

Why A is correct: A slip is an involuntary error caused by inattention, distraction or fatigue, such as a fat-finger or transposition error in routine processing.

Why the others are wrong

- **B:** Rule-based mistakes are voluntary actions based on flawed rules (e.g., incentive-driven mis-selling); no flawed rule was followed here.
- **C:** Knowledge-based mistakes arise from a wrong choice in a new, unfamiliar situation; depositing funds is routine, not novel.
- **D:** A violation is a deliberate deviation from a known rule; the employee had no such intent.

Q134 AnalystPrep Q.5082

To mitigate human errors effectively, they must first be categorized. Which of the following categories of human error is correctly described?

A. Slips: wrong choices made when someone faces a new situation owing to lack of familiarity with a process.
B. Rule-based mistakes: the perpetrator understands the right thing to do but decides to act against the rules.
C. Knowledge-based mistakes: wrong choices made when someone faces a new situation owing to lack of familiarity with a process or lack of training and guidance.
D. Violations: involuntary errors caused by inattention, distraction and fatigue.

Answer: C

Step by step

1. Recall the definitions: slips are involuntary (inattention, distraction, fatigue); rule-based mistakes come from flawed rules or conflicting incentives; knowledge-based mistakes are wrong choices in a new situation; violations are deliberate deviations from known rules.
2. Test A: it describes a knowledge-based mistake, not a slip.
3. Test B: knowingly acting against the rules is a violation, not a rule-based mistake.
4. Test C: matches the definition of a knowledge-based mistake.
5. Test D: involuntary errors from inattention are slips, not violations.

Why C is correct: Knowledge-based mistakes are wrong choices of action in a new situation, due to unfamiliarity and lack of training or guidance, remedied by training, documentation and escalation rules.

Why the others are wrong

- **A:** Swaps slips with knowledge-based mistakes; slips are involuntary errors from inattention, not choices in new situations.
- **B:** Describes a violation (knowing the rule and deviating); rule-based mistakes stem from flawed rules or per-

verse incentives.
- **D:** Describes a slip; a violation is a voluntary misdeed, not an involuntary error.

Q135 AnalystPrep Q.5083

To improve the quality of an operational process and reduce the potential for human error, a risk management unit applies several methods. Which of the following statements is INCORRECT?

A. Lean Six Sigma removes waste and reduces variation by analyzing processes and collaborative tasks.

B. Six Sigma improves quality by identifying and eliminating the causes of errors or defects and minimizing variability in manufacturing and business processes.

C. Quality improvement follows the plan, do, study, act (PDSA) cycle.

D. In the PDSA (Deming) cycle, 'do' refers to analyzing the collected data, comparing it with the set targets and evaluating opportunities for improvement.

Answer: D

Step by step

1. Statements A and B correctly describe Lean Six Sigma (waste and variation) and Six Sigma (defects and variability).
2. Statement C is correct: quality improvement uses the PDSA cycle of plan, do, study, act.
3. In PDSA, 'do' means executing the plan, collecting data and documenting problems.
4. Analyzing the data, comparing it with forecasts and assessing improvement opportunities is the 'study' step, so D is incorrect.

Why D is correct: D misassigns the 'study' activity (analysis and comparison with targets) to 'do', which is only plan execution and data collection.

Why the others are wrong

- **A:** Correct description of Lean Six Sigma, so it is not the incorrect statement.
- **B:** Correct description of Six Sigma, so it is not the incorrect statement.
- **C:** Correct: PDSA stands for plan, do, study, act, so it is not the incorrect statement.

Q136 BT answers P2.T7.23.6.2 · BT notes P2.T7.23.6.2 · BT Mock Q73

Four incidents occurred at MetroScape Bank, one of each major type of human error:
I. Jane, an experienced cashier, was multitasking and handed a customer USD 500 more than she withdrew. Her muscle memory failed in a moment of distraction.
II. After a redesigned incentive plan rewarding cross-selling, branches aggressively sold products to existing customers without sufficient regard for suitability. Customers complained.
III. Tom, an advisor, faced a novel request to combine two products. Without prior experience or training, he made an educated guess that caused significant tax consequences for the client.
IV. Lucy, a senior executive, knew the insider trading policy and traded on confidential merger information anyway, driven by personal gain.
Each of the following statements is true EXCEPT one. Which statement is FALSE?

A. The fourth (IV) is a violation, best mitigated by supervisory controls.

B. The third (III) is a knowledge-based mistake, which can be mitigated by directive controls.

C. The second (II) is a quality-based mistake, for which the best mitigation is typically a Lean Six Sigma response.

D. The first (I) is a skill-based slip, and root-cause analysis may imply process re-engineering, environmental factors and/or workflow changes.

Answer: C

Step by step

1. I: an involuntary lapse from distraction is a skill-based slip; remedies address the root cause (workspace, pace, process redesign), so D is true.
2. II: aggressive mis-selling caused by a flawed incentive scheme is a rule-based mistake, remedied by changing the rules or incentives; 'quality-based mistake' is not a human-error category.
3. III: a wrong choice in a novel situation without training is a knowledge-based mistake; directive controls (training, help files, escalation rules) apply, so B is true.
4. IV: knowingly breaching a known policy for gain is a violation; supervisory controls (monitoring, recordings) apply, so A is true.

5. Only C is false.

Why C is correct: The incentive-driven mis-selling is a rule-based mistake fixed by changing the rules or incentives; there is no 'quality-based' mistake category, and Lean Six Sigma is not the prescribed remedy.

Why the others are wrong

- **A:** True: violations are deliberate and are mitigated by supervisory controls and detection tools.
- **B:** True: knowledge-based mistakes are mitigated by directive controls such as training, documentation and escalation.
- **D:** True: slips are addressed by finding the root cause and redesigning processes, environment or workflow.

Q137 Schweser Module Quiz 46.1 Q3

A trader whose daily trading limit in a particular stock is 10,000 shares mistakenly enters a client order to sell 3,000 shares as an order to sell 30,000 shares. The trader's action is most likely an example of which of the following types of human error?

A. Slip.
B. Violation.
C. Rule-based mistake.
D. Knowledge-based mistake.

Answer: A

Step by step

1. The intended order was 3,000 shares; the trader typed 30,000, an extra zero.
2. An unintended 'fat finger' typo is an involuntary error, i.e., a slip.
3. The trader did not knowingly breach the limit, so it is not a violation; no flawed rule or new unfamiliar situation was involved.

Why A is correct: An extra zero keyed by accident is an involuntary typing error, the classic example of a slip.

Why the others are wrong

- **B:** A violation requires knowingly deviating from a rule; here the limit breach was accidental.
- **C:** Rule-based mistakes come from flawed rules or incentives; nothing suggests the limit rule was flawed.
- **D:** Knowledge-based mistakes occur in new unfamiliar situations; entering a routine order is not one.

LO 46.e

Explain how operational risk can arise with new products, new business initiatives, or mergers and acquisitions, and describe ways to mitigate these risks.

Q138 AnalystPrep Q.5085

REBUILT — Original key (C) was defensible only on a shaky board-vs-ORM argument and option D concerned credit risk; rewrote the options so exactly one statement contradicts the reading.

Bank A plans to acquire Bank B and its risk function is assessing the operational risk implications. Which of the following is NOT a correct description of the risk function's role in the acquisition?

A. Create a risk profile of the operational risk exposures inherited from Bank B.
B. Identify, assess and mitigate the risks of integrating customer and account functions, payroll and other systems, and inter-company communications.
C. Recognise that, unlike credit risk, the operational risk of the acquired firm is usually easy to assess before closing, so a single pre-deal assessment is sufficient.
D. Undertake a complex risk assessment exercise once the acquisition takes place, because the acquirer inherits the target's risks, which are often not known in advance.

Answer: C

Key correction: Source key (AnalystPrep) said C for a different option set (board of directors creating the risk profile); the item was rebuilt, and the answer letter is coincidentally the same.

Step by step

1. The acquirer inherits the target's credit, market and operational risks, and many are not known in advance (D, correct).

2. Credit (and even market) risk assessment is generally straightforward, but operational risk is harder to assess; its level and details may be known only months or years after the deal.
3. ORM helps by creating a risk profile of operational exposures (A, correct).
4. Integration itself creates operational risk (customer/account functions, payroll, communications), which ORM identifies, assesses and mitigates (B, correct).
5. C claims the opposite of the reading: operational risk is difficult to assess and cannot be settled by one pre-deal assessment. C is the NOT-correct statement.

Why C is correct: Operational risk in M&A is hard to assess and often only revealed long after closing, so a one-off pre-deal assessment is not sufficient; C contradicts this.

Why the others are wrong

- **A:** Correct statement: ORM builds a risk profile of the operational exposures; choosing it misreads the NOT in the stem.
- **B:** Correct statement: integration of customer, account, payroll systems and communications is a recognised source of operational risk handled by ORM; choosing it misreads the NOT.
- **D:** Correct statement: inherited risks are often unknown in advance, requiring a complex assessment after the deal; choosing it misreads the NOT.

Q139

AnalystPrep Q.5084

REBUILT — Stem had an OCR/wording error ('unfamiliar or unfamiliar'); corrected the wording and tightened distractor B and D; answer unchanged.

Firms face significant operational risk when they undertake new products and initiatives. Which of the following statements is correct?

A. As a best practice, the owner of each new initiative should present a business case to justify the allocation of resources.

B. When acquiring a firm, operational risk is easier to assess than credit risk.

C. The acquired firm should withhold information, because disclosure makes the acquirer's operational risk assessment harder.

D. In a merger or acquisition, the risks of the acquired business remain with the original firm and are not inherited by the acquirer.

Answer: A

Step by step

1. Best practice: the owner of a new initiative presents a business case (objective, alternatives, expected benefits, commercial aspects, risks) to justify resources, so A is correct.
2. In M&A, credit and market risk are generally straightforward to assess, while operational risk is more challenging, so B reverses the relationship.
3. More information from the target helps the acquirer assess risks, so C is wrong.
4. The acquirer inherits the target's credit, market and operational risks, so D is wrong.

Why A is correct: GARP/Schweser state that the initiative owner should present a business case to justify allocation of resources.

Why the others are wrong

- **B:** Reverses the reading: operational risk is harder to assess than credit risk because its extent is often known only months or years later.
- **C:** Withholding information would worsen the assessment; the acquirer needs as much information as possible.
- **D:** The acquirer takes on (inherits) the target's risks; they do not stay with the original firm.

Q140

Schweser Module Quiz 46.2 Q1

In which of the New Initiative Risk Assessment Process (NIRAP) business case components would a firm most likely find an analysis of project costs and funding arrangements?

A. Objective.

B. Alternatives.

C. Expected benefits.

D. Commercial aspects.

Answer: D

Step by step

1. The NIRAP business case has five components: objective, alternatives, expected benefits, commercial aspects, risks.
2. Costs, investment needs and funding arrangements are the commercial aspects.
3. So the answer is D.

Why D is correct: Commercial aspects cover the costs and funding needs of the initiative.

Why the others are wrong

- **A:** Objective covers what the firm wants to achieve and the rationale, not costs.
- **B:** Alternatives covers other options considered and why the current one was chosen.
- **C:** Expected benefits covers benefits, synergies and drawbacks of the product, not its costs and funding.

LO 46.f

Identify and describe approaches firms should use to mitigate the impact of operational risk events.

Q141

AnalystPrep Q.5086

REBUILT — Option D was grammatically broken/truncated; reworded it as a clear (incorrect) statement without changing the tested point.

Paul Schering, FRM, is a risk manager at ABC Bank. He is presenting approaches firms should use to mitigate the impact of operational risk events. Which of the following statements made by Paul is correct?

A. A contingency plan is a "Plan B", an alternative course of action if the outcome of a future event does not go as expected.

B. The first step in business continuity management (BCM) is to identify threats and risks and link them to the firm's key operational risks.

C. In a crisis, a firm needs only one response team: a technical team that assesses the incident and restores normal processes.

D. In a crisis, a firm needs only one response team: a communication team that assesses the incident and restores normal processes.

Answer: A

Step by step

1. Contingency planning is an alternative action plan ('Plan B') in case the expected outcome does not occur; it specifies who does what and when.
2. Statement A matches this definition.
3. BCM starts with securing senior-level commitment; linking threats to key operational risks comes after the management process is initiated, so B misorders the steps.
4. Firms should have at minimum two incident response teams (technical/recovery and communication), overseen by a senior manager, so C and D (only one team) are wrong.

Why A is correct: Contingency planning is defined as an alternative course of action (Plan B) if events do not go as expected, with clear roles and timing.

Why the others are wrong

- **B:** Wrong order: the first BCM step is senior-level commitment (top-down ownership); linking threats to key operational risks is a later step.
- **C:** A technical team alone is insufficient; at least two teams are required, the second being a communication team for media and stakeholders.
- **D:** A communication team deals with media and stakeholders and does not assess the incident or restore processes (that is the technical/recovery team's job); also at least two teams are needed.

Q142

Mock 1 Q5

In creating a business continuity plan for a major financial institution, provisions need to be made to manage crises and significant operational incidents effectively when they occur. A newly appointed operational risk manager is briefing the crisis-management steering group on the institution's response framework. Which of the following statements is correct for the manager to make?

A. Thoroughness in the response is the quality on which effective crisis management principally depends, because a comprehensive assessment must precede any external communication.

B. Speed is one of three essential qualities the institution must demonstrate, and a thorough business continuity management process is what makes a rapid reaction possible.

C. The recovery team should own both the restoration of normal processes and the management of media and stakeholder communication, in order to ensure a single consistent message.
D. The Recovery Time Objective specifies the volume of data that would need to be re-entered after an outage, and is largely determined by the frequency of data back-ups.

Answer: B
Step by step
1. The three essential qualities in crisis management are speed, competence and transparency.
2. Thorough BCM is the basis for a speedy reaction, so B is correct.
3. Thoroughness is not one of the three qualities, so A is wrong.
4. A technical/recovery team restores processes while a separate communication team handles media and stakeholders, so C is wrong.
5. The data re-entered after an outage and back-up frequency describe the Recovery Point Objective (RPO); RTO is the maximum tolerable downtime, so D is wrong.

Why B is correct: Speed is one of the three essential crisis qualities (with competence and transparency), and a thorough BCM underpins a rapid response.
Why the others are wrong
- **A:** Thoroughness is not one of the three qualities (speed, competence, transparency); crises spread fast, so the response must be swift.
- **C:** Recovery and communication are handled by two separate teams, overseen by a senior manager; combining them is not the prescribed structure.
- **D:** Confuses RTO with RPO: the data to be re-entered and back-up frequency define the RPO, whereas RTO is the maximum tolerable duration of disruption.

Q143 Schweser Module Quiz 46.2 Q2

A disaster recovery plan (DRP) is generally considered to be a form of:
A. event management.
B. contingency planning.
C. business continuity planning (BCP).
D. business continuity management (BCM).

Answer: B
Step by step
1. Contingency planning is the broad category of alternative action plans for when events go wrong.
2. Its two specific forms are business continuity management (BCM) and disaster recovery plans (DRP).
3. So a DRP is a form of contingency planning; BCM and DRP are peers, not parent and child.

Why B is correct: BCM and DRPs are the two specific forms of contingency planning.
Why the others are wrong
- **A:** Event management is not one of the two specific forms of contingency planning (BCM and DRP); a DRP is classified under contingency planning.
- **C:** A BCP is the plan that BCM keeps current; it is a sibling concept, not the broader category containing a DRP.
- **D:** BCM is a peer of the DRP, both being forms of contingency planning; neither is a form of the other.

LO 46.g

Describe methods for the transfer of operational risks and the management of reputational risk, and assess their effectiveness in different situations.

Q144 AnalystPrep Q.5088

REBUILT — Original key D contradicted the reading (insurance suits risks that are easily transferable) and B misstated the basis of compensation; rewrote options so exactly one statement is false.

Operational risk can be transferred through external insurance and outsourcing. Which of the following statements regarding risk transfer is INCORRECT?
A. A firm must weigh the cost of the insurance premium against the loss volatility it would otherwise bear.
B. External insurance does not fully transfer risk, because the insured relies on the insurer's capacity and will-

ingness to pay, and compensation may arrive months after the event.

C. Outsourcing creates incremental third-party risk, because the firm relies on the third party's controls operating properly.

D. Buying an external insurance policy fully transfers the risk, including non-financial consequences such as reputational damage and client detriment.

Answer: D

Key correction: Source key (AnalystPrep) said D for the statement 'It is hard to transfer both risk exposure and consequences'; that wording is actually consistent with GARP (insurance suits risks that are easily transferable) and option B of the source was also questionable, so the item was rewritten with an unambiguous false statement (still D).

Step by step

1. GARP: insurance is a premium-versus-volatility trade-off, so A is true.
2. Insurance is not a perfect transfer: the insured depends on the insurer's capacity and willingness to pay, and payment may be delayed, creating a liquidity gap, so B is true.
3. Outsourcing leaves the firm reliant on the third party's controls (third-party risk), so C is true.
4. Financial consequences are easier to compensate than qualitative impacts such as client detriment or reputational damage, which cannot be insured away, so D is false.

Why D is correct: Insurance mainly compensates financial losses and leaves residual exposure (counterparty, timing, qualitative impacts), so it cannot be described as a full transfer including reputational damage.

Why the others are wrong

- **A:** True statement: the premium-versus-volatility trade-off is a core insurance decision; choosing it means mistaking a correct statement for the false one.
- **B:** True statement: insurer capacity/willingness and delayed payment mean transfer is imperfect.
- **C:** True statement: outsourcing adds third-party risk, so it is often better described as risk sharing.

Q145 BT answers P2.T7.23.6.3 · BT notes P2.T7.23.6.3

According to GARP, operational risk controls play an important role in protecting a firm's reputation, and risk responses include tolerate, treat, transfer and terminate. In regard to treatment, transfer, and the management of reputational damage, which of the following statements is TRUE?

A. Outsourcing always reduces operational risk

B. The three R's of crisis communication are retract, refrain, and redirect

C. The four typical phases of a major operational risk event are crisis, emergency response, recovery, and restoration

D. External insurance policies make the most sense for risks that are both unpredictable and non-transferable

Answer: C

Step by step

1. GARP lists four phases of a major operational risk event: crisis, emergency response, recovery, restoration (back to normal), so C is true.
2. Crisis communication follows the three R's: regret, reason, remedy, so B is false.
3. Outsourcing transfers some risk but can increase or decrease risk depending on the provider and activity, and adds third-party risk (often risk sharing), so A is false.
4. Insurance suits risks that are predictable (priceable) and easily transferable, so D reverses both conditions.

Why C is correct: The four-phase sequence crisis, emergency response, recovery, restoration is the GARP framework for a major event.

Why the others are wrong

- **A:** Outsourcing can increase or decrease risk depending on the activity and provider, and adds third-party risk, so it does not always reduce operational risk.
- **B:** Misremembered mnemonic; the three R's are regret, reason, remedy.
- **D:** Inverted: insurance makes sense for predictable, easily transferable risks so premiums can be priced and the risk is truly mitigated.

Q146 Mock 3 Q42

A bank has completed a risk and control self-assessment (RCSA) and has identified a small number of low-frequency, high-severity operational risks that fall in the tail of its loss distribution. The bank's existing controls reduce the likelihood of these events but cannot eliminate them, and the potential losses would materially impair the bank's capital. Which of the following actions would be most appropriate for transferring this tail risk?

A. Increase the frequency of the RCSA process for the affected business lines.
B. Use a captive insurance subsidiary to cover the bank's tail risk exposure.
C. Acquire insurance against cyber risks and business interruptions from an external insurance company.
D. Allocate additional economic capital against the identified tail risks.

Answer: C

Step by step

1. The exposure is a low-frequency, high-severity tail risk that could impair capital, so the response is to transfer it outside the group.
2. GARP: external insurance is especially attractive for extreme tail events (cyber, business discontinuity, class actions).
3. A captive keeps the loss inside the group and is used to self-insure small losses.
4. More frequent RCSA improves identification, and extra capital is retention; neither transfers risk. So C.

Why C is correct: Only an external insurer takes the tail loss outside the bank, which is true risk transfer for severe events.

Why the others are wrong

- **A:** Assessment frequency improves identification only; it transfers nothing.
- **B:** A captive is group self-insurance for small losses; tail losses remain within the bank's capital.
- **D:** Holding capital is risk retention, not transfer.

Q147 GARP PE1 Q38

An enterprise risk manager at a large bank is reviewing the results of a recently completed firm-wide risk control self-assessment (RCSA). The RCSA indicates that there are multiple classes of risk for which the bank should improve its risk controls, and the manager believes the firm should consider transferring some of these risks. The manager identifies the types of risks the bank should transfer, and also assesses the impact of different risk transfer strategies on the bank's risk profile and its regulatory capital calculation. Which of the following actions should the manager recommend for the bank to take?

A. Use a captive insurance subsidiary to cover the bank's tail risk exposure.
B. Increase the bank's insurance coverage in order to benefit by deducting the cost of the premiums from the bank's required Basel operational risk capital.
C. Acquire insurance against cyber risks and business interruptions from an insurance company.
D. Transfer credit risk and fraud risk by outsourcing core operations such as loan pricing and review of new account applications.

Answer: C

Step by step

1. Tail risks such as cyber and business interruption are the GARP examples where external insurance is most attractive.
2. A: captives self-insure small losses, not tail risk.
3. B: under Basel Standardized Approach insurance recoveries are deducted from gross losses; premiums are not deducted from capital.
4. D: outsourcing does not transfer credit or fraud risk and adds third-party risk. So C.

Why C is correct: External insurance for cyber and business interruption transfers severe, priceable tail losses outside the bank.

Why the others are wrong

- **A:** Captive keeps the risk in the group and suits small, frequent losses.
- **B:** Confuses premiums with recoveries; only recoveries reduce net losses in the Basel loss calculation.
- **D:** Outsourcing cannot transfer credit or fraud risk and creates third-party risk.

Reading 47. Risk Reporting

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 6.

LO	Learning objective	Qs
47.a	identify roles and responsibilities of different organizational committees, and explain how risk reports should be developed for each committee or business function.	6
47.b	describe components of operational risk reports and explain best practices in operational risk reporting.	6
47.c	describe challenges to reporting operational risks, including characteristics of operational loss data, and explain ways to overcome these challenges.	5
47.d	explain best practices for reporting risk exposures to regulators and external stakeholders.	5

LO 47.a

Identify roles and responsibilities of different organizational committees, and explain how risk reports should be developed for each committee or business function.

Q148

AnalystPrep Q.5056

REBUILT — Original keyed an 'operational risk committee reports to executive risk committee' option not supported by the reading; rewritten to test who executes changes in response to board risk committee directives.

A firm's board risk committee reviews its operational risk reports and issues directives to change the firm's risk profile. Which of the following bodies is responsible for executing those changes?

A. The audit committee
B. The central operational risk function
C. The executive committee (ExCo)
D. Business line managers and risk champions

Answer: C

Step by step

1. The board risk committee monitors the risk management framework and sets direction; it does not execute.
2. In response to its directives, the ExCo is responsible for executing the changes needed to better manage the firm's risk profile.
3. The ExCo also oversees the proper and effective execution of the operational risk management framework.
4. The audit committee, central function and business lines have other roles, so C is correct.

Why C is correct: The ExCo is the board's steering committee and is charged with executing changes and overseeing execution of the ORM framework.

Why the others are wrong

- **A:** The audit committee provides third-level oversight and assurance via internal audit; it does not execute risk-profile changes (confusing assurance with execution).
- **B:** The central operational risk function is the second line that aggregates and reports information; it does not execute board directives (confusing reporting with execution).
- **D:** Business line managers monitor their own risks (KRIs, action plans) but are not the body that executes board risk committee directives (confusing the first line with executive-level execution).

Q149

AnalystPrep Q.5097

Which of the following stakeholder groups is authorized by the board to monitor the effectiveness of the firm's risk management framework?

A. The audit committee
B. The risk committee
C. The executive committee
D. Business line managers

Answer: B

Step by step

1. The board delegates monitoring of the risk management framework (all risks, including operational) to its risk

committee.
2. It reviews KRIs tied to risk appetite, event frequency and severity, and large-event investigations.
3. The other groups have different roles, so B is correct.

Why B is correct: The board risk committee is authorized by the board to monitor the effectiveness of the framework and to confirm the firm operates within board-set risk appetite.

Why the others are wrong

- **A:** The audit committee provides third-level oversight/assurance through internal audit, not the framework monitoring role (confusing third line with board risk oversight).
- **C:** The ExCo is the steering committee that oversees execution of the ORM framework and acts on board directives, rather than monitoring its effectiveness for the board.
- **D:** Business line managers monitor their own risks as the first line; they are not authorized by the board for framework-wide monitoring.

Q150

AnalystPrep Q.5098

REBUILT — Reading attributes the collect-and-aggregate role to the central operational risk function (the operational risk committee is the recipient); options changed so exactly one is correct.

Which of the following is responsible for collecting all relevant operational risk information from the business lines to produce aggregated, synthesized reporting and for providing feedback to the business lines?

A. Risk champions
B. The central operational risk function
C. The audit committee
D. The executive committee

Answer: B

Key correction: AnalystPrep key (B) named the operational risk committee, which is the recipient of the reporting; the reading assigns the collection role to the central operational risk function (Bionic Turtle also attributes it to that function).

Step by step

1. The second line of defense centralizes operational risk information.
2. The central operational risk function collects it from the business lines, produces aggregated, synthesized reports for the operational risk committee, and feeds back to the business lines.
3. Risk champions sit in the business lines, and the audit and executive committees have different roles, so B is correct.

Why B is correct: Aggregation, synthesis and feedback is the defining reporting role of the central (second-line) operational risk function.

Why the others are wrong

- **A:** Risk champions are within the business lines and supply data and monitor their own risks; they do not aggregate firm-wide information.
- **C:** The audit committee oversees internal audit (third line) and receives audit findings; it does not compile business-line risk information.
- **D:** The ExCo receives reports and oversees ORM framework execution; it is a recipient, not the compiler.

Q151

GARP PE2 Q52 · Mock 4 Q53

The CRO of a regional mortgage lender has asked an enterprise risk manager to develop a set of policies and procedures for the governance of the firm's risk reporting framework. The manager considers appropriate policies for the framework and also assesses how the firm should structure its risk reports for different stakeholder groups and organizational functions. Which of the following would be most appropriate for the manager to recommend?

A. The firm should report a more detailed and extensive set of key risk indicators to the board of directors than it does to business line managers to support the board's strategic risk review.
B. The operational risk committee should be responsible for executing all necessary changes to the firm's risk exposures after risk reports are reviewed.
C. The central operational risk function should be responsible for aggregating information from each of the business units about operational risk exposures.
D. The firm should emphasize forward-looking risk indicators and avoid the use of backward-looking indicators in its risk reporting.

Answer: C

Step by step

1. Recall the reporting hierarchy: business lines see the full detailed metrics; higher bodies see progressively more aggregated, action-oriented data.
2. Option A reverses this: business line managers need more detail than the board.
3. Option B misplaces execution: the ExCo executes changes after board risk committee directives.
4. Option C matches the reading: the central operational risk function collects and aggregates business-line information. Option D is wrong because both loss-based (backward-looking) and forward-looking indicators are needed.

Why C is correct: The central (second-line) function centralizes operational risk information and produces the aggregated, firm-wide view.

Why the others are wrong

- **A:** Reverses the reporting cake: business line managers get the fullest set of KRIs; the board gets high-level summaries.
- **B:** The operational risk committee reviews reports; the ExCo is responsible for executing changes in response to board risk committee directives.
- **D:** Reports should combine forward-looking indicators with backward-looking data such as loss events; dropping the latter loses essential information.

Q152 Schweser Module Quiz 47.1 Q1

Which of the following statements about the board of directors is most accurate regarding operational risk monitoring?

A. The audit committee serves as the second line of risk oversight.
B. The risk committee is only tasked with monitoring operational-type risks.
C. A committee member cannot serve on both the risk committee and the audit committee.
D. The executive committee oversees the execution of the operational risk management framework.

Answer: D

Step by step

1. Check each statement against the committee roles.
2. The ExCo oversees the proper and effective execution of the ORM framework, so D is true.
3. The audit committee is the third level; the risk committee covers operational and all other risks; members may serve on both committees.

Why D is correct: The ExCo is charged with overseeing execution of the ORM framework and acting on board risk committee directives.

Why the others are wrong

- **A:** Internal audit and the audit committee are the third line; the central operational risk function is the second.
- **B:** The risk committee monitors operational risk and all other risks to the firm.
- **C:** Board members may serve concurrently on both committees to provide continuity between audit findings and mitigation.

Q153 Schweser Module Quiz 47.1 Q2

REBUILT — Option C 'business line committee' is not a body in the reading; changed to business line managers and risk champions.

The internal group most likely to focus on key risk indicators (KRIs) and benchmarking is:

A. the audit committee.
B. the executive committee.
C. business line managers and risk champions.
D. the central operational risk function.

Answer: C

Step by step

1. Business line managers and risk champions are closest to operational risk data.
2. Their reports focus on KRIs, action plan progress and risk types/impacts.
3. They benchmark their risks against other business lines or the firm-wide average, so C is correct.

Why C is correct: The first line monitors its own risks via KRIs and benchmarks them against other lines or firm

averages.

Why the others are wrong

- **A:** The audit committee receives internal audit findings on controls (third line), not KRI benchmarking.
- **B:** The ExCo receives exposures, trends, events, action plans and culture indicators to oversee framework execution, not business-level KRI benchmarking.
- **D:** The central function aggregates and synthesizes firm-wide information; the KRI and benchmarking focus sits with the business lines.

LO 47.b

Describe components of operational risk reports and explain best practices in operational risk reporting.

Q154

AnalystPrep Q.5094

Which of the following components of operational risk reporting involves reporting a list of the top overall risks?

A. Heatmap and risk register.
B. Risk appetite metrics.
C. Top-10 risks and risk outlook.
D. Incidents and near misses.

Answer: C

Step by step

1. The internal ORM report has seven components: prioritized (top-10) risks and outlook, heatmap and risk register, risk appetite metrics, KRIs and issue monitoring, incidents and near misses, action plans, and emerging risks.
2. The component that narrows the full risk inventory to the most important risks, with ranking, description, inherent/residual ratings and outlook, is the top-10 risks and risk outlook.

Why C is correct: The top-10 risks and outlook component is the prioritized list of the most important risks out of the whole inventory, with an outlook for each.

Why the others are wrong

- **A:** The register lists all risks with their controls and the heatmap plots them by likelihood and severity; it is not a short ranked list of the top risks.
- **B:** Risk appetite metrics (risk appetite KRIs) measure compliance with limits, not the most important risks.
- **D:** Incidents and near misses report realized events and avoided losses, not a forward-looking ranked list of risks.

Q155

BT answers P2.T7.23.7.1 · BT notes P2.T7.23.7.1

Sally has just started her new job as Chief Risk Officer (CRO) at a large bank with operations in several major cities on five continents. The board of directors includes an executive committee, an audit committee, and a risk committee. External stakeholders, as usual, include regulators and the public. With respect to risk management, the bank utilizes a traditional three lines of defense (3LoD) model. In terms of risk reporting, the bank's risk reporting function is a bit disorganized because various leaders, over the years, have requested various types of reports. Sally wants to unify and, if possible, streamline the risk reporting function. In regard to risk reporting best practices, which of the following statements is TRUE?

A. Risk reports should be condensed to just a few pages to avoid overwhelming the readers.
B. Averages are better than medians because outliers should be incorporated into summary statistics rather than reported separately.
C. Different management levels and decision-making bodies should receive the same operational risk information (i.e., identical in type and level of detail) to avoid conflicts.
D. The central operational risk function collects and synthesizes operational risk information for the operational risk committee and provides feedback to the business lines based on aggregated operational risk reporting.

Answer: D

Step by step

1. Evaluate each statement against the best practices in operational risk reporting.
2. A: reports must balance size; condensing to a few pages, especially with inappropriate aggregation, makes them

meaningless.
3. B: operational loss data are asymmetric, so averages are misleading; the median and first/third quartiles are preferred.
4. C: different audiences need different type and detail of information (reporting cake).
5. D: matches the role of the second-line central operational risk function.

Why D is correct: The central operational risk function (second line) gathers information from business lines, produces aggregated reporting for the operational risk committee, and gives feedback to the business lines.

Why the others are wrong

- **A:** Over-condensed reports can become meaningless; overly long reports bury key insights. The goal is balance, not a few pages.
- **B:** Reverses the guidance: with skewed loss data driven by a few outliers, medians and quartiles are better than averages.
- **C:** Ignores tailoring: senior management, business departments and units do not all need the same type or level of detail.

Q156

BT answers P2.T7.23.7.2 · BT notes P2.T7.23.7.2

Peter is a certified Financial Risk Manager (FRM) recently hired by a large publicly traded financial services company. The company's governance structure includes the usual board committees, i.e., executive, audit, risk, and compensation. With respect to the operational risk function, there also exists an operational risk committee, business line managers, and risk champions. The primary operational risk management (ORM) report is published internally. Peter notices that this ORM report contains five components: 1. Top-10 risks and risk outlook; 2. Heatmap and risk register; 3. Risk appetite metrics; 4. KRIs and issue monitoring; 5. Action plans and follow-up. Peter further observes that two components are missing from the company's ORM report. Which of the following best describes the two missing components?

A. Combined assurance map; and Inherent plus residual risk plot
B. Conduct metric dashboard; and Risk champions horse race status
C. Actual losses plus near-miss incidents; and horizon scan's emerging risks
D. Cyberattack domain registry; and Three-layer "decide-act-monitor" cake

Answer: C

Step by step

1. The seven standard ORM report components are: top-10 risks and outlook; heatmap and risk register; risk appetite metrics; KRIs and issue monitoring; risk events/incidents and near misses; action plans and remediation; emerging risks and horizon scanning.
2. Peter's report has the first four plus action plans.
3. Missing: incidents/near misses (actual losses and near misses) and emerging risks/horizon scanning.

Why C is correct: The two components absent from Peter's list are incident and near-miss reporting and emerging-risk (horizon scanning) reporting.

Why the others are wrong

- **A:** A combined assurance map and an inherent/residual plot are not standard components (inherent/residual ratings sit inside the top-10 and register).
- **B:** A conduct metric dashboard and a risk champions 'horse race' are not among the seven components.
- **D:** A cyberattack registry is not a component, and the 'cake' is a framework for tiering information by audience, not a report component.

Q157

Mock 2 Q16

Risk reporting is often structured using a "reporting cake" model in which information is presented at different tiers depending on the audience and the decisions being supported. A chief risk officer is preparing to present to the board risk committee for the purpose of finalizing the firm's annual strategic plan. Which tier of the reporting cake is the CRO most likely operating in?

A. Tier 1.
B. Tier 2.
C. Tier 3.
D. Tier 4.

Answer: C

Key correction: Source key (Mock 2) said A; correct is C because in the GARP/Schweser reporting cake Tier 1 is the bottom layer (business lines) and Tier 3 is the top layer (executive and board risk committee).

Step by step

1. Reporting cake tiers: Tier 1 = business line managers and risk champions (full metrics, daily monitoring); Tier 2 = operational risk committee (items requiring action); Tier 3 = executive and board risk committee (high-level information to decide).
2. A board risk committee finalizing strategic plans is a Tier 3 audience.
3. There is no Tier 4.

Why C is correct: The executive and board risk committee are the top layer (Tier 3), receiving the most aggregated information needed to make strategic decisions.

Why the others are wrong

- **A:** Tier 1 is the bottom layer for business line managers and risk champions with the full set of metrics; the numbering is not reversed.
- **B:** Tier 2 is the operational risk committee, which receives only items needing action.
- **D:** The reporting cake has only three tiers.

Q158 Schweser Module Quiz 47.2 Q1

Using a reporting cake model, high-level data such as leading indicators and strategic plan progress will most likely be reported in which tier?

A. Tier 1.
B. Tier 2.
C. Tier 3.
D. Tier 4.

Answer: C

Step by step

1. Tier 1: business line managers/risk champions, full metrics monitored daily.
2. Tier 2: operational risk committee, action-requiring information only.
3. Tier 3: executive and board risk committee, high-level data such as leading indicators, performance trends and progress on strategic plans.
4. Leading indicators and strategic plan progress therefore belong to Tier 3.

Why C is correct: High-level leading indicators, trends and strategic progress are what executives and the board risk committee need to decide, which is the top tier (Tier 3).

Why the others are wrong

- **A:** Tier 1 carries the full, granular metric set for daily monitoring, not high-level data.
- **B:** Tier 2 carries only items requiring action by the operational risk committee.
- **D:** There are only three tiers.

Q159 Schweser Module Quiz 47.2 Q2

A presentation provided to executive leadership on the top 10 risks an entity is facing should include all of the following except:

A. descriptions of the risks.
B. timelines for risk eliminations.
C. inherent and residual elements of each risk.
D. a ranked order based on likelihood and severity.

Answer: B

Step by step

1. For each top-10 risk the presentation includes: ranking, description, inherent components (likelihood, severity, rating), residual components, and outlook (improving, worsening, status quo).
2. Descriptions (A), inherent/residual elements (C) and a ranking driven by likelihood and severity (D) are included.
3. Timelines for eliminating risks are not included; the forward-looking element is an outlook, and risks are mitigated, not eliminated.

Why B is correct: The top-10 template shows ranking, description, inherent and residual ratings and an outlook,

but not timelines for eliminating risks.

Why the others are wrong

- **A:** Risk descriptions are a standard element, so it is included.
- **C:** Inherent and residual likelihood, severity and rating are standard elements, so it is included.
- **D:** Ranking by the magnitude of risk (likelihood and severity) is a standard element, so it is included.

LO 47.c

Describe challenges to reporting operational risks, including characteristics of operational loss data, and explain ways to overcome these challenges.

Q160

AnalystPrep Q.5096

Which of the following is NOT one of the three options for aggregating qualitative risk data for reporting?

A. Conversion and addition
B. Categorization
C. Horizon scanning
D. Worst-case reporting

Answer: C

Step by step

1. The three options for aggregating qualitative data are (1) conversion and addition, (2) categorization, and (3) worst-case reporting.
2. Options A, B and D match these three; horizon scanning is a forward-looking identification technique, not an aggregation method.

Why C is correct: Qualitative data (e.g., ratings, colors, scores) is aggregated by converting to numbers and adding, by categorizing (e.g., red/yellow/green counts), or by reporting the worst case; horizon scanning is none of these.

Why the others are wrong

- **A:** Is one of the three methods (convert qualitative metrics to numeric scores and add them).
- **B:** Is one of the three methods (report counts or proportions by color or score band).
- **D:** Is one of the three methods (the most conservative approach: the worst score in the group is reported).

Q161

AnalystPrep Q.5099

REBUILT — Original stem was vague (options mixed report components with a challenge); reworded so exactly one option is a reporting challenge and the others are report components.

Which of the following is a challenge, rather than a standard component, of non-financial (operational) risk reporting?

A. Risk appetite metrics
B. Action plans and follow-up
C. Asymmetry of operational risk event data
D. Incidents and near misses

Answer: C

Step by step

1. Risk appetite metrics, action plans and follow-up, and incidents and near misses are content items that go into operational risk reports.
2. Operational losses are heavily skewed: a small number of low-frequency, high-severity events drive most of the loss, which makes averages and aggregate reporting difficult.
3. Asymmetry is therefore the reporting challenge.

Why C is correct: Heavily skewed loss data (few large, many small events) complicates assessing and reporting loss experience, whereas the other options are things reported.

Why the others are wrong

- **A:** A report component (how exposures compare with appetite), not a challenge.
- **B:** A report component (tracking remediation of identified issues), not a challenge.
- **D:** A report component (event and near-miss reporting), not a challenge.

Q162 BT Mock Q78 · BT answers P2.T7.23.7.3 · BT notes P2.T7.23.7.3

The central operational risk function (ORF) has primary responsibility for operational risk reporting. Regarding the challenges of reporting operational risks, each of the following statements is true EXCEPT which is false?

A. The ORF should focus scarce resources on preventing and addressing major incidents rather than being distracted by daily volatility.

B. When benchmarking across business units, losses should be reported in dollar terms rather than as percentages or basis points, because low percentages could mask high dollar losses.

C. The ORF should regularly review small, frequent losses to spot potential control breaches or process flaws; if these losses are random and stable, their average cost can be included in the service price.

D. The ORF can address the challenge of compiling qualitative risk data through conversion and addition, categorization, and worst-case reporting.

Answer: B

Key correction: Source key (BT Mock Q78) said A; correct is B because A is true (GARP: focus on large incidents, not daily volatility) and the false statement is B (percentages/basis points, not dollars, are preferred for benchmarking).

Step by step

1. A is true: resources should target the prevention of large incidents, not frequent insignificant events.
2. C is true: small frequent losses are reviewed for control breaches; if random and stable, their average cost can be passed on in pricing.
3. D is true: the three qualitative aggregation methods are listed correctly.
4. B is false: benchmarking uses percentages (of gross income, budget) or basis points of capital because these make entities of different sizes comparable.

Why B is correct: Benchmarking across units of different sizes requires scale-free measures (percent of gross income or budget, basis points of capital), not raw dollar amounts.

Why the others are wrong

- **A:** True statement; choosing it follows the Mock key, which wrongly treats ignoring daily volatility as an error.
- **C:** True statement; GARP recommends reviewing small losses for control flaws and pricing in random, stable ones.
- **D:** True statement; it lists the three recognized aggregation methods.

Q163 Schweser Module Quiz 47.3 Q1

Operational risk distributions do not lend themselves to using averages (mean values) for all of the following reasons EXCEPT:

A. the distribution is asymmetric.

B. the data may have significant variances.

C. there may be a large quantity of outliers.

D. there is minimal skewness in either the right or left tail.

Answer: D

Step by step

1. Averages are meaningful for symmetric, bell-shaped, single-mode distributions with no skewness or outliers and minimal variance.
2. Operational loss data is asymmetric, has large variance and many outliers, and is heavily skewed.
3. Minimal skewness would favor using averages, so D is not a reason against them.

Why D is correct: Operational loss distributions have heavy skew, so a lack of skewness is the opposite of the real reason averages mislead.

Why the others are wrong

- **A:** A valid reason: asymmetry makes the mean unrepresentative.
- **B:** A valid reason: large variance means the mean hides dispersion.
- **C:** A valid reason: outliers pull the mean and are where the real information lies.

Q164 Schweser Module Quiz 47.3 Q2

In a combined assurance approach, the central operational risk function (second line of defense) will perform which of the following activities?

A. Internal audit reviews.

B. Deep dives into specific types of risks.
C. Overall assessments of risks and controls.
D. Oversight of the activities performed by the third line.

Answer: B
Step by step
1. First line: assessment of risks and controls, controls testing, attestation.
2. Second line: oversight of first-line activities, sample control testing, and deep dives into specific risk types.
3. Third line: periodic internal audit reviews.
4. Only B matches the second line.

Why B is correct: Deep dives into specific risk types, together with oversight of the first line and sample testing, are second-line activities.
Why the others are wrong
- **A:** Internal audit is the third line.
- **C:** Risk and control assessment is performed by the first line.
- **D:** The second line oversees the first line, not the third; the third line is independent assurance.

LO 47.d

Explain best practices for reporting risk exposures to regulators and external stakeholders.

Q165 AnalystPrep Q.5091

Which of the following is not a type of information critical in the operational risk requirements?
A. Qualitative information on operational risk management
B. Historical losses
C. Business indicator and subcomponents
D. Risk appetite metrics

Answer: D
Step by step
1. Pillar 3 operational risk disclosure has three required types: qualitative information on ORM, historical losses (10 years of aggregate losses), and the business indicator and its subcomponents.
2. Options A, B and C match these three types.
3. Risk appetite metrics are not one of the prescribed Pillar 3 disclosure categories, so D is the exception.

Why D is correct: Pillar 3 prescribes exactly three operational risk disclosure types; risk appetite metrics are an internal reporting item, not a required disclosure category.
Why the others are wrong
- **A:** This is the first required type (governance, policies, systems, risk transfer); it is required, so it is not the exception.
- **B:** This is the second required type (10 years of aggregate losses); it is required.
- **C:** This is the third required type and the primary input to the capital calculation; it is required.

Q166 AnalystPrep Q.5092

Which of the following types of information pertains to the presentation of an entity's governance and risk management structures that are established to manage and mitigate risk?
A. Qualitative information on operational risk management
B. Historical losses
C. Business indicators and subcomponents
D. Incidents and near misses

Answer: A
Step by step
1. Pillar 3 qualitative information covers the governance and risk management arrangements used to manage, mitigate or transfer operational risk, the ORM and control function structure, and policies, procedures and

guidelines.
2. Historical losses and business indicator items are quantitative templates, and incidents/near misses are internal reporting items, not a Pillar 3 disclosure type.
3. So the answer is A.

Why A is correct: The qualitative disclosure type is the one that describes governance structures, policies and risk management arrangements.

Why the others are wrong

- **B:** Historical losses are quantitative: 10 years of aggregate losses, not a description of governance structures.
- **C:** The business indicator and subcomponents are quantitative inputs to the capital calculation.
- **D:** Incidents and near misses are internal management reporting content, not a Pillar 3 disclosure type, and do not describe structures.

Q167 Mock 1 Q62

An operational risk analyst manages all financial and operational disclosures for the Recharge Bank. In order to meet operational risk disclosure requirements established by the Basel regulatory framework, she will most likely include all of the following in her annual disclosures except:

A. arrangements the bank has in place to transfer operational risks.
B. information on systems and data used to measure operational risk.
C. narratives covering key drivers of operational risk capital calculations.
D. historical loss data covering a minimum of 3 years of aggregate operational losses.

Answer: D

Step by step

1. Qualitative disclosure includes risk transfer arrangements (A) and systems and data used to measure operational risk (B).
2. Narratives on significant changes and key drivers accompany the quantitative data (C).
3. Historical loss disclosure requires 10 years of aggregate losses, not 3 years (3 years applies only to the business indicator component data), so D is the exception.

Why D is correct: Pillar 3 requires 10 years of aggregate operational losses; a 3-year minimum misstates the requirement.

Why the others are wrong

- **A:** Risk transfer arrangements are part of the required qualitative information.
- **B:** Systems and data used to measure operational risk are explicitly part of qualitative disclosure.
- **C:** Narratives on key drivers and significant changes are expected alongside the quantitative data.

Q168 Schweser Module Quiz 47.4 Q1

Significant operational risk event triggers can be categorized under all of the following criteria except:

A. stability.
B. materiality.
C. functional.
D. reputation.

Answer: C

Step by step

1. The notification triggers are materiality, reputation, resilience and stability.
2. Stability, materiality and reputation are on the list.
3. Functional is not a trigger criterion, so C is the exception.

Why C is correct: The four regulator-notification trigger criteria are materiality, reputation, resilience and stability; 'functional' is not one.

Why the others are wrong

- **A:** Stability (events that could result in serious consequences to the financial system) is a trigger.
- **B:** Materiality (events that are significant relative to a loss or materiality threshold) is a trigger.
- **D:** Reputation (events significantly affecting the firm's reputation) is a trigger.

Q169 Schweser Module Quiz 47.4 Q2

To meet operational risk disclosure requirements, the number of years of historical aggregate operational losses that must be disclosed is closest to:

A. 3 years.
B. 5 years.
C. 10 years.
D. 15 years.

Answer: C

Step by step

1. Pillar 3 requires disclosure of aggregate operational losses over the past 10 years.
2. Three years is the window for business indicator component data, not for losses.
3. So the answer is 10 years.

Why C is correct: The historical loss disclosure covers 10 years of aggregate operational losses.

Why the others are wrong

- **A:** Confuses the loss window with the 3-year data window used for the business indicator component.
- **B:** Not a Pillar 3 requirement for the loss history; 10 years is required.
- **D:** Overstates the requirement; no 15-year disclosure is required.

Reading 48. Integrated Risk Management

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 7.

LO	Learning objective	Qs
48.a	describe the role of risk governance, risk appetite, and risk culture in the context of an enterprise risk management (ERM) framework.	4
48.b	explain and differentiate between regulatory capital and economic capital requirements as prescribed in Basel regulations.	4
48.c	describe the elements of a sound stress-testing framework for financial institutions and explain best practices for stress testing.	4
48.d	explain challenges and considerations when developing and implementing models used in stress testing operational risk.	5

LO 48.a

Describe the role of risk governance, risk appetite, and risk culture in the context of an enterprise risk management (ERM) framework.

Q170

BT answers P2.T7.25.1.1

REBUILT — Original had several defensible options (A, C, D all describe valid line roles) and C misassigned independent review; rewrote as a single-answer question on the first line of defense.

Bobber Bank Plc is implementing an enterprise risk management (ERM) framework based on the three lines of defense model, so that risk-taking stays aligned with the bank's risk appetite. Which of the following best describes the role of the first line of defense in this framework?

A. Business line staff and management own and manage risk day to day, with decision-making authority and accountability, within the limits set by the risk appetite.

B. The risk management function independently reviews the effectiveness of risk management activities and reports directly to the board of directors.

C. The operational, market and credit risk departments oversee business-line risk-taking and establish the risk tools, methods and models.

D. Internal auditors and independent third parties evaluate the design of risk activities and report independently to the board.

Answer: A

Step by step

1. The first line of defense is business line staff and management.
2. They hold primary decision-making authority, accountability and responsibility for risk, as designated risk owners who identify, measure, mitigate and report risk, balancing risk and reward.
3. They act within the risk appetite set under the governance framework.
4. Oversight (risk management, compliance, legal) is the second line; independent evaluation reporting to the board is the third line.

Why A is correct: The first line owns the risks; it takes and manages them within appetite limits, which embeds risk management in daily decisions.

Why the others are wrong

- **B:** Describes independent assurance reporting to the board, which is the third line, not the first; the mistake is confusing review with ownership.
- **C:** Describes the second line (oversight, tools, methods, models), mistaking oversight for ownership.
- **D:** Describes the third line (internal audit and independent third parties), confusing assurance with day-to-day risk management.

Q171

Mock 2 Q70

CE Bank began operations only two years ago. As the bank continues to grow, the CEO identifies the need for a risk management department and an enterprise risk management (ERM) framework. Recognizing that they must balance the pursuit of objectives and the incurrence of risk, they have hired an outside consulting firm to help develop the framework. Each of the following statements is an accurate representation of framework components

except:

A. risk governance incorporates three to four standard lines of defense.
B. risk appetite may drive decisions that can lead to refining desired objectives.
C. reporting and monitoring are continuous due to constantly changing exposures to risk.
D. risk culture and corporate culture are independent and should be managed separately.

Answer: D

Step by step

1. Check each statement against the ERM framework.
2. A: three lines of defense, plus external audit often seen as a fourth - accurate.
3. B: appetite dictates how much risk is taken to achieve objectives, which can feed back into refining them - accurate.
4. C: risks and exposures change, so monitoring/reporting is continuous - accurate.
5. D: risk culture is inseparable from corporate culture - inaccurate, so it is the exception.

Why D is correct: Risk culture cannot be separated from corporate culture (shared values, beliefs and behaviors), so statement D is false.

Why the others are wrong

- **A:** True: three lines, with external audit often a fourth.
- **B:** True: appetite and objectives interact.
- **C:** True: exposures change, so ERM is managed continuously.

Q172

Schweser Module Quiz 48.1 Q1

Sovereign Bank has an internal audit department, but the bank CEO hires an external auditor to review its financial statements and assess its internal control system. The audit firm is most likely considered to be which line of defense?

A. First.
B. Second.
C. Third.
D. Fourth.

Answer: D

Step by step

1. First line: business line staff and management.
2. Second line: risk management oversight (risk departments, compliance, legal).
3. Third line: internal auditors and independent third parties, reporting to the board.
4. External audit adds an extra layer of verification and is often referred to as a fourth line of defense.

Why D is correct: An external auditor sits outside the three internal lines and is conventionally treated as a fourth line of defense.

Why the others are wrong

- **A:** First is business line staff and management, which the auditor is not.
- **B:** Second is internal risk oversight; an external firm does not perform it.
- **C:** Third is internal audit, which the bank already has; the external auditor is an additional layer beyond it.

Q173

AnalystPrep Q.5100

REBUILT — Original 'integrated risk management' statements were vague and options B/D were not clearly wrong; rebuilt as a clear question on the roles of governance, culture and appetite in ERM.

Which of the following statements best describes how risk governance, risk culture and risk appetite function within an enterprise risk management (ERM) framework?

A. Governance sets the roles of the three lines of defense and reporting committees, culture reflects the behaviors and values around managing risk, and appetite defines the level of risk the firm is willing to tolerate.
B. Governance defines the level of risk the firm is willing to tolerate, culture assigns the three lines of defense, and appetite reflects employee behaviors and values.
C. Risk appetite applies only to credit and market risk, whereas governance and culture apply to operational risk.
D. Risk culture is determined solely by the second line of defense, so governance and appetite do not influence it.

Answer: A

Step by step

1. ERM priorities are guided by three elements: governance, culture and appetite.
2. Governance: roles and responsibilities of the three lines of defense and reporting/decision committees.
3. Culture: behavior and values associated with managing risk.
4. Appetite: priorities and the level of risk exposure the firm will tolerate; only A assigns each element correctly.

Why A is correct: Each element has a distinct function: governance assigns roles, culture shapes behaviors, appetite sets tolerated risk.

Why the others are wrong

- **B:** Swaps the functions of governance, culture and appetite.
- **C:** Appetite covers credit, market, liquidity and operational risk, not only credit and market.
- **D:** Culture is a firm-wide corporate trait involving executives, leadership and employees, not owned by one line; governance and appetite interact with it.

LO 48.b

Explain and differentiate between regulatory capital and economic capital requirements as prescribed in Basel regulations.

Q174

Mock 2 Q3

REBUILT — Original relied on 'compartmentalized vs unified' terminology not in the curriculum and had a debatable option; rebuilt around the GARP/Schweser capital aggregation concept (inter-risk diversification).

A risk manager at bank XYZ is aggregating stand-alone capital for credit risk, market risk and operational risk. Which of the following statements about the aggregated capital is correct?

A. Total aggregated capital equals the sum of the stand-alone capital amounts because each risk class must be fully covered independently.

B. Total aggregated capital is lower than the sum of the stand-alone amounts because not all risks materialize at the same time, which produces inter-risk diversification benefits.

C. Total aggregated capital is higher than the sum of the stand-alone amounts because interactions between risk classes compound losses.

D. Operational risk provides no diversification benefit because it is perfectly correlated with credit and market risk.

Answer: B

Step by step

1. Capital is first computed stand-alone for each risk class (credit, market, operational).
2. Aggregation uses a correlation matrix across the classes, since the classes have different drivers and losses do not all peak together.
3. Correlation below 1 gives total capital below the simple sum; the reduction is the diversification benefit.
4. Operational risk in particular has low correlation with financial risks, so it adds large diversification benefits.

Why B is correct: Inter-risk diversification means aggregated capital is less than the sum of stand-alone capital.

Why the others are wrong

- **A:** Simple summation assumes perfect correlation and ignores diversification.
- **C:** Reverses the effect of diversification; compounding is a stress consideration, not the aggregation result under imperfect correlation.
- **D:** Operational risk has typically low, not perfect, correlation with other risk types, so it does give diversification.

Q175

BT answers P2.T7.25.1.3

REBUILT — Source numbers were inconsistent with its explanation (EL+UL convention, \$12bn/\$10bn vs \$72bn/\$70bn) and options were vague; restated with consistent figures, EC defined as the 99.9% unexpected loss, and one clearly correct comparison.

A bank estimates economic capital for its credit portfolio as the 99.9% one-year unexpected loss (the 99.9% loss quantile less the expected loss, which is covered by pricing and provisions). Its internal model gives: expected loss (EL) = \$5 billion; 99.9% unexpected loss (UL) = \$72 billion. Its Basel (Pillar 1) regulatory capital requirement for the same portfolio is \$70 billion. Which of the following statements is most accurate?

A. Economic capital is \$72 billion, \$2 billion above the regulatory minimum, which is acceptable because regula-

tory capital is a baseline and economic capital reflects the bank's own risk profile.

B. Economic capital is $77 billion, $7 billion above the regulatory requirement, because it equals EL plus UL.

C. Economic capital is $67 billion, $3 billion below the regulatory requirement, so the bank is in breach of Basel rules.

D. Economic capital is $70 billion, equal to the regulatory requirement, because both are calibrated to the same confidence level.

Answer: A

Key correction: Source key (BT) said A; the answer letter stands, but the source explanation used EC = EL + UL, which conflicts with the convention that economic capital covers unexpected loss only.

Step by step

1. Economic capital covers unexpected loss: $EC = \text{VaR}_{99.9\%} - EL = UL = 72$ billion (EL is covered by pricing/provisions).
2. Regulatory capital (Pillar 1 minimum) = 70 billion.
3. Difference: $72 - 70 = +2$ billion, so EC exceeds the regulatory requirement.
4. Regulatory capital is a minimum baseline and not very risk sensitive; economic capital (closest to Pillars 1 + 2 combined) reflects the firm's own risk profile, so exceeding the minimum is normal.

Why A is correct: Economic capital equals the unexpected loss of $72bn, which is above the $70bn regulatory minimum; regulatory capital is only a baseline while economic capital reflects the bank's own risk profile.

Why the others are wrong

- **B:** Adds EL to UL ($72 + 5 = 77$); expected loss is not part of economic capital.
- **C:** Subtracts EL from UL a second time ($72 - 5 = 67$); UL is already net of EL. Also wrongly treats a shortfall versus a Pillar 1 minimum as the issue.
- **D:** Assumes economic and regulatory capital must coincide; regulatory capital is formula-based and not very risk sensitive, so the two generally differ.

Q176

AnalystPrep Q.5103

REBUILT — Original key said Basel has 'no legal grounds', which is overstated; reworded the correct option to state that Basel standards are not laws or legally binding unless enacted by national authorities.

A newly hired risk manager is presenting to the risk committee on how regulatory and economic capital help ensure an institution's solvency by covering unexpected losses. Which of the following statements is correct?

A. An enterprise risk management framework consists only of regulatory capital and economic capital.

B. Regulatory capital is the internal capital that firms estimate to reflect their own risk profile and potential unexpected losses.

C. Pillar 2 of the Basel framework concerns market discipline through mandatory risk disclosures.

D. Basel standards are not laws and are not legally binding in themselves; they take legal force only where national authorities enact them.

Answer: D

Key correction: Source key (AnalystPrep) said D with the wording 'no legal grounds'; D is still the best answer, but only as restated (non-binding unless enacted nationally).

Step by step

1. Basel standards are issued by the BCBS as guidance; they are not laws, but some jurisdictions have enacted parts of Basel II/III into law, so D is correct.
2. Regulatory capital is the minimum set under Pillar 1 by regulators; the internally estimated capital is economic capital, so B swaps the definitions.
3. Pillar 2 is the supervisory review process; Pillar 3 is market discipline, so C mislabels the pillar.
4. ERM also includes governance, risk appetite, culture and more, so A is too narrow.

Why D is correct: The BCBS issues guidance without legal force; it binds only when national legislation or regulators adopt it, whereas economic capital (not regulatory) is the internal estimate.

Why the others are wrong

- **A:** Treats ERM as only capital measures; it also covers governance, risk appetite, culture and risk-taking processes.
- **B:** Confuses regulatory with economic capital; regulatory capital is the externally mandated minimum.
- **C:** Confuses Pillar 2 (supervisory review) with Pillar 3 (market discipline).

Q177 Schweser Module Quiz 48.1 Q3

Assume that a credit rating of A has a default probability of 0.07%, and a credit rating of AA has a default probability of 0.03%. If a bank is seeking a target rating of A, it will want to ensure that its economic capital will cover unexpected losses at a confidence level of:

A. 93.00%.
B. 97.00%.
C. 99.93%.
D. 99.97%.

Answer: C

Step by step

1. Confidence level = 1 - default probability of the target rating.
2. Target rating A: $100\% - 0.07\% = 99.93\%$.

Why C is correct: Economic capital is sized so the chance of losses exceeding it equals the target rating's default probability, so confidence = 1 - 0.07% = 99.93%.

Why the others are wrong

- **A:** Subtracts 0.07 as whole percentage points (100 - 7) instead of 0.07%.
- **B:** Subtracts 3 as whole percentage points (100 - 3) and uses the AA probability instead of 0.07%.
- **D:** Uses the AA default probability (100% - 0.03%), the confidence for the wrong target rating.

LO 48.c

Describe the elements of a sound stress-testing framework for financial institutions and explain best practices for stress testing.

Q178 AnalystPrep Mock 2 Q48

Which type of stress testing most likely applies qualitative methods, seeks to analyse immeasurable risks, and tries to identify circumstances that might cause a firm to fail?

A. Macroeconomic stress testing.
B. Reverse stress testing.
C. Model stress testing.
D. Parameter stress testing.

Answer: B

Step by step

1. Reverse stress testing is scenario-driven and relies mostly on qualitative approaches.
2. It targets immeasurable risks (unknown unknowns, Knightian uncertainty).
3. It starts from a specific failure outcome and works backward to identify the circumstances that could lead to it.
4. This matches every element of the stem, so the answer is reverse stress testing.

Why B is correct: Reverse stress testing begins with an institutional-failure outcome and asks what circumstances could produce it, using mostly qualitative methods for immeasurable risks.

Why the others are wrong

- **A:** Macro stress testing is holistic, mixing quantitative and qualitative methods and measurable and immeasurable risks; it starts from macro shocks, not from failure.
- **C:** Model (parameter) stress testing is quantitative and applies to measurable risks; it shocks model inputs.
- **D:** Parameter stress testing is the same as model stress testing: quantitative, measurable risks, no search for failure circumstances.

Q179 Schweser Module Quiz 48.2 Q1

The loss of a major client, downgrades in credit ratings, and significant portfolio losses are examples of shocks used as the starting point for which type of stress testing?

A. Model stress testing.
B. Reverse stress testing.
C. Parameter stress testing.

D. Macroeconomic stress testing.

Answer: B

Step by step

1. These shocks are institution-specific failure outcomes (major client loss, rating downgrade, portfolio loss, loss of revenue sources).
2. Reverse stress testing starts from such a specific outcome and identifies the circumstances that could lead to it.
3. Therefore the answer is reverse stress testing.

Why B is correct: Reverse stress testing takes an institution-specific adverse outcome as its starting point and works backward to the causes.

Why the others are wrong

- **A:** Model stress testing changes model parameter values; it does not start from outcomes like client loss.
- **C:** Parameter stress testing is the same quantitative approach of shocking model inputs.
- **D:** Macro stress testing starts from regulator-supplied macroeconomic shocks (GDP, unemployment, inflation), not institution-specific events.

Q180

BT answers P2.T7.25.2.3 Q425

REBUILT — Original asked which analyst's approach 'best addresses' the bank's needs, which is arguable since macro stress testing is also useful; rewrote the analyst statements so each is clearly right or wrong about how the approach works.

Moore Asian Bank has recently experienced two events: (1) a cyber-attack that compromised extensive customer data and sharply lowered its share price, and (2) a compliance failure in trade finance operations that led to large fines and now threatens its licences. Two analysts propose how to identify the circumstances that could cause the bank to fail:

Analyst 1 states that macroeconomic stress testing is the appropriate tool because it starts from a specific institutional failure (such as loss of a licence) and works backward to identify the circumstances that could cause it.

Analyst 2 states that reverse stress testing is the appropriate tool because it starts from a specific failure outcome (such as loss of a licence or massive cyber losses) and works backward to identify the circumstances that could cause it.

Which analyst's statement is correct?

A. Analyst 1 only.
B. Analyst 2 only.
C. Neither Analyst 1 nor Analyst 2.
D. Both Analyst 1 and Analyst 2.

Answer: B

Step by step

1. Reverse stress testing: the starting point is a specific outcome resulting from institutional failure; the bank then identifies circumstances leading to it. Analyst 2 describes this correctly.
2. Macro stress testing starts from regulator-provided macroeconomic shock scenarios (GDP, unemployment, inflation) and examines the impact on bank-wide outputs and solvency; it does not work backward from failure. Analyst 1 is wrong.
3. Only Analyst 2 is correct.

Why B is correct: Working backward from a defined failure outcome is the defining feature of reverse stress testing, not of macro stress testing, which works forward from macroeconomic shocks.

Why the others are wrong

- **A:** Attributes the backward-from-failure logic to macro stress testing; it actually runs forward from macro shocks.
- **C:** Analyst 2 is correct, so 'neither' is wrong.
- **D:** Analyst 1 mislabels the approach; macro testing is not outcome-first.

Q181

GARP PE2 Q2 · Mock 4 Q3

A risk analyst at a bank is designing the bank's stress testing program and wants to identify which activity constitutes macroeconomic stress testing as opposed to other stress testing approaches. Which of the following best describes macroeconomic stress testing?

A. Assessing potential changes in dependencies between different measurable and immeasurable risks faced by the bank under adverse economic conditions.

B. Analyzing the impact of a three-standard-deviation increase in interest rates on the modeled value of the bank's bond portfolio.
C. Determining how severe a financial market crisis would need to be in order to result in the bank's insolvency.
D. Estimating the loss on a single trading desk if its largest counterparty were to default without warning.

Answer: A
Step by step
1. Macro stress testing takes the most holistic view of risk: it stresses both measurable and immeasurable risks and ideally the dependency structure between them, using macroeconomic scenarios.
2. B shocks a single model input (interest rates) to test model sensitivity, which is parameter stress testing.
3. C starts from the outcome of insolvency and asks how severe conditions must be, which is reverse stress testing.
4. D is a single-name, single-desk event, not a bank-wide macroeconomic scenario.

Why A is correct: Macro stress testing is holistic, covering measurable and immeasurable risks and their dependencies under adverse macroeconomic scenarios.
Why the others are wrong
- **B:** Parameter (model) stress testing: a quantitative shock to one model parameter, not a macroeconomic scenario.
- **C:** Reverse stress testing: starts from a defined failure (insolvency) and works backward.
- **D:** An idiosyncratic single-counterparty event on one desk; too narrow to be a macroeconomic, bank-wide scenario.

LO 48.d

Explain challenges and considerations when developing and implementing models used in stress testing operational risk.

Q182

BT answers P2.T7.25.2.2

REBUILT — Source data table was OCR-garbled and the options mixed up parameter stress testing with unrelated data issues; rebuilt as a self-contained question on the key weakness of a plain LDA used for operational risk stress testing.

Mana Capital Partners, an asset manager, is building its operational risk stress-testing framework. The risk team proposes to project stressed losses using a standard loss distribution approach (LDA), with Monte Carlo simulation of frequency and severity distributions fitted to historical loss data. Which of the following is the most significant limitation of using a standard LDA for this purpose?

A. It cannot be combined with Monte Carlo simulation, so stressed losses cannot be projected.
B. It has no risk drivers and assumes the firm's risk exposure is unchanged over time, so it cannot show how losses respond to a changing macroeconomic environment.
C. It models only loss severity, so frequency stresses cannot be reflected.
D. It is acceptable only for regulatory capital and is prohibited by regulators for stress testing.

Answer: B
Step by step
1. Stress testing aims to show how a firm's risk exposure changes over time as the macroeconomic and operating environment changes.
2. A standard LDA fits frequency and severity distributions to history and simulates from them; it contains no explanatory (macro) variables.
3. It therefore treats exposure as constant: a higher observed loss is seen as just a bad draw from the same distribution.
4. This makes it a secondary approach, used mainly when regression models fail; a conditional LDA (regression-based frequency) is a partial fix.

Why B is correct: The LDA lacks risk drivers and assumes immutable risk exposure, which conflicts with the purpose of stress testing.
Why the others are wrong
- **A:** Wrong: Monte Carlo simulation is the core of LDA and is also used in conditional LDA.
- **C:** Wrong: LDA models both frequency and severity; the confusion is with the severity-percentile issue of a conditional LDA.
- **D:** Wrong: regulators do not prohibit LDA for stress testing; they simply impose no percentile requirement.

Fragility and unchanged exposure are the real issues.

Q183 AnalystPrep Q.5109

REBUILT — Original 'unlikely to face' item had a non-sentence fragment as the correct answer; reworded to a positive question with one clearly correct challenge.

A newly hired risk manager is developing a conditional loss distribution approach (conditional LDA) model for operational risk stress testing. Which of the following is a recognised challenge the manager is likely to face?

A. Justifying the percentile chosen for the severity distribution, because the 99.9th percentile used for regulatory capital would leave the bank perpetually projected as undercapitalized.

B. Legal losses occur in the same quarter as an adverse macroeconomic shock, so the forecast must be contemporaneous with the shock.

C. Regulators require the 99.9th percentile to be used for severity in stress tests, which leaves no room for expert judgment.

D. Standard LDA is preferred over regression because it has explicit macroeconomic risk drivers.

Answer: A

Key correction: Original item asked which is NOT a challenge (answer D, the 'approach dimension' fragment); reworded to ask for the actual challenge.

Step by step

1. In a conditional LDA, frequency is forecast by regression on macro variables and severity is held constant, with expert judgment selecting a higher severity percentile.
2. The primary challenge is justifying that percentile.
3. Using the 99.9th percentile (regulatory capital) would be too severe and imply permanent undercapitalization; regulators set no percentile requirement for stress testing.
4. Only A describes this challenge correctly.

Why A is correct: Choosing and justifying the severity percentile is the main challenge of a conditional LDA, since the capital-standard 99.9th percentile is inappropriate for stress testing.

Why the others are wrong

- **B:** Wrong: legal losses lag macro events, sometimes by years.
- **C:** Wrong: regulators impose no percentile requirement for stress testing.
- **D:** Wrong: LDA has no risk drivers and is considered a secondary approach; conditional LDA/regression bring in macro variables.

Q184 BT answers P2.T7.24.5.3 · BT notes P2.T7.24.5.3

REBUILT — Original option A (raise LDA frequency) was arguably defensible and the correct option was vague; rebuilt around the legal-loss lag challenge with one clearly best strategy.

An operational risk manager at Vega Bank is enhancing the bank's operational risk stress-testing framework. Litigation losses are a large share of the bank's operational losses. Which strategy best addresses the practical challenge of forecasting legal losses in the stress test?

A. Forecast legal losses as part of total operational losses without separating them, so they respond immediately to the macroeconomic scenario.

B. Apply the 99.9th percentile to legal loss severity in all scenarios to ensure uniform results.

C. Exclude legal losses from the model because they cannot be linked to the macroeconomic scenario within the stress horizon.

D. Break legal losses out into their own module and build in lags between the macroeconomic drivers and the occurrence of the legal losses.

Answer: D

Step by step

1. Litigation losses are large and should be broken out into a separate legal loss module.
2. The key challenge is the delay between adverse macro conditions and the date legal losses are incurred (e.g., settlements years after a market crash).
3. The forecast must therefore include lags between its drivers (macro variables, firm-specific factors such as assets) and the losses.
4. Only D does both.

Why D is correct: A separate legal loss module with explicit lags captures the delay between macro shocks and litigation losses.

Why the others are wrong

- **A:** Wrong: GARP says legal losses should be broken out separately, and immediate response ignores the lag.
- **B:** Wrong: the 99.9th percentile is a capital standard and is not appropriate for stress testing.
- **C:** Wrong: legal losses are material, so excluding them understates stressed losses.

Q185 Schweser Module Quiz 48.2 Q2

A significant challenge in estimating the legal loss module of an operational risk stress test is that:

A. future litigation cases cannot be estimated.
B. banks are minimally impacted by legal losses.
C. operational risks and legal losses cannot be separated.
D. there is a lag between the macroeconomic event itself and the incurrence of legal losses.

Answer: D

Step by step

1. The legal loss module forecasts immaterial cases, losses above a threshold for current cases, and future cases.
2. The particular challenge is the delay between the onset of adverse macro conditions and when legal losses are incurred.
3. Forecasts must include lags.

Why D is correct: Legal losses can materialize years after the macroeconomic shock, so the module must model the lag.

Why the others are wrong

- **A:** Wrong: the module explicitly forecasts future litigation cases.
- **B:** Wrong: litigation is a large share of operational losses for most banks.
- **C:** Wrong: legal losses should be broken out as a separate subcategory.

Q186 Schweser Module Quiz 48.2 Q3

Risks that are only applicable to each unique bank are best captured using which module in a comprehensive operational risk stress testing framework?

A. Legal loss.
B. Individual risk loss.
C. Idiosyncratic scenario add-on.
D. Expected nonlegal loss forecast.

Answer: C

Step by step

1. The framework has three modules: expected nonlegal loss forecast, legal loss, and idiosyncratic scenario add-on.
2. Bank-specific exposures, based on storylines tied to the bank's own material vulnerabilities, go in the idiosyncratic scenario add-on.

Why C is correct: The idiosyncratic scenario add-on is designed for bank-specific storylines and extreme events.

Why the others are wrong

- **A:** Wrong: the legal loss module covers litigation losses only.
- **B:** Wrong: 'individual risk loss' is not a module of the framework.
- **D:** Wrong: this module forecasts losses by risk type from quantitative models plus expert refinement, not bank-unique storylines.

Reading 49. Cyber-Resilience: Range of Practices

"Cyber-Resilience: Range of Practices" (Basel Committee on Banking Supervision Publication, December 2018).

LO	Learning objective	Qs
49.a	define cyber-resilience and compare recent regulatory initiatives in the area of cyber-resilience.	3
49.b	describe current practices by banks and supervisors in the governance of a cyber-risk management framework, including roles and responsibilities.	7
49.c	explain methods for supervising cyber-resilience, testing and incident response approaches, and cybersecurity and resilience metrics.	4
49.d	explain and assess current practices for the sharing of cybersecurity information between different types of institutions.	3
49.e	describe practices for the governance of risks of interconnected third-party service providers.	3

LO 49.a

Define cyber-resilience and compare recent regulatory initiatives in the area of cyber-resilience.

Q187

AnalystPrep Q.4263

REBUILT — Original keyed answer ('no absolute security', international cooperation) is not stated in the Basel report text and distractors were vague; rebuilt as a clear comparison of national cyber-resilience guidance.

The Basel Committee's 2018 report on cyber-resilience practices compares how jurisdictions have issued specific cyber-risk guidance. Which of the following correctly matches a jurisdiction with the focus of its guidance?

A. Hong Kong: common procedures and methodologies for the assessment of ICT risk
B. Singapore: early detection of cyber intrusions
C. Brazil: the importance of effective cyber-security risk management
D. European Banking Authority: the establishment of a cyber security policy

Answer: B

Step by step

1. The report notes that a few jurisdictions have issued specific cyber guidance, each with a different emphasis.
2. Hong Kong: importance of effective cyber-security risk management.
3. Singapore: early detection of cyber intrusions.
4. Brazil: establishment of a cyber security policy.
5. EBA (EU): common procedures and methodologies for assessing ICT risk.
6. Only option B matches.

Why B is correct: Singapore's guidance emphasizes detecting cyber intrusions early; the other options swap the focus of the other three jurisdictions.

Why the others are wrong

- **A:** This is the EBA's focus; Hong Kong's guidance concerns effective cyber-security risk management.
- **C:** This is Hong Kong's focus; Brazil's guidance concerns establishing a cyber security policy.
- **D:** This is Brazil's focus; the EBA's guidance concerns common procedures and methodologies for ICT risk assessment.

Q188

AnalystPrep Q.4264 · AnalystPrep Q.4481

REBUILT — Two source items tested the same FSB-Lexicon definition with weak 'all/none of the above' options; merged and rebuilt as a single EXCEPT item with plausible distractors.

The Basel Committee's report on cyber-resilience practices adopts the FSB Lexicon definition of cyber-resilience. Which of the following is NOT part of that definition?

A. Anticipating and adapting to cyber threats and other relevant changes in the environment
B. Withstanding, containing and rapidly recovering from cyber incidents
C. Continuing to carry out the organisation's mission
D. Eliminating all cyber vulnerabilities so that incidents cannot occur

Answer: D

Key correction: Source keys (Q.4264 'D: none of the above', Q.4481 'C: all of the above') are consistent with the full definition; the options were reworded so that exactly one element is clearly outside the definition.

Step by step

1. FSB Lexicon: cyber-resilience is the ability of an organisation to continue to carry out its mission by anticipating and adapting to cyber threats and other relevant changes in the environment, and by withstanding, containing and rapidly recovering from cyber incidents.
2. A, B and C are each elements of this wording.
3. Elimination of all vulnerabilities is not part of it; the concept accepts that incidents will occur and focuses on continuity and recovery.
4. The answer is D.

Why D is correct: Cyber-resilience is about continuing the mission, adapting, and withstanding/recovering from incidents, not about preventing every incident.

Why the others are wrong

- **A:** This is part of the definition (anticipating and adapting), so it is not the exception.
- **B:** This is part of the definition (withstanding, containing, rapidly recovering), so it is not the exception.
- **C:** Continuing to carry out the mission is the core of the definition, so it is not the exception.

Q189 AnalystPrep Q.4266

REBUILT — Original 'all of the above' item was vague about the mechanism; rebuilt to test how the report says automation, third parties and cloud change the regulatory perimeter.

According to the Basel Committee's report, regulated institutions' greater use of automation, third-party service providers and cloud services affects cyber-resilience in which of the following ways?

A. The attack surface grows and is accessible from anywhere, the perimeter of interest to regulators becomes bigger with third-party use, and cloud services make that perimeter shared

B. Outsourcing data processing to third parties shrinks the regulators' perimeter of interest because data is held outside the institution

C. Cloud services make the perimeter the sole responsibility of the cloud provider, so the bank's own cyber-resilience duties fall away

D. Greater automation reduces the attack surface, which lowers the incentive for cyber-adversaries to increase their capabilities

Answer: A

Step by step

1. The report states that more automation and integration with third parties and customers produces a growing attack surface accessible from anywhere, which incentivises adversaries to increase capabilities.
2. More third-party use makes the perimeter of interest to regulators bigger; cloud makes it shared.
3. Shared service models require institutions to build resilience in partnership with third parties.
4. Only A states all of this correctly.

Why A is correct: Automation and third-party integration enlarge the attack surface and the regulatory perimeter, and cloud makes the perimeter shared, so institutions must build resilience jointly with providers.

Why the others are wrong

- **B:** Reverses the effect: third-party use enlarges, not shrinks, the perimeter of interest.
- **C:** Perimeter is shared, not transferred; the institution still must build and maintain its cyber-resilience in partnership.
- **D:** Automation and integration enlarge the attack surface and increase adversaries' incentive.

LO 49.b

Describe current practices by banks and supervisors in the governance of a cyber-risk management framework, including roles and responsibilities.

Q190 AnalystPrep Q.4269

With respect to cyber-security strategy as outlined in the Basel Committee report on cyber-resilience, all of the following statements are correct EXCEPT:

A. All regulators expect regulated entities to have a board-approved information security strategy

B. Many jurisdictions expect cyber-risk to be covered by the organisation-wide risk management and/or information security framework

C. Most supervisors review regulated entities' information security strategies, but very few require or evaluate those entities' standalone cyber-security strategies

D. In most jurisdictions the development of a cyber-security strategy is a mandatory requirement anchored in law

Answer: D

Step by step

1. The question asks for the INCORRECT statement.
2. The Basel report states that although most regulators do not require regulated entities to develop a cyber-security strategy, all expect a board-approved information security strategy, policy and procedures.
3. Many jurisdictions expect cyber-risk to be covered by the organisation-wide risk management and/or information security framework, reviewed by senior executives.
4. Supervisors mainly review information security strategies; very few require or evaluate standalone cyber-security strategies.
5. Statement D reverses the first point, so it is the false one.

Why D is correct: Most regulators do NOT mandate a standalone cyber-security strategy; they expect a broader board-approved information security strategy.

Why the others are wrong

- **A:** Correct statement (the report says all regulators expect a board-approved information security strategy), so it is not the EXCEPT answer.
- **B:** Correct statement; many jurisdictions (e.g., Australia, Brazil, Europe) expect cyber-risk to be covered by the broader risk management and/or information security framework, so it is not the EXCEPT answer.
- **C:** Correct statement; it restates the report's finding on supervisory review practice.

Q191 AnalystPrep Q.4270

Which of the following regulatory approaches have jurisdictions adopted to enforce cyber-security strategy requirements among regulated entities?

I. Mandatory sector-specific or cross-sector cyber-security requirements set by the regulator

II. Financial institutions establishing their own cyber-security strategies under principles-based risk management, which regulators review

III. Examining whether institutions have an appropriate IT strategy and accompanying security provisions

A. I and II only

B. II only

C. III only

D. All three

Answer: D

Step by step

1. GARP identifies three non-mutually exclusive regulatory approaches.
2. Approach 1: the regulator sets sector-specific or cross-sector requirements (common in emerging markets with homogeneous banking systems) - matches I.
3. Approach 2: institutions establish their own strategies under principles-based risk management, reviewed by the regulator - matches II.
4. Approach 3: examining whether entities have an IT strategy and security provisions (prevalent in Europe) - matches III.
5. All three statements are valid.

Why D is correct: The three observed approaches are regulator-set requirements, institution-developed strategies, and review of the IT strategy and security provisions.

Why the others are wrong

- **A:** Omits approach III, which is prevalent in Europe.
- **B:** Recognises only the institution-developed approach and ignores the other two.

- **C:** Recognises only the European IT-strategy approach and ignores the other two.

Q192 AnalystPrep Q.4484

According to the Basel Committee's report on cyber-resilience practices, cyber-risk awareness and risk culture at regulated institutions is enhanced through:

A. Cyber training incorporated in all phases of employment, from recruitment to termination
B. Effective processes and controls ensuring that employees, contractors and third-party vendors understand their responsibilities and are suitable for their roles, to reduce the risk of theft, fraud or misuse of facilities
C. Establishing a common risk culture to ensure effective cyber-risk management
D. All of the above

Answer: D

Step by step

1. Check each measure against the report's awareness discussion.
2. Some jurisdictions require cyber-security awareness training during each phase of employment, recruitment to termination - A holds.
3. Regulators assess whether processes and controls ensure staff, contractors and third parties understand responsibilities - B holds.
4. Most regulators encourage a common risk culture sufficient for effective cyber-risk management - C holds.
5. So all are correct.

Why D is correct: Each of training across the employment lifecycle, role-clarity controls for staff and third parties, and a common risk culture is a recognised awareness practice.

Why the others are wrong

- **A:** True but incomplete; B and C are also practices cited.
- **B:** True but incomplete; A and C are also cited.
- **C:** True but incomplete; A and B are also cited.

Q193 BT answers P2.T7.20.14.2 · BT notes P2.T7.20.14.2

The Basel Committee on Banking Supervision (BCBS) borrows the FSB lexicon's definition of cyber-resilience: the ability of an organization to continue to carry out its mission by anticipating and adapting to cyber threats and other relevant changes in the environment and by withstanding, containing, and rapidly recovering from a cyber incident. According to the BCBS, which of the following is a TRUE statement about cyber-resilience?

A. The chief information security officer (CISO) should report to the chief risk officer (CRO)
B. Previously developed national and international standards cannot be leveraged for cyber-resilience because their scope is information technology (IT) or risk management
C. The skills shortage is due to the impracticality and/or inability of regulators to conduct on-site inspections; a lack of any existing, reputable certification or accreditation programs; and a lack of qualified cyber headhunters
D. Although regulators generally do not require a specific cyber strategy, they do expect banks to maintain cyber-resilience capabilities under their broader oversight of technology, information systems (IS), and/or enterprise risk management (ERM)

Answer: D

Step by step

1. Test each statement against the report.
2. A: the report says placing the CISO under the CRO is not ideal (conflicting priorities; CRO tends to favour compliance); some banks now have the CISO report to the CEO or board - false.
3. B: IT and operational risk standards, plus international standards, are used and have relevance to cyber risk - false.
4. C: regulators do assess the workforce through on-site inspections and certification programs exist - false.
5. D: most regulators do not require a cyber-security strategy but expect a board-approved information security strategy under broader technology/risk oversight - true.

Why D is correct: Cyber-resilience is expected within broader IT/ERM oversight rather than through a mandated standalone cyber strategy.

Why the others are wrong

- **A:** Reverses the report's finding: CISO-under-CRO is seen as not ideal, with a trend toward reporting to the CEO or board.

- **B:** Existing IT and operational-risk standards can be and are leveraged; only some jurisdictions lack cyber-specific standards.
- **C:** Invents causes of the skills gap; regulators do use on-site inspections and certification programs exist.

Q194 BT answers P2.T7.20.14.3 · BT notes P2.T7.20.14.3

REBUILT — Original options mixed outsourcing, control taxonomies and information-sharing (49.d/e) with unverifiable claims, so the item was rebuilt around the 49.b three-lines-of-defence (3LD) finding.

According to the Basel Committee report on cyber-resilience practices, each of the following statements about the three-lines-of-defence (3LD) model in cyber-governance is correct EXCEPT:

A. The majority of regulators have adopted the 3LD model to assess cyber-security risk and controls

B. Most regulators require institutions to implement 3LD and prescribe precisely how responsibilities are distributed across the lines

C. There is a greater supervisory focus on the first and second lines than on the third line of defence

D. Only a few jurisdictions have specific expectations on the independent reporting line from the chief audit executive to the audit committee

Answer: B

Key correction: BT sources keyed C (3LD described as a 'three-dimensional holographic' framework) on a mixed-LO item; the item was replaced by a pure 49.b item, so that key no longer applies.

Step by step

1. The question asks for the INCORRECT statement.
2. A: the majority of regulators use 3LD to assess cyber risk and controls - true.
3. B: most regulators do not require 3LD implementation and do not prescribe how responsibilities are distributed; banks must define responsibilities themselves and leave no gaps - false.
4. C: supervision focuses more on lines 1 and 2 than line 3, which may hamper the checks and balances - true.
5. D: only a few jurisdictions set expectations for the chief audit executive's independent line to the audit committee - true.

Why B is correct: Regulators use 3LD as an assessment lens but do not mandate it or prescribe the allocation of responsibilities.

Why the others are wrong

- **A:** Correct statement, so not the EXCEPT answer.
- **C:** Correct statement; third-line supervision receives less attention.
- **D:** Correct statement; few jurisdictions address the audit executive's independent reporting.

Q195 Schweser Module Quiz 49.1 Q1

Country X's banking regulator established sector-specific cybersecurity requirements. This most likely indicates that Country X is a(n):

A. emerging economy with a homogeneous banking system.

B. developed economy with a homogeneous banking system.

C. emerging economy with a fragmented banking system.

D. developed economy with a fragmented banking system.

Answer: A

Step by step

1. Regulator-set sector-specific or cross-sector cybersecurity requirements are one of three approaches to cyber strategy enforcement.
2. This approach is common in emerging market economies with relatively homogeneous banking systems.
3. The European-style approach (reviewing IT strategy) is the one prevalent in developed regions, not this one.
4. Hence emerging economy with a homogeneous banking system.

Why A is correct: The regulator-imposed requirement approach is common where banking systems are relatively homogeneous, as in many emerging markets.

Why the others are wrong

- **B:** Pairs the approach with a developed economy; the European pattern is review of the IT strategy.
- **C:** Fragmented systems are not associated with this approach; homogeneity makes uniform rules practical.
- **D:** Wrong on both attributes.

Q196 AnalystPrep Q.4273

REBUILT — Original relied on an unsourced 'best course of action' judgement with weak distractors; reworded so the answer rests on the report's finding that board and senior management are central to cyber-governance.

Exim Bank has completed a cyber risk assessment and business impact analysis (BIA). The information security manager and the business department manager disagree over who should make the final evaluation of the results and the resulting risk decision. Consistent with the Basel Committee's findings on cyber-governance roles and responsibilities, the best course of action is to:

A. adopt the information security manager's decision, since cyber-risk is a technical matter
B. adopt the business department manager's decision, since the business owns the risk
C. commission a new risk assessment and BIA to remove the disagreement
D. escalate the report to the board and senior management, who are expected to oversee cyber-risk and resolve the dispute

Answer: D

Step by step

1. Regulators in most jurisdictions stress the importance of the board and senior management in overseeing cyber-risk and in cyber-governance.
2. A conflict between two functional managers cannot be settled by either one alone; accountability sits above them.
3. Repeating the assessment (C) does not resolve who owns the decision, since the disagreement concerns authority, not data.
4. Therefore escalate to board/senior management.

Why D is correct: Ultimate accountability for cyber-risk decisions rests with the board and senior management, so unresolved disputes go to them.

Why the others are wrong

- **A:** Lets one function decide unilaterally, ignoring senior management oversight and the business view.
- **B:** Same flaw in reverse; neither manager holds ultimate accountability.
- **C:** Duplicates work and does not settle who has decision authority, so the dispute persists.

LO 49.c

Explain methods for supervising cyber-resilience, testing and incident response approaches, and cybersecurity and resilience metrics.

Q197 Schweser Module Quiz 49.1 Q2

Which of the following is NOT one of the four areas of cyber risk management and incident response and recovery that supervisors review?

A. Architecture and standards
B. Incident response and recovery
C. Supervision of cyber-resilience
D. Cybersecurity and resilience metrics

Answer: A

Step by step

1. The four areas are: (1) supervision of cyber-resilience, (2) information security controls, (3) incident response and recovery, (4) cybersecurity and resilience metrics.
2. Options B, C and D match areas (3), (1) and (4).
3. 'Architecture and standards' is not on the list, so A is the exception.

Why A is correct: The four areas are supervision, information security controls, incident response and recovery, and metrics; architecture and standards is not one.

Why the others are wrong

- **B:** This is area 3 of the four, so it is a valid area; choosing it misreads the EXCEPT wording.
- **C:** This is area 1 of the four; choosing it misreads the EXCEPT wording.
- **D:** This is area 4 of the four; choosing it misreads the EXCEPT wording.

Q198 BT answers P2.T7.20.14.1 · BT notes P2.T7.20.14.1 · BT Mock Q38

According to the Basel Committee on Banking Supervision (BCBS, 2018), no jurisdiction's methodology for assessing cyber-security produces quantitative metrics comparable to those for financial risks; instead, indicators describe an institution's approach to building cyber-resilience, and supervisors also rely on entities' own (not yet mature) management information. With respect to current cyber-security and resilience metrics, each of the following statements is true EXCEPT which is inaccurate?

A. No single cyber-security or resilience metric in isolation is sufficient.
B. Patch ageing (days to patch) is a widespread and comparable metric.
C. Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt.
D. The penetration ratio, the percentage of banks in a jurisdiction that can be penetrated, is a useful metric that compares a supervisor's effectiveness with other jurisdictions.

Answer: D

Key correction: Source key (BT Mock Q38) said B; correct is D because the BCBS report explicitly calls patch ageing a widespread and comparable metric, while the 'penetration ratio' does not exist as a supervisory metric.

Step by step

1. Check each statement against the BCBS report.
2. A: no single item in isolation is seen as a sufficient metric - true.
3. B: the report states patch ageing in particular is a widespread and comparable metric - true.
4. C: backward-looking indicators are useful in stable environments, but adversaries adapt, so forward-looking indicators are needed - true.
5. D: no such 'penetration ratio' supervisory-effectiveness metric exists; penetration testing is a test of an institution's controls, not a cross-jurisdiction comparison of supervisors - inaccurate.

Why D is correct: Only D describes a metric the report does not support; the other three statements reflect the report directly.

Why the others are wrong

- **A:** True per the report (no single metric is sufficient); picking it misreads the EXCEPT wording.
- **B:** True: patch ageing is cited as a widespread and comparable metric. The statement that no jurisdiction has comparable quantitative metrics refers to financial-risk-style standardized metrics, not patch ageing.
- **C:** True: DDoS-type adaptive adversaries make forward-looking indicators necessary; picking it misreads the EXCEPT wording.

Q199 AnalystPrep Q.4271

REBUILT — Original asked which activity is 'most critical' before finalizing a continuity plan, which is ambiguous and not tied to the reading; rebuilt to test how supervisors evaluate incident response and recovery plans.

Under the BCBS (2018) review of cyber-resilience practices, which of the following best describes the focus of supervisory evaluation of regulated institutions' incident response and recovery plans?

A. A standardized quantitative score that ranks institutions' recovery capability against financial-risk benchmarks
B. How plans are triggered, the institution's ability to implement them, preservation of data, and specific actions for critical technology
C. Only whether the institution has a written plan, with no assessment of implementation
D. Solely the volume of cyber-incidents experienced in the prior year

Answer: B

Step by step

1. Recall the BCBS findings on incident response and recovery evaluation.
2. Supervisors focus on how plans are triggered, ability to implement them, data preservation and actions for critical technology; some (e.g., Canada) also test communication protocols.
3. Only B matches this.

Why B is correct: Supervisory evaluation is about whether plans can actually be triggered and executed and data preserved, not about scores, mere existence of a plan, or incident counts.

Why the others are wrong

- **A:** No jurisdiction has standardized quantitative metrics comparable to financial risk; this confuses recovery evaluation with market-risk style metrics.
- **C:** Supervisors assess implementation ability, not just documentation.

- **D:** Incident volume is a backward-looking metric, not the focus of plan evaluation.

Q200

AnalystPrep Q.4265

REBUILT — Original relied on an unsupported claim that DR plans are often untested; rebuilt around the reading's coverage of tests, exercises and the TIBER-EU framework.

Under the ECB's TIBER-EU framework (2018) for controlled testing of cross-border financial entities, which statement is correct?

A. Tests are mandatory for every entity and results are reported as pass or fail
B. Relevant authorities and entities decide if and when tests are performed; tests are tailor-made and give insight into strengths and weaknesses rather than a pass/fail result
C. Tests are conducted and funded solely by the supervisory authority without involving the entity
D. Tests focus exclusively on recovery capability rather than protection and detection

Answer: B

Step by step

1. TIBER-EU was published by the ECB in May 2018 as a Europe-wide framework for controlled tests against cyber-attacks.
2. The authorities and entities themselves determine if and when tests occur.
3. Tests are tailor-made and do not result in pass or fail; they help the entity learn and improve cyber-maturity.

Why B is correct: TIBER-EU is a voluntary-style, bespoke, learning-oriented testing framework with no pass/fail outcome.

Why the others are wrong

- **A:** Tests are not mandatory by default and explicitly produce no pass/fail result.
- **C:** Entities and authorities jointly decide; tests generally are voluntarily funded by the institutions.
- **D:** Tests tend to focus on protection and detection capabilities, not exclusively recovery.

LO 49.d

Explain and assess current practices for the sharing of cybersecurity information between different types of institutions.

Q201

AnalystPrep Q.4272 · Mock 1 Q12 · Schweser Module Quiz 49.2 Q1

According to the Basel Committee report on cyber-resilience practices, which of the following types of cybersecurity information-sharing is the least commonly observed across jurisdictions?

A. Sharing among banks
B. Sharing among regulators
C. Sharing by banks and regulators with security agencies
D. Sharing from banks to regulators

Answer: B

Key correction: Source key (Mock 1 Q12) said D; correct is B because sharing with security agencies is one of the three most common types, while sharing among regulators is the least observed.

Step by step

1. The report identifies five types: among banks, banks to regulators, among regulators, regulators to banks, and banks and regulators to security agencies.
2. The three most commonly observed types are sharing among banks, sharing with security agencies, and sharing from banks to regulators.
3. Sharing among regulators (and regulators to banks) is less frequent; sharing among regulators is the least observed type.
4. Hence the answer is B.

Why B is correct: Sharing among regulators is typically ad hoc, bilateral or within supervisory colleges, with less systematic arrangements, so it is the least observed type.

Why the others are wrong

- **A:** Sharing among banks is one of the three most common types, not the least.

- **C:** Sharing with security agencies is among the three most common types; believing it is rare confuses it with regulator-to-bank sharing.
- **D:** Bank-to-regulator reporting is typically mandated and is among the most common types.

Q202

GARP PE2 Q64 · Mock 4 Q65

The CRO at a bank wants to strengthen the bank's capability to defend itself against emerging cyber-threats. To help achieve this goal, the CRO is assessing the current range of practices regarding the sharing of cybersecurity information between different types of institutions. Which of the following statements would be most appropriate for the CRO to make?

A. The sharing of cybersecurity information among banks is less frequently observed and generally considered to be less effective than other cyber-security information-sharing practices.

B. The scope and depth of information-sharing practices among banks may significantly vary between financial markets, depending on the level of trust among participating banks.

C. Information-sharing among different national regulators has evolved significantly over the past several years and is now a widespread practice at a large majority of jurisdictions.

D. Existing peer-sharing mechanisms among banks focus on the exchange of information related to cyber-security incidents, but such information is generally not shared from banks to regulators.

Answer: B

Step by step

1. Sharing among banks depends heavily on the industry's culture and the level of trust among participants, so scope and depth vary across markets (B is consistent).
2. Sharing among banks is one of the three most common types, so A is wrong.
3. Sharing among regulators is the least observed type, so C is wrong.
4. Banks share incident information with regulators (often mandated) while sharing cyber-threat information among themselves, so D is wrong.

Why B is correct: Bank-to-bank sharing relies on trust; practices therefore differ across financial markets, often starting with a small group before widening.

Why the others are wrong

- **A:** Reverses the facts: sharing among banks is among the most common types.
- **C:** Sharing among regulators is the least observed type and remains ad hoc, not widespread.
- **D:** Banks do share incident information with regulators, frequently by mandate; peer sharing mainly concerns threat information.

Q203

AnalystPrep Q.4267

REBUILT — Original stem was vague and its key (B) was dubious; rewritten around the Basel/GARP finding that regulator-to-bank sharing is less prevalent where bank-to-bank sharing is well established.

According to the Basel Committee report on cyber-resilience practices, in jurisdictions where information sharing among banks is already well established, which type of cybersecurity information sharing tends to be less prevalent?

A. Sharing from banks to regulators

B. Sharing from regulators to banks

C. Sharing among banks themselves

D. Sharing by banks with security agencies

Answer: B

Step by step

1. Where banks already have a well-functioning mechanism to share among themselves, banks have less need for information from regulators.
2. Hence regulator-to-bank sharing tends to be less prevalent in such jurisdictions.
3. Bank-to-regulator reporting is separate and is typically mandated, so it is not displaced.
4. Answer: B.

Why B is correct: A functioning peer-sharing mechanism reduces the need for regulators to pass cyber information to banks.

Why the others are wrong

- **A:** Bank-to-regulator reporting is usually mandated by regulation and is unaffected by peer sharing.

- **C:** This is the established practice stated in the stem, so it cannot be the less prevalent one.
- **D:** Security-agency sharing is less common where bank-to-regulator sharing is established, not where bank-to-bank sharing is; and it is one of the three most common types overall.

LO 49.e

Describe practices for the governance of risks of interconnected third-party service providers.

Q204 Schweser Module Quiz 49.2 Q2

Which of the following risks is least likely to be included in the risk analysis conducted as part of an outsourcing risk management framework?

A. Strategic risk.
B. Liquidity risk.
C. Compliance risk.
D. Vendor lock-in risk.

Answer: B

Step by step

1. Recall the risks that an outsourcing framework's risk analysis is expected to cover: strategic, compliance, security, business continuity, vendor lock-in, counterparty, country, contractual and access risks.
2. Check each option against this list: strategic (A), compliance (C) and vendor lock-in (D) all appear.
3. Liquidity risk is not on the list, so it is the least likely to be included.

Why B is correct: The third-party risk analysis targets risks arising from the service relationship (strategic, compliance, security, continuity, lock-in, counterparty, access); liquidity risk is a financial-balance-sheet risk that is not among them.

Why the others are wrong

- **A:** Strategic risk is explicitly covered; choosing it confuses a listed risk with an unlisted one.
- **C:** Compliance risk is explicitly covered (regulatory and contractual obligations of the outsourced activity).
- **D:** Vendor lock-in risk (ability to withdraw from the provider and absorb or transfer the activity) is explicitly covered; it is easily overlooked because it sounds non-traditional.

Q205 AnalystPrep Q.4478

In the context of cyber-resilience practices, which of the following is/are considered (a) third-party(ies)?

A. Cloud computing services
B. Computer hardware
C. Trading platforms
D. All of the above

Answer: D

Step by step

1. Third parties are defined broadly: all forms of outsourcing (including cloud computing), standardised and non-standardised products and services not typically seen as outsourcing (power supply, telecommunication lines, commercial hardware and software), and interconnected counterparties (payment and settlement systems, trading platforms, CSDs, CCPs).
2. Cloud computing falls in the first group, commercial hardware in the second, and trading platforms in the third.
3. Since all three qualify, the correct choice is 'All of the above'.

Why D is correct: The broad definition of third parties covers outsourcing, non-outsourcing supplies such as commercial hardware, and interconnected counterparties such as trading platforms.

Why the others are wrong

- **A:** Cloud services are a third party, but so are the others; this option is incomplete.
- **B:** Commercial hardware counts as a non-outsourcing third-party supply, but it is not the only one; incomplete.
- **C:** Trading platforms are interconnected counterparties and qualify, but the other items do too; incomplete.

Q206 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A mid-sized bank is moving its payment-processing and customer-data storage to a cloud service provider that itself uses several subcontractors. The CRO asks an operational risk manager to design the bank's governance of this third-party relationship so that it is consistent with typical supervisory expectations on third-party cyber-resilience. Which of the following actions would be most appropriate for the manager to recommend?

A. Run scenario analyses of provider disruptions only once every three years, after the contract is renewed.

B. Treat the cloud arrangement as outside the outsourcing framework because the provider, not the bank, operates the infrastructure.

C. Rely on the provider's own policies for incident response and recovery without aligning them to the bank's policies.

D. Maintain an inventory of outsourced functions and obtain visibility into the provider's sub-outsourcing arrangements.

Answer: D

Step by step

1. Third parties are defined broadly: all forms of outsourcing, including cloud computing, plus services not usually considered outsourcing and interconnected counterparties.
2. Supervisors typically require institutions to disclose their outsourcing arrangements, maintain an inventory of outsourced functions, obtain reports from third parties, and may require visibility into sub-outsourcing.
3. Only the inventory and sub-outsourcing visibility action matches these expectations, so D is correct.

Why D is correct: Supervisory expectations on the visibility of third-party interconnections include disclosing outsourcing arrangements, maintaining an inventory of outsourced functions, obtaining third-party reports, and gaining visibility into sub-outsourcing. Because the bank lacks direct control over the provider and its subcontractors, this visibility is a core governance practice.

Why the others are wrong

- **A:** Wrong. Advanced regulators expect tests of critical activities based on realistic and probable disruptive scenarios, conducted at least yearly, not only once every three years at contract renewal. Mistake: confusing contract-renewal reviews with the expected testing frequency.
- **B:** Wrong. Cloud computing is expressly within the broad definition of third parties and outsourcing, so it falls under the outsourcing framework (roles, scope, risk analysis, monitoring). Mistake: thinking that provider-operated infrastructure removes the bank's responsibility.
- **C:** Wrong. Regulators stress that financial institutions should align the policies of critical suppliers with their own and plan for incident management, response and recovery. Mistake: assuming the provider's own policies are sufficient.

Reading 50. Case Study: Cyberthreats and Information Security Risks

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 9.

LO	Learning objective	Qs
50.a	provide examples of cyber threats and information security risks, and describe frameworks and best practices for managing cyber risks.	18
50.b	describe lessons learned from the Equifax case study.	7

LO 50.a

Provide examples of cyber threats and information security risks, and describe frameworks and best practices for managing cyber risks.

Q207

AnalystPrep Q.4482

REBUILT — Original options were vague and overlapped (B and C both 'develop regulation'); rewrote so the framework-based answer is the single clearly best one.

A risk manager at a bank operates in a country with no cybersecurity regulation. Which course of action is MOST appropriate for protecting the bank against cyber threats?

A. Adopt a globally recognized cybersecurity standard or framework, such as NIST CSF, CIS Critical Security Controls or ISO 27001

B. Draft the bank's own bespoke cybersecurity framework from scratch and apply it internally

C. Wait for the national regulator to issue rules and meanwhile rely on the IT department's existing practices

D. Adopt only a cyber-awareness training program for staff, since behavioral controls are the main defense

Answer: A

Step by step

1. Absent local regulation, the firm still needs a systematic way to mitigate cyber threats and benchmark itself to good market practice.
2. Three standards dominate the market and are globally recognized: NIST CSF, CIS Critical Security Controls and ISO 27001.
3. Adopting one (or a combination) gives expert-developed guidance without waiting for regulators.

Why A is correct: Globally recognized frameworks (NIST CSF, CIS, ISO 27001) provide expert-built, widely accepted guidance and benchmarks, so they substitute for missing local regulation.

Why the others are wrong

- **B:** A bespoke framework is slow, costly and not benchmarked to market practice; it ignores existing proven standards.
- **C:** Waiting leaves the bank exposed; firms should act regardless of regulation.
- **D:** Behavioral controls are only one part of the defense; technical controls and a full framework are also needed.

Q208

AnalystPrep Q.5110

Under the four-quadrant taxonomy of information security risks, which of the following is NOT an example of an involuntary disclosure (loss) incident?

A. Database loss

B. Virus infection

C. System disruptions

D. Loss of printed documents

Answer: B

Step by step

1. The taxonomy splits data incidents into theft or corruption versus loss or involuntary disclosure.
2. Hacking, virus infection and phishing are external, deliberate attacks: theft or corruption.
3. Database loss, system disruptions and loss of paper documents are loss/involuntary-disclosure events.
4. So virus infection is the odd one out.

Why B is correct: A virus infection is a deliberate external attack (theft/corruption), whereas the other three are accidental loss or disruption events.

Why the others are wrong
- **A:** Database loss is an internal loss/involuntary-disclosure event, so it is an example.
- **C:** System disruptions (with disasters and third-party failure) are external loss/involuntary-disclosure events.
- **D:** Loss of printed (non-digital) documents is an accidental loss event, so it is an example.

Q209 AnalystPrep Q.5111

Which of the following is NOT one of the five core functions (steps) of the NIST cybersecurity framework?

A. Identify
B. Protect
C. Mitigate
D. Recover

Answer: C

Step by step
1. The five NIST CSF steps are identify, protect, detect, respond and recover.
2. Identify, Protect and Recover appear in the list; 'Mitigate' does not.
3. Mitigating controls exist in the control taxonomy (prevent/detect/mitigate) but not as a NIST step.

Why C is correct: The NIST CSF steps are Identify, Protect, Detect, Respond, Recover; Mitigate is not among them.

Why the others are wrong
- **A:** Identify is the first NIST step (inventory of assets and policies).
- **B:** Protect is a NIST step (access restriction, encryption, backups, training).
- **D:** Recover is the fifth NIST step (restoring affected equipment and data).

Q210 AnalystPrep Q.5112

Using the typology of information security risks, which of the following is an external cause of information security risk?

A. Database loss
B. Loss of printed documents
C. Systems disruptions
D. Departing employees taking proprietary information

Answer: C

Step by step
1. External causes include third parties: hacking, virus infection, theft, social engineering, disasters, systems disruptions and third-party failure.
2. Database loss and lost paper documents occur inside the firm (internal loss).
3. Departing employees taking information is internal theft/transfer of information.
4. Only systems disruptions is external.

Why C is correct: Systems disruptions (with disasters and third-party failure) sit in the external-cause row of the typology; the other items originate inside the firm.

Why the others are wrong
- **A:** Database loss (and back-up loss) is an internal, unintentional loss.
- **B:** Lost printed documents are an internal loss of non-digital information.
- **D:** Employees leaving with proprietary data are internal theft or transfer of information.

Q211 AnalystPrep Q.5113

Which function of the NIST cybersecurity framework covers reporting an attack to law enforcement and other authorities?

A. Recover
B. Respond
C. Protect
D. Detect

Answer: B

Step by step

1. Respond: keep operations running, notify customers and employees, investigate, contain and report the attack to law enforcement or authorities, then update policies with lessons learned.
2. Recover is about repairing and restoring affected equipment and data.
3. Protect concerns safeguards; Detect concerns monitoring.

Why B is correct: Notifying and reporting to authorities is part of the response plan to an incident, i.e. the Respond function.

Why the others are wrong

- **A:** Recover restores impaired equipment/data and keeps stakeholders informed; it is not the reporting step.
- **C:** Protect covers access control, encryption, backups and training before an attack.
- **D:** Detect covers monitoring for unauthorized access and investigating unusual activity.

Q212 AnalystPrep Q.5114

Which of the following is NOT an action under the Respond function of the NIST cybersecurity framework?

A. Notifying customers, employees and others whose data may be at risk
B. Keeping business operations up and running
C. Investigating unusual activities on the network or by staff
D. Reporting the attack to law enforcement and other authorities

Answer: C

Step by step

1. Respond actions: notify stakeholders, keep operations running, report to authorities, contain the attack.
2. Monitoring and investigating unusual network or staff activity and checking for unauthorized users or connections belong to Detect.
3. Hence C is not a Respond action.

Why C is correct: Investigating unusual activity is part of ongoing monitoring, which is the Detect function.

Why the others are wrong

- **A:** Notification of affected parties is a Respond action.
- **B:** Maintaining operations during an attack is a Respond action.
- **D:** Reporting to law enforcement/authorities is a Respond action.

Q213 AnalystPrep Q.5115

Which category of information controls addresses the actions people take when handling and protecting information?

A. Technical controls
B. Behavioral controls
C. Preventive controls
D. Detective controls

Answer: B

Step by step

1. Information controls are either behavioral or technical.
2. Behavioral controls address human behavior and shortcomings: awareness campaigns, codes of conduct, training, password management, supervision and sanctions.
3. Technical controls address technical aspects of systems for prevention or detection, so preventive and detective are subtypes of technical controls.

Why B is correct: Behavioral controls target human behavior and shortcomings through training, rules of conduct and supervision.

Why the others are wrong

- **A:** Technical controls use technology (firewalls, encryption, DLP, honeypots) rather than addressing people's conduct.
- **C:** Preventive controls are a subtype of technical controls (firewalls, encryption, patching).
- **D:** Detective controls are a subtype of technical controls (e.g. data loss prevention and detection tools).

Q214 AnalystPrep Q.5116

Which of the following is NOT a requirement for an organization to be certified as ISO 27001 compliant?

A. Having an Information Security Management System (ISMS) that manages its information security risks
B. Designing and implementing effective and comprehensive information security controls
C. Adopting an ongoing risk management process
D. Providing guidance on responding to and recovering from cybersecurity incidents

Answer: D

Step by step

1. ISO 27001 is a checklist of what must be achieved for certification, not implementable advice.
2. Requirements: an ISMS that manages information security risks, effective and comprehensive controls, and a comprehensive risk management process with ongoing improvement.
3. Response and recovery guidance is a feature of the NIST framework (Respond/Recover), not an ISO 27001 certification requirement.

Why D is correct: ISO 27001 certification requires an ISMS, comprehensive controls and continual risk management; response/recovery guidance is NIST's contribution.

Why the others are wrong

- **A:** An ISMS is the core certification requirement.
- **B:** Effective, comprehensive controls are required to mitigate risks.
- **C:** An ongoing, improving risk management process is required.

Q215 AnalystPrep Q.5117

A risk manager notes that information controls fall into distinct categories. Which of the following is a category of information control?

A. Protect
B. Recover
C. Behavioral
D. Detect

Answer: C

Step by step

1. Information controls are classified as behavioral or technical.
2. Protect, Detect and Recover are NIST CSF functions (steps), not control categories.
3. Behavioral is the only listed control category.

Why C is correct: Behavioral and technical are the two categories of information controls; the other options are NIST framework steps.

Why the others are wrong

- **A:** Protect is a NIST function, not a control category.
- **B:** Recover is a NIST function, not a control category.
- **D:** Detect is a NIST function (detective controls are a subtype of technical controls, but 'Detect' itself is not a category).

Q216 AnalystPrep Q.5118

Which of the five NIST Cybersecurity Framework functions involves creating and sharing a company cybersecurity policy that covers the roles and responsibilities of employees?

A. Recover
B. Protect
C. Identify
D. Detect

Answer: C

Step by step

1. The NIST CSF has five functions: identify, protect, detect, respond, recover.
2. Identify is the first step: inventory all equipment, software and data, and create and share the cybersecurity policy (roles of employees, vendors, third parties, and steps if an attack occurs).

3. The policy-creation activity therefore maps to Identify.

Why C is correct: Defining and communicating the cybersecurity policy and who is responsible for what is part of the Identify step, alongside the asset inventory.

Why the others are wrong

- **A:** Recover covers repairing and restoring affected equipment/data and keeping stakeholders informed after an attack; it is not about drafting policy.
- **B:** Protect covers access restriction, security software, encryption, backups, safe disposal and training; confusing 'training employees' with 'defining roles' leads here.
- **D:** Detect covers monitoring for unauthorized access, users and connections and investigating unusual activity.

Q217 AnalystPrep Q.5119

Which of the following actions does NOT fall under the 'protect' step of the NIST cybersecurity framework?

A. Controlling who logs onto a company's network
B. Updating security software regularly
C. Creating a company cybersecurity policy that covers roles and responsibilities of employees
D. Having formal policies for safely disposing of electronic files

Answer: C

Step by step

1. Protect actions: restrict who logs on to the network and devices, use and regularly update security software, encrypt and back up data, set policies for safe disposal of files, train employees.
2. A, B and D match these Protect actions.
3. Creating and sharing the cybersecurity policy with employee roles belongs to Identify, so C is the exception.

Why C is correct: The policy defining employee roles is an Identify activity, not a Protect safeguard.

Why the others are wrong

- **A:** Access control is a Protect safeguard, so it does fall under Protect.
- **B:** Regular updating of security software is explicitly a Protect action.
- **D:** Formal policies for safe disposal of electronic files and old devices are part of Protect.

Q218 AnalystPrep Q.5765

Which of the following best describes cyber risk in the context of financial institutions?

A. The potential for reduced operational efficiency due to outdated technology and systems.
B. The likelihood of encountering legal disputes related to breaches of online banking regulations.
C. The risk associated with unauthorized access or attacks on digital systems, compromising data security and integrity.
D. The financial risk stemming from failed IT projects or investments in technology that do not yield the expected returns.

Answer: C

Step by step

1. Cyber risk is an information security risk arising from hacking, malware/virus infection and phishing against digital systems and data.
2. Such attacks threaten the confidentiality, integrity and availability of information.
3. Only C describes this; the others describe efficiency, legal and IT-investment risks.

Why C is correct: Cyber risk is the threat of unauthorized access or attack on digital systems that undermines the security and integrity of data.

Why the others are wrong

- **A:** Describes technology obsolescence/efficiency risk, not an attack on data security.
- **B:** Describes legal/compliance risk; a cyber event may trigger it but it is not the definition.
- **D:** Describes IT project/investment failure risk, unrelated to malicious or unauthorized access.

Q219

BT answers P2.T7.20.13.1

REBUILT — Original option C cited ISO 27007/22307 and B relied on man-machine/A.I. claims outside the reading; rebuilt as a false-statement item on NIST, CIS and ISO 27001 per the reading.

Which of the following statements about cybersecurity frameworks and controls is FALSE?

A. The NIST framework consists of five functions: identify, protect, detect, respond and recover.

B. ISO 27001 supplies detailed, implementable technical fixes, so a firm needs no further guidance to achieve certification.

C. The CIS Critical Security Controls provide detailed guidance that can complement NIST or support ISO 27001 certification.

D. Behavioral controls address human behavior, and technical controls address prevention, detection and mitigation.

Answer: B

Key correction: Source key (BT answers) said B for a different version of the item (man-machine/A.I. statement); that content is outside the reading, so the item was rebuilt around ISO 27001; the false statement remains B.

Step by step

1. A is true: the five NIST functions are identify, protect, detect, respond, recover.
2. C is true: CIS gives detailed, regularly updated controls and can complement NIST or help with ISO 27001 certification.
3. D is true: behavioral controls (awareness, training, conduct rules) address human shortcomings; technical controls are preventive, detective, mitigating.
4. B is false: ISO 27001 does not give implementable advice; it is a checklist of what must be achieved (e.g., an ISMS, controls, risk management process) to be certified.

Why B is correct: ISO 27001 sets requirements for certification rather than providing detailed, implementable guidance, so B is the false statement.

Why the others are wrong

- **A:** True; these are the five NIST CSF functions.
- **C:** True; CIS is detailed guidance and can complement NIST or ISO 27001 efforts.
- **D:** True; this is the standard split of information controls.

Q220

BT answers P2.T7.24.6.1

REBUILT — Ambiguous 'most effectively' framing, a giveaway 'none of the above' option and an uncertain ISO standard number; reframed to ask which framework gives detailed, implementable controls and used ISO 27001 per the reading.

Minance Exchange, a cryptocurrency trading firm, has suffered data breaches from employee errors, including sensitive documents sent to wrong email addresses and misplaced physical files. A consultant is asked to recommend an existing framework that provides detailed, actionable control guidance (including email and web browser protections) that the firm can implement directly. Which framework is most appropriate?

A. CIS Critical Security Controls

B. ISO 27001, which sets a checklist of requirements for certification and provides no directly implementable advice

C. NIST CSF, used solely as a one-time asset inventory exercise

D. A framework limited to external, intentional threats such as hacking

Answer: A

Key correction: Source key (BT answers) said B (CIS) for the original lettering; the rebuilt item places CIS at A.

Step by step

1. The incidents are internal and unintentional (errors), so the firm needs concrete controls.
2. CIS provides detailed, regularly updated, prioritized controls; one is Email and Web Browser Protections.
3. ISO 27001 is a certification checklist (ISMS requirements), not implementable guidance; NIST CSF is a five-function framework, not control-level fixes, and is not just an inventory; D ignores the internal/unintentional nature of the incidents.
4. CIS is therefore the best fit.

Why A is correct: CIS offers detailed, actionable controls (including email and web browser protections) suited to implementing fixes for accidental exposure.

Why the others are wrong

- **B:** ISO 27001 itself does not give implementable advice; it states what must be achieved for certification.

- **C:** NIST CSF covers all five functions, not just an inventory, and is higher-level than CIS's specific controls; the option mischaracterizes it.
- **D:** The firm's incidents are internal and unintentional, so a framework focused on external intentional threats does not address them.

Q221

BT answers P2.T7.24.6.2

REBUILT — Option D also contained behavioral measures (training, code of conduct), making two options defensible; replaced D with a purely technical combination.

A financial services company wants to strengthen its behavioral controls after lapses that allowed unauthorized access to client data due to human error and non-compliance with data governance rules. Which action would be most effective in enhancing behavioral controls?

A. Implementing more robust firewalls and upgrading encryption protocols.
B. Launching a comprehensive awareness campaign, conducting regular fake phishing tests, and enforcing stricter data transfer rules.
C. Increasing the use of honeypots and deploying data loss prevention and detection (DLPD) technologies.
D. Introducing network partitioning and penetration testing of systems.

Answer: B

Step by step

1. Behavioral controls address human behavior and shortcomings: awareness campaigns, training, fake phishing tests, rules of conduct, data transfer rules, sanctions.
2. B consists wholly of such measures.
3. A (firewalls, encryption), C (honeypots, DLPD) and D (network partitioning, penetration testing) are technical controls.
4. So B is the answer.

Why B is correct: Awareness campaigns, phishing tests and data transfer rules directly target employee behavior and data governance.

Why the others are wrong

- **A:** Firewalls and encryption are technical preventive controls.
- **C:** Honeypots and DLPD are technical detective controls.
- **D:** Network partitioning is technical architecture and penetration testing is a technical test; neither changes staff behavior.

Q222

Schweser Module Quiz 50.1 Q1

The four-quadrant approach categorizes information security risks as:

A. internal versus external and intentional versus unintentional.
B. internal versus external and in house versus third party.
C. intentional versus unintentional and digital versus physical.
D. intentional versus unintentional and theft versus corruption.

Answer: A

Step by step

1. Data incidents are classified on two axes: cause (internal vs external) and intent (intentional, i.e. theft/malicious, vs unintentional, i.e. loss/accidental disclosure).
2. Digital vs physical describes the form of the data or incident and appears within quadrants; it is not an axis.
3. In-house vs third party largely restates internal vs external, so B duplicates one axis and omits intent.

Why A is correct: The quadrants are formed by internal/external cause crossed with intentional/unintentional action.

Why the others are wrong

- **B:** Uses two versions of the same axis (internal/external) and omits intent.
- **C:** Digital vs physical describes the form of data, which can occur in any quadrant.
- **D:** Theft is an intentional incident type, but corruption (voluntary or involuntary) is not a second axis; the second axis is intentional vs unintentional (theft vs loss).

Q223 Schweser Module Quiz 50.1 Q2

An employee inadvertently emails a sensitive and confidential file to a third-party recipient. The employee's action would most likely be categorized as:

A. internal and intentional.
B. external and unintentional.
C. internal and unintentional.
D. external and intentional.

Answer: C

Step by step

1. Cause: an employee of the firm, so internal (the third-party recipient is where data ends up, not the source).
2. Intent: 'inadvertently' means unintentional.
3. Internal and unintentional (errors sending documents).

Why C is correct: An employee error with no malicious intent is an internal, unintentional incident.

Why the others are wrong

- **A:** Ignores 'inadvertently'; intentional leaks involve deliberate theft or leaking.
- **B:** Misclassifies the source by focusing on the external recipient; the cause is the firm's own employee.
- **D:** Wrong on both axes; this describes hacking or phishing by outsiders.

Q224 Schweser Module Quiz 50.1 Q3

Following a severe power outage, a firm is unable to recover one of its servers that had significant and sensitive client information. This incident would most likely be categorized as:

A. internal and intentional.
B. external and unintentional.
C. internal and unintentional.
D. external and intentional.

Answer: B

Step by step

1. Cause: a power outage outside the firm's control, so external.
2. Intent: nobody acted deliberately, so unintentional.
3. Disasters, system disruptions and third-party failures are external and unintentional incidents.

Why B is correct: A power outage is an external, non-deliberate event, falling in the external and unintentional quadrant.

Why the others are wrong

- **A:** No employee deliberately caused the loss and the cause is not internal.
- **C:** Would apply to staff errors; the outage was not an employee mistake.
- **D:** There is no attacker; the event was not deliberate.

LO 50.b

Describe lessons learned from the Equifax case study.

Q225 AP Mock 1 Q32

Following the Equifax data breach in 2017, which measure would have been most effective in preventing the incident?

A. Implementing stronger encryption methods for stored data.
B. Regular audits and compliance checks to ensure security protocols are followed.
C. Conducting regular employee training on data security and phishing attacks.
D. Updating and patching security software immediately upon availability of fixes.

Answer: D

Step by step

1. Identify the attack vector: an external intrusion through a known Apache Struts vulnerability in the online

dispute portal.
2. The patch was released on March 7, 2017 and the alert reached Equifax on March 8, but the patch was never applied.
3. The measure that directly closes this vector is timely patching, which Equifax's own policy required within 48 hours for critical vulnerabilities.

Why D is correct: The breach exploited a known, patchable vulnerability, so enforcing prompt patching would have prevented the intrusion itself.

Why the others are wrong

- **A:** Encryption limits the usefulness of stolen data but does not stop the intrusion through the unpatched system.
- **B:** Audits had already found a backlog of 8,500 unpatched vulnerabilities in 2015; auditing without enforcing remediation did not prevent the attack.
- **C:** Phishing was not the attack vector; training addresses a different threat.

Q226

AnalystPrep Q.6531

Which of the following lessons from the Equifax case study is most likely applicable to financial institutions seeking to enhance their cybersecurity risk management framework?

A. A well-developed incident response plan can significantly reduce reputational and financial damage from cyber breaches.
B. Outsourcing cybersecurity functions reduces the risk exposure of financial institutions.
C. Prioritizing cost reduction over cybersecurity investments leads to stronger operational resilience.
D. Delaying disclosure of a cybersecurity breach allows financial institutions to manage public relations better and minimize regulatory penalties.

Answer: A

Step by step

1. Recall the Equifax weaknesses: poor inventory, unenforced patching, inconsistent internal communication, expired SSL certificate, poor external communication.
2. Equifax announced the breach only on September 7, 2017, six weeks after discovery, ignoring the crisis-communication principles.
3. The transferable lesson is therefore to have a tested response and crisis-communication plan to limit damage; only A is consistent with this.

Why A is correct: Equifax's slow, poorly communicated response amplified reputational and financial losses, so a well-developed response plan reduces them.

Why the others are wrong

- **B:** Nothing in the case says outsourcing reduces risk; the failures were internal control and process failures.
- **C:** Underinvesting in security controls is a cause of weakness, not a source of resilience.
- **D:** The delayed disclosure drew criticism and scrutiny; it worsened rather than reduced the damage.

Q227

AnalystPrep Q.6532

Which of the following best explains how poor vulnerability management contributed to the Equifax breach?

A. Equifax failed to identify and patch a known vulnerability in a timely manner, leaving its systems exposed to cyberattacks.
B. Equifax's security team deliberately ignored critical security patches to reduce operational disruptions.
C. Equifax lacked an incident response plan, which prevented it from addressing the breach efficiently.
D. Equifax experienced a zero-day attack, meaning no patch was available to mitigate the risk.

Answer: A

Step by step

1. The exploited Apache Struts flaw had a NIST/CVSS score of 10 and a patch was released March 7, 2017.
2. The alert went to 400 employees on March 8, but the patch was never applied, partly because assets could not be located and communication failed.
3. This is a failure to identify and patch a known vulnerability in time.

Why A is correct: The vulnerability was known and patchable, but weak inventory, policy enforcement and communication left it unpatched.

Why the others are wrong

- **B:** The case shows process and communication failures, not a deliberate decision to ignore patches.
- **C:** This describes a different failure (response) and is not the vulnerability-management issue asked about.
- **D:** A zero-day has no available patch; here the patch existed three days before the attack.

Q228

AnalystPrep Q.6533

Which of the following best describes how ineffective internal controls at Equifax contributed to the breach?

A. Equifax lacked a cybersecurity framework, making it impossible to detect threats.
B. Equifax's monitoring system failed to detect and prevent unauthorized access to sensitive data.
C. Equifax employees intentionally disabled security alerts, allowing hackers to remain undetected.
D. Equifax lacked sufficient financial resources to invest in cybersecurity controls.

Answer: B

Step by step

1. An expired SSL certificate meant Equifax could not inspect encrypted network traffic for months.
2. Hackers therefore remained undetected for about three months, until the certificate was renewed on July 29, 2017.
3. This is a detective-control failure: monitoring did not detect the unauthorized access.

Why B is correct: The expired certificate blinded traffic monitoring, so the intrusion went undetected for over three months.

Why the others are wrong

- **A:** Equifax had frameworks and policies, but they were outdated and poorly maintained and enforced.
- **C:** No intentional disabling is reported; the cause was an expired certificate left unrenewed.
- **D:** The case does not attribute the failures to lack of funds but to weak governance and execution.

Q229

AnalystPrep Q.6534 · AnalystPrep Q.6535

Which of the following best explains how poor breach disclosure practices worsened the impact of the Equifax data breach?

A. Equifax immediately informed the public and regulators, minimizing reputational damage.
B. Equifax delayed disclosing the breach, leading to increased regulatory scrutiny and public backlash.
C. Equifax disclosed the breach before understanding its full impact, which helped reduce penalties.
D. Equifax fully complied with disclosure regulations, avoiding any negative consequences.

Answer: B

Step by step

1. Equifax discovered the breach on about July 29, 2017.
2. It issued a public announcement only on September 7, 2017, six weeks later, and was blamed for lack of transparency.
3. This poor external communication added to reputational damage, investigations, fines and lawsuits.

Why B is correct: The six-week delay breached crisis-communication principles and intensified scrutiny and backlash.

Why the others are wrong

- **A:** Contradicts the facts: disclosure was not immediate.
- **C:** Disclosure was late, not premature, and did not reduce penalties.
- **D:** Equifax faced investigations, fines (up to $700 million) and lawsuits, so negative consequences were not avoided.

Q230

BT answers P2.T7.24.6.3 · BT notes P2.T7.24.6.3

REBUILT — Original 'all of the above' item mapped weaknesses to CIS control names not supported by the reading (training was a stretch); rebuilt around the GARP/Schweser inventory weakness.

Equifax's Global Threats and Vulnerability Management team was expected to patch a critical vulnerability within 48 hours but could not patch Apache Struts. Which Equifax weakness most directly explains this failure?

A. A lack of a comprehensive inventory of IT assets, so the team could not locate where Apache Struts was used.
B. The use of an expired SSL certificate, which blocked inspection of encrypted traffic.
C. Poor public communication, with the announcement delayed until September 7, 2017.

D. The absence of any vendor patch for the vulnerability at the time of the alert.

Answer: A

Step by step

1. The patch was released March 7, 2017, and the alert was distributed March 8, so a patch was available.
2. The team still could not patch because it did not know where Apache Struts ran: Equifax lacked a comprehensive IT asset inventory.
3. The expired certificate and external communication relate to detection and crisis response, not to the ability to patch.

Why A is correct: Without a complete asset inventory the team could not find the vulnerable systems, so the 48-hour patching expectation was missed.

Why the others are wrong

- **B:** The expired certificate caused the breach to go undetected for months; it did not stop patching.
- **C:** Poor public communication occurred after discovery and did not affect patching.
- **D:** A patch was released three days before the attack, so patch unavailability is not the cause.

Q231 Schweser Module Quiz 50.1 Q4

Which of the following issues was not identified as a key weakness in Equifax's information security management?

A. Poor public communication.
B. Using an expired digital certificate.
C. Lack of a comprehensive list of its IT assets.
D. Lack of sufficient internal phishing exercises.

Answer: D

Step by step

1. List the five key weaknesses: insufficient IT asset inventory, failure to enforce patch management policy, inconsistent employee communication, expired SSL certificate, poor external communication.
2. A matches the fifth, B the fourth and C the first.
3. Phishing exercises are not on the list (the attack was a technical exploit, not phishing), so D is the answer.

Why D is correct: The attack exploited an unpatched Struts vulnerability, and phishing training was not among the five identified weaknesses.

Why the others are wrong

- **A:** Poor external/public communication is one of the five weaknesses.
- **B:** The expired SSL certificate is one of the five weaknesses.
- **C:** The lack of a comprehensive IT asset inventory is one of the five weaknesses.

Reading 51. Sound Management of Risks Related to Money Laundering and Financing of Terrorism

"Sound Management of Risks Related to Money Laundering and Financing of Terrorism" (Basel Committee on Banking Supervision Publication, July 2020).

LO	Learning objective	Qs
51.a	explain best practices recommended by the Basel Committee for the assessment, management, mitigation, and monitoring of money laundering and financing of terrorism (ML/FT) risks.	22
51.b	describe recommended practices for the acceptance, verification, and identification of customers at a bank.	5
51.c	explain practices for managing ML/FT risks in a group-wide and cross-border context.	3

LO 51.a

Explain best practices recommended by the Basel Committee for the assessment, management, mitigation, and monitoring of money laundering and financing of terrorism (ML/FT) risks.

Q232

AnalystPrep Q.2991

REBUILT — Original option A (compliance as an employment condition) and D were vague and not directly supported by the Basel text; rebuilt around the written-policy/training requirements of the first line of defense.

Blackest Bank wants to strengthen its AML/CFT culture. Under the Basel Committee's guidance, which action is MOST consistent with the first line of defense?

A. Put AML/CFT policies and procedures in writing, communicate them to all personnel, and provide ongoing training tailored to each role.

B. Have the head of the business line manage the independent testing of the AML/CFT programme.

C. Base employee bonuses on the revenue generated from high-risk customers, regardless of compliance findings.

D. Provide AML/CFT training once, at hiring, and only to front-office staff.

Answer: A

Step by step

1. The business units are the first line of defense in identifying, assessing and controlling ML/FT risks.
2. Basel requires written policies communicated to all personnel, procedures for detecting and reporting suspicious transactions, high ethical standards, and ongoing role-tailored training (including for new hires and refreshers).
3. Option A is the only one that does this.

Why A is correct: A culture of compliance is built through clear written policies, communication, screening and ongoing, tailored training.

Why the others are wrong

- **B:** Independent testing belongs to internal audit (third line); business-line involvement undermines independence.
- **C:** Rewarding revenue from high-risk customers while ignoring compliance incentivises risk-taking and runs against high ethical standards.
- **D:** Basel calls for ongoing training and refreshers for staff across functions, not one-off training of front office only.

Q233

AnalystPrep Q.5453

REBUILT — Original correct option (update protocols per core principles) was weak and correct only by elimination; rebuilt around a clearly stated Basel governance requirement.

Global Trust Bank is revising its ML/FT framework to align with Basel Committee recommendations. Which governance feature best reflects those recommendations?

A. Appoint a qualified chief AML/CFT officer with sufficient stature and authority, a direct reporting line to senior management or the board, and no business-line or internal audit responsibilities.

B. Appoint a chief AML/CFT officer who also heads internal audit, to economize on resources.

C. Have the chief AML/CFT officer report through the head of the largest business line.

D. Assign ML/FT risk management to one department with no cross-collaboration with other units.

Answer: A

Step by step

1. Basel: the board and senior management should appoint a qualified chief AML/CFT officer with stature and authority to get attention of the board, senior management and business lines.
2. To avoid conflicts, the officer should not have business-line, data protection or internal audit responsibilities and needs a direct reporting line to senior management/board.
3. Only A satisfies all of this.

Why A is correct: Independence, authority and direct access to the board are the core governance requirements for the AML/CFT officer.

Why the others are wrong

- **B:** Combining with internal audit is explicitly a conflict of interest.
- **C:** Reporting through a business line subordinates compliance to business interests.
- **D:** The three lines of defense require coordination; siloing ML/FT management is contrary to this.

Q234 AnalystPrep Q.3123

REBUILT — Original referred to 'pass-through accounts' not defined in the Basel reading; restated as unrelated third-party payers inconsistent with the stated account purpose.

A bank in Italy holds a business account for an Italian company that sells suits in Europe and North America. At opening, the stated purpose was to receive payment for sales. A year-long review shows regular wire transfers received from accounts of unrelated third parties with no apparent link to the company's customers, alongside purchases of cotton and garments from China and India. Which factor is MOST important in assessing whether money laundering is a threat?

A. The account is used for both sales receipts and supplier purchases.

B. Incoming payments come from unrelated third-party accounts, inconsistent with the stated purpose of the account.

C. The company buys raw materials rather than finished clothing.

D. Most transactions are wire transfers rather than cash deposits.

Answer: B

Step by step

1. Basel monitoring systems should detect changes in, and inconsistencies with, a customer's expected transactional profile.
2. Receipts from unrelated third parties do not match the stated purpose (payment for the company's own sales) and may obscure the source of funds.
3. Receiving sales proceeds and paying suppliers, buying raw materials, and using wires are normal for a garment business.

Why B is correct: Activity inconsistent with the stated purpose and expected profile, especially funds from unrelated third parties, is the key ML red flag that monitoring should capture.

Why the others are wrong

- **A:** Both receipts and supplier payments are normal for a trading/manufacturing business.
- **C:** Buying raw materials is consistent with a clothing manufacturer's business model.
- **D:** Wires are the normal channel for international trade; they are not suspicious by themselves.

Q235 AnalystPrep Q.3124

Which of the following is the main role of supervisors in banks under the AML/CFT framework put forth by the Basel Committee?

A. Approval and oversight of AML/CFT risk management policies and procedures.

B. Advising banks on the best risk management strategies.

C. Evaluating whether the bank has an appropriate and reasonable risk assessment and an AML/CFT programme that reflects its inherent risks.

D. Helping banks to develop a sound AML/CFT risk management system that can keep track of all customer transactions.

Answer: C

Step by step

1. Supervisors adopt a risk-based approach and must evaluate the adequacy of the bank's risk assessment and

the effectiveness of its controls.
2. Approval and oversight of policies rests with the board.
3. Supervisors review and challenge; they do not design or run the bank's systems.

Why C is correct: Supervision is an independent evaluation of the bank's risk assessment and controls, not management of them.

Why the others are wrong

- **A:** Approving and overseeing policies is the board's responsibility.
- **B:** Advising on strategy would compromise supervisory independence; supervisors set expectations and evaluate.
- **D:** Developing the bank's monitoring system is management's job, not the supervisor's.

Q236 AnalystPrep Q.3125

What is the role of the chief AML/CFT officer in the second line of defense?

A. Conducting customer due diligence checks.
B. Ongoing monitoring of the bank's fulfilment of its AML/CFT duties.
C. Conducting forensic investigations into suspicious transactions.
D. Reviewing and approving high-risk transactions.

Answer: B

Step by step

1. The chief AML/CFT officer is the second line of defense.
2. The officer is responsible for ongoing monitoring of the fulfilment of AML/CFT duties (sample testing of compliance, review of exception reports) and alerting senior management/board.
3. Day-to-day CDD sits with the business units.

Why B is correct: Second-line compliance oversight means monitoring that the bank is meeting its AML/CFT duties.

Why the others are wrong

- **A:** CDD is performed by the business units (first line).
- **C:** Forensic investigation is not the officer's defined second-line role; the officer reports suspicious transactions.
- **D:** Approving transactions would be a business-line responsibility and would compromise the officer's independence.

Q237 AnalystPrep Q.3126

Paul Khan, a risk manager, presents to the board on AML/CFT governance. What is the responsibility of internal audit in the bank's AML/CFT framework?

A. Monitoring customer transactions.
B. Approving new customer accounts.
C. Independently evaluating the effectiveness of risk management and controls.
D. Developing AML/CFT policies and procedures.

Answer: C

Step by step

1. Internal audit is the third line of defense.
2. It independently evaluates risk management and controls and reports to the audit committee/board.
3. Monitoring, account approval and policy development are first/second-line or management functions.

Why C is correct: Independence requires internal audit to evaluate, not operate, the controls.

Why the others are wrong

- **A:** Transaction monitoring is a first/second-line activity.
- **B:** Account acceptance is a business-unit decision.
- **D:** Policy development is management's role; auditing one's own policies would impair independence.

Q238 AnalystPrep Q.3127

All of the following are lines of defense in the context of AML/CFT EXCEPT:

A. The supervisor
B. The internal audit function
C. The chief AML/CFT officer and the compliance department
D. Customer-facing activities

Answer: A

Step by step

1. First line: business units/customer-facing activities.
2. Second line: chief AML/CFT officer/compliance.
3. Third line: internal audit. The supervisor is external oversight.

Why A is correct: The three lines of defense are internal to the bank; supervisors are outside it.

Why the others are wrong

- **B:** Internal audit is the third line.
- **C:** Chief officer/compliance is the second line.
- **D:** Customer-facing units are the first line.

Q239 AnalystPrep Q.3128

REBUILT — Original options were vague/giveaway (profit, hurdle rates); rebuilt to test the Basel risk-assessment dimensions that underpin a risk-based approach.

Under the Basel Committee's guidance, a bank must understand its inherent ML/FT risks in order to apply a risk-based approach. Which set of areas must the bank understand?

A. Its customer base, products and services, delivery channels, and the jurisdictions in which it and its customers operate.
B. Its share price, dividend policy, capital ratios and credit rating.
C. Only its customers' occupations and the size of their deposits.
D. Only the transaction data generated by its own systems.

Answer: A

Step by step

1. Basel requires a thorough understanding of ML/FT risk in: the customer base; products and services; delivery channels (including products in development); and jurisdictions.
2. Controls and monitoring are then made commensurate with those risks.
3. Only A lists these.

Why A is correct: A risk-based approach starts from the four inherent-risk dimensions: customers, products/services, delivery channels, jurisdictions.

Why the others are wrong

- **B:** These are financial-performance measures unrelated to ML/FT risk.
- **C:** Customer background is one input but customers alone are not the full risk picture.
- **D:** Basel uses internal and external data (e.g., national risk assessments), so internal data alone is insufficient.

Q240 AnalystPrep Q.3130

REBUILT — Original stem had OCR damage and 'ALM' typos and an ambiguous audit wording; cleaned and made explicit that the chief officer performs the audit.

A large banking group's AML programme covers suspicious transaction reporting, customer identification, information sharing, account monitoring and red-flag identification. Staff in all 12 branches attend mandatory AML/CFT training twice a year, delivered online, and the board is trained separately by an outside expert. The programme appoints a chief AML/CFT officer, who personally audits the AML/CFT framework twice a year. In what respect does the programme need improvement?

A. The AML/CFT framework should be audited more than twice a year.
B. Employees should be trained in a classroom, not online.
C. The group should consolidate training across branches into a single event.
D. The audit should be performed independently (internal audit), not by the chief AML/CFT officer.

Answer: D

Step by step

1. Basel makes internal audit the third line of defense, independently evaluating AML/CFT risk management and controls.
2. The chief AML/CFT officer is the second line and should not be responsible for internal audit, to avoid conflicts of interest.
3. Training frequency/format and audit frequency are not stated deficiencies; scope and frequency should be risk-based.

Why D is correct: Self-review by the second line lacks independence; the third-line internal audit must test the programme.

Why the others are wrong

- **A:** Audit frequency should be risk-based; twice a year is not shown to be inadequate.
- **B:** Basel requires tailored, ongoing training but does not prescribe classroom delivery.
- **C:** Training should be tailored to function and risk, so a single consolidated event is not required.

Q241 AnalystPrep Q.5452

Global Bank's Chief Risk Officer is revising the bank's ML/FT procedures in line with Basel best practices. Which action should he incorporate for effective management of ML/FT risks?

A. Integrate international risk assessments and country reports to augment the bank's internal monitoring.

B. Prioritize CDD rules as the sole tool for risk identification.

C. Focus mainly on the bank's own risk assessment of ML/FT risks.

D. Delegate responsibility for ML/FT risk management to the board of directors.

Answer: A

Step by step

1. Basel: understanding of inherent risk rests on both internal data (operational and transaction data) and external sources (national risk assessments, country reports from international organizations).
2. So external reports should supplement internal monitoring.
3. The board approves and oversees; it does not run day-to-day management.

Why A is correct: Combining internal and external sources gives a complete view of inherent ML/FT risk.

Why the others are wrong

- **B:** CDD is one control among several (customers, products, channels, jurisdictions), not the primary or only tool.
- **C:** Relying mainly on internal assessment ignores required external inputs.
- **D:** The board oversees and approves; management and the chief AML/CFT officer manage the risks.

Q242 AnalystPrep Q.5454

Regent Bank, a globally renowned financial institution, is aiming to bolster its Anti- Money Laundering and Countering Financing of Terrorism (AML/CFT) protocols. Mrs. Fernandez, the bank's newly appointed Chief AML/CFT officer, is tasked with evaluating and improving the existing frameworks in accordance with global best practices. To achieve this, Mrs. Fernandez considers the importance of the three lines of defense and the role of the chief AML/CFT officer in effective risk management. Based on global best practices, which of the following measures should Mrs. Fernandez prioritize to enhance Regent Bank's AML/CFT risk management?

A. Limit the responsibilities of the Chief AML/CFT officer to solely the operational aspects, excluding any form of executive oversight.

B. Ensure that the Chief AML/CFT officer has a direct reporting line to senior management or the board, clarifying the relationship between the chief officer and other officers.

C. Standardize the AML/CFT training content for all departments to maintain uniformity in knowledge dissemination across the bank.

D. Allocate the Chief AML/CFT officer as the sole authority to modify and adjust bank AML/CFT policies without consulting the board or senior management.

Answer: B

Step by step

1. Under the Basel guidance, the chief AML/CFT officer is the second line of defense.
2. The officer must have the stature to command the attention of the board, senior management and business

lines, with a direct reporting line to senior management or the board.
3. If duties are separated from the CRO or chief compliance officer, their relationship and roles must be clearly defined.
4. Option B matches this; A, C and D each contradict the guidance.

Why B is correct: A direct reporting line gives the officer independence and ensures ML/FT issues reach the board; the relationship to other officers must be clearly defined.

Why the others are wrong

- **A:** Wrong: the officer must be able to monitor and escalate to senior management/board; stripping executive oversight removes the stature and authority the guidance requires.
- **C:** Wrong: training should be tailored to staff function, job responsibility and risk exposure, not identical for all departments.
- **D:** Wrong: policies are approved and overseen by the board and senior management; sole unilateral authority removes governance and checks.

Q243

AnalystPrep Q.5455

Falcon International Bank, known for its expansive global operations, is in the process of overhauling its Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) protocols. As a part of this initiative, Mr. Thompson, the bank's Chief Risk Officer, is collaborating with external consultants to devise a strategy that aligns with international best practices. Their discussions mainly revolve around the roles of the different lines of defense and the importance of human resources in this context. Considering global best practices, which of the following strategies should Mr. Thompson adopt to fortify Falcon International Bank's AML/CFT risk management procedures?

A. Establish a centralized AML/CFT training program that provides identical content to all employees to ensure consistent knowledge across departments.

B. Facilitate robust communication channels between the three lines of defense, ensuring that the first line - business units, are proactive in identifying, assessing, and controlling AML/CFT risks.

C. Assign the internal audit function as the primary decision-maker for amending AML/CFT policies, thereby ensuring an unbiased review and implementation process.

D. Empower the compliance function with advanced technology tools, relegating human intervention to the background, for efficient and error-free monitoring.

Answer: B

Step by step

1. Business units (front office, customer-facing) are the first line and are in charge of identifying, assessing and controlling their ML/FT risks.
2. Effective AML/CFT management needs the three lines (business units, chief AML/CFT officer/compliance, internal audit) to work together with clear roles.
3. B reflects this; A, C and D conflict with the guidance.

Why B is correct: The first line owns the day-to-day identification, assessment and control of ML/FT risk, supported by coordination with the second and third lines.

Why the others are wrong

- **A:** Wrong: training must be tailored to function and risk exposure, not identical for everyone.
- **C:** Wrong: internal audit is the independent third line and evaluates controls; it does not make policy, which would compromise its independence.
- **D:** Wrong: technology supports monitoring, but staff judgment, training and suspicious-transaction reporting remain essential; human intervention cannot be relegated.

Q244

AnalystPrep Q.5456

REBUILT — Original keyed 'continuous real-time auditing', which is not a Basel/GARP recommendation; rewrote stem and options around the GARP internal audit requirements.

Global United Bank's board has asked Mr. Kingston, Senior Internal Auditor, to ensure that internal audit contributes effectively to the bank's AML/CFT framework in line with Basel Committee guidance. Which of the following should he prioritize?

A. Take over responsibility for amending AML/CFT policies so that policy formulation is independent.

B. Conduct periodic, risk-based audits of AML/CFT policies, staff implementation, compliance oversight and training, including bank-wide audits and an evaluation of the IT monitoring system, and proactively follow

up findings.

C. Limit audit work to post-incident reviews so that it does not overlap with the work of the second line.

D. Assume the role of chief AML/CFT officer to provide ongoing monitoring of AML/CFT duties.

Answer: B

Step by step

1. Internal audit is the third line of defense and independently evaluates risk management and controls through periodic evaluations.
2. Basel guidance: audits cover adequacy of policies, effectiveness of staff implementation, compliance oversight and quality control (including alert parameters) and training; frequency is risk-based; bank-wide audits are conducted periodically; auditors proactively follow up findings and evaluate the IT monitoring system.
3. Option B lists exactly these elements; A, C and D blur the independence or purpose of the third line.

Why B is correct: Internal audit provides independent, periodic, risk-based assurance over the whole AML/CFT framework and follows up on its findings.

Why the others are wrong

- **A:** Wrong: policy is approved and overseen by the board/senior management; owning policy would compromise audit independence.
- **C:** Wrong: audits are periodic, risk-based evaluations of controls, not merely post-incident reviews.
- **D:** Wrong: ongoing monitoring is the second-line chief AML/CFT officer's role, and that officer should not be responsible for internal audit; combining the roles creates a conflict of interest.

Q245

BT answers P2.T7.20.5.3 · BT notes P2.T7.20.5.3

After reviewing and updating its money laundering and financing of terrorism (ML/FT) policies, a certain international bank internally propagates the following policies to its staff:
- We classify ML/FT risk as an operational risk but owing to the intensity of the issues and risks, ML/FT risks will receive special attention (and resources) as an emergent sub-field of operational risk.
- With respect to ML/FT risk, we will maintain three lines of defense: the business units; the chief ML/FT officer who reports to the CRO; and internal audit.
- For each customer or business relationship, a profile of normal activity should be built to support the identification of abnormal activity.
- Because we (our bank) have a presence in multiple countries, we should understand and abide by laws and regulations in each country. If a country's laws and regulations prevent adequate management of ML/FT risks, we should consider the cessation of business in the country.
- Subsequent to the KYC process, our customers are entitled to absolute confidentiality and privacy so that neither their activity can be monitored nor can we proactively report suspicious activity to law enforcement authorities (we can only reactively respond, on a case-by-case basis, to specific queries).

Which of these five policies is a problem and needs to be revised; i.e., which of the following statements is TRUE?

A. None are a problem: all five of these policies are sound

B. The bank cannot generate profiles of customers due to discrimination laws

C. The three lines of defense do not apply to ML/FT risks because only two can effectively play a role

D. Absolute confidentiality is incorrect because the bank must monitor (and document) activity and should report suspicious activity

Answer: D

Step by step

1. Treating ML/FT as operational risk with special attention, three lines of defense, activity profiles, and exiting a country that blocks adequate ML/FT management are all consistent with Basel guidance.
2. Basel guidance requires ongoing monitoring of business relationships and operations, and procedures to detect and report suspicious transactions.
3. Absolute confidentiality that prevents monitoring and proactive reporting contradicts this, so that policy must be revised: D.

Why D is correct: Banks must monitor customer activity against its normal profile and report suspicious activity; customer confidentiality cannot override this.

Why the others are wrong

- **A:** Wrong: the confidentiality policy is a clear problem, so not all five are sound.
- **B:** Wrong: building profiles of normal activity is expressly recommended, and nothing in the guidance bars it.
- **C:** Wrong: the three lines (business units, chief AML/CFT officer, internal audit) apply fully to ML/FT; the chief

officer may also be the CRO, so the stated structure is acceptable.

Q246 BT answers P2.T7.24.1.3-341

As a compliance director at Global United Bank, you're overseeing the implementation of the Basel Committee's Three Lines of Defense model in managing ML/FT risks. The bank operates globally, especially in high-risk jurisdictions, and is updating its AML/CFT framework to better meet international standards. Which observation most likely indicates a breach in the effective application of the Three Lines of Defense model in AML/CFT compliance at Global United Bank?

A. Management has started directly conducting customer due diligence for high-risk clients, bypassing the initial checks by less experienced customer-facing staff.

B. The internal audit division performs quarterly AML/CFT reviews and reports directly to the board's audit committee.

C. The chief AML officer leads a team that reviews and strengthens controls established by customer-facing units, with direct access to executive management.

D. Branch-level customer service managers keep detailed transaction records and regularly screen for risk indicators, escalating suspicious cases to the risk management team.

Answer: A

Step by step

1. First line: business/customer-facing units identify, assess and control risks, including carrying out CDD and detecting/reporting suspicious activity.
2. Second line: the chief AML/CFT officer (with a direct reporting line to senior management/board) reviews and monitors. Third line: internal audit reporting to the audit committee.
3. B, C and D describe normal roles; in A management takes over first-line CDD, undermining first-line ownership of risk: a breach.

Why A is correct: The model breaks down when management substitutes for the first line's responsibility for CDD instead of each line performing its own role.

Why the others are wrong

- **B:** Consistent: internal audit is the independent third line reporting to the board's audit committee.
- **C:** Consistent: the second-line chief AML/CFT officer reviews controls and has a direct line to senior management/board.
- **D:** Consistent: the first line keeps records, screens for risk indicators and escalates suspicious cases.

Q247 GARP PE1 Q40

A regional bank follows the three lines of defense approach for managing its operational risk. The CRO of the bank is concerned that several competing banks have recently received regulatory fines resulting from poor management of risks related to money laundering and financing of terrorism (ML/FT), and wants to improve the bank's management of these risks. The CRO reviews the Basel Committee guidelines for the management of ML/FT risks and suggests that the bank adopt industry best practices for different business functions in each line of defense. Which of the following would be most appropriate for the CRO to recommend addressing ML/FT risks?

A. The chief officer in charge of ML/FT risk should be responsible for monitoring and managing these risks within each of the business lines on a daily basis.

B. The CRO should serve as the primary point of contact with regulatory supervisors and financial intelligence units on any issues related to ML/FT risk.

C. Internal and external auditors should evaluate the bank's risk controls to ensure that they are effective in managing ML/FT risk as well as compliant with regulations.

D. The corporate operational risk function should screen current and prospective employees as part of the second line of defense in managing ML/FT risk.

Answer: C

Step by step

1. Third line of defense = internal audit; external auditors can also evaluate AML/CFT policies and controls.
2. They independently evaluate control effectiveness and compliance with AML/CFT policies and regulations: C.
3. A: first-line business units, not the chief officer, manage risk in the business lines. B: the chief AML/CFT officer is the contact for supervisors/FIUs. D: staff screening is a human resources (second-line) function, not the

operational risk function.

Why C is correct: Independent evaluation of control effectiveness and compliance is the role of internal and external audit.

Why the others are wrong

- **A:** Wrong: day-to-day management within business lines is first-line; the chief officer should have no business line responsibilities and does ongoing monitoring.
- **B:** Wrong: the chief AML/CFT officer, not the CRO, is the contact point for supervisors and FIUs (the CRO would only be if also acting as chief officer).
- **D:** Wrong: screening current and prospective staff belongs to human resources within the second line per Basel, not the corporate operational risk function.

Q248

GARP PE2 Q65 · Mock 4 Q66

The board of directors at a large bank wants to improve the bank's practices for managing money laundering and financing of terrorism (ML/FT) risk. The risk committee meets to discuss ways to achieve this objective that conform to best practices. Which of the following actions would be most appropriate for the bank to recommend?

A. Require the bank's business units to screen potential employees as part of the first line of defense in managing ML/FT risk.
B. Establish a threshold transaction value and review all transactions above this threshold for evidence of ML/FT.
C. Exclude politically exposed persons (PEPs) from screening for ML/FT risk due to their much lower ML/FT risk.
D. Give the compliance and legal functions the primary responsibility for managing ML/FT risk.

Answer: A

Step by step

1. Basel guidance: business units are the first line of defense and must identify, assess and control ML/FT risks in their business.
2. The first line follows written policies and procedures and has adequate policies for screening prospective and existing staff.
3. Test each option against this: only A matches; B relies on a threshold, C ignores PEP risk, D misplaces first-line ownership.

Why A is correct: Screening prospective and existing staff is part of the first-line policies, and business units own the ML/FT risk of their activities.

Why the others are wrong

- **B:** A value threshold leaves sub-threshold transactions unmonitored (invites structuring); monitoring must be risk-based and cover all accounts and transactions.
- **C:** PEPs are higher-risk and, especially foreign PEPs, require enhanced due diligence; treating them as low risk reverses the rule.
- **D:** Compliance is part of the second line; the first line (business units) holds primary responsibility, with the chief AMLCFT officer overseeing.

Q249

Mock 1 Q20

Banks have three lines of defense for mitigating money laundering (ML) and the financing of terrorism (FT) risks. Which of the following is NOT one of these lines of defense?

A. Business units.
B. Regulatory authorities.
C. The chief ML/FT officer.
D. Internal and external audits.

Answer: B

Step by step

1. First line: business units (front office, customer-facing activity).
2. Second line: the chief AMLCFT officer, the compliance function (also HR, technology).
3. Third line: the internal audit function.
4. Regulators are external supervisors, not part of the bank's own lines of defense.

Why B is correct: The three lines are all internal to the bank; regulatory authorities supervise from outside.

Why the others are wrong

- **A:** Business units are the first line, so this is a line of defense.

- **C:** The chief AMLCFT officer sits in the second line.
- **D:** Internal audit is the third line (external auditors also evaluate controls), so this belongs to the framework.

Q250

Mock 3 Q40

A compliance manager at a bank is documenting the responsibilities of each line of defense in managing money laundering and financing of terrorism (ML/FT) risk. Which of the following statements most accurately describes a responsibility within this framework?

A. The chief ML/FT officer should perform customer due diligence at account opening for all high-risk customers.

B. Business units should file suspicious activity reports directly with the financial intelligence unit without escalating to the compliance function.

C. Internal and external auditors should evaluate the bank's risk controls to ensure that they are effective in managing ML/FT risk as well as compliant with regulations.

D. The board of directors should review individual customer files to confirm that due diligence has been performed correctly.

Answer: C

Step by step

1. Third line = internal audit (with external auditors), which evaluates whether controls are effective and compliant.
2. A: CDD at onboarding is a first-line task; the chief officer is second line.
3. B: reporting suspicious transactions is the chief AMLCFT officer's responsibility; business units detect and escalate.
4. D: the board approves and oversees the policy, it does not review files.

Why C is correct: Audit is the independent third line that tests both effectiveness and regulatory compliance of ML/FT controls.

Why the others are wrong

- **A:** Confuses the second line with the first; the chief officer oversees and advises rather than doing onboarding CDD.
- **B:** Suspicious transaction reporting to the FIU rests with the chief AMLCFT officer, not with business units bypassing them.
- **D:** The board oversees and approves strategy and policy; file review is operational.

Q251

Schweser Module Quiz 51.1 Q1

Which of the following is an example of external data that the chief Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) officer should analyze and understand in order to manage and mitigate ML/FT risks?

A. Transaction data.

B. Payment message streams.

C. Country reports.

D. Customer passports and identity cards.

Answer: C

Step by step

1. Risk understanding rests on internal data (operational and transaction data) and external sources.
2. External sources named: national risk assessments and country reports from international organisations.
3. Only C is external.

Why C is correct: Country reports from international organisations are an external source of ML/FT risk information.

Why the others are wrong

- **A:** Transaction data is collected by the bank, so it is internal.
- **B:** Payment message streams are generated within the bank's operations, so internal.
- **D:** Identity documents are internal customer information gathered by the bank for CDD.

Q252 Schweser Module Quiz 51.1 Q2

REBUILT — Original options included implausible giveaways (most important line; employees irrelevant); replaced with plausible alternatives from the three-lines structure.

With respect to managing and mitigating money laundering and the financing of terrorism (ML/FT) risks in a bank, bank tellers and branch managers are examples of:

A. the first line of defense.
B. the second line of defense.
C. the third line of defense.
D. staff outside the lines of defense, because ML/FT risk is the sole responsibility of the chief AML/CFT officer.

Answer: A

Step by step

1. Business units (front office, customer-facing activity) are the first line of defense.
2. Tellers and branch managers are customer-facing, so they identify, assess and control ML/FT risk using written policies.
3. Second line is the chief AMLCFT officer and compliance; third is internal audit.

Why A is correct: Customer-facing business units are the first line and own the ML/FT risks of their activities.

Why the others are wrong

- **B:** The second line is the chief AMLCFT officer/compliance (also HR, technology), not customer-facing staff.
- **C:** The third line is the internal audit function.
- **D:** ML/FT risk is not left to one officer; front-line staff must follow policies and detect and report suspicious activity.

Q253 AP Mock 3 Q15

REBUILT — Original 'main driver' wording was ambiguous (FATF vs. banking-system integrity); reworded to the Committee's stated mandate and made options distinct.

Which of the following best describes the primary reason the Basel Committee is committed to sound anti-money laundering and countering the financing of terrorism (AML/CFT) policies in banks?

A. To enforce the Financial Action Task Force recommendations as binding law on banks
B. To strengthen the safety and soundness of banks and preserve the integrity of the international financial system
C. To protect customers from the cost and inconvenience of identification procedures
D. To protect the integrity of securities and capital markets as a whole

Answer: B

Step by step

1. The Committee's AML/CFT work is part of its mandate to enhance worldwide financial stability through stronger regulation, supervision and bank practices.
2. Banks without sound ML/FT risk management face reputational, operational, compliance and concentration risks, which threaten their safety and soundness.
3. Weak controls also undermine the integrity of the international financial system, so B states the core rationale.

Why B is correct: The Basel Committee writes from a prudential mandate: safe and sound banks and a sound international financial system.

Why the others are wrong

- **A:** The Committee builds on FATF standards but does not enforce them or make them binding; it issues supervisory guidance.
- **C:** KYC/CDD adds burden on customers; protecting them is not the purpose (and policies must still avoid denying access to the disadvantaged).
- **D:** Capital-markets integrity is a securities-regulator concern, not the Basel Committee's banking-supervision mandate.

LO 51.b

Describe recommended practices for the acceptance, verification, and identification of customers at a bank.

Q254 AnalystPrep Q.5125

REBUILT — Original ('which justifies comprehensive AML risk management') was vague with several defensible options; rebuilt as a verification/STR scenario from the reading.

A bank processed a first transaction for a new customer before identity verification was complete, to avoid disrupting business. Verification subsequently cannot confirm the customer's identity. According to Basel Committee recommendations, what should the bank do?

A. Treat the matter as closed, since the transaction was completed before the verification failed
B. Reclassify the customer as low risk because no adverse information was found
C. Notify the customer that a suspicious transaction report (STR) is being filed so the customer can supply documents
D. Consider filing an STR, without informing the customer, directly or indirectly, that it has been or will be filed

Answer: D

Step by step

1. Identity could not be confirmed, so CDD is incomplete; the problem cannot be resolved, so the bank should close the account or otherwise block access to it.
2. Where verification fails after proceeding, the bank should consider filing an STR.
3. The customer must not be tipped off, directly or indirectly, so D is correct.

Why D is correct: Failure to complete CDD triggers consideration of an STR, and tipping-off is prohibited.

Why the others are wrong

- **A:** Ignores the unresolved CDD failure; the bank should consider an STR.
- **B:** Absence of information is not evidence of low risk; unverified identity is a red flag.
- **C:** This is tipping off, which is prohibited.

Q255 AnalystPrep Q.5457

During the development of its Customer Acceptance Policy (CAP), MetroBank is focusing on aligning with best practices for the acceptance of new clients while also adhering to the principles of financial inclusion. Based on the recommendations by the Basel Committee, which of the following approaches should MetroBank prioritize to ensure both compliance and inclusivity in its CAP?

A. Implement a universal policy that restricts the opening of accounts to only those customers who can provide a comprehensive financial profile, thereby minimizing the risk of money laundering and terrorist financing.
B. Enforce a stringent CAP that mandates enhanced due diligence for all new customers, irrespective of their risk profile, to maintain a consistent level of scrutiny across all customer interactions.
C. Develop a CAP that incorporates differentiated due diligence measures, considering factors such as the customer's geographical location, type of product desired, and the level of risk associated with their profile, to balance risk management with non-discrimination.
D. Establish a CAP that exclusively targets high-risk individuals, such as politically exposed persons (PEPs) and individuals from high-risk countries, with enhanced due diligence, while simplifying the process for all other customers.

Answer: C

Step by step

1. Basel requires basic due diligence for all customers and commensurate (risk-based) due diligence as risk rises; simplified measures are allowed for proven lower-risk cases.
2. The CAP must not be so restrictive that it denies access to disadvantaged persons.
3. Only C combines differentiated, risk-based measures with inclusion.

Why C is correct: A risk-based, differentiated CAP manages ML/FT risk without excluding the financially or socially disadvantaged.

Why the others are wrong

- **A:** Requiring a comprehensive financial profile excludes low-balance or disadvantaged customers, contradicting financial inclusion.
- **B:** EDD for everyone ignores the risk-based approach and is excessively restrictive.
- **D:** Basic due diligence is still required for all customers and risk assessment uses many factors (products, geography, linked accounts), not only PEPs/high-risk countries.

Q256 BT answers P2.T7.20.5.1 · BT notes P2.T7.20.5.1

REBUILT — Option A said identity 'must' be verified before any transactions; reworded to the reading's 'generally should not ... until' standard so it is not over-absolute.

A person named Jack Ryan has applied to open a new account at Quadstreet International Bank. The bank has a modern, well-established, and compliant customer acceptance policy. Immediately the bank is able to determine two facts: Jack is a high-risk customer, but he previously had an account at another large bank. Further, Quadstreet does conduct business with Jack's previous bank and considers it to be reputable. With respect to money laundering and terrorism financing (ML/FT), which of the following statements is TRUE about Jack Ryan's application?

A. As a general rule, Jack's identity should be established and verified before the bank opens the account or carries out any transactions

B. Because Jack is a high-risk customer, the bank is required to avoid him like any high-risk customer

C. Because Jack already has an account at a reputable bank, he can be classified as low-risk and well-identified

D. If Jack is a politically exposed person (PEP), he can be classified as low-risk on the assumption that the media already vets him and he cannot evade scrutiny

Answer: A

Step by step

1. Basel: a bank should generally not establish a relationship or carry out transactions until the customer's identity is satisfactorily established and verified.
2. High risk does not mean prohibited: the bank sets the risks it will accept and applies EDD, so B fails.
3. A prior account elsewhere does not establish low risk or identity (the other bank may have refused or ejected him), so C fails.
4. PEPs, especially foreign PEPs, require EDD, so D fails.

Why A is correct: Verification precedes the relationship; high risk calls for EDD, not automatic refusal or low-risk labels.

Why the others are wrong

- **B:** Banks may accept high-risk customers with enhanced due diligence and senior-management approval; avoidance is not required.
- **C:** A prior account is not proof of low risk or verified identity; the customer may have been refused by another bank over ML/FT concerns.
- **D:** PEPs are higher risk and need EDD; media scrutiny does not lower the risk.

Q257 BT answers P2.T7.24.1.2

REBUILT — Removed ambiguous 'legal dispute' twist and the vague 'retired UK general' option; clarified the EDD triggers so exactly one trust is lower-risk.

As a compliance manager at a Jersey trust company, you review trust accounts for AML/CFT compliance. Which of the following trusts is LEAST likely to require enhanced due diligence (EDD)?

A. A trust for a foreign politically exposed person (PEP), funded from large and varied overseas sources

B. A trust set up by a local resident for small, regular family distributions, with identity verified through local ID and utility bills

C. A trust for an international businessman with assets in high-risk jurisdictions and a complex ownership structure

D. A trust whose beneficial owner is undisclosed and whose funds come from a secretive jurisdiction

Answer: B

Step by step

1. EDD is called for with foreign PEPs, high-risk jurisdictions, complex structures, and unknown beneficial owners.
2. A, C and D each contain such risk factors.
3. B involves a local customer, small routine flows and reliable verification, which supports simplified/basic due diligence.

Why B is correct: Risk-based CDD: higher-risk features trigger EDD; small, routine, well-verified local activity does not.

Why the others are wrong

- **A:** Foreign PEPs require EDD, and large varied sources add to risk.
- **C:** High-risk jurisdictions and complex ownership are classic EDD triggers.
- **D:** The bank must identify and verify the beneficial owner; secrecy-jurisdiction funds make this high risk.

Q258 Schweser Module Quiz 51.1 Q3

REBUILT — Option B tightened to 'large balances and regular cross-border wires' as in the reading (the combination triggers EDD).

The risk manager of a large U.S. multinational bank is attempting to put in greater risk controls. In keeping with recommendations from the Basel Committee on the sound management of risks related to money laundering and the financing of terrorism (ML/FT), she requires enhanced customer due diligence (CDD) for:

A. all accounts from customers initiated in countries outside the United States.
B. accounts with large balances and regular cross-border wire transfers.
C. individual accounts with balances less than the $250,000 Federal Deposit Insurance Corporation (FDIC) insurance limit.
D. accounts of persons who reside in countries other than their countries of birth.

Answer: B

Step by step

1. Basel says EDD may be essential for an individual planning a large account balance with regular cross-border wires, and for PEPs (required for foreign PEPs).
2. Foreign origin alone, modest balances, or a residence different from birthplace are not by themselves EDD triggers; they are inputs to the risk profile.
3. So B.

Why B is correct: EDD is risk-based and targeted at higher-risk features such as large balances with regular cross-border wires.

Why the others are wrong

- **A:** Too broad: country of origin is one risk factor, not an automatic EDD trigger.
- **C:** Low balances and routine use fit simplified procedures, the opposite of EDD.
- **D:** Country of residence differing from origin is a factor to assess, not an automatic EDD trigger.

LO 51.c

Explain practices for managing ML/FT risks in a group-wide and cross-border context.

Q259 AP Mock 3 Q68 · AnalystPrep Q.3122

REBUILT — Original relied on Annex-1 conditions not in the reading and had weak distractors; rebuilt on the reading's introducer/reliance rules (standards of the introducer, ultimate responsibility).

A bank is considering relying on customer due diligence performed by an introducer in another jurisdiction. Which of the following is consistent with Basel Committee recommendations?

A. Rely on any introducer that is a licensed bank, since it is a regulated institution, and treat the introducer as responsible for any CDD failures
B. Rely on an introducer subject to less strict AML/CFT standards than the bank's own, provided the introducer has a strong written AML policy
C. Rely on an introducer in the same financial group that is subject to the same standards as the bank, with application of those standards supervised at group level, while the bank still obtains the customer information
D. Rely on an introducer in any jurisdiction, because the bank's responsibility for CDD ends once the introducer confirms identification was completed

Answer: C

Key correction: Source keys (AP Mock 3, AnalystPrep) said A (established business relationship); that condition comes from an annex outside the reading, so the item was rebuilt on the reading's rules.

Step by step

1. The bank must not rely on introducers subject to standards less strict than its own, so B fails.
2. Reliance is acceptable where the introducer is subject to equal or stricter standards; an introducer in the same group under the same standards, supervised at group level, qualifies, and the bank should obtain the customer information from it.
3. Ultimate responsibility for customer identification and verification stays with the bank even when a third party performs CDD, so A and D fail.

Why C is correct: Reliance on an introducer requires equivalent or stricter standards, and the bank keeps ultimate responsibility.

Why the others are wrong

- **A:** Being a licensed bank is not enough, since standards may be weaker; and responsibility cannot be transferred to the introducer.
- **B:** Basel says not to rely on introducers subject to less strict standards; a policy on paper does not cure the gap.
- **D:** Ultimate responsibility remains with the bank; outsourcing/reliance does not discharge it.

Q260

BT answers P2.T7.24.1.1

A compliance officer at RevoWise, a fintech company recently granted a UK banking license, is reviewing the company\'s compliance framework. The review focuses on the implementation of "Borderless Monitor," RevoWise\'s proprietary software, within expanded services that now include multicurrency accounts and business banking. The officer is evaluating whether RevoWise\'s practices adhere to the Basel Committee\'s recommendations on combating money laundering (ML) and financing of terrorism (FT). Which of the following practices at RevoWise does NOT align with the Basel Committee\'s recommended best practices?

A. The "QuickVerify" system streamlines customer due diligence with a uniform global approach, eliminating duplication by foregoing country-specific adaptations.

B. "Borderless Monitor" flags atypical transactions in high-risk accounts for further review.

C. The chief compliance officer reports directly to the board and executive management.

D. All wire transfers include detailed sender and recipient information, reviewed manually and automatically.

Answer: A

Step by step

1. Basel expects group-wide AML/CFT policies applied consistently across jurisdictions, but they must remain consistent with and supportive of group policy while respecting local requirements.
2. Where a host jurisdiction's requirements are stricter than the group's, the branch or subsidiary must adopt the host requirements.
3. A uniform global CDD process with no country-specific adaptation cannot meet stricter local rules, so A departs from best practice.
4. B (risk-based monitoring of atypical activity), C (compliance head with access to the board/senior management) and D (complete originator and beneficiary information on wires) are all recommended.

Why A is correct: Group-wide consistency does not mean identical procedures everywhere: local requirements that are stricter than group policy must be implemented, so ignoring country-specific adaptations is non-compliant.

Why the others are wrong

- **B:** Aligned: monitoring that flags unusual transactions in high-risk accounts is the risk-based approach Basel recommends.
- **C:** Aligned: the chief AML/CFT/compliance officer should have direct access to and report to the board and senior management.
- **D:** Aligned: wire transfers should carry full originator and beneficiary information, with monitoring.

Q261

Schweser Module Quiz 51.1 Q4

REBUILT — Original supervisor-role item relied on a supervisor rule not in GARP and option B was also defensible; rewritten around the GARP rule that a branch must adopt stricter host-country requirements and the chief AML/CFT officer's duties.

A banking group headquartered in Country H operates a subsidiary in Country F. Country F's customer due diligence requirements are stricter than the group's AML/CFT policy, and Country F's law restricts transmitting customer names and balances abroad. Which of the following is the most appropriate action for the group under Basel Committee guidance on managing ML/FT risk in a group-wide and cross-border context?

A. Apply the group policy at the subsidiary to keep procedures uniform, since group-wide consistency takes precedence over local rules.

B. Allow the subsidiary to adopt Country F's stricter requirements, and have the group chief AML/CFT officer coordinate information sharing with the head office within legal protections, informing home supervisors if FATF standards cannot be properly implemented.

C. Let the subsidiary adopt Country F's requirements and withhold all customer information from the head office, so the group assesses risk only on a consolidated-balance-sheet basis.

D. Close the subsidiary immediately, because information-sharing restrictions make group-wide risk management impossible.

Answer: B

Key correction: Source key (Schweser) said D (supervisor ensures stricter of two jurisdictions' rules applied); the original item was replaced because that supervisory rule is not stated in the GARP chapter, whose rule is that the branch/subsidiary adopts stricter host requirements.

Step by step

1. Group policies must be applied consistently, yet remain consistent with and supportive of group policy.
2. Where host requirements are stricter than the group's, the branch or subsidiary adopts the host requirements, ruling out A.
3. Banks should still be able, subject to legal protections, to share customer information with the head office for group risk management, so withholding everything (C) is wrong.
4. The group chief AML/CFT officer coordinates group-wide information sharing; if the host country does not permit proper implementation of FATF standards, home supervisors are informed and closure is only a possible last resort (D is premature).
5. Therefore B.

Why B is correct: Stricter host rules apply locally, while the group chief AML/CFT officer maintains legally protected information sharing with the head office and escalates to home supervisors if FATF standards cannot be implemented.

Why the others are wrong

- **A:** Reverses the rule: group consistency does not override stricter host requirements, which must be adopted.
- **C:** Overcorrects: risk management needs head-office access to information on customers (especially high-risk ones), within legal protections, and monitoring on a consolidated basis.
- **D:** Premature: the officer first informs home supervisors; closing operations is only considered in some instances when FATF standards cannot be implemented.

Reading 52. Case Study: Financial Crime and Fraud

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 11.

LO	Learning objective	Qs
52.a	describe elements of a control framework to manage financial fraud risk and money laundering risk.	10
52.b	summarize the regulatory findings and describe the lessons learned from the USAA case study.	5

LO 52.a

Describe elements of a control framework to manage financial fraud risk and money laundering risk.

Q262

AP Mock 3 Q69

On 20 May 2015 the European Parliament and Council issued a directive to prevent the use of the financial system for money laundering or terrorist financing. According to the European Union, which of the following activities constitutes money laundering?

A. Knowingly converting or transferring property derived from criminal activity in order to disguise the illicit origin of the property.

B. The provision or collection of funds to be used, partly or in full, to facilitate any offense considered by the authorities as a terrorism act.

C. Any intentional violation of the law or of internal policies perpetrated by the firm's employees.

D. Getting the money out to use while evading taxes and law enforcement through activities such as fake payments to employees, fake loans, or dividends to accomplices.

Answer: A

Step by step

1. The EU Directive treats as money laundering the conversion or transfer of property known to derive from criminal activity, in order to conceal or disguise its illicit origin.
2. Check each option against that definition: only A matches.
3. B is the IMF description of terrorism financing, C is the Basel 'unauthorized activities' sub-category of internal fraud, and D describes the integration phase of laundering, not the legal definition.

Why A is correct: Money laundering under the EU Directive is organizing, facilitating or concealing criminal activity by converting or transferring funds so as to hide their illicit origin.

Why the others are wrong

- **B:** Describes terrorism financing (provision or collection of funds for terrorist acts), a separate offence.
- **C:** Describes unauthorized activities, a subcategory of internal fraud, committed by employees.
- **D:** Describes the integration (extraction) phase, a stage of the laundering process rather than the Directive's definition of the offence.

Q263

AnalystPrep Q.5120

An operational risk manager at the bank is presenting on financial crimes and fraud. He starts his presentation by defining financial crime. Which of the following is the correct definition of a financial crime?

A. Any criminal conduct relating to money or to financial services or markets

B. Losses due to acts intended to defraud, misappropriate property or circumvent regulations, the law, or company policy excluding diversity/discrimination events involving at least one internal party

C. Losses due to acts of a type intended to defraud, misappropriate property or circumvent the law by a third party

D. Misappropriation of assets, such as extortion, embezzlement, malicious destruction of assets, bribery, and tax evasion

Answer: A

Step by step

1. The UK FCA Handbook defines financial crime as any kind of criminal conduct relating to money or to financial services or markets, including fraud or dishonesty, market misconduct, handling proceeds of crime and financing of terrorism.

2. Match: A.
3. B is the Basel definition of internal fraud, C the Basel definition of external fraud, and D lists examples of the 'theft and fraud' subcategory.

Why A is correct: Financial crime is the broad umbrella (internal and external fraud, money laundering, terrorism financing) defined by the FCA as criminal conduct relating to money, financial services or markets.

Why the others are wrong

- **B:** This is the Basel definition of internal fraud, only one component of financial crime.
- **C:** This is the Basel definition of external fraud, another component only.
- **D:** Lists examples of theft and fraud within internal fraud; too narrow to define financial crime.

Q264

AnalystPrep Q.5122

A risk manager at a large bank states that the bank has zero tolerance for internal fraud. He goes ahead to highlight that the bank has a robust framework of controls and measures to mitigate internal fraud risks. Which of the following is a component of such a framework?

A. Inspections
B. Selection
C. Placement
D. Layering

Answer: B

Step by step

1. The internal fraud control framework has four components: selection, prevention, detection, and deterrents.
2. Selection (screening employees, contractors and third parties at onboarding, tone and culture) is one of them: B.
3. Placement and layering are phases of money laundering; inspections is a function (a subdivision of internal audit at some banks) that performs detection, not a named component.

Why B is correct: The four components of internal fraud management are selection, prevention, detection and deterrents.

Why the others are wrong

- **A:** Inspections is a team/function that supports detection, not one of the four framework components.
- **C:** Placement is the first phase of money laundering, not a fraud control.
- **D:** Layering is the second phase of money laundering, not a fraud control.

Q265

AnalystPrep Q.5123

An operational risk manager at a bank has asked a junior analyst to prepare a presentation on AML risk management to be presented to the board's risk committee. Which of the following controls falls under the "deterrents" step of AML controls?

A. Transaction monitoring system
B. Staff information and training
C. Ethos and values
D. Legal pursuits

Answer: D

Step by step

1. AML controls have four steps: selection, prevention, detection, deterrents.
2. Deterrents are the announced sanctions after an event: escalation to the financial intelligence unit, legal pursuits, and closure of accounts.
3. Legal pursuits therefore fall under deterrents: D.

Why D is correct: Deterrents are punitive actions (legal pursuits, account closure, referral to the FIU) that tip the risk-reward balance against laundering.

Why the others are wrong

- **A:** Transaction monitoring is a detection control.
- **B:** Staff information and training is a prevention control.
- **C:** Ethos and values belongs to selection (customer profiling and culture).

Q266 AnalystPrep Q.5124

A newly hired money laundering risk officer is presenting on AML risk management. He highlights that it is common for criminals to disguise the proceeds of their criminal activities into legitimate sources of funds in phases. Which of the following is a phase of money laundering?

A. Placement
B. Deterrent
C. Detection
D. Protection

Answer: A

Step by step

1. The phases of money laundering are placement, layering and integration (extraction).
2. Placement appears in the list: A.
3. Deterrent and detection are AML control steps, and protection is neither a phase nor a control step.

Why A is correct: Money laundering proceeds through placement, layering and integration; only placement is listed among the options.

Why the others are wrong

- **B:** Deterrents are an AML control step, not a laundering phase.
- **C:** Detection is an AML control step, not a laundering phase.
- **D:** Protection is not part of either the laundering phases or the four-step control framework.

Q267 AnalystPrep Q.5126

REBUILT — Original distractors were giveaways (e.g., banks exempt from AML rules); rewrote options to be plausible while keeping the 2022 FCA challenger-bank finding as the answer.

In its 2022 report, the FCA examined financial crime controls at challenger banks, which are fully digital and offer customers the ability to open accounts very quickly. Which of the following was a key finding of the FCA review?

A. Rapid digital onboarding gives challenger banks more information on new customers than traditional banks hold, so higher-risk customers are identified more reliably.
B. Information gathered at account opening may be insufficient to identify higher-risk customers, so customer identification, verification and risk assessment need to be strengthened.
C. Because challenger banks rely on machine-learning KYC tools, they are exempt from enhanced due diligence for politically exposed persons.
D. Transaction monitoring alerts at challenger banks were consistently well documented, so no changes to oversight frameworks were required.

Answer: B

Step by step

1. The FCA saw a risk that information gathered at fast account opening is insufficient to identify higher-risk customers.
2. It found weaknesses in customer due diligence, inconsistent enhanced due diligence (e.g., for PEPs), and inadequate rationale for ignoring monitoring alerts.
3. It recommended that challenger banks adjust oversight and controls to keep pace with growth. Only B matches.

Why B is correct: The FCA found that quick, digital onboarding can leave too little information to identify higher-risk customers, so identification, verification and risk assessment controls must improve.

Why the others are wrong

- **A:** Reverses the finding: speed of onboarding creates a risk of insufficient information, not better risk identification.
- **C:** AML rules, including enhanced due diligence for PEPs, apply regardless of automation; the FCA found EDD applied inconsistently.
- **D:** The FCA found weaknesses and inconsistent or inadequate rationale in handling alerts, and recommended stronger oversight.

Q268

BT answers P2.T7.24.2.1

REBUILT — Currency symbol was unreadable in OCR; removed the currency and kept the scenario unchanged.

HSBS Bank Plc has implemented a data analytics system that automatically monitors customer transactions for patterns that could indicate fraudulent activity. The system triggers alerts for transactions exceeding a threshold of 5,000 per day to or from high-risk jurisdictions. The alerts are then reviewed by a dedicated team trained to investigate potential financial crimes. What is the most relevant type of control covered in this instance?

A. Selection
B. Prevention
C. Detection
D. Deterrents

Answer: C

Step by step

1. Control steps are selection, prevention, detection and deterrents.
2. The system monitors transactions, raises alerts on abnormal patterns, and a team investigates them: this is transaction monitoring and alerts on abnormal behavior.
3. These are detection controls: C.

Why C is correct: Automated transaction monitoring with alerts on abnormal activity, followed by investigation, is the detection step.

Why the others are wrong

- **A:** Selection concerns screening and KYC/due diligence at onboarding, not ongoing monitoring.
- **B:** Prevention concerns segregation of duties, access rights, training and risk-based controls that limit opportunity, not alerts on activity already occurring.
- **D:** Deterrents are sanctions, legal pursuits and account closures after an event.

Q269

Schweser Module Quiz 52.1 Q1

The risk manager of a regional U.S. commercial bank flags one of the business accounts for potential smurfing. Which of the following descriptions most likely reflects the activity that the manager flagged?

A. The account holder made a series of large withdrawals from the account.
B. The account holder made a series of frequent small deposits into the account.
C. The account was closed and reopened multiple times in a short period of time.
D. The account holder made a very large deposit into the account, well above the anti-money laundering (AML) reporting threshold.

Answer: B

Step by step

1. Smurfing is a placement-phase technique: a series of small transactions instead of one large one.
2. Its purpose is to stay below the AML reporting threshold and avoid detection.
3. So frequent small deposits fit: B.

Why B is correct: Smurfing splits a large amount into many small transactions to remain under AML reporting thresholds.

Why the others are wrong

- **A:** Large withdrawals are not the placement technique of breaking amounts down to avoid reporting.
- **C:** Repeated closing and reopening is not smurfing.
- **D:** A single very large deposit is the opposite of smurfing; it would trigger reporting.

Q270

Schweser Module Quiz 52.1 Q2

Which of the following steps is not one of the phases in money laundering?

A. Conversion.
B. Integration.
C. Layering.
D. Placement.

Answer: A

Step by step

1. Money laundering proceeds through three phases: placement, layering, and integration (also called extraction).
2. Placement introduces illicit cash into the financial system; layering obscures its origin through multiple transactions; integration returns the cleaned funds to the criminal as apparently legitimate wealth.
3. 'Conversion' is not one of these phases, so A is the answer.

Why A is correct: The recognised money laundering sequence is placement, layering, integration; conversion is not a named phase.

Why the others are wrong

- **B:** Integration is the third phase (funds re-enter the economy as legitimate); it is a genuine phase.
- **C:** Layering is the second phase (obscuring the trail through complex transactions); it is a genuine phase.
- **D:** Placement is the first phase (getting illicit cash into the system); it is a genuine phase.

Q271 Schweser Module Quiz 52.1 Q3

Monitoring of abnormal activities and behavior is best associated with which of the following components of the risk management framework?

A. Detection.
B. Deterrents.
C. Prevention.
D. Selection.

Answer: A

Step by step

1. The fraud/AML control framework has four components: selection, prevention, detection, and deterrents.
2. Selection screens employees, contractors, and customers up front; prevention delimits rights and segregates duties; detection monitors and supervises; deterrents are announced sanctions.
3. Monitoring for abnormal activity and behavior is a detective control, so the answer is A.

Why A is correct: Detective controls (supervision, monitoring, alerts) identify abnormal behavior after it occurs so that fraud impact is limited.

Why the others are wrong

- **B:** Deterrents are announced sanctions and disciplinary or legal actions, not monitoring; confusing the deterrent effect of detection with detection itself.
- **C:** Prevention consists of controls such as limiting access rights and segregating duties that stop fraud occurring, not monitoring for it.
- **D:** Selection is the screening/profiling of employees and customers before engagement, not ongoing monitoring.

LO 52.b

Summarize the regulatory findings and describe the lessons learned from the USAA case study.

Q272 AnalystPrep Q.5127

REBUILT — Original had three giveaway distractors and a platitude answer; rebuilt to test the specific lesson on alert management from the case.

USAA Federal Savings Bank was fined $140 million by FinCEN and the OCC in March 2022 for an ineffective Bank Secrecy Act/anti-money laundering (BSA/AML) program. In 2021 the bank introduced a new transaction monitoring system that left roughly 90,000 unreviewed alerts and about 6,900 unreviewed cases by year-end. Which lesson from the case is most directly supported by this feature?

A. A more sensitive monitoring system is always preferable because more alerts mean better detection.
B. Automated monitoring should be calibrated so that alert volumes are manageable and can be reviewed and assessed in a timely way.
C. Regulators impose AML fines only after a specific theft or laundering case has been proven at the institution.
D. Using third-party contractors to review alerts removes the need to train staff in AML matters.

Answer: B

Step by step

1. The 2021 system was 'too sensitive' and generated an unmanageable number of alerts and cases.
2. The resulting backlog (about 90,000 alerts, 6,900 cases) caused unreasonable delays in detecting and reporting

suspicious activity.
3. Lesson: automated detection must keep false positives/negatives and alert volume manageable, with adequate monitoring, assessment and feedback.

Why B is correct: Alerts that cannot be reviewed provide no control; the case shows monitoring systems must be calibrated to produce a manageable, reviewable volume.

Why the others are wrong

- **A:** Opposite of the lesson: excess alerts created the backlog; sensitivity without capacity weakens the control.
- **C:** The fine was imposed for an inadequate AML program, not for a known theft or laundering case.
- **D:** USAA used contractors because of understaffing and failed to train them; this worsened the problem.

Q273

BT answers P2.T7.24.2.2 · BT notes P2.T7.24.2.2

REBUILT — Keyed option relied on a cost claim not made in the case and the source repeated the stem with a conflicting inferred key; rebuilt around the stated lesson that fines follow accumulated failings and delayed response to regulators.

In March 2022, USAA Federal Savings Bank (FSB) was fined $140 million by FinCEN and the OCC for failing to implement and maintain a BSA/AML compliance program. Which of the following represents a key lesson learned from this case study?

A. Outsourcing risk management functions without proper oversight can improve compliance efficiency.
B. Training and staffing are secondary to the technology used in monitoring transactions.
C. Heavy regulatory fines are typically the result of years of accumulated failings and delay in acting on regulatory findings.
D. Regulatory fines are typically unpredictable and not linked to accumulated failings in AML programs.

Answer: C

Step by step

1. Deficiencies at USAA FSB persisted from January 2016 to April 2021 (inadequate controls, reporting, staffing, training, third-party risk management).
2. The case lesson: ignore regulatory findings at one's own peril; heavy fines do not come out of the blue but follow years of postponed improvements.
3. Only C matches; A, B and D contradict the facts.

Why C is correct: The fine reflected a long period of unremedied weaknesses, so timely remediation of regulatory findings is the key lesson.

Why the others are wrong

- **A:** Unsupervised outsourcing worsened USAA's problems (contractors were untrained).
- **B:** Staffing and training were cited deficiencies; the new technology itself created an unmanageable alert backlog.
- **D:** Opposite of the case: the fine was linked to accumulated failings.

Q274

BT answers P2.T7.24.2.3

REBUILT — Option B (prohibit outsourcing) was arguably defensible and OCR error in fine amount was corrected; B replaced with a clearly appropriate remedy.

In response to the USAA FSB case, where the bank was fined $140 million by FinCEN and the OCC due to deficiencies in its BSA/AML compliance program, the board of Nicklewood Trustees is considering enhancing its control environment. Which of the following is NOT an appropriate solution to address the problems identified in the USAA case?

A. Conduct SOC (Service Organization Control) audits of third-party providers on an annual basis.
B. Provide AML training to any third-party contractors and verify their qualifications before they review alerts.
C. Increase the headcount of the compliance department from 2 staff to a team of 5.
D. Procure a transaction monitoring system that is highly sensitive, accepting a backlog of unreviewed alerts.

Answer: D

Step by step

1. USAA's problems: understaffed compliance, untrained third-party contractors, weak third-party risk management, and a system producing ~90,000 unreviewed alerts.
2. A addresses third-party oversight, B addresses contractor training, C addresses understaffing.
3. D reproduces the key failure (unmanageable alert backlog), so it is the inappropriate solution.

Why D is correct: A system that generates a backlog of unreviewed alerts replicates the USAA failure rather than fixing it.

Why the others are wrong
- **A:** Appropriate: third-party risk management was a cited deficiency.
- **B:** Appropriate: USAA failed to train or qualify contractors.
- **C:** Appropriate: the compliance department was significantly understaffed.

Q275
GARP PE1 Q44 · Mock 3 Q36

A risk consultant is meeting with the senior management of a midsize bank to discuss ways in which the bank can improve its management of financial crime and fraud risk. The consultant uses the example of USAA Bank to explain how weaknesses in controls and processes can result in large losses as well as regulatory fines. Which of the following best explains the fine imposed by regulators on USAA Bank?

A. USAA Bank was fined after many retail bankers were found to have opened unwanted accounts for existing customers in order to meet performance incentives.

B. USAA Bank was fined for failing to correct deficiencies identified by regulators in its management and reporting of suspicious transactions after an extended period of time.

C. USAA Bank was fined after several subsidiaries were found to have laundered over USD 1 billion in funds belonging to large criminal organizations.

D. USAA Bank was fined after fraudulently marketing complex interest rate derivatives to large municipal clients by significantly understating their risk.

Answer: B

Step by step
1. The fine ($140 million, March 2022) followed 2016-2021 of an ineffective BSA/AML program.
2. The bank failed to adequately monitor and report suspicious activity and did not remediate regulator-identified deficiencies.
3. Only B matches; A, C, D describe other institutions' scandals.

Why B is correct: USAA was penalised for a prolonged failure to fix AML monitoring and reporting weaknesses, not for a specific crime.

Why the others are wrong
- **A:** Describes the Wells Fargo unauthorized-accounts scandal, not USAA.
- **C:** No laundering of USD 1 billion by subsidiaries was alleged; the fine was for control failures.
- **D:** Describes mis-selling of derivatives, unrelated to USAA.

Q276
Schweser Module Quiz 52.1 Q4

REBUILT — Options C and D were both defensible (unaddressed flagged deficiencies and persistent AML weakness); rebuilt so only the ineffective-program rationale is correct.

Which of the following best describes why USAA Federal Savings Bank was fined $140 million by regulators in 2022?

A. A backlog of unreviewed alerts led to a significant theft of funds in 2021.

B. Systematic money laundering by third parties caused a significant loss over a decade.

C. Retail staff opened accounts without customer consent to meet sales incentives.

D. The bank had an ineffective AML program and inadequate AML controls over several years, even though no specific theft or laundering loss was the basis of the fine.

Answer: D

Step by step
1. The penalty covered at least five years (2016-2021) of an AML program that failed BSA minimum requirements.
2. Regulators cited inadequate controls, monitoring and reporting, not a known theft or loss.
3. D is the only option stating this basis.

Why D is correct: Fines were imposed for lack of adequate AML controls and an ineffective program, not for known thefts or laundering cases.

Why the others are wrong
- **A:** The backlog was cited, but no theft of funds was the basis of the fine.
- **B:** No systematic third-party laundering loss was the basis; fines were for control failure.
- **C:** Describes Wells Fargo's account scandal.

Reading 53. Guidance on Managing Outsourcing Risk

"Guidance on Managing Outsourcing Risk," Board of Governors of the Federal Reserve System, December 2013.

LO	Learning objective	Qs
53.a	explain how risks can arise through outsourcing activities to third-party service providers and describe elements of an effective program to manage outsourcing risk.	9
53.b	explain how financial institutions should perform due diligence on third-party service providers.	5
53.c	describe topics and provisions that should be addressed in a contract with a third-party service provider.	6

LO 53.a

Explain how risks can arise through outsourcing activities to third-party service providers and describe elements of an effective program to manage outsourcing risk.

Q277 AP Mock 3 Q4

A bank based in Texas, USA, is considering outsourcing its human resource activities to an HR agency. Its risk management team is considering all potential risks that could arise from this arrangement, particularly compliance risks. Which of the following would qualify as a compliance risk?

A. Advertising jobs without regard to existing labor laws.
B. Negative public opinion because of poor performance of outsourced activities.
C. Acceptance of fake academic documents.
D. Delays in service delivery.

Answer: A

Step by step

1. Compliance risk = the service provider not operating in compliance with relevant laws and regulations.
2. Classify each option: A is a breach of labor law (compliance); B is negative public perception (reputational); C and D are process or control failures (operational).

Why A is correct: Compliance risk arises when the provider fails to follow applicable laws and regulations, and advertising jobs in disregard of labor law is exactly that.

Why the others are wrong

- **B:** This is reputational risk: substandard performance leading to negative public perception of the bank.
- **C:** Accepting fake documents is a failed internal control or human error, i.e. operational risk.
- **D:** Service delays are an execution failure, i.e. operational risk, not a legal or regulatory breach.

Q278 AnalystPrep Q.2318

Tummers Bank from New York, USA, is considering outsourcing some of its activities to a third party. Which of the following risks (among others) should the bank consider before making a final decision?

A. Credit, market, and operational risks
B. Reputational, operational, and compliance risks
C. Portfolio, counterparty, and market risks
D. Country, legal, and counterparty risks

Answer: B

Step by step

1. The outsourcing risks identified in the guidance are compliance, concentration, reputational, country, operational and legal risks.
2. Check each option: only B lists three risks that are all on this list.
3. A includes credit and market risk; C has none of the listed risks; D includes counterparty risk.

Why B is correct: Reputational, operational and compliance risks are all among the six outsourcing risks named in the guidance, while every other option contains at least one risk that is not.

Why the others are wrong

- **A:** Credit and market risk are not outsourcing risks in the guidance; only operational risk qualifies.
- **C:** Portfolio, counterparty and market risks are financial-risk categories, none of which is a listed outsourcing

risk.
- **D:** Country and legal risks are listed, but counterparty risk is not, so the option is incomplete.

Q279 AnalystPrep Q.2322

New Savings Bank from Texas, USA, recently outsourced its IT services to Novel IT Service company (NIS) from Los Angeles, USA. NIS has a rich history spanning several decades but has recently been the subject of public criticism for various legal violations, as well as poor service delivery, punctuated by costly delays. NIS has most likely exposed the bank to:

A. Outsourcing risk
B. Credit risk
C. Reputational risk
D. Concentration risk

Answer: C

Step by step

1. Public criticism and substandard service delivery by the provider create a negative public perception of the bank.
2. Reputational risk is defined as the provider executing tasks in a substandard manner, harming the institution's public image.
3. No lending is involved (not credit) and nothing suggests few or clustered providers (not concentration).

Why C is correct: Substandard service and public criticism of the provider damage how the public views the bank that chose it, which is reputational risk.

Why the others are wrong

- **A:** Outsourcing risk is the generic umbrella term, not one of the specific risks (reputational, compliance, etc.) the guidance names.
- **B:** Credit risk concerns borrower default; nothing here involves a loan or counterparty default.
- **D:** Concentration risk requires reliance on very few or geographically clustered providers; the stem gives no such facts.

Q280 AnalystPrep Q.2323

Which of the following statements is correct?

A. After the outsourcing of an activity, all responsibility with regard to outsourced activities is transferred to the third party.
B. After the outsourcing of an activity, the third party and senior management have partial responsibility.
C. After the outsourcing of an activity, senior management is still responsible for normal functioning of the bank.
D. It is not possible to outsource accounting services.

Answer: C

Step by step

1. The guidance states that using service providers does not relieve the board and senior management of responsibility for ensuring outsourced activities are safe, sound and compliant.
2. So A (full transfer) and B (shared/partial) are wrong; responsibility stays with the institution.
3. Accounting is explicitly listed among activities that institutions outsource, so D is wrong.

Why C is correct: Outsourcing transfers the task, not accountability; the board and senior management remain responsible for the bank's operations.

Why the others are wrong

- **A:** Responsibility is never fully transferred to the provider.
- **B:** Accountability is not split; the institution's board and senior management retain responsibility for outsourced activities.
- **D:** Accounting is a commonly outsourced activity named in the guidance.

Q281 AnalystPrep Q.2327

REBUILT — Original answer was 'All of the above', a weak format; rebuilt to test the crucial first step of the outsourcing risk management program.

Fort Worth Bank is considering outsourcing its retail loan collection process to ICAP, a service provider in Mexico. Which of the following should the bank's risk management team do FIRST?

A. Perform due diligence on ICAP's financial condition and internal controls.

B. Negotiate contract provisions on confidentiality, audit rights and termination.

C. Perform a risk assessment to decide whether the activity is best executed in-house or by a third party.

D. Establish ongoing monitoring of ICAP's performance and its business continuity plan.

Answer: C

Step by step

1. The program elements are: risk assessments, due diligence, contract provisions, incentive compensation review, oversight and monitoring, and business continuity plans.
2. The crucial first step is the risk assessment, which determines whether the activity should be done in-house or by a third party (and includes cost-benefit and risk analysis).
3. Due diligence, contracting and monitoring only make sense once outsourcing has been judged consistent with the bank's objectives.

Why C is correct: The risk assessment comes first because it decides whether outsourcing is appropriate at all; the other elements follow from that decision.

Why the others are wrong

- **A:** Due diligence is the second element, performed when selecting a provider after the outsourcing decision is made.
- **B:** Contract terms are negotiated only after a provider has been selected.
- **D:** Oversight and contingency planning apply once the relationship is in place.

Q282 BT answers P2.T7.811.2 · BT notes P2.T7.811.2

Cityace Bank is a financial institution that is outsourcing a vital customer-facing function to a third-party service provider. Cityace wants to follow the Board's Guidance on Managing Outsourcing Risk. If Cityace follows the guidance, each of the following is true EXCEPT which one?

A. Cityace should avoid outsourcing risk management activities, especially interest rate risk and model risk, because these are core competencies

B. If the service provider is foreign-based, Cityace should ensure the provider is in compliance with applicable U.S. laws, regulations, and regulatory guidance

C. Cityace should ensure an effective process is in place to review and approve any incentive compensation that may be embedded in service provider contracts

D. Cityace should consider especially the following risks in outsourcing: compliance risks, concentration risks, reputational risks, country risks, operational risks, and legal risks

Answer: A

Step by step

1. The guidance says institutions MAY outsource risk management activities such as aspects of interest rate risk and model risk management, provided evidence and validation are consistent with existing guidance.
2. So A is not true and is the EXCEPT answer.
3. B (foreign providers must comply with U.S. law), C (incentive compensation review) and D (the six outsourcing risks) all match the guidance.

Why A is correct: The guidance permits outsourcing of risk management activities, so a statement that they must be avoided contradicts it.

Why the others are wrong

- **B:** True: foreign-based providers must comply with applicable U.S. laws and guidance, so this is not the exception.
- **C:** True: an effective process to review and approve embedded incentive compensation is required.
- **D:** True: these are the six risks the guidance lists.

Q283 GARP PE1 Q49

REBUILT — Keyed answer (publicly available audit reports) was dubious and weaker than the incentive-compensation point the stem targets; replaced the correct option with an incentive-compensation review and removed the unrealistic wording.

The senior management team of a small regional bank has established a committee to review procedures related to entering into significant contracts with third-party vendors. The committee is reviewing one proposed relationship with a vendor who would have significant responsibility for marketing the bank's financial products to potential customers. In establishing policies to reduce the operational risk associated with this vendor contract, which of the following recommendations would be most appropriate?

A. The bank should ensure an effective process exists to review and approve any incentive compensation embedded in the vendor contract, considering whether it could encourage imprudent selling.
B. The bank should ensure that the vendor's sales representatives are compensated mainly with commissions from the sale of the bank's products.
C. The bank should prevent the third-party vendor from having access to any of its critical processes.
D. The bank should be responsible for developing the vendor's contingency planning process to mitigate risk exposure to the vendor.

Answer: A

Step by step

1. A vendor selling products on the bank's behalf represents the bank, so incentive structures matter.
2. The guidance requires an effective process to review and approve incentive compensation in service provider contracts, because poorly structured incentives (for example commission-heavy pay steering customers to high-margin, unsuitable products) cause reputational damage, litigation and other risks.
3. Check B, C, D against the guidance: B creates the very risk described; C is not required; D misassigns responsibility.

Why A is correct: Reviewing and approving embedded incentive compensation directly addresses the mis-selling and reputational risk of an outsourced sales function.

Why the others are wrong

- **B:** Commission-heavy pay is the example of an inappropriate incentive that encourages imprudent selling.
- **C:** A blanket ban on access to critical processes is not required; risk is managed through due diligence, contract terms and monitoring.
- **D:** The provider is responsible for its own contingency plans; the bank reviews and tests them and develops its own contingency plans for critical services.

Q284 Schweser Module Quiz 53.1 Q1

Bank, Inc., (Bank) operates in the United States and has a service contract in place with Service Co. (Service), which operates in France. Service manages a significant amount of confidential customer data for Bank, and recently a computer glitch at Service resulted in the accidental public disclosure of confidential customer data. As a result of the data breach, which of the following risks is Bank least likely to face?

A. Compliance risk.
B. Country risk.
C. Legal risk.
D. Operational risk.

Answer: B

Step by step

1. Country risk arises from the economic, social and political conditions in the provider's country; the breach was caused by a computer glitch, not by conditions in France.
2. Compliance risk (privacy rules breached), legal risk (customer lawsuits) and operational risk (failed systems) all arise from the breach.
3. So country risk is the least likely.

Why B is correct: A systems failure causing a data leak triggers compliance, legal and operational risk, but is unrelated to France's political or economic conditions.

Why the others are wrong

- **A:** Likely: disclosing confidential customer data may breach privacy laws and regulations.
- **C:** Likely: affected customers may sue the bank, creating legal expense.
- **D:** Likely: the glitch is a failed system or process at the provider, i.e. operational risk.

Q285 Schweser Module Quiz 53.1 Q2

REBUILT — Source key A was weak (an effective program has six elements, not just business continuity) and the correct statement was keyed as wrong; rewrote options so exactly one statement matches the guidance.

Which of the following statements regarding a financial institution's risk management program for outsourcing to third-party service providers is most accurate?

A. The program should focus primarily on business continuity and contingency plans, because service provider failure is the dominant risk.

B. The program should apply the same level of oversight and controls to every outsourced activity, regardless of its materiality.

C. The depth and complexity of the program can be relatively low when there are few outsourced activities and the service providers are established and reliable.

D. Risk assessments should be updated as a result of updated risk mitigation techniques on a sufficiently regular basis.

Answer: C

Key correction: Source key said A; correct is C because the guidance lists six program elements (business continuity is only one) and explicitly scales program depth to the number and reliability of outsourced activities.

Step by step

1. The guidance says an effective program contains six elements: (1) risk assessments, (2) due diligence in selecting providers, (3) contract provisions, (4) incentive compensation review, (5) oversight and monitoring, (6) business continuity and contingency plans.
2. Oversight and controls must be adequate for activities with a material impact on finances and operations, with added emphasis on sensitive customer information and new products and services.
3. Depth and complexity scale with exposure: relatively low if few outsourced activities with established, reliable providers; relatively high if many providers.
4. Risk mitigation techniques are updated as a result of updated risk assessments, not the reverse.
5. Only C matches the guidance.

Why C is correct: The program is risk-based and proportional: depth and complexity are lower when outsourced activities are few and providers are established and reliable, and higher when many providers are involved.

Why the others are wrong

- **A:** Treats one of six elements as the focus; risk assessment is the crucial first step and the program must cover all six elements.
- **B:** Ignores proportionality; oversight is concentrated on activities with material impact, not applied uniformly to all.
- **D:** Reverses the causality; mitigation techniques are updated as a result of updated risk assessments.

LO 53.b

Explain how financial institutions should perform due diligence on third-party service providers.

Q286 AnalystPrep Q.2325

REBUILT — Option C had the typo 'access' and a weak justification; corrected to 'assess ... viability' and reworded option D as a plausible wrong reason.

An American bank is considering outsourcing its IT operations to an Indian IT provider. Is it appropriate for the bank's risk management team to analyze the provider's financial condition before making a final decision?

A. No, because the bank should only be concerned with country risk.

B. No, because it is illegal to outsource IT services in the first place.

C. Yes, in order to assess the financial stability and viability of the service provider.

D. Yes, because it is a requirement under Basel II regulation.

Answer: C

Step by step

1. Due diligence on a prospective service provider covers three areas: business background, reputation and strategy; financial performance and condition; operations and internal controls.
2. Reviewing financial condition (statements, capital strength, liquidity, sustainability, commitment of resources, insurance) is therefore a standard part of due diligence.

3. The purpose is to judge whether the provider can keep delivering the service for the duration of the contract, so the review is appropriate regardless of the provider's country.

Why C is correct: Financial performance and condition is one of the three core due diligence areas, and it reveals whether the provider is viable enough to deliver the service over the contract term.

Why the others are wrong

- **A:** Country risk is only one of several outsourcing risks; due diligence is not limited to it and the financial review is still required.
- **B:** Outsourcing IT services is legal; the guidance asks for risk management, not prohibition.
- **D:** Right action, wrong reason: the review stems from the outsourcing risk guidance on due diligence, not a Basel II mandate.

Q287

AnalystPrep Q.2329

A bank in Palermo, Italy is considering outsourcing its IT services and is preparing a review of the financial condition of IAM Systems, a local service provider. Which of the following would NOT form part of the financial review process?

A. The adequacy of the service provider's insurance coverage.
B. The adequacy of the service provider's review of the financial condition of any subcontractors.
C. The potential financial impact of the bank's business on the provider.
D. The potential impact of the provider's past clients on the bank's financial condition.

Answer: D

Step by step

1. The financial review includes recent financial statements, the provider's sustainability, the impact of the institution's relationship on the provider's finances, commitment of resources, insurance adequacy and review of subcontractors' finances.
2. A, B and C appear in that list.
3. D reverses the direction: the review looks at the provider's financial condition, not at the effect of the provider's past clients on the bank's finances. D is the exception.

Why D is correct: The financial review examines the provider's condition and the effect of the bank's business on the provider, not the effect of the provider's past clients on the bank.

Why the others are wrong

- **A:** Insurance adequacy is an explicit element of the financial review.
- **B:** The provider's review of subcontractors' financial condition is an explicit element.
- **C:** The impact of the bank's relationship on the provider's financial condition is an explicit element.

Q288

AnalystPrep Q.2330

Stroud Bank in Gloucester, UK is executing a major merger and will outsource activities related to relocating facilities. The bank is reviewing the operations and internal controls of several service providers. Which of the following should NOT be included in the review?

A. Training, including compliance training for staff.
B. Compliance with environmental sustainability standards of the business.
C. Business resumption and contingency planning.
D. Service support and delivery.

Answer: B

Step by step

1. The operations and internal controls review covers items such as internal controls, facilities management, training (including compliance training), security of systems, privacy protection, record retention, business resumption and contingency planning, systems development, service support and delivery, employee background checks, and adherence to laws and regulations.
2. A, C and D are on that list.
3. Environmental sustainability compliance is not listed, so B is the exception.

Why B is correct: The guidance lists specific operational and internal control topics; environmental sustainability is not one of them.

Why the others are wrong

- **A:** Compliance training for staff is explicitly listed.

- **C:** Business resumption and contingency planning is explicitly listed.
- **D:** Service support and delivery is explicitly listed.

Q289 BT Mock Q53 · BT answers P2.T7.811.1 · BT notes P2.T7.811.1

According to the Federal Reserve's Guidance on Managing Outsourcing Risk, the overall due diligence process for a third-party service provider should at a minimum include a review of each of the following EXCEPT which?

A. Ensuring the service provider has an appropriate background check program for its employees.
B. A financial review of the service provider's most recent annual report and financial statements.
C. An evaluation of the service provider's performance on environmental, social and governance (ESG) factors.
D. An evaluation of the adequacy of standards, policies and procedures, including adherence to applicable laws, regulations and supervisory guidance.

Answer: C

Step by step

1. Due diligence covers (1) business background, reputation and strategy, (2) financial performance and condition, (3) operations and internal controls.
2. A is in area 1 (employee background check program); B is in area 2 (financial statements, annual report); D is in area 3 (adherence to laws, regulations and supervisory guidance).
3. ESG performance is not among the review elements, so C is the exception.

Why C is correct: The three due diligence areas do not include an ESG evaluation of the provider.

Why the others are wrong

- **A:** Part of business background, reputation and strategy.
- **B:** Part of financial performance and condition.
- **D:** Part of operations and internal controls.

Q290 Schweser Module Quiz 53.1 Q3

When performing due diligence on a service provider, ascertaining the sufficiency of its insurance coverage would most appropriately be covered under which of the following categories?

A. Business background, reputation and strategy.
B. Financial performance and condition.
C. Operations and internal controls.
D. Oversight and monitoring.

Answer: B

Step by step

1. Due diligence has three categories: business background, financial performance and condition, operations and internal controls.
2. The financial review includes the adequacy of the provider's insurance coverage.
3. So insurance sufficiency falls under financial performance and condition.

Why B is correct: Insurance adequacy is listed explicitly as part of the financial review of the provider.

Why the others are wrong

- **A:** This area is a broad look at industry status, references, principals and strategy, not insurance.
- **C:** This area covers controls, training, security and compliance, not insurance coverage.
- **D:** Oversight and monitoring is a separate element of the overall risk management program, not a due diligence category.

LO 53.c

Describe topics and provisions that should be addressed in a contract with a third-party service provider.

Q291 AnalystPrep Q.2326

Sandero bank from Carrington, North Dakota, is considering outsourcing part of its IT services to a third party. Such a move will most likely involve sharing of some nonpublic personal information about the bank's customers with the third party. Should the bank go ahead with its plan?

A. No, all outsourcing activities that can reasonably be expected to expose nonpublic personal information are forbidden.

B. Yes, but the bank should refrain from sharing all nonpublic personal information.

C. Yes, the bank could outsource an activity which requires usage of nonpublic personal information, but the service provider must comply with applicable privacy laws and regulation.

D. No, because such a move would open doors to possible lawsuits by aggrieved customers.

Answer: C

Step by step

1. The guidance does not prohibit outsourcing that involves customer NPPI.
2. Contracts must require the service provider to comply with applicable privacy laws and regulations and to protect the confidentiality of customer information.
3. So the bank may proceed, subject to those contractual privacy obligations.

Why C is correct: Outsourcing involving NPPI is permitted provided the contract obliges the provider to comply with privacy laws and safeguard the data.

Why the others are wrong

- **A:** There is no blanket prohibition on outsourcing that involves NPPI.
- **B:** Refusing to share any NPPI is not required and would often make the service impractical; the safeguard is contractual compliance.
- **D:** Litigation risk is managed through contract terms and oversight, it does not make outsourcing impermissible.

Q292 AnalystPrep Q.5257

REBUILT — Original wrong options used 'only' wording, making them giveaways; reworded so each distractor is a realistic partial or mistaken view of the cost and compensation provision.

Which of the following best describes what a contract with a third-party service provider should address under its cost and compensation provision?

A. A single fixed fee for the service, with all other expenses left to be settled informally as they arise.

B. The compensation, variable charges, and any fees for non-recurring items and special requests, together with which party pays legal, audit and examination fees and equipment-related expenses.

C. The compensation to be paid, with incentive terms left to the provider because the institution should not review how the provider is paid.

D. The compensation and variable charges, with legal and audit fees assumed to be paid by the financial institution without being stated.

Answer: B

Step by step

1. Recall the cost and compensation provision: contracts describe compensation, variable charges and fees for non-recurring items and special requests.
2. The contract should also state which party pays legal, audit and examination fees tied to the activity, and who bears equipment, hardware and software expenses where applicable.
3. Compare options: only B is comprehensive and consistent with this.

Why B is correct: Cost and compensation terms should be comprehensive, covering all forms of compensation and assigning responsibility for associated fees and expenses explicitly.

Why the others are wrong

- **A:** Too narrow: ignores variable charges, non-recurring and special-request fees and the allocation of fees; leaving expenses informal creates disputes.
- **C:** Wrong: the institution should ensure incentives (variable charges, commissions) do not encourage imprudent risk-taking, so it must be able to review them.
- **D:** Incomplete: the contract should explicitly assign responsibility for legal, audit and examination fees and not leave it implied; also omits non-recurring/special fees.

Q293 BT answers P2.T7.811.3 · BT notes P2.T7.811.3

Planetholding International Bank is entering a contract with its third-party service provider, Tristechnology Inc, to outsource the management of its website. Each of the following is a likely contract provision EXCEPT which one?

A. Confidentiality and security of information
B. Business resumption and contingency plan
C. Scope of service (including reference to service level agreement and ability to subcontract)
D. Loss waterfall allocation mechanism (including sufficient credit value adjustment for website downtime)

Answer: D

Step by step

1. Recall the standard provisions: scope, cost and compensation, incentive compensation, right to audit, performance standards, oversight and monitoring, confidentiality and security, ownership and license, indemnification, default and termination, dispute resolution, limits on liability, insurance, customer complaints, business resumption and contingency plan, foreign-based providers, subcontracting.
2. A, B and C appear on this list.
3. A loss waterfall and credit value adjustment belong to CCP/securitization/derivatives credit-risk contexts, not outsourcing contracts.

Why D is correct: Loss waterfalls and CVA are credit and counterparty-risk constructs and are not outsourcing contract provisions; the others are standard.

Why the others are wrong

- **A:** A standard provision: the contract must address confidentiality and security of institution and customer information.
- **B:** A standard provision: the provider's business resumption and contingency plan should be addressed.
- **C:** A standard provision: scope covers rights and responsibilities, service levels and subcontracting.

Q294 GARP PE2 Q41 · Mock 4 Q42

A bank wants to reduce its operating expenses and considers hiring a third-party service provider to offer additional loan origination and credit services to some of the bank's customers. The bank's legal department has begun negotiating the terms of a contract with the provider. Which of the following describes the most appropriate action for the bank to take in the contract?

A. The bank should audit the service provider's operational processes and also give the service provider the similar right to audit the bank's processes.
B. The bank should purchase insurance to cover potential losses resulting from the provider's services and should require that the provider deposit collateral with the bank to mitigate performance risk.
C. The bank should determine the compensation structure for the provider's sales representatives and ensure that it incentivizes their productivity through a high proportion of variable compensation.
D. The bank should define specific events that are considered a default under the contract and give the provider an opportunity to resolve a default before terminating the contract.

Answer: D

Step by step

1. Default and termination provision: the contract should define events of default, list acceptable remedies, and provide opportunities to cure.
2. Test each option against the guidance: audit rights run to the institution; insurance is a provider-side provision; incentive pay must not encourage imprudent risk.
3. Only D matches.

Why D is correct: Guidance says agreements should define events of contractual default, acceptable remedies and opportunities for curing default before termination.

Why the others are wrong

- **A:** The right to audit protects the financial institution; no reciprocal right for the provider is called for.
- **B:** Insurance coverage is a contract topic addressed through the provider's coverage; guidance does not call for the bank to buy its own insurance or require collateral.
- **C:** Incentives must be reviewed so they do not encourage imprudent risk; a high proportion of variable pay on loan origination risks mis-selling, and the provider sets its own staff pay.

Q295 Schweser Module Quiz 53.1 Q4

The use of performance metrics to assist in determining an acceptable level of performance by a service provider would most appropriately be included in which of the following provisions of a contract with a financial institution?

A. Customer complaints.
B. Default and termination.
C. Indemnification.
D. Right to audit.

Answer: B

Step by step

1. Termination rights include failure to meet performance standards.
2. Such failure is detected by measurable performance metrics, so metrics support the default and termination provision.
3. Other options address other matters.

Why B is correct: Default and termination triggers include failure to meet performance standards, which are judged using the measurable metrics.

Why the others are wrong

- **A:** Customer complaints provision assigns which party handles complaints and reporting of them.
- **C:** Indemnification covers the provider holding the institution harmless against claims from the provider's negligence.
- **D:** Right to audit lets the institution audit the provider or access audit reports; it is not about performance thresholds.

Q296 Schweser Module Quiz 53.1 Q5

REBUILT — Original 'least likely to be included' was misleading because a right to audit is commonly included; reworded to ask which provision the guidance describes as optional ('may').

According to the guidance, which of the following contract provisions with a third-party service provider is described as one the agreement 'may' provide, rather than one the agreement 'should' provide?

A. Establishment and monitoring of performance standards.
B. Indemnification.
C. Ownership and license.
D. Right to audit.

Answer: D

Key correction: Source key (Schweser) said D for 'least likely to be included'; D is retained, but the stem is reworded because a right to audit is often included in practice and the guidance treats it as optional.

Step by step

1. The guidance says agreements 'may' provide for the right to audit and/or access to audit reports (optional).
2. Performance standards must be measurable and defined; indemnification for provider negligence should be provided; ownership and license of data and property should be clarified.
3. So D is the optional one.

Why D is correct: The right to audit is framed as optional, whereas the other provisions are expected contract terms.

Why the others are wrong

- **A:** Agreements should define measurable performance standards; they are essential for monitoring.
- **B:** Agreements should provide for indemnification for claims resulting from the provider's negligence.
- **C:** The contract should state when the provider may use the institution's property and who owns data produced.

Reading 54. Case Study: Third-Party Risk Management

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 13.

LO	Learning objective	Qs
54.a	explain how risks related to the use of third parties can arise and describe characteristics of an effective third-party risk management framework.	14
54.b	describe the lessons learned from the presented case studies.	3

LO 54.a

Explain how risks related to the use of third parties can arise and describe characteristics of an effective third-party risk management framework.

Q297

AP Mock 3 Q34

REBUILT — Four labels all arguably applied; pinned the stem to the root-cause source of the loss so only third-party risk fits.

A bank outsourced its customer data analysis to a specialised analytics company. Because of inadequate security protocols at that company, there was unauthorised access to the bank's customer data, exposing sensitive personal and financial information. Classifying the incident by the SOURCE (root cause) of the risk, which risk category does it most closely represent?

A. Operational risk.
B. Third-party risk.
C. Compliance risk.
D. Cyber risk.

Answer: B

Step by step

1. The loss arose because a control failed at an outsourced provider, not inside the bank.
2. Risks arising from vendors, contractors and service providers to which activities are outsourced are third-party (outsourcing) risk.
3. Cyber is the mechanism of the breach and compliance/legal exposure is a consequence; neither identifies where the risk originated.
4. Operational risk is the broad parent category, less specific than third-party risk.

Why B is correct: The root cause was a control failure at an outsourced service provider, which is exactly what third-party risk captures; accountability stays with the bank.

Why the others are wrong

- **A:** Too broad: operational risk is the parent category that contains third-party risk, so it is not the most specific classification.
- **C:** Compliance (data-privacy) consequences follow from the breach but are an outcome, not the origin.
- **D:** Cyber risk describes the attack mechanism, not the source (the vendor's inadequate protocols under an outsourcing arrangement).

Q298

AnalystPrep Q.5137

Which step of the Third-Party Risk Management life cycle involves keeping track of service provision, quality SLAs, and compliance with regulation?

A. Remediation or termination
B. Business model decision
C. Evaluation and risk rating
D. Continuous monitoring

Answer: D

Step by step

1. Monitoring of service provision, quality SLAs and regulatory/contract compliance is the continuous monitoring step.
2. Other steps decide, vet, contract or exit.

Why D is correct: Tracking ongoing service delivery, SLAs and compliance is the definition of continuous monitoring.

Why the others are wrong

- **A:** Remediation/termination handles corrective action and exit.
- **B:** Business model decision is the initial outsource choice.
- **C:** Evaluation and risk rating is the pre-contract assessment.

Q299

AnalystPrep Q.5129

Which of the following is not a step in the Third-Party Risk Management (TPRM) life cycle?

A. Remediation
B. Risk rating & evaluation
C. Shared assessments
D. Continuous monitoring

Answer: C

Step by step

1. The five TPRM stages are: business model decision; evaluation, risk rating and due diligence; contracts, SLAs and contract management; continuous monitoring; remediation or termination.
2. Remediation, risk rating/evaluation and continuous monitoring each match a stage.
3. Shared Assessments is the US certifying organisation (CTPRP) whose life-cycle model is used; it is not itself a step.

Why C is correct: Shared Assessments is the body that proposed the framework, not a stage in it.

Why the others are wrong

- **A:** Remediation (or termination) is the fifth stage.
- **B:** Evaluation, risk rating and due diligence is the second stage.
- **D:** Continuous monitoring is the fourth stage.

Q300

AnalystPrep Q.5130

Which of the five steps in the Third-Party Risk Management cycle involves choosing a third-party service provider after evaluating the risk appetite of the firm?

A. Evaluation, risk rating, due diligence
B. Business model decision
C. Contracts and contract management
D. Continuous monitoring

Answer: B

Step by step

1. The first step decides whether to outsource or keep activities in-house and the provider's quality and price.
2. These strategic choices relate to the firm's risk appetite.
3. Later steps assess, formalise and monitor the chosen provider.

Why B is correct: The business model decision is the strategic step tied to risk appetite, taken before any provider is assessed.

Why the others are wrong

- **A:** Due diligence assesses a provider already under consideration, rather than deciding whether/what to outsource.
- **C:** Contracts formalise responsibilities and SLAs after selection.
- **D:** Monitoring happens once the relationship is running.

Q301

AnalystPrep Q.5131

Which of the following is a good risk management practice of the remediation or termination step of the Third-Party Risk Management life cycle?

A. Defining trigger events for reassessment
B. Establishing limits on the outsourcing by third parties
C. Having an exit strategy

D. Use of standard assessment questionnaires

Answer: C

Step by step

1. Step 5 practices: a grievance procedure, an exit strategy or termination clause, and a planned wind-down.
2. Reassessment triggers are defined in the contract/monitoring stages, outsourcing limits in contracts, and standard questionnaires (e.g. SIG) in due diligence.
3. Only the exit strategy belongs to step 5.

Why C is correct: An exit strategy/termination clause is the core good practice for the remediation or termination stage.

Why the others are wrong

- **A:** Reassessment triggers (breaches, M&A, regulatory change) are defined at the contract/monitoring stage, not at the termination stage.
- **B:** Limits on third-party outsourcing are set in contract terms (stage 3).
- **D:** Standard questionnaires simplify due diligence (stage 2).

Q302 AnalystPrep Q.5132

Which of the following is not an example of an event that can trigger the reassessment of contracts with third parties?

A. Data breaches
B. A merger or acquisition
C. Regulatory change
D. Risk appetite

Answer: D

Step by step

1. Triggers named in the framework: data breaches or incidents, business changes such as mergers/acquisitions, legal or regulatory changes, service changes, acts of God, breach of contract or performance failure.
2. Data breach, merger/acquisition and regulatory change are all listed events.
3. Risk appetite is a standing strategic parameter, not an event.

Why D is correct: A trigger is a specific event; risk appetite is a strategic level of risk tolerance, not an occurrence.

Why the others are wrong

- **A:** A data breach or incident is a listed trigger.
- **B:** A merger or acquisition changes business circumstances and is a listed trigger.
- **C:** Legal or regulatory change is a listed trigger.

Q303 AnalystPrep Q.5133

Which of the five steps in the Third-Party Risk Management cycle requires sound due diligence and verification of third-party service providers?

A. Remediation or termination
B. Continuous monitoring
C. Evaluation and risk rating
D. Business model decision

Answer: C

Step by step

1. Entering a new relationship requires sound due diligence on whom the firm does business with.
2. This is the second step, evaluation, risk rating and due diligence, applied proportionally (more for a cloud host of sensitive data than a one-day trainer).

Why C is correct: Due diligence and verification are the defining tasks of the evaluation and risk rating stage.

Why the others are wrong

- **A:** Remediation/termination deals with exit, wind-down and grievances.
- **B:** Monitoring tracks ongoing performance and compliance, not initial vetting.
- **D:** The business model decision is the strategic outsource/in-house choice before vetting.

Q304

AnalystPrep Q.5134

REBUILT — Original option 'Third parties' was a vague non-risk item; rebuilt with a distinct, clearly non-risk distractor.

Which of the following is not identified as a common risk arising from the use of third parties?

A. Service disruption
B. Espionage and intellectual property theft
C. Accidental data privacy breach or intentional information leak
D. A reduction in the firm's own cost of capital

Answer: D

Step by step

1. Common third-party risks: service disruption, failings in service quality, fraud, accidental privacy breach or intentional leak, compliance breaches, espionage and IP theft, reputational damage.
2. Options A, B and C appear in that list.
3. A lower cost of capital is not a third-party risk.

Why D is correct: A change in the firm's cost of capital is not among the operational risks arising from vendors and outsourced providers.

Why the others are wrong

- **A:** Service disruption is a listed common third-party risk.
- **B:** Espionage and IP theft are listed risks.
- **C:** Privacy breaches and information leaks are listed risks.

Q305

AnalystPrep Q.5135

Which of the following is a good practice when addressing fourth-party risk?

A. Establish standards on outsourcing
B. Define trigger events for reassessment
C. Have an exit strategy
D. Have a termination clause

Answer: A

Step by step

1. Fourth parties are third parties of third parties.
2. Good practice is for the contract to set standards or limits on the outsourcing done by vendors, largely replicating the firm's own vendor rules.
3. Triggers, an exit strategy and a termination clause are general practices, not aimed at fourth parties.

Why A is correct: Contractual standards on vendor outsourcing let the firm's rules cascade down to its vendors' own vendors.

Why the others are wrong

- **B:** Reassessment triggers apply to the third party generally, not specifically to its own outsourcing.
- **C:** An exit strategy is a termination-stage practice.
- **D:** A termination clause covers ending the relationship, not controlling downstream outsourcing.

Q306

AnalystPrep Q.5136

Which of the following is not an action that should be undertaken during the wind-down process of contracts?

A. The transmission of intellectual property
B. Plan to transition to in-house services
C. Having audit rights on their vendors
D. Provide evidence of data transfer or destruction

Answer: C

Step by step

1. Wind-down includes: transmission of IP when necessary, a plan to transition to in-house or another provider, and evidence of data transfer or destruction.
2. Audit rights over vendors are set up in contracts and used in continuous monitoring, not a wind-down action.

Why C is correct: Audit rights verify vendor compliance during the relationship (contracts/monitoring), so they are not a wind-down task.

Why the others are wrong
- **A:** IP transfer is an explicit wind-down item.
- **B:** Transition planning is an explicit wind-down item.
- **D:** Evidence of data transfer/destruction is an explicit wind-down item.

Q307
Mock 2 Q71

The chief risk officer of an Australian bank is evaluating the bank's overall risk appetite and is looking to determine which of the bank's activities should be kept in-house or outsourced to third-party vendors. Within the life cycle stages of third-party risk management (TPRM), these determinations would most likely fall under:

A. business model decisions.
B. contract management.
C. due diligence.
D. evaluation and risk rating.

Answer: A

Step by step
1. The TPRM life cycle has five stages: business model decision; evaluation, risk rating and due diligence; contracts, SLAs and contract management; ongoing monitoring; remediation or termination.
2. Deciding whether and which activities to keep in-house or outsource, in light of the firm's risk appetite, is the first stage: the business model decision.
3. The remaining stages all presuppose that a specific third party has already been chosen.

Why A is correct: Whether to outsource at all is a strategic choice tied to risk appetite and is made before any vendor is assessed, so it is the business model decision stage.

Why the others are wrong
- **B:** Contract management comes after a vendor is selected and sets responsibilities, SLAs and outsourcing limits; it does not decide whether to outsource.
- **C:** Due diligence assesses a specific prospective vendor, in proportion to the relationship's risk; it presupposes the outsourcing decision.
- **D:** Evaluation and risk rating grade a particular candidate vendor; confusing it with the strategic decision ignores that no vendor exists yet.

Q308
Mock 3 Q31

A bank has engaged a third-party vendor whose sales representatives market the bank's investment products directly to retail customers under the bank's brand. The vendor was subject to full due diligence at onboarding two years ago. An operational risk manager is designing the ongoing oversight arrangements for the relationship. Which of the following would be most effective in reducing the bank's operational risk?

A. Relying on the vendor's annual attestation that it remains compliant with the terms of the contract.
B. Monitoring customer complaint and product suitability data arising from the vendor's sales and feeding it into the bank's own conduct risk reporting.
C. Requiring the vendor to indemnify the bank against any regulatory penalties arising from its sales activities.
D. Reviewing the vendor's publicly available audited financial statements each year to confirm its financial soundness.

Answer: B

Step by step
1. The vendor's sales activity under the bank's brand creates conduct (mis-selling) risk that falls on the bank, since accountability for outsourced activities is not transferable.
2. Ongoing monitoring (stage 4 of TPRM) should track service quality and compliance with regulation and contract terms, not only the one-off onboarding due diligence.
3. Complaint and suitability data from the vendor's sales are independent, timely indicators of conduct problems and can be fed into the bank's own risk reporting.
4. The other options rely on self-reporting, loss financing, or solvency, none of which detects or prevents mis-selling.

Why B is correct: Independent monitoring of complaints and suitability outcomes directly detects and reduces the conduct risk, whereas the other options do not address how the vendor actually behaves.

Why the others are wrong

- **A:** A self-attestation is weak assurance from the party being monitored; it does not independently verify behaviour.
- **C:** An indemnity transfers some financial loss but not responsibility; regulatory and reputational damage remain with the bank and the mis-selling is not prevented.
- **D:** Financial statements address solvency, not sales conduct; this is a financial-soundness check that misses the actual risk.

Q309

Schweser Module Quiz 54.1 Q1

Establishing limits on third-party and fourth-party vendor outsourcing would be considered under which of the following third-party risk management (TPRM) steps?

A. Business model decision.
B. Remediation or termination.
C. Evaluation, risk rating, and due diligence.
D. Contracts, service level agreements, and contract management.

Answer: D

Step by step

1. Contract terms are the opportunity to address fourth-party risk: good practice is for the contract to set standards or limits on the outsourcing done by the third-party vendor.
2. Contracts, SLAs and contract management is therefore the stage where limits on third- and fourth-party outsourcing are established (together with audit rights).

Why D is correct: Limits on a vendor's own outsourcing are written into the contract, which is the contracts, SLAs and contract management stage.

Why the others are wrong

- **A:** The business model decision concerns whether to outsource at all and the fit with risk appetite, not contractual limits on vendors' subcontracting.
- **B:** Remediation or termination deals with exit, grievance procedures and wind-down, not setting up outsourcing limits.
- **C:** Evaluation, risk rating and due diligence assess a prospective vendor; limits are a contractual term set afterwards.

Q310

Schweser Module Quiz 54.1 Q2

REBUILT — Original wrong option 'Management of third- and fourth-party vendor relationships' was vague and arguably describes the whole process; replaced with a clearly non-stage item.

Which of the following is not one of the life cycle stages of third-party risk management (TPRM)?

A. Business model decision.
B. Ongoing monitoring.
C. Insurance procurement and claims settlement.
D. Remediation or termination.

Answer: C

Step by step

1. The five TPRM stages are: business model decision; evaluation, risk rating and due diligence; contracts, SLAs and contract management; ongoing monitoring; remediation or termination.
2. Options A, B and D are stages 1, 4 and 5.
3. Insurance procurement and claims settlement is not a stage of the life cycle.

Why C is correct: Insurance procurement and claims settlement is not one of the five stages of the TPRM life cycle.

Why the others are wrong

- **A:** This is stage 1, so it is a stage and not the answer.
- **B:** This is stage 4 (continuous monitoring of service quality and compliance), so it is a stage.
- **D:** This is stage 5 (exit strategy, grievance procedure, wind-down), so it is a stage.

LO 54.b

Describe the lessons learned from the presented case studies.

Q311 GARP PE2 Q30 · Mock 4 Q31

An enterprise risk consultant is presenting about the management of risk associated with third-party vendor relationships at a financial conference. To emphasize the importance of understanding this risk and to illustrate lessons learned, the consultant describes several past examples of large losses and data breaches incurred by different financial institutions due to deficient or fraudulent third-party vendor practices. One example provided is the large loss incurred by Capital One, a US-based bank holding company, that resulted from its relationship with a third-party vendor. Which of the following best describes the circumstances that led to the loss in this case?

A. The vendor provided an inaccurate loan pricing model to Capital One, which incurred far greater default losses than expected.

B. A bill payment system provided by the vendor failed for an extended period of time, resulting in many Capital One customers canceling their accounts and causing severe reputational damage.

C. The vendor's sales manager established extremely high incentives for its representatives to sell Capital One products, resulting in regulatory fines for selling inappropriate products to consumers.

D. A former staff member of the vendor hacked into a database of Capital One's personal customer information that was stored on the vendor's cloud services platform and stole much of this information.

Answer: D

Step by step

1. Recall the Capital One case: in July 2019 the bank announced that data on about 100 million people had been illegally accessed.
2. The attacker was a former employee of AWS, Capital One's cloud services provider, who exploited a misconfigured web application firewall to reach files in an AWS database.
3. The stolen data was unencrypted, so it was immediately usable; the OCC later fined Capital One $80 million for failing to establish effective risk assessment before migrating significant IT operations to the cloud.
4. The circumstances therefore match option D (a former vendor staff member stealing customer data hosted on the vendor's cloud platform).

Why D is correct: The loss arose from a data breach by a former employee of the cloud vendor (AWS), exploiting a misconfigured firewall; this is a third-party data security failure for which Capital One remained accountable.

Why the others are wrong

- **A:** Describes a model risk event (flawed vendor pricing model); nothing like this occurred at Capital One.
- **B:** Describes a service outage with reputational damage from a failed bill payment system; the Capital One case was a data theft, not a prolonged system failure.
- **C:** Describes a conduct/mis-selling event driven by incentives; the OCC fine was for inadequate cloud risk assessment, not product mis-selling.

Q312 Schweser Module Quiz 54.1 Q3

A key conclusion from the Capital One and Morgan Stanley case studies is that accountability for operational risk:

A. is not transferable.

B. ultimately rests with the third-party vendor.

C. ultimately rests with either the third-party vendor or subcontractor, whichever experienced the risk control problem.

D. is a fully shared responsibility between the company using third-party vendors and the third-party vendors themselves.

Answer: A

Step by step

1. Capital One (2019 AWS breach, $80 million OCC fine) and Morgan Stanley (2020 $60 million OCC fine over decommissioning and vendor oversight) both involved third-party failures.
2. In each case the regulator penalised the institution that outsourced, citing its failure to assess and monitor vendor risks.
3. Lesson: functions can be outsourced, but ultimate responsibility and accountability cannot be transferred; the answer is A.

Why A is correct: The regulator held the outsourcing firms themselves responsible, showing that accountability for risk stays with the firm using the vendor and cannot be transferred.

Why the others are wrong

- **B:** Reverses the lesson; the fines fell on the banks, not the vendors, so accountability does not rest ultimately with the vendor.
- **C:** Implies accountability moves to whichever party caused the control failure; accountability stays with the outsourcing firm regardless of where the failure occurred.
- **D:** Vendors may share some blame, but accountability is not divided or shared away; the firm retains ultimate responsibility.

Q313 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank's wealth management division plans to decommission several data servers that hold client records. The bank will hire an IT asset disposal vendor, which in turn will use a subcontractor to wipe and dispose of the hardware. Before approving the arrangement, the CRO asks an operational risk manager to apply the lessons of the OCC's 2020 enforcement action against Morgan Stanley over its weak third-party controls in the decommissioning of data servers. Which of the following actions would best address the deficiencies the OCC identified in that case?

A. Perform due diligence on both the vendor and its subcontractor, monitor their performance during the work, and keep an inventory of the client data stored on each device being decommissioned.

B. Include a contract clause that transfers full liability for any exposure of client data to the vendor, so that the bank is no longer accountable for the risk.

C. Accept the vendor's written certification that all devices have been wiped, given its specialist expertise, and limit oversight of the vendor to the scheduled end-of-contract review.

D. Focus the review on the configuration of the web application firewall protecting the servers, since a misconfigured firewall was the main cause of the data exposure in that case.

Answer: A

Step by step

1. Recall the OCC findings in the Morgan Stanley case: the bank failed to effectively assess or address the risks of decommissioning its hardware, failed to adequately assess the risk of using third-party vendors including subcontractors, and failed to maintain an appropriate inventory of customer data stored on its devices.
2. The OCC also found that the bank did not exercise adequate due diligence in selecting the vendors and did not adequately monitor their performance; disposed hardware was later found to still contain customer data.
3. Choose the action that targets these specific failures: due diligence on the vendor and subcontractor, ongoing performance monitoring, and a data inventory for each device.

Why A is correct: Option A addresses each deficiency the OCC cited in the $60 million Morgan Stanley fine. It covers due diligence on the vendor and the subcontractor, monitoring of their performance, and an inventory of the customer data held on the devices being decommissioned. Without that inventory, the bank cannot confirm that disposed hardware no longer holds client information.

Why the others are wrong

- **B:** Incorrect. A key lesson of both case studies is that outsourcing can transfer some of the operational risk of an activity, but accountability for that risk cannot be transferred. A liability clause does not remove the bank's responsibility to regulators or customers.
- **C:** Incorrect. This repeats Morgan Stanley's failure to monitor vendor performance. Relying only on a vendor certification and an end-of-contract review is weak oversight. Good TPRM practice ties reassessment to trigger events, not only to scheduled reviews.
- **D:** Incorrect. This mixes up the two case studies. The misconfigured web application firewall was the weakness exploited in the Capital One breach by a former AWS employee. The Morgan Stanley deficiencies concerned the decommissioning of hardware, oversight of vendors and subcontractors, and the inventory of customer data.

Reading 55. Case Study: Investor Protection and Compliance Risks in Investment Activities

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 14.

LO	Learning objective	Qs
55.a	summarize important regulations designed to protect investors in financial instruments, including MiFID, MiFID II, and Dodd-Frank.	9
55.b	describe lessons learned from the case studies involving violations of investor protection or compliance regulations.	3

LO 55.a

Summarize important regulations designed to protect investors in financial instruments, including MiFID, MiFID II, and Dodd-Frank.

Q314 AP Mock 3 Q70

Which regulation resulted in the formation of the Consumer Financial Protection Bureau (CFPB) as an independent financial regulator overseeing consumer finance markets?

A. The Markets in Financial Instruments Directive (MiFID).
B. The Dodd-Frank Act.
C. The Financial Industry Regulatory Authority (FINRA).
D. The Volcker Rule.

Answer: B

Step by step

1. The CFPB oversees consumer finance markets such as mortgages, student loans and credit cards.
2. It was created by the Dodd-Frank Wall Street Reform and Consumer Protection Act, the U.S. response to the 2007-2009 crisis.
3. MiFID is EU legislation, FINRA is a self-regulatory body, and the Volcker Rule is only one provision within Dodd-Frank.

Why B is correct: The CFPB was established by the Dodd-Frank Act as an independent regulator of consumer finance markets.

Why the others are wrong

- **A:** MiFID is an EU investor protection directive and has no role in creating a U.S. agency.
- **C:** FINRA is a self-regulatory organization for broker-dealers, not an agency created by legislation as a consumer finance regulator.
- **D:** The Volcker Rule is a provision of Dodd-Frank restricting bank proprietary trading; it did not create the CFPB, so choosing it confuses a part with the act.

Q315 AnalystPrep Q.5139

Which of the following is NOT one of the activities addressed by the Markets in Financial Instruments Regulation (MiFIR)?

A. Definition and independence of investment advice.
B. Best execution of trades for clients.
C. Protection for whistleblowers.
D. Dealings with eligible counterparties.

Answer: C

Step by step

1. MiFID II/MiFIR address trader and advisor remuneration and conflicts of interest, fair and non-misleading communication, definition and independence of investment advice, sales and product governance, best execution, and dealings with eligible counterparties.
2. Whistleblower protection is an element of the U.S. Investor Protection Act within Dodd-Frank, not of MiFIR.

Why C is correct: Whistleblower protection comes from the Dodd-Frank Investor Protection Act; the other three are listed MiFIR topics.

Why the others are wrong
- **A:** Investment advice definition and independence is an explicit MiFIR topic.
- **B:** Best execution of client trades is an explicit MiFIR topic.
- **D:** Dealings with eligible counterparties is an explicit MiFIR topic.

Q316

AnalystPrep Q.5144

Which of the following is an effect of the Volcker Rule?

A. Enhanced protection for witnesses and whistleblowers.
B. Requiring derivatives to be bought and sold through clearinghouses.
C. Stopping commercial banks from engaging in speculative and proprietary trading.
D. Establishing the Consumer Financial Protection Bureau.

Answer: C

Step by step
1. The Volcker Rule is part of Dodd-Frank.
2. It prevents commercial banks from speculative activities and proprietary trading for profit, and limits their investments in hedge funds and private equity funds.
3. The other options are different Dodd-Frank elements.

Why C is correct: The Volcker Rule restricts commercial banks' proprietary trading and fund investments.

Why the others are wrong
- **A:** Whistleblower protection comes from the Investor Protection Act component, not the Volcker Rule.
- **B:** The clearinghouse requirement for OTC derivatives is a separate Dodd-Frank provision.
- **D:** The CFPB was created by another part of Dodd-Frank.

Q317

AnalystPrep Q.5146

REBUILT — Original stem ('key issue addressed') had several defensible answers and the source contradicted its own key; rewritten as a clear MiFID II versus other-regulation distinction.

Which of the following was a key feature of the revision of MiFID into MiFID II (adopted in 2014 together with MiFIR)?

A. Additional requirements on public disclosure of trading data and on disclosure of transaction data to regulators and supervisors.
B. Creation of an independent regulator for consumer finance markets such as mortgages and credit cards.
C. A ban on commercial banks' proprietary trading and investments in hedge funds.
D. A requirement that all OTC derivatives be traded through clearinghouses.

Answer: A

Step by step
1. MiFID II was adopted in 2014 after the 2007-2009 crisis, alongside the regulation MiFIR.
2. It reinforced investor protection and added requirements on public disclosure of trading data and on disclosure of transaction data to regulators.
3. The consumer regulator (CFPB), proprietary trading ban (Volcker Rule) and clearing mandate are all U.S. Dodd-Frank features.

Why A is correct: MiFID II strengthened public trade disclosure and regulatory transaction reporting; the other options are Dodd-Frank features.

Why the others are wrong
- **B:** This describes the CFPB, created by Dodd-Frank.
- **C:** This describes the Volcker Rule, part of Dodd-Frank.
- **D:** The clearinghouse mandate for OTC derivatives is a Dodd-Frank requirement.

Q318

AnalystPrep Q.5147

REBUILT — Original key (Volcker Rule) was questionable for the Investor Protection Act, and options mixed MiFIR and Dodd-Frank items; rebuilt to test the Investor Protection Act's whistleblower feature.

Which of the following is a protection provided to investors by the Investor Protection Act component of the U.S. Dodd-Frank Act?

A. Increased protection for whistleblowers.
B. Best execution of trades for clients.
C. Fair and non-misleading communication with customers.
D. Authorization requirements for regulated markets.

Answer: A
Key correction: Source key (AnalystPrep) said the Volcker Rule (C); the Volcker Rule is a separate Dodd-Frank provision, so the question was rebuilt around whistleblower protection.
Step by step
1. The Investor Protection Act established a committee to consult with the SEC on new products, fees and trading strategies, and gave whistleblowers increased protection.
2. Best execution and fair communication to customers are MiFIR topics; authorization of regulated markets is a MiFID feature.

Why A is correct: Increased whistleblower protection is a feature of the Dodd-Frank Investor Protection Act.
Why the others are wrong
- **B:** Best execution of client trades is a MiFID II/MiFIR topic.
- **C:** Fair and non-misleading communication with customers is a MiFID II/MiFIR topic.
- **D:** Authorization requirements for regulated markets are part of original MiFID.

Q319
Mock 1 Q74

REBUILT — Original key (MiFID II addresses best execution) was not a true difference since MiFID also covered conduct; rebuilt as a statement-accuracy item on MiFID versus MiFID II.

Which of the following statements about the EU's MiFID and MiFID II is most accurate?
A. MiFID prescribes conduct of business and organizational requirements for investment firms and defines regulatory reporting and trade transparency obligations to prevent market abuse.
B. MiFID contains no regulatory reporting requirements; these were introduced for the first time under MiFID II.
C. MiFID II is a U.S. law enacted as part of the Dodd-Frank Act to restrict commercial banks' proprietary trading.
D. MiFID is an investor protection regulation, whereas MiFID II deals specifically with commercial banks.

Answer: A
Key correction: Source key (Mock 1) said MiFID II addresses best execution (C); this was not a defensible differentiator as MiFID also covered conduct, so the item was rebuilt.
Step by step
1. MiFID (2004, applicable from 2007) prescribes conduct and organizational requirements and defines regulatory reporting and trade transparency obligations.
2. MiFID II (2014), with MiFIR, strengthened this with added public and regulatory disclosure and topics such as remuneration, conflicts of interest and best execution.
3. B is wrong since reporting already existed; C confuses MiFID II with the Volcker Rule; D misstates MiFID II's scope.

Why A is correct: MiFID already set conduct, organizational and reporting/transparency rules; MiFID II is an EU strengthening, not a U.S. or bank-specific law.
Why the others are wrong
- **B:** MiFID already defined regulatory reporting and transparency obligations; MiFID II strengthened them.
- **C:** MiFID II is EU legislation; the proprietary trading restriction is the U.S. Volcker Rule.
- **D:** MiFID II is also an investor protection regime for investment activities, not a commercial bank-specific regulation.

Q320
Schweser Module Quiz 55.1 Q1

Which of the following regulations limits commercial banks' investments in hedge funds and private equity funds?
A. The Investor Protection Act.
B. The Markets in Financial Instruments Directive (MiFID).
C. The Markets in Financial Instruments Directive II (MiFID II).
D. The Volcker Rule.

Answer: D

Step by step

1. The Volcker Rule, within Dodd-Frank, prohibits speculative and proprietary trading by commercial banks.
2. It specifically limits their investments in hedge funds and private equity funds.

Why D is correct: Limiting bank investments in hedge and private equity funds is the Volcker Rule's purpose.

Why the others are wrong

- **A:** The Investor Protection Act covers whistleblowers, SEC consultation and OTC derivatives.
- **B:** MiFID is EU conduct and disclosure regulation.
- **C:** MiFID II is the EU revision on disclosure and advisor conduct topics.

Q321

Schweser Module Quiz 55.1 Q2

Which of the following pieces of legislation explicitly deals with whistleblower protection?

A. The Investor Protection Act.
B. The Markets in Financial Instruments Directive (MiFID).
C. The Markets in Financial Instruments Directive II (MiFID II).
D. The Volcker Rule.

Answer: A

Step by step

1. The Investor Protection Act, part of Dodd-Frank, gave whistleblowers increased protection.
2. MiFID, MiFID II and the Volcker Rule do not deal with whistleblowers.

Why A is correct: Whistleblower protection is a feature of the Dodd-Frank Investor Protection Act.

Why the others are wrong

- **B:** MiFID covers conduct, organization and reporting, not whistleblowers.
- **C:** MiFID II covers remuneration, advice, best execution, etc., not whistleblowers.
- **D:** The Volcker Rule restricts bank proprietary trading and fund investments.

Q322

AnalystPrep Q.4852

REBUILT — Original asked for a definition of disclosure that is not in the GARP/Schweser reading; rebuilt to test how investor-protection compliance is classified under the Basel operational risk taxonomy.

A bank is fined for selling a complex product to retail clients without adequate disclosure of its risks and without assessing suitability. Under the Basel operational risk event taxonomy, compliance failures of this kind generally fall into which event type and subcategories?

A. Execution, Delivery and Process Management, covering failed trade settlement and processing errors
B. Clients, Products and Business Practices, covering Suitability, Disclosure and Fiduciary, and Improper Business or Market Practices
C. Internal Fraud, covering unauthorized activity by employees
D. External Fraud, covering systems security breaches and theft by third parties

Answer: B

Step by step

1. Identify the failure: inadequate disclosure of risk and no suitability assessment when dealing with clients.
2. These are breaches of obligations to clients rather than processing errors or fraud by employees or outsiders.
3. Basel event type 4, Clients, Products and Business Practices (CBPB), covers them.
4. Investor-protection compliance falls specifically into subcategory 4.1 (Suitability, Disclosure and Fiduciary) or 4.2 (Improper Business or Market Practices).

Why B is correct: Investor-protection violations concern the firm's duties to clients and market conduct, which is exactly what the CBPB event type and its subcategories 4.1 and 4.2 capture.

Why the others are wrong

- **A:** Execution, Delivery and Process Management covers processing and settlement errors; the failure here is a conduct breach toward clients, not a process error.
- **C:** Internal Fraud requires intentional unauthorized acts by staff; mis-selling without risk disclosure is a business practice failure.
- **D:** External Fraud involves third parties acting against the firm, which is not the situation described.

LO 55.b

Describe lessons learned from the case studies involving violations of investor protection or compliance regulations.

Q323

AnalystPrep Q.5047

REBUILT — Original asked about compliance-risk history (event types, dedicated departments) that is not supported by the GARP case-study chapter and had a conflicting explanation; rewritten as a lesson from the Deutsche Bank Securities best-execution case.

In 2022, FINRA fined Deutsche Bank Securities Inc. $2 million for best execution failures between 2014 and 2018. Which of the following is the most accurate lesson from this case about investor protection and compliance risk?

A. Large fines arise only from deliberate market manipulation such as spoofing, so routing and execution practices carry little regulatory risk.

B. A firm can face regulatory penalties for failing to secure the most favorable execution terms for customers, particularly where it knew of the problems, did not change its order routing, and only vaguely disclosed the rebates it received.

C. The fine was the largest ever imposed on a single organization, showing that best execution breaches are punished more severely than misrepresentation.

D. Because the bank settled without admitting or denying the findings, the case shows that regulators cannot penalize firms absent an admission of responsibility.

Answer: B

Key correction: Source key (AnalystPrep) said C; that option is not supported by the GARP case studies (fines rose, but nothing there addresses dedicated departments), so the item was rebuilt around the case content.

Step by step

1. Recall the case: Deutsche Bank Securities routed orders through a system that caused delays and partial fills, while earning trading rebates only vaguely disclosed.
2. The bank knew of the problems but did not modify its routing or evaluate price-improvement alternatives.
3. The fine ($2 million) was small and did not involve misrepresentation or manipulation; the bank settled without admitting or denying the findings.
4. Lesson: failing to deliver best execution, and not acting on known issues, is itself a compliance breach that is penalized. Only B states this.

Why B is correct: The case shows that regulators penalize best execution failures and inaction on known problems even without manipulation or misrepresentation.

Why the others are wrong

- **A:** Wrong: this case involved no manipulation, yet was still fined; spoofing is the JPMorgan case.
- **C:** Wrong: the largest penalty was UBS's $11.15 billion (2008); the Deutsche Bank fine was small by comparison.
- **D:** Wrong: settling without admission is common and does not prevent a fine; the fine was in fact imposed.

Q324

Schweser Module Quiz 55.1 Q3

Which of the following reasons is least likely a consideration for regulators when imposing fines on financial institutions for investor protection breaches and violations?

A. To cover the cost of litigation.

B. To deter other firms from manipulative banking practices.

C. To ensure that fines cover at least the benefits of breaches.

D. To signal to other firms to change their compliance practices.

Answer: A

Step by step

1. List the rationales in the case studies: punish past violations, deter future misconduct, signal to peers, and set fines high enough to offset breach benefits.
2. Check each option: B is deterrence, C is offsetting benefits, D is signaling.
3. A (recovering litigation costs) is not a direct consideration, so it is the least likely.

Why A is correct: Fines are punitive, deterrent and signaling tools sized to exceed breach benefits; litigation cost recovery is not a direct objective.

Why the others are wrong

- **B:** Valid consideration: fines aim to deter further manipulative practices.
- **C:** Valid consideration: fines are set so the penalty more than offsets benefits gained from the breach.

- **D:** Valid consideration: fines signal to peer firms to adjust their compliance and monitoring practices.

Q325 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Maria Santos, an operational risk analyst at an investment bank, is preparing a training note on the largest individual penalty in US investor protection enforcement data for 2000 to 2022: the July 2008 resolution with UBS totaling \$11.15 billion. Which of the following statements best describes the case and the lesson it offers?

A. The resolution was a \$11.15 billion cash fine imposed by the CFTC for spoofing in precious metals markets, showing that manipulative order entry attracts the largest penalties.

B. The resolution required UBS to buy back \$11 billion of securities and pay \$150 million in penalties, after charges that it misrepresented auction rate securities to investors as safe, cash-equivalent products despite rising liquidity risk.

C. The resolution required UBS to pay \$11 billion in penalties and buy back \$150 million of securities, after charges of best execution failures caused by indirect order routing.

D. The resolution was a settlement for failing to disclose trading rebates received from routing venues, showing that vague disclosure is the costliest investor protection breach.

Answer: B

Step by step

1. Identify the case: the July 2008 UBS resolution is the most severe individual penalty in the 2000-2022 US data (about \$82 billion across 6,612 records).
2. Decompose the total: $11 + 0.15 = 11.15$ billion, i.e., \$11 billion of securities buy-back plus \$150 million of penalties.
3. The charge was misrepresenting auction rate securities as safe, cash-equivalent products when they faced increasing liquidity risk, a suitability/disclosure (misrepresentation) breach in the Clients, Products and Business Practices event type.
4. Lesson: misrepresenting product risk to investors can produce very large penalties; regulators set fines high enough to offset the benefits of the breach, to deter the firm and signal to peer firms.

Why B is correct: The \$11.15 billion consisted of an \$11 billion securities buy-back and \$150 million in penalties, resolving multi-state charges that UBS misrepresented auction rate securities as safe, cash-equivalent products despite rising liquidity risk.

Why the others are wrong

- **A:** Confuses the UBS case with the JP Morgan case: the record spoofing fine was \$920 million, imposed by the CFTC in September 2020, not \$11.15 billion. UBS was about misrepresentation of auction rate securities.
- **C:** Reverses the components (the penalty was \$150 million and the buy-back \$11 billion) and attributes the case to best execution failures, which is the Deutsche Bank Securities case (a \$2 million FINRA fine).
- **D:** Mixes in the Deutsche Bank Securities rebate disclosure finding; that case involved a \$2 million fine, so it is not the largest penalty, and UBS was penalized for misrepresenting securities, not undisclosed rebates.

Reading 56. Supervisory Guidance on Model Risk Management

"Supervisory Guidance on Model Risk Management," Federal Deposit Insurance Corporation (June 2017).

LO	Learning objective	Qs
56.a	describe model risk and explain how it can arise in the implementation of a model.	5
56.b	describe elements of an effective model risk management process.	6
56.c	explain best practices for the development and implementation of models.	5
56.d	describe elements of a strong model validation process and challenges to an effective validation process.	12

LO 56.a

Describe model risk and explain how it can arise in the implementation of a model.

Q326

BT notes P2.T7.604.2

REBUILT — Original relied on an ambiguous 'least likely' EXCEPT format with a contestable keyed answer (empirical distribution) and non-Schweser Crouhy content; rebuilt as a clear classification of the two ways model risk arises.

A bank's derivatives desk adopts a pricing model that was built and approved for plain-vanilla interest rate swaps. Without any modification, the desk begins using it to value a new exotic structured product with embedded optionality. The code is error-free and the vanilla swap valuations remain accurate. Which statement best describes the source of model risk here?

A. Model risk arises because the model contains mathematical or coding errors introduced during design.
B. Model risk arises because the model is being used outside the context and purpose for which it was intended.
C. Model risk arises because the model has no reporting component.
D. There is no model risk, because the model produces correct outputs for the products it was designed for.

Answer: B

Step by step

1. Model risk arises in two ways: (1) the model has fundamental errors and gives faulty outputs; (2) the model is used incorrectly, for an unintended purpose, or out of context.
2. The stem states the code is error-free and vanilla valuations are accurate, so way (1) is ruled out.
3. The model is being applied to a new product it was not designed for, which is way (2): improper use / out of context.

Why B is correct: A model that is sound for its original purpose still creates model risk when applied to new products or changed environments without reassessing its limitations.

Why the others are wrong

- **A:** Describes the first source (faulty design/implementation), but the stem says the code is error-free.
- **C:** Nothing in the stem suggests a missing reporting component; this confuses model structure with the source of model risk.
- **D:** Treats accuracy within the original scope as sufficient; model risk also arises from use outside intended scope.

Q327

AnalystPrep Q.4305

REBUILT — Original asked for an unsupported 'common error across industries' (sample bias) not stated in the reading; rebuilt around the 'garbage in, garbage out' source of model risk covered in the LO.

A risk team builds a credit loss model using only data from a prolonged economic expansion. The model's mathematics and implementation are correct, but it produces badly understated loss estimates when conditions deteriorate. Which principle best explains why the model's outputs are unreliable?

A. The model was used for an unintended purpose.
B. The reporting component converts estimates into the wrong format.
C. The quality of a model's output depends on the quality of its inputs and assumptions (garbage in, garbage out).
D. Models never require assumptions, so the loss understatement must come from user error.

Answer: C

Step by step

1. Math and implementation are stated to be correct, so the fault is not in processing.

2. The model was built from non-representative data (expansion only), so the input/assumption quality is poor.
3. This is the 'garbage in, garbage out' way model risk arises: errors in inputs and assumptions produce faulty outputs.

Why C is correct: Output quality is a function of input data and assumptions; non-representative inputs yield faulty estimates even when the processing is correct.

Why the others are wrong

- **A:** The model is used for its intended purpose (credit losses); the problem is the data, not the use.
- **B:** Nothing in the stem concerns presentation; the reporting component only communicates the estimates.
- **D:** Models necessarily rely on simplifying assumptions (inputs and assumptions are a model component), so this is false.

Q328

AnalystPrep Q.4451

Which of the following is NOT one of the three essential components of a model?

A. A data input component
B. A data processing component
C. A data understanding component
D. A reporting component

Answer: C

Step by step

1. A model has three components: information inputs (assumptions and data), processing (inputs to estimates), reporting (estimates to applied information).
2. Options A, B and D match these three.
3. 'Data understanding' is not one of the three components named in the guidance, so C is the exception.

Why C is correct: The three model components are information input, processing and reporting; data understanding is not a model component.

Why the others are wrong

- **A:** This is a component: it delivers assumptions and data to the model.
- **B:** This is a component: it transforms inputs into estimates.
- **D:** This is a component: it translates estimates into useful business information.

Q329

Schweser Module Quiz 56.1 Q1

Which component of a model deals with converting estimates into useful or applied information?

A. Information inputs.
B. Processing.
C. Reporting.
D. Transformational.

Answer: C

Step by step

1. The information input component handles data and assumptions.
2. The processing component converts inputs into estimates.
3. The reporting component converts estimates into applied, useful information, so C.

Why C is correct: Reporting is the stage that translates model estimates into business information for users.

Why the others are wrong

- **A:** Inputs supply data and assumptions; they do not convert estimates.
- **B:** Processing converts inputs into estimates, which is the step before reporting.
- **D:** 'Transformational' is not one of the three model components.

Q330

AnalystPrep Q.4297

REBUILT — Original had giveaway distractors (cost, timing); rebuilt to test the definition of model risk with plausible alternatives.

Which of the following best describes how model risk can lead to adverse outcomes for a financial institution?

A. Decisions are based on inaccurate outputs of a model that is flawed or is misused, leading to financial and reputational losses.

B. The model's development costs exceed the budget, which is the primary financial loss from model risk.
C. A model that is highly accurate on past data eliminates model risk entirely.
D. Model risk arises only when a model contains coding errors and is unrelated to how the model is used.

Answer: A
Step by step
1. Model risk is the possibility of negative outcomes from decisions made using inaccurate model outputs.
2. Inaccuracy can come from fundamental errors or from improper use; losses can be financial and reputational.
3. Only option A states this.

Why A is correct: Models rely on simplifying assumptions and can never be fully accurate, so poor decisions based on faulty or misused outputs create financial and reputational loss.
Why the others are wrong
- **B:** Development cost is a cost-benefit consideration, not the model risk itself.
- **C:** Models are simplifications and never completely accurate; model risk is always present.
- **D:** Model risk also arises from improper use, use for unintended purposes, or use out of context.

LO 56.b

Describe elements of an effective model risk management process.

Q331

AnalystPrep Q.4742

Which of the following parties is primarily responsible for ensuring a model has undergone appropriate validation and approval processes before being deployed?
A. Internal audit function
B. Model owner
C. Senior management
D. Independent validation team

Answer: B
Step by step
1. The model owner has ultimate accountability for model use and performance within the bank's policies.
2. Duties of the owner include ensuring models are properly developed, implemented and used, and that models in use have undergone appropriate validation and approval.
3. The other parties have different roles, so B is correct.

Why B is correct: Accountability for ensuring a model has been validated and approved sits with the model owner, who also supplies the information validators need.
Why the others are wrong
- **A:** Internal audit verifies that policies and validation records are adequate and followed; it does not ensure validation occurs.
- **C:** Senior management establishes the framework and reports to the board, but delegates ownership duties to model owners.
- **D:** The validation team performs the independent validation; it does not own the model or ensure it is submitted for approval.

Q332

AnalystPrep Q.2235

A BHC is having a supervisory review performed on its modeling practices for capital planning. Following the review, the company receives very positive feedback regarding its model documentation as part of risk management. What could be the reason for the positive feedback?
A. Large-scale collection of relevant historical data for use as input data.
B. Presence of an updated inventory of all models used in the process.
C. Presence of qualified modeling staff.
D. Adoption of large-scale sensitivity testing and stress testing.

Answer: B

Step by step

1. Documentation of model risk management includes a comprehensive, up-to-date model inventory (purpose, usage, restrictions, inputs, components).
2. A, C and D concern data, staff competence and testing, not documentation.
3. So B is correct.

Why B is correct: A comprehensive, current model inventory is a core documentation element, supporting aggregate model risk assessment.

Why the others are wrong

- **A:** Data collection is an input-quality matter, not documentation.
- **C:** Staff competence supports effective challenge but is not documentation.
- **D:** Sensitivity and stress testing are testing practices, not documentation.

Q333 AnalystPrep Q.4299

Which of the following gives a reason why a firm should invest in model risk management?

A. To give incentives to model developers to work faster
B. To cater for losses brought about by a model
C. To provide incentives to management
D. To ensure that the model is used as required

Answer: D

Step by step

1. Sound model risk management depends on substantial investment in supporting systems for data and reporting integrity.
2. It also requires controls and testing to ensure proper implementation, effective systems integration and appropriate use.
3. Hence the reason is to ensure the model is used as required (D).

Why D is correct: Investment in systems, controls and testing is what ensures models are implemented correctly and used appropriately.

Why the others are wrong

- **A:** MRM aims at quality and effective challenge, not developer speed.
- **B:** Providing for losses is not the stated purpose; MRM aims to prevent adverse outcomes from model error or misuse.
- **C:** Incentives matter for effective challenge, but providing management incentives is not the reason for the investment.

Q334 AnalystPrep Q.4450

REBUILT — Original asked which vague term is an 'element' of MRM (not defined in the readings, with a 'none of the above' option); replaced with a testable question on what drives the intensity of model risk management.

According to the supervisory guidance, which of the following is the key factor that should drive the level and comprehensiveness of a bank's model risk management for a particular model?

A. The potential impact of the model on the firm's profits and financial position
B. The number of staff who developed the model
C. Whether the model was built in-house rather than obtained from a vendor
D. How long the model has been in use

Answer: A

Step by step

1. Model risk depends on model complexity, uncertainty of inputs and assumptions, and potential impact on users.
2. Materiality (potential impact on profits/financial position) is the key element driving the level of risk management: the greater the impact, the more detailed the process.
3. B, C and D are not drivers named in the guidance.

Why A is correct: The greater a model's potential impact on profits or financial position, the more detailed and comprehensive its risk management must be.

Why the others are wrong

- **B:** Developer headcount says nothing about exposure.

- **C:** Vendor models still require validation; origin does not set the level of MRM.
- **D:** Length of use is not a materiality measure; an old model may even be out of context and riskier.

Q335 BT answers P2.T7.20.3.1 · BT notes P2.T7.20.3.1

Which of the following is the BEST guiding principle for managing model risk?

A. Model development
B. Effective challenge
C. Regulatory compliance
D. Latest and best technology

Answer: B

Step by step

1. The guiding principle for managing model risk is effective challenge: critical analysis by objective, informed parties who can identify limitations and assumptions and drive changes.
2. It depends on incentives, competence and influence.
3. The other options are not guiding principles.

Why B is correct: Effective challenge by independent, competent and influential parties is the fundamental principle of model risk management.

Why the others are wrong

- **A:** Development is an activity within the process, not the guiding principle.
- **C:** Compliance alone does not ensure sound practice.
- **D:** Newest technology may be unproven and is not the aim.

Q336 Schweser Module Quiz 56.1 Q2

REBUILT — Original option A ('shortcuts and simplifications will increase model risk') is arguably supported by the text, creating two defensible answers; reworded to be clearly false.

Which of the following statements regarding model risk management is correct?

A. Shortcuts and simplifications adopted to manage complexity cannot affect the accuracy of model outputs.
B. Managing model risk requires proper segregation of duties.
C. With appropriate procedures and tools, model risk can be eliminated.
D. Like many other risks, model risk has both an upside and a downside component.

Answer: B

Step by step

1. Effective challenge is fundamental and requires independence from model development, to avoid a self-review threat.
2. Therefore development and critical review must be done by different parties: segregation of duties (B).
3. A, C and D contradict the reading.

Why B is correct: Independent challenge requires that those who build a model are not those who review it.

Why the others are wrong

- **A:** Even innocent simplifications can sacrifice accuracy and introduce error.
- **C:** Models are simplified imitations of reality, so model risk is always present and cannot be eliminated.
- **D:** Model risk is defined by negative outcomes (losses, poor decisions); it has no upside component.

LO 56.c

Explain best practices for the development and implementation of models.

Q337 AnalystPrep Q.4304

Model development and implementation in risk management requires various best practices to ensure that the models are aligned to their intended use. Which of the following statements about these best practices is MOST accurate?

A. The merits and limitations of the model methodologies and processing components should be well explained.

B. Developers should compare their models with alternative approaches and theories.
C. The quality of data used to develop a model should be assessed and documented.
D. All of the above

Answer: D
Step by step
1. Statement A: supporting background for the model must be described, noting strengths and weaknesses - a best practice.
2. Statement B: comparing the model with alternative models serves as a reasonability check on its logic - a best practice.
3. Statement C: data and assumptions must be shown to be robust and relevant, with quality assessed and proxies or adjustments documented - a best practice.
4. All three are best practices, so D is correct.

Why D is correct: Each of the three statements is an explicitly stated development best practice, so the only complete answer is 'all of the above'.

Why the others are wrong
- **A:** True but incomplete; choosing it ignores that B and C are also best practices.
- **B:** True but incomplete; benchmarking against alternatives is only one of the practices.
- **C:** True but incomplete; data-quality assessment is only one of the practices.

Q338
BT Mock Q33 · BT answers P2.T7.20.3.2 · BT notes P2.T7.20.3.2

According to the FDIC, model risk management starts with robust model development, implementation, and use (the first element; the second is a sound validation process, and the third is governance). Which of the following is a TRUE statement about this first element?

A. Model development should be a straightforward and routine technical process.
B. To exercise conservatism, pick an extreme point on a given modeled distribution.
C. Banks should never model output with judgmental or qualitative adjustments.
D. Although the nature of testing varies by the context and type of model, testing is an integral part of model development.

Answer: D
Step by step
1. A: development depends on developer experience and judgment; it is not straightforward or routine - false.
2. B: an extreme point is not necessarily conservative if the distribution is misestimated or misspecified - false.
3. C: quantitative output may be tempered with judgmental or qualitative adjustments if they have a logical basis and are documented - false.
4. D: testing is integral to development, although its form depends on model type and context - true.

Why D is correct: Testing is an integral part of development, and the form of tests varies with the model and its context.

Why the others are wrong
- **A:** Development requires considerable judgment and multidisciplinary skill; treating it as routine understates model risk.
- **B:** Picking an extreme point is arbitrary: if the distribution is misspecified the point may not be conservative.
- **C:** Judgmental or qualitative adjustments are allowed when logical and documented; 'never' is too absolute.

Q339
GARP PE2 Q4 · Mock 4 Q5

A midsize bank specializing in residential mortgages and credit cards is planning to offer a new commercial loan product. The bank has an existing credit rating model for its residential mortgage products which has performed well and has been successfully backtested over several years, but the CRO considers whether the bank should develop and implement a new model for the commercial loan product. In addition, the CRO wants to ensure that the bank follows best practices for the development, validation, and implementation of any models that it uses. Which of the following actions should the CRO recommend that the bank take?

A. Apply the strongly performing existing model to the new commercial loan product given the model's successful track record.
B. Develop a new model for the commercial loan product and avoid the use of qualitative or judgmental adjustments to the model's quantitative output.

C. Make the model development team responsible for validating all of the bank's models that are currently operational.

D. Perform sensitivity analysis on all models used by the bank to identify market conditions under which they might perform poorly.

Answer: D

Step by step

1. A: a model is valid for its intended use; applying a mortgage model to commercial loans is use out of context - a source of model risk.
2. B: judgmental adjustments are acceptable if logical and documented, so avoiding them is not required.
3. C: validation requires effective challenge by independent, unbiased staff; developers validating their own models lacks independence.
4. D: sensitivity analysis and testing identify assumptions and situations where a model performs poorly or becomes unreliable - recommended.

Why D is correct: Sensitivity analysis and testing across conditions reveal where models are weak or unreliable, which is a development and validation best practice.

Why the others are wrong

- **A:** Past performance on mortgages does not make the model appropriate for a different product; this is the out-of-context use that creates model risk.
- **B:** Qualitative adjustments are permitted when well-founded and documented; prohibiting them is not a best practice.
- **C:** Developers lack independence to validate their own work; validation should be done by independent staff.

Q340 Schweser Module Quiz 56.1 Q3

REBUILT — Original asked which item is least likely a key testing consideration with option C 'testing under normal market conditions', but Schweser and GARP require testing in both normal and stressed states, so C was itself a key consideration; reworded to ask which practice is least consistent with sound testing and changed C to restricting testing to normal conditions only.

During model development, a bank's model team is designing its testing program. Which of the following practices is LEAST consistent with sound model testing?

A. Documenting potential weaknesses of the model and circumstances in which it does not work.

B. Using inputs over a wide range of values, including extreme values, to find the boundaries of model effectiveness.

C. Restricting testing to normal market conditions, because stressed conditions are too rare to be informative.

D. Analyzing other models that use the subject model's outputs as their inputs.

Answer: C

Step by step

1. Testing must note potential weaknesses and circumstances where the model does not work, so A is sound practice.
2. Testing should use inputs over a range of values, including extreme values, to establish the acceptable input range and boundaries of performance, so B is sound practice.
3. If the model's outputs feed other models, those other models must also be analyzed, so D is sound practice.
4. Testing should occur in both normal and extreme (stressed) states, including unusual but plausible conditions; limiting it to normal conditions contradicts this, so C is least consistent.

Why C is correct: Sound testing covers a wide variety of conditions, normal and stressed, including scenarios outside ordinary assumptions; excluding stressed conditions hides exactly the situations where a model is most likely to fail.

Why the others are wrong

- **A:** Sound practice: identifying weaknesses and failure circumstances is a core aim of testing; picking A confuses a required step with a flaw.
- **B:** Sound practice: extreme input values reveal the boundaries of model effectiveness; picking B mistakes stress-style testing for over-conservatism.
- **D:** Sound practice: downstream models that rely on the subject model's output must be analyzed; picking D treats this as outside the testing scope.

Q341 AnalystPrep Q.4448

REBUILT — Original options (validation team, data sources, individual users, none) were vague and not grounded in the reading; rebuilt around the documented requirement for data proxies and adjustments.

Which of the following should be clearly documented and made clear to users during model development, so that users can judge the usefulness of the data themselves?

A. The identities of the individual users of the model.
B. The compensation structure of the validation team.
C. Any data proxies, unrepresentative data, and adjustments made to data, such as those from external sources.
D. Only the final model output, since inputs are internal matters.

Answer: C

Step by step

1. Development requires robust, relevant data and assumptions.
2. Data proxies, data that is not representative, and adjustments to data (e.g., from external sources) must be clearly documented.
3. This lets users judge the model's usefulness themselves; so C is correct.

Why C is correct: Documenting data proxies, non-representative data and adjustments is a stated best practice that gives users transparency about data limitations.

Why the others are wrong

- **A:** User identities do not inform data usefulness; documentation covers how the model operates, its limitations and assumptions.
- **B:** Validation team incentives matter for governance and effective challenge, not for documenting data in development.
- **D:** Omitting input data details hides limitations; garbage-in-garbage-out risk requires inputs to be documented.

LO 56.d

Describe elements of a strong model validation process and challenges to an effective validation process.

Q342 AnalystPrep Q.2234

Bank of Elmwood, part of a Bank Holding Company (BHC), is preparing for independent model review and validation. What is included in such a process?

A. An evaluation of conceptual soundness
B. Ongoing monitoring that includes process verification and benchmarking
C. Outcomes analysis
D. All of the above

Answer: D

Step by step

1. Recall the three core elements of a strong validation process: evaluation of conceptual soundness, ongoing monitoring, and outcomes analysis.
2. A is element 1; B is element 2 (process verification and benchmarking are its parts); C is element 3.
3. Since all three are included, the answer is all of the above.

Why D is correct: A strong validation process comprises all three elements, so each listed item belongs in it.

Why the others are wrong

- **A:** Correct element but incomplete: it omits ongoing monitoring and outcomes analysis.
- **B:** Correct element but incomplete: it omits conceptual soundness and outcomes analysis.
- **C:** Correct element but incomplete: it omits conceptual soundness and ongoing monitoring.

Q343 AnalystPrep Q.2238

A bank holding company was recently subjected to a supervisory review of its model risk management and received negative feedback. Which of the following is the most likely cause?

A. Using models without validation, or models with identified weaknesses
B. Using benchmark or challenger models to help assess the reasonableness of the primary model output

C. Employing independent validation staff to critically evaluate the models
D. Clearly disclosing the validation status of all models used for capital planning

Answer: A
Step by step
1. Supervisory guidance expects every model in use to be validated and weaknesses to be addressed or limited.
2. B, C and D are recognized good practices: benchmarking, independent critical review and transparency about validation status.
3. Only A is a lapse that would draw negative feedback.

Why A is correct: Using unvalidated models, or models with known weaknesses that are not remediated, is a core model risk management failure.
Why the others are wrong
- **B:** Benchmarking/challenger models are a recommended validation tool, not a deficiency.
- **C:** Independent, critical review is the very safeguard supervisors want.
- **D:** Transparency about validation status is good practice; supervisors penalize hiding it, not disclosing it.

Q344 AnalystPrep Q.4303

TCC Bank has developed a set of models to analyze liquidity risk, market risk and borrower credit risk. Which of the following model risk management functions is LEAST likely to be handled by the developers of these models?
A. Stating a clear purpose so that development is aligned with the intended use
B. Rigorous assessment of data quality and relevance
C. Model testing
D. Independent model validation

Answer: D
Step by step
1. Developers own the design stage: purpose statement, data quality/relevance assessment and developmental testing.
2. Validation requires a degree of independence from development and use, to avoid bias and conflicts of interest.
3. So validation (D) is the function least likely to be done by the developers.

Why D is correct: Segregation of duties means the people who build a model should generally not be the ones who validate it.
Why the others are wrong
- **A:** Defining the model's objective is a developer responsibility at the outset.
- **B:** Assessing data quality and relevance is part of development.
- **C:** Testing during development is a key developer task; confusing it with independent validation is the error.

Q345 AnalystPrep Q.4443

REBUILT — Original EXCEPT stem was awkward ('None of the above' option; 'involved' too vague given guidance lets developers do some validation work under independent review); rewrote with one clearly inconsistent statement.
Which of the following is NOT consistent with supervisory guidance on the model validation process?
A. Documentation should indicate where the model does not work effectively
B. Validation should be conducted by staff with a degree of independence from model development and use
C. Work done by model developers should be accepted as validation without critical review by an independent party
D. The depth of validation should be commensurate with the complexity and materiality of the bank's models

Answer: C
Step by step
1. A is consistent: limitations of a model must be documented.
2. B and D are consistent: validation requires independence and its extent scales with complexity and materiality.
3. Guidance allows some validation work to be done by developers or users only if it is subject to critical review by an independent party; C removes that review, so it is inconsistent.

Why C is correct: Developer-performed validation work is acceptable only when an independent party critically reviews it.

Why the others are wrong
- **A:** Consistent with guidance; documenting limitations is expected.
- **B:** Consistent with guidance; independence is a core principle.
- **D:** Consistent with guidance; validation rigor is scaled to model complexity and materiality.

Q346
AnalystPrep Q.4449

REBUILT — Original options were poorly worded and 'all of the above' was ambiguous; rebuilt so that the vendor-model challenge is the single clearly correct option.

Which of the following is a recognized challenge to an effective model validation process?

A. Use of vendor models, where the modeling is external and parts of it may be confidential
B. Having all models developed in-house
C. Having extensive documentation of model assumptions
D. Having validation performed independently of model developers

Answer: A

Step by step
1. Vendor products limit access to code and implementation details, and some modeling is confidential.
2. Validation therefore relies more on sensitivity analysis, benchmarking and vendor-provided developmental evidence and testing.
3. B, C and D are features that support, not hinder, validation.

Why A is correct: Vendor models make validation harder because limited transparency forces reliance on indirect tools.

Why the others are wrong
- **B:** In-house development gives validators full access; it is not a challenge.
- **C:** Documentation is what validators examine in the conceptual soundness review.
- **D:** Independence is a requirement of good validation, not an obstacle.

Q347
AnalystPrep Q.4743

REBUILT — Original distractors (good investment, technology advancement, time efficiency) were giveaways; replaced with plausible but incorrect elements.

Which of the following is one of the three core elements of a strong model validation process?

A. Ongoing monitoring
B. Vendor certification
C. Regulatory sign-off
D. Capital add-on calibration

Answer: A

Step by step
1. The three core elements are evaluation of conceptual soundness, ongoing monitoring, and outcomes analysis.
2. Vendor certification, regulatory sign-off and capital add-ons are not validation elements.
3. Hence only A qualifies.

Why A is correct: Ongoing monitoring, including process verification and benchmarking, is one of the three core validation elements.

Why the others are wrong
- **B:** Vendor models need extra validation work, but certification is not a core element.
- **C:** Validation is the bank's responsibility; supervisors review it but are not part of the process.
- **D:** Capital cushions can address model risk but are not a validation element.

Q348
BT answers P2.T7.20.3.3 · BT notes P2.T7.20.3.3

According to the FDIC, an effective model validation framework should include three core elements. What are they?

A. Implementation; ongoing error correction; and reporting
B. Evaluation of sound governance; back-testing; and vendor validation
C. Evaluation of conceptual soundness; ongoing monitoring; and outcomes analysis
D. Evaluation of simulated errors; expert screening; and policy/procedure validation

Answer: C

Step by step

1. The core elements are conceptual soundness evaluation, ongoing monitoring (process verification, benchmarking) and outcomes analysis (including back-testing).
2. C matches exactly.
3. A, B and D mix model-process or governance terms that are not the three elements.

Why C is correct: Conceptual soundness, ongoing monitoring and outcomes analysis are the three core validation elements.

Why the others are wrong

- **A:** Lists model lifecycle/reporting steps, not validation elements.
- **B:** Back-testing is a part of outcomes analysis, not a separate element; governance and vendor validation are not core elements.
- **D:** Invented terms; none are validation elements.

Q349 GARP PE1 Q47

The board of a fast-growing US-based bank is reviewing a regulatory report that identified deficiencies in the bank's model validation process. The CRO consults Federal Reserve guidance to improve validation. Which of the following recommendations is appropriate?

A. Require internal auditors who assess the bank's model risk management framework to also validate and back-test the models

B. Begin monitoring each model's performance only after its backtest has been successfully completed

C. Ensure third-party vendor models are implemented under the assumptions initially provided by the vendor

D. Establish a continuous process of model risk management and validation rather than reviews at specific points in time

Answer: D

Step by step

1. Validation should be ongoing, reflecting changes in products, markets and model use; periodic-only reviews lag.
2. A fails independence: auditors assessing the framework should not also validate the models.
3. B is wrong because monitoring begins once the model is put into use; C is wrong because vendor models must be validated for the bank's own use and may need customization.

Why D is correct: Ongoing, continuous validation is what guidance expects; the other options violate independence, timing or vendor-model expectations.

Why the others are wrong

- **A:** Auditors validating models they assess would compromise independence.
- **B:** Monitoring starts at implementation, not after backtesting.
- **C:** The bank must validate its own use of vendor models; vendor assumptions may not fit its portfolios.

Q350 Schweser Module Quiz 56.2 Q1

Backtesting is most appropriately classified in which element of the validation process?

A. Evaluation of conceptual soundness

B. Ongoing monitoring

C. Outcomes analysis

D. Post-validation review

Answer: C

Step by step

1. Backtesting compares actual results with model estimates over a period different from the development sample.
2. This comparison of outputs with what actually happened is outcomes analysis.
3. Backtesting is a specific form of outcomes analysis.

Why C is correct: Backtesting is one form of outcomes analysis.

Why the others are wrong

- **A:** Conceptual soundness reviews documentation, assumptions and sensitivity tests, not realized outcomes.
- **B:** Ongoing monitoring covers process verification and benchmarking.

- **D:** Post-validation review is not one of the three elements.

Q351 Schweser Module Quiz 56.2 Q2

Comparing a model's inputs and outputs to estimates from other data or models is best described as:

A. benchmarking.
B. ongoing monitoring.
C. process verification.
D. sensitivity analysis.

Answer: A

Step by step

1. Identify the activity: comparing a model's inputs and outputs with other comparable data or models.
2. In the validation framework this is the definition of benchmarking.
3. Significant variances against the benchmark prompt investigation; small variances suggest the model is reliable.

Why A is correct: Benchmarking is the validation activity that compares a model's inputs and outputs with other comparable data or models.

Why the others are wrong

- **B:** Ongoing monitoring is the umbrella element (containing process verification and benchmarking), not the specific comparison activity.
- **C:** Process verification checks that data inputs remain error free and complete, and that the model is implemented as intended; it does not compare against other models.
- **D:** Sensitivity analysis examines how outputs change when inputs or assumptions are varied, not comparison to external estimates.

Q352 AnalystPrep Q.4298

A risk management team at a bank is implementing a comprehensive validation framework for newly developed credit risk models in preparation for regulatory review. According to best practices in model validation, which of the following validation processes would be MOST appropriate for determining whether the mathematical framework of the model is conceptually well-founded?

A. Qualitative review of model documentation, derivation of formulae, and comparison with established theoretical frameworks
B. Data quality checks focusing on the completeness, accuracy, and appropriateness of input data
C. Systems implementation testing including user acceptance testing and code verification
D. Management oversight involving senior leadership in reviewing model outputs and their application in business decisions

Answer: A

Step by step

1. The question concerns conceptual soundness, the first element of strong validation.
2. Conceptual soundness is assessed by reviewing documentation and empirical evidence, the model's design, theory and construction, and comparing with published research and accepted practice.
3. Option A is exactly this qualitative review of theory and derivations.

Why A is correct: Evaluating conceptual soundness requires a qualitative review of the model's design, theory, formulae and supporting evidence against established frameworks.

Why the others are wrong

- **B:** Data quality checks relate to input integrity (process verification/ongoing monitoring), not whether the mathematics is well-founded.
- **C:** Implementation and code testing verify that the model is built as intended, not that its theory is sound.
- **D:** Management oversight concerns governance and use of outputs, not the model's theoretical foundation.

Q353 AnalystPrep Q.2600

REBUILT — Original three-statement item was loosely worded (the 'external audit' statement was not a clear Schweser/GARP validation point); rebuilt as a single-best-answer question on the elements of a strong validation process.

Which of the following statements about a strong model validation process is most accurate?

A. Developmental evidence, including the documentation of assumptions, inputs and data, should be reviewed

before a model is used and again whenever the model undergoes a material change.

B. Once a model has been validated and approved, its inputs and assumptions should be frozen so that results stay comparable over time.

C. To make use of their detailed knowledge, the individuals who developed a model should normally be the ones to validate it.

D. A significant variance between a model's outputs and a benchmark model's outputs proves that the model is faulty.

Answer: A

Key correction: Source key (AnalystPrep) said A for the original statements (I only); that is consistent with the rebuilt answer: inputs and assumptions must be revisited periodically, not frozen.

Step by step

1. Validation has three elements: evaluation of conceptual soundness, ongoing monitoring, and outcomes analysis.
2. Conceptual soundness requires analysing the development documentation (formulation, assumptions, computations, inputs) before use and whenever the model changes significantly, so A is correct.
3. Ongoing monitoring means inputs, assumptions and data are revisited and tested as conditions change (with sensitivity and stress testing), so freezing them (B) is wrong.
4. Segregation of duties means developers should generally not validate their own models (C wrong).
5. Benchmark variances must be judged against an acceptable range; the benchmark may use a different methodology, so a variance is not proof of a faulty model (D wrong).

Why A is correct: Validation must be continuous and documentation-based: developmental evidence is reviewed before use and after any material change, as part of evaluating conceptual soundness.

Why the others are wrong

- **B:** Reverses the requirement: inputs and assumptions must be monitored and revisited periodically (and stress tested) as markets and products change, not frozen.
- **C:** Ignores segregation of duties; developers should generally not validate their own models, with exceptions only for highly technical areas and then under rigorous independent review.
- **D:** Treats a benchmark gap as automatic failure; the variance may simply reflect a different benchmark methodology and should be judged against an acceptable range.

Reading 57. Case Study: Model Risk and Model Validation

Global Association of Risk Professionals. Operational Risk and Resilience. New York, NY: Pearson, 2022. Chapter 16.

LO	Learning objective	Qs
57.a	define a model and describe different ways that financial institutions can become exposed to model risk.	6
57.b	describe the role of the model risk management function and explain best practices in the model risk management and validation processes.	5
57.c	describe lessons learned from the three case studies involving model risk.	6

LO 57.a

Define a model and describe different ways that financial institutions can become exposed to model risk.

Q354

AP Mock 3 Q71

REBUILT — Original options were vague and C ignored that models may use qualitative inputs; rebuilt on the SR 11-7 definition quoted in the reading with clear, single-answer options.

Under the U.S. Federal Reserve guidance SR 11-7, which of the following best defines a model?

A. Any forecasting tool that relies on complex statistical techniques

B. A quantitative method, system, or approach that applies statistical, economic, financial, or mathematical theories, techniques, and assumptions to process input data into quantitative estimates

C. Any estimation approach whose output is a qualitative judgment from experts

D. A computer program that reports data without applying assumptions or theory

Answer: B

Step by step

1. SR 11-7 defines a model as a quantitative method, system, or approach that applies statistical, economic, financial, or mathematical theories, techniques, and assumptions to process input data into quantitative estimates.
2. Test each option: the output must be quantitative, theory/assumptions must be applied, and inputs may be quantitative or qualitative.
3. Only B matches this wording.

Why B is correct: The definition is broad: it requires theory and assumptions applied to input data and a quantitative output, but not statistical complexity and not purely quantitative inputs.

Why the others are wrong

- **A:** Reflects the old, narrow view of models as complex statistical tools; SR 11-7 deliberately broadened the definition and complexity is not required (a simple spreadsheet can be a model).
- **C:** The output must be quantitative; qualitative or expert inputs are allowed, but a purely qualitative output is not a model under SR 11-7.
- **D:** A tool that only reports data applies no theory or assumptions and produces no estimate, so it is not a model.

Q355

AnalystPrep Q.4444

REBUILT — Original options overlapped ('Both B and C') and the model/non-model distinction was vague; rebuilt around the consensus definition (data + assumptions -> uncertain estimate).

The current industry consensus defines a model as any estimation method that uses data and relies on assumptions to generate an uncertain estimate or forecast. Which of the following spreadsheets is most clearly a model?

A. A spreadsheet that sums the trading positions reported by each desk into a firmwide total for reporting

B. A spreadsheet that applies assumed volatilities and correlations to portfolio positions to estimate potential losses under different what-if scenarios each day

C. A spreadsheet that stores historical prices for later retrieval without any calculation

D. A spreadsheet that converts reporting currencies at the published spot rate

Answer: B

Step by step

1. A model needs (1) input data, (2) assumptions/theory, and (3) an uncertain estimate or forecast as output.
2. B uses assumed volatilities and correlations (assumptions) on position data to estimate potential losses (an

uncertain estimate): all three elements are present.
3. A, C and D are aggregation, storage, or deterministic conversion with no assumptions and no uncertain estimate.

Why B is correct: Only B applies assumptions to data to produce an uncertain forecast, which is the defining feature of a model.

Why the others are wrong

- **A:** Simple aggregation of reported figures involves no assumptions and produces no uncertain estimate.
- **C:** Data storage performs no estimation; confusing a spreadsheet with a model.
- **D:** A deterministic conversion at a published rate is a calculation, not an estimate that depends on assumptions.

Q356

AnalystPrep Q.5149

REBUILT — Original distractors were obvious giveaways unrelated to model risk; rebuilt around the two model-risk types (execution risk and conceptual errors).

A newly hired risk manager is asked to explain how a financial institution can become exposed to model risk. Which of the following is an example of exposure arising from a conceptual error rather than an execution error?

A. A coding mistake in the model's pricing routine
B. Use of the wrong input data file when the model is run
C. Relying on a correlation assumption estimated in normal markets that no longer holds when markets become volatile
D. An implementation error when the model is moved to a new system

Answer: C

Step by step

1. Execution risk: the model does not do what it was designed to do (coding errors, implementation errors, wrong data).
2. Conceptual errors: wrong assumptions or modeling techniques are used, so the model may work as coded but not represent reality.
3. A, B and D are coding, data and implementation problems (execution). C is an invalid assumption in changed conditions (conceptual).

Why C is correct: A correlation assumption that fails under stress is an invalid assumption, which is a conceptual error; the others are failures of the model to function as designed.

Why the others are wrong

- **A:** A coding mistake is execution risk, not a conceptual error.
- **B:** Wrong data is an execution error (input data problem).
- **D:** Implementation errors are execution risk; the underlying concept may be sound.

Q357

Schweser Module Quiz 57.1 Q1

Which one of the following items is least likely associated with a model?

A. Qualitative inputs.
B. Mathematical theories.
C. Precise output.
D. Assumptions.

Answer: C

Step by step

1. A model processes inputs using theories and assumptions to produce an estimate.
2. Inputs may be quantitative or qualitative (A), models apply mathematical/statistical theories (B) and rely on assumptions (D).
3. Model output is an uncertain estimate subject to estimation error, so it is not precise (C).

Why C is correct: Model output is an estimate or forecast that is subject to error, never definitive or precise.

Why the others are wrong

- **A:** Qualitative or expert-judgment inputs are permitted under the model definition.
- **B:** Models apply statistical, economic, financial, or mathematical theories.
- **D:** Every model rests on a set of assumptions, which are a central source of conceptual error.

Q358 BT notes P2.T7.605.3

REBUILT — Original relied on out-of-scope items (Tinkerbell phenomenon, third-party VaR comparisons) with a generic correct option; rebuilt on the reading's statements about the nature of model risk.

Which of the following statements about model risk is most accurate?

A. Model risk is inherent to modeling and is often underestimated because its losses are rare and concentrated in the tail

B. Conceptual errors are easier to identify than execution errors because they are always a matter of right or wrong

C. Execution errors are trivial and cannot by themselves contribute to material losses

D. A model that performs well in normal conditions will perform adequately in all economic environments

Answer: A

Step by step

1. Model risk is inherent to modeling; most errors are benign, so tail losses are rare and institutions gain misplaced confidence.
2. Conceptual errors are harder to identify than execution errors because reasonable modelers can disagree on assumptions.
3. Seemingly trivial execution errors, combined with bad luck, have caused catastrophic losses (Barclays/Lehman, NASA Mars orbiter).
4. A model valid in one environment can fail in another (e.g. Gaussian copula when correlations spiked). Only A is accurate.

Why A is correct: Model risk cannot be eliminated and is underestimated precisely because its costs show up rarely, in the tails.

Why the others are wrong

- **B:** Reverses the relationship: conceptual errors are harder to identify, since assumptions are often not simply right or wrong.
- **C:** Trivial-seeming execution errors have produced catastrophic losses in case studies.
- **D:** Models can be valid in one environment and fail in another; assumptions such as correlations change under stress.

Q359 Schweser Module Quiz 57.1 Q2

Which of the following statements describes model execution risk?

A. Inaccurate model inputs.

B. Model coding that is consistent with model assumptions.

C. Inappropriate model assumptions.

D. Incorrect modeling techniques.

Answer: A

Step by step

1. Execution risk is the risk of a model not functioning as intended because of errors in input data or in coding.
2. Inaccurate inputs fit this definition.
3. Assumption and technique problems are conceptual errors.

Why A is correct: Execution risk stems from input-data and coding errors, so inaccurate inputs qualify.

Why the others are wrong

- **B:** Coding consistent with assumptions is correct implementation, not an error.
- **C:** Inappropriate assumptions are a conceptual error.
- **D:** Incorrect techniques are a conceptual error.

LO 57.b

Describe the role of the model risk management function and explain best practices in the model risk management and validation processes.

Q360 AnalystPrep Q.4306

REBUILT — Original asked about model risk reporting infrastructure, which is not in the GARP chapter; rebuilt on the GARP point that continuous MRM is a major challenge, with the same LO.

According to the chapter on model risk and model validation, which of the following is described as one of the biggest challenges many institutions face in model risk management?

A. Moving from periodic, point-in-time validations to a continuous model risk management approach
B. Persuading regulators to accept tiering of models by risk
C. Reviewing every model with the same depth and frequency regardless of tier
D. Locating validators who are independent of model development

Answer: A

Step by step

1. GARP states MRM should be a continuous effort rather than a point-in-time exercise of validations and reviews.
2. It notes that this is still one of the biggest challenges faced by many institutions.
3. Periodic validation is predictable and allows smaller validator teams, so institutions are slow to move away from it, even though growing reliance on models and dynamic environments require continuous management.

Why A is correct: The chapter explicitly identifies the shift from point-in-time to continuous MRM as one of the biggest remaining challenges.

Why the others are wrong

- **B:** Tiering is a standard cost-balancing practice; the chapter does not cite regulator acceptance as a challenge.
- **C:** Uniform depth is the opposite of best practice; tiering exists so that higher-risk models get more scrutiny.
- **D:** MRM is by definition staffed with experts independent of model development; independence is a design feature, not the cited challenge.

Q361 AnalystPrep Q.4446

REBUILT — Original options (cost savings, incentives, all of the above) were not supported by GARP; rebuilt around why MRM is independent of model development.

A bank staffs its model risk management (MRM) function with experts who are independent of model development. Which of the following best describes the purpose of this independence?

A. It lets the first line transfer ownership of model risk to the second line
B. It provides an independent backstop that can catch errors the first line may have missed
C. It removes the need for first-line quality assurance and quality control teams
D. It guarantees that conceptual errors in all models are eliminated

Answer: B

Step by step

1. MRM sits in the second line and is staffed by experts independent of model developers.
2. Its role is to review and challenge models as an independent backstop to catch errors the first line might have missed.
3. It does not relieve the first line (developers and owners), who own the risk and must still mitigate it, nor does it guarantee error-free models.

Why B is correct: Independence gives the second line an objective view, so it can act as a backstop for errors the first line missed.

Why the others are wrong

- **A:** GARP says the existence of MRM must not result in the first line abdicating its responsibilities; developers and owners own the risk.
- **C:** Mature institutions form first-line QA/QC teams, which play a critical role in mitigating execution risk; MRM does not replace them.
- **D:** MRM can reduce and challenge but cannot guarantee elimination of conceptual errors; it mainly improves transparency of assumptions.

Q362 AnalystPrep Q.5150

A junior analyst at a bank wishes to understand more about the role of the model risk management function and best practices in model risk management. What is the role of the model risk management (MRM) function in financial institutions, and how is the frequency of model validation determined?

A. MRM function validates models every year, regardless of their tier, to minimize risks.

B. MRM function specifies the frequency of model validation, but the tier of the model is not taken into consideration.
C. MRM function is responsible for reviewing and challenging models to minimize risks, and models are assigned to different tiers based on their risk level.
D. MRM function monitors the performance of models through reports produced by model owners, but they do not conduct validations.

Answer: C
Step by step
1. MRM reviews and challenges models to minimize model risk.
2. To balance cost and burden, models are tiered by risk (materiality, complexity, client-facing use, regulatory use).
3. The tier dictates thoroughness and frequency of validation; top tiers get comprehensive back-testing, typically every 2-3 years.

Why C is correct: Tiering by risk drives how thoroughly and how often a model is validated, and MRM's core job is to review and challenge models.
Why the others are wrong
- **A:** All models get an annual review, but full validation frequency varies by tier (higher tiers more often/in more detail).
- **B:** Tier is precisely what dictates validation thoroughness and frequency.
- **D:** MRM does review monitoring reports from model owners, but it also performs validations.

Q363 AnalystPrep Q.5151

What is the role of the first line of defense in the three lines of defense model in model risk management, and how do first-line QA/QC teams help mitigate model risk?
A. The first line of defense is responsible for independently assessing the risk and risk management practices of the second line, while the first-line QA/QC teams monitor the performance of models.
B. The first line of defense abdicates its own responsibilities to the second line, while the first-line QA/QC teams ensure models are validated at the appropriate frequency.
C. The first line of defense generates the risk to which the organization is exposed and owns the risk. The first-line QA/QC teams play a pivotal role in mitigating model risk, especially execution risk.
D. The first line of defense is responsible for validating models, while the first-line QA/QC teams conduct comprehensive backtesting.

Answer: C
Step by step
1. Model developers and owners are the first line: they generate and own the model risk and must mitigate it.
2. The second line (MRM) independently assesses the risk and the first line's risk practices.
3. At mature institutions modeling teams form QA/QC teams, the first line's risk team, which play a critical role mitigating model risk, particularly execution risk.

Why C is correct: The first line owns the risk it generates, and its QA/QC teams mainly address execution risk.
Why the others are wrong
- **A:** Reverses roles: independent assessment of the first line is done by the second line (MRM).
- **B:** The first line must not abdicate responsibility to the second line; validation frequency is an MRM matter.
- **D:** Independent validation and comprehensive back-testing of top-tier models are second-line (MRM) activities.

Q364 AnalystPrep Q.5297

What is the primary role of model risk management in dealing with conceptual errors?
A. To identify and correct all conceptual errors.
B. To ensure transparency and clear communication of assumptions to users.
C. To make models always valid in any context.
D. To prevent market conditions from deteriorating.

Answer: B
Step by step
1. Conceptual errors arise from invalid assumptions or inappropriate techniques and are hard to detect.

2. A model valid in one environment may fail in another, so this cannot be eliminated.
3. MRM's role is to shed light on the black box: challenge assumptions and communicate assumptions and limitations clearly to users.

Why B is correct: MRM improves transparency by challenging and clearly communicating the model's assumptions and limitations.

Why the others are wrong

- **A:** 'All' is unrealistic; conceptual errors are hard to identify and cannot all be eliminated.
- **C:** No model is valid in every environment; limitations must be communicated instead.
- **D:** Market conditions are outside MRM's control.

LO 57.c

Describe lessons learned from the three case studies involving model risk.

Q365

AnalystPrep Q.5153

REBUILT — Original keyed C ('reliance solely on quantitative models') which is not the GARP/Schweser lesson; rebuilt around the actual lesson (stale correlation assumption, unchallenged limitations).

A risk manager reviews the Gaussian copula and CDO pricing case study. The model used CDS prices to infer asset correlations in the collateral pool. As housing prices fell in 2008, CDS prices rose sharply, but the model incorporated the changing correlations only with a lag, and its use had become entrenched among market participants. Which lesson from the case is MOST appropriate?

A. Model risk arises mainly from coding errors, so independent replication of the model code is sufficient.
B. Model assumptions and limitations must be challenged and clearly communicated to users, because the model's correlation assumption stopped reflecting reality in stressed markets.
C. Model inputs must be checked for consistency of units of measurement across teams.
D. Simple tools such as spreadsheets are not models and therefore do not need to be reviewed.

Answer: B

Key correction: Source key (AnalystPrep) said C, 'relying solely on quantitative models'; the original options were not tied to the case, and the GARP lesson is the unchallenged assumption/limitations, so the item was rebuilt.

Step by step

1. The copula model inferred correlations from CDS prices and had been adopted widely without regard to its limitations.
2. Historical mortgage-default correlations were low, but in 2008 CDS prices (and implied correlations) shot up while the model updated only with a lag.
3. This is a conceptual (assumption) error, not an execution error.
4. MRM's role: increase transparency, challenge assumptions, and ensure users (often non-quantitative) understand the limitations to prevent misuse.

Why B is correct: The failure came from an invalid, unchallenged correlation assumption and users who did not understand model limitations, so MRM must challenge assumptions and communicate limitations.

Why the others are wrong

- **A:** Replication of code is necessary but the failure was conceptual, not a coding error; this confuses execution risk with conceptual risk.
- **C:** This is the NASA Mars Orbiter lesson (units mismatch), not the copula case.
- **D:** This misreads the Barclays/Lehman lesson, which is that even simple tools must be reviewed.

Q366

AnalystPrep Q.5154

A bank hires an expert to review a newly developed model as part of good model risk management practice. Which case study illustrates losses suffered because a model was widely adopted and entrenched without assessing its assumptions and limitations?

A. Gaussian copula and CDO pricing
B. Barclays' acquisition of Lehman assets and the spreadsheet error
C. NASA Mars Orbiter
D. Lehman Brothers fraud

Answer: A

Step by step

1. Gaussian copula: model adopted by major participants and entrenched without regard to limitations (correlation inferred from CDS prices, updated only with a lag).
2. Barclays: implementation error (hidden rows exposed in PDF).
3. NASA: incorrect input units.
4. Lehman fraud is not one of the three model risk case studies.

Why A is correct: Only the copula case involved a model whose conceptual assumptions were never adequately assessed before entrenched use.

Why the others are wrong

- **B:** This was an implementation error with a spreadsheet, not a model adopted without assessment.
- **C:** This was an input (units) error, not an untested modeling approach.
- **D:** Not one of the three case studies; the Lehman link in the Barclays case is the spreadsheet error, not fraud.

Q367

AnalystPrep Q.5155

REBUILT — Original had overlapping options (A and D are copula lessons, C is the Mars lesson, B vague); rebuilt so exactly one option is the Barclays lesson.

Barclays Capital nearly bought 179 unwanted Lehman Brothers contracts because rows hidden in a spreadsheet became visible when a junior lawyer converted it to a PDF. Which lesson is MOST directly supported by this incident?

A. Even seemingly simple tools should be reviewed and challenged, because implementation errors can create large potential losses.

B. Users must be told that CDS-implied correlations lag market conditions.

C. Inputs measured in different unit systems (English vs. metric) must be reconciled.

D. A model's correlation assumption must be updated when housing prices fall.

Answer: A

Step by step

1. Barclays sent a 1,000-row spreadsheet; 179 unwanted positions were in hidden rows.
2. Conversion to PDF revealed them; the bid was approved and Barclays had to file a motion to exclude the contracts.
3. Classified as an implementation error; the spreadsheet is not a model but a tool that posed large potential loss.
4. Lesson: even simple tools need review and challenge.

Why A is correct: The error was an implementation error in a simple tool, so the lesson is that even simple tools need review.

Why the others are wrong

- **B:** Copula lesson, not Barclays.
- **C:** NASA Mars Orbiter lesson.
- **D:** Copula lesson about the stale correlation assumption.

Q368

Schweser Module Quiz 57.1 Q3

What is the most likely reason for the failure of the Gaussian copula function to price CDOs?

A. Model computation error.

B. Inappropriate model for the task.

C. Invalid model assumption.

D. Invalid use of model output.

Answer: C

Step by step

1. The model used CDS prices to infer a single (constant) correlation among pool assets.
2. In 2008 correlations rose sharply and the model adjusted only with a lag.
3. The failure is therefore a conceptual error: an invalid assumption.

Why C is correct: The static correlation assumption no longer represented reality.

Why the others are wrong

- **A:** Code worked as intended; the flaw was conceptual, not computational.
- **B:** Copulas were a legitimate technique; the problem was the assumption, not the choice of approach.

- **D:** Output was used for its intended purpose (pricing); the assumption failed.

Q369

Schweser Module Quiz 57.1 Q4

The Barclays bid for Lehman assets in bankruptcy court most likely suffered from:

A. ongoing monitoring.
B. improper model use.
C. invalid model assumptions.
D. implementation error.

Answer: D

Step by step

1. Barclays marked 179 unwanted positions in hidden spreadsheet rows.
2. The rows were not deleted, so PDF conversion exposed them.
3. This is an implementation error.

Why D is correct: Failure to delete rows is a mistake in implementation of a tool, not a conceptual one.

Why the others are wrong

- **A:** Monitoring is an MRM activity, not a failure type; the problem was not ongoing monitoring.
- **B:** The spreadsheet was not a model misapplied to the wrong problem.
- **C:** No modeling assumptions were involved.

Q370

Schweser Module Quiz 57.1 Q5

REBUILT — Original options B (model assumptions) and D (incorrect model inputs) were both defensible because GARP calls the unit choice an error in the model's 'assumptions' while Schweser calls it an input error; rebuilt so the key describes the unit mismatch itself (a simple, taken-for-granted unit setting on the inputs, an execution-type error) and the distractors are clearly different error types.

NASA lost a $125 million Mars orbiter because a Lockheed Martin engineering team used English units of measurement while NASA's team used the metric system for a key spacecraft operation. Which description of the failure is MOST consistent with the model risk case study?

A. A conceptual error: the modeling approach rested on a debatable assumption on which reasonable modelers could hold different, equally valid views.
B. A simple, taken-for-granted setting, the unit of measurement (English vs. metric) applied to the model's inputs, was never reconciled between teams; a seemingly trivial execution-type mistake produced a catastrophic loss.
C. The model's output was used for a purpose other than the one it was designed for.
D. A model that had become entrenched among users without regard to its limitations stopped reflecting reality in stressed conditions.

Answer: B

Key correction: Source (Schweser Module Quiz 57.1 Q5) keyed D 'incorrect model inputs' with B 'model assumptions' as a distractor; GARP describes the unit choice as an error in the 'assumptions' of the model, so both original options fit and the item was rebuilt with B as the single defensible key.

Step by step

1. Facts: Lockheed Martin used English units; NASA used metric; the mismatch destroyed a $125 million orbiter.
2. GARP calls this an error in the 'assumptions' of the model, namely the choice of unit (metric vs imperial); the financial parallel is using the wrong currency or discount factor, simple assumptions often taken for granted. Schweser describes the same thing as a mistake in the measurement of model inputs.
3. GARP groups the Barclays and NASA cases as prime examples of execution errors: trivial-seeming mistakes that, with bad luck, produce catastrophic losses.
4. It is not a conceptual error (no debatable modeling judgment), not misuse of output, and not the entrenched-model problem of the Gaussian copula case.
5. Lesson: MRM should rigorously test all assumptions and inputs, because it is impossible to know in advance which small mistakes will be catastrophic.

Why B is correct: The orbiter was lost because a simple unit setting on the inputs, taken for granted, was never checked between teams. GARP calls this an 'assumption' (the unit choice) and Schweser calls it an input-measurement error, and B covers both views. It is a trivial execution-type mistake with a catastrophic outcome.

Why the others are wrong

- **A:** GARP separates conceptual errors, where reasonable modelers can disagree, from execution errors. Using inconsistent units is simply wrong, so nobody could defend it as a reasonable view.

- **C:** Nothing suggests the output was used for an unintended purpose. The calculation itself ran on inconsistent units.
- **D:** This is the Gaussian copula and CDO lesson (an entrenched model with an unchallenged correlation assumption), not the Mars orbiter case.

Reading 58. Stress Testing Banks

Til Schuermann, "Stress Testing Banks," International Journal of Forecasting 30, no. 3 (2014): 717– 728.

LO	Learning objective	Qs
58.a	describe the evolution of the stress testing process and compare the methodologies of historical European Banking Association (EBA), Comprehensive Capital Analysis and Review (CCAR), and Supervisory Capital Assessment Program (SCAP) stress tests.	9
58.b	explain challenges in designing stress test scenarios, including the problem of coherence in modeling risk factors.	4
58.c	explain challenges in modeling a bank's revenues, losses, and its balance sheet over a stress test horizon period.	5

LO 58.a

Describe the evolution of the stress testing process and compare the methodologies of historical European Banking Association (EBA), Comprehensive Capital Analysis and Review (CCAR), and Supervisory Capital Assessment Program (SCAP) stress tests.

Q371

AP Mock 2 Q57

The 2009 Supervisory Capital Assessment Program (SCAP) was an important stress test. Which of the following was least likely a risk factor taken into consideration during SCAP?

A. GDP growth.
B. Unemployment.
C. House price index.
D. Inflation.

Answer: D

Step by step

1. The SCAP scenario state space had only three dimensions: GDP growth, unemployment and the house price index (HPI).
2. Inflation was not one of the three macro variables, so it is the factor least likely considered.

Why D is correct: SCAP used a simple tri-variate macro scenario (GDP growth, unemployment, HPI); inflation was not part of it (inflation appeared later as one of eight macro factors in the 2011 EBA test).

Why the others are wrong

- **A:** GDP growth was one of the three SCAP variables.
- **B:** Unemployment was one of the three SCAP variables.
- **C:** The house price index was one of the three SCAP variables.

Q372

AnalystPrep Q.2307

Stress testing is a simulation technique used in banking to determine the ability of a financial institution to deal with an economic crisis. Which of the following is NOT a stress test exercise conducted by an authoritative body?

A. SCAP 2009 - Supervisory Capital Assessment Program
B. CEBS 2010 - Committee of European Bank Supervisors
C. TCAP 2009 - Treasury's Capital Assistance Program
D. CCAR 2011/2012 - Comprehensive Capital Analysis and Review

Answer: C

Step by step

1. SCAP (2009), the CEBS test (2010) and CCAR (2011/2012) are all macro-prudential stress tests.
2. The Treasury's Capital Assistance Program (CAP) was a funding backstop: the Treasury would borrow and inject equity into banks that failed SCAP and could not raise capital privately. It is not a stress test.

Why C is correct: The CAP was a mechanism to supply equity capital to undercapitalized banks, not a stress test; in the end SCAP banks raised capital privately and did not draw on it.

Why the others are wrong

- **A:** SCAP was the first macro-prudential stress test after the 2007-2009 crisis.
- **B:** The 2010 CEBS test covered 91 European banks.

- **D:** CCAR is the Federal Reserve's annual stress test, run since 2011.

Q373 BT answers P2.T7.24.14.1

REBUILT — Statement 2 contained an unsupported claim about operational risk that made it ambiguous; rewrote it as a clean reversal of the pre-/post-SCAP capital-ratio link.

At a risk committee meeting, a colleague presented the evolution of stress testing since the Supervisory Capital Assessment Program (SCAP):
Statement 1: Initially, the focus was mainly on potential losses within specific business units; now the analysis includes losses, revenues and costs across the entire bank.
Statement 2: Originally, stress test results were explicitly tied to a post-stress capital ratio; now they are usually not tied to capital adequacy.
Which statement is correct?

A. Statement 1 only
B. Statement 2 only
C. Neither statement
D. Both statements

Answer: A

Step by step

1. Pre-SCAP: product or business-unit level, losses only. Post-SCAP: comprehensive and firm-wide, covering losses, revenues and costs. Statement 1 is correct.
2. Pre-SCAP stress tests were not usually tied to capital adequacy; post-SCAP they are tied to an explicit post-stress common equity threshold. Statement 2 reverses this and is incorrect.

Why A is correct: SCAP moved stress testing to a firm-wide view of losses, revenues and costs and linked results to a post-stress capital ratio; Statement 2 states the opposite.

Why the others are wrong

- **B:** Statement 2 is the reverse of the truth, and Statement 1 is correct.
- **C:** Statement 1 correctly describes the shift to firm-wide analysis.
- **D:** Statement 2 is false because post-SCAP tests are explicitly tied to capital ratios.

Q374 BT answers P2.T7.810.1 · BT notes P2.T7.810.1

In regard to the historical evolution of the stress testing process, each of the following is true EXCEPT, which is inaccurate?

A. The U.S. Federal Reserve conducts an annual assessment of bank holding companies (BHC) which includes a Comprehensive Capital Analysis and Review (CCAR) and Dodd-Frank supervisory stress tests
B. The Committee of European Banking Supervisors (CEBS) oversees the most forward-looking and comprehensive regulatory stress tests, in particular with respect to liquidity risks, emerging cyber-risks, and the linkage between macro and intermediate risk factors
C. The Supervisory Capital Assessment Program (SCAP) was the first of the macro-prudential stress tests of the global financial crisis (GFC); its state space had only three dimensions (GDP growth, unemployment, and house price index), and the market risk scenario was based in historical experience
D. For the 2011 European Banking Authority (EBA) test, the supervisors specified over 70 risk factors for the trading book, eight macro-factors for each of 21 countries (GDP growth, inflation, unemployment, residential and commercial real estate price indices, short and long-term government rates, and stock prices), plus sovereign haircuts across seven maturity buckets

Answer: B

Step by step

1. A: the Fed's annual BHC assessment includes CCAR and Dodd-Frank tests - true.
2. C: SCAP was tri-variate (GDP, unemployment, HPI) with a historically based market risk scenario - true.
3. D: the 2011 EBA scenario had over 70 trading-book risk factors, 8 macro factors for 21 countries and sovereign haircuts over 7 maturity buckets - true.
4. B: the CEBS ran the 2010 test of 91 banks (covering market and credit risk only, with less disclosure than SCAP) and was succeeded by the EBA; it did not oversee the most forward-looking and comprehensive tests. B is inaccurate.

Why B is correct: The CEBS 2010 test had relatively limited scope and disclosure and was succeeded by the EBA,

so the claim that it oversees the most forward-looking and comprehensive tests is false.

Why the others are wrong

- **A:** True: the Fed's annual assessment includes CCAR and Dodd-Frank stress tests.
- **C:** True: SCAP's scenario had just three macro dimensions.
- **D:** True: this is the 2011 EBA scenario specification.

Q375

Mock 2 Q74

REBUILT — Option C ('earnings shocks instead of revenues and/or costs') was loosely worded and arguably also incorrect; reworded to 'losses only' per the pre-/post-SCAP comparison.

In 2009, the Supervisory Capital Assessment Program (SCAP) allowed U.S. regulators to assess the capital strength of financial institutions. When comparing stress tests before and after SCAP, which of the following statements is incorrect?

A. Post-SCAP stress tests take a comprehensive, firm-wide look at the effect of the stress scenarios on the institution.

B. Pre-SCAP stress tests were explicitly tied to capital adequacy.

C. Pre-SCAP stress tests focused on losses only, rather than losses, revenues and costs.

D. Post-SCAP stress tests use broad macro scenarios with market-wide, multi-factor stresses.

Answer: B

Step by step

1. Pre-SCAP: mostly single shocks, product/business-unit level, static, not usually tied to capital adequacy, losses only.
2. Post-SCAP: broad macro scenario, comprehensive and firm-wide, dynamic, explicit post-stress common equity threshold, losses, revenues and costs.
3. A, C and D match this; B contradicts it, so B is incorrect.

Why B is correct: Tying results to an explicit post-stress capital threshold is a post-SCAP feature; pre-SCAP tests were not usually linked to capital adequacy.

Why the others are wrong

- **A:** Correct statement of the post-SCAP firm-wide approach.
- **C:** Correct: pre-SCAP tests looked at losses only.
- **D:** Correct: post-SCAP used broad macro, market-wide scenarios.

Q376

GARP PE2 Q54

REBUILT — Original correct option relied on Spanish bank details found only in GARP, outside the Schweser LO; rebuilt around disclosure comparison across SCAP, CEBS, CCAR and EBA, with distractors fixed so exactly one is correct.

A risk consultant presents the evolution of macro-prudential stress testing after the 2007-2009 crisis, comparing the 2009 Supervisory Capital Assessment Program (SCAP), the 2010 CEBS test, the 2011 EBA test and the 2011/2012 Comprehensive Capital Analysis and Review (CCAR). Which statement is correct?

A. The SCAP scenario was less severe than later U.S. tests and did not result in any capital raising by participating banks.

B. The 2011 CCAR disclosed bank-level loss rates, whereas the 2012 CCAR published only the macro scenario.

C. The 2011 EBA test made bank-level results available in downloadable form, including exposures by asset class and geography, which was more detailed than the 2010 CEBS test.

D. SCAP disclosed only overall retail and corporate loss rates, while the EBA test expanded disclosure to asset classes and regions.

Answer: C

Step by step

1. EBA 2011: bank-level results in electronic spreadsheet form, exposures by asset class and geography - considerably more detailed than the CEBS 2010 test, which gave loss rates by firm only for overall retail and corporate. C is correct.
2. A: ten of the 19 SCAP banks had to raise $75 billion within six months - false.
3. B: reversed; the 2011 CCAR published only macro results, the 2012 CCAR disclosed bank-level detail similar to SCAP.
4. D: SCAP released loss rates by major asset class; the 'retail and corporate only' description fits the 2010 CEBS test.

Why C is correct: The 2011 EBA test, following the Irish test, gave detailed bank-by-bank, asset-class and geography data, a step beyond the limited CEBS 2010 disclosure.

Why the others are wrong

- **A:** Wrong: ten of 19 SCAP banks had to raise $75 billion (they raised $77 billion).
- **B:** Wrong: the years are reversed; 2011 CCAR had no bank-level disclosure.
- **D:** Wrong: confuses SCAP with the CEBS test; SCAP disclosed loss rates by major asset class.

Q377

Schweser Module Quiz 58.1 Q1

Which of the following changes in stress testing was not the result of the 2009 Supervisory Capital Assessment Program (SCAP)?

A. Banks are now required to provide the results of their own scenario stress tests.
B. Stress scenarios are now broader in nature.
C. Stress testing now focuses on the whole firm.
D. Stress testing now focuses on revenues, costs, and projected losses.

Answer: A

Step by step

1. SCAP changes: broad macro scenario, firm-wide focus, losses plus revenues and costs, explicit post-stress capital ratio (B, C, D).
2. Requiring banks to submit their own scenario results came later, with the 2011 and 2012 CCAR (SCAP imposed a common supervisory scenario). A is not a SCAP result.

Why A is correct: Banks' own scenario results were introduced by CCAR in 2011, not by SCAP.

Why the others are wrong

- **B:** A genuine SCAP change (broad macro scenario).
- **C:** A genuine SCAP change (firm-wide testing).
- **D:** A genuine SCAP change (losses, revenues and costs).

Q378

Schweser Module Quiz 58.2 Q4

One of the key differences between the 2011 CCAR stress test and the 2011 EBA Irish stress test is that:

A. CCAR did not require banks to provide results from their own stress scenarios.
B. the EBA Irish test did not find any banks in violation of capital adequacy requirements.
C. CCAR required disclosure of macro-level, not bank-level, scenario results.
D. the EBA Irish test allowed for 1-year stress horizons.

Answer: C

Step by step

1. 2011 CCAR: only macro-scenario results were published; the market had to 'do the math' itself.
2. 2011 Irish test: considerable detail, including comparison of bank and third-party loss estimates.
3. So C is the key difference; A, B and D are each false.

Why C is correct: The 2011 CCAR published only macro results, whereas the Irish and EBA tests disclosed extensive bank-level detail.

Why the others are wrong

- **A:** False: CCAR in 2011 and 2012 did require banks to submit their own baseline and stress results.
- **B:** False: the Irish test, run by outside advisors, revealed a capital need of about EUR 24 billion.
- **D:** False: stress horizons in these exercises were typically about two years (SCAP used a two-year scenario); a 1-year horizon is not a feature of the Irish test, and the Irish test's distinguishing balance-sheet feature was allowing for shrinkage.

Q379

AnalystPrep Q.1993

REBUILT — Option C was vague enough to be partly defensible and the VaR-style distractor was missing; reworded C to describe VaR so exactly one option fits the Schuermann definition of stress testing.

Which of the following best describes stress testing as applied by banks and supervisors after the 2007-2009 financial crisis?

A. A simulation of how an institution's results hold up under deteriorating conditions, using adverse scenarios

(e.g., a severe recession) or sensitivities (e.g., a sharp rise in interest rates)

B. A comparison of predicted losses with actual observed losses to judge the reliability of a bank's prediction models

C. A statistical estimate of the maximum loss over a given horizon at a given confidence level under normal market conditions

D. The administering of hypothetical credit assessments to a counterparty to estimate its probability of default

Answer: A

Step by step

1. Stress testing asks: what is the institution's resilience to deteriorating conditions?
2. It simulates financial results under adverse conditions, which take the form of scenarios (e.g., a severe recession) or sensitivities (e.g., a significant rise in interest rates).
3. After 2007-2009 supervisors used it to judge whether banks had enough capital and liquidity to support their risk taking.
4. Only option A matches this description.

Why A is correct: Stress testing simulates results under adverse scenarios or sensitivities to assess whether capital and liquidity are adequate; this is what option A states.

Why the others are wrong

- **B:** This describes backtesting, which compares predicted with realized outcomes to validate a model, not a forward-looking adverse simulation.
- **C:** This describes value at risk, which is calibrated to normal market conditions and a confidence level, whereas stress tests look at adverse conditions.
- **D:** This confuses stress testing with counterparty credit assessment; stress tests are applied to the institution's own portfolio, balance sheet and results.

LO 58.b

Explain challenges in designing stress test scenarios, including the problem of coherence in modeling risk factors.

Q380

BT answers P2.T7.24.14.2

REBUILT — Source key (statistical volatility/correlation model) contradicts the reading, which says coherent outcomes in a high-dimensional market-risk space are hard to find short of historical realizations; rebuilt so the historical-scenario approach is the single correct answer.

A bank's trading book consists of hundreds of thousands of marked-to-market positions mapped to tens of thousands of risk factors in its VaR system. A risk team must design a market-risk stress scenario whose risk-factor moves are jointly coherent. According to Schuermann, which approach is most practical for achieving coherence in this setting?

A. Specify an extreme move in the single most volatile risk factor and hold all other risk factors constant.

B. Set every risk factor to its own worst-case value independently, so that the scenario is as severe as possible.

C. Use historical realizations of risk-factor changes, such as the market moves from June 30 to December 31, 2008 used in the 2009 SCAP.

D. Estimate volatilities and correlations from VaR data and shock each risk factor to a fixed extreme quantile of its own distribution.

Answer: C

Key correction: Source key (BT answers) said C (an advanced statistical volatility/correlation model); the reading says coherent outcomes in this high-dimensional space are hard to find short of historical realizations, so the question was rebuilt around the historical-scenario option.

Step by step

1. The scenario must be multi-factor: when one risk factor moves, others do not stay fixed, so the joint outcome must be plausible.
2. With tens of thousands of risk factors, finding coherent joint outcomes by judgment or by model is 'daunting'.
3. Schuermann notes the practical route is historical realizations: the observed moves occurred together, so they are coherent by construction.
4. The 2009 SCAP market-risk scenario did exactly this (June 30 to December 31, 2008: flight to quality, Lehman failure), achieving a loose coherence; the cost is that it tests nothing new.

Why C is correct: Historical realizations embed the joint behaviour of all risk factors that actually occurred, so

they are coherent without having to specify tens of thousands of interdependent moves.

Why the others are wrong

- **A:** Holding other factors constant ignores that factors move together (e.g., a volatility spike freezes credit markets); it is the one-factor-at-a-time error the reading warns against.
- **B:** Independent worst cases ignore dependencies and can be mutually impossible (e.g., all currencies cannot depreciate, no safe haven); severity is confused with coherence.
- **D:** Shocking each factor to its own quantile still treats factors one by one; correlations estimated in calm times need not hold in stress, and the reading does not offer this as the route to coherence.

Q381 BT answers P2.T7.810.2 · BT notes P2.T7.810.2

Although the problem of coherence is generic to scenario design, which of the following elements is most likely to support the goal of designing risk factors that meet the goal of COHERENCE in a stress test?

A. Use historical scenarios
B. Exclude safe havens (i.e., flight to quality)
C. Use sub-additive metrics in a higher-dimensional space
D. Depreciate all foreign exchange rates simultaneously

Answer: A

Step by step

1. Coherence means the joint values of risk factors are extreme yet possible.
2. Finding coherent outcomes in a high-dimensional space is difficult short of resorting to historical realizations.
3. Historical scenarios (e.g., the 2009 SCAP market-risk scenario) were actually observed together, so they support coherence, though they test nothing new.
4. Options B, C and D either remove features a coherent scenario needs or are unrelated to coherence.

Why A is correct: Historical realizations of risk factors occurred jointly, so a historical scenario is coherent by construction.

Why the others are wrong

- **B:** A flight to quality requires safe-haven assets; excluding them makes the scenario incoherent.
- **C:** Sub-additivity is a property of risk measures and is unrelated to making risk-factor moves jointly plausible; a higher-dimensional space makes coherence harder.
- **D:** Not all currencies can depreciate at once, since exchange rates are relative prices; this is an incoherent assumption.

Q382 GARP PE1 Q63 · Mock 3 Q18

An internal auditor at a national bank is reviewing the assumptions of the scenarios that are used in the bank's stress tests. The auditor wants to ensure that the different sets of assumptions are plausible and that they exhibit the property of coherence. Which of the following combinations of assumptions used in a stress scenario would provide the auditor the greatest cause for concern about their coherence?

A. The USD, EUR, and JPY all depreciate at the same time against a basket of world currencies.
B. Correlations between returns of commodities used for industrial production and returns of industrial sector equities increase at the same time.
C. Yields on BBB-rated corporate debt increase as yields on AAA-rated short-term sovereign debt decrease.
D. Monetary authorities respond to higher consumer prices by raising central bank lending rates.

Answer: A

Step by step

1. Coherence asks whether the joint assumptions can occur together.
2. Exchange rates are relative prices: if some currencies depreciate, others must appreciate.
3. USD, EUR and JPY are major components of a world basket, so all three cannot depreciate against it simultaneously.
4. B (correlations rising in stress), C (flight to quality) and D (standard policy response to inflation) are mutually consistent.

Why A is correct: Not everything can go bad at once; a simultaneous depreciation of the dominant currencies against a basket of currencies is internally inconsistent.

Why the others are wrong

- **B:** Correlations typically rise in stress, so this is plausible; the error is mistaking a severe assumption for an incoherent one.
- **C:** Widening credit spreads alongside falling safe-haven yields is a classic flight to quality, which is coherent.
- **D:** Raising rates in response to higher inflation is the normal monetary reaction; the reading notes high inflation needs a policy response.

Q383 Schweser Module Quiz 58.2 Q1

Piper Hook, a bank examiner, is trying to make sense of stress tests done by one of the banks she examines. The stress tests are multi-factored and complex. The bank's scenarios specify extreme values for many risk factors at once, and Hook struggles to judge whether those joint values could really occur together. One of the key stress test design challenges that Hook must deal with in her examination of stress tests is:

A. multiplicity.
B. efficiency.
C. coherence.
D. efficacy.

Answer: C

Step by step

1. The challenge described is multi-factor scenarios that must be extreme yet reasonable.
2. That challenge is named coherence: the joint values of all risk factors must be possible together.
3. Multiplicity, efficiency and efficacy are not stress-test design challenges named in the reading.

Why C is correct: Stress scenarios are inherently multi-factor, and the key design challenge is specifying joint outcomes that are extreme but coherent.

Why the others are wrong

- **A:** 'Multiplicity' is not a design challenge named in the reading; it is a distractor built from the many risk factors in the stem.
- **B:** Efficiency concerns cost or speed, not the plausibility of joint risk-factor moves.
- **D:** Efficacy is a generic term and not the named challenge; the issue is joint plausibility.

LO 58.c

Explain challenges in modeling a bank's revenues, losses, and its balance sheet over a stress test horizon period.

Q384 BT answers P2.T7.24.14.3

REBUILT — John's claim that the model 'should use' a dynamic balance sheet is not supported (the challenge exists under static and dynamic assumptions), so 'all three correct' was indefensible; John's statement was reworded to a false claim and options rebuilt.

Three analysts at Largo Bank are developing a stress test model to assess capital adequacy over a two-year horizon.
John: Balance sheet modeling is difficult only if a dynamic balance sheet is assumed; under a static balance sheet the modeler need not consider capital actions.
Greta: The denominator of the capital ratio is typically risk-weighted assets, computed using the risk weights of the prevailing regulatory capital regime (e.g., Basel II).
Dace: The modeler must consider capital actions such as acquisitions, spin-offs, dividend changes and share repurchases or issuance, because they deplete or conserve capital.
Which of the analysts' statements is/are correct?

A. All three are correct.
B. John only.
C. Greta only.
D. Greta and Dace only.

Answer: D

Key correction: Source key (BT answers) said A (all three correct); John's original statement was a prescription not supported by the reading, so it was reworded as false and the correct answer is D.

Step by step

1. Greta: capital adequacy is capital over assets; the denominator is typically RWA under the prevailing regime

(Basel I/II). Correct.
2. Dace: after each quarter the modeler must consider asset sales/originations, acquisitions, spin-offs, dividend changes, buybacks and issuance. Correct.
3. John: the balance sheet modeling problem exists under both static and dynamic assumptions. Incorrect.
4. Only Greta and Dace are correct: D.

Why D is correct: Balance sheet modeling requires RWA under the prevailing regime and judgments on capital actions, and the difficulty exists whether the balance sheet is static or dynamic.

Why the others are wrong

- **A:** Accepts John's claim; the problem exists under both static and dynamic assumptions, not only dynamic.
- **B:** Selects the one false statement and rejects the two correct ones.
- **C:** Omits Dace; capital actions are explicitly part of the balance sheet modeling challenge.

Q385 BT answers P2.T7.810.3 · BT notes P2.T7.810.3

In regard to the challenges in modeling a bank's revenues, losses, and its balance sheet over a stress test horizon period, which of the following statements is TRUE?

A. Compared to the banking book, stress testing the trading book is more difficult because it is a newer discipline

B. Implementing stress scenarios for revenues is much more developed and easier than stress testing for losses

C. The most robust stress tests assume loss severity (i.e., loss given default, LGD) rates that are invariant to geography, business cycle, or industry and uniformly represent the worst-case scenarios

D. Regardless of whether the denominator in the capital ratio is common equity or risk-weighted assets (RWA), determining post-stress capital adequacy requires modeling both the income statement and the balance sheet over the course of the stress test horizon, which is typically about two years

Answer: D

Step by step

1. Revenue modeling is far less developed than loss modeling (SCAP gave little detail on revenues), so B is false.
2. LGD varies with geography, business cycle and industry (e.g., airline collateral vs. healthcare), so uniform worst-case LGDs are not robust: C is false.
3. Stress testing of the banking book (credit risk) is the newer discipline, and implementing stress scenarios in the trading book is more straightforward; A reverses this and is false.
4. Post-stress capital adequacy needs both the income statement (flows) and the balance sheet (stocks) over a horizon typically about two years: D is true.

Why D is correct: Capital ratios after stress depend on the evolution of both the income statement and the balance sheet over the horizon, typically about two years.

Why the others are wrong

- **A:** Reverses the facts: formal stress testing of the banking book (dominated by credit risk) is the more recent discipline, and trading book implementation is more straightforward.
- **B:** Reverses the facts: revenue modeling is less developed and harder than loss modeling.
- **C:** Ignores that LGD differs by geography, industry and cycle; one-size-fits-all LGDs make mapping macro factors to bank results less meaningful.

Q386 Schweser Module Quiz 58.2 Q2

Greg Nugent, a regulator with the Office of the Comptroller of the Currency, is presenting research on stress tests to a group of regulators. He is explaining that macro-variable stress testing can be misleading for some banks because of geographical differences in macro risk factors. He gives the example of the wide range of unemployment rates across the United States following the 2007-2009 financial crisis. Which type of loan did Nugent most likely identify as having losses tied to unemployment rates?

A. Residential real estate loans.

B. Credit card loans.

C. Commercial real estate loans.

D. Industrial term loans.

Answer: B

Step by step

1. The example contrasts July 2011 unemployment of 12.9% (Nevada), 3.3% (North Dakota) and 9.1% nationally.
2. Credit card losses are particularly sensitive to unemployment, and card loss rates varied widely by region.

3. Answer: credit card loans (B).

Why B is correct: Credit card losses are especially sensitive to unemployment, so regional unemployment differences produced very different loss rates.

Why the others are wrong

- **A:** Mortgages are affected by unemployment too, but borrowers quit paying cards before mortgages; geographic effects here stem more from house prices.
- **C:** Commercial real estate losses are tied to property markets and geography, not primarily unemployment.
- **D:** Industrial loan losses depend on industry cycles and collateral, not specifically unemployment rates.

Q387

Schweser Module Quiz 58.2 Q3

A risk modeler has to make assumptions about acquisitions and spinoffs, if dividend payments will change, and if the bank will buy back stock or issue stock options to employees. These factors make it especially challenging to:

A. get a CAMELS rating of 2 or better.

B. determine if the bank has enough liquidity to meet its obligations.

C. meet the Tier 1 equity capital to risk-weighted assets ratio.

D. model a bank's balance sheet over a stress test horizon.

Answer: D

Step by step

1. The starting balance sheet generates quarter-1 income and loss, which gives the quarter-end balance sheet.
2. The modeler must then assume asset sales/originations, acquisitions, spin-offs, dividend changes, buybacks and issuance each quarter.
3. These assumptions make modeling the balance sheet over the horizon difficult: D.

Why D is correct: Capital actions change both capital and assets each quarter, so the balance sheet path over the horizon requires many judgment-based assumptions.

Why the others are wrong

- **A:** CAMELS ratings are supervisory assessments, not the modeling issue described.
- **B:** Liquidity is a constraint to maintain, but these capital actions are cited as complicating balance sheet projection, not liquidity determination.
- **C:** Meeting the ratio is the outcome tested; the stated assumptions complicate the modeling itself.

Q388

AnalystPrep Q.2308

REBUILT — Original 'which measures are important' with 'all of the above' was vague and generic; replaced with a specific question on revenue modeling challenges from the reading.

Which of the following statements about modeling a bank's revenues over a stress test horizon is most accurate?

A. Revenue modeling is more developed than loss modeling because the 2009 SCAP specified a detailed revenue methodology.

B. Total income is split into interest and noninterest income; noninterest income (fees, service charges) is especially hard to model, even though its share of bank income has been rising.

C. Interest income is easy to model because the yield curve fully determines how changing rates affect bank profitability.

D. Noninterest income can be ignored because it is a small and declining share of U.S. bank income.

Answer: B

Step by step

1. Revenue modeling is much less developed than loss modeling; SCAP gave little clarity: A false.
2. Main approach: split income into interest and noninterest. The yield curve helps estimate interest income, but the net effect of rates on profitability remains unclear: C false.
3. Noninterest income (fees, service charges) is even harder to model, and its share has risen steadily: D false, B true.

Why B is correct: Revenues are modeled as interest plus noninterest income, and noninterest income is both increasingly important and the hardest to model.

Why the others are wrong

- **A:** Reverses the facts; SCAP offered little guidance on revenue and the area is less developed than losses.
- **C:** The yield curve can estimate interest income, but how rate changes affect profitability is unclear, so it is not 'fully determined'.

- **D:** Noninterest income’s share has been steadily increasing, so ignoring it would be a serious omission.

Reading 59. Risk Capital Attribution and Risk-Adjusted Performance Measurement

Michel Crouhy, Dan Galai, and Robert Mark, The Essentials of Risk Management, 2nd Edition (New York, NY: McGraw-Hill, 2014). Chapter 17.

LO	Learning objective	Qs
59.a	define, compare, and contrast risk capital, economic capital, and regulatory capital and explain methods and motivations for using economic capital approaches to allocate risk capital.	10
59.b	describe the RAROC (risk-adjusted return on capital) methodology and its use in capital budgeting.	5
59.c	calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance.	21
59.d	explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level.	4
59.e	calculate the hurdle rate and apply this rate in making business decisions using RAROC.	4
59.f	calculate the adjusted RAROC for a project to determine its viability.	11
59.g	explain challenges in modeling diversification benefits, including aggregating a firm's risk capital and allocating economic capital to different business lines.	3
59.h	explain best practices in implementing an approach that uses RAROC to allocate economic capital.	4

LO 59.a

Define, compare, and contrast risk capital, economic capital, and regulatory capital and explain methods and motivations for using economic capital approaches to allocate risk capital.

Q389 AnalystPrep Q.2211

The difference between risk capital and regulatory capital is that:

A. Regulatory capital only applies to a few closely monitored industries like banking and insurance.

B. While risk capital depends on an institution's individual characteristics and investment choices, regulatory capital is calculated based on industry-wide rules.

C. Aggregate risk capital and regulatory capital may be equal at firm level, but different at business lines level.

D. All of the above.

Answer: D

Step by step

1. Statement A: regulatory capital applies only to regulated industries such as banking and insurance, whereas risk capital is relevant to any firm. True.
2. Statement B: risk capital reflects the firm's own risk profile; regulatory capital follows industry-wide rules and formulas. True.
3. Statement C: even if firm-wide totals coincide, the split across divisions can differ (hence allocate the greater of the two to a division). True.
4. All three are true, so D.

Why D is correct: Each statement is one of the standard contrasts between risk capital and regulatory capital, so 'all of the above' is correct.

Why the others are wrong

- **A:** True but incomplete; it omits the other two valid differences.
- **B:** True but incomplete; it omits the other two valid differences.
- **C:** True but incomplete; it omits the other two valid differences.

Q390 AnalystPrep Q.2213

An Indian bank is in the process of calculating its risk capital. The main purpose of risk capital calculation is:

A. To show the level of expected losses that the bank could absorb.

B. To find differences between regulatory and risk capital.

C. To show the level of unexpected losses the bank could absorb.

D. All of the above.

Answer: C

Step by step

1. Risk capital is a buffer against unexpected losses up to a chosen confidence level (e.g., 99.9%).
2. Expected losses are priced into spreads and covered by reserves, not by risk capital, so A is wrong.
3. Comparing with regulatory capital is a by-product, not the purpose, so B is wrong and therefore D is wrong.

Why C is correct: Risk capital exists to absorb unexpected losses so the firm stays solvent and a going concern at the target confidence level.

Why the others are wrong

- **A:** Confuses expected with unexpected losses; expected losses are a cost of doing business covered by pricing and loss reserves.
- **B:** A comparison with regulatory capital may be useful but is not the purpose of the calculation.
- **D:** Includes the incorrect statements A and B.

Q391

AnalystPrep Q.2223

REBUILT — Original 'best defined' options were vague and C and D were both defensible; rebuilt around the alternative definition of economic capital in the reading.

In the alternative definition used when linking risk capital to RAROC, economic capital is best described as:

A. Risk capital plus strategic risk capital (burned-out capital and goodwill).
B. Risk capital minus expected losses.
C. The greater of regulatory capital and risk capital.
D. Risk capital plus regulatory capital.

Answer: A

Step by step

1. Risk capital covers unexpected losses (credit, market, operational) at a chosen confidence level.
2. Strategic risk capital covers the risk of strategic failure and is measured as burned-out capital + goodwill.
3. Economic capital = risk capital + strategic risk capital.

Why A is correct: Economic capital is usually identical to risk capital, but the broader definition adds a capital allocation for strategic risk.

Why the others are wrong

- **B:** Risk capital already excludes expected losses (they are covered by pricing and reserves); subtracting them again is wrong.
- **C:** This is the allocation rule for a division (greater of regulatory and risk capital), not a definition of economic capital.
- **D:** Double counts; regulatory capital is a separate, rule-based minimum, not a component of economic capital.

Q392

BT answers P2.T7.24.11.1

REBUILT — Statement 2 described investments with a 'high probability of success', inconsistent with strategic capital, and the answer wording mixed 'Economic' with 'Risk'; reworded to uncertain returns and clean option labels.

A board plans to raise $75 million of capital. The CRO's report states:
Statement 1: $10 million will serve as a cushion to absorb unexpected losses from the firm's activities, sized so the firm remains solvent up to a 99.9% confidence level.
Statement 2: $5 million will cover potential write-offs on investments and acquisitions whose success is uncertain, proxied by spent (burned-out) capital and goodwill.
Which type of capital does each statement describe?

A. Both are strategic risk capital.
B. Statement 1: strategic risk capital; Statement 2: risk capital.
C. Both are risk capital.
D. Statement 1: risk capital; Statement 2: strategic risk capital.

Answer: D

Step by step

1. Statement 1: a buffer for unexpected losses at a high confidence level (99.9%) is risk capital.
2. Statement 2: write-off risk on start-ups/acquisitions, measured by burned-out capital plus goodwill, is strategic risk capital.
3. Statement 1 = risk capital, Statement 2 = strategic risk capital: D.

Why D is correct: Risk capital absorbs unexpected losses from business activities; strategic risk capital (burned-

out capital + goodwill) covers strategic failure.

Why the others are wrong

- **A:** Treats the loss-absorbing cushion in Statement 1 as strategic.
- **B:** Reverses the two assignments.
- **C:** Ignores that goodwill and burned-out capital measure strategic, not risk, capital.

Q393 BT answers P2.T7.24.11.2

A leveraged fund needs capital to maintain solvency at a 99.5% confidence level. Its portfolio value is $2 billion and expected loss is $50 million. The worst-case loss at that confidence level is expected to rise from $500 million to $1 billion. Which is closest to the increase in the fund's economic capital requirement?

A. $950m
B. $450m
C. $500m
D. $1bn

Answer: C

Step by step

1. Economic capital = worst-case loss - expected loss (unexpected loss).
2. Before: 500 - 50 = $450m.
3. After: 1,000 - 50 = $950m.
4. Increase = 950 - 450 = $500m.

Why C is correct: Expected loss is the same before and after, so the increase equals the rise in worst-case loss.

Why the others are wrong

- **A:** This is the new capital level, not the increase.
- **B:** This is the old capital level.
- **D:** Uses the new worst-case loss alone, ignoring expected loss and the starting capital.

Q394 BT answers P2.T7.606.1 · BT notes P2.T7.606.1

Peter Smith works at Bland Bancorp, a large international bank. At a board committee meeting he is asked to explain the difference between risk capital, economic capital, and regulatory capital. Which of the following is the TRUE statement that is part of his response?

A. "Economic capital is less than risk capital because risk capital is the capital required to guarantee shareholders with 100.0% confidence that it holds enough risk capital to ride out any eventuality"
B. "Regulatory capital is almost certainly higher than economic capital in our securitization business; therefore, we do not need to know its economic capital; in other words, whenever regulatory capital exceeds economic capital, there is no reason or need to know economic capital"
C. "We need enough risk capital to absorb unexpected losses and remain solvent over a time horizon with a selected confidence level, but with respect to risk capital, (unlike regulatory capital) the horizon and confidence belong to us as policy parameters."
D. "For capital budgeting purposes, we should use ex-ante RAROC with economic capital in the numerator; for performance evaluation, we should use ex-post RAROC with regulatory capital in the numerator"

Answer: C

Step by step

1. A: risk and economic capital are generally treated as identical, and no firm can offer a 100% confidence level. False.
2. B: even where regulatory capital is higher (e.g., securitization), economic capital must still be computed to judge the economic viability of the activity. False.
3. C: risk capital absorbs unexpected losses at a confidence level and horizon set by senior management/board, unlike rule-based regulatory capital. True.
4. D: economic capital belongs in the denominator of RAROC, not the numerator. False.

Why C is correct: Risk capital's confidence level and horizon are internal policy parameters, whereas regulatory capital is set by industry-wide rules.

Why the others are wrong

- **A:** Risk and economic capital are normally identical, and a 100% confidence level is unattainable.
- **B:** Economic capital is still needed to assess economic viability even when regulatory capital is larger.

- **D:** Economic capital is the RAROC denominator, and it is not replaced by regulatory capital for performance evaluation.

Q395 BT notes P2.T7.1.1

Consider the following statements about economic capital (EC):
I. Economic capital must equal or exceed regulatory capital.
II. Economic capital should exceed equity capital.
III. Economic capital should equal the discounted stream of future excess return on capital.
IV. Economic capital absorbs expected losses (EL) and should equal the firm's reserves.
V. Economic capital absorbs unexpected losses (UL) up to a certain confidence level and over a specified time horizon.
Which of the above statements is (are) TRUE?

A. I only
B. V only
C. I, II, and III
D. All are true

Answer: B

Step by step

1. I: regulatory capital is an industry-wide minimum formula-based amount; it can be above or below a firm's economic capital (the gap can be wide in either direction), so EC need not equal or exceed it. False.
2. II: nothing requires EC to exceed equity capital; EC is the buffer sized to the risk taken. False.
3. III: a discounted stream of future excess return describes value added / NPV, not capital. False.
4. IV: expected losses are covered by pricing and reserves/provisions; EC is the buffer for unexpected losses. False.
5. V: EC is the cushion against unexpected losses (worst-case loss minus expected loss) at a chosen confidence level (e.g., 99.9%) over a horizon (usually one year). True. Only V is true.

Why B is correct: Economic capital is a VaR-type buffer against unexpected losses at a stated confidence level and horizon; expected losses are handled by pricing and reserves.

Why the others are wrong

- **A:** Treats regulatory capital as a floor for EC; regulatory capital is a generic minimum that may be higher or lower than EC.
- **C:** Confuses EC with equity capital (II) and with the discounted excess-return (value added) concept (III), and keeps the false regulatory-floor claim in I.
- **D:** Includes IV, which wrongly assigns expected losses to EC; expected losses are covered by reserves/pricing.

Q396 BT notes P2.T7.3.3

REBUILT — Original relied on the Crouhy loan-equivalent/AECE-vs-PFE methodology outside the 2026 LOs; kept only the in-scope unexpected-loss capital calculation and made it self-contained.

A bank holds a credit portfolio with a loan-equivalent exposure of $480 million. The expected loss is 1.0% of exposure, and the worst-case loss at the bank's chosen confidence level is 3.0% of exposure. Assuming expected losses are covered by pricing and loan loss reserves, how much economic capital should be attributed to credit risk?

A. $4.8 million
B. $9.6 million
C. $14.4 million
D. $19.2 million

Answer: B

Step by step

1. Economic capital = unexpected loss = worst-case loss − expected loss.
2. Unexpected loss = 3.0% − 1.0% = 2.0% of exposure.
3. Economic capital = 0.02 × $480 million = $9.6 million.

Why B is correct: Economic (risk) capital covers only the excess of the worst-case loss over the expected loss, because expected loss is already provided for through pricing and reserves.

Why the others are wrong

- **A:** Uses the 1.0% expected loss (0.01 × 480 = $4.8m) instead of the unexpected loss.
- **C:** Uses the full 3.0% worst-case loss (0.03 × 480 = $14.4m) without subtracting expected loss.

- **D:** Adds expected and worst-case loss (4.0% × 480 = $19.2m) instead of subtracting.

Q397 BT notes P2.T7.1.2

Each of the following is true about the purpose of economic capital at a bank EXCEPT:

A. Economic capital, if properly calibrated, guarantees the bank will never default
B. Economic capital is designed to keep a bank solvent; i.e., it is a solvency measure
C. Economic capital provides a means for capital allocation within the bank
D. Economic capital is a metric (a risk currency) that can be aggregated across and up and down the organization

Answer: A

Step by step

1. Economic capital is sized to a confidence level tied to a target credit rating (e.g., about 99.9%), so a small probability (about 0.1%) remains that losses exceed it.
2. Hence it cannot guarantee that default never occurs; A is false.
3. B, C and D are true: EC answers how much capital is needed to remain solvent, is used to allocate capital and its cost to business lines, and serves as a common risk measure across units.

Why A is correct: Economic capital covers losses only up to a chosen confidence level, so some residual probability of default always remains.

Why the others are wrong

- **B:** True statement: EC's traditional role is to answer how much capital keeps the firm solvent.
- **C:** True statement: EC is used to allocate capital and capital costs to business lines.
- **D:** True statement: EC is a common risk currency that can be aggregated across and within business units.

Q398 BT notes P2.T6.505.1

REBUILT — Original asked about the Schroeck 'senior uninsured debt' threshold, which is not in the GARP/Schweser text; rebuilt to test the covered link between the economic capital confidence level and the target credit rating.

A bank is computing economic capital and wants to maintain a AA credit rating. Which of the following best describes how the confidence level for the economic capital calculation should be chosen?

A. At 95%, the level typically used for trading desk VaR limits, because the AA rating is based on day-to-day losses
B. At 99%, the level used for regulatory capital, because regulatory and economic capital must share the same confidence level
C. Above 99.95%, because the confidence level should correspond to the default probability implied by the target credit rating
D. At whatever lower level minimizes the capital allocated, because a lower confidence level always improves risk-adjusted performance without affecting solvency

Answer: C

Step by step

1. Economic (risk) capital is the buffer for unexpected losses over the horizon (usually one year) at a chosen confidence level.
2. The confidence level is tied to the target credit rating, because a rating is explicitly associated with a probability of default.
3. A high rating such as AA or AAA corresponds to a very low default probability, so the confidence level must exceed 99.95%.
4. Therefore the confidence level is set by the rating target, not by trading limits or regulatory minimums.

Why C is correct: The confidence level must be consistent with the default probability implied by the desired rating; for AA/AAA this requires more than 99.95%.

Why the others are wrong

- **A:** A 95% level is a short-horizon trading limit convention and implies a far higher default probability than a AA rating allows.
- **B:** Regulatory capital uses general industry benchmarks and is usually far below risk capital; economic capital's confidence level is driven by the firm's target rating, not the regulatory level.
- **D:** A lower confidence level does reduce allocated capital and changes RAROC, but it does so by weakening the solvency standard and no longer matches the target rating.

LO 59.b

Describe the RAROC (risk-adjusted return on capital) methodology and its use in capital budgeting.

Q399

Schweser Module Quiz 48.1 Q2

REBUILT — Source key relied on a claim (RAROC applied more to credit than operational risk) not supported by GARP; rewrote the options around the RAROC definition so exactly one is correct.

Which of the following statements about the risk-adjusted return on capital (RAROC) measure is most accurate?

A. The numerator is pretax risk-adjusted net income.
B. Regulatory capital is the denominator.
C. Expected losses are set to zero in the numerator when the activity is a credit exposure.
D. The numerator is after-tax expected risk-adjusted net income and the denominator is economic capital.

Answer: D

Key correction: Source key (Schweser) said D under the original wording (RAROC applied more to credit than to operational risk); GARP does not support that claim, so the options were rewritten and the correct choice is now the definition statement, D.

Step by step

1. GARP defines RAROC = after-tax expected risk-adjusted net income / economic capital.
2. Numerator: expected revenues less costs, expected losses and taxes, plus return on risk capital, +/- transfers; so it is after tax and net of expected loss.
3. Denominator: economic capital (risk capital plus strategic risk capital), a proxy for risk, not regulatory capital.
4. Only D matches this.

Why D is correct: RAROC divides after-tax expected income, net of expected losses, by economic capital, which proxies for risk.

Why the others are wrong

- **A:** Taxes are deducted in the numerator, so it is after tax, not pretax.
- **B:** Regulatory capital is a rule-based amount; RAROC uses economic capital.
- **C:** For credit exposures, expected default losses (loan loss reserve) are the main expected loss deducted; they are not set to zero.

Q400

AnalystPrep Q.6012

REBUILT — Option B described the denominator as exposure at default; rewrote so the correct choice is economic capital covering unexpected losses and distractors are realistic confusions.

In the RAROC formula, which of the following best describes the denominator and its role in assessing the risk of a loan?

A. The loan amount, which proxies for the revenue the loan can generate.
B. Economic capital, the buffer held against unexpected losses at the bank's chosen confidence level, which serves as the proxy for risk.
C. The bank's cost of capital, used as the benchmark that the loan's return must exceed.
D. The interest rate spread, which measures profitability relative to market rates.

Answer: B

Step by step

1. RAROC = after-tax expected risk-adjusted net income / economic capital.
2. The numerator measures reward; the denominator, economic capital, measures risk.
3. Economic capital (risk capital plus strategic risk capital) covers worst-case loss above expected loss at a stated confidence level, similar to one-year VaR.
4. So B.

Why B is correct: The denominator is economic capital, which captures unexpected-loss risk and is the risk proxy in the risk-reward trade-off.

Why the others are wrong

- **A:** Loan size is not the denominator; confuses exposure/revenue with risk capital.
- **C:** The cost of equity or hurdle rate is the benchmark RAROC is compared with, not the denominator.
- **D:** The spread is part of the revenue in the numerator, not the denominator.

Q401

BT answers P2.T7.606.2 · BT notes P2.T7.606.2

REBUILT — Original relied on RORAC/RAROA/RORAA variants that are outside the Schweser LOs; kept the same EXCEPT structure and false NPV statement but tested RAROC versus ROE/ROA and NPV.

Which of the following statements about RAROC and related performance measures is NOT correct?

A. RAROC adjusts the numerator by deducting expected loss from the return and adjusts the denominator by substituting economic capital for accounting capital.

B. Return on equity and return on assets rely on accounting book values and therefore do not capture the underlying risk.

C. NPV uses a CAPM-beta-based discount rate that reflects systematic risk only, while the RAROC earnings figure reflects both systematic and unsystematic risk.

D. NPV is incompatible with the CAPM and has no convenient way of incorporating risk.

Answer: D

Step by step

1. A is the two RAROC adjustments per the text: true.
2. B: ROE/ROA use accounting book values, so they miss risk: true.
3. C: NPV discounts at a CAPM risk-adjusted rate (beta, systematic risk), whereas RAROC's risk measure captures total volatility: true.
4. D: NPV is built on CAPM and incorporates risk through the discount rate, so D is false, i.e., the EXCEPT answer.

Why D is correct: NPV uses a CAPM-based risk-adjusted discount rate, so claiming it cannot incorporate risk is false.

Why the others are wrong

- **A:** True statement, so not the exception.
- **B:** True statement; choosing it means doubting that accounting measures ignore risk.
- **C:** True statement; choosing it confuses the CAPM beta (systematic only) with RAROC's total-risk treatment.

Q402

BT notes P2.T7.1.5

REBUILT — Statement I referred to cost of equity and II had awkward wording; restated with hurdle rate and clean statements, keeping all four true.

Consider the following statements about RAROC:

I. If a project's RAROC exceeds the hurdle rate, the project should be accepted.

II. Because the denominator is economic capital, which covers unexpected losses, the numerator deducts expected losses.

III. RAROC can be used for capital allocation on an ex ante basis, using expected revenues and losses.

IV. RAROC can be used for performance evaluation on an ex post basis, using realized revenues and losses.

Which statements are true?

A. I and III only

B. II and IV only

C. III and IV only

D. All four

Answer: D

Step by step

1. I: RAROC > hurdle rate implies value creation, accept: true.
2. II: economic capital covers unexpected losses, so expected loss is deducted from return: true.
3. III: RAROC was proposed for ex ante capital allocation with expected figures: true.
4. IV: ex post, realized revenues and losses are used: true.
5. All four are true.

Why D is correct: Each statement matches the RAROC decision rule, its construction and its ex ante / ex post uses.

Why the others are wrong

- **A:** Omits II and IV, which are also true.
- **B:** Omits I and III, which are also true.
- **C:** Omits I and II, which are also true.

Q403 Schweser Module Quiz 59.1 Q1

Which of the following statements regarding the risk-adjusted return on capital (RAROC) methodology is correct?

A. In the context of performance measurement, RAROC uses accounting profits.

B. In the numerator of the RAROC equation, expected loss is added to the return.

C. If a business unit's cost of equity is greater than its RAROC, then the business unit is not adding value to shareholders.

D. RAROC is useful for determining incentive compensation, but it lacks the flexibility to consider deferred or contingent compensation.

Answer: C

Step by step

1. Cost of equity is the minimum return shareholders require.
2. If RAROC is below it, the unit earns less than required and destroys value.
3. A, B, D contradict the stated benefits and formula.

Why C is correct: A RAROC below the required return (cost of equity/hurdle rate) means value destruction.

Why the others are wrong

- **A:** RAROC uses economic profits, not accounting profits.
- **B:** Expected loss is deducted, not added.
- **D:** RAROC is flexible enough to allow deferred or contingent compensation and clawbacks.

LO 59.c

Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance.

Q404 AnalystPrep Q.2214

A certain bank is calculating RAROC for some of its business lines. The available data gives information on: expected revenues, costs, taxes, return on risk capital, transfers, and economic capital. What type of data is missing?

A. Sharpe ratio

B. Expected losses

C. Net present value

D. VaR (Value-at-risk)

Answer: B

Step by step

1. Capital budgeting RAROC = (expected revenues - costs - expected losses - taxes + return on risk capital +/- transfers) / economic capital.
2. Compare to the data listed: revenues, costs, taxes, return on risk capital, transfers, economic capital are all present.
3. Expected losses are absent.

Why B is correct: Expected losses are the only component of the detailed RAROC formula not provided.

Why the others are wrong

- **A:** The Sharpe ratio is a comparable risk-return measure, not an input to RAROC.
- **C:** NPV is another related measure, not an input.
- **D:** VaR may be used to estimate economic capital, which is already given; it is not a separate numerator or denominator term.

Q405 AP Mock 2 Q38

REBUILT — Original options mixed in adjusted RAROC and risk-free comparisons that cannot be evaluated without beta data; rebuilt to ask for RAROC and the decision against the stated hurdle rate.

Smart Bank is reviewing the Small and Medium Businesses Lending division (SMBL) of AC Bank, which holds a USD 150 million loan portfolio funded by deposits. PD = 7%, LGD = 60% and EAD = 100% of exposure. Over a one-year horizon:

Item	Value
Economic capital	USD 120 million
Operating direct costs	USD 1.25 million
Interest costs	USD 6 million
Return on risk capital (on economic capital)	2%
Return on the loan portfolio	12%
Hurdle rate	4.6%

Ignore taxes and transfers. What is the division's RAROC and what should management do?

A. RAROC = 5.71%; keep the division because RAROC exceeds the hurdle rate
B. RAROC = 3.71%; close the division because RAROC is below the hurdle rate
C. RAROC = 2.21%; close the division because RAROC is below the hurdle rate
D. RAROC = 10.96%; keep the division because RAROC exceeds the hurdle rate

Answer: A

Step by step

1. Expected revenue = 12% × 150 = USD 18.00 million.
2. Expected loss = PD × LGD × EAD = 7% × 60% × 150 = USD 6.30 million.
3. Return on economic capital = 2% × 120 = USD 2.40 million.
4. Numerator = 18.00 − 1.25 − 6.00 − 6.30 + 2.40 = USD 6.85 million.
5. RAROC = 6.85/120 = 5.71%, which exceeds the 4.6% hurdle, so keep the division.

Why A is correct: RAROC deducts operating costs, funding cost and expected loss (PD × LGD × EAD) and adds the risk-free return on economic capital; a RAROC above the hurdle rate creates value.

Why the others are wrong

- **B:** Omits the 2% return on economic capital: 4.45/120 = 3.71%, leading to a wrong 'close' decision.
- **C:** Uses expected loss = PD × EAD = 10.5 (forgets LGD): 2.65/120 = 2.21%.
- **D:** Omits expected loss altogether: 13.15/120 = 10.96%; overstates return.

Q406

AnalystPrep Q.2207

A Catalonian bank will invest only in the activity with the highest RAROC, provided it exceeds the cost of equity of 6.2%. The options are:

I. Tarragona Construcciones: expected risk-adjusted net income EUR 3,000,000 per year, economic capital EUR 50,000,000.

II. Valencia Bonos: expected risk-adjusted net income EUR 1,500,000 per year, economic capital EUR 22,000,000.

Based on RAROC, the bank would most likely invest in:

A. Tarragona Construcciones
B. Valencia Bonos
C. Both projects
D. Neither project

Answer: B

Step by step

1. $\text{RAROC}_I = 3.0/50 = 6.00\%$.
2. $\text{RAROC}_{II} = 1.5/22 = 6.82\%$.
3. Cost of equity = 6.2%: only Valencia (6.82%) exceeds it, so only Valencia creates shareholder value.

Why B is correct: An activity adds value only if its RAROC exceeds the cost of equity; Tarragona's 6.00% falls short while Valencia's 6.82% clears it.

Why the others are wrong

- **A:** Tarragona has the larger absolute profit (3.0m) but a lower return per unit of capital, 6.00% < 6.2%.
- **C:** Ignores that Tarragona's RAROC is below the cost of equity.
- **D:** Wrongly concludes both fail; Valencia's 6.82% exceeds 6.2%.

Q407

AnalystPrep Q.2210

A bank will choose among four factories by RAROC (risk-adjusted return on capital). Data (USD):

Factory	Expected revenues	Expected losses	Economic capital
A	150,000	8,000	1,400,000
B	175,000	15,000	1,500,000
C	200,000	15,000	1,800,000
D	250,000	10,000	2,000,000

Ignore costs, taxes and other items. Based on RAROC, the bank will most likely pick:

A. Factory A
B. Factory B
C. Factory C
D. Factory D

Answer: D

Step by step

1. RAROC = (revenues − expected losses)/economic capital.
2. A: 142,000/1,400,000 = 10.14%; B: 160,000/1,500,000 = 10.67%.
3. C: 185,000/1,800,000 = 10.28%; D: 240,000/2,000,000 = 12.00%.
4. Factory D has the highest RAROC.

Why D is correct: RAROC measures risk-adjusted return per unit of economic capital, so the highest ratio, not the highest profit or lowest capital, wins.

Why the others are wrong

- **A:** A has the smallest capital and lowest losses but only 10.14% RAROC.
- **B:** B's 10.67% is second best; ranks wrongly.
- **C:** C has the second-largest revenue but its capital is high; RAROC 10.28%.

Q408

AnalystPrep Q.2734

REBUILT — Original distractors were not tied to realistic errors; options replaced with results of omitting or mis-signing items.

Determine the RAROC for the following loan.

Item	Value
Loan value	USD 2 million
Gross revenue	USD 250,000
Expected loss	300 bps of loan value
Interest expense	USD 100,000
Operating costs	USD 60,000
Return on invested economic capital	USD 10,000
Economic capital required	USD 400,000

Ignore taxes and transfers.

A. 10.00%
B. 7.50%
C. 5.00%
D. 35.00%

Answer: A

Step by step

1. Expected loss = 0.03 × 2,000,000 = USD 60,000.
2. Numerator = 250,000 − 60,000 (EL) − 60,000 (operating) − 100,000 (interest) + 10,000 (return on capital) = USD 40,000.
3. RAROC = 40,000/400,000 = 10.00%.

Why A is correct: The return on economic capital is added and expected loss, costs and funding cost are subtracted before dividing by economic capital.

Why the others are wrong

- **B:** Omits the USD 10,000 return on economic capital: 30,000/400,000 = 7.5%.
- **C:** Subtracts the return on economic capital instead of adding it: 20,000/400,000 = 5%.
- **D:** Omits the USD 100,000 interest expense: 140,000/400,000 = 35%.

Q409

AnalystPrep Q.2964

REBUILT — Original distractors were arbitrary; replaced with results of realistic errors (omitting return, forgetting EL, forgetting tax).

For a loan, the following information is available (USD millions): expected revenue 89.5; operating cost 8.89; tax expense 50.98; expected loss 10; return on risk capital 24.01; economic capital 69.5. What is the RAROC for the loan?

A. 0.2824
B. 0.7718
C. 0.6279
D. 1.3614

Answer: C

Step by step

1. Numerator = 89.5 − 8.89 − 50.98 − 10 + 24.01 = 43.64.
2. RAROC = 43.64/69.5 = 0.6279.

Why C is correct: RAROC = (revenues − costs − expected losses − taxes + return on risk capital)/economic capital.

Why the others are wrong

- **A:** Omits the 24.01 return on risk capital: 19.63/69.5 = 0.2824.
- **B:** Forgets to deduct the expected loss of 10: 53.64/69.5 = 0.7718.
- **D:** Forgets to deduct the 50.98 tax expense: 94.62/69.5 = 1.3614.

Q410

AnalystPrep Q.6014

REBUILT — Source asked only for a loan-revenue numerator using a non-GARP model; rebuilt as a GARP-style full RAROC computation with economic capital and return on capital.

A bank holds a EUR 1,500,000 loan funded by deposits. Annual expected revenue is EUR 52,500 (interest spread plus origination fee), expected loss is EUR 20,000 and direct operating costs are EUR 15,000. Economic capital is EUR 90,000 and earns a risk-free return of 2%. The effective tax rate is 30%; ignore transfers. What is the RAROC?

A. 15.01%
B. 13.61%
C. 21.44%
D. 30.57%

Answer: A

Step by step

1. Return on economic capital = 2% × 90,000 = 1,800.
2. Pre-tax numerator = 52,500 − 20,000 − 15,000 + 1,800 = 19,300.
3. After tax = 19,300 × 0.70 = 13,510.
4. RAROC = 13,510/90,000 = 15.01%.

Why A is correct: RAROC is after-tax risk-adjusted net income (including the return on economic capital) divided by economic capital.

Why the others are wrong

- **B:** Omits the 1,800 return on economic capital: 12,250/90,000 = 13.61%.
- **C:** Forgets to apply the 30% tax: 19,300/90,000 = 21.44%.
- **D:** Forgets to deduct the expected loss: (52,500 − 15,000 + 1,800) × 0.7/90,000 = 30.57%.

Q411

Schweser Module Quiz 23.2 Q4

REBUILT — Source used a duration-based capital-at-risk approach outside the reading; rebuilt as a GARP-style portfolio RAROC question.

A bank has an USD 800 million loan portfolio yielding 8%, funded by deposits with a transfer-priced interest charge of 5%. Operating costs are USD 10 million, expected loss is 0.6% of principal, and economic capital is 9% of principal, invested at the risk-free rate of 3%. The effective tax rate is 25%; ignore other transfers. What is the portfolio's RAROC?

A. 11.83%
B. 9.58%
C. 15.78%
D. 16.83%

Answer: A

Step by step

1. Revenue = 64; interest expense = 40; expected loss = 4.8; economic capital = 72; return on capital = 3% × 72 = 2.16 (USD m).
2. Pre-tax = 64 − 10 − 40 − 4.8 + 2.16 = 11.36; after tax = 11.36 × 0.75 = 8.52.
3. RAROC = 8.52/72 = 11.83%.

Why A is correct: RAROC divides after-tax expected revenue net of costs, funding and expected loss, plus return on capital, by economic capital.

Why the others are wrong

- **B:** Omits the return on economic capital: 6.9/72 = 9.58%.
- **C:** Forgets the tax: 11.36/72 = 15.78%.
- **D:** Forgets the expected loss: (64 − 10 − 40 + 2.16) × 0.75/72 = 16.83%.

Q412

AnalystPrep Q.3206

REBUILT — Source left the units of economic capital unstated and gave no tax or transfer information; added units and 'no taxes or transfers', and replaced arbitrary distractors with realistic-error ones.

A bank is evaluating a loan using RAROC. Assume no taxes and no transfer costs. The following annual information is available:

Item	Amount
Gross revenue	$12.0 million
Operating costs of making the loan	$2.3 million
Expected loss on the loan	$1.6 million
Economic capital	$80 million
Return on economic capital	$0.7 million

Based on this information, RAROC is closest to:

A. 15.0%

B. 13.0%

C. 10.1%

D. 11.0%

Answer: D

Step by step

1. Risk-adjusted return = revenue − costs − expected loss + return on economic capital (no taxes or transfers).
2. 12.0 − 2.3 − 1.6 + 0.7 = 8.8 million.
3. RAROC = 8.8/80 = 0.11 = 11.0%.

Why D is correct: RAROC divides expected net income after deducting expected loss (and adding the return earned on the economic capital) by economic capital.

Why the others are wrong

- **A:** Adds the expected loss instead of deducting it: (12.0 − 2.3 + 1.6 + 0.7)/80 = 15.0%.
- **B:** Ignores the expected loss: (12.0 − 2.3 + 0.7)/80 = 13.0%. Expected loss is the risk adjustment that distinguishes RAROC from ROE.
- **C:** Omits the return on economic capital: (12.0 − 2.3 − 1.6)/80 = 10.1%.

Q413

AnalystPrep Q.6016

A bank is assessing two different loans using the Risk-Adjusted Return on Capital (RAROC) model. Both loans have identical principal amounts, interest rate spreads, and origination fees. However, Loan X has expected loan losses of €30,000 and operating costs of €5,000, while Loan Y has expected loan losses of €20,000 and operating costs of €10,000. All other factors constant, how do the differences in expected loan losses and operating costs between Loan X and Loan Y affect their respective RAROC values?

A. Loan X will have a higher RAROC as lower operating costs outweigh higher expected loan losses.

B. Loan Y will have a higher RAROC as lower expected loan losses outweigh higher operating costs.

C. Both loans will have the same RAROC as the differences in expected losses and operating costs cancel each other out.

D. It is impossible to determine which loan will have a higher RAROC without knowing the interest rate spreads and origination fees.

Answer: B

Step by step

1. Revenue (spread plus fees) and economic capital are identical, so the numerators differ only by expected loss plus operating costs.
2. Loan X deductions: $30{,}000 + 5{,}000 = 35{,}000$.
3. Loan Y deductions: $20{,}000 + 10{,}000 = 30{,}000$.
4. Loan Y deducts 5,000 less, so its risk-adjusted return, and hence its RAROC, is higher.

Why B is correct: With identical revenue and capital, RAROC is higher for the loan with the lower sum of expected loss and operating costs; Loan Y's 10,000 lower expected loss exceeds its 5,000 higher operating cost.

Why the others are wrong

- **A:** Reverses the comparison: X's 5,000 saving on operating costs is smaller than its 10,000 higher expected loss.
- **C:** Costs do not cancel: the totals are 35,000 vs 30,000, not equal.
- **D:** Because revenue is identical for both loans, it drops out of the comparison; the ranking depends only on the differing deductions.

Q414

BT notes P2.T6.24.7.3 · BT answers P2.T6.24.7.3

REBUILT — BT answers copy had OCR-damaged inputs (7-year, 7% operating costs) that could not reproduce its key; used the consistent 1-year version and stated that the transaction cost is deducted before tax.

A commercial bank is evaluating a 1-year loan of EUR 420 million fully funded by deposits. Terms:
- Interest rate on the loan: 6% per annum
- Cost of funding (interest paid on deposits): 0.5% per annum
- Expected loss: 3% of the loan's face value
- Operating costs: 1% of the loan's face value
- Economic capital: 5% of the loan amount
- Pre-tax return on economic capital: 2%
- Tax rate: 35%
- Transaction costs: EUR 1 million (deducted before tax)

The after-tax RAROC is closest to:

A. 42%
B. 21%
C. 18%
D. 1%

Answer: C

Step by step

1. Economic capital $= 0.05 \times 420 = 21.0$ million.
2. Revenue $0.06 \times 420 = 25.2$; funding cost $0.005 \times 420 = 2.1$; expected loss 12.6; operating costs 4.2; return on EC $0.02 \times 21 = 0.42$; transaction costs 1.0.
3. Pre-tax $= 25.2 - 2.1 - 12.6 - 4.2 + 0.42 - 1.0 = 5.72$ million.
4. After tax $= 5.72 \times 0.65 = 3.718$ million.
5. RAROC $= 3.718/21.0 = 17.7\% \approx 18\%$.

Why C is correct: After-tax expected net income (after expected loss, costs and the return credited on the capital) is divided by economic capital, not by the loan amount.

Why the others are wrong

- **A:** Applies the 2% return to the whole loan (8.4 instead of 0.42): $(5.72 - 0.42 + 8.4) \times 0.65/21 = 42\%$.
- **B:** Omits the EUR 1m transaction costs: $6.72 \times 0.65/21 = 20.8\% \approx 21\%$.
- **D:** Divides by the loan amount instead of economic capital: $3.718/420 = 0.9\% \approx 1\%$.

Q415

BT answers P2.T7.24.11.3 · BT notes P2.T7.24.11.3

REBUILT — Duplicate copies had inconsistent OCR figures (3.0% vs 8.23%; 0.5% vs 1%); restored the set that reproduces 8.23% (EL omitted) and -1.40% (EL included, costs doubled), and made the stem self-contained.

A bank's Audit and Risk Committee (ARC) is reviewing the performance of $100 million of new loans. The RAROC projected at initiation was 8.23%. By year-end the actual RAROC was -1.40%. The ARC asked internal audit and finance to explain the variance, giving both the following information (percentages of the $100 million of loans):
- Economic capital: 40%
- Expected revenue: 7%

- Return on economic capital: 0.5%
- Expected credit loss on loans: 3%
- Operating costs (budgeted): 2.5%
- Tax rate: 30%
- Sector-wide credit spreads had narrowed.

Finance department: The RAROC was miscalculated because the initial estimates did not factor in expected losses. In addition, actual operating costs were double the anticipated amount.
Internal Audit: The initial RAROC calculation was accurate; the variance was primarily due to macroeconomic and systematic sector-wide factors, not budgeting errors.
Which statement is most likely correct?

A. Finance department
B. Internal Audit department
C. Both are correct
D. Both are incorrect

Answer: A

Step by step

1. Economic capital = 40 million; return on EC = $0.005 \times 40 = 0.2$ million.
2. Initial RAROC ignoring EL: $(7 + 0.2 - 2.5) \times 0.70/40 = 3.29/40 = 8.23\%$, exactly the projected figure, so the projection omitted expected loss.
3. Correct initial RAROC including EL: $(7 + 0.2 - 2.5 - 3) \times 0.70/40 = 1.19/40 = 2.98\%$.
4. Actual with operating costs doubled to 5: $(7 + 0.2 - 5 - 3) \times 0.70/40 = -0.56/40 = -1.40\%$, exactly the reported actual.
5. The variance is fully explained by the omitted expected loss and the doubled operating costs, so Finance is right and Audit's claim that the initial calculation was accurate is false.

Why A is correct: The projected 8.23% reproduces only if expected loss is left out of the numerator, and doubling costs reproduces the actual -1.40%; these are internal estimation and budgeting errors, not market-wide effects.

Why the others are wrong

- **B:** The initial calculation was not accurate (8.23% vs 2.98% with EL), and the actual result is explained by an internal cost overrun, so Audit is wrong.
- **C:** Audit's statement is incorrect, so both cannot be correct.
- **D:** Finance's two claims (EL omitted, operating costs doubled) are both confirmed by the numbers.

Q416

BT answers P2.T7.24.12.1

REBUILT — Clarified that the 'other transfer costs' of 0.25% are on the face value of the loan (needed to reproduce the answer) and that the loan is 1-year.

Ant Commercial Bank is considering a 1-year loan fully funded by deposits:
- Loan amount: $700 million
- Average annual interest rate paid on deposits: 0.5%
- Annual interest rate received on loan: 6.5%
- Expected loss: 1% of face value of loan
- Annual operating costs: 2% of face value of loan
- Economic capital required: 20% of the loan amount
- Average pre-tax return on economic capital: 4%
- Effective tax rate: 39%
- Other transfer costs: 0.25% of face value of loan

What is the after-tax RAROC for this loan?

A. 10.83%
B. 17.75%
C. 2.17%
D. 13.88%

Answer: A

Step by step

1. Revenue $0.065 \times 700 = 45.5$; deposit interest 3.5; EL 7.0; operating costs 14.0; transfer costs 1.75; economic capital $0.20 \times 700 = 140$; return on EC $0.04 \times 140 = 5.6$ (millions).
2. Pre-tax $= 45.5 - 3.5 - 7.0 - 14.0 - 1.75 + 5.6 = 24.85$.
3. After tax $= 24.85 \times 0.61 = 15.1585$.

4. RAROC = 15.1585/140 = 10.83%.

Why A is correct: After-tax risk-adjusted net income divided by economic capital (20% of the loan).

Why the others are wrong

- **B:** Ignores tax: 24.85/140 = 17.75%.
- **C:** Divides by the loan amount instead of economic capital: 15.1585/700 = 2.17%.
- **D:** Leaves out the expected loss: 31.85 × 0.61/140 = 13.88%.

Q417

GARP PE1 Q73 · Mock 3 Q8

REBUILT — Replaced the arbitrary distractor 10.73% with the result of a realistic error (omitting expected loss).

A regional commercial bank is considering a 1-year loan fully funded by deposits, with the following parameters:

Parameter	Value
Loan amount	JPY 4.2 billion
Average annual interest rate paid on deposits	0.4%
Annual interest rate received on loan	3.2%
Expected loss	2.0% of face value of loan
Annual operating costs	0.5% of face value of loan
Economic capital required to support the loan	10.0% of the loan
Average pre-tax return on economic capital	1.4%
Effective tax rate	38%
Other transfer costs	JPY 0

What is the after-tax RAROC for this loan?

A. 0.27%
B. 2.73%
C. 4.40%
D. 15.13%

Answer: B

Step by step

1. In JPY million: revenue 0.032 × 4200 = 134.4; deposit interest 16.8; EL 84.0; operating costs 21.0; EC 420; return on EC 0.014 × 420 = 5.88.
2. Pre-tax = 134.4 − 16.8 − 84.0 − 21.0 + 5.88 = 18.48.
3. After tax = 18.48 × 0.62 = 11.4576.
4. RAROC = 11.4576/420 = 2.73%.

Why B is correct: After-tax expected risk-adjusted income is measured against the economic capital of JPY 420 million.

Why the others are wrong

- **A:** Divides by the loan amount (4,200) instead of economic capital: 11.4576/4200 = 0.27%.
- **C:** Pre-tax RAROC: 18.48/420 = 4.40%; forgets the 38% tax.
- **D:** Omits expected loss: (18.48 + 84) × 0.62/420 = 15.13%.

Q418

Mock 4 Q18

REBUILT — Replaced the arbitrary distractor 10.07% with the result of a realistic error (omitting expected loss).

A regional commercial bank is considering a 1-year loan fully funded by deposits, with the following parameters:

Parameter	Value
Loan amount	CNY 3.8 billion
Average annual interest rate paid on deposits	0.6%
Annual interest rate received on loan	4.1%
Expected loss	3.0% of face value of loan
Annual operating costs	0.3% of face value of loan
Economic capital required to support the loan	15.0% of the loan
Average pre-tax return on economic capital	2.0%
Effective tax rate	38%
Other transfer costs	CNY 0

What is the after-tax RAROC for this loan?

A. 0.31%
B. 2.07%

C. 3.33%
D. 14.47%

Answer: B
Step by step
1. In CNY million: revenue $0.041 \times 3800 = 155.8$; deposit interest 22.8; EL 114.0; operating costs 11.4; EC $0.15 \times 3800 = 570$; return on EC $0.02 \times 570 = 11.4$.
2. Pre-tax $= 155.8 - 22.8 - 114.0 - 11.4 + 11.4 = 19.0$.
3. After tax $= 19.0 \times 0.62 = 11.78$.
4. RAROC $= 11.78/570 = 2.07\%$.

Why B is correct: After-tax expected risk-adjusted income divided by economic capital of CNY 570 million.
Why the others are wrong
- **A:** Divides by the loan amount: $11.78/3800 = 0.31\%$.
- **C:** Pre-tax RAROC $19.0/570 = 3.33\%$; forgets the tax.
- **D:** Omits expected loss: $(19.0 + 114) \times 0.62/570 = 14.47\%$.

Q419

BT answers P2.T7.606.3 · BT notes P2.T7.606.3

REBUILT — Distractors were unexplainable random values; replaced with values produced by realistic component omissions (answer unchanged).
A $1.0 billion corporate loan portfolio earns 5.0% per annum. The bank has direct operating costs of $6.0 million per annum and an effective tax rate of 25.0%. The portfolio is funded by $1.0 billion of retail deposits with a transfer-priced interest charge of 1.40%. Economic capital is $80.0 million (8.0% of the loan amount) and must be invested in risk-free securities yielding 1.0%. Expected loss is 1.0% per annum ($10 million). Ignoring other transfer-pricing issues, which is NEAREST to the after-tax RAROC?

A. 18.75%
B. 19.50%
C. 26.00%
D. 28.88%

Answer: B
Step by step
1. Revenue = 5.0% x $1,000m = $50.0m; interest expense = 1.40% x $1,000m = $14.0m; expected loss = $10.0m; operating costs = $6.0m.
2. Return on economic capital = 1.0% x $80m = $0.8m.
3. Pre-tax risk-adjusted income = 50 - 6 - 14 - 10 + 0.8 = $20.8m.
4. After tax = 20.8 x (1 - 0.25) = $15.6m.
5. RAROC = 15.6 / 80 = 19.50%.

Why B is correct: RAROC = after-tax expected risk-adjusted net income (including the risk-free return on economic capital) divided by economic capital: 15.6/80 = 19.5%.
Why the others are wrong
- **A:** Omits the $0.8m risk-free return on economic capital: 14.0 x 0.75 / 80 = 18.75%.
- **C:** Omits the tax: 20.8/80 = 26.0%.
- **D:** Omits the expected loss: (50 - 6 - 14 + 0.8) x 0.75 / 80 = 28.88%.

Q420

BT notes P2.T7.1.3 (Q1.3)

REBUILT — Stem did not say taxes are ignored and distractors were unexplained; stated no taxes, cleaned stem and rebuilt options.
A $1.0 billion loan portfolio pays an average 5.0% and is funded by $1.0 billion of deposits costing 2.0%. Economic capital is $80 million (8.0% of assets), invested in Treasury bills paying 1.0%. Expected loss is 1.0% of the portfolio and allocated operating costs are $6.0 million. Ignoring taxes, what is the RAROC?

A. 17.50%
B. 18.50%
C. 26.00%
D. 31.00%

Answer: B
Step by step

1. Revenue = $50m; interest expense = 2.0% x $1,000m = $20m; EL = $10m; operating costs = $6m.
2. Return on EC = 1.0% x $80m = $0.8m.
3. Risk-adjusted income = 50 - 20 - 10 - 6 + 0.8 = $14.8m (no tax).
4. RAROC = 14.8 / 80 = 18.50%.

Why B is correct: Net income after funding cost, EL and operating cost, plus the risk-free return on capital, is divided by economic capital.

Why the others are wrong

- **A:** Omits the return on economic capital: 14.0/80 = 17.50%.
- **C:** Omits operating costs: 20.8/80 = 26.0%.
- **D:** Omits expected loss: 24.8/80 = 31.0%.

Q421

BT notes P2.T7.1.3 (Q1.4)

REBUILT — Source gave no answer or explanation and did not say taxes are ignored; stated no taxes, recomputed and rebuilt distractors.

A $10.0 billion loan portfolio earning 4.0% is funded by $10.0 billion of deposits paying 2.0%. Expected loss is 1.0% of the loans. Economic capital equals 12.0% of loan assets and is invested safely to earn 2.0%. Operating costs are $22.0 million. Ignoring taxes, what is the RAROC?

A. 6.50%
B. 8.50%
C. 10.33%
D. 16.83%

Answer: B

Step by step

1. Revenue = 4% x $10,000m = $400m; interest = 2% x $10,000m = $200m; EL = $100m; opex = $22m.
2. EC = 12% x $10,000m = $1,200m; return on EC = 2% x 1,200 = $24m.
3. Income = 400 - 200 - 100 - 22 + 24 = $102m.
4. RAROC = 102 / 1,200 = 8.50%.

Why B is correct: Risk-adjusted income of $102m on $1.2bn of economic capital.

Why the others are wrong

- **A:** Omits the $24m return on EC: 78/1,200 = 6.50%.
- **C:** Omits operating costs: 124/1,200 = 10.33%.
- **D:** Omits expected loss: 202/1,200 = 16.83%.

Q422

Mock 1 Q35

A risk analyst is computing the after-tax risk-adjusted return on capital (RAROC) for a business line at a commercial bank. Annual figures (USD million): expected revenues 32.0; operating costs 14.0; expected loss 5.0; economic capital allocated 60.0. The bank earns a 3% risk-free return on the economic capital it holds, and its effective tax rate is 30%. What is the after-tax RAROC for this business line?

A. 15.17%
B. 17.27%
C. 23.10%
D. 24.67%

Answer: B

Step by step

1. Return on EC = 3% x 60 = 1.8.
2. Pre-tax income = 32.0 - 14.0 - 5.0 + 1.8 = 14.8.
3. After tax = 14.8 x 0.70 = 10.36.
4. RAROC = 10.36 / 60 = 17.27%.

Why B is correct: After-tax expected risk-adjusted income including the risk-free return on capital, divided by economic capital.

Why the others are wrong

- **A:** Omits the return on economic capital: 13.0 x 0.7 / 60 = 15.17%.
- **C:** Omits the expected loss: 19.8 x 0.7 / 60 = 23.10%.
- **D:** Omits tax: 14.8/60 = 24.67%.

Q423 Schweser Module Quiz 59.1 Q2

REBUILT — Two distractors (12.33%, 14.66%) did not correspond to identifiable errors; replaced with error-derived values (answer unchanged).

Information for a commercial loan portfolio: $1.2 billion principal; 6% pre-tax expected return; direct annual operating costs of $8 million; funded by $1.2 billion of retail deposits at 4%; expected loss 0.4% of principal per annum; unexpected loss (economic capital) of 7% of principal; risk-free rate 1%; effective tax rate 30%; no transfer-pricing issues. Which amount is closest to the RAROC?

A. 9.33%
B. 10.03%
C. 14.03%
D. 14.33%

Answer: B

Step by step

1. Revenue = 6% x 1,200 = $72.0m; interest = 4% x 1,200 = $48.0m; EL = 0.4% x 1,200 = $4.8m; opex = $8.0m.
2. EC = 7% x 1,200 = $84.0m; return on EC = 1% x 84 = $0.84m.
3. Pre-tax = 72 - 8 - 48 - 4.8 + 0.84 = $12.04m.
4. After tax = 12.04 x 0.70 = $8.428m.
5. RAROC = 8.428 / 84 = 10.03%.

Why B is correct: RAROC divides after-tax risk-adjusted income by economic capital, here equal to unexpected loss.

Why the others are wrong

- **A:** Omits the return on economic capital: 11.2 x 0.7 / 84 = 9.33%.
- **C:** Omits expected loss: 16.84 x 0.7 / 84 = 14.03%.
- **D:** Omits tax: 12.04/84 = 14.33%.

Q424 AnalystPrep Q.2553

REBUILT — Source divided by risk-weighted assets (not economic capital), omitted expected losses and taxes, and gave a pre-tax figure; rebuilt as a standard RAROC calculation with economic capital in the denominator.

A bank project generates revenues of $20,000 and incurs expenses of $10,000 and expected losses of $2,000. The tax rate is 30%, and the economic capital allocated to the project is $40,000. The project's RAROC is closest to:

A. 14.0%
B. 17.5%
C. 20.0%
D. 28.0%

Answer: A

Key correction: Source key (AnalystPrep) said B (0.33) by dividing pre-tax income by risk-weighted assets; the original item mixed risk-weighted assets with risk capital and ignored expected losses and tax, so it was rebuilt rather than re-keyed.

Step by step

1. RAROC = after-tax expected risk-adjusted net income / economic capital.
2. Pre-tax risk-adjusted income = 20,000 - 10,000 - 2,000 = 8,000.
3. After tax: 8,000 x (1 - 0.30) = 5,600.
4. RAROC = 5,600 / 40,000 = 0.14 = 14.0%.

Why A is correct: RAROC measures after-tax income net of expected losses per dollar of economic (risk) capital, so 5,600 / 40,000 = 14.0%.

Why the others are wrong

- **B:** 17.5% = 7,000/40,000: subtracts tax from 10,000 income but forgets to deduct expected losses.
- **C:** 20.0% = 8,000/40,000: deducts expected losses but forgets the 30% tax.
- **D:** 28.0% = 5,600/20,000: divides by revenues instead of economic capital.

LO 59.d

Explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level.

Q425

BT answers P2.T7.24.12.2

REBUILT — Statement 1 (longer horizon 'dilutes' short-term actions) is not supported by the text and the option 'All three are NOT correct' was awkward; restated statements to match the reading's challenges and cleaned options.

Leo, a CRO at Inquisitive Bank, summarizes challenges in using RAROC for performance measurement:
Statement 1: Risk and return data for periods beyond one year tend to be of low quality, which is a practical difficulty when a longer time horizon is chosen for RAROC.
Statement 2: Using point-in-time (PIT) default probabilities makes RAROC and economic capital more volatile, because ratings and PDs respond to current economic conditions.
Statement 3: Choosing a lower confidence level makes the economic capital assessment more conservative.
Which of the statements is NOT correct?

A. Statement 1 only
B. Statement 3 only
C. Statement 2 only
D. Statements 1 and 3 only

Answer: B

Step by step

1. Statement 1: GARP notes that although a longer horizon (5-10 years) could capture the business cycle, risk and return data beyond one year may be of low quality. Correct.
2. Statement 2: a firm is less likely to keep its rating under PIT than TTC, so PIT increases the volatility of economic capital and RAROC. Correct.
3. Statement 3: the confidence level expresses risk appetite/target rating; a lower confidence level reduces the capital allocated (especially for operational, credit and settlement risks), so it is less, not more, conservative. Incorrect.
4. Only Statement 3 is not correct.

Why B is correct: A lower confidence level cuts the tail loss that economic capital must cover, so it lowers capital and is less conservative; the other two statements match the reading.

Why the others are wrong

- **A:** Treats Statement 1 as false by assuming longer horizons are preferable; but data beyond one year is low quality.
- **C:** Treats PIT as stabilizing; in fact PIT PDs are more volatile than TTC.
- **D:** Combines the Statement 1 error with the Statement 3 error; Statement 1 is actually correct.

Q426

BT answers P2.T7.25.1.4

REBUILT — OCR-damaged table figures were reconstructed from arithmetic, the data was made internally consistent (shortfall $3.4B), and the options were re-worked so exactly one is defensible and tests the confidence-level/target-rating link.

ETA Bank targets an AA rating (about 3 bps one-year default probability, 99.97% confidence), but its economic capital model uses a 99.9% VaR (about 10 bps, closer to an A rating). Diversification across credit, market and operational risk reduces the sum of stand-alone VaRs by 20% at both confidence levels.

Metric	99.9% VaR	99.97% VaR
Credit risk VaR	$10B	$12B
Market risk VaR	$6B	$7.2B
Operational risk VaR	$4B	$5B
Capital held	$16B	$16B

Which of the following is the correct assessment and response for the bank's risk manager?

A. Required economic capital at 99.97% is about $19.4B, a shortfall of about $3.4B; the bank should raise capital or reduce risk exposures.
B. Required economic capital at 99.97% is $24.2B, a shortfall of $8.2B; the bank should raise capital or reduce risk exposures.
C. Required economic capital at 99.97% is about $19.4B, a shortfall of about $3.4B; the bank should assume lower correlations between risk types to remove the gap.
D. There is no shortfall because the bank holds $16B, which covers the 99.9% requirement; the bank should accept an A-equivalent standard.

Answer: A

Step by step

1. Sum of stand-alone VaRs at 99.97%: 12 + 7.2 + 5 = 24.2 ($B).
2. After the 20% diversification benefit: 24.2 x 0.8 = 19.36, about $19.4B.
3. Shortfall = 19.4 - 16 = about $3.4B.
4. The confidence level must match the target rating (AA: 99.95-99.97%), so the gap is closed by raising capital or cutting risk, not by weakening assumptions or lowering the target.

Why A is correct: Economic capital at the confidence level implied by the AA target is $19.4B after diversification; the shortfall must be closed with real capital or lower risk.

Why the others are wrong

- **B:** Ignores the 20% diversification benefit (24.2 - 16 = 8.2).
- **C:** Right numbers but artificially lowers correlations to shrink required capital; this is unjustified and defeats the AA objective.
- **D:** Compares capital to the 99.9% requirement, abandoning the AA target the bank set.

Q427 Schweser Module Quiz 59.2 Q1

Which of the following statements regarding the computation of economic capital is correct?
I. Selecting a longer time horizon for RAROC calculations is preferable because risk and return data is more reliable with more time.
II. Choosing a lower confidence level is not likely to reduce the amount of risk capital required if the firm has little exposure to operational, credit, and settlement risks.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: B

Step by step

1. Statement I: practitioners use a one-year horizon (business planning cycle, recapitalization time); risk and return data beyond one year is of low quality. False.
2. Statement II: a lower confidence level sharply reduces risk capital mainly when the firm is dominated by operational, credit and settlement risks, where large losses are rare. With little such exposure the reduction is small. True.
3. Only II is correct.

Why B is correct: The confidence-level effect is large only for rare-large-loss risk types, while longer-horizon data is less reliable, not more.

Why the others are wrong

- **A:** Assumes more data means better data; beyond one year the quality is questionable.
- **C:** Accepts I, which is false.
- **D:** Rejects II, missing that the capital reduction depends on exposure to rare large-loss risks.

Q428 Schweser Module Quiz 59.2 Q2

Which of the following statements regarding the choice of default probability approaches in computing economic capital is correct?

A. A through-the-cycle (TTC) approach should be used to price financial instruments with credit risk exposure.
B. A point-in-time (PIT) approach is more commonly used for computations involving profitability and strategic decisions.
C. A TTC approach is more likely to result in a lower volatility of capital compared to the PIT approach.
D. A firm's rating will not change when analyzed under the PIT approach versus the TTC approach.

Answer: C

Step by step

1. PIT PDs respond to current conditions, so a firm's rating is more likely to change under PIT than TTC.
2. Hence TTC gives lower volatility of economic capital.
3. PIT suits pricing credit-risk instruments and near-term expected loss; TTC suits economic capital, current profitability and strategic decisions.

Why C is correct: TTC PDs, used by rating agencies, smooth the cycle, so ratings and economic capital are more stable than under PIT.

Why the others are wrong
- **A:** Reverses the uses: PIT, not TTC, is used for pricing credit-risk instruments.
- **B:** Reverses the uses: TTC is more common for profitability and strategic decisions.
- **D:** Ratings do change; the probability of staying in the same rating is lower under PIT than TTC.

LO 59.e

Calculate the hurdle rate and apply this rate in making business decisions using RAROC.

Q429

BT answers P2.T7.24.12.3 · BT notes P2.T7.24.12.3

The CFO of Avenues Lending has requested the calculation of the hurdle rate for loan portfolio managers as part of the Loan Portfolio department's annual appraisal. The department's RAROC is 6.8%. The market value of common equity is $200 million and the market value of preferred equity is $800 million. A market study provides: equity beta 1.25; market return 8%; risk-free rate 5%; dividend on preferred shares $48 million. The Board is evaluating whether performance fees should be based on meeting the risk-adjusted hurdle rate.
What is the hurdle rate, and what is the implication?

A. 8.75%, the department is generating sufficient returns to cover the cost of capital
B. 6.55%, the department is not generating sufficient returns to cover the cost of capital
C. 8.75%, the department is not generating sufficient returns to cover the cost of capital
D. 6.55%, the department is generating sufficient returns to cover the cost of capital

Answer: D

Step by step
1. Cost of common equity (CAPM): $r_{CE} = 5\% + 1.25(8\% - 5\%) = 8.75\%$.
2. Cost of preferred equity = dividend / market value = $48/800 = 6.0\%$.
3. Hurdle rate: $h_{AT} = \frac{200(8.75\%)+800(6.0\%)}{200+800} = \frac{17.5+48}{1000} = 6.55\%$.
4. Decision rule: RAROC 6.8% > hurdle 6.55%, so the department creates value and covers its cost of capital.

Why D is correct: The hurdle rate is the market-value-weighted average cost of common and preferred equity; RAROC above it signals value creation.

Why the others are wrong
- **A:** Uses the cost of common equity alone (8.75%) and ignores the 80% preferred weight; the stated conclusion is also inconsistent with 6.8% < 8.75%.
- **B:** Correct hurdle (6.55%) but wrong implication: 6.8% exceeds 6.55%, so returns are sufficient.
- **C:** Uses the cost of common equity only (8.75%) as the hurdle, ignoring the preferred equity; the weighted hurdle is 6.55%.

Q430

BT notes P2.T5.507.1

REBUILT — Original risk-free-project funding-cost item relied on non-syllabus reasoning (comparing a risk-free project to the risk-free rate); rebuilt as a hurdle-rate calculation with a RAROC decision rule.

A bank is funded by common equity with a market value of USD 600 million and preferred equity with a market value of USD 400 million. The risk-free rate is 4%, the expected market return is 9%, the bank's equity beta is 1.2, and the preferred shares yield 5%. A business unit proposes to continue an activity whose after-tax RAROC is 7.8%.
What is the bank's hurdle rate, and what should the bank do with the activity?

A. 8.0%; discontinue or reject the activity because RAROC is below the hurdle rate
B. 8.0%; continue the activity because RAROC exceeds the hurdle rate
C. 10.0%; discontinue or reject the activity because RAROC is below the hurdle rate
D. 7.5%; continue the activity because RAROC exceeds the hurdle rate

Answer: A

Step by step
1. Cost of common equity: $r_{CE} = 4\% + 1.2(9\% - 4\%) = 10.0\%$.
2. Cost of preferred equity = yield = 5.0%.
3. Hurdle rate: $h_{AT} = \frac{600(10\%)+400(5\%)}{1000} = 8.0\%$.
4. RAROC 7.8% < 8.0%, so the activity destroys value and should be rejected or discontinued.

Why A is correct: The market-value-weighted cost of equity capital is 8.0%, and RAROC below the hurdle rate indicates value destruction.

Why the others are wrong

- **B:** Correct hurdle but reverses the decision rule; 7.8% is below 8.0%.
- **C:** Uses only the CAPM cost of common equity (10.0%), ignoring the cheaper preferred equity in the capital structure.
- **D:** Takes a simple unweighted average of 10% and 5% (7.5%) instead of weighting by market value, which wrongly makes RAROC exceed the hurdle.

Q431

GARP PE2 Q17

A risk analyst in the RAROC group at Firm QZK is calculating the hurdle rate to be used in assessing whether some of the firm's business units are adding value. Firm QZK is funded only with common equity and preferred equity.
Market value of common equity: GBP 20 billion
Market value of preferred equity: GBP 8 billion
Dividend on preferred equity as a percentage of par value: 6.0%
Current yield on preferred equity: 4.5%
Expected return of the market portfolio: 8%
Volatility of market returns: 20%
Volatility of Firm QZK's common equity returns: 14%
Correlation between market return and QZK common equity return: 0.65
The analyst uses the CAPM to determine the cost of common equity and assumes the risk-free rate remains constant at 3%. What is the correct hurdle rate?

A. 5.05%
B. 5.28%
C. 5.48%
D. 7.64%

Answer: A

Step by step

1. Beta $= \rho\,\sigma_{CE}/\sigma_M = 0.65 \times 0.14/0.20 = 0.455$.
2. Cost of common equity: $3\% + 0.455(8\% - 3\%) = 5.275\%$.
3. Cost of preferred equity = current yield 4.5% (not the 6.0% coupon).
4. $h_{AT} = \frac{20(5.275\%)+8(4.5\%)}{28} = \frac{105.5+36}{28} = 5.05\%$.

Why A is correct: Beta comes from correlation times the volatility ratio, and the preferred cost is its market yield; the two costs are then weighted by market value.

Why the others are wrong

- **B:** Cost of common equity alone (5.275%), ignoring the preferred equity weight.
- **C:** Uses the 6.0% coupon as the preferred cost: $(20 \times 5.275 + 8 \times 6)/28 = 5.48\%$; the cost of preferred is its current yield.
- **D:** Inverts the volatility ratio in beta (0.65 x 0.20/0.14 = 0.929), giving a common cost of 7.64% and ignoring preferred equity.

Q432

Mock 1 Q36

Following the calculation of a business line's RAROC, a risk analyst must determine the hurdle rate against which it should be compared. The bank's capital structure and market data are:

Item	Value
Common equity (market value)	USD 800 million
Preferred equity (market value)	USD 200 million
Risk-free rate	3.0%
Expected market return	9.0%
Equity beta	1.30
Preferred dividend yield	6.0%

What is the bank's hurdle rate?

A. 6.00%
B. 8.40%
C. 9.84%

D. 10.80%

Answer: C
Step by step
1. Cost of common equity: $3\% + 1.30(9\% - 3\%) = 10.80\%$.
2. Cost of preferred equity = 6.0%.
3. Weights: 800/1000 = 80% common, 20% preferred.
4. $h_{AT} = 0.80(10.80\%) + 0.20(6.0\%) = 8.64\% + 1.20\% = 9.84\%$.

Why C is correct: The hurdle rate is the market-value-weighted average of the CAPM cost of common equity and the preferred yield.
Why the others are wrong
- **A:** Preferred cost alone, ignoring common equity.
- **B:** Unweighted average of 10.80% and 6.0% (8.40%), ignoring the 80/20 market-value weights.
- **D:** Cost of common equity alone, ignoring the preferred equity.

LO 59.f

Calculate the adjusted RAROC for a project to determine its viability.

Q433
AP Mock 2 Q52

An entrepreneur is considering a project with an estimated RAROC of 12%. The company has an equity beta of 1.2, the return on risk-free Treasury bills is 3%, and the expected market return is 10% per annum.
Using the adjusted RAROC criterion, the company should:

A. Accept the project because the ARAROC is lower than the market expected excess return.
B. Reject the project because the ARAROC is lower than the market expected excess return.
C. Accept the project because the ARAROC is higher than the risk-free rate of return.
D. Reject the project because the ARAROC is higher than the market expected excess return.

Answer: C
Step by step
1. $\text{ARAROC} = \text{RAROC} - \beta_E(R_M - R_F)$ with market premium $10\% - 3\% = 7\%$.
2. $\text{ARAROC} = 12\% - 1.2 \times 7\% = 12\% - 8.4\% = 3.6\%$.
3. Benchmark is the risk-free rate: $3.6\% > 3\%$, so accept.

Why C is correct: Once RAROC is adjusted for systematic risk, the hurdle is the risk-free rate, and 3.6% exceeds 3%.
Why the others are wrong
- **A:** Right decision, wrong reason: ARAROC is not compared with the market excess return (7%), and being lower than a benchmark would not justify accepting.
- **B:** Uses the wrong benchmark (7% market excess return) to reach reject; the correct benchmark is the risk-free rate.
- **D:** ARAROC (3.6%) is not above the 7% market excess return, and that is not the benchmark anyway.

Q434
GARP PE2 Q80 · Mock 4 Q81

A packaging materials manufacturer is considering a project that has an estimated RAROC of 12%. Suppose that the risk-free rate is 4% per year, the expected market rate of return is 10% per year, and the company's equity beta is 1.6. The manufacturer uses the adjusted RAROC metric as the criterion to decide whether or not to accept the project. Which of the following correctly describes the decision the company should make and the rationale for making that decision?

A. Reject the project because the adjusted RAROC is higher than the market expected excess return.
B. Accept the project because the adjusted RAROC is higher than the market expected excess return.
C. Reject the project because the adjusted RAROC is lower than the risk-free rate.
D. Accept the project because the adjusted RAROC is lower than the risk-free rate.

Answer: C

Step by step

1. ARAROC = RAROC $-\beta_E(R_M - R_F)$, market premium $10\% - 4\% = 6\%$.
2. ARAROC $= 12\% - 1.6 \times 6\% = 12\% - 9.6\% = 2.4\%$.
3. $2.4\% < R_F = 4\%$, so reject.

Why C is correct: Adjusted RAROC must exceed the risk-free rate; 2.4% falls short of 4%.

Why the others are wrong

- **A:** ARAROC (2.4%) is below, not above, the 6% market premium, and that is not the decision benchmark.
- **B:** Wrong benchmark and wrong decision: unadjusted RAROC of 12% exceeds 6%, which tempts acceptance.
- **D:** Correctly finds ARAROC below the risk-free rate but accepts, contradicting the rule.

Q435

AnalystPrep Q.2735

Given that the RAROC on a project is 12%, the risk-free rate is 4%, the return on the market portfolio is 10%, and the firm's equity beta is 1.25, calculate the adjusted RAROC for the project and determine whether it should be accepted or rejected.

A. 6.4%; rejected
B. 4.5%; accepted
C. 6.0%; accepted
D. 6.0%; rejected

Answer: B

Step by step

1. ARAROC = RAROC $-\beta_E(R_M - R_F)$.
2. ARAROC $= 12\% - 1.25 \times (10\% - 4\%) = 12\% - 7.5\% = 4.5\%$.
3. $4.5\% > R_F = 4\%$, so accept.

Why B is correct: Subtracting beta times the market premium removes systematic risk; the result (4.5%) exceeds the 4% risk-free rate.

Why the others are wrong

- **A:** 6.4% does not follow from the formula, and rejecting is also wrong since the correct ARAROC exceeds 4%.
- **C:** 6.0% is the market premium alone (or 12% - 6%), omitting beta; accept is the right decision but the figure is wrong.
- **D:** Same 6.0% error (ignoring beta), and rejecting a figure above 4% is inconsistent with the rule.

Q436

AnalystPrep Q.3136

REBUILT — Original asked how much ARAROC must 'increase' (answer 0, junk options); rebuilt as the minimum RAROC needed for acceptance.

Sigma Inc. has an equity beta of 1.18. The risk-free rate is 2% and the expected market return is 7.932%. A proposed project has the same beta as Sigma Inc. What is the minimum RAROC the project must offer for it to be accepted under the adjusted RAROC criterion?

A. 2.00%
B. 5.93%
C. 7.00%
D. 9.00%

Answer: D

Step by step

1. Accept if RAROC $-\beta_E(R_M - R_F) > R_F$.
2. Market premium $= 7.932\% - 2\% = 5.932\%$; $\beta_E \times 5.932\% = 1.18 \times 5.932\% = 7.00\%$.
3. Required RAROC $> R_F + 7.00\% = 2\% + 7.00\% = 9.00\%$.

Why D is correct: ARAROC exceeds the risk-free rate only if RAROC covers the risk-free rate plus the beta-scaled market premium.

Why the others are wrong

- **A:** Treats the risk-free rate as the hurdle for unadjusted RAROC, ignoring the systematic risk charge.
- **B:** Uses the market premium (5.932%) without multiplying by beta and without adding the risk-free rate.
- **C:** Computes only the systematic risk charge (7.00%) and forgets to add the risk-free rate.

Q437

AnalystPrep Q.3207

REBUILT — Distractors did not correspond to identifiable errors; replaced with realistic mistakes.

Larry Sing is evaluating an investment in Hudson Petroleum. Its RAROC is 17%, its beta is 1.2, the return on the market is 12%, and the risk-free rate is 8%. What is the adjusted RAROC for Hudson?

A. 13.0%
B. 9.0%
C. 12.2%
D. 4.8%

Answer: C

Step by step

1. ARAROC = RAROC $-\ \beta_E(R_M - R_F)$.
2. Market premium = 12% − 8% = 4%; systematic charge = 1.2 × 4% = 4.8%.
3. ARAROC = 17% − 4.8% = 12.2%.

Why C is correct: Adjusted RAROC deducts beta times the market premium from RAROC.

Why the others are wrong

- **A:** Subtracts the market premium (4%) without multiplying by beta.
- **B:** Subtracts the risk-free rate (17% - 8%) instead of the systematic risk charge.
- **D:** Reports the systematic risk charge (4.8%) instead of subtracting it from RAROC.

Q438

AnalystPrep Q.3209

REBUILT — Distractors replaced with realistic mistakes; unused required-return detail retained as a red herring.

Henry Campbell is analyzing a new market-expansion project. The project's RAROC is 13%, the risk-free rate is 6%, the market return is 14%, the firm's required return on equity is 12%, and the firm's beta is 1.5. What is the ARAROC and should the project be accepted?

A. 7.0%; accept
B. 4.7%; reject
C. 1.0%; accept
D. 1.0%; reject

Answer: D

Step by step

1. ARAROC = RAROC $-\ \beta_E(R_M - R_F)$; market premium = 14% − 6% = 8%.
2. ARAROC = 13% − 1.5 × 8% = 13% − 12% = 1.0%.
3. 1.0% $< R_F =$ 6%, so reject (the 12% required return is not used).

Why D is correct: ARAROC of 1.0% is far below the 6% risk-free hurdle.

Why the others are wrong

- **A:** Subtracts only the risk-free rate (13% - 6%) and accepts because 7% exceeds 6%.
- **B:** Computes (13% − 6%)/1.5 = 4.7%, the beta-scaled variant, which is not ARAROC.
- **C:** Correct 1.0% but accepts; it is below 6% (merely positive is not enough).

Q439

BT answers P2.T7.24.13.1

REBUILT — Cost items were ambiguous (bases for deposits, transfer cost, EC); restated every item in dollars.

A bank is considering a $200 million loan portfolio. Economic capital is $100 million (50% of the loan), invested to earn a pre-tax return of 5% ($5 million). Annual figures: interest earned on loans $20 million; interest paid on deposits $8 million; direct operating costs $1.6 million; transfer costs $2 million; expected loss $4 million. The effective tax rate is 32%. Market data: expected market return 7%, risk-free rate 3%, equity beta 1.05. The board's stated hurdle rate is 6%. Based on adjusted RAROC, does the project add value?

A. Yes, because adjusted RAROC is greater than the hurdle rate of 6%.
B. No, because adjusted RAROC is less than the risk-free rate of 3%.
C. Yes, because adjusted RAROC is greater than the risk-free rate of 3%.
D. No, because adjusted RAROC is less than the hurdle rate of 6%.

Answer: B

Step by step

1. Pre-tax profit $= 20 + 5 - 8 - 1.6 - 2 - 4 = 9.4$ million; after tax $= 9.4 \times 0.68 = 6.392$.
2. RAROC $= 6.392/100 = 6.39\%$.
3. Systematic charge $= 1.05 \times (7\% - 3\%) = 4.2\%$.
4. ARAROC $= 6.39\% - 4.2\% = 2.19\%$.
5. $2.19\% < R_F = 3\%$, so reject.

Why B is correct: After removing the systematic risk charge, 2.19% does not beat the 3% risk-free rate.

Why the others are wrong

- **A:** Compares unadjusted RAROC (6.39%) with the 6% hurdle, skipping the beta adjustment.
- **C:** Uses the right benchmark but 2.19% is below 3%, not above.
- **D:** Right conclusion for the wrong benchmark: ARAROC is compared with the risk-free rate, not the 6% hurdle.

Q440

BT answers P2.T7.607.1 · BT notes P2.T7.607.1

REBUILT — Options listed only RAROC while the stem asked for adjusted RAROC; options restated to give ARAROC and decision.

A bank's \$2.0 billion corporate loan portfolio earns 6.0% per annum (\$120 million). Expected loss is 1.5% (\$30 million). The portfolio is funded by \$2.0 billion of deposits with a transfer-priced charge of 2.00% (\$40 million). Direct operating costs are \$16.0 million and the effective tax rate is 25.0%. Economic capital of \$200 million (10% of the loan amount) is invested in risk-free securities yielding 1.0% (\$2 million). The expected market return is 8.0%, the risk-free rate is 1.0%, and the equity beta is 1.60. What is the adjusted RAROC and is the project advisable?

A. 13.50%; advisable
B. 2.30%; advisable
C. 2.30%; not advisable
D. 0.70%; not advisable

Answer: B

Step by step

1. After-tax income $= (120 - 30 - 40 - 16 + 2) \times 0.75 = 36 \times 0.75 = 27$ million.
2. RAROC $= 27/200 = 13.50\%$.
3. ARAROC $= 13.50\% - 1.60 \times (8\% - 1\%) = 13.50\% - 11.20\% = 2.30\%$.
4. $2.30\% > R_F = 1.0\%$, so the project is advisable.

Why B is correct: After deducting the systematic risk charge of 11.2%, ARAROC of 2.30% still exceeds the 1.0% risk-free rate.

Why the others are wrong

- **A:** Reports unadjusted RAROC rather than adjusted RAROC.
- **C:** Right ARAROC but rejects, though 2.30% is above 1.0%.
- **D:** Uses $R_M = 8\%$ instead of the premium: $13.5\% - 1.6 \times 8\% = 0.70\%$.

Q441

BT notes P2.T7.24.12.3

REBUILT — Source was an open question truncated before its prompt, options and answer; rebuilt as a self-contained MCQ using the given data and the ARAROC decision rule.

A bank is considering a \$200 million investment in a loan portfolio and will judge it using adjusted RAROC (ARAROC). Economic capital (EC) is 50% of the loan amount, and the board is concerned with systematic risk. The internal hurdle rate for the unadjusted RAROC is 6%.

Forecast Data	Value
Interest earned from loans	10%
Pre-tax return on economic capital	5.0%
Interest paid on deposits	4%
Operating direct costs	0.8%
Transfer costs	1%
Expected loss	2.00%
Effective tax rate	32%
Market Data	Value
Equity market return	7%
Risk-free rate	3%
Equity beta	1.05

(All percentages other than EC are on the $200 million loan amount; transfer costs are a charge against the activity.) Which of the following is the correct conclusion?

A. Reject; ARAROC is 2.19%, below the 3% risk-free rate
B. Accept; RAROC is 6.39%, above the 6% hurdle rate
C. Accept; ARAROC is 3.55%, above the 3% risk-free rate
D. Reject; ARAROC is 2.39%, below the 3% risk-free rate

Answer: A

Step by step

1. Amounts on $200m: revenue $0.10 \times 200 = 20$; interest expense $0.04 \times 200 = 8$; operating cost $0.008 \times 200 = 1.6$; transfer cost $0.01 \times 200 = 2$; expected loss $0.02 \times 200 = 4$.
2. EC $= 0.50 \times 200 = 100$; return on EC $= 0.05 \times 100 = 5$.
3. Pre-tax income $= 20 - 8 - 1.6 - 2 - 4 + 5 = 9.4$; after tax $= 9.4 \times (1 - 0.32) = 6.392$.
4. RAROC $= 6.392/100 = 6.392\%$.
5. ARAROC $=$ RAROC $- \beta_E(R_M - R_F) = 6.392\% - 1.05(7\% - 3\%) = 6.392\% - 4.20\% = 2.192\%$.
6. Decision rule: accept only if ARAROC $> R_F = 3\%$. $2.19\% < 3\%$, so reject.

Why A is correct: ARAROC subtracts the CAPM charge for systematic risk, $\beta_E(R_M - R_F)$, so the hurdle becomes the risk-free rate. After this adjustment the project earns 2.19%, less than 3%, so it is rejected even though unadjusted RAROC clears the 6% hurdle.

Why the others are wrong

- **B:** Uses unadjusted RAROC against the 6% hurdle; this ignores the systematic-risk adjustment the board asked for.
- **C:** Omits the 1% transfer cost (after-tax income 7.752, RAROC 7.752%) and then subtracts 4.20%, giving 3.55% and a wrong accept.
- **D:** Subtracts $R_M - R_F = 4\%$ without multiplying by beta (6.392% - 4% = 2.39%); the conclusion matches but the ARAROC figure is wrong.

Q442 BT notes P2.T7.3.4

REBUILT — Original correct option was a clumsy Crouhy-specific paraphrase and one distractor was a near-miss; reworded so the answer follows the GARP/Schweser ARAROC rationale (systematic-risk adjustment, common risk-free hurdle).

Which of the following best describes the advantage of adjusted RAROC (ARAROC) over unadjusted RAROC when evaluating projects?

A. ARAROC is computed from market values, whereas RAROC relies on book values
B. ARAROC is a multi-period measure, whereas RAROC is a single-period measure
C. ARAROC deducts a charge for the project's systematic risk, so that the same hurdle rate (the risk-free rate) applies across all projects and business lines
D. ARAROC replaces economic capital in the denominator with regulatory capital

Answer: C

Step by step

1. Unadjusted RAROC ignores systematic risk, so high-return, high-risk projects tend to pass a hurdle and low-risk projects tend to fail it.
2. ARAROC $=$ RAROC $- \beta_E(R_M - R_F)$ removes the return that merely compensates for nondiversifiable risk.
3. After the adjustment the hurdle becomes R_F for every project: accept if ARAROC $> R_F$.
4. Only option C describes this.

Why C is correct: The adjustment subtracts the CAPM-required excess return for systematic risk, which makes the decision rule consistent across projects with a single risk-free hurdle.

Why the others are wrong

- **A:** ARAROC is not a book-versus-market-value correction; both use the same economic capital framework.
- **B:** Both RAROC and ARAROC are single-period measures here; the adjustment concerns systematic risk, not multiple periods.
- **D:** ARAROC changes the numerator (beta charge), not the capital definition; it does not use regulatory capital.

Q443 BT notes P2.T7.3.5

REBUILT — Source key (and all options) used a ratio form (RAROC - Rf)/beta; rebuilt with GARP/Schweser formula ARAROC = RAROC - beta(Rm - Rf) and fresh options.

The risk-free rate is 2.0% and the expected market return is 8.0%. A bank estimates that a risky project with a beta of 1.60 (with respect to the market) has a RAROC of 11.0%. Is the project advisable if the bank uses adjusted RAROC (ARAROC)?

A. Not advisable; ARAROC is 1.40%, below the 2.0% risk-free rate
B. Not advisable; ARAROC is -1.80%
C. Advisable; ARAROC is 5.00%, above the 2.0% risk-free rate
D. Not advisable; ARAROC is 5.625%, below the 6.0% market excess return

Answer: A

Key correction: Source key (BT notes) said B (ARAROC 5.625%, not advisable); that uses a ratio form (RAROC - Rf)/beta not in GARP/Schweser. The reasoning ends in 'not advisable', but the correct ARAROC is 1.40% vs Rf of 2.0% (option A).

Step by step

1. Market excess return $R_M - R_F = 8\% - 2\% = 6\%$.
2. Systematic-risk charge $= \beta(R_M - R_F) = 1.60 \times 6\% = 9.6\%$.
3. ARAROC $= 11.0\% - 9.6\% = 1.4\%$.
4. Rule: accept if ARAROC $> R_F = 2.0\%$. $1.4\% < 2.0\%$, so the project is not advisable.

Why A is correct: ARAROC $=$ RAROC $- \beta(R_M - R_F)$ and is compared with the risk-free rate; 1.4% is below 2.0%, so reject.

Why the others are wrong

- **B:** Applies beta to the full market return ($11\% - 1.6 \times 8\% = -1.8\%$) instead of the excess return over the risk-free rate.
- **C:** Omits beta ($11\% - 6\% = 5\%$), understating the risk charge and wrongly accepting.
- **D:** Uses a ratio form $(11\% - 2\%)/1.6 = 5.625\%$ and compares with the market excess return; not the ARAROC formula.

LO 59.g

Explain challenges in modeling diversification benefits, including aggregating a firm's risk capital and allocating economic capital to different business lines.

Q444 AnalystPrep Q.5105

REBUILT — Option B (units viewed individually) is actually the usual stand-alone practice described in the text, making it defensible; replaced B with a clearly false statement.

A bank's risk manager wants the bank to look beyond regulatory and economic capital and consider aggregate capital needs. Which statement about risk capital aggregation and diversification is correct?

A. Diversification can be achieved only across different risk classes, never within one risk class.
B. The diversification benefit of the firm can be allocated back to business units in an objective way, with no subjectivity.
C. A diversification benefit exists whenever the correlation between two risks is exactly +1.
D. Large diversification benefits can arise when operational risk is aggregated with other risks.

Answer: D

Step by step

1. Aggregate capital lies between the zero-correlation figure (square root of sum of squares) and the perfect-correlation figure (simple sum).
2. Operational risk is relatively weakly related to market and credit risk, so correlation is well below +1 and the aggregate is well below the sum: a large benefit (D).
3. With $\rho = +1$ the aggregate equals the sum, so no benefit (C wrong).
4. Benefits also arise within a risk class and across business activities (A wrong); allocation of the benefit to units is highly subjective (B wrong).

Why D is correct: Diversification benefit requires correlation below +1; operational risk's low correlation with other risks gives large benefits when aggregated.

Why the others are wrong

- **A:** Diversification also arises across business lines and activities within a risk type; it is not limited to different risk classes.
- **B:** Allocating the benefit is subjective and affects unit performance measures; it is not objective.
- **C:** At correlation +1 the aggregate equals the simple sum, so there is zero benefit; the mistake is reversing the relationship.

Q445

BT answers P2.T7.24.13.2

REBUILT — OCR-damaged data and a muddled 'increase allocation' framing; restated the data (rho = -0.1162) and asked for the pro rata fully diversified allocation with distinct, realistic distractors.

Two departments have 99% stand-alone VaRs of $90 million (Consumer Credit Cards) and $110 million (Consumer Purchase Loans), with a correlation of -0.1162. The firm computes the combined VaR with the square-root-of-variance formula and allocates the diversification benefit pro rata to stand-alone capital. What is the fully diversified capital of each department (Cards, Loans)?

A. $29.80 million and $36.42 million
B. $60.20 million and $73.58 million
C. $23.79 million and $43.79 million
D. $90.00 million and $110.00 million

Answer: B

Step by step

1. Combined VaR = $\sqrt{90^2 + 110^2 + 2(-0.1162)(90)(110)} = \sqrt{17{,}899.24} = 133.79$.
2. Diversification benefit = 90 + 110 - 133.79 = 66.21.
3. Pro rata share: Cards $66.21 \times 90/200 = 29.80$; Loans $66.21 \times 110/200 = 36.42$.
4. Fully diversified capital: Cards 90 - 29.80 = 60.20; Loans 110 - 36.42 = 73.58 (sum 133.78, equal to 133.79 apart from rounding).

Why B is correct: Fully diversified capital is stand-alone capital minus the pro rata share of the diversification benefit; the allocations sum to the combined VaR.

Why the others are wrong

- **A:** These are the benefit shares themselves, not the capital left after subtracting them.
- **C:** These are marginal (incremental) capital figures: 133.79 - 110 = 23.79 and 133.79 - 90 = 43.79; they do not sum to the combined VaR.
- **D:** Stand-alone capital ignores diversification entirely, summing to $200 million.

Q446

BT answers P2.T7.607.2 · BT notes P2.T7.607.2

A business unit has two activities, X and Y, with the following risk capital measures:

Activities within BU	X	Y	Sum
Stand-alone capital	$70.0	$80.0	
Fully diversified capital	56.0	64.0	120.0
Incremental (marginal) capital	40.0	50.0	

The diversification effect of $30 (= 70 + 80 - 120) is allocated pro rata to stand-alone capital. Each of the following statements is true EXCEPT:

A. Given that BU risk capital is $120, the implied correlation between the activities must be zero.
B. Fully diversified capital ($56 for X and $64 for Y) should be used to assess the firm's solvency and minimum risk pricing.
C. Incremental capital ($40 for X and $50 for Y) should be used for active portfolio management and business mix decisions.
D. Stand-alone capital can be used for incentive compensation, while fully diversified capital can be used to assess the extra performance generated by diversification.

Answer: A

Step by step

1. Zero correlation would give BU capital $\sqrt{70^2 + 80^2} = 106.3$, not 120.
2. Solving $120^2 = 70^2 + 80^2 + 2\rho(70)(80)$ gives $\rho = 3{,}100/11{,}200 = 0.277$: positive, so A is false.
3. B, C, D match the standard uses: fully diversified for solvency/minimum pricing, marginal for business mix decisions, stand-alone for incentive pay with fully diversified showing the diversification effect.

Why A is correct: BU capital of 120 exceeds the zero-correlation value 106.3, so the correlation is about +0.28, not zero.

Why the others are wrong

- **B:** True statement of the recommended use of fully diversified capital.
- **C:** True; marginal capital fully reflects diversification and suits business mix decisions.
- **D:** True; stand-alone for incentive pay, fully diversified for measuring diversification-driven performance.

LO 59.h

Explain best practices in implementing an approach that uses RAROC to allocate economic capital.

Q447 AnalystPrep Q.2212

Kimberley Excavations, a diamond-mining company from South Africa, has implemented a RAROC (risk-adjusted return on capital) system for future strategic investments. Its diamond mines have shown decreasing yield with sharp rises and drops. Mine management is deeply dissatisfied with the new system, complaining that RAROC unfairly attributes too much economic capital (EC) to their businesses. What is the correct course of action for senior management?

A. Neglect dissatisfaction in the local management and enforce the RAROC system at all costs.
B. Display commitment to RAROC, expand in-house communication and education regarding the system.
C. Disregard RAROC and return to the old, tried and proven risk management system.
D. Allocate resources to each of the businesses for purpose of stricter employee monitoring.

Answer: B

Step by step

1. Identify the issue: business-unit complaint that economic capital attribution is excessive.
2. Best practice requires senior management commitment AND communication and education so the capital allocation process is seen as fair and transparent.
3. Therefore management should stay committed to RAROC while explaining the methodology and opening dialogue with unit leaders.

Why B is correct: Best practice pairs visible top-management commitment with transparent communication and education to win buy-in and address concerns about excessive EC attribution.

Why the others are wrong

- **A:** Ignoring concerns breeds resistance; best practice calls for open dialogue and transparency, not forced enforcement.
- **C:** Abandoning RAROC loses risk-adjusted insight; the problem is acceptance, not the methodology's existence.
- **D:** Stricter monitoring of employees does not address the fairness-of-EC-attribution complaint.

Q448 BT answers P2.T7.24.13.3

Financial Frontier Bank (FFB) has implemented a RAROC system to align risk and capital across its business units. After a year, which outcome is LEAST likely to result from the RAROC implementation?

A. Increase in disputes over economic capital allocations
B. Reallocation of resources from lower to higher RAROC units
C. Increased activity in the parameter review group
D. Decrease in the frequency of risk parameter reviews

Answer: D

Step by step

1. RAROC makes capital allocation explicit, so units dispute their economic capital (A likely).
2. RAROC acts as an internal capital market, shifting scarce resources toward higher-RAROC units (B likely).
3. Key parameters (PD, migration, LGD, volatility, correlation) drive EC, so a parameter review group is used more, not less (C likely, D unlikely).

Why D is correct: RAROC makes economic capital depend on risk parameters, so these must be reviewed periodically; reviews become more, not less, frequent.

Why the others are wrong

- **A:** Likely: explicit EC charges provoke debate over fairness of allocations.
- **B:** Likely: RAROC works like an internal capital market that rewards higher-RAROC units.
- **C:** Likely: the parameter review group gains importance as EC parameters are scrutinized.

Q449 BT answers P2.T7.607.3 · BT notes P2.T7.607.3

According to Crouhy, Galai, and Mark, best practice in the implementation of a RAROC system includes each of the following elements EXCEPT which one?

A. Given the strategic nature of the decisions steered by a RAROC system, the marching orders must come from the top management of the firm; specifically, the CEO and his or her executive team should sponsor the implementation of a RAROC system and should be active in its diffusion

B. In order to preserve the integrity and transparency of the RAROC system, there should be an exclusively quantitative decision rule for activities: if the ex-ante RAROC does not exceed the firm's hurdle rate (if the RAROC return is "low"), the firm should exit the activity

C. The value at risk (VaR) methodologies for measuring market risk and credit risk that underpin RAROC calculations are "generally well-accepted" by business units (although this is not yet true for operational risk); in practice, disagreements concern the setting of the parameters that feed into these models which determine the size of economic capital

D. Balance sheet requests from the business units, such as economic capital, leverage ratio, liquidity ratios, and risk-weighted assets, should be channeled to the RAROC group every quarter; limits are then set for economic capital, leverage ratio, liquidity ratios, and risk-weighted assets

Answer: B

Step by step

1. Recall the six recommendations: senior management commitment; communication and education; ongoing consultation (parameter review group); data quality control; combine RAROC with qualitative factors; active capital management.
2. A matches recommendation 1, C matches the ongoing-consultation discussion, D matches active capital management (quarterly requests, limits set).
3. B contradicts recommendation 5: besides RAROC vs. the hurdle rate, qualitative factors (quality of earnings, strategic importance, growth potential, synergies) must be considered, so an exclusively quantitative exit rule is not best practice.

Why B is correct: Best practice requires combining RAROC (quantity of earnings) with qualitative quality-of-earnings factors; an exclusively quantitative exit rule ignores strategic value, so B is the incorrect element.

Why the others are wrong

- **A:** True: CEO and executive team sponsorship is recommendation 1.
- **C:** True: VaR for market/credit risk is well accepted; disputes center on parameters, and operational risk is less accepted.
- **D:** True: quarterly balance-sheet requests to the RAROC group and subsequent limit-setting is the active capital management process.

Q450 Schweser Module Quiz 59.3 Q1

Which of the following statements regarding best practices in implementing a RAROC approach is correct?

A. A successful RAROC approach is focused on maximizing profits earned by the firm.

B. A restriction on the firm's growth due to leverage limitations may result in higher profits.

C. The data collection process throughout the firm should be decentralized to allow the various business units to ensure the utmost accuracy of data.

D. Metrics involving credit risk, market risk, and operational risk to compute economic capital are generally clearly defined and may be computed objectively.

Answer: B

Step by step

1. A: GARP stresses that what counts is not how much income is generated but how well the firm is compensated for the risks taken, so A is false.
2. C: the RAROC group is accountable for centralized data collection, calculation and reporting; units are only accountable for the data they produce, so C is false.

3. D: credit and market metrics are reasonably defined, but operational risk metrics are subjective and debated, so D is false.
4. B: leverage limits that restrain growth force the firm to work every dollar of capital hard (RAROC analysis helps), which can improve profitability, so B is correct.

Why B is correct: Leverage limits constrain growth but push the firm to allocate scarce capital to the best risk-adjusted uses, which can raise profits.

Why the others are wrong

- **A:** Confuses RAROC with profit maximization; the focus is return relative to risk taken.
- **C:** Data collection should be centralized with edit and reasonability checks, with the RAROC group accountable.
- **D:** Operational risk metrics are not well defined and involve significant subjectivity; only market/credit are comparatively accepted.

Reading 60. Range of Practices and Issues in Economic Capital Frameworks

"Range of Practices and Issues in Economic Capital Frameworks" (Basel Committee on Banking Supervision Publication, March 2009).

LO	Learning objective	Qs
60.a	within the economic capital implementation framework, describe the challenges that appear in: defining and calculating risk measures risk aggregation validation of models dependency modeling in credit risk evaluating counterparty credit risk assessing interest rate risk in the banking book	35
60.b	describe the recommendations by the Bank for International Settlements (BIS) that supervisors should consider in order to make effective use of internal risk measures, such as economic capital, that are not designed for regulatory purposes.	5
60.c	explain benefits and impacts of using an economic capital framework within the following areas: credit portfolio management risk-based pricing customer profitability analysis management incentives	6
60.d	describe best practices and assess key concerns for the governance of an economic capital framework.	3

LO 60.a

Within the economic capital implementation framework, describe the challenges that appear in: defining and calculating risk measures risk aggregation validation of models dependency modeling in credit risk evaluating counterparty credit risk assessing interest rate risk in the banking book

Q451

BT notes P2.T5.22.13.3

A risk analyst must aggregate potential loss distributions across several business lines, each with its own marginal loss distribution, and plans to use a copula. Each of the following is a true advantage of the copula approach EXCEPT:

A. The copula approach allows simulation of the entire distribution of aggregate losses, not only the tails.

B. Given the same marginal distributions, she can test (simulate) different copula functions to see the effect of the assumed dependence structure.

C. The marginals may be skewed or heavy-tailed, because a copula places no requirement on the form of the univariate distributions it joins.

D. Pearson linear correlation is the only dependence input available, but all major copulas automatically ensure it accurately describes the true dependence structure even when dependence is dynamic.

Answer: D

Step by step

1. A copula separates the marginal distributions from the dependence structure, so marginals can take any form (C true).
2. The same marginals can be joined with different copulas (e.g., Gaussian, t) and simulated to compare aggregate outcomes (B true), and simulation yields the full distribution (A true).
3. D is false: linear correlation is not the only dependence measure (rank correlation, tail dependence exist), and a copula does not guarantee a correct or time-varying dependence structure; its parameters still have to be estimated and validated.

Why D is correct: Copulas give flexibility in marginals and dependence, but they do not automatically make a correlation input accurate or capture changing dependence.

Why the others are wrong

- **A:** True statement; simulation from the copula gives the whole loss distribution.
- **B:** True statement; swapping copulas with fixed marginals is a standard sensitivity test.
- **C:** True statement; copulas impose no restriction on marginals, so non-normal marginals are fine.

Q452

BT answers L2.T7.4.1 · BT notes P2.T7.4.1

Which of the following is the best summary of how economic capital (EC) is most often parameterized?

A. Capital a bank needs to absorb expected losses (EL) over a certain time horizon at a given confidence level

B. Capital a bank needs to absorb unexpected losses (UL) over a certain time horizon at a given confidence level

C. Capital a bank needs to absorb both expected losses (EL) and unexpected losses (UL) over a certain time horizon at a given confidence level

D. Capital a bank needs to absorb unexpected losses (UL) over a one-year horizon with 99.0% confidence

Answer: B

Step by step

1. EC is a common measure across all risks and businesses, set as capital to absorb losses over a horizon at a confidence level.
2. Expected losses are already covered through product pricing and loan loss provisioning, so only unexpected losses require capital.
3. Horizon and confidence level are bank-specific choices (not fixed at one year and 99.0%), so the general statement in B is the best summary.

Why B is correct: EC is parameterized as the capital needed to absorb unexpected losses over a chosen horizon at a chosen confidence level, because expected losses are handled by pricing and provisioning.

Why the others are wrong

- **A:** EL is covered by pricing and provisioning, not by capital; this confuses EC with provisions.
- **C:** Adds EL to the capital need, double counting what pricing and provisions already cover.
- **D:** Too specific: horizon and confidence level are internal choices that vary by bank (and 99.9% or higher is more typical for solvency targets than 99.0%).

Q453 BT answers L2.T7.6.2 · BT notes P2.T7.6.2

Consider the following three statements about dependency modeling in credit risk:
I. The majority of banks use one of three credit models: Moody's/KMV (MKMV), CreditMetrics, and CreditRisk+.
II. The three credit models tend to produce significantly different economic capital estimates in both default-only mode and mark-to-market mode.
III. Most in-use models estimate asset correlations among obligors in terms of common dependence on systematic risk factors; consequently, banks are implicitly accounting for concentration (both single name and sectoral) because large parts of their books are subject to the same underlying risk factors or to multiple risk factors.
According to the BIS, which of the above statements is TRUE?

A. I only
B. II and III only
C. I, II and III
D. III only

Answer: C

Key correction: Source keys differed only because option order differed (BT answers C = all three; BT notes D = all three); content agrees.

Step by step

1. I: the majority of banks use MKMV, CreditMetrics or CreditRisk+ (or in-house versions of them) - true.
2. II: the IACPM/ISDA study found significant differences in EC estimates across the models in both default-only and mark-to-market modes - true.
3. III: factor-based asset correlations mean books exposed to common factors implicitly capture single-name and sector concentration - true.
4. All three are true.

Why C is correct: Each statement restates a BIS finding on dependency modeling, so all three hold.

Why the others are wrong

- **A:** Ignores that II and III are also BIS findings.
- **B:** Wrongly treats the model-type statement I as false.
- **D:** Wrongly treats I and II as false; II in particular reflects the study showing large differences between models.

Q454 AnalystPrep Q.5388

REBUILT — Original stem never stated which approach the migration matrix used, so options were not answerable; rebuilt around the point-in-time vs through-the-cycle issue described in the BIS paper.

A bank uses point-in-time (PIT) default probabilities for day-to-day credit portfolio management, but through-the-cycle (TTC) default probabilities for its economic capital calculation. The economy is now entering a severe recession. Which statement is most accurate?

A. PIT default probabilities will rise sharply, while TTC probabilities will change little, so TTC-based economic capital understates near-term default risk relative to a PIT view.
B. TTC default probabilities will rise sharply, while PIT probabilities will change little, so PIT-based measures understate near-term default risk.

C. PIT and TTC default probabilities must be identical because both use the same time horizon.
D. PIT default probabilities will fall in a recession because rating migrations are smoothed over the cycle.

Answer: A
Step by step
1. PIT PDs reflect current conditions, so they move up quickly as the cycle deteriorates.
2. TTC PDs are averaged across the cycle and are relatively stable, so they respond little to the downturn.
3. Hence EC built on TTC PDs does not reflect the elevated near-term risk that PIT PDs show.
4. The BIS notes that even with the same default horizon, PIT PDs for day-to-day use and TTC estimates for EC can coexist - they are different estimates.

Why A is correct: PIT PDs are cyclical and TTC PDs are smoothed, so in a severe recession TTC-based EC understates near-term risk relative to PIT-based measures.
Why the others are wrong
- **B:** Reverses the definitions of PIT and TTC.
- **C:** Same horizon does not make PIT and TTC equal; they differ in how cyclical conditions enter the estimate.
- **D:** Describes smoothing (a TTC feature) and gets the direction wrong; PIT PDs rise in a recession.

Q455 BT answers L2.T7.5.4 · BT notes P2.T7.5.4

According to the BIS and their survey, most banks use some flavor of the variance-covariance approach to aggregate risk. They tend to employ conservative correlations with an upward bias, with many using stressed values or even setting them equal to unity. Consider the following three inter-risk correlations:
I. Correlation between market and credit risk
II. Correlation between business risk and credit/market risk
III. Correlation between operational risk and all other risks
From LOWEST to HIGHEST correlation, which is the most common assumption made by banks?
A. I, II, III
B. II, III, I
C. II, I, III
D. III, II, I

Answer: D
Step by step
1. BIS survey: high correlation between market and credit risk (I).
2. Lower correlation between business risk and credit/market risk (II).
3. Very low correlation between operational risk and all other risks (III).
4. Lowest to highest: III, II, I.

Why D is correct: Banks assume operational risk is nearly uncorrelated with others, business risk moderately correlated, and market-credit highly correlated.
Why the others are wrong
- **A:** Exact reverse of the survey ordering.
- **B:** Places operational risk above business risk, contradicting the 'very low' operational risk assumption.
- **C:** Puts business risk lowest and operational risk highest, wrong on both ends.

Q456 BT notes P2.T7.6.1 · BT answers L2.T7.6.1

In regard to the validation of internal economic capital models, the BIS asserts each of the following EXCEPT:
A. Validation refers broadly to all processes that provide evidence-based assessments about a model's fitness for its purpose, including both "before deployment" (ex-ante) and afterward as ex-post statistical tests
B. Validation is a quantitative process by definition, but it is NOT a qualitative process
C. Benchmarking is a common and useful validation process, but its drawback is that it does not test against reality
D. There are a wide range of validation techniques making possible a "layered approach," and generally, the more layers, the better

Answer: B
Key correction: BT notes had no answer key; BT answers key B is correct.

Step by step
1. A: BIS defines validation broadly (ex-ante and ex-post) - asserted.
2. C: benchmarking compares one model with another but does not test against reality - asserted.
3. D: a layered approach; more layers give more comfort - asserted.
4. B: validation must encompass both qualitative elements (use test, qualitative review, data quality, management oversight) and quantitative ones - so B is not asserted.

Why B is correct: BIS stresses that validation combines qualitative and quantitative elements, so calling it purely quantitative is the exception.

Why the others are wrong
- **A:** Is asserted by the BIS; broad definition covering ex-ante and ex-post work.
- **C:** Is asserted by the BIS; benchmarking only compares models, not model vs reality.
- **D:** Is asserted by the BIS; more layers give more evidence on performance.

Q457
AnalystPrep Q.2219

REBUILT — Original distractors (lack of algorithms, scarce personnel) were weak giveaways; replaced with more plausible distractors.

The BIS notes that the main validation concern for economic capital continues to be the accuracy and stability of correlation estimates, particularly in times of stress. Correlation estimates from current models are inaccurate and unstable mainly because they:

A. depend heavily on explicit or implicit model assumptions
B. are computed from market data that are always too short to use
C. cannot be incorporated into any aggregation approach
D. are fixed by supervisors rather than estimated by the bank

Answer: A

Step by step
1. BIS: correlation estimates from current models depend heavily on explicit or implicit model assumptions.
2. Such assumptions (e.g., stable correlations, Gaussian copula, factor structure) may fail, especially in stress, making estimates inaccurate and unstable.
3. Only A states this mechanism.

Why A is correct: Correlations are largely driven by model assumptions rather than observed data, so they are fragile when assumptions fail.

Why the others are wrong
- **B:** Data scarcity matters, but the BIS states the key issue is dependence on assumptions, and 'always too short' is an overstatement.
- **C:** Correlations are routinely used in aggregation (variance-covariance); this is false.
- **D:** Banks estimate or assume correlations themselves; supervisors do not set them.

Q458
AP Mock 2 Q7 · AnalystPrep Q.2222

Which of the following financial products would pose the greatest challenge to the determination of the economic capital of a bank?

A. Ordinary stocks.
B. Preference shares.
C. Bonds with embedded options.
D. Fixed-rate interest rate loans.

Answer: C

Step by step
1. Economic capital requires measuring risk over a long holding period and projecting cash flows on assets and liabilities.
2. Embedded options (e.g., prepayment or call features) make cash flows indeterminate and the payoff nonlinear.
3. Their exercise depends on borrower/depositor behavior and future rates, so a simple linear sensitivity measure understates risk.
4. Ordinary stocks, preference shares and plain fixed-rate loans have comparatively direct or scheduled cash flows, so they are less problematic.

Why C is correct: Embedded optionality creates uncertain cash flows and nonlinear risk that requires complex stochastic-path evaluation, and the resulting risk can be significantly greater than the risk measure suggests.

Why the others are wrong

- **A:** Ordinary stocks have a linear payoff and no embedded optionality, so their risk is comparatively easy to capture.
- **B:** Preference shares have bond-like, relatively predictable payments; mistaking 'hybrid' for 'option-laden' leads here.
- **D:** Plain fixed-rate loans have scheduled cash flows and no assumed optionality; confusing long maturity with uncertain cash flows leads here.

Q459

AnalystPrep Q.1984

REBUILT — Source answer (uninformative historical data) is not the challenge given in the reading; rebuilt around the stated difficulty of identifying and quantifying wrong-way risk.

Which of the following best describes a main challenge in capturing wrong-way risk in a bank's economic capital calculation for counterparty credit risk?

A. Exposure at default and PD/LGD are always estimated jointly, so the interaction is double counted.

B. Identifying it requires understanding the counterparty's market risk factor sensitivities (often opaque, e.g., a hedge fund) and comparing them with the bank's own exposures, and its magnitude is hard to quantify over a one-year horizon at a high confidence level.

C. Regulators do not permit wrong-way risk to be reflected in economic capital.

D. Exposure and counterparty creditworthiness are statistically independent, so there is no relationship to model.

Answer: B

Key correction: Source key (AnalystPrep) said D (uninformative historical data); that is not the challenge stated in the reading, which emphasises identification and quantification of magnitude.

Step by step

1. Wrong-way risk: PD and LGD tend to rise at the same time as exposure to the counterparty rises.
2. Many systems estimate EAD (market-risk driven) and PD/LGD (credit-risk driven) separately, so the interaction is not captured directly (an add-on may be used).
3. Identifying it needs knowledge of the counterparty's risk factor sensitivities and a comparison with the bank's own, which is hard for opaque counterparties.
4. Even when direction is known, magnitude is hard to quantify over a one-year horizon at a high confidence level.

Why B is correct: The difficulty lies in identifying the link between counterparty credit quality and exposure and in quantifying its magnitude at long horizon and high confidence.

Why the others are wrong

- **A:** Reverses the problem: EAD and PD/LGD are frequently estimated separately, which is why the interaction is missed rather than double counted.
- **C:** Invented restriction; wrong-way risk may be incorporated, e.g., via an add-on in the economic capital process.
- **D:** Wrong-way risk is defined precisely by dependence between exposure and counterparty credit quality; assuming independence ignores the risk.

Q460

AnalystPrep Q.2216

The main challenge faced by financial institutions while choosing the risk measure to use for economic capital purposes is that:

A. There are no generally accepted properties of a good risk measure.

B. Most risk measures are too complex, which means implementation and eventual communication to stakeholders can be quite difficult.

C. There is a general lack of relevant and reliable data that can be used to assess risks.

D. There is no singularly preferred risk measure for economic capital purposes.

Answer: D

Step by step

1. Desirable properties (intuitive, stable, easy to compute and understand, coherent, interpretable) are broadly agreed.
2. Standard deviation, VaR, ES and spectral/distorted measures each have strengths and weaknesses (e.g., VaR is not subadditive).
3. Regulators likewise show no clear preference, so no one measure is preferred for all uses; banks must choose and understand limitations.

Why D is correct: Every measure in use has advantages and disadvantages, and no single measure is preferred for economic capital purposes, so the choice must fit the intended application.

Why the others are wrong

- **A:** There is general agreement on the desirable properties of a risk measure; the problem is that none satisfies all of them.
- **B:** Complexity is a concern for some measures (spectral) but not the main, general challenge; VaR is widely used and simple to communicate.
- **C:** Data availability is one factor in the choice, not the central challenge in choosing a measure.

Q461

AnalystPrep Q.2217

REBUILT — Original had two defensible 'errors' (diversification recognition versus ignored correlations) and was not tied to the reading; rebuilt around the unit-of-account problem.

A bank is aggregating its market, credit, operational and business risk capital into a single economic capital figure. Which of the following is most likely to make the aggregate figure inaccurate?

A. Combining component measures that were calculated at different confidence levels and time horizons without first expressing them on a common basis.

B. Aggregating first by business unit rather than by risk type.

C. Using a copula approach rather than simple summation.

D. Expressing every risk component with the same risk metric, confidence level and time horizon.

Answer: A

Step by step

1. Before aggregation, risk components must be expressed in comparable units (risk metric, confidence level, time horizon).
2. Different confidence levels and horizons make the components non-comparable, and combining them gives inaccurate comparisons.
3. Business-unit and risk-type silo approaches are both used; there is no universally accepted method.
4. Copulas are more flexible than simple summation, and consistent units are what aggregation requires.

Why A is correct: Aggregation requires a common risk currency; mixing confidence levels and horizons distorts the combined figure.

Why the others are wrong

- **B:** Both risk-type and business-unit silos are used in practice; each has advantages, so this alone is not a source of error.
- **C:** Copulas capture nonlinear dependence better than simple summation; their drawback is parameterization, not inaccuracy relative to summation.
- **D:** This is exactly what comparable units mean and is desirable, not a cause of error.

Q462

AnalystPrep Q.2220

REBUILT — Source answer (strain on human resources) is not the stated aggregation challenge; rebuilt around netting-set level measurement across many counterparties and holding periods.

A bank-wide view of counterparty credit risk for economic capital purposes is difficult to build mainly because:

A. Market risk VaR models allow gains on one counterparty to offset losses on another, which makes results incomparable.

B. Counterparty exposure is measured over a single short holding period, which is too short for derivatives.

C. Exposure cannot be netted across counterparties, so it must be calculated at the netting-set level for thousands of counterparties and over many future holding periods, greatly increasing computational intensity.

D. It relies heavily on the independent opinions of credit rating agencies, some of which can be compromised.

Answer: C

Key correction: Source key (AnalystPrep) said A (strain on human resources); the reading attributes the aggregation difficulty to netting-set level calculations and multiple holding periods.

Step by step

1. VaR models combine all positions in one simulation so gains offset losses; counterparty exposure cannot be netted across counterparties.
2. Each counterparty may have several legally enforceable netting agreements, so calculation is at the netting-set level.

3. Exposure must be simulated over multiple future holding periods (years or decades) rather than one day or ten days.
4. Many counterparties times many holding periods makes the computation far more intensive.

Why C is correct: No cross-counterparty netting plus netting-set calculations over many counterparties and long horizons makes aggregation computationally demanding.

Why the others are wrong

- **A:** Reverses the point: VaR models net across positions, but counterparty exposure cannot be netted across counterparties.
- **B:** Reverses the point: market VaR uses a short holding period, while counterparty exposure needs multiple, long future holding periods.
- **D:** Rating agencies are not identified as the main aggregation challenge.

Q463 AnalystPrep Q.2221

One of the main challenges in the calculation of economic capital for interest rate risk in the banking book relates to:

A. The long holding period of balance sheet assets and liabilities.
B. Varying market forces of supply and demand.
C. The unpredictable nature of regulatory action by central banks.
D. The presence of a large bouquet of products, all priced differently.

Answer: A

Step by step

1. IRRBB capital must reflect the long holding period assumed for the banking book.
2. Cash flows on assets and liabilities are indeterminate because of embedded optionality (prepayment, non-maturity deposits).
3. Together these make the computation challenging; the other options are not the stated challenge.

Why A is correct: The long holding period of banking-book positions, combined with embedded optionality, is the stated computational challenge for IRRBB.

Why the others are wrong

- **B:** General market forces affect all risks and are not the specific IRRBB challenge.
- **C:** Central bank actions are not cited as the main calculation challenge.
- **D:** Product variety is not the stated problem; optionality and holding period are.

Q464 AnalystPrep Q.2967

Copulas combine the marginal probability distributions into a joint distribution. Which of the following is an advantage of copulas as a form of risk aggregation methodology?

A. The effect of fixed diversification is sensitive to underlying interactions between the different components.
B. The method is easy to use as it easily estimates inter-risk correlations and does not capture nonlinearities.
C. Simulation of common drivers provides for calculating the distribution of outcomes and economic capital risk measure.
D. Is more flexible than a covariance matrix and allows for nonlinearities and higher-order dependencies.

Answer: D

Step by step

1. Copulas join marginal distributions into a joint distribution.
2. Advantage: more flexible than the variance-covariance matrix and can capture nonlinear and higher-order dependencies.
3. Disadvantages: parameterization is hard to validate and building the joint distribution is difficult.

Why D is correct: Copulas relax the linear-correlation limitation of the covariance matrix, capturing nonlinear and higher-order dependence.

Why the others are wrong

- **A:** Constant diversification applies a fixed percentage and is not sensitive to interactions; this is not a copula property.
- **B:** Describes the variance-covariance matrix, whose correlations are difficult to estimate and which misses nonlinearities.

- **C:** Describes full modeling/simulation of common risk drivers, not copulas.

Q465 AnalystPrep Q.2968

Broadways Bank uses the unit of account as a component of risk aggregation methodology. Which of the following is NOT a characteristic of the unit of risk accounting?

A. Risk metric
B. Confidence level
C. Complex simulation
D. Time horizon

Answer: C

Step by step

1. The unit of account puts risk types in comparable units.
2. Its three characteristics are risk metric, confidence level and time horizon.
3. Complex simulation is an aggregation methodology, not a unit-of-account characteristic.

Why C is correct: Risk metric, confidence level and time horizon define the common risk currency; simulation is a method for combining risks.

Why the others are wrong

- **A:** Risk metric is one of the three characteristics.
- **B:** Confidence level is one of the three characteristics.
- **D:** Time horizon is one of the three characteristics.

Q466 BT answers L2.T7.4.3 · BT notes P2.T7.4.3

According to the BIS, each of the following is a desirable characteristic of a risk measure within the economic capital framework EXCEPT:

A. Coherent
B. Intuitive and ”easy to understand”
C. Precise
D. Easy to compute

Answer: C

Step by step

1. The ideal risk measure is intuitive, stable, easy to compute, easy to understand, coherent and interpretable in economic terms, with simple and meaningful risk decomposition.
2. Coherent, intuitive/easy to understand and easy to compute are all on the list.
3. Precision is not on the list (no single measure is perfect, and estimates depend on model assumptions), so C is the exception.

Why C is correct: The desirable-characteristics list contains no requirement of precision.

Why the others are wrong

- **A:** Coherence is an explicit desirable characteristic (monotonicity, positive homogeneity, translation invariance, subadditivity).
- **B:** Intuitive and easy to understand are both on the list.
- **D:** Ease of computation is on the list; complex measures need evidence that the accuracy gain outweighs the added complexity.

Q467 BT answers L2.T7.4.4 · BT notes P2.T7.4.4

Which of the following is a weakness of standard deviation as a risk measure within the economic capital framework?

A. It is not coherent because it violates the monotonicity condition
B. It is not intuitive
C. It is not easy to compute
D. It is neither simple nor easy to understand

Answer: A

Step by step

1. Standard deviation is rated sufficiently intuitive, easy to compute and easy to understand.
2. Its weaknesses are that it is not stable (depends on loss-distribution assumptions) and not coherent: it violates monotonicity.
3. Only A states a genuine weakness.

Why A is correct: Standard deviation treats upside and downside dispersion alike, so it can rank a portfolio that is always worth at least as much as another as riskier, violating monotonicity.

Why the others are wrong

- **B:** Standard deviation is sufficiently intuitive; the 'not intuitive' label belongs to spectral/distorted measures.
- **C:** It is easy to compute (only an estimate of the loss distribution is needed).
- **D:** It is simple and easy to understand; the confusion is with measures such as ES or spectral measures.

Q468

BT answers L2.T7.4.5 · BT notes P2.T7.4.5

VaR is not coherent because it can violate subadditivity. Aside from its lack of coherence, which of the following is a weakness of VaR as a risk measure within the economic capital framework?

A. It is not intuitive
B. It is not stable because it depends on assumptions about the loss distribution
C. It is insufficiently easy to compute
D. It is insufficiently easy to understand

Answer: B

Step by step

1. VaR is rated intuitive, sufficiently easy to compute and easy to understand.
2. It is not stable: results depend on assumptions about the loss distribution.
3. So B is the remaining weakness.

Why B is correct: VaR is a single quantile estimate, so small changes in distributional assumptions or simulation runs can move it materially.

Why the others are wrong

- **A:** VaR is rated 'Yes' on intuitive.
- **C:** VaR is sufficiently easy to compute (needs a loss-distribution estimate).
- **D:** VaR is easy to understand and easier to communicate to senior management than ES.

Q469

BT answers L2.T7.4.6 · BT notes P2.T7.4.6

REBUILT — Source asked for a 'glaring weakness' of ES with 'none' as the key; the reading states ES is hard to interpret and lacks a clear link to the target rating, so options were rewritten to have one defensible answer.

Which of the following is a recognized weakness of expected shortfall (ES) as a risk measure within the economic capital framework?

A. It is not coherent because it violates subadditivity
B. It cannot be decomposed into contributions of business lines
C. It does not lend itself to easy interpretation and has no clear link to the bank's desired target rating
D. It is unstable under every loss distribution

Answer: C

Key correction: BT sources said 'none of the above: no glaring weaknesses'; the reading identifies difficulty of interpretation and lack of link to target rating as ES weaknesses.

Step by step

1. ES is coherent, so it satisfies subadditivity (A is false).
2. ES supports capital allocation, and is more stable than VaR for allocating portfolio capital (B and D are false).
3. The stated weakness is that ES is not easily interpreted and gives no clear link to the target rating, unlike VaR.

Why C is correct: ES averages losses over the tail, which is harder to interpret and does not map directly to a target-rating confidence level.

Why the others are wrong

- **A:** This describes VaR; ES is coherent.
- **B:** ES is increasingly used for capital allocation and decomposes consistently with the portfolio measure.

- **D:** ES stability depends on the loss distribution and it is more stable than VaR for allocation, not unstable in all cases.

Q470 BT answers L2.T7.4.7 · BT notes P2.T7.4.7

REBUILT — Original option D ('Not stable') was partly defensible since stability depends on the distribution; reworded it to 'always unstable' so only B is correct.

According to the BIS, which of the following is a weakness of spectral and distorted risk measures within the economic capital framework?

A. They are not coherent
B. They are not intuitive
C. They are not easy to compute
D. They are always unstable

Answer: B

Step by step

1. Spectral and distorted measures are coherent (A false).
2. Computing them is sufficiently easy: loss distribution weighted by the spectrum or distortion function (C false).
3. Stability depends on the loss distribution, not 'always unstable' (D false).
4. They are not intuitive, since they require choosing a spectrum or distortion function; they are also rarely used in practice.

Why B is correct: Choosing a spectrum or distortion function has no natural intuitive interpretation, so these measures are not intuitive.

Why the others are wrong

- **A:** They are coherent.
- **C:** Rated 'sufficiently easy' to compute.
- **D:** Stability depends on the loss distribution; it is not uniformly poor.

Q471 BT answers L2.T7.4.8 · BT notes P2.T7.4.8

According to BIS, which of the following perspectives is likely to lead to the highest confidence level in calibrating economic capital?

A. Perspective of creditors, rating agencies, and supervisors
B. Perspective of management in order to allocate capital
C. Perspective of management in order to identify exposures that are critical to profit (or profit centers)
D. None of the above: in a robust EC framework, confidence level should not vary according to constituent perspective

Answer: A

Step by step

1. Economic capital confidence levels are linked to the bank's target rating and to the purpose of the measurement.
2. Creditors, rating agencies and supervisors want capital sufficient to minimise bankruptcy risk, so they favour very high confidence levels.
3. Management uses lower confidence levels to allocate capital to business lines/exposures and to identify exposures critical to profit objectives in a normal business environment.
4. Hence banks typically use different confidence levels for different purposes, and the highest level reflects the external (creditor/rating agency/supervisor) perspective: A.

Why A is correct: High confidence levels serve the solvency perspective of external stakeholders, whereas internal management uses (lower) levels suited to capital allocation and profit analysis.

Why the others are wrong

- **B:** Capital allocation is an internal management use for which lower confidence levels are typically used.
- **C:** Identifying profit-critical exposures is a normal-business-environment use, so a lower confidence level is used.
- **D:** Confusing consistency with robustness: banks explicitly use different confidence levels for different purposes.

Q472 BT answers L2.T7.5.1 · BT notes P2.T7.5.1

REBUILT — Merged two versions; the answers-file version had inconsistent figures (USD 70M + USD 20M shown as USD 30M), so the consistent version (USD 10M + USD 20M = USD 30M) was used and the garbled option text was repaired.

A bank holds a portfolio of loans denominated in a foreign currency. The bank separately measures the credit risk and market risk of the portfolio, then determines the portfolio's economic capital by adding (aggregating) the two risk components. Specifically, the bank determines the portfolio's economic capital is USD 30 million because the market risk component is USD 10 million per a value at risk (VaR) method and the credit risk component is USD 20 million per a CVaR method. Consider four statements about this aggregation:

I. As VaR is not subadditive, it is technically possible for the portfolio's VaR to exceed USD 10 plus USD 20 million

II. As this summation implicitly assumes zero correlation (and zero covariance) between market and credit risk, if their correlation is actually positive, USD 30 million understates economic capital

III. There may be "wrong way" risk between the credit and market risk components, in which case the portfolio's economic capital may be higher than USD 30 million

IV. VaR is a quantile, not a tail risk measure. Expected shortfall (ES) should be used, in which case, by definition, the portfolio's economic capital (EC) must be higher than USD 30 million

Which of the above are TRUE?

A. I. only
B. I. and III.
C. II. and IV.
D. All four

Answer: B

Step by step

1. I: VaR is not subadditive in general, so aggregate risk can exceed the sum of the components (here USD 30 million). TRUE.
2. II: Simple summation implicitly assumes perfect correlation (equal to 1), not zero correlation, and gives no diversification benefit. With any correlation below 1 the sum overstates capital. FALSE.
3. III: Wrong-way risk (e.g., FX-denominated loans whose default probability rises as the currency moves against the borrower) makes market and credit risk compound, so aggregate capital can exceed USD 30 million. TRUE.
4. IV: Switching to ES does not by definition raise capital above USD 30 million; the result depends on the confidence levels used (e.g., a 99.9% VaR can exceed a 95% ES). FALSE.
5. True statements: I and III, so B.

Why B is correct: Summation is only a conservative upper bound if risks are measured correctly and subadditivity holds; non-subadditivity of VaR and wrong-way risk can push true aggregate risk above the sum.

Why the others are wrong

- **A:** Omits III; wrong-way risk between credit and market components is a valid reason why capital may exceed the sum.
- **C:** II reverses the assumption (summation implies correlation of 1, not 0, so it overstates rather than understates), and IV wrongly claims ES must yield a higher figure.
- **D:** Includes II and IV, which are both false for the reasons above.

Q473 BT answers L2.T7.5.2 · BT notes P2.T7.5.2

BIS says, "The use of a variance-covariance matrix (or correlation matrix), which summarizes the interdependencies across risk types, provides a more flexible framework for recognizing diversification benefits while still maintaining the desirable features of being intuitive and easy to communicate." But BIS also says that EACH of the following is a disadvantage of the variance-covariance matrix EXCEPT:

A. It depends greatly on correlation pairs which are difficult to estimate, in part because they are time-varying
B. Operational inter-risk correlations can be costly and difficult to estimate because operational loss "data are scarce and do not cover long time periods"
C. It is inferior to approaches that "isolate on marginal risk contribution," such as marginal VaR and component VaR
D. By focusing on average covariance between risks, the approach will tend to underestimate tail risks

Answer: C

Step by step

1. Recall the cited drawbacks of the variance-covariance approach: inter-risk correlations are hard to estimate and may be time-varying; data are scarce, especially for operational risk; dependencies are assumed linear and

fixed, so nonlinearities and tail dependence are missed.
2. A, B and D each restate one of these drawbacks.
3. C is not a cited disadvantage: marginal and component VaR rely on a covariance structure and are not presented as superior alternatives for aggregation.
4. Therefore C is the EXCEPT answer.

Why C is correct: The variance-covariance method's weaknesses are estimation difficulty, time-varying/linear dependence and understated tail dependence; being inferior to marginal/component VaR is not one of them.

Why the others are wrong

- **A:** A genuine cited disadvantage (correlations hard to estimate, time-varying), so not the exception.
- **B:** A genuine cited disadvantage (scarce operational loss data make correlations hard to estimate), so not the exception.
- **D:** A genuine drawback: linear, average-based dependence does not capture tail dependence, which is why banks use stressed correlations.

Q474 BT answers L2.T7.5.3 · BT notes P2.T7.5.3

A serious limitation of some risk aggregation methodologies is their inability to capture non-linear dependencies. Which risk aggregation method(s) overcome this limitation and do not necessarily assume linear dependencies?

A. Summation
B. Constant diversification
C. Variance-covariance
D. Copulas and Simulations

Answer: D

Step by step

1. Simple summation, fixed (constant) diversification and the variance-covariance matrix all assume linear, fixed dependence and do not capture nonlinearities.
2. Copulas join marginal distributions into a joint distribution with flexible functional forms, allowing nonlinear and higher-order dependence.
3. Full simulation of common risk drivers also does not require linearity.
4. Answer: D.

Why D is correct: Only copulas and full simulation approaches can represent nonlinear dependence between risk types.

Why the others are wrong

- **A:** Summation implicitly assumes correlation of one and captures no nonlinearity.
- **B:** Constant diversification just subtracts a fixed percentage from the sum, so it is insensitive to underlying interactions.
- **C:** The correlation matrix still assumes linear, time-invariant interactions.

Q475 BT answers L2.T7.5.5 · BT notes P2.T7.5.5

BIS cites three supervisory concerns relating to risk aggregation. Each of the following is a supervisory concern EXCEPT:

A. Popular methods are overly dependent on subjective judgment and insufficiently quantitative
B. Economic capital frameworks are inherently difficult to validate
C. Popular methodologies, in presuming or over-estimating diversification benefits, may under-estimate true aggregate risk
D. Sophisticated methods, in producing apparently accurate outcomes, may lead to a (behavioral) sense of over- or false-confidence

Answer: A

Step by step

1. The supervisory concerns on aggregation relate to validation of inputs, methodology and outputs.
2. B: economic capital frameworks are very difficult to validate (holistic measures, bank-specific, short data history, backtesting impracticable).
3. C: diversification cannot be taken as given; mis-measurement can lead to under-estimation of overall risk.
4. D: greater sophistication can create unwarranted confidence in accuracy.
5. A is not a concern: BIS states that meaningful aggregation necessarily involves compromises and judgment to

augment quantitative methods. Answer: A.

Why A is correct: BIS accepts judgment as a necessary complement to quantitative methods; its concerns are validation difficulty, over-stated diversification and false confidence.

Why the others are wrong

- **B:** A stated concern (validation difficulty).
- **C:** A stated concern (overstated diversification leading to under-estimated aggregate risk).
- **D:** A stated concern (behavioral false confidence from sophisticated methods).

Q476 BT answers L2.T7.6.3 · BT notes P2.T7.6.3

BIS asserts that each of the following is a difficulty ("challenge") in the modeling and/or evaluation of counterparty credit risk in an economic capital framework EXCEPT:

A. Trying to measure current exposure
B. Trying to measure potential exposure
C. Difficult to identify wrong-way risk between credit risk (i.e., increase in PD and LGD) and market risk (increase in EAD)
D. Differences in risk profiles between margined and non-margined counterparties

Answer: A

Step by step

1. Current exposure is what the counterparty would owe if it defaulted today; it is observable from current mark-to-market values, so it is not a modeling challenge.
2. Potential exposure is the possible increase in exposure up to a future horizon; it needs long-horizon simulation of market factors for the whole portfolio, which is hard.
3. Wrong-way risk (PD/LGD rising when exposure rises) is hard to identify and quantify, so it is a challenge.
4. Margined counterparties have short forecasting (close-out) periods while non-margined ones have long ones, creating differing risk profiles, so it is a challenge.
5. The only item that is not a challenge is measuring current exposure: A.

Why A is correct: Current exposure is known today, whereas potential exposure, wrong-way risk and margining differences are genuine modeling difficulties cited by the Basel Committee.

Why the others are wrong

- **B:** A true challenge: potential exposure requires simulating risk factors over multiple, long holding periods for the whole portfolio; choosing it confuses it with current exposure.
- **C:** A true challenge: wrong-way risk requires understanding the counterparty's market risk factor sensitivities and is hard to quantify over a one-year, high-confidence horizon.
- **D:** A true challenge: margined counterparties have short forecasting periods and non-margined ones long periods, and gap risk remains even when margined.

Q477 BT answers L2.T7.6.4 · BT notes P2.T7.6.4

According to BIS, each of the following is a MAIN SOURCE of interest rate risk in the banking book EXCEPT:

A. Repricing risk
B. Yield curve risk
C. Earnings at risk (EaR)
D. Basis risk

Answer: C

Step by step

1. The three main sources of IRRBB are repricing risk, yield curve risk and basis risk.
2. Repricing risk arises from maturity/repricing-term differences between assets and liabilities.
3. Yield curve risk arises from asymmetric movements in rates along the curve; basis risk from imperfect correlation of rates earned and paid on instruments with similar repricing characteristics.
4. Earnings at risk (EaR) measures the loss of net interest income from rate changes; it is a measurement approach, not a source: C.

Why C is correct: EaR is a metric for quantifying IRRBB, whereas repricing, yield curve and basis risk are the underlying sources.

Why the others are wrong

- **A:** Repricing risk is a main source (maturity/repricing mismatch); it is not the exception.

- **B:** Yield curve risk is a main source (asymmetric movements along the curve); it is not the exception.
- **D:** Basis risk is a main source (imperfect correlation between rates earned and paid); it is not the exception.

Q478 BT answers P2.T5.406.2

In contrasting a compartmentalized approach versus a unified (integrated) approach to risk measurement, the Basel Committee says each of the following is true EXCEPT, which is not?

A. A compartmentalized approach sums risks measured separately, while a unified approach considers the interaction between these risks explicitly

B. From a theoretical perspective, an integrated approach is needed to capture potential compounding effects that are ignored in compartmentalized risk measurement approaches

C. Studies suggest the magnitude of diversification benefits (the amount by which aggregate risk is below the sum of individual risks) depends upon the level at which risks are measured; at higher levels of aggregation, such as the holding company level, the benefits are more often detected

D. The most conservative estimate of risk is always a simple aggregation of compartmentalized measures

Answer: D

Step by step

1. Simple summation of separately measured risks is usually viewed as conservative because it ignores diversification and gives an upper bound.
2. However, risk interactions can be such that the combined risk is higher, not lower, than the sum: e.g., separately measuring market and credit risk of foreign-currency loans understates risk because FX moves also raise default probabilities (a compounding effect).
3. So simple aggregation is not ALWAYS the most conservative; the word 'always' makes D false.

Why D is correct: Compounding interactions between risk types can make true integrated risk exceed the simple sum, so summation is not always conservative.

Why the others are wrong

- **A:** True description of the two approaches; not the exception.
- **B:** True: only an integrated approach captures compounding effects; not the exception.
- **C:** True per the studies cited: diversification is more evident at higher aggregation levels; not the exception.

Q479 BT answers P2.T7.24.7.1 · BT notes P2.T7.24.7.1

In the wake of Lehman Brothers' collapse in 2008, the banking sector faced heightened counterparty credit risk, particularly in the derivatives market. This situation underscored the necessity for rigorous risk assessments and enhanced collateral management to mitigate losses linked to counterparty failures.

Which aspect of economic capital integration is most directly related to the scenario described?

A. Risk aggregation

B. Validation of models

C. Dependency modeling in credit risk

D. Evaluating counterparty credit risk

Answer: D

Step by step

1. The scenario concerns losses from derivative counterparties failing and collateral (margin) management.
2. Measuring exposure, PD/LGD, wrong-way risk and margining for derivative counterparties is the 'evaluating counterparty credit risk' challenge: D.

Why D is correct: Losses from failing derivative counterparties and collateral management map directly to counterparty credit risk evaluation.

Why the others are wrong

- **A:** Risk aggregation is about combining risk types into a total figure, not assessing a counterparty's default exposure.
- **B:** Model validation concerns testing model performance, which the scenario does not discuss.
- **C:** Dependency modeling in credit risk concerns correlation among credit exposures in the portfolio, not exposure to derivative counterparties and collateral.

Q480 BT answers P2.T7.24.7.2 · BT notes P2.T7.24.7.2

REBUILT — Original stem named 'risk aggregation frameworks', giving away the answer; reworded the stem so the mechanism, not the label, points to the answer.

During the Eurozone debt crisis around 2011-2012, many banks found that adding up separately measured risk figures failed to capture how stress in sovereign credit risk spread to private sector credit and market risks, both within and across countries. The resulting total capital figures understated the systemic impact of these interconnected risks.

Which challenge within the economic capital implementation framework does this scenario most directly relate to?

A. Risk aggregation
B. Validation of models
C. Evaluating counterparty credit risk
D. Assessing interest rate risk in the banking book

Answer: A

Step by step

1. The problem is that separately measured risks were combined without capturing their interdependence, giving a misleading total.
2. Combining different risk types (and their dependencies under stress) into a single capital figure is risk aggregation: A.

Why A is correct: Failing to reflect cross-dependencies between risk types when combining them into a total is the core risk aggregation challenge.

Why the others are wrong

- **B:** Model validation concerns testing a model's performance and assumptions; the scenario is about combining risks, not testing.
- **C:** Counterparty credit risk concerns exposure to derivative/financing counterparties, not sovereign-private interdependence.
- **D:** IRRBB concerns interest rate effects on the banking book (repricing, yield curve, basis), not sovereign-credit contagion.

Q481 BT answers P2.T7.24.7.3 · BT notes P2.T7.24.7.3

REBUILT — Original London Whale item was ambiguous (model validation and risk-measure definition both defensible) and had no clean tie to the reading; replaced with a direct test of the VaR limitation for capital allocation and limit setting.

A bank uses value at risk (VaR) both to measure absolute risk and to allocate economic capital and set limits for its sub-portfolios. Which weakness of VaR most directly undermines this allocation and limit-setting process?

A. VaR is not subadditive, so the sum of sub-portfolio VaRs can understate the bank-wide VaR
B. VaR violates monotonicity, so a portfolio with higher future returns can show greater risk
C. VaR has no clear link to the bank's desired target credit rating
D. VaR requires the choice of a spectrum or distortion function that is not intuitive

Answer: A

Step by step

1. Allocation and limit setting require that risk measured for sub-portfolios be consistent with risk for the whole bank.
2. The reading notes VaR is not coherent because it violates subadditivity.
3. Hence $\text{VaR}(A+B) > \text{VaR}(A) + \text{VaR}(B)$ is possible, so stand-alone sub-portfolio VaRs can understate total risk and distort allocations and limits.
4. The other options describe weaknesses of other measures (standard deviation, expected shortfall, spectral measures).

Why A is correct: Because VaR can violate subadditivity, sub-portfolio VaRs may not add up to a conservative measure of total risk, which causes problems in internal capital allocation and limit setting.

Why the others are wrong

- **B:** Monotonicity violation is the reading's criticism of standard deviation, not VaR.
- **C:** An unclear link to the target rating is the reading's criticism of expected shortfall.
- **D:** Choice of spectrum or distortion function is the criticism of spectral and distorted risk measures.

Q482 MK-GARP PE2-Q34 · MK-Mock Mock 4 Q35

The CRO of a regional bank wants to ensure that the modeling assumptions used in the bank's economic capital models are sound. The CRO asks a member of the validation team to review the bank's process of assessing the interest rate risk in its banking book and to validate the assumptions used in its interest rate models. Which of the following assumptions would be most appropriate for the bank to make?

A. The bank changes the interest rate it offers to depositors by the full amount of any change in market interest rates.
B. The bank models its retail non-maturity deposits as floating-rate, putable bonds.
C. The bank assumes that its residential mortgages exhibit positive convexity as interest rates decrease.
D. The bank models its interest rate risk in the banking book independently from its credit risk.

Answer: B

Step by step

1. Non-maturity deposits embed two options: the bank sets (and resets) the deposit rate, so the deposit behaves like a floating-rate bond.
2. The depositor can withdraw at par, i.e. put the bond back to the bank.
3. So they are modeled as floating-rate, putable bonds (B).
4. Check the other assumptions against the reading: deposit rates adjust with a lag and by less than the full market change; mortgages show negative convexity when rates fall; IRRBB interacts with credit risk.

Why B is correct: Non-maturity deposits carry a bank option to reset the rate and a depositor option to withdraw at par, which is exactly a floating-rate, putable bond.

Why the others are wrong

- **A:** Banks adjust deposit rates with a lag and by less than the full change in market rates, particularly when rates rise.
- **C:** Mortgages are negatively convex when rates fall because prepayments accelerate (contraction risk); positive convexity is the wrong sign.
- **D:** IRRBB interacts with credit risk (e.g., borrowers' prepayment and default options), so modeling them independently is not appropriate.

Q483 Schweser Module Quiz 60.1 Q1

Which of the following risk measures is the least commonly used measure in the practice of risk management?

A. Value at risk.
B. Standard deviation.
C. Expected shortfall.
D. Spectral risk measures.

Answer: D

Step by step

1. VaR and expected shortfall are the most commonly used measures in practice.
2. Standard deviation is a simple, familiar measure still used.
3. Spectral and distorted risk measures are not intuitive (they require choosing a spectrum or distortion function) and are largely of academic interest.
4. So spectral measures are least used (D).

Why D is correct: Spectral and distorted risk measures are not widely used and are largely of academic interest.

Why the others are wrong

- **A:** VaR is the most commonly used measure.
- **B:** Standard deviation is simple and widely understood, so it is not the least used.
- **C:** Expected shortfall is, with VaR, one of the two most commonly used measures.

Q484 Schweser Module Quiz 60.1 Q2

Which of the following aggregation methodologies is characterized by great difficulty in validating parameterization and building a joint distribution?

A. Copulas.
B. Constant diversification.
C. Variance-covariance matrix.

D. Full modeling/simulation.

Answer: A

Step by step

1. Copulas combine marginal distributions through copula functions and allow nonlinearities and higher-order dependencies.
2. Their disadvantages are that parameterization is very difficult to validate and building a joint distribution is very difficult.
3. So the answer is copulas (A).

Why A is correct: Copulas are flexible but their parameterization is hard to validate and the joint distribution is hard to build.

Why the others are wrong

- **B:** Constant diversification is simple but ignores differing correlations; it is not criticized for these difficulties.
- **C:** The variance-covariance matrix is simple but does not capture nonlinearities; its weakness is linearity, not these difficulties.
- **D:** Full simulation is the most demanding in inputs, IT and time, but its weakness is practical demands rather than joint-distribution validation.

Q485 Schweser Module Quiz 60.2 Q1

Which of the following model validation processes is specifically characterized by the limitation that it provides little comfort that the model actually reflects reality?

A. Backtesting.
B. Benchmarking.
C. Stress testing.
D. Qualitative review.

Answer: B

Step by step

1. Benchmarking compares one model (or calibration) with another, not with actual outcomes.
2. Hence it may give little assurance the model reflects reality or about absolute levels of output; it only confirms broad comparability.
3. So the answer is benchmarking (B).

Why B is correct: Benchmarking can only compare one model to another, so it provides little assurance that the model reflects reality.

Why the others are wrong

- **A:** Backtesting does test forecasts against actual outcomes, so it speaks directly to reality.
- **C:** Stress testing compares outputs with stress losses and can give a modest degree of comfort on absolute capital levels.
- **D:** Qualitative review examines documentation and assumptions; the 'little comfort about reality' limitation is attributed to benchmarking.

LO 60.b

Describe the recommendations by the Bank for International Settlements (BIS) that supervisors should consider in order to make effective use of internal risk measures, such as economic capital, that are not designed for regulatory purposes.

Q486 AnalystPrep Q.3211 · Schweser Module Quiz 60.3 Q1

Which of the following categories of BIS recommendations specifically refers to the need to consider using additional methods, such as stress testing, to help cover all exposures?

A. Risk aggregation.
B. Counterparty credit risk.
C. Dependency modeling in credit risk.
D. Interest rate risk in the banking book.

Answer: B

Step by step

1. Recall the ten BIS recommendations and the supplementary tool each one names.
2. Recommendation 9 (counterparty credit risk): banks must understand trade-offs among measurement methods and use complementary methods such as stress testing, because no single method covers all counterparty exposures.
3. Match: only counterparty credit risk pairs stress testing with covering all exposures.

Why B is correct: The counterparty credit risk recommendation says measurement methods involve trade-offs and must be supplemented (e.g., stress testing) to cover all exposures.

Why the others are wrong

- **A:** Risk aggregation concerns harmonising confidence level and horizon and reflecting the bank's business mix; stress testing to cover exposures is not its theme.
- **C:** Dependency modeling calls for sensitivity and scenario analysis to address credit-model limitations, a different supplementary tool and purpose.
- **D:** IRRBB focuses on embedded options and earnings-based versus economic-value approaches, not stress testing to cover all exposures.

Q487

BT answers L2.T7.6.5 · BT notes P2.T7.6.5

REBUILT — Options C and D were near-duplicates after OCR; replaced C with the transparency recommendation so each option is distinct and exactly one is false.

Each of the following is a BIS recommendation that supervisors should consider in order to make effective use of internal risk measures not designed for regulatory purposes, EXCEPT:

A. A bank should be able to demonstrate how its economic capital model has been integrated into the business decision-making process.

B. All risks should be directly quantified within the economic capital model.

C. Economic capital model results should be transparent and taken seriously so they are useful to senior management.

D. Economic capital model validation should be rigorous and comprehensive, aimed at demonstrating the model is fit for purpose.

Answer: B

Step by step

1. Test each option against the BIS recommendations.
2. A: recommendation 1 (use in assessing capital adequacy) requires demonstrating integration into decision-making. True.
3. C: recommendation 3 (transparency and integration) requires transparent, credible results. True.
4. D: recommendation 7 (validation) requires rigorous, comprehensive validation showing fit for purpose. True.
5. B: recommendation 4 states that not all risks can be directly quantified; material hard-to-quantify risks (funding liquidity, reputational) are captured through compensating controls such as sensitivity analysis, stress testing or scenario analysis. B is therefore not a recommendation.

Why B is correct: BIS explicitly acknowledges that not all risks can be directly quantified and prescribes compensating controls instead, so 'all risks should be quantified' is the exception.

Why the others are wrong

- **A:** This is a genuine recommendation (use in capital adequacy assessment), so it is not the exception.
- **C:** This is a genuine recommendation (transparency and integration), so it is not the exception.
- **D:** This is a genuine recommendation (validation), so it is not the exception.

Q488

BT answers P2.T7.24.8.1 · BT notes P2.T7.24.8.1

Q-Capita Bank has implemented an economic capital model to better understand its capital adequacy and support strategic decision-making. During a supervisory review, the bank's board of directors is questioned about their awareness of how the economic capital model impacts business decisions.

Which of the following responses would most likely satisfy the supervisor's expectations regarding the use of economic capital models?

A. We focus primarily on regulatory capital requirements, and our risk department only uses the economic capital model for internal risk measurement purposes.

B. Our economic capital model helps guide key decisions, such as capital allocation, by assessing the difference between standalone and diversified risk.

C. While we use the economic capital model for assessing capital adequacy, its results do not significantly influence our business strategies or risk management practices.

D. Our board does not directly engage with the economic capital model; instead, we rely entirely on external consultants to ensure its proper application.

Answer: B

Step by step

1. BIS recommendation 1: the bank must demonstrate that the model is integrated into business decision-making, and the board must understand the gap between gross (stand-alone) and net (diversified) risk.
2. B shows both: decisions guided by the model and awareness of stand-alone versus diversified risk.
3. A, C and D each describe limited use or no board engagement, which contradicts the recommendation.

Why B is correct: Only B shows the model feeding decisions and the board understanding stand-alone versus diversified risk, as recommendation 1 requires.

Why the others are wrong

- **A:** Confines the model to the risk department and ignores integration into business decisions and board understanding.
- **C:** Admits results do not influence strategy, the opposite of demonstrated integration into decision-making.
- **D:** Board outsourcing the understanding fails the expectation that the board itself grasps gross versus net risk.

Q489

BT answers P2.T7.24.8.2 · BT notes P2.T7.24.8.2

REBUILT — Option A (backtesting and stress testing) was defensible as good practice, creating two plausible answers; replaced with a clearly non-BIS statement that a single risk measure is preferred.

Zedra Bank uses internal Value at Risk (VaR) models to manage its economic capital, focusing on credit risk and counterparty credit risk. During a supervisory review, the bank is asked to demonstrate how it ensures the effectiveness of these internal risk measures. Based on BIS recommendations, which of the following actions would best align with effective use of economic capital models that are not designed purely for regulatory purposes?

A. The bank adopts VaR as the single preferred risk measure for all risk types, on the basis that BIS identifies it as the best measure for economic capital.

B. The bank assesses whether the dependency structures in its credit portfolio model are appropriate and uses supplementary approaches such as sensitivity and scenario analysis to address model limitations.

C. The bank conducts periodic reviews of its counterparty credit risk models but focuses primarily on meeting minimum regulatory requirements.

D. The bank emphasizes market risk in its economic capital models, with plans to incorporate credit and counterparty risks at a later stage.

Answer: B

Step by step

1. Recommendation 8 (dependency modeling in credit risk): assess whether dependency structures suit the credit portfolio, identify model limitations, and address them with supplementary approaches such as sensitivity analysis and scenario analysis.
2. B matches this directly.
3. A contradicts recommendation 5 (no singularly preferred risk measure); C targets only minimum regulatory requirements; D defers material risks, contradicting the thorough risk identification of recommendation 4.

Why B is correct: B applies the BIS dependency-modeling recommendation: scrutinise dependency structures and cover model limitations with sensitivity and scenario analysis.

Why the others are wrong

- **A:** BIS states there is no singularly preferred risk measure; each has advantages and disadvantages that must be understood.
- **C:** Focusing on regulatory minimums misses the purpose of internal measures; the recommendations ask for understanding trade-offs and using complementary methods such as stress testing.
- **D:** Deferring credit and counterparty risk leaves material risks unmeasured, contrary to the risk identification recommendation.

Q490 BT notes P2.T7.4.2

REBUILT — Original EXCEPT item (economic capital as sole determinant of required capital) was not a BIS supervisory recommendation and had no source answer; rewritten to test the BIS risk-measures recommendation.

Regarding the BIS recommendation on risk measures used in economic capital frameworks, which of the following statements is most accurate?

A. BIS identifies Value at Risk at a 99.9% confidence level as the preferred measure for economic capital.

B. There is no singularly preferred risk measure, and a bank should understand the limitations of the measures it uses and the implications of its choice.

C. A bank should use the same risk measure as the regulatory framework so that economic and regulatory capital are directly comparable.

D. Expected shortfall should replace VaR because it is free of limitations and so needs no further validation.

Answer: B

Step by step

1. BIS recommendation 5 (risk measures): all measures in use have advantages and disadvantages to be understood in the context of their intended application.
2. There is no singularly preferred measure; banks must understand limitations and the implications of their choice.
3. Only B restates this; A, C and D each assert a single mandated or flawless measure.

Why B is correct: Recommendation 5 states that no single risk measure is preferred and that banks must understand the limitations of the measures they choose.

Why the others are wrong

- **A:** BIS names no preferred measure or confidence level; this invents a prescription.
- **C:** Nothing requires adopting the regulatory measure; the recommendation concerns internal measures not designed for regulatory purposes.
- **D:** No measure is limitation-free; the recommendation is to understand limitations, not to replace one measure with a supposedly perfect one.

LO 60.c

Explain benefits and impacts of using an economic capital framework within the following areas: credit portfolio management risk-based pricing customer profitability analysis management incentives

Q491 BT answers L2.T7.4.2 (q0587)

REBUILT — Original 'sole determinant of required capital' item was not supported by the reading and mixed LOs; rebuilt as a constraints/opportunities true-false item on credit portfolio management, pricing and incentives.

Which of the following statements about the use of an economic capital framework is INCORRECT?

A. In credit portfolio management, the creditworthiness of each borrower is assessed in a portfolio context rather than on a stand-alone basis.

B. In risk-based pricing, deals whose expected RAROC falls below a given threshold are rejected, so the allocated economic capital sets a floor on the interest rate.

C. Compensation schemes rank quite low among the actual uses of economic capital measures at the business-unit level.

D. A loan with higher stand-alone risk always contributes more risk to the portfolio, so ranking loans by stand-alone risk is sufficient to assess concentration.

Answer: D

Step by step

1. A: the reading states borrower credit quality is judged in a portfolio setting - true.
2. B: pricing decisions are driven by a floor (minimum RAROC or minimum interest rate) computed from allocated economic capital - true.
3. C: studies show compensation ranks low among actual uses of EC at business-unit level - true.
4. D: a loan with higher stand-alone risk does NOT necessarily contribute more portfolio risk; the marginal (incremental) contribution is what matters for concentration - false, so D is the incorrect statement.

Why D is correct: Under an economic capital framework, a loan's incremental contribution to portfolio risk, not its stand-alone risk, drives concentration assessment.

Why the others are wrong
- **A:** A true statement of the portfolio-context constraint, so not the exception.
- **B:** A true statement of the RAROC hurdle / minimum rate in risk-based pricing.
- **C:** A true statement: the actual use of compensation linkage is low.

Q492 BT answers L2.T7.6.6 · BT notes P2.T7.6.6

Which is nearest to the Basel Committee's position on management incentives (compensation) and economic capital (EC)?

A. Linking incentive pay to EC does NOT tend to motivate participation in the EC allocation process; such linkage need not be attempted, and actual linkage (EC metrics to pay) in practice is quite low

B. Linking incentive pay to EC does NOT tend to motivate participation in the EC allocation process; some such linkage should be attempted, and actual linkage (EC metrics to pay) in practice is prevalent

C. Linking incentive pay to EC DOES tend to motivate participation in the EC allocation process; some such linkage should be attempted, but actual linkage (EC metrics to pay) in practice is quite low

D. Linking incentive pay to EC DOES tend to motivate participation in the EC allocation process; some such linkage should be attempted, and actual linkage (EC metrics to pay) in practice is already prevalent

Answer: C

Step by step
1. Opportunity: incentives are the most sensitive element for most managers and motivate their involvement in the technical EC allocation process, so EC should influence business-unit incentive structures.
2. Constraint: evidence shows compensation schemes rank quite low among actual uses of EC at business-unit level.
3. Both parts together: motivating effect, linkage should be attempted, actual use low - option C.

Why C is correct: Incentives motivate managers to engage with EC allocation, so linkage is encouraged, yet in practice compensation is a minor use of EC.

Why the others are wrong
- **A:** Wrong on motivation (incentives do motivate) and on 'need not be attempted'; only the 'low use' part is right.
- **B:** Wrong that incentives do not motivate, and wrong that linkage is prevalent.
- **D:** Gets the motivating effect right but wrongly says linkage is already prevalent; in practice compensation ranks low.

Q493 BT answers P2.T7.24.9.1 · BT notes P2.T7.24.9.1

REBUILT — Options A and C were both defensible (marginal impact of each loan); rewrote A as the stand-alone-risk fallacy so exactly one option is correct.

Alpha Bank is analyzing the risk and return profile of its credit portfolio and wants to reduce concentration risk by transferring some high-risk loans. How can the economic capital framework assist the bank in making this decision?

A. By assuming that a loan with higher stand-alone risk always adds more risk to the portfolio, so the loans with the highest stand-alone risk are transferred first

B. By evaluating the individual credit risk of each borrower in isolation

C. By quantifying the incremental risk each loan adds to the total portfolio, guiding rebalancing and hedging decisions

D. By estimating potential future losses from default without considering portfolio correlations

Answer: C

Step by step
1. Credit portfolio management assesses each borrower in a portfolio setting.
2. A loan's marginal (incremental) contribution to portfolio risk determines concentration and which hedging or risk-transfer actions to use.
3. Only C measures incremental portfolio risk.

Why C is correct: Economic capital captures each loan's marginal contribution to portfolio risk, so transfers can target true concentration rather than stand-alone risk.

Why the others are wrong
- **A:** Confuses stand-alone with marginal risk; a high stand-alone-risk loan does not necessarily contribute more to the portfolio.
- **B:** Stand-alone, isolated assessment ignores the portfolio context the framework requires.

- **D:** Ignoring correlations misses diversification and concentration effects, the core of portfolio-level EC.

Q494 BT answers P2.T7.24.9.2 · BT notes P2.T7.24.9.2

REBUILT — Options A and C were both defensible; replaced C with a clearly flawed pricing approach so only A is correct.

Amero Bank is setting the interest rate for a new loan product for corporate clients. The bank wants pricing that reflects the risk of the product and supports shareholder value. How does the economic capital framework influence this risk-based pricing decision?

A. The minimum rate covers the cost of funding, expected loss, the cost of the allocated economic capital, and the extra return required by shareholders

B. The rate accounts only for operational costs and expected loan losses

C. The rate is set to match prevailing peer-bank rates, with allocated economic capital used only after the loan is booked

D. The same rate is set across all loan products to simplify internal processes

Answer: A

Step by step

1. Risk-based pricing incorporates (1) cost of funding, (2) expected loss, (3) allocated economic capital and (4) extra return required by shareholders.
2. This yields a minimum interest rate (or minimum RAROC) that at least does not reduce shareholder value.
3. Only A lists these components.

Why A is correct: Allocated economic capital sets a floor on the loan rate together with funding cost, expected loss and the shareholders' required extra return.

Why the others are wrong

- **B:** Omits funding cost and the capital charge; expected loss alone does not compensate for unexpected-loss risk.
- **C:** Treating EC as an afterthought ignores that it determines the minimum rate/RAROC hurdle before the deal.
- **D:** Uniform pricing ignores risk differences across products, contrary to risk-based pricing.

Q495 BT answers P2.T7.24.9.3 · BT notes P2.T7.24.9.3

REBUILT — Source answer missing for one copy and A/C both defensible; removed the incentive twist and rewrote C so only A is correct.

11th Avenue Bank is analyzing its customer base to improve profitability and resource allocation, balancing customer relationships against the economic capital each one absorbs. How does the economic capital framework support this customer profitability analysis?

A. By segmenting customers by net return per unit of risk, it identifies unprofitable or marginally profitable customers so capital can be reallocated to more profitable relationships

B. By prioritizing customer segments with the highest revenue, independent of the capital they require

C. By ranking customers on absolute profit, dropping the lowest, without regard to the economic capital each absorbs

D. By concentrating resources on low-risk customers, regardless of their overall profitability

Answer: A

Step by step

1. Customer profitability analysis segments customers by ranges of net return per unit of risk.
2. Economic capital provides the risk base, giving relative risk-adjusted profitability of relationships.
3. Unprofitable customers can be dropped and capital reallocated; only A describes this.

Why A is correct: EC makes returns comparable per unit of risk, supporting reallocation of capital to the most profitable relationships (subject to the difficulty of measuring risks and costs per customer).

Why the others are wrong

- **B:** Revenue ignores the capital and risk absorbed.
- **C:** Absolute profit ignores capital consumed, so a large but capital-hungry customer would look better than it is.
- **D:** Low risk alone is not profitability; return per unit of risk is what matters.

Q496 Schweser Module Quiz 60.3 Q2

The use of which of the following items is meant more for protecting against risk deterioration?

A. Risk-based pricing.
B. Management incentives.
C. Credit portfolio management.
D. Customer profitability analysis.

Answer: C

Step by step

1. Credit portfolio management is described as a means of protecting against risk deterioration.
2. Other items serve different purposes: pricing (profitability), customer analysis (identify unprofitable customers), incentives (motivate participation).

Why C is correct: Credit portfolio management (hedging and risk transfer to control concentration) is the tool framed as protection against risk deterioration.

Why the others are wrong

- **A:** Risk-based pricing maximizes profitability and sets a minimum rate.
- **B:** Management incentives motivate managers to take part in EC allocation.
- **D:** Customer profitability analysis identifies unprofitable customers for reallocation of capital.

LO 60.d

Describe best practices and assess key concerns for the governance of an economic capital framework.

Q497 BT answers P2.T7.24.8.3 · BT notes P2.T7.24.8.3

DEF Bank has implemented an economic capital framework to manage its risk and capital adequacy. To align with best practices in governance, senior management is reviewing the governance structure to ensure it supports strategic decision-making and risk management.

Which of the following actions would most likely reflect best practices for the governance of an economic capital framework, according to BIS recommendations?

A. Assigning economic capital oversight to the finance department, with minimal involvement from senior management.
B. Establishing governance controls, including formalized change processes and active senior management engagement.
C. Performing economic capital calculations annually, with senior management reviewing the results at year-end.
D. Focusing on regulatory capital requirements and using economic capital models only for internal reporting.

Answer: B

Step by step

1. Recall that the reliability of economic capital rests on strong controls: tight control over changes to risk measurement techniques, thorough documentation, policies ensuring procedures are followed, and use of the measures in day-to-day decisions.
2. Senior management must ensure robust controls and governance over the entire process and is a main driver of the framework.
3. Option B matches both elements (formal change control and active senior management engagement), so B is correct.

Why B is correct: Effective economic capital governance requires formal change controls and active senior management commitment across the whole process.

Why the others are wrong

- **A:** Delegating to one function with minimal senior involvement contradicts the requirement that senior management be committed and responsible for governance of the entire process.
- **C:** Most banks compute economic capital monthly or quarterly (data quality and manual effort limit it further); annual calculation with a year-end review is too infrequent to support decisions.
- **D:** Economic capital should be applied meaningfully to daily business decisions, not confined to internal reporting; it is not just a regulatory-capital substitute.

Q498

MK-GARP PE1-Q69

REBUILT — Original answer (escalation procedures in a contingency plan) is not clearly supported by the reading and distractors mixed in aggregation; rebuilt around the change-control and validation governance practices described in GARP.

An internal auditor at a large bank finds that changes to the aggregation methodology in the economic capital model are implemented without any formal approval, no diagnostics are run afterwards, and there is no procedure to follow when the reported economic capital moves unexpectedly by a large amount. Which of the following actions should the CRO recommend to bring the governance framework in line with best practice?

A. Require responsible parties to sign off on methodology changes, run diagnostics and formal validation after each change, and set internal escalation procedures for unexpectedly large changes in economic capital.

B. Allow the economic capital model to be changed at the discretion of the model owner, since the model is used only internally and need not follow change-control processes.

C. Reduce the calculation frequency to annual so that fewer model changes need to be reviewed.

D. Remove the post-change diagnostics because the owner of the model already mechanically aggregates all component reports.

Answer: A

Key correction: Source key (MK-GARP PE1-Q69) said C (escalation procedures in the contingency plan); the rebuilt item tests the same governance theme with a single clearly supported answer.

Step by step

1. Identify the deficiencies: no change control, no post-change diagnostics, no escalation for large unexplained changes.
2. GARP describes strong controls over changes to risk measurement techniques, sign-off by responsible parties on methodology changes, and diagnostics run after a model change; it notes that formalised validation after changes and internal escalation procedures for unexpectedly large differences in economic capital are uncommon, so adding them closes the gaps.
3. Only A addresses all three deficiencies, so A is correct.

Why A is correct: Sign-off, post-change diagnostics and validation, and escalation procedures strengthen the controls over changes that the reliability of economic capital depends on.

Why the others are wrong

- **B:** Being internal does not remove the need for formal controls; reliability of economic capital depends on strong controls over changes.
- **C:** Most banks compute monthly or quarterly; reducing frequency weakens monitoring and does not fix the control gaps.
- **D:** Mechanical aggregation by the owner is not a substitute for diagnostics and validation; removing them worsens the weakness.

Q499

MK-Mock Mock 3 Q12

REBUILT — Original options were mostly unrelated to governance and the key (escalation procedures) is weakly supported; rebuilt on the key governance concern that economic capital should not be the only determinant of required capital.

A newly appointed CRO at a regional bank is reviewing the governance of the bank's economic capital framework. The CFO proposes that required capital be set solely by the economic capital model output because it is the most risk-sensitive measure. Which response best reflects a key governance concern?

A. Economic capital should not be the only measure determining required capital; banks typically also align capital with target external credit ratings, balancing shareholders' preference for lower capital against rating agencies' preference for higher capital.

B. The proposal is appropriate because economic capital is risk-sensitive and therefore removes the need to consider external credit ratings.

C. The proposal is appropriate provided available capital is defined as the standard Tier 1 figure, since all banks define available capital resources identically.

D. The proposal is appropriate because a high-confidence economic capital model makes integrated stress testing unnecessary.

Answer: A

Key correction: Source key (MK-Mock 3 Q12) said C (escalation procedures in the contingency plan); the rebuilt item tests a different, clearly documented governance concern.

Step by step

1. Key governance concern: economic capital should not be the sole determinant of required capital.

2. Most banks align economic capital with external credit ratings; shareholders want profitability via lower capital while rating agencies encourage solvency via higher capital.
3. B ignores rating considerations, C falsely assumes a uniform definition of available capital, D dismisses stress testing. Only A is correct.

Why A is correct: Required capital reflects the tension between shareholder and rating-agency expectations, so economic capital cannot be the only determinant.

Why the others are wrong

- **B:** Risk sensitivity does not remove rating-based targets; banks align economic capital with external ratings.
- **C:** There is no common definition of available capital across banks; most adjust Tier 1 capital in different ways.
- **D:** Integrated stress testing is encouraged to assess the impact of scenarios on economic capital; a high confidence level does not replace it.

Reading 61. Capital Planning at Large Bank Holding Companies: Supervisory Expectations and Range of Current Practice

Board of Governors of the Federal Reserve System, August 2013

LO	Learning objective	Qs
61.a	describe the Federal Reserve's Capital Plan Rule and explain the seven principles of an effective capital adequacy process for bank holding companies (BHCs) subject to the Capital Plan Rule.	9
61.b	describe practices that can result in a strong and effective capital adequacy process for a BHC in the following areas: risk identification internal controls, including model review and valuation corporate governance capital policy, including setting of goals and targets and contingency planning stress testing and stress scenario design estimating losses, revenues, and expenses, including quantitative and qualitative methodologies assessing the impact of capital adequacy, including risk-weighted asset (RWA) and balance sheet projections	15

LO 61.a

Describe the Federal Reserve's Capital Plan Rule and explain the seven principles of an effective capital adequacy process for bank holding companies (BHCs) subject to the Capital Plan Rule.

Q500

AnalystPrep Q.2224

Oak Creek Bank, a bank holding company (BHC), is preparing for the annual CCAR. Which of the following is NOT one of the seven principles of an effective capital adequacy process under the Federal Reserve's Capital Plan Rule?

A. Effective loss-estimation methodologies
B. Sufficient capital adequacy impact assessment
C. Adequate IT resources
D. Robust internal controls

Answer: C

Step by step

1. The seven principles are: (1) sound foundational risk management, (2) effective loss-estimation methodologies, (3) solid resource-estimation methodologies, (4) sufficient capital adequacy impact assessment, (5) comprehensive capital policy and capital planning, (6) robust internal controls, (7) effective governance.
2. A is principle 2, B is principle 4, D is principle 6.
3. 'Adequate IT resources' appears nowhere in the list, so C is the exception.

Why C is correct: The CAP principles cover risk management, loss estimation, resource estimation, impact assessment, capital policy, controls and governance; IT resourcing is not a stand-alone principle.

Why the others are wrong

- **A:** This is principle 2, so it is part of the CAP; choosing it misreads the NOT wording.
- **B:** This is principle 4 (combining losses and resources to assess capital adequacy), so it is a genuine principle.
- **D:** This is principle 6 (model validation, documentation, internal audit), so it is a genuine principle.

Q501

AnalystPrep Q.2225

During its annual CCAR, Great Falls Bank, a BHC, discovers that one of its models used in the capital planning process has been neither appropriately validated nor independently reviewed. Which principle of an effective capital adequacy process has been violated?

A. Robust internal controls
B. Sufficient capital adequacy impact assessment
C. Effective loss-estimation methodologies
D. Effective governance

Answer: A

Step by step

1. Principle 6 (robust internal controls) covers policies and procedures, change control, model validation and independent review, documentation and internal audit.
2. The failure is a missing model validation / independent review, which falls squarely under principle 6.

3. Therefore A.

Why A is correct: Model validation and independent review are explicitly listed components of the internal-controls principle.

Why the others are wrong

- **B:** Impact assessment is about combining loss and resource estimates against capital goals, not about validating models.
- **C:** Loss-estimation concerns turning risk measures into stressed loss estimates; the problem here is lack of validation, which is a control issue.
- **D:** Governance concerns board and senior management oversight and approval of capital decisions; it is the tempting 'oversight' answer, but validation sits with controls.

Q502 AnalystPrep Q.2226

Minnetonka Bank, a BHC, finds that its process for translating risk measures into estimates of potential losses does not cover a satisfactory range of stressful scenarios and environments. Which principle of an effective capital adequacy process has been violated?

A. Sound foundational risk management
B. Sufficient capital adequacy impact assessment
C. Effective governance
D. Effective loss-estimation methodologies

Answer: D

Step by step

1. Principle 2 requires effective processes for translating risk measures into loss estimates over a range of stressful scenarios and aggregating them across the BHC.
2. The described failure (insufficient range of stress scenarios in loss estimation) is a breach of principle 2.
3. Therefore D.

Why D is correct: The wording in the stem mirrors the definition of the loss-estimation principle.

Why the others are wrong

- **A:** Principle 1 concerns the infrastructure to identify, measure, assess and control material risks, not the translation of risk measures into stressed losses.
- **B:** Principle 4 combines loss and resource estimates to assess capital adequacy; the flaw here arises earlier, in producing the loss estimates.
- **C:** Governance is board and senior management oversight; nothing in the stem concerns oversight.

Q503 AnalystPrep Q.2232

Fetterman Bank, a BHC subject to the Capital Plan Rule, is projecting its revenue and expenses for the capital plan. Over what planning horizon does the Federal Reserve's Capital Plan Rule require such estimates?

A. Nine quarters
B. Eight quarters
C. Twelve quarters
D. There is no explicit length of time defined by the Federal Reserve

Answer: A

Step by step

1. The Capital Plan Rule requires BHCs to project revenues, expenses, losses and capital over a nine-quarter planning horizon.
2. Nine quarters corresponds to option A.

Why A is correct: The rule specifies a nine-quarter planning horizon, not a flexible or longer/shorter period.

Why the others are wrong

- **B:** Eight quarters (two years) is a common round-number mistake; the rule uses nine quarters.
- **C:** Twelve quarters (three years) overstates the horizon.
- **D:** The horizon is explicitly defined as nine quarters.

Q504

AnalystPrep Q.2236

Campbell Bank, a BHC, has not had its risk infrastructure or its loss-estimation methodologies periodically reviewed by its board and senior management for more than a year. Which principle of an effective capital adequacy process does this most directly violate?

A. Robust internal controls
B. Sound foundational risk management
C. Effective governance
D. Sufficient capital adequacy impact assessment

Answer: C

Step by step

1. Principle 7 (effective governance) requires board and senior management oversight, including periodic review of the BHC's risk infrastructure and loss- and resource-estimation methodologies.
2. The stem describes a lapse in periodic review of exactly these items.
3. Therefore C.

Why C is correct: Periodic board/senior management review of the risk infrastructure and loss-estimation methods is a named element of the governance principle.

Why the others are wrong

- **A:** Internal controls cover validation, documentation, change control and internal audit; periodic senior-level review of infrastructure and methodologies is assigned to governance.
- **B:** Principle 1 is about having sound risk-measurement and management infrastructure, not about the periodic oversight review of it.
- **D:** Impact assessment concerns combining losses and resources against capital goals.

Q505

BT answers P2.T7.806.1 · BT notes P2.T7.806.1

REBUILT — Options A and B reworded to match the GARP chapter wording (CCAR as supervisory program; DFA stress tests with adverse and severely adverse scenarios).

Each of the following statements about capital planning at large BHCs supervised by the Federal Reserve is true EXCEPT:

A. The Comprehensive Capital Analysis and Review (CCAR) is the Federal Reserve's supervisory program for assessing the capital plans of large BHCs.
B. The Dodd-Frank Act supervisory stress tests play an important role in CCAR, using supervisory adverse and severely adverse scenarios.
C. Once a BHC meets the minimum regulatory capital requirements in normal, non-stressed times, the Federal Reserve cannot restrict its freedom to pay dividends or repurchase shares.
D. The seven principles of an effective capital adequacy process are: sound foundational risk management; effective loss-estimation methodologies; solid resource-estimation methodologies; sufficient capital adequacy impact assessment; comprehensive capital policy and capital planning; robust internal controls; and effective governance.

Answer: C

Step by step

1. A is true: CCAR is the Federal Reserve's supervisory program for evaluating the capital plans of large BHCs.
2. B is true: the Dodd-Frank supervisory stress tests play an important role in CCAR and use supervisory adverse and severely adverse scenarios.
3. D lists the seven principles correctly.
4. C is false: the Federal Reserve assesses planned capital actions (dividends, repurchases) under stress and can object to a capital plan, so meeting minimum ratios in normal times does not free a BHC from supervisory limits on payouts.

Why C is correct: Capital distributions are assessed under stress and the Fed can object to a capital plan, so meeting minimum ratios in good times does not remove supervisory limits on payouts.

Why the others are wrong

- **A:** True; CCAR is the Federal Reserve's supervisory program for assessing capital plans.
- **B:** True; the supervisory stress tests feed into CCAR and use adverse and severely adverse scenarios.
- **D:** True; this is the correct list of the seven principles.

Q506 MK Mock 2 Q66

Kimco, a BHC, has its capital resource management and allocation process assessed by the Federal Reserve. Which of the following statements does NOT accurately describe one of the seven principles applied in the assessment?

A. Internal controls should be documented and subjected to internal audit.
B. A process should be in place to estimate potential losses and aggregate them on a firm-wide basis.
C. Oversight should be managed by a third-party risk assessor appointed by the BHC's board of directors.
D. Loss estimates and capital resources should be considered in determining the impact on capital adequacy.

Answer: C

Step by step

1. A matches principle 6 (documentation and internal audit).
2. B matches principle 2 (estimate losses and aggregate across the BHC).
3. D matches principle 4 (combine losses and resources to assess capital adequacy).
4. Principle 7 places oversight with the board and senior management, not a third-party assessor, so C is inaccurate.

Why C is correct: The governance principle assigns oversight to the board and senior management of the BHC itself.

Why the others are wrong

- **A:** Accurate: principle 6 requires comprehensive documentation and internal audit review.
- **B:** Accurate: this is principle 2.
- **D:** Accurate: this is principle 4.

Q507 Schweser Module Quiz 61.1 Q1

Which of the following is among the seven principles of an effective capital adequacy process for BHCs subject to the Capital Plan Rule?
I. Oversight from peer BHCs.
II. Annual reporting to the stock exchange on which the BHC's stock is listed.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: D

Step by step

1. Principle 7 assigns oversight to the BHC's own board and senior management, not to peer BHCs, so I is not a principle.
2. Exchange reporting is not among the seven principles, so II is not one either.
3. Neither applies: D.

Why D is correct: The seven principles concern internal risk management, estimation, impact assessment, policy, controls and governance; peer oversight and exchange reporting are not included.

Why the others are wrong

- **A:** Peer oversight is not a principle; oversight is by the BHC's own board and senior management.
- **B:** Exchange reporting is not a principle.
- **C:** Neither statement is a principle.

Q508 Schweser Module Quiz 61.1 Q2

The Federal Reserve's Capital Plan Rule requires which of the following BHCs to develop a capital plan supported by an effective process for assessing capital adequacy?

A. All BHCs, whether U.S. or non-U.S. domiciled.
B. BHCs with more than five years of operational history.
C. BHCs with net annual income of more than $5 billion.
D. U.S.-domiciled BHCs with total consolidated assets of $50 billion or more.

Answer: D

Step by step

1. The rule applies to U.S.-domiciled top-tier BHCs with total consolidated assets of $50 billion or more.
2. A ignores the domicile and size criteria; B and C use criteria (age, income) that are not in the rule.
3. Therefore D.

Why D is correct: Coverage is defined by U.S. domicile and total consolidated assets of at least $50 billion.

Why the others are wrong

- **A:** The rule covers only U.S.-domiciled BHCs above the asset threshold, not every BHC.
- **B:** Operating history is not a criterion.
- **C:** The test is based on assets, not net income.

LO 61.b

Describe practices that can result in a strong and effective capital adequacy process for a BHC in the following areas: risk identification internal controls, including model review and valuation corporate governance capital policy, including setting of goals and targets and contingency planning stress testing and stress scenario design estimating losses, revenues, and expenses, including quantitative and qualitative methodologies assessing the impact of capital adequacy, including risk-weighted asset (RWA) and balance sheet projections

Q509

AnalystPrep Q.2231

Highlands Bank forms part of a Bank Holding Company (BHC). The bank is computing loss estimates on a number of its business lines. What are the components that the bank should take into account when estimating losses?

A. Probability of default (PD), time value of money (TM), and loss given default (LGD).
B. Probability of default (PD), loss given default (LGD), and exposure at default (EAD).
C. Probability of default (PD), time value of money (TM), and exposure at default (EAD).
D. Loss given default (LGD), exposure at default (EAD), and credit rating (CR).

Answer: B

Step by step

1. Under the expected loss approach, loss is a function of three components: PD, LGD and EAD.
2. Expected loss = PD × LGD × EAD, estimated at segment or individual-loan level.
3. Option B lists exactly these three; every other option includes a non-component (time value of money or credit rating).

Why B is correct: The expected loss approach decomposes credit loss into probability of default, loss given default and exposure at default.

Why the others are wrong

- **A:** Replaces EAD with 'time value of money', which is not a loss component; it omits exposure.
- **C:** Replaces LGD with time value of money; without LGD the severity of loss is missing.
- **D:** Credit rating is an input that may help estimate PD, not a separate component; PD is omitted.

Q510

AnalystPrep Q.2227

A certain bank based in New York is assessing risks as part of its preparation for the annual CCAR (Comprehensive Capital Analysis and Review). During the process of stress-testing, several risk categories are defined, particularly those that are difficult to quantify or not directly attributable to any of the specific integrated firm-wide risk categories. Which of the following risks would not fall under such a category?

A. Compliance risk
B. Credit risk
C. Reputational risk
D. Strategic risk

Answer: B

Step by step

1. Fed expectations: BHCs must account for 'other risks' that are hard to quantify or not directly attributable to a specific firm-wide risk category.

2. Examples of other risks: reputational, strategic and compliance risk.
3. Credit risk is a core, quantifiable, firm-wide risk category (models, loss estimation), so it is not an 'other risk'.

Why B is correct: Credit risk is a well-defined, quantifiable risk category, whereas the other three are the standard examples of hard-to-quantify other risks.

Why the others are wrong

- **A:** Compliance risk is a listed example of an other risk, so it does fall in the category (the question asks for the one that does not).
- **C:** Reputational risk is a listed example of an other risk and is hard to quantify.
- **D:** Strategic risk is a listed example of an other risk, arising from changes in strategy, and is hard to quantify.

Q511

GARP PE1 Q34

Risk analysts at a large bank have been asked to implement a stress testing procedure for a newly mandated regulatory stress test. The stress test requires that the bank project its revenues, losses, and capital ratios for each quarter over a nine-quarter horizon. Which of the following actions or assumptions should the analysts recommend for the stress test that is consistent with best practices?

A. Use a full-revaluation approach to model the bank's complex derivative exposures.

B. Use minimum required regulatory capital ratios as the initial trigger for potential contingency capital actions to be taken.

C. Assume that the bank will exit its riskiest lines of business and reduce its expenses during the stress test horizon period.

D. Model the loss given default of the bank's loan portfolio using a weighted-average approach at the aggregate portfolio level.

Answer: A

Step by step

1. Stress moves are very large and positions such as derivatives are nonlinear, so approximations (e.g. delta-gamma) can mismeasure losses.
2. GARP/Fed: BHCs should generally use full-revaluation methods for stress testing.
3. B: triggers based only on minimum regulatory ratios are weak (too late). C: mitigating actions must be conservative and are more limited under stress. D: aggregate weighted-average LGD is a lagging practice.

Why A is correct: Full revaluation captures nonlinear payoffs under the very large risk-factor moves of a stress scenario.

Why the others are wrong

- **B:** Triggers set at regulatory minima leave no time to act; contingency triggers should be set above minima and consider stakeholder expectations.
- **C:** Assuming exit of risky businesses and expense cuts is an aggressive mitigating-action assumption; such actions are harder in stress and must be conservative and executable.
- **D:** A weighted-average LGD at aggregate level hides segment variation and ignores LGD rising under stress; leading practice segments LGD and ties it to scenario severity.

Q512

AnalystPrep Q.2229

Fairgrounds Bank forms part of a Bank Holding Company (BHC). The bank has been very successful in a business line that was established about 6 months ago. The bank intends to stress test models for the business line for a longer period. As part of best practice during stress testing, the bank should:

A. Ensure minimal variation from established internal data patterns.

B. Test a wide range of adverse effects reaching outside the established data patterns.

C. Only use the data which reflects the most positive outcomes.

D. Only use the data which reflects the most negative outcomes.

Answer: B

Step by step

1. The Fed expects generally conservative assumptions in stress testing.
2. Models must use data with sufficiently adverse outcomes; a business that has done well in a short history (6 months) should not be assumed to keep its prior patterns.
3. So the bank should test a wide range of adverse effects, including those outside established data patterns.

Why B is correct: Stress tests should challenge historical patterns with severe adverse outcomes rather than assume past performance persists.

Why the others are wrong

- **A:** Staying close to internal data patterns assumes past (benign) behavior continues, which the Fed warns against.
- **C:** Using only the most positive data makes the test the opposite of conservative.
- **D:** Using only the most negative data is an extreme and unrepresentative input; the requirement is to include sufficiently adverse outcomes and test a wide range, not to restrict data to the worst.

Q513 AnalystPrep Q.2230

Clayton bank forms part of a Bank Holding Company (BHC) and has been requested, by the Federal Reserve, to compile documentation regarding its estimation practices. What are the main guidelines that should be followed by the bank while documenting its estimates?

A. Extremely detailed explanations of key methodologies with every bit of data available being presented.
B. Limited documentation emphasizing the importance of management's judgments in estimates.
C. Limited documentation encompassing theoretical assumptions and quantitative estimates.
D. Concisely explained key methodologies and assumptions presented in a well-organized manner.

Answer: D

Step by step

1. The Fed expects clear documentation of key methodologies and assumptions.
2. Stronger practice: documentation that concisely explains methodologies, drivers and their relationship to estimates, and delineates model outputs, qualitative overlays and purely qualitative estimates.
3. Only D describes concise, well-organized documentation.

Why D is correct: Leading practice documents key methodologies and assumptions concisely and in an organized way.

Why the others are wrong

- **A:** Presenting every bit of data is not concise; the focus is on key methodologies and assumptions.
- **B:** Limited documentation leaning on management judgment is characteristic of weaker practices.
- **C:** Limited documentation is also weak; documentation must be adequate to demonstrate the drivers and quantitative estimates, not just theoretical assumptions.

Q514 AnalystPrep Q.2237

What do you understand by "feeder models" as used in modeling by BHCs?

A. Models outsourced from external sources for the purpose of performance comparison with internal models.
B. Models used to produce projections or estimates that can then be used in another model to generate final figures for expected losses, expenses and revenue.
C. Models whose outcome has been disputed by experts and analysts at firm level.
D. Models used to generate the final projected figures for losses, expenses, and revenues.

Answer: B

Step by step

1. BHCs should maintain an inventory of all models used in capital planning, including input or 'feeder' models.
2. A feeder model produces projections or estimates that are used as inputs into another model.
3. The downstream model generates the final projected losses, revenues and expenses; this matches B.

Why B is correct: Feeder models supply intermediate estimates that are inputs to the models producing final loss, revenue and expense projections.

Why the others are wrong

- **A:** Describes vendor or benchmark models, not feeder models.
- **C:** Describes challenged or disputed models; unrelated to a feeder role.
- **D:** Describes the final (end) models that produce the final projections, i.e. what feeder models feed into.

Q515 AnalystPrep Q.2971

Internal controls in bank holding companies (BHCs) should ensure that there is integrity of reported results and the documentation, review, and approval of all material changes to the capital planning process and its components. Such controls as ensured by BHCs should exist at all levels of the capital planning process, with specific control measures to perform all the following roles apart from:

A. Making sure that there is sufficient robustness in MIS for capital analysis and decision making to be supported, with sufficient flexibility to run ad-hoc analysis whenever necessary.

B. Provide for reconciliation and data integrity process for all key reports.

C. Enable the addressing of presentation of aggregate, enterprise-wide capital planning results that gives the description of manual adjustments created in the aggregation process and identified weaknesses are compensated by these adjustments.

D. Ensure that the documentation provides evidence that results and recommendations can be challenged by the Board.

Answer: D

Step by step

1. GARP lists specific control measures: MIS robust enough to support analysis and flexible for ad hoc analysis; reconciliation and data integrity for key reports; presentation of aggregate results describing manual adjustments and how they compensate for weaknesses; and accurate, timely reports to senior management and board.
2. Options A, B and C restate the first three.
3. D, evidence enabling the Board to challenge results, is a board/governance responsibility (the board receives enough information to challenge reported results), not one of the specific control measures listed.

Why D is correct: A, B and C are control measures named in the internal controls guidance; D relates to board oversight, not an internal control measure.

Why the others are wrong

- **A:** This is a listed control measure (robust, flexible MIS), so it is not the exception.
- **B:** Reconciliation and data integrity for key reports is a listed control measure.
- **C:** Presenting aggregate results with a description of manual adjustments is a listed control measure.

Q516 BT answers P2.T7.806.2 · BT notes P2.T7.806.2

In regard to practices that can result in a strong and effective capital adequacy process, according to "Capital Planning at Large Bank Holding Companies: Supervisory Expectations and Range of Current Practice," each of the following is true EXCEPT which is false?

A. Most BHCs use some forms of expert judgment for some purposes, often as a management adjustment overlay to modeled outputs

B. With respect to Expected Loss Approaches, BHCs with leading practices were able to break down losses into PD, LGD, and EAD components, separately identifying key risk drivers for each of those components

C. Although best practices in operational-risk models are still evolving, common approaches to operational loss estimation for stress scenarios include regression models, modified loss-distribution-approach (LDA), and scenario analysis

D. With respect to Market and Counterparty Credit Risk, the Federal Reserve much prefers probabilistic approaches to deterministic approaches because deterministic approaches do not link stress scenarios to risk factor shocks

Answer: D

Step by step

1. A is true: most BHCs use expert judgment, often as a management overlay to modeled output.
2. B is true: leading practice under the expected loss approach breaks loss into PD, LGD, EAD with separate drivers.
3. C is true: operational-risk stress loss estimation uses regression models, modified LDA and scenario analysis.
4. D is false: the Fed does not 'much prefer' probabilistic approaches. Both have strengths and weaknesses (probabilistic: complex, lacks transparency, hard to tie to scenarios; deterministic: easier to communicate), and most BHCs used deterministic approaches for CCAR. Deterministic approaches do translate scenarios into risk-factor shocks.

Why D is correct: The Fed does not favor probabilistic over deterministic approaches; most BHCs used deterministic approaches, which do tie scenarios to risk-factor shocks.

Why the others are wrong

- **A:** True statement, so not the exception.
- **B:** True statement, so not the exception.
- **C:** True statement, so not the exception.

Q517 BT answers P2.T7.806.3 · BT notes P2.T7.806.3

Betaplanet International is a large bank holding company (BHC) with a high dependence on customer deposits. Betaplanet recently submitted its capital plan and stress scenarios to the Federal Reserve with the following features:
I. Betaplanet projected estimated revenue and expenses forward over a nine-quarter planning horizon
II. Betaplanet separately estimated losses, revenues, and expenses under hypothetical stressed conditions for business lines that are sensitive to different risk drivers
III. The adverse scenarios assume that Betaplanet will gain, respectively, +2.0% and +5.0% market share due to its competitive advantage and reputation for safety
IV. For some portfolios where it has limited expertise, Betaplanet relied on commercial, third-party models
V. For its wholesale portfolios, where it has limited history, Betaplanet supplemented internal data with external data
VI. Betaplanet projected net interest income (NII) and its components but (due to its high dependence on deposits) excluded projections of non-interest income and non-interest expense
Which of the features above are UNLIKELY to meet the regulatory expectations of the capital planning process?

A. None, all of the mentioned features are explicitly permitted
B. III. and VI., Betaplanet should not assume market share gains under stress scenario, nor should it exclude non-interest income and expenses in projections
C. II. and IV., Betaplanet should not separate losses/revenues/expenses by business line, nor should rely on third-party models whatsoever (versus internal models)
D. I. and V., Betaplanet must project revenue and expenses at least five years, and it cannot rely on external data

Answer: B

Step by step

1. I: the Capital Plan Rule requires projections over a nine-quarter planning horizon, so I meets expectations.
2. II: estimating losses, revenues and expenses separately for portfolios sensitive to different risk drivers is expected practice.
3. III: assuming market share gains in a stress scenario (because the BHC is viewed as strong) is contrary to the intent of stress testing, so III fails.
4. IV: third-party/vendor models are acceptable if staff understand them and they are validated per SR 11-7, so IV is fine.
5. V: external data may supplement limited internal history, provided it is a reasonable proxy, so V is fine.
6. VI: non-interest income and expense must be projected consistently with scenario conditions, so VI fails. Items III and VI fail: B.

Why B is correct: Stress scenarios must not embed assumptions that specifically benefit the BHC (such as market share gains), and all components of revenue and expense, including non-interest income and expense, must be projected consistently with the scenario.

Why the others are wrong

- **A:** Wrong because III and VI clearly contradict supervisory expectations; it ignores the market share and omitted-projection problems.
- **C:** Wrong: separate estimation by risk-driver-sensitive business line is expected, and vendor models are allowed when validated and understood; it flags acceptable practices.
- **D:** Wrong: the horizon is nine quarters, not five years, and external data may supplement limited internal history; it confuses the rule's horizon and data requirements.

Q518 MK GARP PE2-Q13

REBUILT — Original keyed B ('OTTI recognized in the first few months'), which GARP does not state; rewrote option B to the GARP-supported practice (projected ratings migration for OTTI testing) so exactly one option is correct.

The CRO of a large US-based bank holding company has asked a financial officer to develop a capital plan for the next 3-year period and to recommend a process for stress testing the capital plan over the same period. The officer wants to ensure that the plan conforms with the relevant guidelines for modeling revenues and losses across the entire planning period under both normal and stressed market conditions. Which of the following approaches will be appropriate for the officer to recommend?

A. Rely on scalar adjustments to baseline loss estimates for large loan portfolios to project losses during adverse market conditions.
B. Test credit-sensitive securities for other-than-temporary impairment using ratings that are projected to migrate under the stress scenario, rather than holding ratings static.
C. Model better-than-average revenue performance during times of stress to reflect the bank's competitive advantage.
D. Remove the impact of seasonality by dividing annual revenue and loss estimates for different types of products evenly across the year.

Answer: B

Key correction: Source key (MK) said B for the original wording ('non-temporary impairment losses recognized in the first few months'); the rebuilt B is the GARP-supported practice and remains the correct choice.

Step by step

1. A: scalar adjustments lack transparency and sensitivity to portfolio composition and scenario variables; use should be limited to immaterial portfolios, not large ones.
2. B: stronger practice projects ratings migration under the scenario and applies a ratings-based OTTI threshold; weaker practice kept ratings static. B is appropriate.
3. C: assuming relative strength or market share gains in stress is contrary to supervisory expectations.
4. D: spreading annual estimates evenly over quarters ignores the expected quarterly path of losses and revenues.

Why B is correct: Credit-sensitive AFS/HTM securities should be tested for OTTI using ratings that migrate with the stress scenario; the other approaches are explicitly identified as weak practices.

Why the others are wrong

- **A:** Scalars are acceptable at most for immaterial portfolios; applying them to large portfolios ignores their lack of sensitivity to scenario variables.
- **C:** Assuming outperformance because of competitive strength benefits the BHC and defeats the purpose of a stress test.
- **D:** Evenly allocating annual figures across quarters does not reflect seasonality or the true quarterly loss path.

Q519 MK Mock 4-Q14

An external auditor is reviewing the modeling processes used by a US-based bank to model operational losses as part of the bank's capital planning process. Using guidelines set by the Federal Reserve with respect to capital planning, which of the following processes or assumptions would the auditor find most appropriate?

A. Assuming a high positive correlation between operational risk severity and equity index movements during normal market conditions.
B. Using a net charge-off model to predict shorter-term credit losses and a roll-rate model to predict losses over a longer time horizon.
C. Modeling operational losses by projecting an annual loss estimate and then evenly distributing the losses across the four quarters of the year.
D. Incorporating forward-looking factors and idiosyncratic risk exposures into stressed operational loss estimates.

Answer: D

Step by step

1. Stronger BHC practice combines historical loss experience with forward-looking elements and idiosyncratic risks in stressed operational loss projections.
2. A has no supervisory support and concerns normal, not stressed, conditions.
3. B: roll-rate models are useful near term but have weak predictive power over longer horizons, and the statement is about credit, not operational, losses.
4. C: evenly distributing annual estimates over four quarters is criticized as not reflecting a careful quarterly loss path.
5. So D.

Why D is correct: Operational loss projections for capital planning should blend historical experience with forward-looking elements and idiosyncratic risks, which is what D describes.

Why the others are wrong

- **A:** Unsupported assumption, and tied to normal conditions rather than a stressed projection.
- **B:** Mixes up credit models with operational loss modeling, and reverses roll-rate models' strength (near term, weak at longer horizons).

- **C:** Even quarterly allocation is a weak practice that ignores the likely timing of losses.

Q520 Schweser Module Quiz 61.2 Q1

How many of the following statements are most likely correct? BHCs should have risk identification processes that evaluate:
I. on- and off-balance sheet positions.
II. risk transfer and/or risk mitigation techniques.
III. changes in institutions' risk profile due to portfolio quality.
IV. reputational risk.

A. One statement.
B. Two statements.
C. Three statements.
D. Four statements.

Answer: D

Step by step

1. I: risk identification must cover on- and off-balance sheet exposures. Correct.
2. II: assumptions about risk reduction from transfer/mitigation must be critically scrutinized. Correct.
3. III: processes should detect changes in the risk profile, including from changes in portfolio quality or concentration. Correct.
4. IV: reputational risk is one of the 'other risks' (with strategic and compliance risk) that must be identified and accounted for. Correct.
5. All four are correct: D.

Why D is correct: Risk identification must cover all exposures that bear on capital needs, including off-balance sheet items, mitigation assumptions, risk-profile shifts and 'other risks' such as reputational risk.

Why the others are wrong

- **A:** Undercounts; assumes only one item qualifies, but all four are supported.
- **B:** Undercounts; probably treats the hard-to-quantify items (e.g., reputational risk) as outside the process.
- **C:** Undercounts by one; likely excludes reputational risk as unquantifiable, yet it falls under 'other risks' that must still be identified.

Q521 Schweser Module Quiz 61.2 Q2

Which of the following statements is most likely correct?

A. The internal controls policy of BHCs requires that senior management should furnish the board of directors with sufficient information to comprehend the BHC risk exposures.
B. A governance policy offers fundamental guidelines and principles to BHCs for capital issuance, use, distribution, and planning purposes.
C. Suspension or reduction in dividends or repurchase programs do not fall under the capital policy of BHCs.
D. Designing and testing a scenario-related default of a major counterparty is an example of BHC stress testing and a stress scenario design policy.

Answer: D

Step by step

1. A: informing the board of risk exposures is a governance requirement, not an internal controls one.
2. B: guidelines for capital issuance, use, distribution and planning describe the capital policy, not governance.
3. C: suspending or reducing dividends/repurchases are capital contingency actions within capital policy.
4. D: counterparty default scenarios are typical stress scenario design elements. Correct.

Why D is correct: Counterparty-default scenarios are a standard part of stress testing and scenario design; the other statements attribute features to the wrong policy or deny that contingency actions belong to capital policy.

Why the others are wrong

- **A:** Attributes a governance requirement to internal controls.
- **B:** Attributes the capital policy's purpose to governance.
- **C:** Dividend/repurchase suspension is a key capital contingency action under capital policy, so the negation is wrong.

Q522 Schweser Module Quiz 61.2 Q3

Which of the following statements is most likely correct?
I. Under the expected losses methodologies, loss estimation involves three elements: probability of default, loss given default, and exposure at default.
II. Net interest income projections should incorporate changing conditions for balance sheet positions, including embedded options, prepayment rates, loan performance, and repricing rates.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: C

Step by step

1. I: the expected loss (economic loss) approach categorizes losses into PD, LGD and EAD. Correct.
2. II: NII projections must reflect changing balance sheet conditions, including embedded options, prepayments, loan performance and repricing. Correct.
3. Both correct: C.

Why C is correct: Expected loss estimation uses PD, LGD and EAD, and NII projections must reflect changing balance sheet conditions; both statements restate GARP/Schweser guidance.

Why the others are wrong

- **A:** Overlooks that II is also correct about NII inputs.
- **B:** Overlooks that I is also correct about PD/LGD/EAD.
- **D:** Both statements are correct, so rejecting them both is wrong.

Q523 Schweser Module Quiz 61.3 Q1

An analyst is discussing net interest income projections with a colleague. Which of the following items should not be incorporated into net interest income projections?

A. Prepayment rates.
B. Balance sheet positions.
C. Forward earnings guidance.
D. Embedded options.

Answer: C

Step by step

1. NII projections should reflect changing conditions in balance sheet positions, embedded options, prepayment rates, loan performance and repricing rates.
2. A, B and D are on this list.
3. Forward earnings guidance is not a driver of NII under scenario conditions: C.

Why C is correct: NII must be driven by scenario-consistent balance sheet and rate-behavior drivers; management earnings guidance is not one of them.

Why the others are wrong

- **A:** Prepayment rates are an explicit NII input.
- **B:** Balance sheet positions determine interest-earning assets and liabilities.
- **D:** Embedded options affect cash flows and repricing, so they belong in NII projections.

Reading 62. Capital Regulation Before the Global Financial Crisis

Mark Carey, "Capital Regulation Before the Global Financial Crisis," GARP Risk Institute (April 2019).

LO	Learning objective	Qs
62.a	explain the motivations for introducing the Basel regulations, including key risk exposures addressed, and explain the reasons for revisions to Basel regulations over time.	4
62.b	explain the calculation of risk-weighted assets and the capital requirement per the original Basel I guidelines.	25
62.c	describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book.	9
62.d	describe changes to the Basel regulations made as part of Basel II, including the three pillars.	10
62.e	compare the standardized internal ratings-based (IRB) approach, the foundation IRB approach, and the advanced IRB approach for the calculation of credit risk capital under Basel II.	16
62.f	calculate credit risk capital under Basel II utilizing the IRB approach.	12
62.g	compare the basic indicator approach, the standardized approach, and the advanced measurement approach for the calculation of operational risk capital under Basel II.	9
62.h	summarize elements of the Solvency II capital framework for insurance companies.	8

LO 62.a

Explain the motivations for introducing the Basel regulations, including key risk exposures addressed, and explain the reasons for revisions to Basel regulations over time.

Q524

AnalystPrep Q.4216

REBUILT — Original option A was garbled and the 'all/none of the above' format gave away the answer; rewrote as a single-best-answer question on the two events that motivated Basel I.

According to Carey, which of the following pairs BEST describes the two developments that motivated the creation of the original Basel I accord?

A. The need to capture operational and market risk in capital requirements, and the need to let banks use internal models to set capital.

B. The growth of cross-border finance, which gave the G10 nations a common interest in banks having enough equity to absorb large losses, and the perception that banks in countries with low minimum capital had a competitive advantage.

C. The need to give the Basel Committee legal authority over banks worldwide, and the need to impose minimum liquidity coverage ratios.

D. The need to create a lender of last resort to replace private clearinghouses, and the need to stimulate bank lending in emerging economies.

Answer: B

Step by step

1. Carey identifies two events behind Basel I: (1) continued growth of cross-border finance after the 1974 Herstatt failure, so the G10 nations shared an interest in banks holding enough equity to absorb large losses (solvency).
2. (2) Banks competed vigorously in each other's home countries while national minimum capital levels differed widely, so banks in low-minimum countries were perceived to have a competitive advantage (level playing field).
3. Only option B states both; A describes Basel II and the 1996 amendment, C describes things Basel I lacked or that came later, D describes pre-Basel history and an unrelated goal.

Why B is correct: Basel I responded to cross-border finance (need for adequate bank equity, solvency) and to unequal national capital minimums that distorted competition.

Why the others are wrong

- **A:** Operational risk capital and internal models arrived with Basel II (and market risk with the 1996 amendment); Basel I covered credit risk only with standardized risk weights.
- **C:** Basel I has no legal authority of its own (nations adopt it through domestic law), and liquidity ratios came with Basel III.
- **D:** Central banks replacing private clearinghouses as lenders of last resort is earlier history that led to official-sector regulation; Basel I raised capital requirements rather than aiming to boost lending.

Q525

AnalystPrep Q.4217

REBUILT — Original had a silly distractor (common currency) and 'none of the above'; rewrote distractors to realistic misconceptions about Basel I.

Which of the following BEST describes the main goal of the original Basel I accord?

A. To eliminate settlement (Herstatt) risk in foreign exchange transactions by synchronizing delivery times across time zones.

B. To make Basel standards legally binding on all banks worldwide so that national regulators lose discretion.

C. To ensure internationally active banks hold enough capital, tied to risk-weighted assets, to remain solvent in times of distress while creating a more level competitive playing field.

D. To raise the capital ratios of small banks until they equal those of large banks.

Answer: C

Step by step

1. Basel I (1988) set international risk-based capital standards: Tier 1 capital/RWA above 4% and total capital/RWA above 8%.
2. Its purpose was solvency of internationally active banks under stress plus a level playing field across countries.
3. A confuses Basel I with the settlement problem that prompted creation of the BCBS in 1974; B is wrong because Basel I has no legal standing; D is not an objective since requirements are ratios applied to all banks.

Why C is correct: Basel I was designed to keep banks adequately capitalized relative to risk and to reduce competitive inequality from differing national minimums.

Why the others are wrong

- **A:** Settlement risk from Herstatt led to creation of the BCBS in 1974, not to Basel I's capital ratios.
- **B:** Basel I has no legal authority; countries adopt it via domestic law and regulation.
- **D:** Basel I applied common minimum ratios to all internationally active banks; equalizing small and large banks was never a goal.

Q526

BT Mock Q48 · BT answers P2.T7.20.7.1 · BT notes P2.T7.20.7.1

The Basel regulatory framework has undergone several major renovations but started in the late 1980s when the Basel Committee on Banking Supervision published the first Basel accord, now called Basel I, implemented in 1992. According to Carey (GARP Chapter 19), which BEST summarizes the motivation(s) of the original Basel?

A. To formalize regulatory capital requirements with a set of regulations that would become legal requirements and therefore carry the force of law; i.e., the motivation was legal.

B. Regulators realized that specific measures of short- and long-term liquidity were necessary given that liquidity crunches had been the common denominator of recent financial disasters; i.e., the motivation was liquidity.

C. The growth of cross-border finance required both the solvency of banks during stressful periods and fair competition (a level playing field) between banks competing in each other's home countries; i.e., the motivations were bank safety and competitive concerns.

D. In the wake of a global recession, there was a motivation to incentivize banks to increase their lending in order to grow developed and emerging economies; i.e., the motivation was economic recovery.

Answer: C

Step by step

1. Carey cites two events: cross-border finance kept growing after Herstatt, so G10 nations wanted banks with enough equity to absorb large losses (solvency).
2. International banks competed in each other's home countries while national minimum capital differed, so a global minimum was set to level the playing field.
3. Only C captures both safety and competitive motives.

Why C is correct: Basel I addressed bank solvency in an era of cross-border finance and the competitive distortion from differing national capital minimums.

Why the others are wrong

- **A:** Basel I has no legal standing; countries adopt it through domestic law, so legal force was not the motivation.
- **B:** Liquidity measures (LCR, NSFR) came with Basel III; Basel I was a solvency (capital) standard.
- **D:** Basel I raised capital requirements; it was not designed to stimulate lending or economic recovery.

Q527

AnalystPrep Q.1553

REBUILT — Original stem wrongly tied the Basel Accords to the 2007-2009 crisis and had a vague, giveaway key; rebuilt to test the actual motivations for Basel I (1988) per the reading.

Which of the following best describes the main motivations for the Basel Committee's introduction of the first international capital standard (Basel I, 1988)?

A. Capital rules differed widely across countries and were based on total assets, ignoring the riskiness of individual banks and off-balance-sheet exposures; internationally active banks in low-requirement countries were seen to have a competitive advantage.

B. A need to impose capital charges for market risk and operational risk, which banks had previously left unmeasured.

C. A need to replace regulatory capital rules with banks' internal models for credit risk, so that capital would reflect each bank's own estimates of default probability.

D. A need to introduce supervisory review and market discipline to complement minimum capital requirements.

Answer: A

Key correction: Source stem referred to the Basel Accords as a response to the 2007-2009 crisis; the Accords (Basel I) date from 1988, so the stem was rebuilt. The source key A (incentives for better risk measurement) was only loosely correct.

Step by step

1. Before 1988, capital requirements were inconsistent across countries and stated as ratios to total assets, which ignored the riskiness of a bank's assets.
2. Off-balance-sheet OTC derivatives were growing but did not affect total assets, so they did not affect required capital.
3. Growth of cross-border finance (e.g., after the Herstatt failure) gave the G10 a common interest in adequate bank capital, while differing national minimums created a perception that banks in low-requirement countries had a competitive advantage; the BCBS wanted to level the playing field.
4. Basel I responded with risk-weighted assets and a minimum capital ratio of 8%, covering credit risk only.
5. Only option A lists these motivations.

Why A is correct: Basel I was motivated by inconsistent national rules, capital measured against total assets rather than risk, growth of off-balance-sheet exposures, and competitive inequality among international banks.

Why the others are wrong

- **B:** Market risk came in with the 1996 amendment and operational risk with Basel II; Basel I covered credit risk only.
- **C:** Internal ratings-based models arrived with Basel II (advanced approaches), not Basel I, which used fixed regulatory risk weights.
- **D:** Supervisory review and market discipline are Pillars 2 and 3 of Basel II, not motivations for Basel I.

LO 62.b

Explain the calculation of risk-weighted assets and the capital requirement per the original Basel I guidelines.

Q528

MK AP Mock 1-Q41

REBUILT — Original distractor values (USD 1,125,000 and 3,500,000) did not correspond to realistic errors; replaced with error-based distractors, stem and answer unchanged.

Family Bank has entered a USD 100 million interest rate swap with a corporation. The swap has a remaining maturity of five years and a current value of USD 4.5 million. Add-on factors as a percentage of principal are:

Time remaining to maturity	Interest rate	Equity
< 1 year	0.0	6.0
1 to 5 years	0.5	8.0
> 5 years	1.5	10.0

Based on the table, the equivalent risk-weighted asset under Basel I is closest to:

A. USD 5,000,000.

B. USD 2,250,000.

C. USD 3,000,000.

D. USD 2,500,000.

Answer: D

Step by step

1. Current exposure: $\max(V, 0) = \max(4.5, 0) = 4.5$ million.
2. A maturity of exactly five years falls in the '1 to 5 years' bucket, so the interest rate add-on is 0.5%: $0.005 \times 100 = 0.5$ million.
3. Credit equivalent amount $= 4.5 + 0.5 = 5.0$ million.
4. Under Basel I the risk weight for a derivative counterparty follows the counterparty type but is capped at 50%, so a corporation gets 50%.
5. RWA $= 5.0 \times 0.5 = 2.5$ million.

Why D is correct: RWA for an OTC derivative is the credit equivalent amount (current exposure plus add-on) times the counterparty risk weight, which for derivatives is capped at 50%.

Why the others are wrong

- **A:** Stops at the credit equivalent amount (5.0 million) and never applies the 50% counterparty risk weight.
- **B:** Omits the add-on and applies the 50% weight to the current value only ($4.5 \times 0.5 = 2.25$).
- **C:** Uses the >5 year add-on of 1.5% ($(4.5 + 1.5) \times 0.5 = 3.0$) although a five-year swap is in the 1 to 5 year bucket.

Q529

AnalystPrep Q.3102

REBUILT — Original used non-Basel 'riskiness' weights (30%, 50%) and an implausible 4.61x ratio; rebuilt with Basel I risk weights and a realistic capital ratio.

Bank A has \$6 million of Tier 1 capital and \$4 million of Tier 2 capital. Its assets are a \$50 million claim on an OECD bank, \$100 million of corporate loans, and \$40 million of uninsured residential mortgages. Using Basel I risk weights, the bank's ratio of total capital to risk-weighted assets is closest to:

A. 4.6%
B. 5.3%
C. 7.7%
D. 6.7%

Answer: C

Step by step

1. OECD bank claim: 50 x 20% = 10.
2. Corporate loans: 100 x 100% = 100.
3. Uninsured residential mortgages: 40 x 50% = 20.
4. RWA $= 10 + 100 + 20 = 130$ million.
5. Total capital $= 6 + 4 = 10$ million, so ratio $= 10/130 = 7.7\%$ (below the 8% minimum).

Why C is correct: The Basel I ratio uses total capital (Tier 1 + Tier 2) over risk-weighted, not gross, assets.

Why the others are wrong

- **A:** Uses only Tier 1 capital (6/130 = 4.6%).
- **B:** Divides by unweighted assets (10/190 = 5.3%) instead of RWA.
- **D:** Weights the mortgages at 100% (RWA = 150, 10/150 = 6.7%).

Q530

MK AP Mock 3-Q21

REBUILT — Replaced distractors that did not map to identifiable errors (0.517 kept as 4% error, 1.535 replaced) with ones from realistic mistakes; stem and correct answer unchanged.

Jinshi&Houshi Corporation has adopted the Basel I framework and must maintain at least 8% capital to risk-weighted assets. The add-on factors (percent of principal) are:

Remaining maturity	Interest rate	FX and gold	Equity	Precious metals ex gold	Other commodities
< 1 yr	0.0	1.0	6.0	7.0	10.0
1 to 5 yr	0.5	5.0	8.0	7.0	12.0
> 5 yr	1.5	7.5	10.0	8.0	15.0

The bank made the following transactions:
(a) A seven-year interest rate swap, notional USD 400 million, current market value -USD 3 million.
(b) A three-year interest rate swap, notional USD 170 million, current value +USD 7 million.
(c) A four-month commodity derivative, principal USD 80 million, currently worth USD 4 million.
What is the capital requirement if the counterparty is a corporation (risk weight 0.5), assuming no netting?

A. USD 1.034 million.
B. USD 2.068 million.
C. USD 0.517 million.

D. USD 0.914 million.

Answer: A

Step by step

1. (a) Negative value floored at zero: $0 + 400 \times 1.5\% = 6.00$ million (over 5 years, interest rate).
2. (b) $7 + 170 \times 0.5\% = 7.85$ million (1 to 5 years).
3. (c) $4 + 80 \times 10\% = 12.00$ million (under 1 year, other commodities).
4. Total credit equivalent $= 6.00 + 7.85 + 12.00 = 25.85$ million.
5. RWA $= 25.85 \times 0.5 = 12.925$ million.
6. Capital $= 12.925 \times 8\% = 1.034$ million.

Why A is correct: Each derivative's credit equivalent is max(V,0) plus notional times add-on; with no netting the negative value in (a) cannot offset the others, and capital is 8% of the 50%-weighted total.

Why the others are wrong

- **B:** Applies a 100% risk weight instead of the stated 50% (25.85 x 8% = 2.068).
- **C:** Uses a 4% capital ratio (the Tier 1 level) instead of 8% (12.925 x 4% = 0.517).
- **D:** Lets the -3 million market value of (a) offset its add-on ($-3 + 6 = 3$), which is netting; total 22.85 x 0.5 x 8% = 0.914.

Q531 AnalystPrep Q.2044

Which of the following statements is true under Basel I?

A. All banks must hold at least \$8 of capital for every \$100 of risk-weighted assets.
B. All banks must hold at most \$10 of capital for every \$100 of risk-weighted assets.
C. All banks must hold at least \$10 of capital for every \$100 of risk-weighted assets.
D. All banks must hold at most \$8 of capital for every \$100 of risk-weighted assets.

Answer: A

Step by step

1. Basel I (1988) set a minimum ratio of total capital (Tier 1 plus Tier 2) to risk-weighted assets of 8%.
2. With RWA of \$100, minimum capital is $0.08 \times 100 = \$8$.
3. A regulatory requirement is a floor, not a cap, so only 'at least \$8' is correct.

Why A is correct: The Basel I capital requirement is a minimum of 8% of RWA (with Tier 1 at least 4%).

Why the others are wrong

- **B:** Treats the requirement as a ceiling and uses 10%, which is not a Basel I figure.
- **C:** Correct direction (minimum) but wrong level; 10% is not the Basel I minimum (some countries required more, but the accord set 8%).
- **D:** Right percentage but wrong direction: Basel sets a floor, not a maximum.

Q532 AnalystPrep Q.2334

In 1992, Germany was under Basel I regulations. Eintracht Bank from Frankfurt had the following portfolio (in Deutsche Marks, DM):

Asset	Amount (DM)
Loans to corporations	1 billion (of which 600 million is uninsured residential real estate)
OECD government exposures	500 million
Cash and balances with a central bank	200 million

Basel I risk weights: 0% for cash and claims on OECD governments; 20% for claims on OECD banks and public sector entities; 50% for uninsured residential mortgage loans; 100% for all other claims. The risk-weighted assets of Eintracht Bank are closest to:

A. 700 million
B. 1 billion
C. 500 million
D. 1.2 billion

Answer: A

Step by step

1. Uninsured residential real estate: 600 x 50% = 300.

2. Remaining corporate loans: 1,000 - 600 = 400, x 100% = 400.
3. OECD government exposures: 500 x 0% = 0. Cash: 200 x 0% = 0.
4. RWA = 300 + 400 + 0 + 0 = 700 million DM.

Why A is correct: RWA is the sum of exposure times Basel I risk weight; only the 600 residential portion gets 50%, the rest of the corporate book is 100%, and sovereign claims and cash are 0%.

Why the others are wrong

- **B:** Applies 100% to all 1 billion of loans, ignoring the 50% weight on the residential real estate.
- **C:** Applies 50% to the whole 1 billion loan book (500) instead of only the 600 million residential part.
- **D:** Adds the 200 million of cash at 100% to the 1 billion of loans (1,200), ignoring the 0% weights and the 50% mortgage weight.

Q533

AnalystPrep Q.2335

In 1992, Italy was under Basel I regulations. Scala Bank from Milan had the following portfolio (in USD):

Asset	Amount
Loans to corporations	$1.5 billion (of which $600 million is commercial real estate)
OECD government exposures	$300 million
Cash and balances with a central bank	$500 million

Basel I risk weights: 0% for cash and claims on OECD governments; 20% for claims on OECD banks and public sector entities; 50% for uninsured residential mortgage loans; 100% for all other claims. The risk-weighted assets of Scala Bank are closest to:

A. $1.2 billion
B. $1.5 billion
C. $2.3 billion
D. $1.65 billion

Answer: B

Step by step

1. Commercial real estate is not a residential mortgage, so it falls in 'all other claims': 100%.
2. Corporate loans: 1,500 x 100% = 1,500 million.
3. OECD government: 300 x 0% = 0; cash: 500 x 0% = 0.
4. RWA = 1,500 million = $1.5 billion.

Why B is correct: Only residential mortgages get 50% under Basel I; commercial real estate and corporate loans are 100%, and sovereign claims and cash are 0%.

Why the others are wrong

- **A:** Gives the 600 million commercial real estate a 50% weight (900 + 300 = 1,200) as if it were a residential mortgage.
- **C:** Adds all assets (1,500 + 300 + 500 = 2,300) with no risk weighting.
- **D:** Applies 50% to the OECD government exposure (1,500 + 150 = 1,650) instead of 0%.

Q534

AnalystPrep Q.2336

REBUILT — Original stem used a 100% counterparty risk weight, which GARP says Basel I did not allow for derivatives (no derivative risk weight could exceed 50%); stem changed to a corporate counterparty at the 50% cap and options re-keyed (RWA EUR 3m).

Arrenberg Bank in Rotterdam must calculate risk-weighted assets under Basel I (current exposure method) for an over-the-counter FX swap with a corporate counterparty and a remaining maturity of less than one year. Add-on factor: 1%. Notional amount: EUR 500 million. Current value: EUR 1 million. Counterparty risk weight: 50% (the maximum Basel I allowed for derivative counterparties). The RWA is closest to:

A. EUR 250.5 million
B. EUR 3 million
C. EUR 6 million
D. EUR 2.5 million

Answer: B

Key correction: Source key (AnalystPrep Q.2336) said EUR 6m using a 100% counterparty weight; under GARP's Basel I treatment derivative risk weights cannot exceed 50%, so the stem was rebuilt at 50% and the correct answer is EUR 3m.

Step by step

1. Current exposure = $\max(V, 0) = \max(1, 0) = 1$ million.
2. Add-on = $D \times L = 1\% \times 500 = 5$ million.
3. Credit equivalent amount = $1 + 5 = 6$ million.
4. RWA = $6 \times 50\% = 3$ million.

Why B is correct: RWA = (max(V,0) + D x notional) x counterparty risk weight; under Basel I the derivative counterparty risk weight is capped at 50%, so EUR 6m x 0.5 = EUR 3m.

Why the others are wrong

- **A:** Uses notional plus current value (500 + 1 = 501) as the exposure instead of applying the add-on factor, then weights at 50%.
- **C:** Stops at the credit equivalent amount of EUR 6m (equivalently, applies a 100% weight), ignoring the 50% cap on derivative counterparty risk weights.
- **D:** Omits the current exposure of EUR 1m and weights only the EUR 5m add-on (5 x 0.5).

Q535

AnalystPrep Q.2337

REBUILT — Original stem used a 100% counterparty risk weight, which GARP says Basel I did not allow for derivatives (cap of 50%); stem changed to a corporate counterparty at the 50% cap and options re-keyed (RWA EUR 7.5m).

Calc Bank in Frankfurt must calculate risk-weighted assets under Basel I (current exposure method) for an over-the-counter interest rate swap with a corporate counterparty and a remaining maturity of more than five years. Add-on factor: 1.5%. Notional amount: EUR 1 billion. Current value: EUR -2 million. Counterparty risk weight: 50% (the maximum Basel I allowed for derivative counterparties). The RWA is closest to:

A. EUR 6.5 million
B. EUR 0
C. EUR 7.5 million
D. EUR 15 million

Answer: C

Key correction: Source key (AnalystPrep Q.2337) said EUR 15m using a 100% counterparty weight; under GARP's Basel I treatment derivative risk weights cannot exceed 50%, so the stem was rebuilt at 50% and the correct answer is EUR 7.5m.

Step by step

1. A negative value means no current exposure: $\max(-2, 0) = 0$.
2. Add-on = $1.5\% \times 1{,}000 = 15$ million.
3. Credit equivalent amount = $0 + 15 = 15$ million.
4. RWA = $15 \times 50\% = 7.5$ million.

Why C is correct: The current value is floored at zero, so only the add-on (EUR 15m) contributes to the credit equivalent amount, which is then weighted at the 50% maximum Basel I allowed for derivative counterparties.

Why the others are wrong

- **A:** Lets the -2 million value reduce the add-on (15 - 2 = 13) instead of flooring it at zero, then weights at 50% (6.5).
- **B:** Assumes a negative market value means no exposure at all and ignores the add-on for potential future exposure.
- **D:** Stops at the credit equivalent amount of EUR 15m (equivalently, applies a 100% weight), ignoring the 50% cap on derivative counterparty risk weights.

Q536

AnalystPrep Q.2737

REBUILT — Original distractors (850, 700) mostly did not arise from identifiable errors; replaced with error-based distractors. Stem unchanged.

A bank's balance sheet shows the following assets:

Asset	$ million
Cash	50
Treasury bills	100
Loans to corporations	750
Uninsured residential mortgages	100

Calculate the bank's risk-weighted assets based on the Basel I guidelines.

A. $850 million
B. $750 million
C. $900 million
D. $800 million

Answer: D

Step by step

1. Cash and Treasury bills (OECD government claims): 0%, so they contribute 0.
2. Corporate loans: 750 x 100% = 750.
3. Uninsured residential mortgages: 100 x 50% = 50.
4. RWA = 0 + 0 + 750 + 50 = 800 million.

Why D is correct: Basel I risk weights: 0% for cash and OECD sovereign claims, 50% for uninsured residential mortgages, 100% for corporate loans.

Why the others are wrong

- **A:** Applies 100% to the uninsured mortgages (750 + 100 = 850).
- **B:** Omits the mortgages altogether, counting only the corporate loans.
- **C:** Gives Treasury bills a 100% weight (100 + 750 + 50 = 900) instead of 0%.

Q537

AnalystPrep Q.2994

REBUILT — Stem said 20% applies to loans to government agencies and banks while the asset was OECD government bonds (0%); restated assets and weights consistently with Basel I and replaced inconsistent options.

G&R Bank's assets consist of $267 million of corporate loans, $17 million of bonds issued by an OECD central government, and $79 million of uninsured residential mortgages. Using the original Basel I risk weights, the bank's total risk-weighted assets are closest to:

A. $309.9 million
B. $306.5 million
C. $363.0 million
D. $346.0 million

Answer: B

Step by step

1. Corporate loans: 267 x 100% = 267.0.
2. OECD central government bonds: 17 x 0% = 0.
3. Uninsured residential mortgages: 79 x 50% = 39.5.
4. RWA = 267.0 + 0 + 39.5 = 306.5 million.

Why B is correct: OECD sovereign claims carry 0%, uninsured residential mortgages 50%, and corporate loans 100%.

Why the others are wrong

- **A:** Applies 20% to the OECD government bonds (+3.4); 20% is the weight for OECD banks and public sector entities, not central governments.
- **C:** Applies 100% to every asset (267 + 17 + 79 = 363), i.e. no risk weighting of the mortgages or bonds.
- **D:** Gives mortgages 100% and the sovereign bonds 0% (267 + 79 = 346).

Q538

AnalystPrep Q.4218

REBUILT — Original had an 'all of the above' / 'none of the above' format with trivial distractors; replaced with four plausible capital measures.

Which capital measure did the original Basel I accord use to judge the capital sufficiency of banks?

A. Capital divided by the book value of total on-balance-sheet assets
B. Capital divided by risk-weighted assets, including off-balance-sheet and derivative exposures
C. Capital divided by total customer deposits
D. Tier 1 capital divided by total assets, with no minimum for Tier 2 capital

Answer: B

Step by step

1. Basel I needed a standard that works for banks of very different size, so it used a ratio rather than a dollar amount.
2. A plain ratio of capital to total assets ignores the differing riskiness of assets and of off-balance-sheet positions, so total assets did not reflect a bank's total risk.
3. Instead the BCBS adopted a risk-based capital ratio: capital divided by risk-weighted assets (RWA).
4. RWA covers on-balance-sheet assets plus off-balance-sheet exposures (e.g., loan commitments) and derivatives, each weighted by risk.
5. Minimums: total capital > 8% of RWA and Tier 1 capital > 4% of RWA.

Why B is correct: Basel I deliberately chose a risk-sensitive ratio (capital / RWA) over a leverage ratio so that capital requirements reflect the riskiness of assets, including off-balance-sheet items.

Why the others are wrong

- **A:** This is a leverage-type ratio that ignores asset risk and off-balance-sheet exposures; Basel I's innovation was to risk-weight instead.
- **C:** Deposits are a funding measure, not a measure of asset risk; Basel I never used a deposit-based denominator.
- **D:** Basel I required both a Tier 1 minimum (4% of RWA) and a total capital minimum (8% of RWA), and used RWA, not total assets, as the denominator.

Q539

AnalystPrep Q.4222

Under Basel I, risk-weighted assets are used as the denominator of the capital ratio. Which of the following exposures is assigned a risk weight of 0%?

A. Uninsured residential mortgages
B. Commercial and consumer loans
C. Claims on OECD central governments (e.g., government bonds)
D. Claims on OECD banks and OECD public sector entities

Answer: C

Step by step

1. Recall the Basel I risk-weight buckets: 0%, 20%, 50%, 100%.
2. 0%: cash and claims on OECD governments (and instruments fully guaranteed by an OECD government).
3. 20%: claims on OECD banks and OECD public sector entities.
4. 50%: uninsured residential mortgages.
5. 100%: all other exposures, such as commercial and consumer loans.
6. Only OECD government claims fall in the 0% bucket, so the answer is C.

Why C is correct: Basel I assumed no OECD government would default on its obligations, so those claims carry a 0% weight.

Why the others are wrong

- **A:** Uninsured residential mortgages are weighted 50%.
- **B:** Commercial and consumer loans fall in the residual 100% bucket.
- **D:** Claims on OECD banks and public sector entities are weighted 20%, not 0%.

Q540

AnalystPrep Q.4223

REBUILT — Original described 'residential mortgages issued by the public sector', which is ambiguous between the 50% mortgage and 20% public sector weights; restated as claims on OECD public sector entities.

A US bank holds $200 million of US Treasury bonds, $500 million of loans to corporations, $300 million of uninsured residential mortgages, and $250 million of claims on OECD public sector entities (e.g., municipalities). What are the bank's risk-weighted assets under the original Basel I accord?

A. $1,250 million
B. $600 million
C. $700 million
D. $850 million

Answer: C

Step by step

1. Treasury bonds (OECD government): 0% x 200 = 0.
2. Corporate loans: 100% x 500 = 500.
3. Uninsured residential mortgages: 50% x 300 = 150.
4. OECD public sector entities: 20% x 250 = 50.
5. RWA = 0 + 500 + 150 + 50 = $700 million.

Why C is correct: Each exposure is multiplied by its Basel I risk weight (0/20/50/100%) and the products are summed: RWA = 0 + 500 + 150 + 50 = 700.

Why the others are wrong

- **A:** $1,250m is the unweighted total book value (200+500+300+250); no risk weights were applied.
- **B:** $600m does not follow from any consistent application of the Basel I weights (correct is 0 + 500 + 150 + 50 = 700).

- **D:** $850m applies 100% to the mortgages instead of 50% (0 + 500 + 300 + 50).

Q541 AnalystPrep Q.5958

A regional bank is assessing its capital adequacy ratio (CAR) under the Basel framework. It has measured its Tier 1 and Tier 2 capital and assigned risk weights to its assets. Which of the following correctly describes how the CAR is calculated?

A. Sum of Tier 1 and Tier 2 capital divided by total (unweighted) assets
B. Sum of Tier 1 and Tier 2 capital divided by risk-weighted assets (RWA)
C. Sum of the risk weights of all assets divided by the sum of Tier 1 and Tier 2 capital
D. Tier 1 capital minus Tier 2 capital divided by total liabilities

Answer: B

Step by step

1. The Basel capital ratio is a risk-based ratio: capital in the numerator, RWA in the denominator.
2. RWA = sum of (risk weight x exposure) over all on- and off-balance-sheet exposures.
3. Total capital = Tier 1 + Tier 2.
4. CAR = (Tier 1 + Tier 2) / RWA, which must exceed 8% (Tier 1 alone must exceed 4% of RWA).

Why B is correct: The capital adequacy ratio compares total regulatory capital with risk-weighted assets, so assets are scaled by their riskiness.

Why the others are wrong

- **A:** Using unweighted total assets gives a leverage ratio, which Basel rejected because it ignores asset risk.
- **C:** This inverts the ratio and sums risk weights (percentages) rather than weighting exposures.
- **D:** Tiers are added, not subtracted, and the denominator is RWA, not liabilities.

Q542 AnalystPrep Q.5961

REBUILT — Original asked a vague 'which factor impacts CAR' with a defensible alternative (dividend policy changes capital); rebuilt as a concrete Basel I RWA reallocation question with a single correct answer.

Holding regulatory capital constant, which of the following portfolio shifts of $100 million would LOWER a bank's capital ratio under the original Basel I accord?

A. Moving from uninsured residential mortgages into OECD government bonds
B. Moving from commercial loans into uninsured residential mortgages
C. Moving from OECD government bonds into commercial loans
D. Moving from claims on OECD banks into OECD government bonds

Answer: C

Step by step

1. Capital ratio = capital / RWA; with capital fixed, the ratio falls only if RWA rises.
2. A: 50% -> 0%: RWA falls by $50m.
3. B: 100% -> 50%: RWA falls by $50m.
4. C: 0% -> 100%: RWA rises by $100m, so the ratio falls.
5. D: 20% -> 0%: RWA falls by $20m.
6. Only C increases RWA.

Why C is correct: Capital is fixed, so the ratio falls only when RWA rises; moving from a 0% asset to a 100% asset adds $100m of RWA.

Why the others are wrong

- **A:** Moves from a 50% to a 0% weight, lowering RWA and raising the ratio.
- **B:** Moves from 100% to 50%, lowering RWA and raising the ratio.
- **D:** Moves from 20% to 0%, lowering RWA and raising the ratio.

Q543 BT answers P2.T7.20.7.2 · BT notes P2.T7.20.7.2

With respect to the original Basel I accord, each of the following is true EXCEPT which is false?

A. There were only four risk weights: 0%, 20%, 50%, and 100%
B. Total regulatory capital had to exceed 8.0% of risk-weighted assets (RWA), and Tier 1 capital had to exceed 4.0% of RWA

C. Traditional off-balance-sheet exposures were converted to a credit equivalent that was then assigned a risk weight based on the counterparty

D. The credit equivalent for derivatives was their current market value, so out-of-the-money positions reduced the bank's RWA

Answer: D

Step by step

1. A true: the weights were 0%, 20%, 50% and 100%.
2. B true: the Cooke ratios required total capital > 8% of RWA and Tier 1 > 4% of RWA.
3. C true: off-balance-sheet items (e.g., a $100m five-year commitment to an OECD municipality) were multiplied by a credit conversion factor (20%, giving a $20m credit equivalent) and then weighted by counterparty (20%), contributing $4m to RWA.
4. D false: for derivatives the current exposure method set V = max(market value, 0), then added an add-on D for potential future changes. Negative values are floored at zero, so they cannot reduce RWA.

Why D is correct: Derivative credit equivalent = max(V, 0) + add-on, so out-of-the-money positions contribute zero and never subtract from RWA.

Why the others are wrong

- **A:** True statement, so not the exception.
- **B:** True statement (8% total, 4% Tier 1 of RWA), so not the exception.
- **C:** True statement (credit conversion then counterparty weight), so not the exception.

Q544

BT answers P2.T7.24.10.1 · BT notes P2.T7.24.10.1

REBUILT — Original combined the Basel I calculation with a qualitative, source-dependent Basel II comparison and a garbled table; restated as a pure Basel I RWA calculation with numeric options.

Bank DeNiro holds the following assets, with Basel I risk weights as given:

Asset	Book value	Basel I risk weight
Corporate loans	$100 million	100%
OECD government loans	$10 million	0%
Uninsured residential mortgages	$50 million	50%

What are the bank's risk-weighted assets under Basel I?

A. $125 million

B. $135 million

C. $150 million

D. $160 million

Answer: A

Step by step

1. Corporate loans: 100% x 100 = 100.
2. OECD government loans: 0% x 10 = 0.
3. Uninsured mortgages: 50% x 50 = 25.
4. RWA = 100 + 0 + 25 = $125 million.

Why A is correct: RWA is the sum of each exposure times its risk weight: 100 + 0 + 25 = 125.

Why the others are wrong

- **B:** $135m wrongly gives the OECD government loans a 100% weight (100 + 10 + 25).
- **C:** $150m wrongly gives the mortgages a 100% weight (100 + 0 + 50).
- **D:** $160m is the unweighted book value (100 + 10 + 50); no risk weights applied.

Q545

BT answers P2.T7.515.1 · BT notes P2.T7.515.1

REBUILT — Original asked a conceptual question whose keyed option ignored the max(V,0) rule and gave no add-on data; rebuilt as a numeric current exposure method calculation with the add-on table supplied.

A bank under the original Basel I guidelines enters an over-the-counter interest rate swap with an OECD bank. The notional principal is $100 million, the remaining maturity is three years, and the swap currently has a value of -$3.0 million to the bank. Use the current exposure method with these add-on factors for interest rate contracts (percentage of principal) and the 20% risk weight for claims on OECD banks.

Remaining maturity	Add-on factor
Less than 1 year	0.0%
1 to 5 years	0.5%
More than 5 years	1.5%

What is the swap's contribution to risk-weighted assets?

A. $0.5 million
B. -$0.5 million
C. $0.1 million
D. $0

Answer: C

Key correction: Source key (BT) said C (notional x add-on x counterparty risk weight); the same conclusion holds in the rebuilt version, where the negative current value is floored at zero.

Step by step

1. Current exposure method: CEA $= \max(V, 0) + D \times L$.
2. The swap has negative value, so $\max(-3.0, 0) = 0$: the bank loses nothing today if the counterparty defaults.
3. Add-on for a 3-year interest rate swap: 0.5% × $100 million = $0.5 million.
4. CEA $= 0 + 0.5 =$ $0.5 million.
5. RWA $= 0.5 \times 20\% =$ $0.1 million.

Why C is correct: Negative current value is floored at zero, so only the potential future exposure (add-on) enters the credit equivalent amount, which is then multiplied by the counterparty risk weight.

Why the others are wrong

- **A:** $0.5m is the credit equivalent amount; it omits the 20% counterparty risk weight.
- **B:** Nets the -$3.0m against the add-on, (-3.0 + 0.5) x 20%; Basel I floors V at zero and never creates negative exposure.
- **D:** Assumes a negative-value swap carries no exposure at all; this ignores the add-on for possible future increases in value.

Q546

Schweser Module Quiz 62.1 Q1

REBUILT — Add-on table exhibit was missing and the corporate 100% risk weight conflicts with GARP's 50% cap on derivative counterparty weights; rebuilt with the table supplied and an OECD bank counterparty (20%).

A bank has entered into a $200 million interest rate swap with an OECD bank. The remaining maturity of the swap is six years and its current value is $3.5 million to the bank. Using the Basel I current exposure method, the add-on factors below, and a 20% risk weight for claims on OECD banks, the swap's risk-weighted asset (RWA) amount is closest to:

Remaining maturity	Add-on factor (% of principal)
Less than 1 year	0.0%
1 to 5 years	0.5%
More than 5 years	1.5%

A. $6.5 million
B. $0.6 million
C. $1.3 million
D. $0.9 million

Answer: C

Key correction: Source key (Schweser) said $6.5m using a 100% corporate counterparty weight; GARP caps derivative counterparty risk weights at 50%, so the counterparty is changed to an OECD bank (20%) and the answer is $1.3m.

Step by step

1. Add-on factor: six years is more than 5 years, so $D = 1.5\%$.
2. Add-on amount = 1.5% × $200 million = $3.0 million.
3. $V = +3.5$, so $\max(V, 0) = 3.5$.
4. CEA $= 3.5 + 3.0 =$ $6.5 million.
5. RWA $= 6.5 \times 20\% =$ $1.3 million.

Why C is correct: CEA is current exposure plus the maturity-based add-on; multiplying by the OECD bank's 20% weight gives the RWA.

Why the others are wrong

- **A:** This is the credit equivalent amount; it forgets to apply the counterparty risk weight.
- **B:** Applies 20% to the add-on only (3.0 x 20%), leaving out the positive current value of 3.5m.
- **D:** Uses the 0.5% add-on for 1-5 years instead of 1.5% for over 5 years: (3.5 + 1.0) x 20% = 0.9m.

Q547 AnalystPrep Q.2338

Kediray Bank is calculating its regulatory capital under Basel I regulations. It has the following capital instruments: equity, noncumulative perpetual preferred stock, and subordinated debt with an original maturity of over 5 years. What is the structure of its regulatory capital?

A. Tier 1 capital includes equity, noncumulative perpetual preferred stock, and subordinated debt.

B. Tier 1 capital includes equity, and Tier 2 capital includes noncumulative perpetual preferred stock and subordinated debt.

C. Tier 1 capital includes equity, and Tier 2 includes noncumulative perpetual preferred stock; subordinated debt is not included in regulatory capital.

D. Tier 1 capital includes equity and noncumulative perpetual preferred stock, and Tier 2 includes subordinated debt.

Answer: D

Step by step

1. Tier 1 (core) capital: equity (less goodwill) and noncumulative perpetual preferred stock.
2. Tier 2 (supplementary) capital: cumulative perpetual preferred, certain 99-year debentures, and subordinated debt with original maturity over five years.
3. Equity and noncumulative perpetual preferred are therefore Tier 1; the subordinated debt is Tier 2.

Why D is correct: Only permanent, loss-absorbing instruments that preserve solvency are Tier 1; subordinated debt protects depositors and is Tier 2.

Why the others are wrong

- **A:** Puts subordinated debt in Tier 1; it only protects depositors, so it is Tier 2.
- **B:** Moves noncumulative perpetual preferred stock to Tier 2; because dividends need not accumulate it counts as Tier 1.
- **C:** Excludes subordinated debt from capital altogether; debt with original maturity over five years qualifies as Tier 2.

Q548 AnalystPrep Q.2339

Banat Bank is calculating its regulatory capital under Basel I regulations. It has the following capital instruments (in EUR; the preferred stock is perpetual):

Equity	150m
Subordinated debt (original maturity over 5 years)	50m
Cumulative preferred stock	20m

What are its Tier 1 and Tier 2 capital?

A. Tier 1: 170m; Tier 2: 50m

B. Tier 1: 150m; Tier 2: 70m

C. Tier 1: 150m; Tier 2: 20m

D. Tier 1: 170m; Tier 2: 0

Answer: B

Step by step

1. Tier 1: equity 150m (no noncumulative preferred is held).
2. Tier 2: cumulative perpetual preferred 20m + subordinated debt 50m = 70m.
3. Check: Tier 2 (70m) is below half of total capital (110m of 220m), so it all counts.

Why B is correct: Equity is Tier 1; cumulative preferred and long-dated subordinated debt are Tier 2.

Why the others are wrong

- **A:** Treats cumulative preferred as Tier 1; only noncumulative perpetual preferred is Tier 1.
- **C:** Leaves out the 50m subordinated debt, which is Tier 2.
- **D:** Puts cumulative preferred in Tier 1 and ignores the subordinated debt.

Q549 AnalystPrep Q.2340

Osijek Commercial Bank has to calculate its Tier 1 and Tier 2 capital under Basel I regulations. It has the following capital structure (in EUR; the preferred stocks are perpetual):

Equity	50m
Subordinated debt (original maturity over 5 years)	30m
Cumulative preferred stock	5m
Noncumulative preferred stock	10m

What are the bank's Tier 1 and Tier 2 capital?

A. Tier 1: 60m; Tier 2: 35m
B. Tier 1: 55m; Tier 2: 40m
C. Tier 1: 55m; Tier 2: 10m
D. Tier 1: 60m; Tier 2: 5m

Answer: A

Step by step

1. Tier 1 = equity 50m + noncumulative perpetual preferred 10m = 60m.
2. Tier 2 = subordinated debt 30m + cumulative preferred 5m = 35m.
3. Tier 2 (35m) is below Tier 1 (60m), so the half-of-total limit is met.

Why A is correct: Noncumulative perpetual preferred joins equity in Tier 1; cumulative preferred and subordinated debt are Tier 2.

Why the others are wrong

- **B:** Swaps the two preferred stocks: puts cumulative (5m) in Tier 1 and noncumulative (10m) in Tier 2.
- **C:** Same swap, and also omits the 30m subordinated debt from Tier 2.
- **D:** Gets Tier 1 right but counts only the cumulative preferred in Tier 2, omitting the 30m subordinated debt.

Q550 AnalystPrep Q.4219

Under the Basel I framework, what is the required value of the ratio of Tier 1 capital to risk-weighted assets (RWA)?

A. Greater than 4%
B. Less than 4%
C. Greater than 8%
D. Less than 8%

Answer: A

Step by step

1. Basel I requires Tier 1 capital / RWA > 4%.
2. Total capital (Tier 1 + Tier 2) / RWA must exceed 8%, which is the separate total-capital requirement.

Why A is correct: Basel I sets a minimum, not a maximum, of 4% for Tier 1 to RWA (half of the 8% total).

Why the others are wrong

- **B:** Reverses the inequality; the 4% is a floor.
- **C:** Quotes the total capital ratio (8%) rather than the Tier 1 ratio.
- **D:** Reverses the inequality and uses the 8% total-capital figure.

Q551 AnalystPrep Q.4220

Based on the Basel I framework, which of the following expressions is incorrect?

A. Tier 1 capital > 4% x RWA
B. Total capital > 8% x RWA
C. Tier 1 capital + Tier 2 capital = Total capital
D. Tier 1 capital + Tier 2 capital > Total capital

Answer: D

Step by step

1. A and B are the Basel I minimum requirements (Tier 1/RWA > 4%, Total/RWA > 8%).
2. Total capital is defined as Tier 1 plus Tier 2, so C is an identity.
3. D claims the sum exceeds total capital, which contradicts that definition.

Why D is correct: Total capital is by definition Tier 1 plus Tier 2, so the sum can never exceed it.
Why the others are wrong
- **A:** A correct Basel I requirement, so not the incorrect expression.
- **B:** A correct Basel I requirement, so not the incorrect expression.
- **C:** A true definition, so not the incorrect expression.

Q552

AnalystPrep Q.4221

According to Basel I classification of capital, which of the following is NOT a constituent of Tier 2 capital?

A. Undisclosed reserves
B. Common equity
C. Hybrid instruments
D. Loan loss reserves not allocated to non-performing assets

Answer: B
Step by step
1. Tier 1 (core) capital is common equity and disclosed reserves, less goodwill.
2. Tier 2 includes undisclosed reserves, hybrid instruments and general (unallocated) loan loss reserves.
3. Common equity is therefore the item that is not Tier 2.

Why B is correct: Common equity is the core, loss-absorbing Tier 1 component.
Why the others are wrong
- **A:** Undisclosed reserves are Tier 2 (supplementary) capital.
- **C:** Hybrid instruments (subordinated, deferrable payments) are Tier 2.
- **D:** General loan loss reserves not tied to particular impaired assets are Tier 2 (subject to a cap).

LO 62.c

Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book.

Q553

AnalystPrep Q.2995

REBUILT — Stem did not say the trades share one netted counterparty and two distractors (0.63, 1.31) matched no realistic error; replaced them with error-based options.

A bank has three derivatives with a single counterparty, covered by an enforceable bilateral netting agreement (EUR million):

Transaction	Principal	Current value
2-year interest rate swap	1000	95
5-year foreign exchange forward	1000	-35
8-month option on a stock	700	80

What is the net replacement ratio (NRR) under the 1995 Amendment?

A. 0.74
B. 1.25
C. 0.80
D. 0.34

Answer: C
Step by step
1. NRR = current exposure with netting / current exposure without netting = $\max(\sum V_i, 0) / \sum \max(V_i, 0)$.
2. With netting: $95 - 35 + 80 = 140$ (positive, so no floor applies).
3. Without netting only positive values count: $95 + 0 + 80 = 175$.
4. NRR $= 140/175 = 0.80$.

Why C is correct: Netting lets the -35 forward offset the positive positions, but without netting a negative value is floored at zero, so the denominator is the sum of positive values only.
Why the others are wrong
- **A:** 0.74 = 140/190: the denominator used the absolute value of the -35 (95 + 35 + 80) instead of flooring nega-

tives at zero.
- **B:** 1.25 = 175/140: the ratio was inverted (without netting over with netting); netting can only reduce exposure so NRR cannot exceed 1.
- **D:** 0.34 = 60/175: the numerator omitted the 80 option (95 - 35), so the netted exposure was understated.

Q554

AnalystPrep Q.3234

REBUILT — Added that all three trades are with one counterparty under an enforceable netting agreement, which is needed for the netting calculation to be unambiguous.

A bank applying Basel I with the 1995 netting amendment has three derivatives with the same counterparty, covered by an enforceable bilateral netting agreement: (a) a seven-year interest rate swap, notional $400 million, current value -$3 million; (b) a three-year interest rate swap, notional $170 million, current value +$7 million; (c) a four-month commodity derivative, notional $80 million, current value +$4 million. What is the net replacement ratio (NRR)?

A. 1.375
B. 1.000
C. 0.727
D. 0.636

Answer: C

Step by step

1. Current exposure with netting $= \max(-3 + 7 + 4, 0) = 8$.
2. Current exposure without netting $= 0 + 7 + 4 = 11$ (the negative value is floored at zero).
3. NRR $= 8/11 = 0.727$.

Why C is correct: NRR compares the netted exposure (8) to the sum of the positive exposures (11), showing that netting removes about 27% of current exposure.

Why the others are wrong

- **A:** 1.375 = 11/8: the ratio was inverted; NRR is netted over gross and cannot exceed 1.
- **B:** 1.000 treats the portfolio as having no netting benefit (e.g., ignoring the negative swap value in the numerator).
- **D:** 0.636 = 7/11: the numerator counted only the +7 swap and dropped the +4 derivative (and the -3), rather than netting all three values.

Q555

AnalystPrep Q.4227

REBUILT — Stem said 'Basel I framework' although market risk capital came from the 1996 Amendment; reworded and relabelled option C as the standardized measurement method.

The 1996 Amendment to the Basel I Accord added capital for market risk on trading book positions. Which of the following was one of the two methods it allowed for measuring market risk?

A. Current exposure method
B. Original exposure method
C. Standardized measurement method
D. Foundation internal ratings-based approach

Answer: C

Step by step

1. The 1996 Amendment offered two market risk methods: the standardized measurement method and the internal model-based approach (bank VaR model).
2. The current and original exposure methods are Basel I ways of calculating credit equivalent amounts for derivatives (credit risk).
3. The foundation IRB approach is a Basel II credit risk method.
4. Only the standardized measurement method is a market risk method.

Why C is correct: Market risk capital under the 1996 Amendment is calculated by either the standardized measurement method or the internal model-based approach.

Why the others are wrong

- **A:** The current exposure method estimates the credit equivalent amount of derivatives for credit risk, not market risk.
- **B:** The original exposure method is the alternative credit-risk add-on method for derivatives, not a market risk method.

- **D:** Foundation IRB is a Basel II credit risk approach and did not exist under the 1996 Amendment.

Q556 BT answers P2.T7.20.7.3 · BT notes P2.T7.20.7.3

The 1995 Amendment to Basel I allowed reductions in credit equivalent amounts when enforceable bilateral netting agreements were in place. Shortly afterwards the 1996 Amendment added capital for market risk. Which of the following statements about the 1996 Amendment is TRUE?

A. It introduced the third pillar, which was praised for giving supervisors a consistent rule-enforcing playbook

B. Conditional on meeting several criteria, including daily backtesting, banks could now employ their own internal value at risk (VaR) model to determine capital requirements for market risk

C. To reduce regulatory arbitrage, it removed all internal approaches so that only standardized approaches could be applied to credit, market, and operational risk

D. It removed Tier 2 capital so that only Tier 1 capital remained, with Tier 1 split into common equity and non-common equity

Answer: B

Step by step

1. The 1996 Amendment offered a standardized measurement method and an internal model-based approach for market risk.
2. Under the internal approach a bank uses its own 10-day 99% VaR, multiplied by a factor of at least 3, provided qualitative criteria are met and the one-day 99% VaR is backtested over the previous 250 days.
3. The three pillars came with Basel II; operational risk capital was also a Basel II innovation.
4. The Amendment added Tier 3 capital for market risk; it did not remove Tier 2. So only B is true.

Why B is correct: The 1996 Amendment's key innovation was allowing qualifying banks to use internal VaR models, subject to a multiplier and backtesting, for market risk capital.

Why the others are wrong

- **A:** The three pillars (including Pillar 3 market discipline) were introduced with Basel II in 2004, not in 1996.
- **C:** The Amendment did the opposite: it added an internal model-based approach, and it covered only market risk; operational risk capital arrived with Basel II.
- **D:** The Amendment created Tier 3 capital for market risk and kept Tier 2; the common equity emphasis is a Basel III development.

Q557 BT answers P2.T7.24.10.2 · BT notes P2.T7.24.10.1

REBUILT — Source key (A) netted market values across both counterparties (NRR 0.25, CEA 33.05); netting is allowed only within a counterparty, so net exposure is 15, NRR 0.75 and CEA 58.35. Question rebuilt to ask for the CEA with corrected options.

A bank has the following derivatives with two counterparties, each covered by an enforceable bilateral netting agreement ($ millions):

Counterparty	Derivative	Maturity (yrs)	Notional	Market value	Add-on factor
1	Interest rate	2	100	20	0.5%
1	Interest rate	3	100	0	0.5%
1	Wheat	2	200	-5	5%
2	Equity	6	100	0	10%
2	Foreign exchange	0.5	300	-10	10%

Using the 1995 Amendment, CEA $= \sum_i \max(\sum_j V_{ij}, 0) + \sum_j L_j[0.4D_j + 0.6D_j \times \text{NRR}]$, what is the total credit equivalent amount?

A. $33.05 million

B. $43.35 million

C. $58.35 million

D. $66.00 million

Answer: C

Key correction: Source key (BT answers) said A ($33.05 million, NRR 0.25); correct is C because netting applies within each counterparty only: the -10 on counterparty 2 cannot offset the positive exposure to counterparty 1.

Step by step

1. Net exposure by counterparty: CP1 $= \max(20 + 0 - 5, 0) = 15$; CP2 $= \max(0 - 10, 0) = 0$. Total $= 15$.
2. Gross exposure (positive values only) $= 20$, so NRR $= 15/20 = 0.75$.

3. Add-ons (notional x factor): 0.5 + 0.5 + 10 + 10 + 30 = 51.
4. Add-on after netting $= 51 \times (0.4 + 0.6 \times 0.75) = 51 \times 0.85 = 43.35$.
5. CEA $= 15 + 43.35 = 58.35$.

Why C is correct: Netting is recognised only within a counterparty; the net current exposure (15) is added to add-ons that are reduced to 85% of gross through the NRR of 0.75.

Why the others are wrong

- **A:** 33.05 nets all five values across both counterparties (total 5, NRR 0.25): 5 + 51 x 0.55 = 33.05. Netting across counterparties is not allowed.
- **B:** 43.35 is only the netted add-on; the net current exposure of 15 was omitted.
- **D:** 66.00 = 15 + 51: add-ons were not reduced by the netting factor (NRR treated as 1).

Q558 BT answers P2.T7.515.2 · BT notes P2.T7.515.2

The 1996 Amendment to the 1988 Basel Accord (implemented in 1998) required capital for the market risks of trading activities and distinguishes between the banking book and the trading book. About these books, each of the following is true EXCEPT, which is false?

A. The trading book consists of instruments that are actively traded and marked to market daily; the banking book consists primarily of loans that are held to maturity and not marked to market daily

B. The 1996 Amendment introduced a capital charge for the market risk associated with all items in the trading book

C. Under the 1996 Amendment, the credit risk capital charge in the 1988 Accord continued to apply to all on-balance-sheet and off-balance-sheet items in the trading and banking book, except positions in the trading book that consisted of (i) debt and equity traded securities and/or (ii) positions in commodities and foreign exchange

D. The 1996 Amendment eliminated the standardized measurement method for the calculation of market risk and abolished the distinction between the banking and trading books due to widespread regulatory arbitrage between the two

Answer: D

Step by step

1. The 1996 Amendment required fair-value (mark-to-market) accounting for trading book positions held for trading; the banking book (loans, some debt securities held for investment) is not marked to market. A is true.
2. It added a market risk capital charge on trading book items, using the standardized measurement method or the internal model-based approach. B is true.
3. Credit risk capital under Basel I continued to apply, except to trading book debt/equity securities and FX/commodity positions, which fall under the market risk charge. C is true.
4. The Amendment created two methods (standardized and internal model-based) and relied on the banking/trading book distinction; it did not eliminate either. D is false.

Why D is correct: D contradicts the Amendment, which introduced the standardized measurement method alongside the internal model-based approach and depended on the trading/banking book split.

Why the others are wrong

- **A:** True statement: the trading book is marked to market, the banking book is not; it is not the exception.
- **B:** True statement: market risk capital is charged on trading book items; it is not the exception.
- **C:** True statement: Basel I credit risk charges stayed in force except for the listed trading book positions, which move to the market risk charge; it is not the exception.

Q559 BT answers P2.T7.515.3 · BT notes P2.T7.515.3

About the calculation of market risk under the 1996 Amendment, each of the following is true EXCEPT, which is false?

A. The standardized approach assigns capital separately to each of debt securities, equity securities, foreign exchange risk, commodities risk, and options; but no account is taken for correlations between different types of instruments

B. If a bank elects to use the historical simulation (HS) approach to calculating its value-at-risk (VaR), then a backtest is not required because an HS is effectively already a backtest

C. In the internal model-based approach (IMA), the value-at-risk (VaR) measure used is calculated with a 10-day time horizon and a 99.0% confidence level

D. Most large banks preferred to use the internal model-based approach because it better reflected the benefits

of diversification and led to lower capital requirements

Answer: B

Step by step

1. Test each statement against the 1996 Amendment.
2. A: the standardized method charges each category separately and ignores correlations - true.
3. B: the amendment requires daily backtesting of the one-day 99% VaR over the most recent 250 days for each model, regardless of how VaR is computed - so the claim is false.
4. C: IMA uses a 10-day, 99% VaR - true.
5. D: IMA reflects diversification, gives lower capital, and was preferred by banks with material trading books - true.

Why B is correct: Backtesting is mandatory for every internal VaR model; the choice of VaR method (historical simulation or otherwise) does not exempt a bank, and the number of exceptions drives the multiplier m.

Why the others are wrong

- **A:** True statement; choosing it means wrongly believing the standardized method recognises cross-category correlations.
- **C:** True statement; confusing the regulatory 10-day/99% horizon with the one-day 99% used for backtesting.
- **D:** True statement; overlooks that diversification benefits make IMA capital generally lower than standardized.

Q560

Schweser Module Quiz 62.1 Q2

Saugatuck National Bank uses the internal model-based approach to set market risk capital as prescribed by the 1996 Amendment to the 1988 Basel Accord. The bank has backtested its 99% one-day VaRs against the actual losses over the last 250 trading days and recorded 11 exceptions. Based on these results, the multiplicative factor (m) should be set:

A. less than 3.
B. equal to 3.
C. between 3.1 and 3.9.
D. equal to 4.

Answer: D

Step by step

1. Backtest window: 250 days; exceptions = 11.
2. Rule (GARP): five or fewer exceptions in 250 days allow m = 3; more than five but fewer than ten can lead to a larger multiplier at the supervisor's discretion; 10 or more require m = 4.
3. 11 >= 10, so m = 4.

Why D is correct: With 10 or more exceptions in 250 days the 1996 Amendment requires the maximum multiplier of 4.

Why the others are wrong

- **A:** m can never be below 3; this ignores the floor on the multiplier.
- **B:** m = 3 applies only to a low number of exceptions (green zone); 11 is far beyond it.
- **C:** A multiplier above 3 but below 4 is at the supervisor's discretion for a moderate number of exceptions (more than five but fewer than ten); 11 exceeds this band.

Q561

BT answers L2.T5.62.3

REBUILT — Original said 'Basel II' for a 1996 Amendment topic and used a vague 'All of the above' stem; reworded to the 1996 Amendment and made the specific-risk requirement explicit.

Under the 1996 Amendment to Basel I, for which of the following trading book positions is a capital charge for specific risk required?

A. Only fixed income positions measured under the standardized method
B. Only equity positions measured under the standardized method
C. Only positions measured under the internal model-based approach
D. Fixed income and equity positions under the standardized method, and positions under the internal model-based approach

Answer: D

Step by step

1. Under both approaches, capital is computed separately for specific risk (SR) and general market risk (MR) in each of the five categories.
2. Standardized method: SR charges apply to fixed income, equity and derivatives, so A and B are each incomplete.
3. Internal models: SR may be modelled but the multiplier is 4 rather than 3 and SR capital cannot be less than half of the standardized SR capital.
4. Hence SR capital is required in all cases: D.

Why D is correct: The VaR model captures only general market movements, so company-specific risk (e.g., credit spread, issuer-specific price moves) needs its own charge under every approach.

Why the others are wrong

- **A:** Correct in part but incomplete: equity positions and internal-model banks also carry an SR charge.
- **B:** Correct in part but incomplete: it omits fixed income and internal-model positions.
- **C:** Wrongly assumes only IMA banks face SR; standardized-method banks also have SR charges, and the IMA SR charge has a multiplier of 4 and a 50% floor.

LO 62.d

Describe changes to the Basel regulations made as part of Basel II, including the three pillars.

Q562

BT answers P2.T7.24.5.2 · BT notes T7-24.5.2

REBUILT — Statement 2 was loosely worded (banks adjusting their own requirements); reworded to the ICAAP and supervisory-review mechanism so that only Statement 4 is clearly false.

A junior risk analyst reviews the Basel II framework and summarizes her understanding as follows:

1. Pillar 1 sets minimum capital requirements, equal to 8% of risk-weighted assets covering credit, market and operational risk.
2. Under Pillar 2, banks must have an internal capital adequacy assessment process (ICAAP) that reflects their risk profile, and supervisors review it and may require capital above the minimum.
3. Pillar 3 requires banks to disclose qualitative and quantitative information on their risks and capital, so that market discipline can operate.
4. Basel II is designed to encourage banks to hold as much capital as possible, so that regulatory requirements align with investment objectives.

Which of the above statements is NOT correct?

A. Statement 1
B. Statement 2
C. Statement 3
D. Statement 4

Answer: D

Step by step

1. Statement 1: total capital $= 0.08 \times$ (credit RWA + market RWA + operational RWA); correct.
2. Statement 2: Pillar 2 requires an ICAAP and supervisory evaluation, and supervisors can require capital above the minimum; correct.
3. Statement 3: Pillar 3 (market discipline) increases transparency through qualitative and quantitative disclosures; correct.
4. Statement 4: Basel II sets minimum capital tied to risk; it does not aim to maximize capital holdings. Incorrect.

Why D is correct: Basel II ties minimum capital to risk and relies on supervision and disclosure; it never seeks to maximize bank capital, so Statement 4 is the false one.

Why the others are wrong

- **A:** Statement 1 is true: the 8% minimum applies to credit + market + operational RWA; choosing it confuses the Pillar 1 scope with Basel I (credit only).
- **B:** Statement 2 is true: ICAAP and supervisory review (with power to require excess capital) are the core of Pillar 2; choosing it misreads supervisory discretion as a flaw.
- **C:** Statement 3 is true: Pillar 3 is market discipline through disclosure; choosing it overlooks that disclosure is the purpose of the pillar.

Q563 BT answers P2.T7.518.1 · BT notes P2.T7.518.1

The Second Pillar (Pillar 2) of Basel II is concerned with the supervisory review process. Four key principles of supervisory review are specified. Each of the following is a principle EXCEPT, which is inaccurate?

A. Banks should have a process for assessing their overall capital adequacy in relation to their risk profile and a strategy for maintaining their capital levels.

B. Supervisors should review and evaluate banks' internal capital adequacy assessments and strategies, as well as their ability to monitor and ensure compliance with regulatory capital ratios, and take appropriate supervisory action if not satisfied with the result of this process.

C. Supervisors should discourage banks from operating above the minimum regulatory capital limits because this tends to be associated with regulatory arbitrage.

D. Supervisors should seek to intervene at an early stage to prevent capital from falling below the minimum levels required to support the risk characteristics of a particular bank and should require rapid remedial action if capital is not maintained or restored.

Answer: C

Step by step

1. Pillar 2 principles: (1) banks have an ICAAP and capital strategy; (2) supervisors review ICAAPs and act if unsatisfied; (3) banks are expected to hold capital above the minimum, and supervisors can require it; (4) supervisors intervene early to prevent capital falling below the minimum.
2. A, B and D restate principles 1, 2 and 4.
3. C reverses principle 3: supervisors expect (not discourage) banks to operate above the minimum. C is inaccurate.

Why C is correct: Pillar 2 requires supervisors to ensure banks hold more than the minimum capital, so discouraging excess capital contradicts it.

Why the others are wrong

- **A:** True; this is the ICAAP principle. Picking it means mistaking a bank-side duty for a supervisor-side one.
- **B:** True; the supervisory review principle. Picking it comes from reading the long wording as an overreach.
- **D:** True; early intervention / prompt corrective action. Picking it confuses it with ex-post enforcement only.

Q564 BT answers P2.T7.518.2 · BT notes P2.T7.518.2

In regard to the Third Pillar, Hull writes, "Pillar 3 of Basel II is concerned with increasing the disclosure by a bank of its risk assessment procedures and capital adequacy. The extent to which regulators can force banks to increase their disclosure varies from jurisdiction to jurisdiction." According to Hull, each of the following is true about this Third Pillar, EXCEPT, which is inaccurate?

A. Banks should disclose the structure of the risk management function and how it operates.

B. Banks should disclose the terms and conditions of the main features of all capital instruments.

C. If a bank publishes an annual report with GAAP-compliant financial reports, then additional Third Pillar disclosure is waived due to redundancies.

D. Banks should disclose the entities in the banking group to which Basel II is applied and adjustments made for entities to which it is not applied.

Answer: C

Step by step

1. Required Pillar 3 disclosures include: entities to which Basel II applies; terms of all capital instruments; Tier 1 and Tier 2 instruments; capital requirements for credit, market and operational risk; other risks; risk management function structure.
2. A, B, D are on this list.
3. C is false: regulatory disclosures differ from accounting disclosures, so GAAP statements do not waive Pillar 3.

Why C is correct: Pillar 3 is a separate regulatory disclosure requirement; accounting statements do not substitute for it.

Why the others are wrong

- **A:** True; risk management function disclosure is required. Choosing it reflects not recalling this item.
- **B:** True; terms and conditions of capital instruments are required.
- **D:** True; the scope of application of Basel II within the group must be disclosed.

Q565 MK AP Mock 2-Q66

Pillar 1 of Basel II allows banks to use a range of approaches to compute capital for operational, credit and market risk. Which of the following approaches allows a bank to explicitly recognise diversification benefits?

A. The Foundation Internal Ratings Based approach for credit risk.
B. The Advanced IRB approach for credit risk.
C. The Standardized approach for operational risk.
D. The internal models-based approach for market risk.

Answer: D

Step by step

1. IRB approaches (foundation and advanced) use a supervisory formula with prescribed copula correlation; no bank-specific diversification.
2. Operational standardized approach sums capital across business lines (implicitly perfect correlation).
3. The market risk internal models-based approach use the bank's own VaR model, which reflects estimated correlations across positions, so diversification is recognized.

Why D is correct: Only the internal models-based approach for market risk uses bank-estimated correlations in its VaR, thereby recognizing diversification.

Why the others are wrong

- **A:** Foundation IRB uses supervisory-prescribed correlations, so no bank-specific diversification; the error is assuming internal PD estimates imply internal correlations.
- **B:** Advanced IRB lets the bank estimate LGD and EAD but correlation remains fixed by the formula.
- **C:** The standardized approach simply adds business-line charges (no diversification credit).

Q566 AnalystPrep Q.2341

Basel II introduced a capital requirement for one new risk in Pillar 1. Which one?

A. Interest rate risk in the banking book
B. Market risk in the trading book
C. Operational risk
D. Credit risk

Answer: C

Step by step

1. Credit risk was covered by Basel I (1988).
2. Market risk was added by the 1996 Amendment.
3. Basel II added operational risk to Pillar 1: total capital $= 0.08 \times$ (credit RWA + market RWA + operational RWA).

Why C is correct: Operational risk was the only new risk given a Pillar 1 charge in Basel II.

Why the others are wrong

- **A:** Interest rate risk in the banking book is addressed under Pillar 2, not a Pillar 1 charge.
- **B:** Market risk capital came from the 1996 Amendment and was unchanged in Basel II.
- **D:** Credit risk dates to Basel I; Basel II only refined its measurement.

Q567 AnalystPrep Q.2740

All of these are pillars of the Basel II framework, except:

A. Minimum capital requirements
B. Sound corporate governance
C. Supervisory review
D. Market discipline

Answer: B

Step by step

1. Basel II pillars: (1) minimum capital requirements, (2) supervisory review, (3) market discipline.
2. Corporate governance is not a pillar.

Why B is correct: The three pillars are minimum capital, supervisory review and market discipline; corporate governance is not among them.

Why the others are wrong

- **A:** Pillar 1 is a genuine pillar; selecting it overlooks the 8% minimum.
- **C:** Pillar 2 is supervisory review; selecting it overlooks ICAAP.
- **D:** Pillar 3 is market discipline via disclosure.

Q568 BT Mock Q13

Peter wants to understand the difference between the original Basel I accord and Basel II, published in 2004. His colleague Mary says that, in comparison to Basel I (including the 1995/1996 Amendments), the significant innovations in Basel II included:
I. In addition to credit risk and market risk, Basel II required capital for operational risk.
II. Basel II eliminated risk weights and risk-weighted assets (RWA).
III. Basel II introduced the three-pillar structure, adding supervisory review and market discipline.
IV. Basel II allowed banks to use internal ratings-based approaches for credit risk capital.
Is Mary correct?

A. Yes, all four statements are correct.
B. I. and II. are correct but III. and IV. are inaccurate.
C. Only II. is inaccurate but I., III. and IV. are correct.
D. Unfortunately, none of her statements are correct.

Answer: C

Step by step

1. I is true: Basel II added a capital requirement for operational risk alongside credit and market risk.
2. II is false: Basel II kept risk-weighted assets; it made credit risk weights more risk sensitive (external ratings or internal estimates) rather than eliminating them.
3. III is true: Pillar 1 (minimum capital), Pillar 2 (supervisory review) and Pillar 3 (market discipline through disclosure).
4. IV is true: risk weight formulas based on banks' internal risk measures (IRB approaches) were introduced.
5. Only II is inaccurate, so the answer is C.

Why C is correct: Basel II refined, rather than removed, risk-weighted assets; the other three statements are genuine Basel II innovations.

Why the others are wrong

- **A:** Accepts II, but Basel II did not eliminate RWA.
- **B:** Reverses the truth values: II is false while III and IV are true.
- **D:** Would require every statement to be false, but I, III and IV are true.

Q569 BT answers P2.T7.20.8.1 · BT notes P2.T7.20.8.1

Peter is studying for the FRM exam and wants to understand the difference between the original Basel I accord (the 1988 accord, which includes the 1995/1996 Amendments) and Basel II, published in 2004. His colleague Mary says that significant innovations in Basel II, compared with the original accord, included:
I. In addition to credit risk and market risk, Basel II required capital for operational risk.
II. Basel II eliminated risk weights and risk-weighted assets (RWA) and replaced them with direct calculation of risk charges.
III. Basel II contained specific requirements for supervision related to capital and risk management (Pillar 2) and required public disclosures (Pillar 3).
IV. To fine-tune the accord's design, Basel II made repeated use of Quantitative Impact Studies (QIS), to which banks contributed data that was analyzed by supervisors.
Is Mary correct?

A. Yes, all four statements are correct.
B. I. and II. are correct, but III. and IV. are inaccurate.
C. Only II. is inaccurate, but I., III., and IV. are correct.
D. Unfortunately, none of her statements are correct.

Answer: C

Step by step

1. I is true: operational risk capital was a new Basel II requirement.
2. II is false: Basel II kept RWA as the basis of the capital ratio and introduced more sophisticated risk weight formulas for credit risk.

3. III is true: Pillar 2 (supervisory review, e.g. ICAAP) and Pillar 3 (public disclosure) were added to Pillar 1 minimum capital.
4. IV is true: repeated QIS rounds, in which banks supplied data analyzed by supervisors, were used to tune the design.
5. Only II is inaccurate, so the answer is C.

Why C is correct: Basel II changed how risk weights are computed but did not eliminate RWA; the remaining statements match the four innovations of Basel II.

Why the others are wrong

- **A:** Includes II, but RWA were retained.
- **B:** Treats II as true and III and IV as false; III and IV are genuine innovations.
- **D:** Would need all four statements to be false, but I, III and IV are true.

Q570 AnalystPrep Q.5959

A large commercial bank is calculating its capital adequacy ratio (CAR) in line with Basel II guidelines. Tier 1 capital is $200 million, Tier 2 capital is $100 million, and risk-weighted assets (RWA) are $2 billion. What is the bank's CAR?

A. 10%
B. 15%
C. 5%
D. 20%

Answer: B

Step by step

1. Total capital = Tier 1 + Tier 2 = 200 + 100 = $300 million.
2. CAR = total capital / RWA = 300 / 2,000 = 15%.
3. This exceeds the Basel II minimum total capital ratio of 8%.

Why B is correct: The CAR uses total capital (Tier 1 plus Tier 2) in the numerator: $300/2000 = 15\%$.

Why the others are wrong

- **A:** Used Tier 1 only: 200/2000 = 10%.
- **C:** Used Tier 2 only: 100/2000 = 5%.
- **D:** Double counted Tier 1 (400/2000) instead of adding Tier 2.

Q571 AnalystPrep Q.5960

A mid-sized investment bank reviews its capital adequacy ratio (CAR) under Basel II. Tier 1 capital is $150 million, Tier 2 capital is $50 million, and total risk-weighted assets (RWA) are $1.5 billion. What is the bank's CAR, and does it meet the minimum requirement set by Basel II?

A. 10% and it meets the minimum requirement.
B. 13.33% and it does not meet the minimum requirement.
C. 13.33% and it meets the minimum requirement.
D. 10% and it does not meet the minimum requirement.

Answer: C

Step by step

1. Total capital = 150 + 50 = $200 million.
2. CAR = 200 / 1,500 = 13.33%.
3. The Basel II minimum total capital ratio is 8% of RWA; 13.33% > 8%, so the requirement is met.

Why C is correct: CAR $= (150 + 50)/1500 = 13.33\%$, above the 8% minimum, so the bank complies.

Why the others are wrong

- **A:** Used Tier 1 only (150/1500 = 10%); the ratio is wrong even though the conclusion happens to be 'meets'.
- **B:** Correct ratio but wrong conclusion: 13.33% is above the 8% minimum.
- **D:** Used Tier 1 only (10%) and also concluded it fails, although 10% would still exceed 8%.

LO 62.e

Compare the standardized internal ratings-based (IRB) approach, the foundation IRB approach, and the advanced IRB approach for the calculation of credit risk capital under Basel II.

Q572

Schweser Module Quiz 23.1 Q3

The internal ratings-based (IRB) approach for calculating a bank's capital adequacy ratio (CAR) has several underlying elements. Which of the following statements relates to the IRB approach?

A. Idiosyncratic risk is explicitly factored into the model.
B. Estimates of probability of default are made by regulators.
C. Maturity adjustments should be higher for borrowers with high probabilities of default.
D. The maturity parameter reflects that capital requirements should increase with maturity.

Answer: D

Step by step

1. The IRB formula (Gordy one-factor model) assumes a large, well-diversified portfolio, so only systematic risk drives capital; idiosyncratic risk is diversified away.
2. Under both IRB variants the bank supplies the PD estimate (subject to a 0.03% floor); regulators set LGD, EAD and M only under foundation IRB.
3. The maturity adjustment $MA = \dfrac{1 + (M - 2.5)b}{1 - 1.5b}$, with $b = [0.11852 - 0.05478 \ln(PD)]^2$, rises with M because longer exposures have more time to deteriorate.
4. Because b is larger for low PD, the adjustment is larger for low-PD borrowers, not high-PD ones. Only D is correct.

Why D is correct: Longer-maturity exposures face greater downgrade/default risk, so the IRB capital requirement increases with maturity via the maturity adjustment.

Why the others are wrong

- **A:** IRB assumes a well-diversified portfolio, so idiosyncratic risk is not modelled; confuses the model with a concentrated-portfolio model.
- **B:** Banks, not regulators, estimate PD under IRB; regulators set LGD/EAD/M only in the foundation approach.
- **C:** Reversed: b (and hence the adjustment) is larger for low-PD borrowers, which have more room to be downgraded.

Q573

AnalystPrep Q.2342

REBUILT — Original answer offered all three approaches to a US bank, contradicting GARP (US did not implement the standardized approach; only large internationally active banks used IRB); rewritten to test the US implementation.

NYC Bank is a large, internationally active US bank. When Basel II was implemented in the United States, which statement best describes the credit risk capital approaches that applied to such a bank?

A. It could freely choose among the standardized, foundation IRB and advanced IRB approaches.
B. It was required to use IRB approaches, because the US chose not to implement the Basel II standardized approach.
C. It could only use the standardized approach, with risk weights based on external ratings.
D. It continued to use the Basel I risk weights, because the US did not adopt Basel II at all.

Answer: B

Step by step

1. Basel II offered three credit risk approaches: standardized, foundation IRB and advanced IRB.
2. The US chose not to implement the standardized approach.
3. Internationally active US banks were required to use IRB approaches; other US banks used an updated version of the Basel I requirements.
4. NYC Bank, as a large internationally active bank, therefore had to use IRB. Answer B.

Why B is correct: US implementation required IRB for internationally active banks and left other banks on updated Basel I rules; the standardized approach was not implemented.

Why the others are wrong

- **A:** This is the generic Basel II menu, but the US did not implement the standardized approach and required IRB for internationally active banks.

- **C:** Reverses the US choice: the standardized approach was the one not implemented.
- **D:** The US did adopt Basel II for internationally active banks; only the other banks stayed on updated Basel I.

Q574

AnalystPrep Q.2343

REBUILT — Original relied on EU-specific regulation and Basel I/II overlap not in the reading; rewritten as a clean comparison of the Basel II approaches by sophistication.

PSV Bank, a small regional bank, has relatively unsophisticated risk management and no internal rating system. Under Basel II, which approach for credit risk capital is designed for such a bank, and how are its risk weights determined?

A. Standardized approach; risk weights depend on asset class and external credit ratings.
B. Standardized approach; risk weights depend on the bank's own estimates of PD, LGD and EAD.
C. Foundation IRB approach; the bank supplies LGD and EAD, regulators supply PD.
D. Advanced IRB approach; the bank supplies all inputs, with no supervisory approval needed.

Answer: A

Step by step

1. Basel II offers three credit risk approaches: standardized, foundation IRB, advanced IRB.
2. The standardized approach is for banks with less sophisticated risk management; like Basel I it uses prescribed risk weights, but now sensitive to external credit ratings of countries, banks and corporates (OECD status no longer matters).
3. IRB approaches require internal estimates (at least PD) and supervisory approval, which PSV Bank is unlikely to meet. Answer A.

Why A is correct: The standardized approach applies regulator-set risk weights that vary with asset class and external ratings, needing no internal models.

Why the others are wrong

- **B:** Own PD/LGD/EAD estimates are features of the IRB approaches, not the standardized approach.
- **C:** Reversed: under foundation IRB the bank supplies PD, while regulators set LGD, EAD and M; and it needs internal ratings the bank lacks.
- **D:** Advanced IRB is the most demanding approach and requires supervisory approval; it is not suited to an unsophisticated bank.

Q575

AnalystPrep Q.3240

REBUILT — Answer was correct; replaced distractors that could not be traced to a realistic error (1.792) with plausible mistakes (unweighted exposure, multiplying by 12.5).

Python Commercial Bank uses the standardized approach to estimate total risk-weighted credit risk exposure. External credit ratings give the following risk weights.

Exposure	Risk weight
$24 million	80%
$12 million	120%
$18 million	70%
$17 million	30%
$3 million	10%

As a rough approximation under Basel II, the minimum capital the bank must hold is closest to:

A. $51.600 million
B. $5.920 million
C. $4.128 million
D. $645.000 million

Answer: C

Step by step

1. RWA = $0.8(24) + 1.2(12) + 0.7(18) + 0.3(17) + 0.1(3) = 19.2 + 14.4 + 12.6 + 5.1 + 0.3 = 51.6$ million.
2. Minimum capital = $8\% \times 51.6 = 4.128$ million.

Why C is correct: Basel II requires capital of at least 8% of risk-weighted assets, so capital = 0.08 x 51.6 = $4.128 million.

Why the others are wrong

- **A:** This is the RWA itself; the 8% capital ratio was not applied.
- **B:** Applied 8% to the unweighted exposure total ($0.08 \times 74 = 5.92$), ignoring risk weights.
- **D:** Multiplied RWA by 12.5 (the RWA-to-capital conversion in the wrong direction) instead of multiplying by 8%.

Q576

AnalystPrep Q.5962

REBUILT — Original was ambiguous (both IRB variants use internal models) and had giveaway operational-risk distractors; rewritten to test the foundation vs advanced IRB parameter split.

Under Basel II, a bank uses the foundation IRB approach for its corporate loans. Which statement correctly describes the inputs to the capital calculation?

A. The bank estimates PD, while LGD, EAD and maturity are set by supervisors.
B. The bank estimates PD, LGD, EAD and maturity.
C. Supervisors set PD, while the bank estimates LGD and EAD.
D. All inputs are set by supervisors according to external credit ratings.

Answer: A

Step by step

1. Foundation IRB: the bank supplies PD only (floor 0.03% for bank and corporate exposures).
2. LGD (45% senior, 75% subordinated), EAD and M (usually 2.5) are supervisory values.
3. Advanced IRB is the approach in which the bank supplies PD, LGD, EAD and M. Answer A.

Why A is correct: Foundation IRB gives banks only the PD estimate; all other inputs are prescribed by the Basel Committee.

Why the others are wrong

- **B:** Describes the advanced IRB approach, not foundation.
- **C:** Reverses the split: PD is the bank's estimate, LGD/EAD are supervisory in foundation IRB.
- **D:** Describes the standardized approach with external ratings, not an IRB approach.

Q577

AnalystPrep Q.5963

REBUILT — Original had giveaway operational-risk distractors (AMA, BIA); rewritten to test which inputs the bank supplies under advanced IRB.

A large bank with extensive default and loss data receives supervisory approval to adopt the advanced IRB approach for its corporate exposures. Which inputs does the bank itself estimate?

A. PD only.
B. PD and LGD only; EAD and maturity are set by supervisors.
C. PD, LGD, EAD and maturity.
D. LGD, EAD and maturity only; PD is set by supervisors.

Answer: C

Step by step

1. Advanced IRB: the bank provides its own PD, LGD, EAD and M (subject to supervisory approval).
2. PD can reflect credit mitigants (floor 0.03%), LGD reflects collateral and seniority, EAD may use own credit conversion factors.
3. Only the foundation approach limits the bank to PD. Answer C.

Why C is correct: Advanced IRB lets the bank supply all four inputs, giving the most risk-sensitive credit capital measure.

Why the others are wrong

- **A:** This is foundation IRB.
- **B:** Wrongly leaves EAD and M with supervisors; in advanced IRB they are bank estimates.
- **D:** PD is estimated by the bank in both IRB variants; supervisors never set it.

Q578

AnalystPrep Q.5964

A regional bank has chosen the Basel II standardized approach for credit risk capital. What is the primary action it needs to take to calculate its capital requirement?

A. Develop sophisticated internal risk assessment models.
B. Obtain regulatory approval for its internal risk models.
C. Assign predefined risk weights to its assets as set by regulatory authorities.

D. Calculate the probability of default (PD) for each type of asset.

Answer: C

Step by step

1. Under the standardized approach, supervisors prescribe the risk weights (varying with asset class and external ratings).
2. The bank classifies exposures, applies the weights to get RWA, and holds capital of 8% of RWA.
3. No internal models, approvals or own PD estimates are needed. Answer C.

Why C is correct: The standardized approach uses regulator-specified risk weights, so the bank's task is to map exposures to those weights.

Why the others are wrong

- **A:** Internal models are for IRB approaches; the standardized approach is for banks without them.
- **B:** Supervisory approval is required for IRB, not for the standardized approach.
- **D:** Own PD estimates are an IRB input; the standardized approach uses external ratings-based weights.

Q579

AnalystPrep Q.5971

A regional bank applying the advanced IRB approach under Basel II examines the maturity adjustment in its capital calculation for corporate exposures. Which statement is correct?

A. Capital requirements decrease with longer loan maturities and are higher for high-PD borrowers.

B. Maturity adjustments are unrelated to PD and depend solely on the loan term.

C. Capital requirements increase with longer loan maturities, and the maturity adjustment is larger for low-PD borrowers.

D. Maturity adjustments are the same for all borrowers regardless of loan term and PD.

Answer: C

Step by step

1. $MA = \dfrac{1 + (M - 2.5)b}{1 - 1.5b}$, $b = [0.11852 - 0.05478 \ln(PD)]^2$; MA = 1 when M = 1.
2. Numerically at M = 5: PD = 0.1% gives $b = 0.2469$ and MA = 2.57; PD = 5% gives $b = 0.0799$ and MA = 1.36.
3. So MA rises with maturity, and rises more for low-PD borrowers (more room to be downgraded). Answer C.

Why C is correct: Longer exposures stay on the books longer and may deteriorate; b falls as PD rises, so the adjustment is larger for low-PD borrowers.

Why the others are wrong

- **A:** Both parts are reversed: capital rises with maturity and the adjustment is larger for low PD.
- **B:** MA depends on PD through b as well as on M.
- **D:** MA varies with M (equal to 1 only at M = 1) and with PD.

Q580

BT answers P2.T7.517.1 · BT notes P2.T7.517.1

Consider the following four statements which attempt to summarize the approach to credit risk capital under Basel II:

I. For credit risk, Basel II specified three approaches: the Standardized Approach, the Foundation Internal Ratings Based (IRB) Approach, and the Advanced IRB Approach.

II. For the IRB approach, regulators base the capital requirement on the value at risk calculated using a one-year time horizon and a 99.9% confidence level; they recognize that expected losses are usually covered by the way a financial institution prices its products. The capital required is, therefore, the value at risk minus the expected loss.

III. Under the Foundation IRB approach, banks supply PD while LGD, EAD, and M are supervisory values set by the Basel Committee. PD is subject to a floor of 0.03% for bank and corporate exposures. LGD is set at 45% for senior claims and 75% for subordinated claims. When there is eligible collateral, LGD is reduced using the comprehensive approach. The EAD is calculated in a manner similar to the credit equivalent amount in Basel I and includes the impact of netting. M is set at 2.5 in most circumstances.

IV. Under the Advanced IRB approach, banks supply their own estimates of PD, LGD, EAD, and M for corporate, sovereign, and bank exposures. PD can be reduced by credit mitigants such as credit triggers (subject to the same 0.03% floor for bank and corporate exposures). The two main factors influencing LGD are the seniority of the debt and the collateral. In calculating EAD, banks can, with regulatory approval, use their own estimates of credit conversion factors.

Which of the above statements is (are) accurate?

A. None are accurate
B. Only I and II
C. Only I and IV
D. All are accurate

Answer: D
Step by step
1. I: Basel II offers exactly three credit risk approaches (Standardized, Foundation IRB, Advanced IRB) - accurate.
2. II: IRB capital is the 1-year, 99.9% VaR minus expected loss (EL is covered by pricing), so capital covers unexpected loss - accurate.
3. III: Foundation IRB - bank supplies PD (floor 0.03%); supervisor sets LGD (45% senior / 75% subordinated, reduced for collateral via the comprehensive approach), EAD (Basel I-style credit equivalent with netting) and M (2.5) - accurate.
4. IV: Advanced IRB - bank estimates PD, LGD, EAD and M; PD floor still 0.03%; LGD driven by seniority and collateral; own credit conversion factors with approval - accurate.
5. All four statements are accurate, so the answer is D.

Why D is correct: Each statement restates the Basel II credit-risk framework correctly: three approaches, 1-year 99.9% VaR minus EL, and the Foundation versus Advanced split of who supplies PD, LGD, EAD and M.

Why the others are wrong
- **A:** Wrong: it treats at least one accurate statement as false; none of the four contains an error.
- **B:** Wrong: it wrongly rejects III and IV, e.g. doubting the 45%/75% LGD and M = 2.5 supervisory values or the Advanced IRB own-estimate features.
- **C:** Wrong: it wrongly rejects II (confusing capital, VaR minus EL, with total VaR) and III.

Q581 BT answers P2.T7.519.1 · BT notes P2.T7.519.1

Under the Basel II internal ratings-based (IRB) approach, the capital required is the excess of the 99.9% worst-case loss over the expected loss. The key parameters are the one-year probability of default (PD), the exposure at default (EAD), the loss given default (LGD), and the worst-case probability of default (WCDR), defined so that the bank is 99.9% certain it will not be exceeded next year for the counterparty. Consider the following three statements:
I. For corporate, sovereign, and bank exposures under the Foundation IRB approach, banks supply PD while LGD, EAD, and maturity (M) are supervisory values set by the Basel Committee. PD is subject to a floor of 0.03% for bank and corporate exposures. LGD is 45% for senior claims and 75% for subordinated claims. M is set at 2.5 in most circumstances.
II. For corporate, sovereign, and bank exposures under the Advanced IRB approach, banks supply their own estimates of PD, LGD, EAD, and M. PD can be reduced by credit mitigants such as credit triggers. The two main factors influencing LGD are the seniority of the debt and the collateral. In calculating EAD, banks can, with regulatory approval, use their own estimates of credit conversion factors.
III. For retail exposures, the Foundation and Advanced IRB approaches are merged, and all banks using the IRB approach provide their own estimates of PD, EAD, and LGD. There is no maturity adjustment.
Which of the above statements is (are) true?

A. None are true
B. Only I is true
C. Only II is true
D. All are true

Answer: D
Step by step
1. I: Foundation IRB - bank gives PD (0.03% floor for bank/corporate); LGD 45%/75%, EAD and M (2.5) are supervisory - true.
2. II: Advanced IRB - bank estimates PD, LGD, EAD and M; credit triggers can reduce PD; LGD depends on seniority and collateral; own CCFs with approval - true.
3. III: Retail - the two IRB methods are merged, banks estimate PD, EAD and LGD themselves, and there is no maturity adjustment - true.
4. All three are true, so the answer is D.

Why D is correct: The statements correctly distinguish who supplies LGD, EAD and M under Foundation versus Advanced IRB, and note that retail exposures use a merged, advanced-style treatment without a maturity adjustment.

Why the others are wrong

- **A:** Wrong: it rejects every statement, e.g. by thinking the supervisor supplies PD under Foundation IRB; each statement is in fact correct.
- **B:** Wrong: it accepts only the Foundation description and wrongly doubts that Advanced IRB banks estimate M and CCFs, and that retail has no maturity adjustment.
- **C:** Wrong: it accepts only the Advanced description and wrongly doubts the Foundation supervisory values (45%/75%, M = 2.5) and the retail treatment.

Q582

BT answers P2.T7.517.2 · BT notes P2.T7.517.2

REBUILT — The stem fixed a comprehensive-approach convention (whole net exposure weighted at the counterparty's 150%) that contradicts GARP, under which the collateral's risk weight applies to the haircut collateral and the counterparty's weight to the remaining exposure; the stem, question, options and solution were rebuilt to the GARP convention (answer $27.0 million).

Under the Basel II standardized approach, banks can adjust for collateral using the simple approach or the comprehensive approach. Under the simple approach, the collateral's risk weight replaces the counterparty's risk weight for the covered portion of the exposure, and the counterparty's risk weight applies to the rest. Under the comprehensive approach, the exposure is adjusted upward and the collateral value downward for possible changes in value; the collateral's risk weight is then applied to the reduced amount of collateral and the counterparty's risk weight to the remaining exposure.

A bank has a $100.0 million exposure to a counterparty rated B+, secured by $80.0 million of bonds issued by an A-rated company. The counterparty's risk weight is 150% and the collateral's risk weight is 50%. Under the comprehensive approach, the exposure is adjusted by +10% and the collateral value by -15%. By how much do the risk-weighted assets (RWA) under the comprehensive approach exceed the RWA under the simple approach?

A. $7.0 million
B. $12.0 million
C. $27.0 million
D. $97.0 million

Answer: C

Key correction: Source key (BT 517.2) said simple minus comprehensive = $7.0 million, weighting the whole net exposure of $42 million at 150%; GARP applies the collateral's 50% weight to the $68 million haircut collateral and 150% to the $42 million remainder, so comprehensive RWA is $97.0 million and exceeds simple RWA by $27.0 million.

Step by step

1. Simple approach: RWA = 50% × $80 + 150% × ($100 - $80) = 40 + 30 = $70.0 million.
2. Comprehensive approach: adjusted exposure = 100 × 1.10 = $110.0 million; adjusted collateral = 80 × 0.85 = $68.0 million.
3. Remaining (uncovered) exposure = 110 - 68 = $42.0 million.
4. Comprehensive RWA = 50% × 68 + 150% × 42 = 34 + 63 = $97.0 million.
5. Difference = 97.0 - 70.0 = $27.0 million (comprehensive exceeds simple).

Why C is correct: The comprehensive approach grosses up the exposure and haircuts the collateral, so less of the exposure receives the collateral's 50% weight and more receives the counterparty's 150% weight: $0.5 \times 68 + 1.5 \times 42 = 97$ versus $0.5 \times 80 + 1.5 \times 20 = 70$ under the simple approach.

Why the others are wrong

- **A:** Wrong: weights the entire net exposure of $42 million at 150% (= $63 million) and gives no RWA for the collateral, so it gets 70 - 63 = $7.0 million; GARP applies the collateral's 50% weight to the $68 million of haircut collateral.
- **B:** Wrong: applies only the -15% collateral haircut and forgets the +10% exposure adjustment: 0.5 × 68 + 1.5 × 32 = 82, and 82 - 70 = $12.0 million.
- **D:** Wrong: reports the comprehensive-approach RWA itself ($97.0 million) rather than the difference from the simple approach.

Q583

GARP book PE2 Q47 · Mock 4 Q48

A researcher at a national regulatory agency is examining the use of the standardized approach (SA) and the advanced internal ratings-based (A-IRB) approach in determining credit risk capital. The researcher evaluates the implications of applying these approaches to two banks, Global Bank and Resource Bank.

Global Bank lends to large global corporations that have highly diverse business lines by providing sizable, long-dated unsecured credit facilities.

Resource Bank lends to oil and gas producers in its region, most of which have small-scale operations.
Which of the following is the most appropriate conclusion for the researcher to reach?

A. The A-IRB approach does not adjust for maturity, which means that A-IRB could undercapitalize the long-dated exposures at Global Bank.
B. The SA gives lower risk weights to unrated credits than to credits rated B+ or lower, which may incentivize Resource Bank to lend to unrated companies with poor credit quality.
C. Model parameters used in the SA effectively capture the high default correlation that is likely to exist between Resource Bank's lending counterparties.
D. The A-IRB approach would require Global Bank to model the exposure at default while using a probability of default specified by the Basel Committee.

Answer: B

Step by step

1. Under the SA, corporate risk weights are 100% for unrated and 150% for B+ to B- (and below), so unrated corporates are cheaper than weak rated ones.
2. Resource Bank's small producers are likely unrated, so the bank can hold poor-quality unrated borrowers at 100% rather than 150%, creating a perverse incentive - B is true.
3. A-IRB includes an explicit maturity adjustment (MA) for exposures beyond one year, so A is false.
4. The SA uses fixed risk weights with no correlation parameter, so C is false.
5. Under A-IRB the bank estimates PD, LGD, EAD and M itself; a supervisory PD is not used, so D is false.

Why B is correct: Because unrated corporate exposures get a 100% risk weight while B+ or lower rated ones get 150%, the SA rewards lending to low-quality but unrated borrowers.

Why the others are wrong

- **A:** Wrong: the IRB formula includes a maturity adjustment that raises capital for long-dated exposures; it is not ignored.
- **C:** Wrong: the SA is a fixed risk-weight table with no model or correlation parameters; correlation appears in the IRB formula.
- **D:** Wrong: this mixes up the approaches; under A-IRB the bank estimates PD itself (and LGD, EAD, M), whereas in Foundation IRB the bank supplies PD and the supervisor sets the others.

Q584

AnalystPrep Q.2738

Which of the following approaches is NOT appropriate for calculating credit risk capital under Basel II?

A. Standardized Approach
B. Foundation IRB Approach
C. Advanced IRB Approach
D. Advanced Measurement Approach

Answer: D

Step by step

1. Basel II offers three credit risk approaches: standardized, foundation IRB and advanced IRB.
2. The Advanced Measurement Approach (AMA) is one of three operational risk approaches (with basic indicator and standardized).
3. So AMA is the one not used for credit risk capital.

Why D is correct: AMA is an operational risk capital method, not a credit risk method.

Why the others are wrong

- **A:** Standardized approach is a valid credit risk approach (risk weights from external ratings).
- **B:** Foundation IRB is valid: bank estimates PD, regulator supplies LGD and EAD.
- **C:** Advanced IRB is valid: bank estimates PD, LGD and EAD.

Q585

AnalystPrep Q.3094

Which of the following is NOT an approach for calculating credit risk capital under Basel II?

A. Standardized approach
B. Internal ratings-based approach - Foundation
C. Internal ratings-based approach - Advanced
D. Standardized approach - Advanced

Answer: D

Step by step

1. Basel II credit risk approaches: standardized, foundation IRB, advanced IRB.
2. There is no 'advanced standardized' approach; 'advanced' labels only the IRB and (for operational risk) AMA methods.
3. D is therefore not a Basel II credit risk approach.

Why D is correct: The standardized approach has a single version based on external ratings; only IRB splits into foundation and advanced.

Why the others are wrong

- **A:** Valid approach using external-rating-based risk weights.
- **B:** Valid: own PD, supervisory LGD and EAD.
- **C:** Valid: own PD, LGD and EAD.

Q586

AnalystPrep Q.5968

A regional bank, following Basel II, has chosen the Foundation Internal Ratings-Based (F-IRB) Approach for its capital adequacy calculation. Which of the following is a key action the bank must take to comply with the F-IRB Approach?

A. Assigning a uniform risk weight to all types of assets

B. Developing internal models for Loss Given Default (LGD) and Exposure at Default (EAD)

C. Estimating the Probability of Default (PD) for each type of credit exposure

D. Seeking external credit ratings for all its credit exposures

Answer: C

Step by step

1. Under F-IRB the bank supplies only its own PD estimates.
2. LGD and EAD are set by the regulator (supervisory values).
3. Hence the key requirement is internally estimating PD for each exposure type.

Why C is correct: F-IRB requires internal PD estimates only; LGD and EAD are prescribed.

Why the others are wrong

- **A:** Uniform risk weights are not IRB; IRB weights depend on PD (and LGD, EAD, maturity).
- **B:** Internal LGD and EAD models are required only under advanced IRB.
- **D:** Reliance on external ratings characterizes the standardized approach.

Q587

BT answers P2.T6.605.2 · BT notes P2.T6.605.2

REBUILT — Original option B was loosely worded and option C misstated the standardized weight in a confusing way; reworded options so exactly one is clearly true per GARP (retail 75%, mortgage 35%, correlations 0.15/0.04).

Which of the following is TRUE about the Basel regulatory approach to retail exposures?

A. Retail exposures must use the standardized approach to credit risk

B. Retail exposures must use the foundation IRB approach, with supervisory LGD and EAD

C. Under the standardized approach, "regulatory retail" exposures are assigned a risk weight of 25%

D. Under the IRB approach for retail exposures, residential mortgages are assigned a higher asset correlation than qualifying revolving retail exposures

Answer: D

Step by step

1. Retail under IRB uses an advanced-style treatment: bank supplies PD, LGD and EAD; no maturity adjustment.
2. Asset correlations: 0.15 for residential mortgages, 0.04 for qualifying revolving exposures, formula-based 0.03 to 0.16 for other retail.
3. 0.15 > 0.04, so D is true.
4. Standardized retail weight is 75% (unrated mortgages 35%), not 25%.

Why D is correct: Mortgages are assigned a correlation of 0.15 versus 0.04 for qualifying revolving retail, so D is true.

Why the others are wrong

- **A:** False: banks with supervisory approval may use IRB for retail; the standardized approach is not mandatory.
- **B:** False: for retail, foundation and advanced are merged and the bank estimates PD, LGD and EAD itself.

- **C:** False: standardized retail weight is 75% (35% for mortgages), not 25%.

LO 62.f

Calculate credit risk capital under Basel II utilizing the IRB approach.

Q588

AnalystPrep Q.3238 · AnalystPrep Q.3239

REBUILT — Merged the two Kimpala items (same data); options rebuilt because the exact WCDR is 9.14% (source used rounded z-values giving 9.07%) and distractors were unrealistic.

Kimpala Leasing Bank holds $690 million of retail loans. The bank's actuary estimates PD = 1% and LGD = 40%, and the Basel II copula correlation is $\rho = 0.1216$. Under the Basel II IRB approach, what are the worst-case default rate at the 99.9% confidence level (DR99.9) and the expected loss (EL)?

A. DR99.9 = 9.14%; EL = $2.76 million
B. DR99.9 = 9.14%; EL = $6.90 million
C. DR99.9 = 90.86%; EL = $2.76 million
D. DR99.9 = 9.14%; EL = $22.46 million

Answer: A

Key correction: Source key for Q.3239 said 9.07% (rounded z-values); exact answer is 9.14%. Q.3238 key (0.0907 and $2.76m) has the correct EL.

Step by step

1. $N^{-1}(0.01) = -2.3263$, $N^{-1}(0.999) = 3.0902$, $\sqrt{\rho} = 0.3487$, $\sqrt{1-\rho} = 0.9372$.
2. $\text{DR}_{99.9} = N\left[\frac{-2.3263+0.3487\times3.0902}{0.9372}\right] = N(-1.3324) = 0.0914$ (9.14%; rounding the z-values to -2.33 and 3.09 gives about 9.1%).
3. $\text{EL} = \text{EAD} \times \text{LGD} \times \text{PD} = 690 \times 0.40 \times 0.01 = \$2.76$ million.
4. (For reference, IRB capital = $690 \times 0.40 \times (0.0914 - 0.01) = \$22.46$ million.)

Why A is correct: The Vasicek/Gordy formula stresses the PD up to the 99.9th percentile of the default-rate distribution, while EL uses the unstressed PD.

Why the others are wrong

- **B:** EL omits LGD (690 x 0.01 = 6.90).
- **C:** Took N(+1.3324) = 0.9086 instead of N(-1.3324); the negative z-score gives the small left-tail probability.
- **D:** 22.46 is the IRB capital (unexpected loss), EAD x LGD x (DR - PD), not the expected loss.

Q589

BT answers P2.T7.24.10.3 · BT notes P2.T7.24.10.3

REBUILT — Cleaned stem (removed irrelevant maturity-adjustment formula note and made it a retail exposure explicitly).

Bank Fortuna is computing risk-weighted assets (RWA) for a retail loan portfolio under the Basel II IRB approach.

Parameter	Value
Exposure at default (EAD)	$400 million
Loss given default (LGD)	35%
Worst-case default rate (DR99.9)	10%
Probability of default (PD)	2%

What is the RWA? (Retail exposures carry no maturity adjustment.)

A. $140 million
B. $35 million
C. $400 million
D. $2.8 million

Answer: A

Step by step

1. Retail: no maturity adjustment, so capital = $\text{EAD} \times \text{LGD} \times (\text{DR}_{99.9} - \text{PD})$.
2. Capital = $400 \times 0.35 \times (0.10 - 0.02) = \$11.2$ million.
3. $\text{RWA} = 12.5 \times 11.2 = \$140$ million.

Why A is correct: RWA is 12.5 times the unexpected-loss capital, EAD x LGD x (DR99.9 - PD), with no maturity

adjustment for retail.

Why the others are wrong

- **B:** Used PD (2%) instead of (DR - PD): 12.5 x 400 x 0.35 x 0.02 = 35.
- **C:** Assumed RWA equals the exposure (100% risk weight, Basel I style).
- **D:** Computed expected loss 400 x 0.35 x 0.02 = 2.8; not capital and not multiplied by 12.5.

Q590

AnalystPrep Q.6038

A bank applies Vasicek's model under Basel II to a portfolio of commercial loans. How does the credit correlation parameter ρ influence the capital requirement derived from the model?

A. A higher ρ lowers capital because defaults are more diversified across the portfolio.
B. A higher ρ lowers capital because it signals better management of default risk.
C. A lower ρ raises capital because default patterns are more unpredictable.
D. A higher ρ raises capital because defaults become more likely to occur together.

Answer: D

Step by step

1. $\text{DR}_{99.9} = N\left[\frac{N^{-1}(PD)+\sqrt{\rho}\,N^{-1}(0.999)}{\sqrt{1-\rho}}\right]$; at the 99.9th percentile $N^{-1}(0.999) > 0$.
2. Raising ρ increases DR99.9 (if $\rho = 0$, DR99.9 = PD).
3. Capital = EAD x LGD x (DR99.9 - PD) therefore rises with ρ.

Why D is correct: Higher correlation means defaults cluster, fattening the tail of the loss distribution and raising the worst-case default rate and capital.

Why the others are wrong

- **A:** Reverses the effect: higher correlation means less diversification, not more.
- **B:** Correlation is not a measure of risk-management quality; the effect is reversed.
- **C:** Reverses direction: lower correlation lowers DR99.9 and capital.

Q591

AnalystPrep Q.6040

A risk manager explains Vasicek's model under the Basel II IRB approach with PD = 2%, $\rho = 12\%$ and confidence level X = 99.9%. What role does the confidence level X play?

A. It sets the threshold for the expected loss that capital must cover.
B. It determines the worst-case default rate (WCDR) that the bank must be able to withstand.
C. It is the minimum percentage of capital that must be held against future losses.
D. It dictates the degree of diversification the bank must achieve in its loan portfolio.

Answer: B

Step by step

1. $\text{WCDR} = N\left[\frac{N^{-1}(PD)+\sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right]$.
2. X enters through $N^{-1}(X)$ (3.09 for 99.9%), selecting the percentile of the default-rate distribution.
3. Capital then equals EAD x LGD x (WCDR - PD), covering unexpected loss up to that percentile.

Why B is correct: X picks the percentile of the default-rate distribution, i.e. the worst-case default rate the bank is 99.9% sure will not be exceeded in a year.

Why the others are wrong

- **A:** Expected loss uses PD and is covered by pricing/provisions; X relates to unexpected loss.
- **C:** Capital is not a fixed percentage set by X; it is computed from WCDR (the 8% factor links RWA to capital).
- **D:** Diversification is not set by X; the model assumes a granular portfolio and uses the correlation input.

Q592

AnalystPrep Q.6041

REBUILT — Original options were all plausible supervisory requests; restated the question as the inputs needed to recompute the WCDR so only one option is correct.

A regulatory auditor wants to independently recompute a bank's worst-case default rate (WCDR) from Vasicek's model under Basel II. Which set of inputs does the WCDR formula require?

A. PD, LGD, EAD and maturity
B. PD, the asset (copula) correlation and the confidence level

C. EAD, LGD and the confidence level
D. PD, LGD and the expected loss

Answer: B
Step by step
1. WCDR $= N\left[\frac{N^{-1}(PD)+\sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right]$.
2. The formula contains only PD, ρ and X.
3. LGD, EAD and M enter later when WCDR is converted into dollar capital.

Why B is correct: WCDR depends only on PD, the correlation and the confidence level.
Why the others are wrong
- **A:** These are the inputs to dollar capital/RWA (with the maturity adjustment), not to WCDR itself.
- **C:** EAD and LGD scale the loss in dollars; PD and correlation, which the formula needs, are missing.
- **D:** LGD and EL do not enter WCDR, and the correlation is missing.

Q593 Mock AP Mock 3 Q65

A financial institution holds USD 120 million of commercial loans. The average probability of default is 2%, the average recovery rate is 40%, and the copula correlation is 0.25. Assume a maturity adjustment of 1. What is the IRB capital requirement (USD million) at the 99.9% confidence level?
A. 43.2
B. 20.05
C. 18.61
D. 33.42

Answer: C
Step by step
1. LGD = 1 - 0.40 = 0.60. $N^{-1}(0.02) = -2.0537$, $N^{-1}(0.999) = 3.0902$.
2. WCDR $= N\left[\frac{-2.0537+0.5\times3.0902}{\sqrt{0.75}}\right] = N(-0.5873) = 0.2785$.
3. Capital = $120 \times 0.60 \times (0.2785 - 0.02) = 18.61$.

Why C is correct: Capital is the unexpected loss: EAD x LGD x (WCDR - PD).
Why the others are wrong
- **A:** 120 x 0.6 x 0.6 = 43.2 misuses LGD and ignores the WCDR calculation.
- **B:** 20.05 = 120 x 0.6 x 0.2785 is the 99.9% loss without subtracting expected loss.
- **D:** 33.42 = 120 x 0.2785 omits both LGD and the expected-loss subtraction.

Q594 AnalystPrep Q.2346

Astoria Bank has chosen the IRB approach to calculate its credit risk capital. In line with the Basel II approach, the capital is based on a calculation of:
A. Value at risk with a one-year horizon and 99.9% confidence.
B. Value at risk with a one-year horizon and 99% confidence.
C. Value at risk with a one-month horizon and 99.9% confidence.
D. Value at risk with a 10-day horizon and 99% confidence.

Answer: A
Step by step
1. Basel II IRB credit risk capital uses a one-year time horizon.
2. The confidence level is 99.9%.
3. Capital is this VaR minus expected loss (unexpected loss).

Why A is correct: The IRB approach sets capital to cover credit losses over one year with 99.9% certainty.
Why the others are wrong
- **B:** 99% with one year is not the IRB standard; 99.9% is.
- **C:** The horizon is one year, not one month.
- **D:** 10-day, 99% VaR is the 1996 market risk standard, not credit risk.

Q595

AnalystPrep Q.2965

REBUILT — Replaced the vague option C and the 'all of the above' option D with plausible but incorrect statements.

Fidelity Bank uses the asymptotic single risk factor (ASRF) model underlying the Basel II IRB risk weights. Which statement about the capital charge for an exposure is correct?

A. The capital charge depends on the composition of the portfolio to which the exposure is added.
B. The capital charge depends only on the risk characteristics of the exposure itself.
C. The capital charge rises with the concentration of the portfolio in a single obligor.
D. The capital charge is independent of the exposure's PD.

Answer: B

Step by step

1. The ASRF (Gordy) model assumes a large, granular portfolio, so idiosyncratic risk diversifies away.
2. Only one systematic factor remains, so each exposure's contribution to capital depends only on its own EAD, LGD, PD (and correlation).
3. Hence capital is portfolio-invariant and additive across exposures.

Why B is correct: In the ASRF model capital is portfolio-invariant: it depends only on the exposure's own risk characteristics.

Why the others are wrong

- **A:** This describes a portfolio-dependent model; ASRF is portfolio-invariant.
- **C:** Single-name concentration risk is ignored in ASRF (granularity assumed).
- **D:** Capital rises with PD via WCDR and the (DR99.9 - PD) term, so it depends on PD.

Q596

AnalystPrep Q.4405

REBUILT — Original gave a maturity adjustment of 2.5 (unrealistic; MA is near 1); rebuilt with MA = 1.26 and an explicit DR99.9 = 0.14.

A Canadian bank has assets consisting of CAD 300 million of BB-rated drawn corporate loans. Under the Basel II IRB approach the one-year probability of default is PD = 0.01, LGD = 30%, the 99.9th percentile default rate is DR = 0.14, and the maturity adjustment factor is MA = 1.26. What are the risk-weighted assets (RWA) for these loans?

A. CAD 14.74 million
B. CAD 146.25 million
C. CAD 184.28 million
D. CAD 198.45 million

Answer: C

Key correction: Source key (AnalystPrep) said D (CAD 253.13 million) using MA = 2.5, which is not a realistic maturity adjustment; item rebuilt with MA = 1.26.

Step by step

1. RWA = $12.5 \times \text{EAD} \times \text{LGD} \times (\text{DR} - \text{PD}) \times \text{MA}$.
2. Unexpected-loss rate: $0.14 - 0.01 = 0.13$.
3. $12.5 \times 300 \times 0.30 = 1125$.
4. $1125 \times 0.13 = 146.25$.
5. $146.25 \times 1.26 = 184.275 \approx$ CAD 184.28 million.

Why C is correct: Capital is the 99.9% worst-case loss minus expected loss, EAD × LGD × (DR − PD), scaled by MA; multiplying by 12.5 (= 1/8%) converts capital into RWA.

Why the others are wrong

- **A:** Omits the 12.5 multiplier, giving the capital-type figure 300 x 0.30 x 0.13 x 1.26 = 14.74 instead of RWA.
- **B:** Forgets the maturity adjustment (MA = 1), giving 146.25.
- **D:** Uses DR instead of (DR - PD), i.e. fails to subtract expected loss: 1125 x 0.14 x 1.26 = 198.45.

Q597

BT answers P2.T7.517.3 · BT notes P2.T7.517.3

REBUILT — Merged BT answers/notes duplicates; original distractors (66.0, 85.5, 368.9) did not correspond to realistic mistakes and were replaced with error-based options.

Under the internal ratings-based (IRB) approach, the worst-case default rate is $\text{WCDR} = N\left[\frac{N^{-1}(PD)+\sqrt{\rho}\,N^{-1}(0.999)}{\sqrt{1-\rho}}\right]$, where for corporate, sovereign and bank exposures $\rho = 0.12\,(1 + e^{-50\,PD})$. A bank holds $100.0 million of loans to A-rated corporations. PD = 0.90%, LGD = 75%, and the average maturity is 2.50 years, such that the maturity adjustment is 1.270. Which is nearest to the risk-weighted assets under the IRB approach?

A. $11.84 million
B. $116.55 million
C. $148.0 million
D. $158.74 million

Answer: C
Step by step
1. $\rho = 0.12(1 + e^{-0.45}) = 0.12 \times 1.6376 = 0.1965$.
2. $N^{-1}(0.009) = -2.366$; $N^{-1}(0.999) = 3.090$.
3. Argument $= (-2.366 + 0.4433 \times 3.090)/\sqrt{0.8035} = -1.1108$, so WCDR $= N(-1.1108) = 13.33\%$.
4. RWA $= 12.5 \times 100 \times 0.75 \times (0.13332 - 0.009) \times 1.270$.
5. $= 937.5 \times 0.12432 \times 1.270 = 148.02$ million.

Why C is correct: RWA applies 12.5 to EAD x LGD x (WCDR - PD) x MA; the unexpected-loss rate is 12.43% and MA scales the result to about $148.0 million.

Why the others are wrong
- **A:** Omits the 12.5 multiplier (100 x 0.75 x 0.12432 x 1.27 = 11.84), giving a capital-type figure rather than RWA.
- **B:** Omits the maturity adjustment (MA = 1): 937.5 x 0.12432 = 116.55.
- **D:** Uses WCDR instead of (WCDR - PD), i.e. does not subtract expected loss: 937.5 x 0.13332 x 1.27 = 158.74.

Q598

GARP PE1 Q19 · Mock 3 Q61

A regulatory analyst at a large bank is preparing a report of the bank's credit risk capital and uses the Basel II IRB approach. The analyst identifies a portfolio of credit exposures of equal size held by borrowers with the same probability of default, with the following data:

Exposure at default	EUR 200 million
1-year expected loss on the portfolio	EUR 4.2 million
Expected recovery rate on a defaulted credit	30.00%
1-year portfolio default rate at the 95th percentile	5.66%
1-year portfolio default rate at the 99.9th percentile	9.87%

What is the correct estimate of the Basel II credit risk capital that the bank should reserve for this portfolio?

A. EUR 3.72 million
B. EUR 5.92 million
C. EUR 9.62 million
D. EUR 13.82 million

Answer: C
Step by step
1. IRB uses the 99.9th percentile default rate: DR = 9.87%.
2. LGD $= 1 - 0.30 = 0.70$.
3. Worst-case loss $= 200 \times 0.70 \times 0.0987 = 13.818$ million.
4. Capital = worst-case loss minus EL $= 13.818 - 4.2 = 9.618 \approx$ EUR 9.62 million.

Why C is correct: IRB capital covers unexpected loss only: the 99.9% worst-case loss less expected loss, which is covered by pricing and provisions.

Why the others are wrong
- **A:** Uses the 95th percentile rate: 200 x 0.70 x 0.0566 - 4.2 = 3.72.
- **B:** Uses the recovery rate (30%) as LGD and omits EL: 200 x 0.30 x 0.0987 = 5.92.
- **D:** Correct worst-case loss but does not subtract expected loss (13.82).

Q599

AnalystPrep Q.4406

REBUILT — Original options 0.1562, 0.1453, 0.2341 did not reflect realistic mistakes; replaced with error-based options and stated the correlation formula.

Under Basel II, the asset correlation for bank, corporate and sovereign exposures is $\rho = 0.12\frac{1-e^{-50PD}}{1-e^{-50}} + 0.24\left(1 - \frac{1-e^{-50PD}}{1-e^{-50}}\right)$. For a borrower with PD = 0.01, what is the approximate asset correlation?

A. 0.1673
B. 0.1800
C. 0.1928

D. 0.2400

Answer: C

Step by step

1. $e^{-50\times0.01} = e^{-0.5} = 0.6065$; $e^{-50} \approx 0$.
2. Weight $w = 1 - 0.6065 = 0.3935$.
3. $\rho = 0.12 \times 0.3935 + 0.24 \times 0.6065$.
4. $= 0.0472 + 0.1456 = 0.1928$.

Why C is correct: The weight on 0.12 is 0.3935 and on 0.24 is 0.6065 at PD = 1%, giving 0.1928; correlation falls as PD rises.

Why the others are wrong

- **A:** Swaps the weights (0.24 x 0.3935 + 0.12 x 0.6065 = 0.1673).
- **B:** Takes the simple average of 0.12 and 0.24, ignoring PD.
- **D:** Uses the 0.24 upper bound (the limit as PD approaches 0), ignoring the PD adjustment.

LO 62.g

Compare the basic indicator approach, the standardized approach, and the advanced measurement approach for the calculation of operational risk capital under Basel II.

Q600

Mock AP Mock 1-Q40

REBUILT — Source key divided by three after flooring a negative year at zero; per GARP negative-total years are ignored in the average, so the answer changes, and distractors C and D were replaced with traceable errors.

Jacque's & Josh Bank reports the following gross income (USD millions). The standardized approach multipliers for corporate finance and retail banking are 17.99% and 12.11% respectively.

Business line	Multiplier	Year 1	Year 2	Year 3
Corporate finance	17.99%	11.9	19.3	29.7
Retail banking	12.11%	-26.2	29.4	-31.7

How much operational risk capital should the bank hold under Basel II applying the standardized approach?

A. USD 2.845 million
B. USD 4.268 million
C. USD 3.516 million
D. USD 2.502 million

Answer: B

Key correction: Source key (Mock) said A (2.845), which floors Year 1 at zero but still divides by 3; correct is B because years with negative total capital are ignored in the average (divide by 2).

Step by step

1. Year 1 capital: $0.1799 \times 11.9 + 0.1211 \times (-26.2) = 2.141 - 3.173 = -1.032$ (negative, so the year is ignored).
2. Year 2 capital: $0.1799 \times 19.3 + 0.1211 \times 29.4 = 3.472 + 3.560 = 7.032$.
3. Year 3 capital: $0.1799 \times 29.7 + 0.1211 \times (-31.7) = 5.343 - 3.839 = 1.504$.
4. Negative business-line capital offsets positive within a year, but a year with a negative total is dropped from the average, leaving two years.
5. Capital $= (7.032 + 1.504)/2 = 4.268$ million.

Why B is correct: Under the standardized approach, business-line capital is netted within each year and years with a negative total are excluded from (not just zeroed in) the three-year average.

Why the others are wrong

- **A:** Sets Year 1 to zero but still divides by 3: (0 + 7.032 + 1.504)/3 = 2.845; the negative year should be dropped from the denominator.
- **C:** Uses only Year 2 (7.032/2 = 3.516), ignoring Year 3 and halving a single year.
- **D:** Averages all three years including the negative one with no exclusion: (-1.032 + 7.032 + 1.504)/3 = 2.502.

Q601

Mock AP Mock 1-Q64

REBUILT — Original correct option relied on an Alternative Standardised Approach not covered by the reading; rebuilt as an 'incorrect statement' item on the three Basel II operational risk approaches.

Which of the following statements about the Basel II approaches for calculating operational risk capital is incorrect?

A. Under the basic indicator approach, capital is 15% of average annual gross income over the past three years, ignoring years with negative gross income.

B. Under the standardized approach, gross income is split by business line and a different multiplier is applied to each line.

C. Under the advanced measurement approach, a bank cannot recognize insurance or other risk mitigants in its capital calculation.

D. Under the advanced measurement approach, capital is based on a one-year operational loss measure at the 99.9% confidence level from internal models.

Answer: C

Step by step

1. Check each statement against the Basel II rules.
2. A is correct: BIA uses 15% of three-year average gross income, excluding negative-income years.
3. B is correct: the standardized approach uses line-specific multipliers (e.g., 12%, 15%, 18%).
4. D is correct: AMA uses internal models for a one-year, 99.9% VaR-type loss measure.
5. C is incorrect: AMA allows recognition of risk mitigants such as insurance (subject to conditions), which is one of its advantages.

Why C is correct: The AMA is the only approach that lets banks reflect risk mitigants like insurance, so saying it cannot is false.

Why the others are wrong

- **A:** A true statement of the BIA rule; choosing it reflects misreading the question as 'which is correct'.
- **B:** A true description of the standardized approach; line-specific multipliers are what distinguish it from the BIA.
- **D:** A true description of the AMA's internal-model, 99.9% one-year basis; the figure is easily confused with 99% market-risk VaR but is correct here.

Q602

AnalystPrep Q.4403

A bank operates four business lines with the following multipliers and annual gross income (USD millions):

Business line	Multiplier	Year 1	Year 2	Year 3
Retail banking	13%	6	18	8
Asset management	14%	8	10	18
Trading and sales	19%	9	18	28
Corporate finance	18%	42	25	20

Under Basel II, what is the required operational risk capital under the basic indicator approach?

A. 7.2

B. 4.0

C. 10.2

D. 10.5

Answer: D

Step by step

1. Total gross income: Year 1 = 6+8+9+42 = 65; Year 2 = 18+10+18+25 = 71; Year 3 = 8+18+28+20 = 74.
2. All three years are positive, so the average = $(65 + 71 + 74)/3 = 70$.
3. BIA capital = $0.15 \times 70 = 10.5$ million; business-line multipliers are not used.

Why D is correct: The BIA applies a single 15% factor to total average gross income, regardless of business line.

Why the others are wrong

- **A:** Matches no valid BIA calculation; the BIA uses a flat 15% of total income, not business-line weights.
- **B:** Implausibly low; matches no valid use of the 15% factor and three-year average.
- **C:** Not equal to 0.15 x 70; arises from an arithmetic or averaging slip rather than the BIA formula.

Q603 Mock AP Mock 1-Q37

A bank operates four business lines with the following multipliers and annual gross income (USD millions):

Business line	Multiplier	Year 1	Year 2	Year 3
Retail banking	13%	6	18	8
Asset management	14%	8	10	18
Trading and sales	19%	9	-48	28
Corporate finance	18%	42	25	20

Based on Basel II, what is the required operational risk capital under the standardized approach?

A. 11.83
B. 11.17
C. 12.48
D. 7.59

Answer: A

Key correction: Source key (Mock) said D (7.59), averaging all three years including the negative one; correct is A because a year with negative total capital is ignored in the average.

Step by step

1. Year 1: $0.13(6) + 0.14(8) + 0.19(9) + 0.18(42) = 0.78 + 1.12 + 1.71 + 7.56 = 11.17$.
2. Year 2: $0.13(18) + 0.14(10) + 0.19(-48) + 0.18(25) = 2.34 + 1.40 - 9.12 + 4.50 = -0.88$ (negative, so ignored).
3. Year 3: $0.13(8) + 0.14(18) + 0.19(28) + 0.18(20) = 1.04 + 2.52 + 5.32 + 3.60 = 12.48$.
4. Average over the two positive years $= (11.17 + 12.48)/2 = 11.825 \approx 11.83$.

Why A is correct: Negative business-line capital offsets positive capital within a year, but a year whose total is negative is dropped from the three-year average.

Why the others are wrong

- **B:** Takes Year 1 only.
- **C:** Takes Year 3 only.
- **D:** Averages all three years including the negative -0.88: 22.77/3 = 7.59; the negative year should be excluded, not included.

Q604 BT answers P2.T7.518.3 · BT notes P2.T7.518.3

You are helping your bank choose an approach to calculating operational risk capital under Basel II. A colleague summarizes the three approaches:
I. In the basic indicator approach, total capital is 25% of last year's annual revenue.
II. In the standardized approach, gross income is calculated for different business lines, and the capital percentage applied differs by business line.
III. In the advanced measurement approach, the bank uses internal models to determine a 99.9% one-year VaR.
Which of the statements is (are) correct?

A. None are correct
B. I only
C. II and III only
D. All are correct

Answer: C

Step by step

1. I is wrong: the BIA charge is 15% (not 25%) of the average gross income over the past three years (not last year's revenue).
2. II is correct: the standardized approach applies different multipliers (e.g., 12%, 15%, 18%) to business-line gross income.
3. III is correct: the AMA uses internal models for a one-year, 99.9% loss measure.
4. Only II and III are correct.

Why C is correct: Only the BIA description contains wrong parameters (25%, last year only).

Why the others are wrong

- **A:** Wrongly rejects II and III; the standardized and AMA descriptions are accurate.
- **B:** Selects the only false statement as true.
- **D:** Accepts statement I, missing that BIA uses 15% of a three-year average.

Q605 Mock Mock 1 Q39

DuPont Partners is a Paris-based firm whose only activity is payments and settlement. With regulatory approval, it changes from the basic indicator approach to the standardized approach for operational risk capital. Payments and settlement carries an 18% multiplier under the standardized approach. Which of the following is most likely to be the change in operational risk capital?

A. It will decrease
B. It will increase
C. It will stay the same
D. It will decrease for a year, then increase again

Answer: B

Step by step

1. BIA capital = 15%× average gross income.
2. Standardized capital for a single payments and settlement line = 18%× the same gross income.
3. Since 18% > 15%, capital rises by a factor of 18/15 = 1.2, about 20%.

Why B is correct: With all income in a higher-multiplier business line, the standardized charge exceeds the flat 15% BIA charge.

Why the others are wrong

- **A:** Assumes the standardized approach is always cheaper; that holds only for firms concentrated in lower-multiplier lines (12% or 15%).
- **C:** Ignores that the multiplier differs (18% vs 15%).
- **D:** Invents a transition pattern; Basel II specifies none.

Q606 Schweser Module Quiz 62.2 Q2

Which of the following accords first required banks to hold capital for operational risk?

A. Basel I
B. The 1996 Amendment to Basel I
C. Basel II
D. Solvency II

Answer: C

Step by step

1. Basel I addressed credit risk only.
2. The 1996 Amendment added market risk capital for the trading book.
3. Basel II added operational risk capital (BIA, standardized, AMA).
4. Solvency II applies to insurers, not banks.

Why C is correct: Basel II was the first bank capital accord to introduce an explicit operational risk charge.

Why the others are wrong

- **A:** Basel I covered credit risk only.
- **B:** The 1996 Amendment covered market risk, not operational risk.
- **D:** Solvency II is the insurance capital framework, not a bank accord.

Q607 Mock Mock 1 Q57

Which of the following is correct regarding the operational risk methodologies allowed under Basel II?
I. Standardized approach
II. Internal ratings advanced approach
III. Internal models approach
IV. Basic indicator approach
V. Internal ratings foundation approach
VI. Supervisory formula approach
VII. Advanced measurement approach

A. I and II
B. I, II, and V
C. VI and VII
D. I, IV, and VII

Answer: D

Step by step

1. Basel II operational risk approaches: basic indicator (IV), standardized (I), advanced measurement (VII).
2. II and V are credit risk IRB approaches; III is a market risk approach; VI is a securitization method.

Why D is correct: The three operational risk approaches are the basic indicator, standardized and advanced measurement approaches.

Why the others are wrong

- **A:** II is an IRB credit risk approach.
- **B:** II and V are credit risk IRB approaches.
- **C:** VI is a credit risk securitization method; only VII is an operational risk approach.

Q608

BT answers P2.T7.20.8.2 · BT notes P2.T7.20.8.2

Which of the following statements is TRUE about Basel II?

A. Under the Internal Ratings Based (IRB) approach to credit risk, default probability (PD) increases as the correlation parameter increases

B. Expected losses are included in the credit risk charge but excluded from the operational risk charge

C. Two popular approaches to the operational risk Advanced Measurement Approach (AMA) are Scenario Analysis and the Loss Distribution Approach (LDA; i.e., parametric and Monte Carlo)

D. The operational risk Advanced Measurement Approach (AMA) was one of Basel II's most successful features such that it persisted into Basel III and IV

Answer: C

Step by step

1. AMA banks estimate a distribution of operational losses (frequency and severity) and read the 99.9th percentile.
2. GARP names two popular AMA methods: a parametric and Monte Carlo approach (loss distribution approach) and detailed scenario analysis; many banks combine them. So C is true.
3. Check A: PD is an input to the IRB formula, not something driven by correlation; the correlation parameter is itself a decreasing function of PD. A is false.
4. Check B: IRB credit capital covers unexpected loss only (expected loss is covered by reserves), whereas operational risk capital under the AMA is set at the 99.9th percentile of total loss. B reverses this.
5. Check D: AMA required internal loss data that included large penalties for compliance failures and misbehavior; it lost favor and is no longer permitted, so it did not persist.

Why C is correct: The AMA is built on a simulated or scenario-based loss distribution at the 99.9th percentile, and the loss distribution (parametric/Monte Carlo) and scenario analysis approaches are the two most widely used implementations.

Why the others are wrong

- **A:** PD is an input to the IRB formula, not a function of the correlation parameter; higher correlation raises the worst-case default rate, not PD (confuses WCDR with PD).
- **B:** Reversed: IRB credit capital excludes expected loss (it is covered by loan loss reserves), while AMA operational risk capital includes expected loss as part of the 99.9th percentile loss.
- **D:** False: the AMA lost favor because of large penalty losses in its internal data and is no longer permitted under the post-Basel II framework.

LO 62.h

Summarize elements of the Solvency II capital framework for insurance companies.

Q609

Mock 2 Q17

Pillar 1 of Solvency II specifies a minimum capital requirement (MCR) and a solvency capital requirement (SCR). Under the internal models approach to calculating the SCR, the internal model must satisfy several tests, EXCEPT:

A. Use test.

B. Calibration test.

C. Validity test.

D. Statistical quality test.

Answer: C

Step by step

1. Recall the three tests for Solvency II internal models: statistical quality, calibration, and use.
2. Compare each option with this list.
3. Use, calibration and statistical quality all appear; 'validity test' does not.

Why C is correct: Solvency II internal models must pass exactly three tests (statistical quality, calibration, use); there is no 'validity test'.

Why the others are wrong

- **A:** A real requirement: the model must be genuinely used in business decision making.
- **B:** A real requirement: risks must be measured in line with regulator-set SCR target criteria.
- **D:** A real requirement: data and methodology must be sound.

Q610

AnalystPrep Q.4401

REBUILT — Original options ('size of data should be small', 'all/none of the above') were giveaways; rewrote options with realistic misstatements of the internal-model conditions.

Solvency II allows insurers to compute the solvency capital requirement (SCR) with either the standardized approach or an internal model. Which of the following is a condition that an internal model must satisfy?

A. Risks must be calibrated to a target standard chosen by the insurer rather than the regulator.

B. The model must be used in actual business decision making, not just for regulatory reporting.

C. The model must be based on a 99.9% confidence level over a one-year horizon.

D. The model must be validated by external auditors but need not be used by management.

Answer: B

Step by step

1. Internal models must pass three tests: sound data and methodology, calibration to regulator target criteria, and actual use in business decisions.
2. Option B is the 'use test'.
3. The other options misstate calibration, the confidence level, or the use test.

Why B is correct: The use test requires the model to be embedded in real management decisions, not merely a compliance exercise.

Why the others are wrong

- **A:** Calibration is to target criteria set by the regulator (a common SCR standard), not the insurer's own choice.
- **C:** Solvency II uses a one-year VaR at 99.5%; 99.9% is the Basel II credit/operational level.
- **D:** The use test explicitly requires management use of the model.

Q611

BT answers P2.T7.20.8.3 · BT notes P2.T7.20.8.3

Solvency II is a risk-based regulatory capital requirement that applies to insurance companies in the European Union. GARP explains, "In Europe, regulation of insurance companies is done by the European Union's (EU) European Insurance and Occupational Pensions Authority. The first capital regulations at the EU level were known colloquially as Solvency I, which has recently been replaced by Solvency II. More than 10 years in the making, Solvency II resembles Basel II in that many elements of its capital requirements are based on a one-year VaR concept (at the 99.5th percentile) ..." In regard to a comparison between Solvency II and Basel II, which of the following statements is TRUE?

A. Similar to Basel II, Solvency II is an international regulatory standard.

B. Similar to Basel II, Solvency II has three pillars and three tiers of capital.

C. Similar to Basel II, Solvency II considers three risks: credit, market and operational risk.

D. Similar to Basel II, Solvency II ignores balance sheet liabilities but requires a minimum percentage of risk-weighted assets.

Answer: B

Step by step

1. Solvency II pillars: quantitative requirements, internal governance and supervision, disclosure. Same three as Basel II.
2. Capital is Tier 1 (equity, retained earnings), Tier 2 (subordinated liabilities available in liquidation), Tier 3 (subordinated, other criteria not met): three tiers.

3. Check A, C, D: A is EU-only, C omits underwriting risk, D ignores that liabilities are the main insurer risk.

Why B is correct: Both frameworks have three pillars and three tiers of capital.

Why the others are wrong

- **A:** Solvency II is an EU regime; no international insurance standard exists yet.
- **C:** Solvency II charges underwriting, investment (market and credit) and operational risk; this option omits underwriting, the key insurer risk.
- **D:** Liabilities are a major risk source for insurers and are central to Solvency II; it is not a risk-weighted-assets ratio.

Q612

BT answers P2.T7.519.2 · BT notes P2.T7.519.2

Solvency II has three pillars analogous to Basel II's three pillars. Pillar 1 concerns the calculation of capital requirements and eligible capital, Pillar 2 the supervisory review process, and Pillar 3 disclosure of risk management information to the market. Pillar 1 specifies both a minimum capital requirement (MCR) and a solvency capital requirement (SCR). Each of the following is true about the SCR or MCR EXCEPT which is false?

A. If capital drops below the SCR level, the supervisor might require the insurance company to take particular measures to correct the situation.

B. If capital drops below the SCR level, an insurance company should, at minimum, deliver to the supervisor a plan to restore capital to above the SCR level.

C. If capital drops below the MCR level, supervisors are likely to insist that the insurance company add new businesses and underwrite additional policies to increase earned premiums.

D. If capital drops below the MCR level, supervisors might force the insurance company into liquidation, transferring its policies to another company.

Answer: C

Step by step

1. SCR breach: firm presents a capital restoration plan and the supervisor may impose additional requirements (A and B true).
2. MCR breach (absolute minimum): supervisors may prevent new policies being written or put the firm into resolution, e.g. sale or liquidation (D true).
3. C says supervisors push for more new business, the opposite of the real response, so C is false.

Why C is correct: A breach of the MCR leads supervisors to restrict new business or resolve the firm, not to encourage more underwriting.

Why the others are wrong

- **A:** True statement, so not the exception; SCR breach allows supervisory measures.
- **B:** True statement; a restoration plan is the minimum response to an SCR breach.
- **D:** True statement; resolution (sale or liquidation) is a possible MCR response.

Q613

BT answers P2.T7.519.3 · BT notes P2.T7.519.3

REBUILT — Source option D listed stress percentages garbled by OCR and option A added a claim not in GARP; replaced them with GARP-consistent statements while keeping the false Tier 1 statement as the exception. Option D changed to a GARP-supported Tier statement (QIS for Solvency II is not in GARP).

Pillar 1 of Solvency II specifies a minimum capital requirement (MCR) and a solvency capital requirement (SCR). The SCR involves capital charges for investment risk (market and credit), underwriting risk (life, non-life, health) and operational risk. The SCR can be calculated with the standardized approach or the internal models approach. Regarding capital and the internal models approach, each of the following is true EXCEPT which is false?

A. The internal models approach uses a VaR with a one-year horizon and a 99.5% confidence level, lower than the 99.9% level used for credit and operational risk under Basel II.

B. Internal models must satisfy three tests: statistical quality (sound data and methodology), calibration (risks assessed to the regulator's SCR target criteria) and use (the model is used in actual business decisions).

C. All qualifying capital must be Tier 1, which can consist only of liabilities that are subordinated to policyholders and satisfy availability criteria in wind-down scenarios.

D. Capital requirements may be met by a combination of Tier 1, Tier 2 and Tier 3 capital, similar to Basel II.

Answer: C

Step by step

1. Check A: GARP gives a 99.5% one-year VaR; Basel II uses 99.9% for credit and operational risk. True.
2. Check B: the three tests are statistical quality, calibration and use. True.

3. Check D: GARP says requirements may be satisfied by a combination of Tier 1, Tier 2 and Tier 3 capital, as under Basel II. True.
4. C: Tier 1 is equity, retained earnings and equivalents; liabilities subordinated to policyholders are Tier 2/Tier 3. Capital can be Tiers 1, 2 and 3, so C is false.

Why C is correct: Solvency II capital has three tiers, and Tier 1 is equity-type capital, not subordinated liabilities.

Why the others are wrong

- **A:** True; this is the correct confidence level and horizon.
- **B:** True; these are the three internal-model tests.
- **D:** True; capital can be a combination of Tier 1, Tier 2 and Tier 3.

Q614

Mock 1 Q73

REBUILT — Source keyed investment risk as having no charge, which contradicts the reading (investment risk is charged); changed options so exactly one excluded risk remains.

An analyst at an insurance company is reviewing the capital requirements under the Solvency II directive. The analyst identifies the risk categories for which the firm must hold a specific capital charge in the solvency capital requirement (SCR). Which of the following risks does NOT carry a specific SCR capital charge?

A. Underwriting risk.
B. Investment (market and credit) risk.
C. Operational risk.
D. Reputational risk.

Answer: D

Key correction: Source key (Mock 1) said B (investment risk); this contradicts GARP/Schweser, where investment risk (market and credit) carries an SCR charge. Correct answer in the rebuilt item is D.

Step by step

1. The SCR has charges for three categories: underwriting risk (life, non-life, health), investment risk (market and credit) and operational risk.
2. A, B and C are all on this list.
3. Reputational risk is not a separate charge, so D is the answer.

Why D is correct: Solvency II capital charges cover underwriting, investment and operational risk only.

Why the others are wrong

- **A:** Has its own charge (life, non-life, health).
- **B:** Has its own charge, subdivided into market and credit risk; the source key wrongly treated it as excluded.
- **C:** Has its own charge as in Basel II.

Q615

Mock 2 Q75

REBUILT — Original options B, C, D were vague or irrelevant (Level 1 diversification, consolidated Tier 1 ratio); replaced with plausible misstatements while keeping the same tested fact.

Basel II was revised into Basel III. Somewhat similar to the Basel banking regulations is Solvency II, a European Union framework for insurance companies. When compared to Basel II/III, which of the following statements correctly reflects Solvency II?

A. It specifies both a solvency capital requirement (SCR) and a minimum capital requirement (MCR).
B. It sets capital charges for credit, market and operational risk only, with no charge for underwriting risk.
C. It uses a one-year VaR at the 99.9% confidence level for all risk types.
D. It requires a single minimum ratio of capital to risk-weighted assets, with no internal models allowed.

Answer: A

Step by step

1. Solvency II Pillar 1 has two thresholds: the MCR (absolute floor) and the SCR (buffer above MCR).
2. B is wrong: underwriting risk is charged. C is wrong: the VaR is 99.5% one-year. D is wrong: an internal models approach is allowed.

Why A is correct: Solvency II has two capital thresholds, the SCR and the MCR, with the SCR set above the MCR.

Why the others are wrong

- **B:** Omits underwriting risk, the insurer-specific charge (alongside investment and operational risk).
- **C:** Confuses with Basel II's 99.9% credit/operational level; Solvency II uses 99.5%.

- **D:** Solvency II permits internal models for the SCR and is not a risk-weighted-assets ratio.

Q616 Schweser Module Quiz 62.2 Q3

Which of the following statements is correct regarding capital requirements for insurance companies?

A. Basel II includes the regulation of banks and insurance companies in the three pillars.

B. The minimum capital requirement is likely to be higher than the solvency capital requirement for insurance companies.

C. The repercussion for violating the solvency capital requirement is likely liquidation and the transfer of company insurance policies to another firm.

D. The internal models approach to calculating the solvency capital requirement is similar to internal ratings-based approach under Basel II in that the firm must calculate a VaR with a one-year time horizon.

Answer: D

Step by step

1. Insurer capital is governed by Solvency II (EU), which resembles Basel II: three pillars and a one-year VaR concept (99.5th percentile for Solvency II; the Basel II IRB approach also uses a one-year horizon, at 99.9%).
2. The SCR is the required buffer above the MCR, so MCR < SCR.
3. Breaching the MCR lets supervisors bar new policies or put the firm into resolution (sale or liquidation); breaching the SCR only calls for a capital restoration plan.
4. Only statement D is consistent with these facts.

Why D is correct: Solvency II internal models, like Basel II IRB, rest on a one-year VaR; the confidence level differs (99.5% vs 99.9%) but the horizon is the same.

Why the others are wrong

- **A:** Basel II regulates banks only; insurers fall under Solvency II. Confuses the two three-pillar frameworks.
- **B:** The MCR is a floor below the SCR (buffer = SCR less MCR), so it is lower, not higher.
- **C:** Liquidation or transfer of policies is the consequence of breaching the MCR; breaching the SCR leads to a capital restoration plan. The two thresholds are swapped.

Reading 63. Solvency, Liquidity, and Other Regulation After the Global Financial Crisis

Mark Carey, "Solvency, Liquidity, and Other Regulation After the Global Financial Crisis," GARP Risk Institute (April 2019).

LO	Learning objective	Qs
63.a	describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge.	17
63.b	explain the process of calculating the incremental risk capital charge for positions held in a bank's trading book.	6
63.c	describe the comprehensive risk (CR) capital charge for portfolios of positions that are sensitive to correlations between default risks.	3
63.d	define in the context of Basel III and calculate where appropriate: Tier 1 capital and its components Tier 2 capital and its components required Tier 1 equity capital, total Tier 1 capital, and total capital	18
63.e	describe the motivations for and calculate the capital conservation buffer and the countercyclical buffer, including special rules for globally systemically important banks (G-SIBs).	8
63.f	describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio.	20
63.g	describe the mechanics of contingent convertible bonds (CoCos) and explain the motivations for banks to issue them.	12
63.h	provide examples of legislative and regulatory reforms that were introduced after the 2007–2009 financial crisis.	3

LO 63.a

Describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge.

Q617

AnalystPrep Q.2348

Cosomora Bank, a large European bank with a big trading book, has operated under Basel II and is now implementing Basel II.5. What will be the main effect of shifting from Basel II to Basel II.5?

A. Capital charges for credit risk will increase.
B. Capital charges for credit risk will be reduced.
C. Capital charges for market risk will increase.
D. Sweeping changes in liquidity measurement techniques.

Answer: C

Step by step

1. Basel II.5 (implemented by end-2011) revised the trading-book rules: stressed VaR, incremental risk charge, comprehensive risk capital.
2. All three add capital against trading-book (market) risk.
3. Hence the main effect is higher market risk capital charges.

Why C is correct: Basel II.5 amended the market risk framework, and its additions (stressed VaR, IRC, comprehensive risk charge) are added to existing charges, so market risk capital rises.

Why the others are wrong

- **A:** Credit risk (banking book) rules were not the focus of II.5; confuses with Basel II/III credit changes.
- **B:** Credit risk charges were not reduced by II.5; no basis for this.
- **D:** Liquidity measures (LCR, NSFR) came with Basel III, not II.5.

Q618

AnalystPrep Q.2741

REBUILT — Typo in option B (incremental risk change) corrected to charge.

All of the following were changes implemented through Basel 2.5, EXCEPT:

A. Calculation of a stressed VaR.
B. Implementation of a new incremental risk charge (IRC).
C. A comprehensive risk measure (CRM) for instruments sensitive to correlations between default risks.
D. Calculation of the net stable funding ratio (NSFR) and the liquidity coverage ratio (LCR).

Answer: D

Step by step

1. Basel 2.5 changes: stressed VaR, IRC, comprehensive risk capital.

2. NSFR and LCR are Basel III liquidity ratios.
3. So D is the exception.

Why D is correct: Liquidity ratios were introduced by Basel III, not Basel 2.5.

Why the others are wrong

- **A:** Is a Basel 2.5 change.
- **B:** Is a Basel 2.5 change.
- **C:** Is a Basel 2.5 change.

Q619

BT notes P2.T7.20.9.1

A bank uses the VaR and stressed VaR (SVaR) framework of Basel 2.5. All VaR figures are 10-day and in USD millions; both multipliers m_c and m_s are 3.0. The bank's internal models give:

Confidence level	Previous day VaR	Previous day SVaR	60-day average VaR	60-day average SVaR
95.0%	110.0	190.0	127.0	238.0
99.0%	185.0	360.0	140.0	270.0
99.9%	283.0	740.0	228.0	490.0

What is the market risk capital requirement under Basel 2.5?

A. $630.0 million
B. $810.0 million
C. $1,230.0 million
D. $2,154.0 million

Answer: C

Step by step

1. Basel 2.5 uses 10-day, 99% VaR: $MR = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$.
2. VaR term: $\max(185, 3.0 \times 140 = 420) = 420$.
3. SVaR term: $\max(360, 3.0 \times 270 = 810) = 810$.
4. Total = 420 + 810 = $1,230.0 million.

Why C is correct: The 99% row is the regulatory level; in each term the multiplied 60-day average exceeds the previous-day figure, so it binds.

Why the others are wrong

- **A:** 360 + 270 = 630: adds SVaR figures without the multiplier and omits the VaR term.
- **B:** 810 is only the SVaR component; forgot to add the VaR component.
- **D:** Used the 99.9% row (684 + 1,470 = 2,154) instead of 99%.

Q620

AnalystPrep Q.4285

After the global financial crisis it was recognized that minimum capital charges under the market risk amendment were insufficient for trading book risks. Which of the following was a significant change implemented by 2011 to address this, later known as Basel 2.5?

A. VaR computation was expanded to include a stressed VaR component
B. A portion of operational risk capital was required on top of credit and market risk
C. Credit risk weights were to be based on modern credit risk models and banks' internal measures
D. Tier 1 capital was designated for going-concern loss absorption, while Tier 2 capital was to be used for recapitalizing an institution in resolution

Answer: A

Step by step

1. Basel 2.5 made three trading-book changes: stressed VaR, incremental risk charge, comprehensive risk capital.
2. Only option A is one of these.
3. B, C and D relate to Basel II (operational risk, IRB) and Basel III (capital tiers).

Why A is correct: Adding stressed VaR to the traditional VaR is the first of the three Basel 2.5 changes.

Why the others are wrong

- **B:** Operational risk capital was introduced under Basel II (2004), not Basel 2.5.
- **C:** Internal-ratings-based credit risk weights are a Basel II banking-book feature.
- **D:** Describes Basel III's capital tier philosophy, not trading book changes.

Q621

AnalystPrep Q.4287

REBUILT — Original distractors (1,850; 1,160; 2,460) did not correspond to identifiable errors; replaced with options from realistic mistakes.

The previous-day 10-day 99% VaR for ABC Bank is $800. The average 10-day 99% VaR over the last 60 days is $360. Over the past seven years, the most stressful previous-day-equivalent 10-day 99% VaR is $950 and the most stressful 60-day average 99% VaR is $370. The multiplier on the recent 60-day average VaR is 3.0, and the multiplier on the stressed 60-day average VaR is 3.5. What is the market risk capital charge under Basel 2.5?

A. $1,750
B. $2,030
C. $2,375
D. $2,095

Answer: C

Step by step

1. $MR = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$.
2. VaR term: $\max(800, 3.0 \times 360 = 1{,}080) = 1{,}080$.
3. SVaR term: $\max(950, 3.5 \times 370 = 1{,}295) = 1{,}295$.
4. Total = 1,080 + 1,295 = $2,375.

Why C is correct: Both multiplied 60-day averages exceed the previous-day values, so they set each component.

Why the others are wrong

- **A:** 800 + 950 = 1,750: used previous-day VaR and SVaR only, ignoring the multiplied averages.
- **B:** 1,080 + 950 = 2,030: applied the multiplier to VaR but took the stressed previous-day value.
- **D:** 800 + 1,295 = 2,095: applied the multiplier only to the stressed component.

Q622

AP Mock 2 Q60

REBUILT — Original distractors (3,000; 3,452; 3,563) were not tied to realistic errors; replaced.

An American bank reports the following (all 10-day, 99% confidence):

Measure	Value
Previous day VaR	USD 1,000
Average VaR over the recent 60 days	USD 480
Most stressful previous-day-equivalent VaR over the past seven years	USD 1,100
Most stressful 60-day average VaR over the past seven years	USD 500
Multiplier on the recent 60-day average VaR	3.9
Multiplier on the stressed 60-day average VaR	3.4

What is the market risk capital charge under Basel 2.5?

A. USD 3,572
B. USD 2,972
C. USD 2,700
D. USD 3,822

Answer: A

Step by step

1. $MR = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$.
2. VaR term: $\max(1{,}000, 3.9 \times 480 = 1{,}872) = 1{,}872$.
3. SVaR term: $\max(1{,}100, 3.4 \times 500 = 1{,}700) = 1{,}700$.
4. Total = 1,872 + 1,700 = USD 3,572.

Why A is correct: Each multiplied average exceeds the corresponding previous-day figure, so those bind.

Why the others are wrong

- **B:** 1,872 + 1,100 = 2,972: did not multiply the stressed average.
- **C:** 1,000 + 1,700 = 2,700: did not multiply the regular average.
- **D:** 3.9 x (480 + 500) = 3,822: applied the VaR multiplier to the stressed average too, instead of 3.4.

Q623

AP Mock 3 Q20

REBUILT — Original distractors (9.2m, 5.0m, 4.987m) were unrealistic; replaced with error-based options.

A financial institution has a trading portfolio with the following (all 10-day, 99% confidence):

Item	Value
Previous day VaR	USD 1,200,000
Average VaR over the past 60 days	USD 1,400,000
Previous day stressed VaR	USD 2,200,000
Average stressed VaR over the past 60 days	USD 2,000,000
Multiplier on VaR	3
Multiplier on stressed VaR	3

Calculate the market risk capital charge under Basel 2.5.

A. USD 10,200,000
B. USD 3,600,000
C. USD 6,400,000
D. USD 3,400,000

Answer: A

Step by step

1. $MR = \max(VaR_{t-1}, 3\,VaR_{avg}) + \max(SVaR_{t-1}, 3\,SVaR_{avg})$.
2. VaR term: $\max(1.2, 3 \times 1.4 = 4.2) = 4.2$ million.
3. SVaR term: $\max(2.2, 3 \times 2.0 = 6.0) = 6.0$ million.
4. Total = 4.2 + 6.0 = USD 10.2 million.

Why A is correct: The multiplied averages dominate both terms.

Why the others are wrong

- **B:** 1.4 + 2.2 = 3.6m: ignored the multipliers (compared unmultiplied averages with previous-day values).
- **C:** 4.2 + 2.2 = 6.4m: multiplied only the VaR average, not the stressed average.
- **D:** 1.2 + 2.2 = 3.4m: used previous-day VaR and SVaR only.

Q624 AnalystPrep Q.2347

BastaBank has just adopted Basel II.5 after years of Basel II compliance. The risk team prepares a report on the main changes under Basel II.5. These most likely relate to:

A. Capital for liquidity risk, stressed VaR, and a new methodology of capital calculation.
B. Stressed VaR, a new incremental risk charge, and a comprehensive risk measure for instruments dependent on credit correlation.
C. A new incremental risk charge, a comprehensive risk measure for correlation-dependent instruments, and a new methodology of capital calculation.
D. A new incremental risk charge, new requirements for IRB parameters, and new requirements for liquidity measurement.

Answer: B

Step by step

1. Basel II.5 made three changes: stressed VaR, incremental risk charge, comprehensive risk capital for securitizations/correlation-sensitive positions.
2. Only B lists exactly these.
3. Each other option includes a non-II.5 item.

Why B is correct: The three Basel II.5 changes are stressed VaR, IRC and the comprehensive risk measure.

Why the others are wrong

- **A:** Liquidity risk capital and a new capital methodology are not II.5 items (liquidity ratios are Basel III).
- **C:** Omits stressed VaR and adds a non-existent new capital methodology.
- **D:** IRB parameter changes and liquidity measurement are not part of II.5.

Q625 AnalystPrep Q.2349

Under Basel II.5 banks must calculate stressed VaR. Stressed VaR was introduced mainly because of:

A. very high capital requirements caused by high volatility of market variables.
B. VaR that was too low as a result of low volatility of market variables.
C. increased capital charges for credit risk.
D. None of the above.

Answer: B

Step by step

1. Historical-simulation VaR drew on recent data; in calm years (2003-2006) measured VaR drifted down.
2. When volatility rose in 2007, VaR was slow to respond because most observations came from the low-volatility period.
3. Stressed VaR uses the most stressful 250-day period of the past seven years, so it is large and stable.

Why B is correct: Stressed VaR counters the tendency of VaR to be too low after prolonged low volatility.

Why the others are wrong

- **A:** Reverses the problem; the concern was insufficient, not excessive, capital.
- **C:** Credit risk charges are unrelated to the VaR motivation.
- **D:** B is correct.

Q626

AnalystPrep Q.2362

Over what length of period of stressed market conditions must a bank draw observations to calculate stressed VaR under Basel II.5?

A. 90 days
B. 150 days
C. 250 days
D. 360 days

Answer: C

Step by step

1. Banks must identify the one-year period from the most recent seven years that was most stressful for the current portfolio.
2. One year = 250 trading days.

Why C is correct: Stressed VaR is based on the most stressful 250-day (one-year) window within the past seven years.

Why the others are wrong

- **A:** 90 days is not a Basel window length (could be confused with the 60-day averaging).
- **B:** 150 days has no basis.
- **D:** 360 is a calendar-year approximation; the rule uses 250 trading days.

Q627

AnalystPrep Q.3242

REBUILT — Stem asked for 'the correct stressed VAR amount' ambiguously; reworded to ask for the stressed VaR component of the Basel 2.5 capital charge.

Under Basel 2.5, a bank's previous-day stressed VaR is \$20 million and its average stressed VaR over the previous 60 trading days is \$7 million. The multiplication factor m_s is 4. What is the stressed VaR component of the market risk capital charge?

A. \$28 million
B. \$20 million
C. \$48 million
D. \$8 million

Answer: A

Step by step

1. Stressed component = $\max(\text{SVaR}_{t-1},\ m_s \times \text{SVaR}_{avg})$.
2. $m_s \times \text{SVaR}_{avg} = 4 \times 7 = 28$.
3. $\max(20, 28) = 28$ million.

Why A is correct: The stressed component is the larger of yesterday's stressed VaR and the multiplier times the 60-day average stressed VaR; here the scaled average (28) exceeds 20.

Why the others are wrong

- **B:** Takes the previous-day stressed VaR only, ignoring that the scaled 60-day average is larger.
- **C:** Adds the previous-day figure and the scaled average (20 + 28) instead of taking the maximum.
- **D:** Subtracts the previous-day figure from the scaled average (28 - 20); no such operation exists in the formula.

Q628 AnalystPrep Q.4286

According to Basel 2.5, how is the stress period used to calculate stressed VaR identified?

A. Banks must use the most recent one-year period of historical data to calculate stressed VaR.
B. Banks must identify the most stressful one-year (250-day) period from the most recent seven years.
C. Banks can use any one-year period from their historical data that shows the highest volatility.
D. Banks must use a fixed one-year period from the 2007-2008 financial crisis for all portfolios.

Answer: B

Step by step

1. Stressed VaR uses a 250-day window of stressed market conditions.
2. The window is chosen as the most stressful period within the most recent seven years for the bank's own portfolio.
3. Hence it is portfolio-specific and can differ across banks.

Why B is correct: Basel 2.5 requires each bank to find the 250-day period in the last seven years that is most stressful for its actual portfolio.

Why the others are wrong

- **A:** The most recent year is typically calm; the stressed period is by definition the worst, not the latest.
- **C:** Not restricted to the last seven years and tied to market volatility rather than the bank's portfolio, so it is not the rule.
- **D:** Regulators initially thought 2008 would work but a fixed common period was dropped in favour of a bank-specific one.

Q629 BT Mock Q23

A bank uses the VaR and stressed VaR framework under Basel 2.5. Its internal models produce the following measures (USD millions, 10-day 99%):

Measure	Value
Previous day's VaR (VaRt-1)	400
Average VaR over the past 60 days	120
Previous day's stressed VaR (SVaRt-1)	260
Average stressed VaR over the past 60 days	220

The supervisor has set both multiplication factors, m and ms, to 3.0. What is the market risk capital requirement?

A. USD 660.0 million.
B. USD 1,020.0 million.
C. USD 1,060.0 million.
D. USD 1,980.0 million.

Answer: C

Step by step

1. VaR leg = $\max(400, 3 \times 120 = 360) = 400$.
2. Stressed leg = $\max(260, 3 \times 220 = 660) = 660$.
3. Total = 400 + 660 = 1,060 million.

Why C is correct: Each leg takes the larger of yesterday's value and the multiplier times the 60-day average, and the two legs are added.

Why the others are wrong

- **A:** Counts only the stressed leg (660) and omits the VaR leg.
- **B:** Uses 3 x 120 = 360 for the VaR leg although 400 is larger (360 + 660).
- **D:** Multiplies the sum of the previous-day figures by 3: 3 x (400 + 260) = 1,980.

Q630 BT answers P2.T7.520.1 · BT notes P2.T7.520.1

A risk manager must compute the Basel 2.5 market risk capital requirement for a bank's trading book. The 95% one-day VaR on the previous day is \$20,000; the average 95% one-day VaR over the last 60 trading days is \$12,000; the 95% one-day stressed VaR on the previous day is \$188,000; the average 95% one-day stressed VaR over the last 60 days is \$50,000. Both multiplication factors are 3.0, and returns are assumed normal, so a one-day 95% VaR converts to a 10-day 99% VaR by multiplying by $2.326/1.645 \times \sqrt{10}$. Which is nearest to the market risk capital requirement?

A. $161,000
B. $188,000
C. $840,800
D. $1.0 million

Answer: D
Step by step
1. Scaling factor $= 2.326/1.645 \times \sqrt{10} = 4.4714$.
2. VaR leg: $\max(20{,}000,\ 3 \times 12{,}000 = 36{,}000) = 36{,}000$; scaled $= 36{,}000 \times 4.4714 \approx 160{,}971$.
3. Stressed leg: $\max(188{,}000,\ 3 \times 50{,}000 = 150{,}000) = 188{,}000$; scaled $\approx 840{,}624$.
4. Total = 160,971 + 840,624 = 1,001,594, about $1.0 million.

Why D is correct: Both legs use the larger of the previous-day and multiplied-average figures, are converted to 10-day 99% VaR, and are summed.

Why the others are wrong
- **A:** Reports only the scaled VaR leg (about $161,000).
- **B:** Uses the unscaled stressed VaR of $188,000 only, ignoring the VaR leg and the 10-day 99% conversion.
- **C:** Reports only the scaled stressed leg (about $840,800) and omits the VaR leg.

Q631

Mock 2 Q17

Bank XYZ has calculated a market risk VaR for the previous day equal to USD 25 million. The average VaR over the last 60 days is USD 6 million. The stressed VaR for the previous day is USD 28 million and the average stressed VaR over the last 60 days is USD 31 million. What is the total market risk capital charge, assuming both multiplicative factors are 3?

A. USD 43 million.
B. USD 75 million.
C. USD 109 million.
D. USD 118 million.

Answer: D
Step by step
1. VaR leg $= \max(25, 3 \times 6 = 18) = 25$.
2. Stressed leg $= \max(28, 3 \times 31 = 93) = 93$.
3. Total = 25 + 93 = 118 million.

Why D is correct: Take the larger of yesterday's and scaled-average value in each leg, then add the legs.

Why the others are wrong
- **A:** Adds 25 + 18, i.e. uses only VaR figures and drops the stressed leg.
- **B:** Triples the previous-day VaR (3 x 25) and ignores the stressed VaR.
- **C:** Uses 3 x 28 = 84 (previous-day stressed VaR scaled) instead of the scaled 60-day average 93: 25 + 84.

Q632

Schweser Module Quiz 63.1 Q1

Which of the following statements about the stressed value at risk (VaR) required under Basel 2.5 is correct?

A. Basel 2.5 has established the year 2008 as the "stress" period. All banks use data from 2008 to calculate the stressed VaR.
B. The stressed VaR replaces the "normal" VaR for the purpose of calculating capital for credit risks.
C. Market risk capital under Basel 2.5 should be close to double that of market risk capital under Basel II due to the addition of the stressed VaR.
D. The stressed VaR must be calculated using a 99.9% confidence interval.

Answer: C
Step by step
1. Basel 2.5 market risk capital = usual VaR leg + stressed VaR leg, with equal multipliers.
2. Stressed VaR is at least as large as ordinary VaR since the stress period is the worst in the last seven years.
3. So the charge is at least roughly double the pre-Basel 2.5 charge.

Why C is correct: Because the stressed VaR is added to (not substituted for) the normal VaR and is at least as large, the market risk charge is about twice as large or more.

Why the others are wrong
- **A:** Each bank chooses its own most stressful 250-day period in the last seven years; 2008 was only an early idea.
- **B:** Stressed VaR is added to normal VaR and applies to market risk, not credit risk.
- **D:** Both VaRs use 99% confidence over 10 days; 99.9% is the IRB/incremental risk percentile.

Q633

Mock 3 Q11

REBUILT — Flattened table restored and the required 99% confidence level stated via the question; original distractors (1,248 and 4,799) did not correspond to realistic errors and 1,533 mixed rows, so options were replaced with realistic mistakes.

A bank uses the VaR and stressed VaR framework of Basel II.5 for its trading book. Its internal models produced the following 10-day risk measures (USD million):

Confidence level	Latest 10-day VaR	Latest 10-day stressed VaR	Average 10-day VaR, previous 60 days	Average 10-day stressed Va
95.0%	238	484	252	546
99.0%	451	995	413	1,106
99.9%	578	1,281	528	1,372

The supervisor has set the multiplication factors for both VaR and stressed VaR at 3. Using the confidence level required by Basel II.5, what is the market risk capital requirement?

A. USD 1,446 million.
B. USD 2,234 million.
C. USD 4,557 million.
D. USD 3,769 million.

Answer: C

Step by step
1. Basel II.5 uses 99% confidence, 10-day VaR: $MR = \max(\text{VaR}_{t-1}, m_c\ \text{VaR}_{avg}) + \max(\text{SVaR}_{t-1}, m_s\ \text{SVaR}_{avg})$.
2. Use the 99.0% row only: latest VaR 451, average VaR 413, latest stressed VaR 995, average stressed VaR 1,106.
3. VaR leg: $\max(451,\ 3 \times 413) = \max(451,\ 1{,}239) = 1{,}239$.
4. Stressed VaR leg: $\max(995,\ 3 \times 1{,}106) = \max(995,\ 3{,}318) = 3{,}318$.
5. Capital = $1{,}239 + 3{,}318 = 4{,}557$ USD million.

Why C is correct: Each leg of the Basel II.5 charge is the larger of the latest measure and the multiplier times the 60-day average; here the multiplied averages dominate, and the two legs are added.

Why the others are wrong
- **A:** Takes only the latest values with no multiplier or averaging: $451 + 995 = 1{,}446$.
- **B:** Applies the multiplier to the VaR average but not to the stressed VaR average: $3 \times 413 + 995 = 2{,}234$.
- **D:** Applies the multiplier to the stressed VaR average only and uses the latest VaR: $451 + 3 \times 1{,}106 = 3{,}769$.

LO 63.b

Explain the process of calculating the incremental risk capital charge for positions held in a bank's trading book.

Q634

AnalystPrep Q.4035

Which of the following risks is specifically recognized by the incremental risk charge (IRC)?

A. Jump-to-default risk
B. Interest rate risk
C. Foreign exchange risk
D. Equity price risk

Answer: A

Key correction: Source explanation cited a 99% confidence level and Basel III counterparty credit risk; the IRC is a Basel 2.5 charge measured at 99.9% over one year.

Step by step
1. The IRC was added in Basel 2.5 to capture credit-related losses in the trading book that the 10-day VaR and specific risk charge missed.
2. It is a one-year, 99.9% VaR on credit-sensitive positions, covering default and rating-migration losses.
3. Of the listed risks, only jump-to-default is a credit event captured by the IRC.

Why A is correct: The IRC roughly captures jump-to-default risk (together with rating-migration risk) on credit-sensitive trading book positions.

Why the others are wrong

- **B:** Interest rate risk is captured by the regular and stressed VaR, not the IRC.
- **C:** FX risk is a market risk captured by VaR/stressed VaR; the IRC is credit-focused.
- **D:** Equity price risk is a market risk captured by VaR/stressed VaR, not the IRC.

Q635

BT answers P2.T7.20.9.2 · BT notes P2.T7.20.9.2

In regard to the incremental risk charge (IRC) in Basel 2.5, each of the following is true EXCEPT, which is false?

A. The IRC is calculated as the one-year 99.9% value at risk (VaR) for losses on credit instruments in the trading book

B. The IRC is designed to account for correlation risks in the correlation book, which includes securitizations and re-securitizations

C. The IRC (itself an amended IDRC) was motivated by regulatory arbitrage; consequently, required capital in the trading book is the greater of market risk capital and banking book capital

D. In addition to default risk, via its constant level of risk rebalancing assumption, the IRC measures credit deterioration (ratings downgrades and/or credit spread widening) and/or loss of liquidity

Answer: B

Step by step

1. Check each statement against the Basel 2.5 treatment.
2. A is true: IRC is a 99.9%, one-year VaR on credit-sensitive trading book positions.
3. C is true: the IDRC/IRC removed the incentive to move default-risky assets to the trading book, so trading book capital is effectively the greater of market risk and banking book capital.
4. D is true: downgrades are included and positions are rebalanced to the original rating at the liquidity horizon.
5. B is false: securitizations and re-securitizations in the correlation book are covered by the comprehensive risk (CR) charge, not the IRC.

Why B is correct: Correlation-sensitive positions (securitizations, re-securitizations) fall under the comprehensive risk charge, so B is the false statement.

Why the others are wrong

- **A:** True statement: the IRC is a one-year 99.9% VaR for credit losses.
- **C:** True statement: the IRC grew out of the 2005 IDRC that responded to regulatory arbitrage between books.
- **D:** True statement: the IRC includes migration and spread/liquidity effects and uses a constant level of risk assumption.

Q636

BT answers P2.T7.520.2 · BT notes P2.T7.520.2

Basel 2.5 introduced three changes to market risk capital: a stressed VaR, an incremental risk charge (IRC), and a comprehensive risk measure for instruments dependent on credit correlation. In regard to the IRC, each of the following is true EXCEPT which statement is untrue?

A. The IRC was introduced because instruments in the trading book often required less regulatory capital than equivalent instruments in the banking book

B. The IRC is calculated as the one-year 99.9% value at risk (VaR) for losses from credit instruments in the trading book

C. The IRC excludes ratings changes or migrations because a rating change does not itself lead to default within the one-year horizon

D. The constant level of risk assumption assumes that the bank will have the opportunity to rebalance its portfolio during the year so that default risk is mitigated

Answer: C

Step by step

1. A: true. Trading book capital was lower than banking book capital, which created regulatory arbitrage.
2. B: true. The IRC is a one-year, 99.9% VaR on credit instruments.
3. D: true. Downgraded or defaulted positions are replaced at the liquidity horizon with positions of the original rating, which keeps credit quality constant and lessens default risk.
4. C: untrue. After the crisis (losses came from downgrades, spread widening, liquidity loss) the IDRC was

widened into the IRC to include rating changes/migrations.

Why C is correct: The IRC explicitly includes rating migration in addition to default, so the statement that it excludes migrations is false.

Why the others are wrong

- **A:** True: lower trading book capital was the motivation.
- **B:** True: this is the IRC definition.
- **D:** True: the constant level of risk assumption is the rebalancing at each liquidity horizon.

Q637

AnalystPrep Q.4288

REBUILT — Original used a weak 'all of the above' format and Basel I wording; rebuilt as a single-best-answer item on the 2005 IDRC variants (GARP wording).

In 2005 the Basel Committee proposed an incremental default risk charge (IDRC) for trading book instruments. Which description of the proposal is most accurate?

A. An internal default model calibrated to the 99.9th percentile at a one-year horizon, as in the IRB approach, or, absent such a model, a standardized or current exposure approach similar to Basel I specific risk charges

B. An internal default model calibrated to a 99% confidence level over a 10-day horizon, multiplied by a factor of at least three

C. Only a standardized approach based on external ratings, with no internal model permitted

D. A stressed VaR using a 250-day period of market stress, added to the usual VaR

Answer: A

Step by step

1. The IDRC aimed to remove the incentive to hold default-risky illiquid assets in the trading book, where capital was lower.
2. Two variants were proposed: (1) an internal model of default risk at the 99.9th percentile, one-year horizon, same as the IRB approach; (2) absent such a model, a standardized or current exposure approach resembling Basel I specific risk charges.
3. This matches A; as a result trading book capital would be the greater of market risk capital and banking book capital.

Why A is correct: The IDRC used an IRB-style 99.9%, one-year internal model, with a Basel I-like standardized/current exposure fallback.

Why the others are wrong

- **B:** Describes the old trading book VaR (10-day, 99%, multiplier), the very measure that produced low capital.
- **C:** Ignores the internal model variant; both variants were proposed.
- **D:** Describes the stressed VaR, a separate Basel 2.5 change.

Q638

Schweser Module Quiz 63.1 Q2

Banks are required to rebalance their portfolios as the creditworthiness of bonds declines, leading to losses over time but generally not to outright default. This requirement to specify a liquidity horizon for each instrument in the portfolio and rebalance at the end of the liquidity horizon is part of the:

A. incremental risk charge calculation.

B. net stable funding charge formula.

C. countercyclical buffer estimation.

D. comprehensive risk charge calculation.

Answer: A

Step by step

1. The IRC requires a liquidity horizon for each instrument (minimum three months).
2. At the end of each horizon a defaulted or downgraded bond is assumed replaced by a bond of the original rating (constant level of risk), recording a loss on the sale.
3. Losses from downgrades are taken but outright default is generally avoided, which describes the IRC.

Why A is correct: Liquidity horizons with rebalancing to the original credit quality (constant level of risk) are core features of the IRC.

Why the others are wrong

- **B:** The net stable funding ratio is a Basel III liquidity measure, unrelated to trading book rebalancing.
- **C:** The countercyclical buffer is an extra equity buffer in credit booms, not a VaR-type charge.

- **D:** The comprehensive risk charge covers the correlation book (securitizations), not liquidity-horizon rebalancing of credit positions.

Q639 BT answers P2.T7.520.2 · BT notes P2.T7.520.2

About Basel 2.5, Hull says, "During the credit crisis, it was recognized that some changes were necessary to the calculation of capital for market risk in the Basel II framework. These changes are referred to as 'Basel 2.5' ... There are three changes involving: 1. The calculation of a stressed VaR; 2. A new incremental risk charge; and 3. A comprehensive risk measure for instruments dependent on credit correlation." In regard to the incremental risk charge (IRC), each of the following is true EXCEPT which statement is untrue about the IRC?

A. The IRC was introduced because instruments in the trading book often required less regulatory capital than equivalent instruments in the banking book

B. The IRC is calculated as the one-year 99.9% value at risk (VaR) for losses from credit instruments in the trading book

C. The IRC excludes ratings changes or migrations because a rating change does not itself lead to default within the one-year horizon

D. The constant level of risk assumption assumes that the bank will have the opportunity to rebalance its portfolio during the course of the year so that default risk is mitigated

Answer: C

Step by step

1. A: true. Illiquid, default-prone instruments were shifted to the trading book to get lower capital; the IRC (originally IDRC) removes that incentive.
2. B: true. The IRC is a one-year, 99.9% VaR, roughly aligning trading-book with banking-book capital.
3. C: false. Because many crisis losses came from downgrades and spread widening rather than defaults, the IRC covers credit-sensitive instruments and explicitly includes rating migrations as well as defaults.
4. D: true. Under the constant level of risk assumption, positions are rebalanced back to their initial risk at the end of each liquidity horizon (minimum three months), which reduces the 99.9% VaR.
5. The untrue statement is C.

Why C is correct: The IRC was widened from default risk only to include ratings changes (migrations), so a statement saying it excludes them is false.

Why the others are wrong

- **A:** True statement (regulatory arbitrage motivation), so not the exception.
- **B:** True statement (one-year horizon, 99.9% confidence), so not the exception.
- **D:** True statement (constant level of risk assumption with rebalancing at each liquidity horizon), so not the exception.

LO 63.c

Describe the comprehensive risk (CR) capital charge for portfolios of positions that are sensitive to correlations between default risks.

Q640 BT answers P2.T7.520.3 · BT notes P2.T7.520.3

According to Hull, each of the following is true about the comprehensive risk measure (CRM) EXCEPT, which is not accurate?

A. The CRM is a single capital charge replacing the incremental risk charge and the specific risk charge for instruments dependent on credit correlation

B. The CRM can be calculated using a standardized approach which determines a charge based on the credit rating (e.g., BBB+ to BBB-) of the securitization or re-securitization

C. The maximum capital charge under the standardized approach is 28.0% which applies securitization with the lowest credit rating

D. The Basel Committee allows banks with supervisory approval to use their internal models to calculate the CRM for unrated positions, but the models developed by banks have to be sophisticated in order to be approved by bank supervisors

Answer: C

Step by step

1. Recall the standardized CR charge table (capital as a fraction of exposure): securitizations are 1.6% (AAA/AA), 4% (A), 8% (BBB), 28% (BB), and 100% (below BB- or unrated).
2. Option C says the maximum is 28%, applied to the lowest rating. But 28% is the charge for BB-rated securitizations only; the maximum is 100% (dollar-for-dollar capital) for unrated or below-BB- positions. So C is false.
3. A is true: the CR charge replaces the IRC and specific risk charge for the correlation book.
4. B is true: the standardized approach depends only on the rating, with higher charges for re-securitizations (e.g., 18% vs 8% at BBB).
5. D is true: with supervisory approval banks may use internal models, which must be sophisticated.

Why C is correct: The standardized charge peaks at 100% for unrated or below-BB- securitizations; 28% is only the BB-rated charge, so the statement about a 28% maximum is the false one.

Why the others are wrong

- **A:** True statement; the CR charge replaces both the incremental risk charge and the specific risk charge for correlation-dependent instruments. Choosing it reflects misremembering what was replaced.
- **B:** True statement; the standardized approach is purely rating-based, with re-securitizations charged more. Choosing it confuses the approach with an internal model.
- **D:** True statement; internal models need supervisory approval and must be sophisticated (capture multiple defaults, correlation volatility, hedge rebalancing costs, recovery-rate volatility).

Q641

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Maria Lopez, a market risk analyst at a bank that uses the Basel 2.5 standardized approach for its correlation book, holds three positions: a $60 million AA-rated securitization tranche, a $25 million BBB-rated re-securitization tranche, and an $8 million unrated securitization tranche. The standardized capital charges are 1.6%, 4%, 8%, 28%, and 100% for securitizations rated AAA to AA, A, BBB, BB, and below BB or unrated, respectively, and 3.2%, 8%, 18%, 52%, and 100% for re-securitizations in the same rating buckets. What is the total comprehensive risk capital charge for the three positions?

A. $6.10 million
B. $10.96 million
C. $13.46 million
D. $7.70 million

Answer: C

Step by step

1. AA securitization: $60 \times 1.6\% = 0.96$ million.
2. BBB re-securitization uses the re-securitization schedule: $25 \times 18\% = 4.50$ million.
3. Unrated tranche is charged dollar-for-dollar: $8 \times 100\% = 8.00$ million.
4. Total: $0.96 + 4.50 + 8.00 = 13.46$ million.

Why C is correct: Each position is charged by its rating bucket and its type; re-securitizations use the higher schedule and unrated tranches must be fully capitalized, giving $13.46 million.

Why the others are wrong

- **A:** Wrong: treats the unrated tranche as 8% of its value ($0.64 million) instead of 100%; total $0.96 + 4.50 + 0.64$ is about $6.10 million.
- **B:** Wrong: applies the ordinary securitization 8% to the BBB re-securitization ($2.00 million) instead of the 18% re-securitization charge; total about $10.96 million.
- **D:** Wrong: charges the unrated tranche at the BB rate of 28% ($2.24 million) instead of 100%; total $0.96 + 4.50 + 2.24$ is about $7.70 million.

Q642

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager is explaining the Basel 2.5 comprehensive risk (CR) charge to the trading desk heads. Which of the following statements is correct?

A. Under the standardized approach, a re-securitization carries the same capital charge as a securitization with the same rating because the charge depends only on the rating.

B. Highly rated securitization tranches are excluded from the CR charge because they are not exposed to losses even when default correlations rise.
C. A bank approved to use an internal model may set the CR charge at whatever level the model produces, since approval removes any link to the standardized approach.
D. The CR charge applies to the correlation book and replaces both the incremental risk charge and the specific risk charge for those positions.

Answer: D
Step by step
1. The CR charge was introduced to cover instruments sensitive to default correlation (ABS, CDOs, re-securitizations).
2. It replaces the IRC and specific risk charge for the correlation book, so D is correct.
3. Standardized charges are higher for re-securitizations; internal-model charges are subject to a floor tied to the standardized charge; and even top-rated tranches suffered losses in the crisis.

Why D is correct: The CR charge is a single charge for the correlation book that takes the place of the IRC and specific risk charge for those positions.
Why the others are wrong
- **A:** Wrong: charges depend on both rating and type; re-securitizations are more vulnerable to correlation changes and have higher charges (e.g., 18% versus 8% for BBB).
- **B:** Wrong: when correlations rise in stress, even the highest-rated tranches can lose value, which is why they are included.
- **C:** Wrong: the model-based charge may not fall below a specified fraction of the standardized charge, and the model needs supervisory approval and must be unusually robust.

LO 63.d

Define in the context of Basel III and calculate where appropriate: Tier 1 capital and its components Tier 2 capital and its components required Tier 1 equity capital, total Tier 1 capital, and total capital

Q643

BT notes P2.T6.24.6.3

REBUILT — 2017 Operational Risk RWA was OCR-damaged (203,825) and did not sum to the printed total; recovered as 293,825 (components then sum to 1,298,688) and the answer is now derived.

Assess Company XYZ's capital adequacy over the last three years, as measured by the three key capital ratios (CET1, Tier 1 and total capital, each divided by total RWAs).
Company XYZ - Excerpt from Annual Report Disclosure

Item ($m)	2017	2016	2015
Common Equity Tier 1 Capital	146,424	142,367	137,100
Additional Tier 1 Capital	22,639	20,443	17,600
Tier 2 Capital	22,456	27,564	38,200
Total Regulatory Capital	191,519	190,374	192,900
Credit Risk RWAs	960,763	989,639	968,600
Market Risk RWAs	44,100	36,910	49,600
Operational Risk RWAs	293,825	256,300	224,300
Total RWAs	1,298,688	1,282,849	1,242,500

Based on the above, the capital adequacy of Company XYZ over the last three years signals which of the following conditions?
A. Mixed
B. Declining
C. Improving
D. Not enough data to determine

Answer: A
Step by step
1. CET1 ratio = CET1 / total RWAs: 137,100/1,242,500 = 11.03% (2015), 142,367/1,282,849 = 11.10% (2016), 146,424/1,298,688 = 11.27% (2017): rising.

2. Tier 1 ratio = (CET1 + AT1) / RWAs: 12.45% (2015), 12.69% (2016), 13.02% (2017): rising.
3. Total capital ratio = total regulatory capital / RWAs: 192,900/1,242,500 = 15.53%, 190,374/1,282,849 = 14.84%, 191,519/1,298,688 = 14.75%: falling.
4. Two ratios improve while the total capital ratio declines (Tier 2 capital shrank from 38,200 to 22,456), so the signal is mixed.

Why A is correct: The Tier 1 ratios improve because higher-quality capital (CET1 and AT1) grew faster than RWAs, but total capital fell as Tier 2 shrank, so the three ratios do not point in one direction.

Why the others are wrong

- **B:** Looks only at the total capital ratio (15.53% to 14.75%) and ignores the rising CET1 and Tier 1 ratios.
- **C:** Looks only at the CET1 and Tier 1 ratios and ignores the falling total capital ratio.
- **D:** All data needed (capital and RWAs for three years) are given; the ratios can be computed directly.

Q644

AnalystPrep Q.4292

The estimated risk-weighted assets of a bank stand at $400 million. In the context of Basel III, what is the bank's minimum Tier 1 capital?

A. $18 million
B. $12 million
C. $16 million
D. $24 million

Answer: D

Step by step

1. Minimum total Tier 1 = 6% of RWA (4.5% CET1 + 1.5% AT1).
2. $0.06 \times 400 = 24$ ($ million).

Why D is correct: Total Tier 1 must be at least 6% of RWA.

Why the others are wrong

- **A:** 4.5% x 400 = 18: that is the CET1 minimum, not total Tier 1.
- **B:** 3% x 400 = 12: not a Basel III Tier 1 ratio.
- **C:** 4% x 400 = 16: the old pre-Basel III Tier 1 minimum.

Q645

AP Mock 2 Q8

Rosenborg Bank from Trondheim, Norway, is preparing for Basel III implementation. Its risk management team has established that as of 31 March 2018, the bank's Tier 1 capital amounts to 5% of risk-weighted assets. Is this acceptable?

A. Yes, total Tier 1 capital should be at least 2% of risk-weighted assets (RWA).
B. No, total Tier 1 capital should be at least 8% of RWA.
C. Yes, total Tier 1 capital should be at least 4% of RWA.
D. No, total Tier 1 capital should be at least 6% of RWA.

Answer: D

Step by step

1. Basel III minimums: CET1 4.5%, total Tier 1 (CET1 + AT1) 6%, total capital (Tier 1 + Tier 2) 8% of RWA.
2. Bank's Tier 1 = 5% < 6% minimum.
3. Not acceptable; the required minimum is 6%.

Why D is correct: Basel III raised the minimum total Tier 1 ratio to 6% of RWA, so 5% is a shortfall.

Why the others are wrong

- **A:** 2% is not a Basel III minimum for total Tier 1 (it is the old Basel II core-equity level); Basel III requires 6%.
- **B:** 8% is the minimum for total capital (Tier 1 + Tier 2), not Tier 1.
- **C:** 4% was the Basel II Tier 1 minimum, replaced by 6% in Basel III.

Q646

AnalystPrep Q.2353

Katerini Bank from Greece is in the process of implementing Basel III regulations. One of the first assignments of its risk management team is to calculate the required regulatory capital. In line with Basel III, the bank should have the following categories of capital, except:

A. Tier 1 capital
B. Tier 2 capital
C. Tier 3 capital
D. Additional Tier 1 capital

Answer: C
Step by step
1. Basel III capital: Tier 1 = Common Equity Tier 1 (core) + Additional Tier 1; plus Tier 2.
2. Tier 3 capital (short-term subordinated debt covering market risk) was eliminated by Basel III.
3. So Tier 3 is the category that does not exist.

Why C is correct: Basel III eliminated Tier 3 capital and split Tier 1 into Core (CET1) and Additional Tier 1.
Why the others are wrong
- **A:** Tier 1 capital is a Basel III category (CET1 plus AT1).
- **B:** Tier 2 capital remains a Basel III category (e.g. subordinated debt, general loan loss reserves).
- **D:** Additional Tier 1 is a new Basel III subdivision of Tier 1.

Q647 AnalystPrep Q.2354

Which of the following is a component of Tier 1 capital under Basel III?
A. Subordinated debt
B. Share capital
C. Goodwill
D. General loan loss reserves

Answer: B
Step by step
1. Tier 1 equity capital = common equity (share capital), retained earnings, and a limited amount of minority interest and unrealized gains/losses.
2. Subordinated debt and general loan loss reserves are Tier 2; goodwill is deducted from Tier 1.
3. Share capital is the Tier 1 component.

Why B is correct: Common equity (share capital) is the highest-quality, permanent loss-absorbing capital and forms the core of Tier 1.
Why the others are wrong
- **A:** Qualifying subordinated debt (5+ years original maturity) is Tier 2.
- **C:** Goodwill and other intangibles are deducted from Tier 1, not added.
- **D:** General loan loss reserves are Tier 2 (capped at 1.25% of standardized RWAs or 0.6% of IRB RWAs).

Q648 AnalystPrep Q.3107

The initial phase of the Basel III framework focused, in part, on increasing the quality of bank regulatory capital to cover unexpected losses. As such, the minimum Tier 1 capital requirement:
A. Rose from 4% to 6% of risk-weighted assets.
B. Rose from 5% to 6% of risk-weighted assets.
C. Rose from 5% to 7% of risk-weighted assets.
D. Rose from 4% to 7% of risk-weighted assets.

Answer: A
Step by step
1. Basel I/II minimum Tier 1 = 4% of RWA.
2. Basel III minimum Tier 1 (CET1 4.5% + AT1 1.5%) = 6% of RWA.
3. The rise is 4% to 6%.

Why A is correct: Basel III raised the Tier 1 minimum from 4% to 6% of RWA.
Why the others are wrong
- **B:** 5% is not the old Tier 1 minimum (4%).
- **C:** Wrong starting point (5%) and 7% is not a minimum; it is 4.5% CET1 plus the 2.5% conservation buffer.
- **D:** 7% is CET1 including the 2.5% conservation buffer, not the Tier 1 minimum.

Q649 AnalystPrep Q.3246

Consider a bank balance sheet with:
1. Common stock of $730,000,000;
2. Unrealized long-term marketable equity securities gain: $33,000,000;
3. Five-year subordinated debt: $28,000,000;
4. Goodwill: $92,000,000.

Based on Basel III capital requirements and solely on the above information, the Tier 1 and Tier 2 capital amounts are, respectively:

A. $730,000,000 and $0
B. $730,000,000 and $61,000,000
C. $822,000,000 and $33,000,000
D. $671,000,000 and $28,000,000

Answer: D

Step by step
1. Tier 1 = common stock + unrealized gains - goodwill = $730 + 33 - 92 = 671$ ($ million).
2. Goodwill is deducted from Tier 1.
3. Tier 2 = qualifying subordinated debt (five-year) = $28 million.
4. Answer: $671m and $28m.

Why D is correct: Tier 1 equity includes common equity and unrealized gains but nets out goodwill; subordinated debt is Tier 2.

Why the others are wrong
- **A:** Ignores the gain and goodwill and omits subordinated debt from Tier 2.
- **B:** Ignores the goodwill deduction and puts the gain (33) in Tier 2 with the debt (33 + 28 = 61).
- **C:** Adds goodwill (730 + 92) instead of deducting it, and treats the 33 gain as Tier 2.

Q650 AnalystPrep Q.4289

Which of the following are components of Tier 1 capital in the context of the Basel III capital definition?

A. Common equity and retained earnings
B. General loan loss reserves
C. Subordinated debt
D. Goodwill and other intangibles

Answer: A

Step by step
1. Tier 1 equity capital = common equity, retained earnings, and limited minority interest and unrealized gains/losses.
2. General loan loss reserves and subordinated debt belong to Tier 2.
3. Goodwill and intangibles are deducted.

Why A is correct: Common equity and retained earnings are the core of Tier 1 equity capital.

Why the others are wrong
- **B:** General loan loss reserves are Tier 2.
- **C:** Subordinated debt (5+ years) is Tier 2.
- **D:** Goodwill and other intangibles are deducted from Tier 1, not included.

Q651 AnalystPrep Q.4290

Assume that a bank has common equity of $100 million, retained earnings of $80 million, minority interest and unrealized gains and losses of $20 million, and goodwill and other intangibles of $5 million. What is the value of Tier 1 equity capital in the context of the Basel III accord?

A. $190 million
B. $195 million
C. $205 million
D. $100 million

Answer: B

Step by step

1. Tier 1 equity = common equity + retained earnings + minority interest/unrealized gains - goodwill and intangibles.
2. $100 + 80 + 20 - 5 = 195$ ($ million).

Why B is correct: Goodwill and intangibles are deducted from the sum of the equity components.

Why the others are wrong

- **A:** Deducted 10 (double the goodwill) or otherwise mis-subtracted.
- **C:** Added goodwill (+5) instead of deducting it.
- **D:** Counted common equity only, omitting retained earnings and minority interest/gains.

Q652 AnalystPrep Q.4291

The estimated risk-weighted assets of a bank are $200 million. In the context of Basel III, the Core Tier 1 (Tier 1 equity) capital of the bank must be at least:

A. $10 million
B. $4.5 million
C. $9 million
D. $12 million

Answer: C

Step by step

1. Minimum Core Tier 1 (CET1) = 4.5% of RWA.
2. $0.045 \times 200 = 9$ ($ million).

Why C is correct: The Basel III CET1 minimum is 4.5% of RWA.

Why the others are wrong

- **A:** 5% of RWA; not a Basel III minimum.
- **B:** Used 4.5 as a dollar figure without multiplying by RWA.
- **D:** 6% of RWA, which is the total Tier 1 minimum, not Core Tier 1.

Q653 AnalystPrep Q.4293

In the context of Basel III, Tier 2 capital is designed to absorb losses after failure and thus protects the depositors and other creditors of the bank. Which of the following is a component of Tier 2 capital?

A. Common equity
B. General loan loss reserves
C. Retained earnings
D. Approved minority interest not included in Core Tier 1

Answer: B

Step by step

1. Tier 2 (supplementary) capital consists of subordinated debt (original maturity of five years or more, callable only after five years) and general loan loss reserves not allocated to specific positions.
2. General loan loss reserves count only up to 1.25% of standardized-approach RWA (or 0.6% of IRB RWA).
3. Common equity and retained earnings are Tier 1 equity capital; approved minority interest not in Core Tier 1 is Additional Tier 1.
4. Only B is Tier 2.

Why B is correct: Tier 2 is gone-concern capital made up of long-dated subordinated debt and general (unallocated) loan loss reserves, the latter subject to a cap on RWA.

Why the others are wrong

- **A:** Common equity is the core of Tier 1 equity capital (going-concern capital), not Tier 2.
- **C:** Retained earnings are part of Tier 1 equity capital, not Tier 2.
- **D:** Approved minority interest not included in Core Tier 1 is classified as Additional Tier 1 capital; the mistake is treating any non-core item as Tier 2.

Q654

AnalystPrep Q.5938

REBUILT — Original compared two bare Tier 1 ratios with no calculation and a subjective 'better positioned' conclusion; rebuilt as a self-contained calculation of Basel III minimum capital tiers (excluding buffers) for one bank.

A bank has risk-weighted assets (RWA) of $250 million. Its capital consists of Tier 1 equity capital of $10 million, Additional Tier 1 capital of $4 million, and Tier 2 capital of $8 million. Considering only the Basel III minimum ratios (4.5% Tier 1 equity, 6.0% total Tier 1, 8.0% total capital) and ignoring all buffers, which of the following is correct?

A. The bank meets all three minimum requirements.
B. The bank meets the total capital requirement but falls short on both the Tier 1 equity and the total Tier 1 requirements.
C. The bank meets the Tier 1 equity requirement but falls short on the total Tier 1 and total capital requirements.
D. The bank falls short on all three minimum requirements.

Answer: B

Step by step

1. Tier 1 equity ratio = 10 / 250 = 4.00% < 4.5% (required 11.25m): fails.
2. Total Tier 1 = 10 + 4 = 14; ratio = 14 / 250 = 5.60% < 6.0% (required 15m): fails.
3. Total capital = 14 + 8 = 22; ratio = 22 / 250 = 8.80% >= 8.0% (required 20m): passes.
4. So the bank passes only the total capital test.

Why B is correct: Each tier is tested against RWA separately; total capital (8.80%) clears 8%, but Tier 1 equity (4.00%) and total Tier 1 (5.60%) do not clear 4.5% and 6.0%.

Why the others are wrong

- **A:** Judges the bank on total capital alone (8.80% > 8%) and ignores that Tier 1 equity and total Tier 1 have their own minimums.
- **C:** Misreads 4.00% as meeting the 4.5% test and leaves Tier 2 out of total capital (14/250 = 5.60% < 8%).
- **D:** Leaves Tier 2 out of total capital (14/250 = 5.60%); including the $8 million of Tier 2 gives 22/250 = 8.80%, which meets the 8% minimum.

Q655

BT answers P2.T7.20.9.3 · BT notes P2.T7.20.9.3

REBUILT — Removed the 2017/2018 phase-in columns (the 2018 values were inconsistent with base plus buffers) and the unused, uncertain total assets figure; kept only the fully phased-in 2019 requirements. Merged the two duplicate sources.

The CRO of LGX is reporting on the bank's capital adequacy. Capital and exposure data (USD billions) are: Common equity Tier 1 (CET1) capital 2.40; noncumulative preferred stock (Additional Tier 1) 0.90; Tier 2 capital 1.70; risk-weighted assets (RWA) 36.00; total leverage exposure 65.00. The fully phased-in requirements are:

Ratio	Requirement
CET1 ratio (4.5% + 2.5% CCB + 1.5% G-SIB)	8.50%
Tier 1 ratio (6.0% + 2.5% CCB + 1.5% G-SIB)	10.00%
Total capital ratio (8.0% + 2.5% CCB + 1.5% G-SIB)	12.00%
Leverage ratio (Tier 1 / total exposure)	4.00%

For which of the four ratios does the bank meet the requirement?

A. Leverage and total capital ratios are sufficient, but CET1 and Tier 1 ratios are too low
B. Tier 1 and leverage ratios are sufficient, but CET1 and total capital ratios are too low
C. Only the CET1 ratio is sufficient; Tier 1, total capital and leverage ratios are too low
D. All four ratios are sufficient

Answer: A

Step by step

1. CET1 ratio = 2.40 / 36.00 = 6.67% < 8.50%: fails.
2. Tier 1 = 2.40 + 0.90 = 3.30; ratio = 3.30 / 36.00 = 9.17% < 10.00%: fails.
3. Total capital = 3.30 + 1.70 = 5.00; ratio = 5.00 / 36.00 = 13.89% >= 12.00%: passes.
4. Leverage ratio = Tier 1 / exposure = 3.30 / 65.00 = 5.08% >= 4.00%: passes.

Why A is correct: Only the total capital ratio (13.89%) and leverage ratio (5.08%) exceed their requirements; CET1 (6.67%) and Tier 1 (9.17%) fall short of the buffer-inclusive requirements.

Why the others are wrong

- **B:** Calls Tier 1 sufficient, but 9.17% is below 10.00%; and calls total capital insufficient, though 13.89% exceeds 12.00%.

- **C:** Calls CET1 sufficient, but 2.40/36.00 = 6.67% is below 8.50%; the total capital and leverage ratios in fact pass.
- **D:** Compares the ratios with the bare minimums (4.5%, 6%) and so passes CET1 and Tier 1; against the buffer-inclusive 8.50% and 10.00% requirements, 6.67% and 9.17% fail.

Q656

BT answers P2.T7.25.1.2

Boer Bank of South Africa has provided its capital adequacy measures for two consecutive years. The table shows risk-weighted assets (RWA) for credit, market and operational risk, along with the bank's Common Equity Tier 1 (CET1), Tier 1 and Tier 2 capital.

Item	2023	2024
Credit Risk Weighted Assets	140	200
Market Risk Weighted Assets	60	50
Operational Risk Weighted Assets	10	30
CET-1 Capital	5	20
Tier 1 Capital	40	25
Tier 2 Capital	25	30

Did Boer Bank meet the Basel III minimum capital requirements (4.5% CET1, 6.0% Tier 1, 8.0% total capital) in 2023 and 2024?

A. Bank was compliant with minimum capital requirements in 2023 and not in 2024.
B. Bank was not compliant with minimum capital requirements in 2023 and complied in 2024.
C. Bank was compliant with minimum capital requirements in both 2023 and 2024.
D. Bank was compliant with minimum capital requirements in neither 2023 nor 2024.

Answer: B

Step by step

1. Total RWA 2023 = 140 + 60 + 10 = 210; 2024 = 200 + 50 + 30 = 280.
2. 2023: CET1 = 5/210 = 2.38% < 4.5% (fails). Tier 1 = 40/210 = 19.05%; total = (40+25)/210 = 30.95%.
3. 2024: CET1 = 20/280 = 7.14% >= 4.5%; Tier 1 = 25/280 = 8.93% >= 6%; total = (25+30)/280 = 19.64% >= 8%.
4. 2023 fails the CET1 test; 2024 passes all three.

Why B is correct: Capital is compared with total RWA (credit + market + operational); the 2023 CET1 ratio of 2.38% is below 4.5% while all 2024 ratios exceed their minimums.

Why the others are wrong

- **A:** Judges 2023 on the large Tier 1 and total figures and ignores the CET1 shortfall; 2024 in fact passes every test.
- **C:** Overlooks the 2023 CET1 ratio of only 2.38%, which is below 4.5%.
- **D:** Wrongly fails 2024, e.g. by using only credit RWA or misreading the capital rows; with total RWA of 280 the ratios are 7.14%, 8.93% and 19.64%, all above the minimums.

Q657

BT answers P2.T7.521.1 · BT notes P2.T7.521.1

Under Basel III, a bank's total regulatory capital consists of Tier 1 equity capital, Additional Tier 1 capital and Tier 2 capital (Tier 3 capital was eliminated). About the bank's total regulatory capital, each of the following is true EXCEPT which is false?

A. Tier 1 equity capital (core Tier 1) includes share capital and retained earnings
B. Tier 1 equity capital (core Tier 1) excludes goodwill and deferred tax assets
C. Additional Tier 1 capital (non-core) consists of items such as noncumulative preferred stock that are not common equity
D. Tier 2 capital is "going-concern capital" that must be equity, cannot be debt, ranks above depositors in a liquidation, and includes noncumulative preferred stock

Answer: D

Step by step

1. A true: Tier 1 equity = common equity (share capital) plus retained earnings.
2. B true: goodwill, other intangibles and deferred tax assets are deducted from core Tier 1.
3. C true: Additional Tier 1 includes noncumulative perpetual preferred stock and similar non-common instruments.

4. D false: Tier 2 is gone-concern capital (absorbs losses after failure), includes subordinated debt with original maturity of 5+ years, and ranks below depositors.

Why D is correct: Tier 2 capital absorbs losses after failure, is largely subordinated debt, and ranks below depositors, so D contains several errors.

Why the others are wrong

- **A:** True statement: share capital and retained earnings are Tier 1 equity.
- **B:** True statement: goodwill and deferred tax assets are deducted from core Tier 1.
- **C:** True statement: noncumulative perpetual preferred stock is Additional Tier 1.

Q658

BT answers P2.T7.521.2 · BT notes P2.T7.521.2

As of full implementation in 2019, each of the following regulatory capital requirements under Basel III is accurate EXCEPT which is not?

A. Core Tier 1 equity capital must be at least 4.5% of risk-weighted assets; including the capital conservation buffer, the core Tier 1 requirement is 7.0%

B. Total Tier 1 capital (Tier 1 equity capital plus Additional Tier 1 capital) must be at least 6.0% of risk-weighted assets at all times

C. Tier 2 capital must be at least 5.5% of risk-weighted assets; the capital conservation buffer can be counted toward Tier 2 capital

D. Total capital (total Tier 1 plus Tier 2) must be at least 8.0% of risk-weighted assets at all times; including the capital conservation buffer, the minimum total capital requirement is 10.5%

Answer: C

Step by step

1. A: 4.5% + 2.5% CCB = 7.0% core Tier 1 - true.
2. B: total Tier 1 minimum is 6.0% of RWA - true.
3. C: Basel III sets no stand-alone 5.5% Tier 2 minimum (8% total less 6% Tier 1 leaves only 2%), and the CCB must be met with common equity, not Tier 2 - false.
4. D: 8.0% + 2.5% = 10.5% - true.

Why C is correct: The conservation buffer is held in common equity and Basel III sets no stand-alone Tier 2 minimum, so statement C is wrong.

Why the others are wrong

- **A:** True: 4.5% + 2.5% = 7.0%.
- **B:** True: Tier 1 (equity plus Additional Tier 1) must be 6.0% of RWA.
- **D:** True: 8.0% + 2.5% CCB = 10.5%.

Q659

GARP PE1 Q76 · Mock 3 Q5

A treasury analyst at a bank is calculating the bank's Basel III regulatory capital ratios for its year-end financial reporting. The analyst considers the following items on the bank's balance sheet:

Item	Value (EUR millions)
Common equity	2,010
Non-callable preferred equity	500
Subordinated debt	1,500
Retained earnings	3,210
Goodwill from prior acquisitions	850

The bank has total risk-weighted assets of EUR 49,700 million. What is the Core Tier 1 (CET1) capital ratio that the analyst should calculate?

A. 8.79%

B. 9.80%

C. 10.50%

D. 11.51%

Answer: A

Step by step

1. Core Tier 1 (Tier 1 equity) capital includes common equity and retained earnings; goodwill and other intangibles are deducted.

2. Preferred equity is Additional Tier 1 and subordinated debt is Tier 2, so neither counts toward Core Tier 1.
3. CET1 = 2,010 + 3,210 - 850 = 4,370 (EUR millions).
4. CET1 ratio = 4,370/49,700 = 8.79%.

Why A is correct: Core Tier 1 contains only common equity and retained earnings, net of deductions such as goodwill, giving 4,370 / 49,700 = 8.79%.

Why the others are wrong

- **B:** Adds the 500 of non-callable preferred equity to CET1 (4,870 / 49,700); preferred is Additional Tier 1, not Core Tier 1.
- **C:** Forgets to deduct goodwill (5,220 / 49,700 = 10.50%); goodwill has no loss-absorbing value and must be deducted.
- **D:** Includes the preferred equity and omits the goodwill deduction (5,720 / 49,700); two errors combined.

Q660

Schweser Module Quiz 63.2 Q1

Under Basel III, capital must be adjusted downward to reflect which of the following?

A. Planned bonuses for managers.
B. Deficits in defined benefit pension plans.
C. Corporate convertible debt.
D. There is no requirement under Basel III to adjust capital downward for anything.

Answer: B

Step by step

1. Basel III requires several deductions from capital beyond goodwill and intangibles.
2. These include defined-benefit pension plan deficits, certain cross-holdings within a group, and mortgage servicing rights exceeding 10% of common equity.
3. Of the choices, only defined benefit pension deficits is such a deduction.

Why B is correct: Basel III deducts defined-benefit pension plan deficits (along with cross-holdings and excess mortgage servicing rights) from capital.

Why the others are wrong

- **A:** Planned bonuses are not a listed Basel III capital deduction.
- **C:** Convertible debt with conversion/write-down triggers can qualify as Additional Tier 1 capital, so it is not deducted.
- **D:** Incorrect: Basel III specifies several deductions, including goodwill, deferred tax assets and pension deficits.

LO 63.e

Describe the motivations for and calculate the capital conservation buffer and the countercyclical buffer, including special rules for globally systemically important banks (G-SIBs).

Q661

AnalystPrep Q.3096

REBUILT — Stem said December 2018, when the capital conservation buffer was still being phased in (not yet the full 2.5%); changed to the fully phased-in Basel III requirement so the answer is unambiguous.

Prime Bank has risk-weighted assets (RWA) of $200 million. Under fully phased-in Basel III, what is the minimum Common Equity Tier 1 (CET1) capital the bank must hold to satisfy the CET1 minimum plus the capital conservation buffer (ignoring any G-SIB or countercyclical buffer)?

A. $9,000,000
B. $14,000,000
C. $12,000,000
D. $16,000,000

Answer: B

Step by step

1. Basel III minimum CET1 ratio = 4.5% of RWA.
2. Capital conservation buffer = 2.5% of RWA, met with Tier 1 equity (CET1).
3. Required CET1 ratio = 4.5% + 2.5% = 7.0%.

4. Required CET1 = 7.0% × \$200m = \$14m.

Why B is correct: The buffer is added on top of the 4.5% CET1 minimum, giving a 7% CET1 requirement that applies to RWA.

Why the others are wrong

- **A:** 4.5% × 200 = 9m: ignored the conservation buffer.
- **C:** 6.0% × 200 = 12m: used the minimum total Tier 1 ratio (6%) instead of CET1 plus buffer.
- **D:** 8.0% × 200 = 16m: used the minimum total capital ratio (8%), which includes Tier 2.

Q662

AnalystPrep Q.3103

Four European banks, A, B, C, and D have the following capital amounts and risk-weighted assets (in \$m):

Bank	A	B	C	D
Tier 1 capital	5	8	15	25
Tier 2 capital	3	3	5	10
Risk-weighted assets	30	40	240	230

Using the fully phased-in Basel III requirements including the capital conservation buffer, which of the four banks fails the capital adequacy requirements?

A. Bank A
B. Bank B
C. Bank C
D. Bank D

Answer: C

Step by step

1. With the 2.5% conservation buffer, minimum ratios are: total Tier 1 8.5% and total capital (Tier 1 + Tier 2) 10.5% of RWA.
2. Total capital ratios: A 8/30 = 26.7%; B 11/40 = 27.5%; C 20/240 = 8.3%; D 35/230 = 15.2%.
3. Tier 1 ratios: A 16.7%; B 20.0%; C 15/240 = 6.25%; D 10.9%.
4. Only Bank C is below 10.5% (and below 8.5% on Tier 1); all others pass both tests.

Why C is correct: Bank C's total capital ratio (8.3%) is below the 10.5% requirement including the buffer, and its Tier 1 ratio (6.25%) is below 8.5%.

Why the others are wrong

- **A:** Ratios are 26.7% total and 16.7% Tier 1, well above 10.5% and 8.5%; picking it reflects dividing by capital instead of RWA or a miscalculation.
- **B:** Ratios are 27.5% total and 20.0% Tier 1, the strongest of the four.
- **D:** Ratios are 15.2% total and 10.9% Tier 1; comfortably above the requirements.

Q663

AnalystPrep Q.4295

Which of the following statements correctly describes Systemically Important Financial Institutions (SIFIs)?

A. They are entities subject to less supervision and regulation.
B. They are entities whose failure or distress would cause substantial problems for the whole financial system or economy.
C. They are entities whose failure affects only their own stakeholders but not the broader financial system or economy.
D. They are entities whose failure can be reversed by government financing without affecting their stakeholders.

Answer: B

Step by step

1. A SIFI is defined by the consequences of its failure: failure or distress would cause substantial problems in the financial system or real economy (e.g., Lehman).
2. G-SIBs are the subset of SIFIs designated globally by the FSB; they face extra capital buffers (1% to 3.5%) and TLAC requirements, i.e., more regulation.
3. Only B matches this definition.

Why B is correct: Systemic importance means that failure or distress spills over to the wider financial system or the real economy.

Why the others are wrong

- **A:** The opposite: SIFIs/G-SIBs face more, not less, regulation (additional buffers, TLAC).
- **C:** Describes a non-systemic institution whose failure is contained; the defining feature of a SIFI is spillover.
- **D:** Government bailouts are what post-crisis rules try to avoid; the definition concerns systemic impact, not reversibility of failure.

Q664

BT Mock Q8 · BT answers P2.T7.20.10.1 · BT notes P2.T7.20.10.1

The central feature of Basel III's first pillar is the regulatory capital (CET1, Tier 1, and total capital) requirements, which require minimum fractions of risk-weighted assets (RWA). In addition to these minimums, as of early 2019 (when the buffers were fully phased in), Basel specifies three buffers: the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB). Your colleague Mary makes the following five points:
I. For all three, the additional buffer must be CET1.
II. For all three, breach of the buffer requirements implies the bank's ability to pay dividends will be restricted.
III. The CCB requires an additional 2.5% of CET1 capital and is meant to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred.
IV. The additional G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, 3.5%} and is meant to reduce the likelihood and severity of the failure of a global systemically important financial institution.
V. The CCyB requirement varies between zero and 2.5% and is meant to protect the banking sector from periods of excess aggregate credit growth associated with the build-up of system-wide risks.
Is Mary correct in her summary?

A. No, none of her statements is correct.
B. I. and II. are correct, but III., IV. and V. are wrong.
C. III., IV. and V. are correct, but I. and II. are wrong.
D. Yes, all five statements are correct.

Answer: D

Step by step

1. I: CCB, G-SIB buffer and CCyB must all be met with CET1 (Tier 1 equity) - true.
2. II: breaching any buffer restricts dividends/bonuses (distribution constraints) - true.
3. III: CCB = 2.5% of RWA, a usable cushion drawn down in losses (PCA-style rationale) - true.
4. IV: G-SIB buckets are 1, 1.5, 2, 2.5 and 3.5% - true.
5. V: CCyB ranges 0% to 2.5% at national supervisors' discretion, to restrain excess credit growth - true.
6. All five are correct.

Why D is correct: Each statement matches the Basel III buffer framework: CET1-only buffers with distribution restrictions, a 2.5% CCB, 1-3.5% G-SIB buckets and a 0-2.5% CCyB.

Why the others are wrong

- **A:** Wrong: every statement is accurate, so 'none correct' fails.
- **B:** Accepts I and II but wrongly rejects III-V; the CCB size, G-SIB buckets and CCyB range are all correctly stated.
- **C:** Wrongly rejects I and II; the buffers must be CET1 and breach does restrict dividends.

Q665

BT answers P2.T7.521.3 · BT notes P2.T7.521.3

REBUILT — Original relied on Hull retention tables that were not reproduced in the stem; supplied the quartile retention schedule so the question is self-contained, and replaced the weak distractors.

A bank has a CET1 (Tier 1 equity) ratio of 6.0%. Distributions are constrained when the CET1 ratio lies inside the buffer zone above the 4.5% minimum. The zone is divided into four equal quartiles, and the minimum share of earnings that must be retained is:

Quartile of buffer zone (lowest to highest)	Minimum earnings retained
1st	100%
2nd	80%
3rd	60%
4th	40%
Above the buffer zone	0%

With no countercyclical buffer (CCyB), the buffer zone is the 2.5% conservation buffer (CET1 from 4.5% to 7.0%). With a 2.5% CCyB, the zone is 5.0% wide (CET1 from 4.5% to 9.5%). What is the maximum dividend, as a percent of earnings, that can be paid (i) with no CCyB and (ii) with a 2.5% CCyB?

A. (i) 60%; (ii) 80%
B. (i) 40%; (ii) 40%

C. (i) 40%; (ii) 20%
D. (i) 0%; (ii) 0%

Answer: C
Step by step
1. (i) Zone 4.5%-7.0%, width 2.5%, each quartile 0.625%: Q1 4.5-5.125, Q2 5.125-5.75, Q3 5.75-6.375, Q4 6.375-7.0.
2. 6.0% is in the 3rd quartile: retain at least 60%, so maximum dividend = 100% - 60% = 40%.
3. (ii) Zone 4.5%-9.5%, width 5.0%, each quartile 1.25%: Q1 4.5-5.75, Q2 5.75-7.0, Q3 7.0-8.25, Q4 8.25-9.5.
4. 6.0% is in the 2nd quartile: retain at least 80%, so maximum dividend = 20%.
5. Answer: 40% and 20%; a higher required buffer pushes the same ratio into a lower quartile and tightens payouts.

Why C is correct: The wider buffer zone with a CCyB places the same 6.0% CET1 ratio in a lower quartile, raising required retention from 60% to 80%.

Why the others are wrong
- **A:** Reports the retention rates (60%, 80%) rather than the maximum dividend (40%, 20%).
- **B:** Ignores the CCyB, so the zone is left at 4.5-7.0% for part (ii) as well.
- **D:** Assumes any ratio below the 7% target bars all dividends; in fact dividends are only limited, since 6.0% is above the 4.5% minimum and inside the buffer zone.

Q666
Schweser Module Quiz 63.2 Q2

The capital conservation buffer:
A. is intended to protect banks from the countercyclical nature of bank earnings.
B. can be set between 0.0% and 2.5% of risk-weighted assets, and is at the discretion of the regulators in individual countries.
C. causes the Tier 1 equity capital ratio requirement to increase to 7% of risk-weighted assets in normal economic periods.
D. requires that total capital to risk-weighted assets must be 10.5% at all times.

Answer: C
Step by step
1. The CCB is a fixed 2.5% of RWA Tier 1 equity buffer built in normal times and drawn down in stress.
2. Normal-time requirements: Tier 1 equity 4.5% + 2.5% = 7.0%; total Tier 1 8.5%; total capital 10.5%.
3. Hence C is right; A and B describe the CCyB, and D overstates 'at all times'.

Why C is correct: Adding the 2.5% buffer to the 4.5% CET1 minimum lifts the normal-period Tier 1 equity requirement to 7%.

Why the others are wrong
- **A:** Confuses the CCB with the countercyclical buffer (CCyB), which addresses cyclicality.
- **B:** Describes the CCyB (0-2.5%, national discretion); the CCB is a fixed 2.5% requirement.
- **D:** 10.5% is the normal-period requirement, but the buffer is meant to be drawn down in stress, so it is not required at all times.

Q667
AnalystPrep Q.2360

REBUILT — Original had joke distractors (invented acronym meanings) that gave away the answer; rewrote options as realistic confusions between G-SIB, CCB and CCyB features.

CIB Bank is designated a global systemically important bank (G-SIB) by the Financial Stability Board. Which statement best describes the designation and its consequences?
A. Its distress would cause substantial problems for the global financial system or economy, so it must hold an additional capital buffer of 1% to 3.5% of RWA, depending on its systemic importance score.
B. Its distress would cause substantial problems for the global financial system, so its G-SIB buffer replaces the 2.5% capital conservation buffer.
C. It is designated because its total assets exceed a fixed threshold, and national supervisors set an additional buffer of 0% to 2.5% at their discretion.
D. Because governments are expected to rescue it, it is exempt from the capital conservation buffer and dividend restrictions.

Answer: A

Step by step

1. G-SIBs are the FSB-designated subset of SIFIs: failure or distress would cause substantial problems in the financial system or real economy.
2. The FSB scores banks on size, interconnectedness, complexity, international activity and other factors; the score determines the bucket.
3. The G-SIB buffer is 1%, 1.5%, 2%, 2.5% or 3.5% of RWA, in addition to the 2.5% CCB (and any CCyB).
4. Only A states this correctly.

Why A is correct: G-SIB status reflects systemic importance, and bears an extra 1-3.5% buffer on top of the other Basel III requirements.

Why the others are wrong

- **B:** The G-SIB buffer is additional to the CCB, not a substitute.
- **C:** Mixes in the CCyB (0-2.5%, national discretion); designation uses a multi-factor score, not a single size threshold.
- **D:** Reverse of the intent: reforms aim to end 'too big to fail' bailouts, and G-SIBs face more, not less, restriction.

Q668

BT answers P2.T7.20.11.3 · BT notes P2.T7.20.11.3

The Basel Committee developed a methodology for assigning global systemically important banks (G-SIBs) to one of five buckets:

Bucket	Score Range	Higher loss absorbency requirement (CET1 as % of RWA)
5	530-629	3.50%
4	430-529	2.50%
3	330-429	2.00%
2	230-329	1.50%
1	130-229	1.00%

As of November 2019, Bank of America (ticker: BAC) was a $2.4 trillion G-SIB assigned to Bucket 2. There were no banks in Bucket 5, and JP Morgan Chase was the only bank in Bucket 4. According to Basel III, if we include the buffer requirements due to its status as a G-SIB, what is Bank of America's minimum common equity tier 1 (CET1) risk-weighted ratio, and its minimum leverage ratio (excluding the countercyclical buffer, CCyB, which can anyhow be zero)?

A. BAC requires a minimum CET1 risk-weighted ratio of 7.00% and a minimum leverage ratio of 3.00%

B. BAC requires a minimum CET1 risk-weighted ratio of 8.50% and a minimum leverage ratio of 3.75%

C. BAC requires a minimum CET1 risk-weighted ratio of 10.50% and a minimum leverage ratio of 4.50%

D. BAC requires a minimum CET1 risk-weighted ratio of 10.50% and a minimum leverage ratio of 5.25%

Answer: B

Step by step

1. Minimum CET1 ratio in normal times = 4.5% + 2.5% capital conservation buffer = 7.0% of RWA.
2. Bucket 2 G-SIB surcharge (higher loss absorbency) = 1.50% of RWA, met with CET1.
3. CET1 requirement = 7.0% + 1.5% = 8.5% (the CCyB is excluded as it can be zero).
4. The G-SIB leverage ratio buffer = one-half of the risk-based G-SIB surcharge (excluding the CCB and CCyB) = 0.5 × 1.5% = 0.75%.
5. Minimum leverage ratio = 3.00% + 0.75% = 3.75%. The answer is B.

Why B is correct: A G-SIB must add its bucket surcharge (1.5%) to the 7.0% CET1 minimum (4.5% + 2.5% CCB), and its leverage buffer is half the risk-based surcharge, giving 8.5% and 3.75%.

Why the others are wrong

- **A:** Gives the standard non-G-SIB figures (4.5% + 2.5% = 7.0% and 3.0% leverage), ignoring the G-SIB surcharge entirely.
- **C:** Reaches 10.5% by using the 10.5% total-capital (Tier 1 + Tier 2) ratio or by adding the 3.5% Bucket 5 surcharge to 7.0% instead of the 1.5% Bucket 2 surcharge; the 4.5% leverage figure then adds a 1.5% buffer (the full surcharge rather than half).
- **D:** Repeats the incorrect 10.5% CET1 figure, and computes leverage as half of 10.5% (5.25%) instead of 3% plus half of the 1.5% surcharge.

LO 63.f

Describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio.

Q669

AP Mock 2 Q21

REBUILT — Original distractors (92%, 98%) were not tied to any realistic error; options replaced with results of specific mistakes.

A bank has the following balance sheet (all figures in millions):

Assets	Amount	RSF factor
Cash	20	0%
Treasury bonds (> 1 yr)	20	5%
Mortgages	50	65%
Small business loans	180	85%
Fixed assets	30	100%
Total	300	
Liabilities & capital	Amount	ASF factor
Retail deposits (stable)	80	90%
Wholesale deposits	100	50%
Tier 2 capital	40	100%
Tier 1 capital	80	100%
Total	300	

Which of the following is closest to the bank's net stable funding ratio?

A. 80.7%.
B. 89.5%.
C. 100.0%.
D. 111.8%.

Answer: D

Step by step

1. ASF = 80(0.90) + 100(0.50) + 40(1.00) + 80(1.00) = 72 + 50 + 40 + 80 = 242.
2. RSF = 20(0) + 20(0.05) + 50(0.65) + 180(0.85) + 30(1.00) = 0 + 1 + 32.5 + 153 + 30 = 216.5.
3. NSFR = 242/216.5 = 111.8%, above the 100% minimum.

Why D is correct: NSFR is weighted available stable funding divided by weighted required stable funding; both sides are weighted, not taken at book value.

Why the others are wrong

- **A:** Divides ASF by total unweighted assets (242/300).
- **B:** Inverts the ratio (216.5/242).
- **C:** Assumes a balanced balance sheet (300 = 300) implies an NSFR of 100%; the factors make ASF and RSF differ.

Q670

AnalystPrep Q.4296

The liquidity coverage ratio (LCR) of a bank is approximated to be 1.30. Under Basel III liquidity requirements, does the bank fulfill the required LCR?

A. No, because LCR > 1
B. Yes, because LCR > 1
C. No, because LCR < 2
D. Yes, because LCR < 2

Answer: B

Step by step

1. Basel III requires LCR = HQLA/net 30-day outflows ≥ 100% (i.e. 1.0).
2. 1.30 > 1, so the requirement is met; there is no upper limit.

Why B is correct: The LCR minimum is 100%; any ratio above 1 satisfies it.

Why the others are wrong

- **A:** Right test (LCR > 1) but the wrong conclusion; exceeding 1 is compliant.
- **C:** Invents a minimum of 2.
- **D:** Reaches the right conclusion for the wrong reason; there is no cap of 2.

Q671

AP Mock 3 Q16

REBUILT — Source balance sheet did not balance (funding 100 vs assets 117.5); wholesale deposits changed from 50 to 67.5 so the sheet balances, and answer/options recomputed.

Goodwill Bank's balance sheet contains the following items (in millions), with available stable funding (ASF) and required stable funding (RSF) factors shown:

Funding source	Amount	ASF factor
Retail deposits	35	90%
Wholesale deposits	67.5	50%
Tier 2 capital	5	100%
Tier 1 capital	10	100%
Asset	Amount	RSF factor
Cash	7	0%
Mortgages	38	65%
Treasury bonds	6.5	5%
Small business loans	54	85%
Fixed assets	12	100%

Which of the following is closest to the bank's net stable funding ratio?

A. 68.3%.
B. 96.8%.
C. 103.3%.
D. 141.7%.

Answer: B

Step by step

1. ASF = 35(0.90) + 67.5(0.50) + 5(1.00) + 10(1.00) = 31.5 + 33.75 + 5 + 10 = 80.25.
2. RSF = 7(0) + 38(0.65) + 6.5(0.05) + 54(0.85) + 12(1.00) = 0 + 24.7 + 0.325 + 45.9 + 12 = 82.925.
3. NSFR = 80.25/82.925 = 96.8%, below the 100% minimum.

Why B is correct: NSFR = weighted available stable funding / weighted required stable funding; here ASF is smaller than RSF so the ratio is below 100%.

Why the others are wrong

- **A:** Divides ASF by total assets (80.25/117.5) instead of by RSF.
- **C:** Inverts the ratio (RSF/ASF = 82.925/80.25).
- **D:** Uses unweighted funding (117.5) over RSF (82.925), ignoring the ASF factors.

Q672

AnalystPrep Q.2358

Berthold Bruhne, a risk manager for the bank of Salzburg, presented the results of the liquidity coverage ratio (LCR) calculation to the board. According to him, the bank's LCR stood at 152% as of December 31st, 2016, safely above the required minimum. His conclusion was that the bank could survive liquidity disruptions in the next:

A. 1 year
B. 60 days
C. 30 days
D. 15 days

Answer: C

Step by step

1. LCR = HQLA / net cash outflows over a 30-day stress period, with a minimum of 100%.
2. 152% > 100%, so HQLA cover the stressed net outflows over that 30-day horizon.

Why C is correct: The LCR is defined over a 30-day severe stress period.

Why the others are wrong

- **A:** One year is the horizon of the NSFR, not the LCR.
- **B:** The LCR has no 60-day horizon.
- **D:** The LCR horizon is 30 days, not 15.

Q673 AnalystPrep Q.2359

In line with Basel III, the LCR is calculated as the:

A. Ratio between stable funding and high-quality liquid assets.
B. Ratio between high-quality liquid assets and total assets.
C. Ratio between high-quality liquid assets and net cash outflows in a 30-day period.
D. Ratio between stable funding and net cash outflows in a 30-day period.

Answer: C

Step by step

1. $\text{LCR} = \dfrac{\text{HQLA}}{\text{net cash outflows over 30 days}} \geq 100\%$.

Why C is correct: The LCR compares the stock of high-quality liquid assets to stressed 30-day net cash outflows.

Why the others are wrong

- **A:** Mixes the NSFR's stable funding with HQLA and is not a Basel ratio.
- **B:** Total assets are not the denominator; that resembles a leverage-type measure.
- **D:** Stable funding over outflows is not a Basel ratio; the NSFR compares available to required stable funding.

Q674 AnalystPrep Q.2743

Which of the following correctly describes the time horizon considered by the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR)?

A. LCR: Focuses on a 30-day period; NSFR: Focuses on a 2-year period.
B. LCR: Focuses on a 1-year period; NSFR: Focuses on a 30-day period.
C. LCR: Focuses on a 2-year period; NSFR: Focuses on a 30-day period.
D. LCR: Focuses on a 30-day period; NSFR: Focuses on a 1-year period.

Answer: D

Step by step

1. LCR: ability to withstand a 30-day period of disrupted liquidity.
2. NSFR: ability to manage liquidity over a period of one year.

Why D is correct: The LCR is a short-term (30-day) measure; the NSFR is a longer-term (one-year) structural measure.

Why the others are wrong

- **A:** Correct LCR horizon but the NSFR horizon is one year, not two.
- **B:** Swaps the two horizons.
- **C:** Swaps the horizons and uses a 2-year period that appears nowhere in Basel III.

Q675 AnalystPrep Q.3237

REBUILT — Distractor options 0.875, 1.1 and 1.4 were not tied to realistic errors; options replaced with results of plausible misuse of the given data.

Exim Bank estimates its stable funding to be $100 million. Net cash outflows over the coming 30 days are estimated at $155 million. Exim Bank has capital of $10 million and total exposure of $150 million. The bank's high-quality liquid assets are valued at $140 million. What is the bank's liquidity coverage ratio (LCR) as stipulated in Basel III?

A. 0.645
B. 0.903
C. 0.933
D. 1.107

Answer: B

Step by step

1. LCR = HQLA/net 30-day cash outflows.
2. = 140/155 = 0.9032, i.e. 90.3%, below the 100% minimum.

Why B is correct: Only HQLA and 30-day net outflows enter the LCR; stable funding, capital and total exposure are irrelevant.

Why the others are wrong

- **A:** Uses stable funding in the numerator (100/155).
- **C:** Uses total exposure as the denominator (140/150).

- **D:** Inverts the ratio (155/140).

Q676

AnalystPrep Q.3250

REBUILT — Source sign convention was sloppy (net outflow shown as 214-487) and options C and D were not tied to realistic errors; options rebuilt around genuine mistakes.

A bank is checking its Basel III liquidity ratios. Given the following information (in $), what is the bank's liquidity coverage ratio?

High-quality liquid assets = 236
Marketable securities = 107
Required amount of stable funding (RSF) = 320
Cash inflows over the next 30 days = 214
Cash outflows over the next 30 days = 487
Long-term economic capital = 640
Available amount of stable funding (ASF) = 305

A. 48.46%
B. 86.45%
C. 110.28%
D. 125.64%

Answer: B

Key correction: Source printed net outflow as 214-487 = -273; the correct net outflow is +273 (outflows minus inflows). Final 86.45% is unchanged.

Step by step

1. Net 30-day cash outflows = 487 - 214 = 273.
2. LCR = 236/273 = 86.45%, below the 100% minimum.
3. RSF, ASF, capital and marketable securities are irrelevant to the LCR as given.

Why B is correct: LCR = HQLA / (outflows - inflows) over 30 days; only those three items are needed.

Why the others are wrong

- **A:** Ignores inflows (236/487).
- **C:** Divides HQLA by inflows (236/214).
- **D:** Adds marketable securities to HQLA ((236+107)/273); they are not stated as HQLA.

Q677

AnalystPrep Q.3950

REBUILT — Source table was garbled beyond reliable recovery; rebuilt as a clean, self-contained NSFR calculation with a consistent balance sheet.

A bank has the following balance sheet (in millions) with ASF and RSF factors:

Funding	Amount	ASF factor
Capital	100	100%
Stable retail deposits	250	90%
Wholesale funding (< 1 year)	550	50%
Long-term debt (> 1 year)	100	100%
Assets	Amount	RSF factor
Cash	50	0%
Liquid securities	100	5%
Mortgages	350	65%
Corporate loans	380	85%
Fixed assets	120	100%

What is the bank's net stable funding ratio?

A. 70.0%.
B. 96.5%.
C. 103.6%.
D. 148.0%.

Answer: C

Step by step

1. ASF = 100(1.00) + 250(0.90) + 550(0.50) + 100(1.00) = 100 + 225 + 275 + 100 = 700.
2. RSF = 50(0) + 100(0.05) + 350(0.65) + 380(0.85) + 120(1.00) = 0 + 5 + 227.5 + 323 + 120 = 675.5.

3. NSFR = 700/675.5 = 103.6%, above the 100% minimum.

Why C is correct: NSFR = weighted available stable funding / weighted required stable funding; ASF of 700 exceeds RSF of 675.5.

Why the others are wrong

- **A:** Divides ASF by total unweighted funding (700/1000).
- **B:** Inverts the ratio (675.5/700).
- **D:** Uses unweighted total funding over RSF (1000/675.5).

Q678

AnalystPrep Q.3952

REBUILT — Source options did not correspond to realistic errors; options rebuilt so each wrong choice reflects a specific mistake.

The following table is extracted from ABC bank's balance sheet. Determine the Liquidity Coverage Ratio of the bank for 2018 and 2019, respectively.

Description	2018	2019
High-quality liquid assets	2100	1458
Average monthly withdrawals	1200	2458
Expected monthly outflows in a stressed scenario	1600	1700

A. 175% and 59%

B. 131% and 86%

C. 76% and 117%

D. 75% and 35%

Answer: B

Step by step

1. LCR = HQLA / stressed 30-day net outflows.
2. 2018: 2100/1600 = 131.25%.
3. 2019: 1458/1700 = 85.76%.
4. Average historical withdrawals are not the stress-scenario outflows.

Why B is correct: The LCR denominator is the expected net outflows in a stressed 30-day scenario, not average historical withdrawals.

Why the others are wrong

- **A:** Uses average withdrawals as the denominator (2100/1200, 1458/2458).
- **C:** Inverts the ratio (1600/2100, 1700/1458).
- **D:** Adds withdrawals and stressed outflows in the denominator (2100/2800, 1458/4158).

Q679

BT answers P2.T7.20.10.2 · BT notes P2.T7.20.10.2

Below is a summary balance sheet of Acme Bank, but in addition to the typical accounts (assets, liabilities, and equities), there are four additional columns to assist in ascertaining whether the bank satisfies Basel III's liquidity risk requirements:

Assets (millions)	Amount	HQLA factor (1)	HQLA	RSF factor	Required stable funding
Cash	$50.0	100.0%	$50.0	0.0%	$0.0
Sovereign debt	40.0	100.0%	40.0	5.0%	$2.0
BBB securities	110.0	50.0%	55.0	50.0%	$55.0
Loans	800.0	0.0%	0.0	85.0%	$680.0
Sum	$1,000.0	| $145.0	| $737.0		

(1) HQLA factor = 1 - haircut

Liabilities (millions)	Amount	Runoff rate	Net cash outflows	ASF factor	Available stable funding
Retail deposits	$450.0	5.0%	22.5	95.0%	427.5
Wholesale CDs (2)	300.0	100.0%	100.0	0.0%	0.0
Senior bonds	150.0	0.0%	0.0	100.0%	150.0
Common equity	100.0	0.0%	0.0	100.0%	100.0
Sum	$1,000.0	| $122.5	| $677.5		

(2) Run-off rate applied to 1/3rd of wholesale CDs that mature in the next month.

For purposes of estimating the bank's liquidity coverage ratio (LCR), the following columns are included: HQLA factors, HQLA, runoff rate, and net cash outflows. For purposes of estimating the net stable funding ratio (NSFR), the following columns are included: RSF factor, required stable funding, ASF factor, and available stable funding.

Does Acme Bank meet the LCR and NSFR requirements?

A. Neither LCR nor NSFR is met
B. LCR is met, but NSFR fails to meet the minimum
C. NSFR is met, but LCR fails to meet the minimum
D. Both LCR and NSFR are met

Answer: B

Step by step

1. LCR = HQLA / net 30-day cash outflows = 145.0/122.5 = 1.184, which is at least 100%, so the LCR is met.
2. NSFR = available stable funding / required stable funding = 677.5/737.0 = 0.919, below 100%, so the NSFR fails.

Why B is correct: Both Basel III liquidity ratios must be at least 100%. The LCR is 118.4% (met) but the NSFR is 91.9% (shortfall).

Why the others are wrong

- **A:** LCR is actually met (1.184); saying it fails reflects inverting the ratio (122.5/145 = 0.845).
- **C:** Reverses the result: LCR passes and NSFR fails, not the other way round.
- **D:** NSFR is 677.5/737.0 = 0.919 < 1; this answer inverts the NSFR (737.0/677.5 = 1.088) or ignores the 100% floor.

Q680

BT answers P2.T7.25.2.1

REBUILT — Original options were management actions that did not answer the stem's question and the source key was inconsistent; rebuilt as a metric-identification question.

Apex Global Bank is evaluating its resilience under economic stress. Baseline and stressed metrics are:

Financial Metric	Baseline	Stressed
Common Equity Tier 1 ratio (CET1)	12%	9.7%
Liquidity coverage ratio (LCR)	120%	92%
Commercial real estate exposure	$1.2 billion	$0.9 billion
Short-term wholesale funding (share of funding)	25%	20%

According to Basel III, which stressed outcome indicates that the bank has breached a minimum regulatory liquidity requirement?

A. CET1 ratio falling from 12% to 9.7%
B. LCR falling from 120% to 92%
C. Short-term wholesale funding share falling from 25% to 20%
D. Commercial real estate exposure falling from $1.2 billion to $0.9 billion

Answer: B

Key correction: Source key (BT answers) said C, but its own explanation describes a different option and the item was ill-posed; the correct finding is the stressed LCR of 92% < 100%.

Step by step

1. Basel III requires LCR = HQLA / 30-day net cash outflows ≥ 100%.
2. Stressed LCR = 92% < 100%, a breach of a liquidity minimum.
3. CET1 of 9.7% is a capital ratio and is above the 4.5% minimum (and the 7% minimum including the conservation buffer); wholesale funding share and CRE exposure have no Basel minimum and both declined.

Why B is correct: Only the LCR has a Basel III liquidity minimum (100%), and the stressed 92% falls below it.

Why the others are wrong

- **A:** CET1 is a capital, not liquidity, measure and 9.7% still exceeds the 4.5% minimum and 7% with the buffer.
- **C:** Basel III sets no minimum on wholesale funding share; a lower share actually reduces liquidity risk.
- **D:** No regulatory liquidity minimum applies to CRE exposure; a lower exposure is not a breach.

Q681

BT answers P2.T7.522.1 · BT notes P2.T7.522.1

What is the key difference between the Basel III leverage ratio and the other regulatory ratios (i.e., core Tier 1 equity capital ratio, total Tier 1 capital ratio, and total capital ratio)?

A. The numerator of the leverage ratio is not regulatory capital; rather, it is adjusted for the risk of the funding source
B. The denominator of the leverage ratio is assets on the balance sheet without risk weighting (plus some off-

balance-sheet items such as loan commitments)

C. The leverage ratio is defined in terms of cash inflows and cash outflows and therefore provides an additional buffer against liquidity risk

D. There is no difference in the definition of the ratio; instead, the leverage ratio is simply the ”placeholder ratio” which applies until the other regulatory ratios are fully phased-in in order to guard against systemic risk

Answer: B

Step by step

1. Core Tier 1, total Tier 1 and total capital ratios divide capital by risk-weighted assets.
2. The Basel III leverage ratio = capital / total exposure with a 3% minimum; exposure includes all balance sheet items unweighted plus some off-balance-sheet items such as loan commitments.

Why B is correct: The leverage ratio is a non-risk-based backstop: its denominator is unweighted exposure rather than risk-weighted assets.

Why the others are wrong

- **A:** The numerator is still capital; no adjustment is made for the risk of the funding source (that describes ASF factors in the NSFR).
- **C:** Describes the LCR (cash inflows and outflows), not the leverage ratio.
- **D:** The leverage ratio is a permanent Basel III requirement with a different denominator, not a placeholder.

Q682

BT answers P2.T7.522.2 · BT notes P2.T7.522.2

REBUILT — Option D covered CVA, which is outside LO 63.f; replaced with a statement on the leverage ratio.

Regarding the leverage ratio, liquidity coverage ratio (LCR) and net stable funding ratio (NSFR), each of the following is true EXCEPT which is not accurate?

A. The LCR is the ratio of high-quality liquid assets to net cash outflows over a stressed 30-day period.

B. The NSFR is the ratio of weighted stable funding sources (liabilities and equity) to weighted assets requiring stable funding.

C. The NSFR will increase if retail deposits are replaced by wholesale deposits and marketable securities are replaced by residential mortgages.

D. The Basel III leverage ratio is capital divided by total exposure, which includes unweighted balance sheet items and some off-balance-sheet items, with a 3% minimum.

Answer: C

Step by step

1. A: LCR = HQLA / 30-day net outflows, true.
2. B: NSFR = ASF (liability and equity side) / RSF (asset side), true.
3. C: wholesale deposits have lower ASF factors than retail deposits, and mortgages have higher RSF factors than marketable securities, so ASF falls and RSF rises; the NSFR decreases, so C is false.
4. D: matches the Basel III definition, true.

Why C is correct: Replacing stable funding with less stable funding and liquid assets with less liquid assets lowers the NSFR.

Why the others are wrong

- **A:** True statement of the LCR.
- **B:** True statement of the NSFR.
- **D:** True statement of the leverage ratio.

Q683

BT answers P2.T8.20.3.2 · BT notes P2.T8.20.3.2

REBUILT — Original stem asked which liquidity ratio has ’regulatory capital in its numerator’, which made B (capital is an ASF component) and D (neither numerator is capital itself) both defensible; reworded to ask in which ratio’s numerator Tier 1 and Tier 2 capital is counted as a funding source with a 100% factor, so only B is defensible.

Under Basel III, in which liquidity ratio’s numerator is the bank’s Tier 1 and Tier 2 capital counted as a funding source, receiving a 100% weighting factor?

A. The liquidity coverage ratio only

B. The net stable funding ratio only

C. Both the liquidity coverage ratio and the net stable funding ratio

D. Neither ratio; capital appears only in the numerator of the leverage ratio

Answer: B

Step by step

1. LCR = HQLA / net cash outflows over 30 days ≥ 100%: the numerator is a stock of high-quality liquid assets (e.g., cash, central bank reserves, zero-risk-weight sovereign debt, corporate debt after a 50% haircut), so no funding source such as capital enters it.
2. NSFR = available stable funding / required stable funding ≥ 100%: the numerator multiplies each funding source (retail deposits, wholesale funding, capital, and so on) by an ASF factor.
3. Tier 1 and Tier 2 capital (with preferred stock and debt maturing beyond 1 year) receive a 100% ASF factor, so capital is counted in full in the NSFR numerator only.

Why B is correct: The NSFR numerator is built from the liability-and-equity side of the balance sheet, and capital is the most stable funding source, so its ASF factor is 100%. The LCR numerator is built from liquid assets, so capital never enters it.

Why the others are wrong

- **A:** Confuses the LCR numerator (HQLA, a stock of liquid assets) with the funding side of the balance sheet, where capital sits.
- **C:** The LCR numerator contains only liquid assets after haircuts and does not count funding sources such as capital.
- **D:** Capital is the numerator of the leverage ratio (minimum 3%), but it is also counted, with a 100% ASF factor, in the NSFR's available stable funding, so 'neither' is wrong.

Q684

Mock 1 Q10

REBUILT — Original options (117.0% vs exact 116.8%) were imprecise and weak; options recomputed from realistic errors.

Bank BCCA has the following balance sheet:

Assets	Amount
Cash	15
Central bank reserves	12
Treasury bonds > 1 year	15
Mortgages	30
Retail loans < 1 year	32
Small-business loans < 1 year	94
Fixed assets	22
Total assets	220
Liabilities & equity	Amount
Retail deposits (less stable)	130
Wholesale deposits	65
Tier 2 capital	5
Tier 1 capital	20
Total liabilities & equity	220

Available stable funding (ASF) factors: 100% Tier 1 and Tier 2 capital, preferred stock, debt > 1 year; 95% "Stable" demand and term deposits from individuals/businesses < 1 year; 90% "Less stable" demand and term deposits from individuals/businesses < 1 year; 50% Wholesale funding (demand and term deposits) < 1 year; 0% All other liability and equity categories.

Required stable funding (RSF) factors: 0% Cash and short-term instruments; 5% Marketable securities (> 1 year, sovereign 0% risk weight); 15% Corporate bonds rated AA- or higher; 50% Bonds rated A+ to A-; 65% Residential mortgages; 85% Loans to small businesses or retail customers < 1 year; 100% All other assets.

What is Bank BCCA's net stable funding ratio (NSFR)?

A. 85.6%
B. 95.1%
C. 121.2%
D. 116.8%

Answer: D

Key correction: Source key (Mock 1) said D at 117.0%; the value is kept but the option is made exact at 116.8%.

Step by step

1. ASF = 20 + 5 + 130 x 0.90 + 65 x 0.50 = 20 + 5 + 117 + 32.5 = 174.5.
2. RSF = 15 x 0 + 12 x 0 + 15 x 0.05 + 30 x 0.65 + (32 + 94) x 0.85 + 22 x 1.00 = 0.75 + 19.5 + 107.1 + 22 = 149.35.
3. NSFR = 174.5/149.35 = 116.8%.

Why D is correct: NSFR = ASF / RSF using the given factors; ASF 174.5 exceeds RSF 149.35, so the ratio is 116.8%.

Why the others are wrong

- **A:** Inverted the ratio (RSF/ASF = 149.35/174.5 = 85.6%).
- **B:** Omitted the wholesale deposits' 50% ASF (32.5), giving 142/149.35.
- **C:** Applied 95% (stable) rather than 90% (less stable) to retail deposits: 1.212.

Q685

Mock 1 Q50

REBUILT — Original used a Level 2A 15% haircut and a 75% inflow cap not covered by the reading; replaced with the reading's convention (corporate debt 50% haircut) and a net-outflow LCR.

A bank is calculating its liquidity coverage ratio (LCR) using Basel III conventions as presented in the reading. Data (USD million):

Item	Amount
Treasury securities (zero risk weight)	90
Corporate debt (50% haircut for HQLA)	40
Stable retail deposits	500
Unsecured wholesale funding from non-financial corporates	300
Expected cash inflows over 30 days	60

Assume a 5% run-off rate on stable retail deposits and a 40% run-off rate on the wholesale funding. What is the bank's LCR?

A. 75.9%.
B. 105.9%.
C. 129.4%.
D. 152.9%.

Answer: C

Key correction: Source key (C, 145.9%) relied on a 15% Level 2A haircut and a 75% inflow cap outside the reading's conventions; the item was rebuilt, so the key changes with the data.

Step by step

1. HQLA: Treasuries count in full = 90; corporate debt 40 Ã— (1 â˜ 0.50) = 20; HQLA = 110.
2. Outflows: 500 Ã— 0.05 = 25; 300 Ã— 0.40 = 120; total 145.
3. Net 30-day cash outflows = 145 â˜ 60 = 85.
4. LCR = 110/85 = 129.4%, above the 100% minimum.

Why C is correct: LCR = HQLA (after haircuts) divided by net cash outflows over a 30-day stress period; both numerator and denominator must be adjusted before dividing.

Why the others are wrong

- **A:** 110/145: used gross outflows and ignored the 60 of inflows.
- **B:** 90/85: excluded the corporate debt from HQLA altogether instead of counting it at 50%.
- **D:** 130/85: counted the corporate debt at face value, forgetting the 50% haircut.

Q686

Mock 4 Q74

REBUILT — Original mixed credit, FX and liquidity effects with a hedged correct answer; rebuilt to test the LCR effect of swapping HQLA government bonds for loans.

A bank sells a large portion of its holdings of zero-risk-weight government bonds and uses all the proceeds to extend new corporate loans, with no change in its funding. Which of the following statements about the effect on its liquidity ratios is most accurate?

A. The LCR will increase because loans earn a higher return than government bonds.
B. The LCR will decrease because government bonds count as HQLA without a haircut, whereas corporate loans do not count as HQLA.
C. The NSFR will improve because loans are short-term assets that need little stable funding.
D. Neither ratio is affected because total assets are unchanged.

Answer: B

Step by step

1. Zero-risk-weight government bonds are HQLA with no haircut, so they count in full in the LCR numerator.
2. Loans are not HQLA (factor of 0%), so converting bonds into loans reduces HQLA.
3. Net 30-day cash outflows are unchanged because funding is unchanged, so LCR = HQLA / net outflows falls.

4. In the NSFR, loans carry higher required stable funding factors than government bonds, so the NSFR also worsens.

Why B is correct: The swap removes HQLA from the LCR numerator while leaving the stressed outflows unchanged, so the LCR falls.

Why the others are wrong

- **A:** Confuses yield with liquidity; the LCR depends on HQLA quality, not return.
- **C:** Reverses the logic: loans require more stable funding than government bonds, so the NSFR worsens.
- **D:** Treats asset composition as irrelevant; the ratios depend on the liquidity of assets, not total size.

Q687

Schweser Module Quiz 63.2 Q3

Highlands Bank has estimated stable funding in the bank to be $100 million. The bank estimates that net cash outflows over the coming 30 days will be $137 million. The bank has capital of $5 million and a total exposure of $140 million. The bank estimates that it has high-quality liquid assets of $125 million. What is the bank's liquidity coverage ratio (LCR)?

A. 89.3%.
B. 91.2%.
C. 73.0%.
D. 3.6%.

Answer: B

Step by step

1. LCR = HQLA / net cash outflows over 30 days.
2. 125/137 = 0.912 = 91.2%.
3. This is below the 100% Basel III minimum.

Why B is correct: Only HQLA and 30-day net outflows enter the LCR; the other figures are for other ratios.

Why the others are wrong

- **A:** 125/140: divided HQLA by total exposure instead of net outflows.
- **C:** 100/137: used stable funding (an NSFR input) as the numerator.
- **D:** 5/140: this is the leverage ratio (capital / exposure), not the LCR.

Q688

AnalystPrep Q.3245

REBUILT — Original added derivative and repo exposures to total assets (likely double counting) and had no clearly stated exposure measure; rebuilt with explicit on- and off-balance-sheet items.

Bank ABC is subject to Basel III. Balance sheet information (in millions):

Item	Amount
Tier 1 capital	$250
Tier 2 capital	$150
Total on-balance-sheet assets (all items, not risk-weighted)	$2,000
Off-balance-sheet loan commitments	$300

Assume the loan commitments enter the leverage exposure measure at full face value (100% conversion factor) and Tier 1 capital is the capital measure. What is the bank's Basel III leverage ratio?

A. 10.87%.
B. 12.50%.
C. 17.39%.
D. 20.00%.

Answer: A

Key correction: Source key (B, 10.20%) added overlapping exposure items to total assets; the rebuilt data give 10.87%.

Step by step

1. Leverage ratio = capital / total exposure, with capital measured as Tier 1.
2. Total exposure = 2,000 + 300 = 2,300 (on-balance-sheet assets in full plus loan commitments at the stated 100% conversion factor, no risk weighting).
3. 250/2300 = 10.87%, above the 3% minimum.

Why A is correct: The leverage ratio divides Tier 1 capital by unweighted total exposure including off-balance-sheet items such as loan commitments.

Why the others are wrong
- **B:** 250/2,000: omitted the off-balance-sheet loan commitments from exposure.
- **C:** 400/2,300: used total capital (Tier 1 + Tier 2) instead of Tier 1.
- **D:** 400/2,000: both errors, total capital and no off-balance-sheet items.

LO 63.g

Describe the mechanics of contingent convertible bonds (CoCos) and explain the motivations for banks to issue them.

Q689

AnalystPrep Q.5611

REBUILT — Original option C was loosely worded and its source explanation was off-topic; restated around the Schweser/GARP motivation (debt in normal times, equity in stress, private-sector capital).

Which of the following best describes the main motivation for a bank to issue contingent convertible bonds (CoCos)?

A. They are debt in normal periods, so they do not dilute equity or drag down return on equity, but they become equity in stress and so absorb losses using private-sector rather than government funds.

B. They give the bank a higher return on equity in stress periods because bondholders can choose when to convert into shares.

C. They are cheaper than common equity because CoCo holders accept losses only after all equity holders have been wiped out.

D. They let the bank defer all loss recognition because the bonds are treated as equity from the date of issue.

Answer: A

Step by step
1. Under Schweser/GARP, a CoCo is a debt instrument when issued, so it does not appear in the equity account and does not reduce ROE in normal times.
2. When a trigger is hit, the bond converts to equity (or is written down), raising equity just when losses occur.
3. The capital cushion comes from private bondholders, which helps the bank avoid a government bailout.
4. Only A states both the normal-period and stress-period features.

Why A is correct: CoCos combine a debt-like accounting treatment in good times with automatic equity creation in distress, so the loss cushion is funded privately.

Why the others are wrong
- **B:** Conversion is automatic via a trigger, not at the holder's option (that describes traditional convertibles).
- **C:** CoCo holders bear losses similar to equity holders but get little upside, so CoCos should be expensive, not cheaper than equity; also they are not junior to equity.
- **D:** CoCos are not equity until converted; the accounting advantage arises precisely because they are not in the equity account before conversion.

Q690

AnalystPrep Q.5616

REBUILT — Removed the vague 'systematic risk component' wording and made distractors cleaner.

Which feature makes contingent convertible bonds (CoCos) attractive to some investors despite their risks?

A. They guarantee repayment of principal even if the issuing bank fails.

B. They pay a risk-free return that does not depend on the bank's condition.

C. They offer higher yields than ordinary bonds as compensation for bearing equity-like losses if converted or written down.

D. They rank ahead of senior debt in bankruptcy.

Answer: C

Step by step
1. CoCo holders can suffer conversion into equity or a write-down when a trigger is hit, so they bear losses similar to equity holders in failure.
2. They receive little of equity's upside, but investors demand a higher coupon for this contingent loss risk.
3. Hence the attraction is the higher yield; the other options contradict the loss-absorbing nature.

Why C is correct: The yield premium compensates investors for contingent loss absorption.

Why the others are wrong
- **A:** CoCos carry no principal protection; they absorb losses in distress.
- **B:** Returns depend on the bank staying above the trigger; they are not risk free.
- **D:** CoCos are subordinated loss-absorbing debt, not senior to other debt.

Q691

AnalystPrep Q.5617

REBUILT — Original keyed answer (management incentive alignment) is not supported by Schweser/GARP; rebuilt on the GARP statement that equity rises via conversion or write-down.

According to GARP, how do contingent convertible bonds (CoCos) cause a bank's equity to increase when distress occurs?

A. Bondholders exercise a conversion option when the share price rises above a threshold set in the indenture.
B. Triggers written into the indenture lead the bond either to convert into equity or to be written down, regardless of the holder's wishes.
C. The bank must repurchase the bonds at par, funding the repurchase with new government equity.
D. Interest on the bonds is suspended permanently and the principal becomes a deposit.

Answer: B

Step by step
1. GARP calls CoCos the mirror image of traditional convertibles.
2. Traditional convertibles convert at the holder's option when the stock price is high.
3. CoCos increase equity in distress through triggers in the indenture, not at the holder's option, via conversion or write-down of the bond's value.
4. So B is correct.

Why B is correct: Equity rises automatically on a trigger, either via conversion into shares or via a write-down of the debt.

Why the others are wrong
- **A:** This is the traditional convertible bond mechanism (holder's option, high stock price).
- **C:** A repurchase would drain capital and rely on government money; CoCos avoid the bailout.
- **D:** Nothing in the CoCo mechanics converts principal into a deposit.

Q692

AnalystPrep Q.5619

REBUILT — Reframed on the GARP threshold rule for CoCos to count as Additional Tier 1 vs Tier 2.

Under the treatment described in GARP, a CoCo whose trigger is a Core Tier 1 capital to risk-weighted assets ratio of 5.125% or higher may be included in a bank's:

A. Common equity Tier 1 capital.
B. Additional Tier 1 capital.
C. Tier 2 capital only.
D. No regulatory capital category.

Answer: B

Step by step
1. GARP: CoCos may be included in Additional Tier 1 capital if the trigger threshold is 5.125% or higher.
2. If the threshold is lower, the CoCo is included in Tier 2 capital instead.
3. The stated trigger is 5.125%, so the CoCo is Additional Tier 1.

Why B is correct: A trigger at or above 5.125% qualifies the CoCo as Additional Tier 1 capital.

Why the others are wrong
- **A:** CoCos are debt until converted, so they are not common equity Tier 1.
- **C:** Tier 2 treatment applies only when the trigger threshold is below 5.125%.
- **D:** CoCos can qualify as regulatory capital.

Q693

AnalystPrep Q.5628

REBUILT — Credit Suisse 2023 write-down facts are not in the Schweser/GARP text; rebuilt on the supervisory-judgment trigger covered by Schweser.

Which of the following is a potential CoCo conversion trigger based on the judgment of the bank's supervisor?

A. Conversion occurs if the bank's stock price exceeds a threshold specified in the indenture.

B. Conversion occurs automatically if supervisors determine the bank needs public-sector aid (equity capital) to avoid insolvency.
C. Conversion occurs only if bondholders vote to convert at their option.
D. Conversion occurs when the ratio of market capitalization to assets rises above a set level.

Answer: B
Step by step
1. Schweser lists three triggers: Tier 1 equity to RWA below a threshold, supervisors' judgment on solvency, and a minimum market capitalization to assets ratio.
2. The Credit Suisse 2011 CoCos converted automatically if supervisors determined the bank needed public-sector aid to avoid insolvency.
3. Only B is a supervisory-judgment trigger.

Why B is correct: A supervisory-judgment trigger converts the bond when the supervisor judges the bank non-viable without public aid.
Why the others are wrong
- **A:** A high stock price is the traditional convertible's holder-option condition, not a CoCo stress trigger.
- **C:** CoCo conversion is automatic, not at the holder's option.
- **D:** The market-value trigger activates when the ratio falls below a minimum, not when it rises.

Q694

AnalystPrep Q.5630 · AnalystPrep Q.5631

REBUILT — Original items asked about the 2023 Credit Suisse write-down and regulator strategy, which is not in the reading; rebuilt to test the CoCo loss-absorption mechanics (conversion or write-down on a trigger) that the case illustrates.

A bank issues contingent convertible bonds (CoCos) with an indenture trigger tied to its capital ratio and to a regulatory nonviability determination. Which statement best describes how the CoCo's loss-absorption mechanism works?

A. On a trigger, the bond either converts into equity or is written down, so the bank's equity increases; the trigger is set in the indenture and is not at the holder's option.
B. The holder may choose, at any time before maturity, to convert the bond into equity once the bank's share price exceeds a threshold.
C. On a trigger, the bond is converted into short-term debt, which improves the bank's liquidity but leaves its equity unchanged.
D. The bond is always converted into equity at a fixed rate; a write-down is never possible because it would breach the creditor hierarchy.

Answer: A
Step by step
1. A CoCo is the bank-side mirror image of a traditional convertible: equity rises when the bank is in distress.
2. The trigger (e.g., Core Tier 1 capital to RWA below a threshold, or the regulator declaring the bank nonviable) is written into the indenture.
3. Equity increases either because the bond converts to equity or because its value is written down; the holder has no option.
4. Hence A.

Why A is correct: CoCos absorb losses automatically on indenture triggers, through conversion or write-down, raising the bank's equity cushion without a government bailout.
Why the others are wrong
- **B:** Describes a traditional convertible, which converts at the holder's option when the share price is high; CoCos are triggered by distress, not by the holder.
- **C:** Converting into short-term debt would not add loss-absorbing equity; invented mechanism.
- **D:** Assumes conversion is the only route; GARP states equity increases either by conversion or by write-down.

Q695

BT answers P2.T7.20.10.3 · BT notes P2.T7.20.10.3

The Basel regulations were hardly the only high-level effort by regulators in response to the global financial crisis (GFC) of 2007-08. For example, as Mark Carey explains, "Banks will fail in the future in spite of Basel I, II, III and later reforms. To limit the disruptions caused by such failures, the FSB agreed in 2014 that national resolution regimes for G-SIBs would have 12 key attributes and that each G-SIB should have sufficient total loss absorbing capacity (TLAC) to enable it to recapitalize itself. Recapitalization might be accomplished by causing convertible bonds to

become equity or by bail-in, in which certain wholesale debt liabilities are either written down or converted to equity."
In regard to this and other reforms, which of the following statements is TRUE?

A. CoCo bonds contain an embedded put option that gives the buyer the right, but not the obligation, to purchase equity at a predetermined conversion price
B. CoCo bonds are attractive to banks because they do not decrease return on equity (ROE) prior to conversion, and they are also attractive to regulators because they absorb losses when triggered if distress occurs
C. Gold plating is when a national regulator applies a less onerous (lower) regulatory capital ratio to its domestic banks, giving them a competitive advantage against international banks
D. The TLAC minimums developed by the FSB will be gradually phased out and replaced by the leverage and liquidity ratios in Basel IV as they are phased in

Answer: B

Step by step

1. Before conversion a CoCo is debt, so it does not appear in the equity account and does not dilute ROE.
2. On a trigger it converts or is written down, increasing equity and absorbing losses without a bailout, which regulators value.
3. A: conversion is automatic on a trigger, not at the holder's option. C: gold plating means national rules that are superequivalent (stricter) than Basel. D: TLAC is an additional loss-absorbency requirement, not being replaced.
4. Hence B.

Why B is correct: CoCos are debt in good times (no ROE drag) and automatically add equity in distress, which suits both banks and regulators.

Why the others are wrong

- **A:** Treats a CoCo as a holder option to buy equity; the conversion is mandatory and trigger-based, not a holder option.
- **C:** Reverses the meaning: gold plating is imposing stricter (superequivalent) national rules than Basel, not more lenient ones.
- **D:** TLAC is a complement to capital, leverage and liquidity requirements and is not being phased out.

Q696 BT answers P2.T7.522.3 · BT notes P2.T7.522.3

Which of the following is TRUE about contingent convertible (CoCo) bonds?

A. A drawback of CoCo bonds is that, prior to conversion, they hurt the bank's return on equity (ROE)
B. A popular trigger for CoCo bonds is the ratio of Tier 1 equity capital to risk-weighted assets
C. Regulators do not like CoCo bonds because they encourage regulatory arbitrage
D. CoCo bonds are NOT attractive to banks because they increase the odds of a public sector bailout if the bank experiences financial difficulty

Answer: B

Step by step

1. Typical CoCo triggers: Tier 1 (Core Tier 1) equity capital to RWA below a threshold; supervisor's nonviability judgment; market-capitalization-to-assets ratio.
2. A: before conversion CoCos are debt and do not drag ROE. C: regulators like them as loss absorbers. D: they reduce bailout odds because private bondholders supply the capital.
3. Hence B.

Why B is correct: A capital ratio such as Tier 1 equity to risk-weighted assets is a standard CoCo conversion trigger.

Why the others are wrong

- **A:** Confuses CoCos with equity; as debt they do not reduce ROE before conversion.
- **C:** Regulators favor CoCos because they add loss-absorbing capital in stress; no such regulatory-arbitrage objection is made.
- **D:** Inverts the motivation: CoCos recapitalize the bank from private funds and help avoid a bailout.

Q697 GARP PE1 Q70

A junior analyst at a banking regulatory agency is reviewing a report on a failed bank that had issued a high volume of contingent convertible bonds (CoCos), some still outstanding at failure. The analyst researches CoCos, including their mechanics, accounting, and treatment under Basel guidelines. Which of the following will the analyst find correct about CoCos?

A. CoCos give their holders the option to convert them into the equity of the issuing bank either before maturity or at maturity, depending on the type of CoCo.

B. CoCos give their holders the option to convert their medium- or long-term debt to short-term debt if certain interest rate conditions specified in the indenture are met.

C. CoCos may be included in Tier 2 capital but not in Tier 1 or Additional Tier 1 capital due to the risks they pose to the issuing bank and the difficulties in pricing them.

D. CoCos can be more expensive for banks to issue than equity, but they allow the issuing bank to report a higher return on equity until they are converted.

Answer: D

Step by step

1. CoCo holders get little upside but bear losses similar to equity holders in failure, so CoCos should be expensive to issue.
2. They are not in the equity account until converted, so reported ROE is higher before conversion.
3. A: conversion is triggered by indenture terms, not at holder's option. B: no such feature. C: CoCos count as Additional Tier 1 if the trigger is 5.125% or higher, Tier 2 otherwise.
4. Hence D.

Why D is correct: CoCos are costly (equity-like downside) but sit outside equity until conversion, which flatters ROE.

Why the others are wrong

- **A:** Applies the traditional convertible's holder option; CoCo conversion is trigger-driven.
- **B:** Invents a debt-maturity conversion; CoCos convert to equity or are written down.
- **C:** Ignores that CoCos with a trigger of at least 5.125% qualify as Additional Tier 1; only lower-trigger ones are Tier 2.

Q698 Schweser Module Quiz 63.3 Q1

Which of the following statements is correct regarding the mechanics and motivations of contingent convertible bonds (CoCos)?

I. During normal financial periods, CoCos are debt and do not drag down return on equity.

II. During periods of financial stress, CoCos convert to equity, providing a cushion against loss, which helps prevent insolvency.

A. Statement I only.

B. Statement II only.

C. Both Statements I and II.

D. Neither Statement I nor II.

Answer: C

Step by step

1. Statement I: a CoCo is a debt instrument when issued and does not appear in the equity account until converted, so the bank can report a higher return on equity (the accounting advantage of CoCos). True.
2. Statement II: when the trigger in the indenture is hit (e.g., Core Tier 1 / RWA falls below a threshold, or the regulator declares the bank nonviable), the CoCo converts to equity or is written down, raising the bank equity cushion and helping to avoid insolvency. True.
3. Both statements are true, so the answer is C.

Why C is correct: CoCos sit outside equity (no ROE drag) while the bank is healthy, but convert to equity (or are written down) at the trigger, automatically recapitalizing the bank in distress.

Why the others are wrong

- **A:** Treats Statement II as false; it overlooks that conversion in stress adds equity and loss-absorbing capacity.
- **B:** Treats Statement I as false; it wrongly assumes CoCos dilute equity or ROE before a trigger, when they are debt until conversion.
- **D:** Rejects both statements; it confuses CoCos with traditional convertibles, which convert at the holder option when the share price is high, not on distress.

Q699 AnalystPrep Q.5616

REBUILT — Original asked about investor appeal, which is outside the LO (bank mechanics and issuance motivation); rebuilt to test the bank's motivation for issuing CoCos.

A bank is considering issuing contingent convertible bonds (CoCos) rather than ordinary equity. Which of the following best describes the bank's motivation for issuing CoCos?

A. CoCos are debt in normal times and so do not dilute return on equity, but convert automatically to equity in stress, providing a loss-absorbing cushion funded by private investors rather than the government.

B. CoCos convert to equity at the option of the bondholder once the share price exceeds a threshold, allowing the bank to avoid issuing equity until its performance improves.

C. CoCos are senior to deposits in liquidation, which lowers the bank's funding cost and reduces its required Tier 1 capital in normal times.

D. CoCos convert to equity only when the bank applies for public-sector support, so the government supplies the capital needed to avoid insolvency.

Answer: A

Step by step

1. CoCos are issued as debt, so in normal periods they do not drag down return on equity as new equity would.
2. Conversion (or write-down) is automatic once a trigger in the indenture is hit (for example, a Tier 1 equity-to-RWA ratio below a threshold, supervisory judgment of nonviability, or a market-capitalization-to-assets floor).
3. On conversion, equity rises exactly when the bank is stressed, cushioning losses and helping prevent insolvency.
4. The cushion is supplied by private bondholders, so a taxpayer bailout can be avoided. This matches A.

Why A is correct: CoCos give debt-like treatment in good times and automatic equity-like loss absorption in bad times, with private rather than public capital providers.

Why the others are wrong

- **B:** Describes a traditional convertible bond, which converts at the holder's option when the share price is high; CoCos convert automatically on a stress trigger.
- **C:** CoCos are subordinated, not senior to deposits, and they are not a way to cut required capital in normal times.
- **D:** Conversion is triggered automatically by contractual triggers (although a supervisory nonviability determination is one trigger); the point is that private holders, not government, supply capital.

Q700 AnalystPrep Q.5623

REBUILT — Original tested a 2023 Credit Suisse event not covered in the Schweser/GARP reading; rebuilt to test CoCo trigger mechanics covered by the LO.

Which of the following is a recognized trigger for a contingent convertible bond (CoCo) to convert to equity (or be written down)?

A. The bondholder's decision to convert when the bank's share price rises above a threshold set in the indenture.

B. The bank's Tier 1 equity capital to risk-weighted assets ratio falling below a specified threshold.

C. A rise in the bank's credit rating by one notch.

D. The bank's liquidity coverage ratio rising above 100%.

Answer: B

Step by step

1. CoCos convert (or are written down) automatically when a contractual trigger is breached, not at the holder's option.
2. Typical triggers: Tier 1 (or Core Tier 1) equity to RWA falling below a threshold; supervisors' judgment that the bank is nonviable or needs public aid; a minimum market capitalization to assets ratio.
3. Only B is one of these triggers.

Why B is correct: A capital-ratio breach signals stress and is a standard automatic CoCo trigger.

Why the others are wrong

- **A:** Holder-option conversion on a high share price describes a traditional convertible bond, not a CoCo.
- **C:** A rating upgrade signals improvement, not distress, and is not a trigger.
- **D:** A rising LCR indicates stronger liquidity; it is not a conversion trigger.

LO 63.h

Provide examples of legislative and regulatory reforms that were introduced after the 2007–2009 financial crisis.

Q701 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Daniel Okafor is the compliance officer of a large U.S. bank holding company that accepts insured deposits. Its head of proprietary trading proposes launching a desk that takes directional positions in commodities using the bank's own capital to profit from short-term price moves. Which of the following is the most likely regulatory response under post-crisis reforms?

A. The Volcker Rule would generally prohibit the desk because it restricts proprietary trading at deposit-taking firms.

B. The Volcker Rule would permit the desk provided the bank retains at least 5% of each position it takes.

C. The desk would be permitted if all its trades are cleared through a central counterparty.

D. The desk would be permitted if the bank's board risk committee includes a member with risk management experience.

Answer: A

Step by step

1. The Volcker Rule (Dodd-Frank) restricts proprietary trading and investment in hedge funds and private equity at deposit-taking firms.
2. Rationale: banks funded by insured deposits should not speculate.
3. The 5% retention, CCP clearing and board risk committee rules address other issues and do not authorize proprietary trading.

Why A is correct: A desk taking directional positions with the bank's own capital is proprietary trading, which the Volcker Rule is designed to curtail at insured-deposit-funded banks.

Why the others are wrong

- **B:** Wrong: the 5% retention requirement applies to issuers of securitizations, not to trading positions.
- **C:** Wrong: central clearing is required for standardized OTC derivatives between financial institutions; it does not convert proprietary trading into a permitted activity.
- **D:** Wrong: the risk-expert board member requirement is a governance reform and has no bearing on whether proprietary trading is allowed.

Q702 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Two large banks plan to enter a plain-vanilla interest rate swap with standard terms. Under the post-crisis reforms in the United States and the European Union, which of the following is most likely required for this transaction?

A. The swap must be traded bilaterally and kept off exchange to preserve confidentiality, with the banks holding extra capital instead.

B. The swap may be traded on a swap execution facility and must be cleared through a central counterparty.

C. The swap must be rated by an agency overseen by the Office of Credit Ratings before it can be executed.

D. The swap must be reported to the Consumer Financial Protection Bureau, which approves its pricing.

Answer: B

Step by step

1. Relatively standard OTC derivatives must be traded on swap execution facilities (electronic platforms promoting price transparency).
2. Derivatives between financial institutions must be cleared by central counterparties.
3. Other options confuse the OTC reforms with rating agency oversight and consumer protection.

Why B is correct: Reforms moved standardized OTC derivatives onto SEFs and required central clearing for trades between financial institutions.

Why the others are wrong

- **A:** Wrong: the reforms push standardized derivatives toward transparent venues and central clearing, not opaque bilateral trading.
- **C:** Wrong: the Office of Credit Ratings oversees rating agencies; swaps are not rated and have no rating prerequisite.
- **D:** Wrong: the CFPB protects consumers in products such as mortgages and credit cards and does not approve interbank derivative prices.

Q703 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank in the European Union securitizes a loan pool into a $320 million senior tranche, a $50 million mezzanine tranche, and a $30 million equity tranche. The issuer must comply with the post-crisis risk retention rule requiring it to retain at least 5% of each tranche. What is the minimum total amount the bank must retain?

A. $1.5 million
B. $4.0 million
C. $20.0 million
D. $16.0 million

Answer: C

Step by step

1. Senior: $320 \times 5\% = 16.0$ million.
2. Mezzanine: $50 \times 5\% = 2.5$ million.
3. Equity: $30 \times 5\% = 1.5$ million.
4. Total: $16.0 + 2.5 + 1.5 = 20.0$ million.

Why C is correct: Retaining 5% of every tranche requires 5% of the total $400 million, or $20.0 million, aligning issuer and investor incentives.

Why the others are wrong

- **A:** Wrong: retains only 5% of the equity tranche, ignoring the other tranches.
- **B:** Wrong: retains 5% of the mezzanine and equity tranches only ($2.5 + $1.5 million), omitting the senior tranche.
- **D:** Wrong: retains 5% of the senior tranche only; the rule applies to each tranche.

Reading 64. High-Level Summary of Basel III Reforms

"High-Level Summary of Basel III Reforms" (Basel Committee on Banking Supervision Publication, December 2017).

LO	Learning objective	Qs
64.a	explain the motivations for revising the Basel III framework and the goals and impacts of the December 2017 reforms to the Basel III framework.	4
64.b	summarize the December 2017 revisions to the Basel III framework in the following areas: the standardized approach to credit risk the internal ratings-based (IRB) approaches for credit risk the CVA risk framework the operational risk framework the leverage ratio framework	13
64.c	describe the revised output floor introduced as part of the Basel III reforms and approaches to be used when calculating the output floor.	3

LO 64.a

Explain the motivations for revising the Basel III framework and the goals and impacts of the December 2017 reforms to the Basel III framework.

Q704

AnalystPrep Q.2356

REBUILT — Source key (discretion over RWA) is not the GARP/BCBS rationale and option wording was loose; rewrote the options around the GARP purpose of the leverage ratio.

Within the initial phase of the Basel III reforms, the Basel Committee specified a minimum leverage ratio requirement. What was its main purpose?

A. To constrain excess leverage in the banking system and complement the risk-weighted capital requirements
B. To replace risk-weighted capital requirements with a simple non-risk-based measure
C. To build up capital in good times that can be drawn down in times of stress
D. To place a floor on the capital benefits banks obtain from internal models

Answer: A

Key correction: Source key (AnalystPrep) said B (banks had too much discretion in RWA); the GARP text states the leverage ratio's purpose as constraining excess leverage and complementing risk-weighted requirements.

Step by step

1. The BCBS lists the leverage ratio among the initial-phase Basel III measures: a minimum leverage ratio requirement.
2. Its stated aim is to constrain excess leverage in the banking system and to complement (not replace) risk-weighted requirements.
3. Capital buffers and the output floor are separate reform elements with different aims.

Why A is correct: The leverage ratio is a non-risk-based backstop that limits excess leverage and sits alongside the risk-weighted capital framework.

Why the others are wrong

- **B:** It complements rather than replaces risk-weighted requirements; this confuses a backstop with a substitute.
- **C:** This describes capital buffers built in good times to limit procyclicality, not the leverage ratio.
- **D:** This describes the output floor, a different reform element.

Q705

AnalystPrep Q.3092

REBUILT — Original distractors misdescribed the motivations (e.g. LCR, new default definition); rebuilt around the GARP list of initial-phase Basel III measures.

The Basel III framework was revised because the pre-crisis framework failed to prevent systemic shocks. Which of the following was NOT among the initial-phase Basel III measures to address those shortcomings?

A. Improving capital quality through a greater focus on Common Equity Tier 1 (CET1) capital
B. Introducing capital buffers built up in good times that can be drawn down in times of stress
C. Specifying a minimum leverage ratio requirement to constrain excess leverage
D. Relaxing the large exposures regime to allow more concentrated exposures to individual counterparties

Answer: D

Step by step

1. Recall the GARP list: CET1 focus, higher capital levels, better risk capture, macroprudential elements, minimum

leverage ratio.
2. A, B and C are all on that list.
3. The macroprudential elements include a large exposures regime that mitigates systemic risk from interlinkages and concentrated exposures, so relaxing it is the opposite of the reform.

Why D is correct: Basel III tightened the large exposures regime to limit concentration; relaxing it was not a reform.

Why the others are wrong

- **A:** Genuine measure: focus on going-concern, loss-absorbing CET1.
- **B:** Genuine measure: buffers limit procyclicality.
- **C:** Genuine measure: minimum leverage ratio.

Q706

AnalystPrep Q.3109

REBUILT — Ambiguous stem (Basel III dates from 2010; the 2017 reforms were finalised in December 2017); reframed to ask when the 2017 reforms are implemented from.

The December 2017 Basel III reforms finalised the post-crisis framework. Most of the revisions (revised standardised approach for credit risk, IRB, CVA, operational risk) are to be implemented beginning in which year?

A. 2018
B. 2020
C. 2022
D. 2027

Answer: C

Step by step

1. The reforms were announced in December 2017.
2. GARP's implementation table gives 1 January 2022 for the revised SA, IRB, CVA and operational risk frameworks.
3. 2018 applies only to the existing leverage ratio exposure definition, and 2027 is when the output floor phase-in reaches 72.5%.

Why C is correct: The post-crisis revisions take effect from 1 January 2022, with the output floor phased in to 2027.

Why the others are wrong

- **A:** 2018 is the date for the existing leverage ratio exposure definition, not the main revisions.
- **B:** No Basel III reform milestone falls on 2020; plausible only as a 'two years after announcement' guess.
- **D:** 2027 is when the output floor reaches 72.5%, the end of the phase-in, not the start.

Q707

Mock 2 Q49

Revisions to Basel III aim to address the failures in preventing systemic shocks that severely weakened the global economy during the financial crisis of 2007-2009. Which of the following statements is incorrect regarding the main goals and impacts of the December 2017 Basel III reforms?

A. Introduce a leverage ratio buffer for global systemically important banks (G-SIBs).
B. Create an output floor that is more robust and risk sensitive than the current Basel II floor.
C. Expand the use of internal model approaches for credit risk, credit valuation adjustment (CVA) risk, and operational risk.
D. Expand the robustness and sensitivity of the standardized approaches (SA) for measuring credit risk, CVA risk, and operational risk.

Answer: C

Step by step

1. List the goals: expand robustness and sensitivity of the SA; restrict internal models; leverage ratio buffer for G-SIBs; more robust output floor.
2. A, B and D match goals.
3. C says expand internal models, the opposite of the stated goal to restrict them.

Why C is correct: The reforms restrict, not expand, internal model approaches (e.g. A-IRB limited, internal models removed for CVA and operational risk).

Why the others are wrong

- **A:** Correct statement: the G-SIB leverage ratio buffer is a goal.
- **B:** Correct statement: the output floor replaces the Basel II floor with a more robust one.

- **D:** Correct statement: SA is made more granular and risk sensitive.

LO 64.b

Summarize the December 2017 revisions to the Basel III framework in the following areas: the standardized approach to credit risk the internal ratings-based (IRB) approaches for credit risk the CVA risk framework the operational risk framework the leverage ratio framework

Q708

AnalystPrep Q.3095

REBUILT — Replaced the implausible option D with a realistic distractor; facts unchanged.

Capital Bank, a hypothetical global systemically important bank (G-SIB), is subject to a 5% risk-weighted higher-loss absorbency (HLA) requirement. Under the Basel III reforms, the bank's leverage ratio buffer requirement is:

A. 5%
B. 10%
C. 2.5%
D. 7.5%

Answer: C

Step by step

1. The G-SIB leverage ratio buffer is set at 50% of the G-SIB's risk-weighted higher-loss absorbency requirement.
2. Buffer = 50% × 5% = 2.5%.
3. It must be met with Tier 1 capital.

Why C is correct: The leverage ratio buffer is half of the bank's risk-weighted HLA requirement, giving 2.5%.

Why the others are wrong

- **A:** Sets the leverage buffer equal to the risk-weighted HLA requirement, ignoring the 50% scaling.
- **B:** Doubles the HLA requirement instead of halving it.
- **D:** Adds 50% to the HLA requirement (5% x 1.5) rather than taking 50% of it.

Q709

AnalystPrep Q.3097

REBUILT — Original EXCEPT item with an 'All of the above' option was ambiguous; recast as a single-best-answer statement on the CVA framework.

Which of the following best describes the December 2017 revisions to the CVA risk framework?

A. The internal model approach was removed; banks must use the standardized approach (SA-CVA) or the basic approach (BA-CVA).
B. The internal model approach was retained as the primary method, with the standardized approach as a fallback.
C. All banks must use a single basic approach (BA-CVA) regardless of size or sophistication.
D. CVA risk is no longer capitalized because it is already covered by the market risk framework.

Answer: A

Step by step

1. The revisions aim to raise risk sensitivity (include the exposure component and eligible hedges), robustness, and consistency with the revised market risk framework (fair value sensitivities).
2. To improve robustness and comparability, the internal model option was removed.
3. Banks use SA-CVA or the simpler BA-CVA, so A is correct.

Why A is correct: The internal model approach was eliminated, leaving the standardized (SA-CVA) and basic (BA-CVA) approaches.

Why the others are wrong

- **B:** Reverses the reform: the internal model option was removed, not retained.
- **C:** Ignores SA-CVA; two approaches remain, not a single one.
- **D:** CVA risk remains a separately capitalized risk; the framework was made consistent with market risk, not merged away.

Q710 AnalystPrep Q.3098

The new standardized approach for determining a bank's operational risk capital requirement assumes that:
I. Operational risk increases at a decreasing rate with a bank's income.
II. Banks that have experienced greater operational risk losses historically are more likely to experience operational risk losses in the future.

A. I only
B. II only
C. Both I and II
D. Neither I nor II

Answer: B

Step by step

1. The approach has two inputs: a measure of income and a measure of historical losses.
2. Assumption I: operational risk increases at an increasing rate with income (not decreasing), so I is false.
3. Assumption II: past losses are positively related to future losses, so II is true.
4. Answer: II only.

Why B is correct: Risk rises at an increasing rate with income and past losses predict future losses, so only statement II is correct.

Why the others are wrong

- **A:** Statement I reverses the relationship (increasing, not decreasing, rate), and it omits true statement II.
- **C:** Wrongly accepts I; the rate of increase with income is increasing.
- **D:** Wrongly rejects II; historical losses are assumed positively related to future losses.

Q711 AnalystPrep Q.3101

REBUILT — Original treated the HLA requirement as a dollar amount, which is not how it is defined; rebuilt with a percent-of-RWA HLA and a leverage exposure measure.

A G-SIB is subject to a 2.0% risk-weighted higher-loss absorbency (HLA) requirement. Its leverage ratio exposure measure is $800 billion. Under the Basel III reforms, the additional Tier 1 capital required to meet the leverage ratio buffer is closest to:

A. $16 billion
B. $8 billion
C. $4 billion
D. $24 billion

Answer: B

Step by step

1. Leverage buffer = 50% of HLA = $0.5 \times 2.0\% = 1.0\%$.
2. Required Tier 1 = $1.0\% \times \$800\text{ bn} = \$8\text{ bn}$.

Why B is correct: The buffer is 50% of the HLA requirement (1.0%) applied to the leverage exposure measure.

Why the others are wrong

- **A:** Applied the full 2.0% HLA to exposure without the 50% scaling.
- **C:** Halved twice (0.5% buffer).
- **D:** Applied 150% of the HLA (3.0%).

Q712 AnalystPrep Q.3105

In the 2007/2008 crisis, banks suffered large losses from CVA risk, i.e., losses from deterioration of derivative counterparties' creditworthiness. In response, the Basel Committee enhanced the CVA framework with a view to:

A. Keep losses associated with CVA risk below 10% of the total value of derivatives.
B. Eliminate CVA losses entirely by requiring due diligence on all counterparties before contracts begin.
C. Enhance the risk sensitivity of the framework by recognizing more risk drivers.
D. Limit derivative contracts to no more than 20% of a bank's total capital.

Answer: C

Step by step

1. The enhanced framework pursues risk sensitivity (exposure component and hedges), robustness (internal

models removed) and consistency with the market risk framework.
2. Only C describes one of these objectives.

Why C is correct: The reform's stated aim is greater risk sensitivity by recognizing more risk drivers such as the exposure component and eligible hedges.

Why the others are wrong

- **A:** Invents a numerical loss cap; the framework sets capital requirements, not loss ceilings.
- **B:** Capital rules cannot eliminate CVA losses and impose no such due diligence mandate.
- **D:** Invents a limit on derivative volume; no such 20% cap exists in the reforms.

Q713

BT Mock Q3 · BT answers P2.T7.20.11.1 · BT notes P2.T7.20.11.1

In December 2017 the Basel Committee revised the Basel III framework, including the standardized approach (SA) and the internal ratings-based (IRB) approaches for credit risk. Which of the following statements regarding these revisions is TRUE?

A. In the SA for residential real estate exposures, risk weights are lower when the loan-to-value (LTV) ratio falls into a lower band.

B. The SA eliminated the treatment of unrated exposures because all exposures are now required to have an external credit rating.

C. The IRB framework removed the foundation IRB (F-IRB) option because the advanced IRB (A-IRB) must be used for all asset classes.

D. The IRB framework removed the parameter input floors (PD, LGD, EAD) to increase RWA variability across banks.

Answer: A

Step by step

1. SA: residential real estate risk weights now depend on LTV band (e.g., whole-loan weights rising from 20% below 50% LTV to 70% above 100%), so lower LTV means a lower weight: A true.
2. SA: reliance on ratings was reduced and a more granular treatment of unrated exposures was kept: B false.
3. IRB: A-IRB was removed for large corporates, banks and other financial institutions (F-IRB required): C reversed.
4. IRB: input floors were introduced to reduce, not increase, RWA variability: D false.

Why A is correct: The revised SA ties residential mortgage risk weights to LTV, so a lower LTV band gives a lower risk weight.

Why the others are wrong

- **B:** Reduced reliance on ratings does not mean unrated exposures were eliminated; a granular unrated treatment remains.
- **C:** Reverses the reform: A-IRB was removed for certain exposures and F-IRB retained.
- **D:** Reverses the reform: input floors were added, and their purpose is to reduce RWA variability.

Q714

BT answers P2.T7.20.11.2 · BT notes P2.T7.20.11.2

The December 2017 revisions to Basel III covered credit risk approaches, CVA risk, operational risk, the leverage ratio, and a revised output floor. Each of the following statements is true EXCEPT:

A. The CVA framework was revised to enhance its risk sensitivity, strengthen its robustness, and improve its consistency with the market risk framework.

B. For operational risk, the previous standardized approaches and the advanced measurement approach were replaced by a single risk-sensitive standardized approach for all banks.

C. For operational risk, the new standardized approach eliminated the bank's historical internal loss experience as an input because it was uncorrelated with actual operational losses.

D. Under the revised output floor, a bank's RWA under internal models cannot be lower than 72.5% of the RWA calculated using only standardized approaches.

Answer: C

Step by step

1. A is true: risk sensitivity, robustness and consistency were the CVA goals.
2. B is true: AMA and the three older SAs were replaced by a single SA.
3. C is false: the new SA uses income plus historical losses, with past losses assumed positively correlated with future losses.

4. D is true: the floor is 72.5% of standardized RWA.

Why C is correct: Historical losses remain an input to the new standardized approach, being assumed positively related to future losses, so C is the false statement.

Why the others are wrong

- **A:** True statement, so not the exception.
- **B:** True statement, so not the exception.
- **D:** True statement (72.5% floor), so not the exception.

Q715

GARP PE2 Q35 · Mock 4 Q36

The CRO of a large bank has asked a group of risk managers to prepare a plan to implement the revisions to the Basel III guidelines finalized in December 2017. The bank expects to qualify as a global systemically important bank (G-SIB) by the time the revisions are implemented. Which of the following actions is most appropriate for the bank to take to comply with the revised Basel III guidelines?

A. Introduce an internally created model to determine the bank's operational risk capital.

B. Assign different credit risk weights for residential mortgages based on each mortgage's loan-to-value ratio.

C. Apply the advanced internal rating-based approach for certain types of credit exposures, such as exposures to banks and large corporations.

D. Decrease the bank's target level of its countercyclical buffer and its Basel III Tier 1 leverage ratio.

Answer: B

Step by step

1. The revised standardized approach for credit risk increases the granularity of risk weights.
2. Under Basel II one risk weight applied to all residential mortgages; under the revisions the weight depends on the loan-to-value (LTV) ratio.
3. Check the others: internal operational risk models are removed, A-IRB is withdrawn for large corporates and banks, and G-SIBs gain a leverage ratio buffer.

Why B is correct: The revised SA makes residential mortgage risk weights depend on LTV, so assigning LTV-based weights complies with the reforms.

Why the others are wrong

- **A:** The AMA/internal models for operational risk were eliminated; a single standardized approach based on income and historical losses is the only one allowed.
- **C:** A-IRB is no longer permitted for exposures to banks, financial institutions and large (and mid-size) corporates; F-IRB must be used.
- **D:** The reforms add a leverage ratio buffer for G-SIBs, so the leverage requirement rises rather than falls; reducing it is non-compliant.

Q716

Schweser Module Quiz 64.1 Q1

The Basel III reforms restricted the use of internal model approaches for all of the following risk categories except:

A. credit risk.

B. systemic risk.

C. operational risk.

D. credit valuation risk.

Answer: B

Step by step

1. The reforms restrict internal models for credit risk (A-IRB limits), CVA risk (IRB approach removed) and operational risk (AMA removed).
2. Systemic risk has no internal model approach to restrict; it is addressed through the G-SIB leverage ratio buffer.

Why B is correct: Internal model approaches were restricted for credit, CVA and operational risk, but systemic risk is not a modeled risk category.

Why the others are wrong

- **A:** Credit risk: A-IRB was removed for large/mid-size corporates and banks, so it was restricted.
- **C:** Operational risk: the AMA was eliminated in favor of one standardized approach.
- **D:** CVA risk: the IRB approach to CVA was removed, leaving SA-CVA or the basic approach.

Q717 Schweser Module Quiz 64.1 Q2

The Basel III reforms introduced a leverage buffer ratio for all:

A. global banks.
B. unregulated global banks.
C. global systemically important banks.
D. banks regulated by the U.S. Federal Reserve and the European Banking Authority.

Answer: C

Step by step

1. The revised leverage ratio framework adds a leverage ratio buffer for G-SIBs, met with Tier 1 capital.

Why C is correct: The buffer applies specifically to G-SIBs, not to banks generally.

Why the others are wrong

- **A:** Too broad: only systemically important global banks, not all global banks, face the buffer.
- **B:** Unregulated banks are outside Basel standards; this is not a category in the reforms.
- **D:** Regulator identity does not define the buffer; G-SIB status does.

Q718 Schweser Module Quiz 64.1 Q3

The advanced internal ratings-based (A-IRB) approach for credit risk will no longer be permissible to use under the Basel III reforms for each of the following except:

A. large corporate exposures.
B. mid-size corporate exposures.
C. banks and other financial institutions.
D. residential and commercial real estate.

Answer: D

Step by step

1. A-IRB was removed for large and mid-size corporates and for banks and other financial institutions; F-IRB is required.
2. Real estate exposures are not on the removal list, so A-IRB remains available.

Why D is correct: Only corporate (large and mid-size) and bank/financial institution exposures lost A-IRB; real estate did not.

Why the others are wrong

- **A:** A-IRB was removed for large corporates.
- **B:** A-IRB was removed for mid-size corporates.
- **C:** A-IRB was removed for banks and other financial institutions.

Q719 Schweser Module Quiz 64.1 Q4

The new operational risk capital requirements under the Basel III reforms are determined by measures of the bank's:

A. leverage and income.
B. income and historical operational risk losses.
C. income and expected operational risk losses.
D. leverage and expected operational risk losses.

Answer: B

Step by step

1. The single standardized approach uses a bank's income and its historical operational risk losses.
2. Both are assumed positively correlated with future operational risk.

Why B is correct: Capital depends on income and historical losses, replacing internal models.

Why the others are wrong

- **A:** Leverage is not an input to operational risk capital, and historical losses are omitted.
- **C:** Expected losses are modeled, forward-looking figures; the framework uses actual historical losses.
- **D:** Neither input is correct: leverage is irrelevant and expected losses are not used.

Q720 AnalystPrep Q.3106

The Basel Committee has agreed on various additional enhancements to the IRB approaches to further reduce unwarranted RWA variability. Which of the following correctly outlines a measure that has been put forth for adoption by banks?

A. Secured exposures: increasing the haircuts that apply to the collateral; Unsecured exposures: reducing the LGD parameter from 45% to 40% for exposures to non-financial corporates.

B. Secured exposures: reducing the LGD parameters; Unsecured exposures: reducing the LGD parameter from 25% to 20% for exposures to non-financial corporates.

C. Secured exposures: decreasing the haircuts that apply to the collateral; Unsecured exposures: reducing the LGD parameter from 45% to 40% for exposures to non-financial corporates.

D. Secured exposures: increasing the LGD parameters; Unsecured exposures: increasing the LGD parameter from 40% to 50% for exposures to non-financial corporates.

Answer: A

Step by step

1. F-IRB supervisory parameters were adjusted: for exposures secured by non-financial collateral, haircuts increase (and LGD parameters fall).
2. For unsecured exposures to non-financial corporates, the LGD parameter falls from 45% to 40%.
3. Only A has haircuts increasing and the correct 45% to 40% unsecured change.

Why A is correct: The F-IRB adjustments raise collateral haircuts and cut the unsecured corporate LGD from 45% to 40%; A is the only option consistent with both.

Why the others are wrong

- **B:** Misstates the unsecured LGD change as 25% to 20%; the actual starting level is 45%.
- **C:** Haircuts were increased, not decreased.
- **D:** Both directions are reversed: secured LGDs fell, and unsecured LGD fell from 45% to 40% rather than rising to 50%.

LO 64.c

Describe the revised output floor introduced as part of the Basel III reforms and approaches to be used when calculating the output floor.

Q721 Schweser Module Quiz 64.1 Q5

The Basel III output floor sets the minimum level of:

A. leverage.

B. functional capital.

C. credit valuation adjustment (CVA) risk exposure.

D. risk-weighted assets.

Answer: D

Step by step

1. The output floor is a backstop on how far internal models can reduce capital requirements relative to the standardized approaches (SAs).
2. Banks' risk-weighted assets (RWA) must be at least the higher of (1) RWA from all approved approaches (standardized and internal) and (2) 72.5% of RWA computed using only the SAs.
3. The quantity being floored is therefore risk-weighted assets.

Why D is correct: The output floor puts a lower bound (72.5% of standardized-approach RWA) on total RWA, limiting the capital benefit of internal models.

Why the others are wrong

- **A:** Leverage is governed by the separate leverage ratio framework (with a G-SIB buffer), not by the output floor.
- **B:** 'Functional capital' is not a Basel III concept; the floor applies to risk-weighted assets, not a type of capital.
- **C:** CVA risk is only one component whose SA is used when computing the floored RWA; the floor is not a floor on CVA exposure itself.

Q722 Schweser Module Quiz 64.1 Q6

Under the Basel III reforms, the approaches that can be used when calculating the output floor for CVA risk include all of the following EXCEPT:

A. the basic (BA-CVA) approach.
B. the standardized (SA-CVA) approach.
C. the internal ratings-based (IRB-CVA) approach.
D. 100% of the bank's counterparty credit risk capital requirement.

Answer: C

Step by step

1. The output floor is computed using only standardized (non-internal-model) approaches.
2. For CVA risk the permitted choices are SA-CVA, BA-CVA, or 100% of the bank's counterparty credit risk capital requirement.
3. An IRB-type approach is an internal-model approach and is not permitted for the floor; it is also not a CVA approach in the framework.
4. So the option that cannot be used is the IRB-CVA approach (C).

Why C is correct: The floor relies on standardized approaches only, so an internal ratings-based approach cannot be used for CVA risk in the floor calculation.

Why the others are wrong

- **A:** BA-CVA is a permitted approach for CVA risk in the floor calculation, so it is not the exception.
- **B:** SA-CVA is a permitted approach for CVA risk in the floor calculation, so it is not the exception.
- **D:** Using 100% of the counterparty credit risk capital requirement is an explicitly permitted alternative for CVA risk.

Q723 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A large bank's regulator has fully phased in the Basel III revised output floor. A risk analyst at the bank calculates total risk-weighted assets (RWA) of $600 billion using the internal models the bank has supervisory approval to use (combined with standardized approaches where models are not approved). The same exposures would produce total RWA of $900 billion if only the standardized approaches were applied. Which of the following is the bank's RWA for regulatory capital purposes?

A. $600.0 billion
B. $900.0 billion
C. $720.0 billion
D. $652.5 billion

Answer: D

Step by step

1. Output floor RWA = higher of (i) RWA under all approved approaches = $600 billion and (ii) 72.5% of RWA under standardized approaches only.
2. Floor = $0.725 \times 900 = 652.5$ billion.
3. $\max(600, 652.5) = 652.5$, so the floor binds and the bank's RWA is $652.5 billion.

Why D is correct: Under the revised floor, RWA must be the higher of the RWA from approved approaches (including internal models) and 72.5% of RWA from the standardized approaches alone. Here 72.5% of $900 billion is $652.5 billion, which exceeds the internally modelled $600 billion, so the floor applies.

Why the others are wrong

- **A:** Uses the internally modelled RWA and ignores the floor; the floor exists precisely to prevent models from reducing RWA below 72.5% of the standardized result.
- **B:** Treats the floor as 100% of standardized RWA; the floor limits the benefit of internal models to a minimum of 72.5% of standardized RWA, not 100%.
- **C:** Applies 80%, the calibration of the old Basel II floor (based on Basel I), instead of the 72.5% calibration of the revised Basel III floor.

Reading 65. Basel III: Finalizing Post-Crisis Reforms

"Basel III: Finalizing Post-Crisis Reforms" (Basel Committee on Banking Supervision Publication, December 2017): 128–136.

LO	Learning objective	Qs
65.a	explain the elements of the new standardized approach to measure operational risk capital, including the business indicator, internal loss multiplier, and loss component, and calculate the operational risk capital requirement for a bank using this approach.	17
65.b	compare the Standardized Measurement Approach (SMA) to earlier methods of calculating operational risk capital, including the Advanced Measurement Approaches (AMA).	3
65.c	describe general and specific criteria recommended by the Basel Committee for the identification, collection, and treatment of operational loss data.	3

LO 65.a

Explain the elements of the new standardized approach to measure operational risk capital, including the business indicator, internal loss multiplier, and loss component, and calculate the operational risk capital requirement for a bank using this approach.

Q724 BT answers P2.T7.23.2.1 · BT answers T7-23.2.1

Although operational risk was a concept of practical importance prior to 2004, that's the year Basel II introduced it formally into prudential regulation. Along with Basel II's three Pillars (i.e., regulatory capital, supervisory review, and market discipline), the BCBS published Principles for the Sound Management of Operational Risk (PSMOR), which were revised in 2011 and again in March 2021. The most recent Principles, for example, added a 12th principle on information and communication technology (ICT) risk, in addition to high-level implementation guidance around tools and controls. In regard to Basel's First Pillar, the original capital calculation allowed for a choice between two standardized approaches or an internal (aka, advanced) approach. However, few banks were able to successfully adopt the advanced approach, while the standardized approaches were criticized for the lack of risk sensitivity. Consequently, Basel III replaced them with a single approach that is now called the Standardized Approach (SA). Which of the following best summarizes the Standardized Approach to the operational risk capital (ORC) charge under the Basel regulations?

A. The ORC is a multiple (i.e., 12x|15x|18x scalar multiplier) of the average annual losses incurred over the ten previous years

B. The ORC is a percentage (i.e., 12|15|18% marginal coefficient) of the Business Indicator (BI) scaled by the Internal Loss Multiplier (ILM), where the BI is the sum of various income statement and/or balance sheet items

C. The ORC is the minimum of either (i) 2.25% of interest-earning assets or (ii) the absolute value of the net interest income

D. The ORC is 15% +/- 3% of the Internal Loss Indicator (ILI) scaled by the Business Segment Multiplier (BSM), where the ILI is the sum of the absolute values of the trading and banking books

Answer: B

Step by step

1. The standardized approach has three building blocks: the Business Indicator (BI), the BI Component (BIC) and the Internal Loss Multiplier (ILM).
2. BIC = BI multiplied by marginal coefficients of 12% (BI up to EUR 1bn), 15% (EUR 1bn to 30bn) and 18% (above EUR 30bn).
3. The BI is built from income statement and balance sheet items (ILDC + SC + FC).
4. ORC = BIC × ILM, so choice B is the only accurate summary.

Why B is correct: Capital equals the BI-driven component (marginal coefficients of 12/15/18% applied to the BI) scaled by an internal loss multiplier built from the bank's own loss history.

Why the others are wrong

- **A:** Losses enter only through the loss component (15 times average annual loss over 10 years) inside the ILM; capital is not a 12/15/18x multiple of losses.
- **C:** This is the cap on the net interest term (min of 2.25% of interest-earning assets and absolute net interest income) within the interest, lease and dividend component of the BI; it is not the capital charge.
- **D:** ILI and BSM are invented terms; the real approach uses the BI and ILM, and the coefficients are 12/15/18%,

not 15% +/- 3%.

Q725

Schweser Module Quiz 43.1 Q1

REBUILT — Exhibit with income statement data was missing; reconstructed consistent three-year average data that reproduce the source answer (BI = EUR 1,070m) and rebuilt the distractors as realistic component errors.

Rosedale Community Bank (RCB) has the following average annual figures (EUR million) over the past three years:

Item	Average annual amount
Interest income	1,100
Interest expense	380
Interest-earning assets	40,000
Dividend income	20
Fee and commission income	185
Fee and commission expense	120
Other operating income	45
Other operating expense	30
Net profit (loss) on trading book	70
Net profit (loss) on banking book	(30)

Assume the figures are already three-year averages of year-by-year values. Using only the information provided, what is RCB's business indicator (BI) under the standardized approach?

A. EUR870 million
B. EUR920 million
C. EUR1,010 million
D. EUR1,070 million

Answer: D

Step by step

1. ILDC = min[|1,100 - 380| ; 2.25% x 40,000] + dividends = min[720 ; 900] + 20 = 740.
2. SC = max(other operating income, expense) + max(fee income, fee expense) = 45 + 185 = 230.
3. FC = |trading| + |banking| = 70 + 30 = 100.
4. BI = 740 + 230 + 100 = EUR1,070 million.

Why D is correct: BI sums the interest/lease/dividend component, the services component (taking the larger of income and expense in each pair) and the financial component (absolute values of trading and banking book results).

Why the others are wrong

- **A:** 870 comes from subtracting the financial component (740 + 230 - 100) instead of adding it.
- **B:** 920 comes from netting income and expense in the services component (65 + 15 = 80) instead of taking the larger of each pair: 740 + 80 + 100.
- **C:** 1,010 nets the trading and banking results (70 - 30 = 40) instead of adding their absolute values: 740 + 230 + 40.

Q726

Schweser Module Quiz 43.1 Q2

REBUILT — Original options relied on a wrong claim (a BIC of EUR900m implies a 12% rate); rebuilt as a numeric ORC question using the loss component and ILM formula.

A bank's average annual operational losses over the last 10 years (high-quality data) were EUR80 million, and its business indicator component (BIC) is EUR900 million. Assume no supervisory discretion to set the ILM to 1. Using the standardized approach, the bank's operational risk capital requirement is closest to:

A. EUR900 million
B. EUR982 million
C. EUR1,200 million
D. EUR870 million

Answer: B

Step by step

1. Loss component LC = 15 x 80 = EUR1,200 million.
2. ILM $= \ln\left[e - 1 + (\text{LC}/\text{BIC})^{0.8}\right] = \ln[1.7183 + (1.3333)^{0.8}] = \ln[1.7183 + 1.2589] = 1.0909$.
3. ILM > 1 because LC > BIC, so losses raise capital.

4. ORC = BIC x ILM = 900 x 1.0909 = EUR982 million.

Why B is correct: Losses are high relative to the BIC (LC > BIC), so the ILM exceeds 1 and capital is above the BIC.

Why the others are wrong

- **A:** Sets ILM = 1 (ORC = BIC), ignoring the loss history; that applies only to bucket 1 or national discretion.
- **C:** Uses LC/BIC (1.333) or the LC itself as the multiplier/charge instead of the logarithmic ILM formula (900 x 1.333 = 1,200).
- **D:** Uses 10 instead of 15 as the loss multiple (LC = 800, ILM = 0.966, ORC = 870).

Q727

AP Mock 1 Q33

REBUILT — Options were loosely rounded (and 'closest to' left several defensible); restated with precise values built from realistic errors (four-year average, latest year only, no averaging).

The following has been extracted from the P&L of a European bank (EUR billions; each line is the already-netted component as defined for the standardized approach):

Year ended	20X6	20X7	20X8	20X9
Interest, leases and dividends (ILDC)	0.95	1.3	1.8	1.6
Services (SC)	1.6	2.2	2.6	1.8
Financial (FC; absolute net P/L)	0.5	1.1	1.3	1.5

Using the standardized approach, the bank's business indicator at 31 December 20X9 is:

A. EUR4.56 billion
B. EUR5.07 billion
C. EUR4.90 billion
D. EUR15.2 billion

Answer: B

Step by step

1. BI uses the most recent three years: 20X7, 20X8, 20X9.
2. ILDC average = (1.3 + 1.8 + 1.6)/3 = 1.567; SC average = (2.2 + 2.6 + 1.8)/3 = 2.200; FC average = (1.1 + 1.3 + 1.5)/3 = 1.300.
3. BI = 1.567 + 2.200 + 1.300 = EUR5.07 billion.

Why B is correct: The BI is the sum of the three components, each averaged over the latest three years.

Why the others are wrong

- **A:** Averages all four years of data (total of 18.25 / 4 = 4.56) instead of the latest three.
- **C:** Uses only the 20X9 values (1.6 + 1.8 + 1.5) instead of a three-year average.
- **D:** Adds the three years of all components without dividing by 3.

Q728

AnalystPrep Q.3099

REBUILT — Source used a non-Basel business-line formula (sum of line coefficients x indicators); rebuilt around the actual BI bucket structure with marginal coefficients.

A bank has a business indicator (BI) of EUR2.5 billion. The bank's supervisor has set the internal loss multiplier (ILM) to 1. Under the standardized approach (marginal coefficients 12% for BI up to EUR1bn, 15% for BI between EUR1bn and EUR30bn, 18% above EUR30bn), the bank's operational risk capital requirement is:

A. EUR300 million
B. EUR345 million
C. EUR375 million
D. EUR495 million

Answer: B

Step by step

1. Bucket 1 slice: 12% x 1,000 = EUR120 million.
2. Bucket 2 slice: 15% x (2,500 - 1,000) = EUR225 million.
3. BIC = 120 + 225 = EUR345 million; ORC = BIC x ILM = 345 x 1 = EUR345 million.

Why B is correct: Marginal coefficients apply only to the slice of BI within each bucket, then the BIC is scaled by the ILM (1 here).

Why the others are wrong

- **A:** Applies the 12% bucket-1 rate to the whole BI (12% x 2,500).

- **C:** Applies the bucket-2 rate of 15% to the whole BI, ignoring that the first EUR1bn is charged at 12%.
- **D:** Applies 12% to the first EUR1bn but then 15% to the full EUR2.5bn (120 + 375), double counting the first EUR1bn.

Q729

AnalystPrep Q.3104

REBUILT — Source used a non-Basel business-line formula; rebuilt as a bucket-3 BIC calculation with marginal coefficients.

A bank has a business indicator (BI) of EUR40 billion and the supervisor has set the internal loss multiplier to 1. Under the standardized approach (marginal coefficients 12% up to EUR1bn, 15% from EUR1bn to EUR30bn, 18% above EUR30bn), the operational risk capital requirement is:

A. EUR6,000 million
B. EUR6,150 million
C. EUR6,270 million
D. EUR7,200 million

Answer: C

Step by step

1. Slice 1: 12% x 1,000 = 120.
2. Slice 2: 15% x (30,000 - 1,000) = 4,350.
3. Slice 3: 18% x (40,000 - 30,000) = 1,800.
4. BIC = 120 + 4,350 + 1,800 = 6,270; ORC = 6,270 x 1 = EUR6,270 million.

Why C is correct: Each marginal coefficient applies only to the BI falling in its bucket; the totals are summed and scaled by the ILM of 1.

Why the others are wrong

- **A:** Applies 15% to the whole EUR40bn.
- **B:** Omits the 12% slice for the first EUR1bn (4,350 + 1,800).
- **D:** Applies the top 18% coefficient to the entire BI.

Q730

AnalystPrep Q.3111

A bank holding company based in Germany has two subsidiaries, A and B. The business indicator values of each are given in the table below:

Bank	A	B
BI	EUR 800 million	EUR 1.2 billion

In light of this information, which of the following statements is correct?

A. Bank A would be expected to calculate operational risk capital based on the Advanced Measurement Approach while Bank B would employ the standardized measurement approach.
B. Only Bank B would be expected to set aside capital for operational risk.
C. Bank B would be expected to use loss experience in the standardized approach calculations.
D. Neither Bank A nor Bank B would be expected to set aside some capital for operational risk.

Answer: C

Step by step

1. Bank A: BI = 800m <= EUR1bn, bucket 1, ILM = 1, capital = 12% x 800 = EUR96m.
2. Bank B: BI = 1.2bn > EUR1bn, bucket 2, so it must use its own loss data via the ILM.
3. The standardized approach replaced AMA for all banks, and both banks hold capital.

Why C is correct: Banks (including subsidiaries) with BI above EUR1bn are in bucket 2 or higher and must incorporate their own loss experience in the calculation.

Why the others are wrong

- **A:** The standardized approach replaces all earlier approaches including AMA; no bank uses AMA under the new framework.
- **B:** Bank A also holds capital: ORC = BIC = 12% of BI in bucket 1.
- **D:** Every bank must hold operational risk capital; bucket 1 simply sets ILM = 1.

Q731

AnalystPrep Q.3115

REBUILT — Distractor 0.9288 had no identifiable error; replaced with the ILM obtained by forgetting the factor of 15 in the loss component.

The following are verified historical annual loss data for an international bank over a 10-year period (EUR billions): 3.8, 2.9, 2.8, 2.8, 0.6, 0.4, 0.1, 0.2, 0.1, 0.2. The bank's business indicator component (BIC) is EUR18 billion. The bank's internal loss multiplier (ILM) under the standardized approach is closest to:

A. 1.39
B. 0.6137
C. 1.0
D. 1.0449

Answer: D

Step by step

1. Total losses = 13.9; average annual loss = 1.39.
2. LC = 15 x 1.39 = 20.85.
3. LC/BIC $= 20.85/18 = 1.1583$; $1.1583^{0.8} = 1.1248$.
4. ILM $= \ln[e - 1 + 1.1248] = \ln[2.8431] = 1.0449$.

Why D is correct: LC slightly exceeds BIC, so the ILM is a little above 1.

Why the others are wrong

- **A:** This is the average annual loss, not the multiplier.
- **B:** Uses the average loss (1.39) instead of 15 x average in the ratio: ln[1.7183 + (1.39/18)^0.8] = 0.614.
- **C:** Treats ILM as 1, ignoring that LC (20.85) exceeds BIC (18).

Q732

AnalystPrep Q.3116

REBUILT — Stem lacked the no-national-discretion assumption and a distractor mixed units; clarified and cleaned the options.

The following are verified annual loss data for a large established bank over a 10-year period (EUR billions): 0.8, 0.9, 0.7, 0.8, 0.06, 0.04, 0.10, 0.09, 0.03, 0.00. The bank's business indicator is EUR960 million. Assuming the default Basel rules (no national discretion), the bank's operational risk capital under the standardized approach is:

A. EUR115.2 million
B. EUR5,280 million
C. EUR361 million
D. EUR144 million

Answer: A

Step by step

1. BI = EUR960m <= EUR1bn, so the bank is in bucket 1 and BIC = 12% x 960 = EUR115.2m.
2. In bucket 1 the ILM is set to 1, so loss data do not enter.
3. ORC = 115.2 x 1 = EUR115.2 million.

Why A is correct: For BI of EUR1bn or less the ILM equals 1, so capital is simply 12% of BI.

Why the others are wrong

- **B:** Uses the loss component (15 x 0.352bn = 5,280m) as the capital charge.
- **C:** Applies the ILM formula (LC/BIC = 45.8, ILM = 3.14) even though bucket 1 banks have ILM = 1: 115.2 x 3.14 = 361.
- **D:** Uses 15% instead of 12%; the bucket-1 coefficient is 12%.

Q733

AnalystPrep Q.3119

REBUILT — Replaced the implausible option 'EUR 35bn' with 18% applied to the whole BI, so every distractor is a realistic bucketing error.

An international lender based in Dubai has a Business Indicator of EUR 34.5 billion. Using the table below, determine the Business Indicator Component (BIC) for the bank.

BI bucket	1	2	3
BI range	BI <= EUR 1bn	EUR 1bn < BI <= EUR 30bn	BI > EUR 30bn
Marginal BI coefficient	12%	15%	18%

A. EUR 0.12 billion
B. EUR 4.35 billion
C. EUR 6.21 billion
D. EUR 5.28 billion

Answer: D

Step by step

1. BI = 34.5 > 30, so the bank is in bucket 3 and marginal coefficients apply slice by slice.
2. Bucket 1: $1 \times 12\% = 0.12$ bn.
3. Bucket 2: $(30 - 1) \times 15\% = 4.35$ bn.
4. Bucket 3: $(34.5 - 30) \times 18\% = 0.81$ bn.
5. BIC $= 0.12 + 4.35 + 0.81 = 5.28$ bn.

Why D is correct: The BIC is built from marginal coefficients: each slice of BI is charged at its own bucket's rate, and the slices are summed.

Why the others are wrong

- **A:** Only the bucket 1 slice (0.12bn) was counted; the higher slices were ignored.
- **B:** Only the bucket 2 slice (4.35bn) was counted; buckets 1 and 3 were omitted.
- **C:** Applied the top 18% coefficient to the entire BI (34.5 x 18% = 6.21bn) instead of marginal slices.

Q734

AnalystPrep Q.3120

REBUILT — OCR-garbled ILM exponent restored and formula stated in stem; weak distractors replaced with realistic calculation errors.

A bank has a Business Indicator (BI) of EUR 40 billion and a Loss Component (LC) of EUR 1.2 billion. The Internal Loss Multiplier is ILM $= \ln\left[e - 1 + (\text{LC/BIC})^{0.8}\right]$. Using the table below, the bank's operational risk capital requirement is closest to:

BI bucket	1	2	3
BI range	BI <= EUR 1bn	EUR 1bn < BI <= EUR 30bn	BI > EUR 30bn
Marginal BI coefficient	12%	15%	18%

A. EUR 6.27 billion

B. EUR 4.30 billion

C. EUR 4.06 billion

D. EUR 4.80 billion

Answer: B

Step by step

1. BIC = $1(12\%) + 29(15\%) + 10(18\%) = 0.12 + 4.35 + 1.80 = 6.27$ bn.
2. LC/BIC $= 1.2/6.27 = 0.1914$; raised to 0.8 gives 0.2664.
3. ILM $= \ln(1.7183 + 0.2664) = \ln(1.9847) = 0.6855$.
4. Bucket 3 bank, so ORC $=$ BIC $\times$ ILM $= 6.27 \times 0.6855 = 4.30$ bn.

Why B is correct: ORC equals BIC times ILM for banks above bucket 1; a low loss history (LC below BIC) gives ILM below 1 and reduces capital below the BIC.

Why the others are wrong

- **A:** This is the BIC alone; it ignores the ILM (equivalent to ILM = 1).
- **C:** Dropped the 0.8 exponent, using $\ln(e - 1 + 0.1914) = 0.647$, so ORC = 4.06bn.
- **D:** Applied 12% to the whole BI (40 x 12% = 4.80bn); neither bucketing nor ILM was used.

Q735

BT answers P2.T7.20.12.1 · BT notes P2.T7.20.12.1

Under Basel III's standardized approach (SMA) to operational risk, capital is the Business Indicator Component (BIC) multiplied by an Internal Loss Multiplier (ILM). The BIC is a function of the Business Indicator (BI), a financial-statement proxy built from components, each averaged over three years. Each of the following is a component of the BI EXCEPT:

A. The financial component (FC) includes net profit/loss on the trading and banking books

B. The services component (SC) includes other operating income/expenses and fee income/expenses

C. The administrative component (AC) includes insurance premiums paid/received, recognized changes in goodwill, and the maximum of (staff expenses; or outsourcing fees)

D. The interest, leases, and dividend component (ILDC) adds average dividend income to the minimum of (average net interest income; or a percentage of interest-earning assets)

Answer: C

Step by step

1. BI = ILDC + SC + FC; there are exactly three components.

2. ILDC = min[abs(interest income - interest expense); 2.25% x interest-earning assets] + dividend income (D is true).
3. SC = max[other operating income; other operating expense] + max[fee income; fee expense] (B is true).
4. FC = abs(net P&L trading book) + abs(net P&L banking book) (A is true).
5. No 'administrative component' exists in the BI, so C is the exception.

Why C is correct: The BI has only the ILDC, SC and FC components; an administrative component (a remnant of older approaches) is not part of it.

Why the others are wrong

- **A:** True description of the financial component, so it is not the exception.
- **B:** True description of the services component, so it is not the exception.
- **D:** True description of the ILDC (interest capped at 2.25% of interest-earning assets, plus dividends), so it is not the exception.

Q736

BT answers P2.T7.20.12.3 · BT notes P2.T7.20.12.3

REBUILT — Chart reference removed (ILM given in stem), OCR-garbled coefficients restored, and distractors changed to specific calculation errors.

Under the standardized measurement approach, Acme Bank has a Business Indicator of EUR 56.0 billion and an Internal Loss Multiplier (ILM) of 0.8050. Using the table below, Acme's operational risk capital (ORC) is closest to:

BI bucket	1	2	3
BI range	BI <= EUR 1bn	EUR 1bn < BI <= EUR 30bn	BI > EUR 30bn
Marginal BI coefficient	12%	15%	18%

A. EUR 6.72 billion
B. EUR 7.37 billion
C. EUR 45.08 billion
D. EUR 9.15 billion

Answer: B

Step by step

1. BIC = 1(12%) + 29(15%) + 26(18%) = 0.12 + 4.35 + 4.68 = 9.15 bn.
2. ORC = BIC × ILM = 9.15 × 0.8050 = 7.366 bn, about EUR 7.37 bn.

Why B is correct: ORC is the bucket-based BIC scaled by the ILM; ILM below 1 reflects favorable loss experience and reduces capital.

Why the others are wrong

- **A:** Applied 12% to the entire BI (56 x 12% = 6.72bn), ignoring higher buckets and the ILM.
- **C:** Multiplied the BI itself by the ILM (56 x 0.805 = 45.08bn), skipping the BIC calculation.
- **D:** This is the BIC; the ILM was not applied.

Q737

Mock 2 Q15

Titan Bank is planning to move to the standardized approach for operational risk capital and wants to incorporate the loss component into its calculation for the coming fiscal year. The bank opened 8 years ago and only has reliable loss data for the last 6 years. Will it still be able to use the loss component and the standardized approach?

A. No, as the bank must have at least a 10-year observation period overall and specifically for loss data.
B. No, as the bank must be in existence for at least 10 years, with reliable loss data for the last 5 years.
C. Yes, as reliable loss data is not required as long as the bank has been in existence for at least 5 years.
D. Yes, as the bank will be eligible as long as the loss data covering at least the last 5 years is deemed good quality and reliable.

Answer: D

Step by step

1. The loss component ideally uses 10 years of high-quality annual loss data.
2. As a transition, a bank first moving to the standardized approach may use a minimum of 5 years of high-quality data.
3. Titan has 6 years of reliable data, which exceeds the 5-year minimum, so it can use the loss component.
4. The bank's age is not the test; only the quality and length of loss data matters (a bank with under 5 years of good data just uses BIC alone).

Why D is correct: The transition rule lets a bank with at least five years of good-quality loss data compute the loss

component with fewer than ten years.

Why the others are wrong

- **A:** Wrongly requires a full 10 years; the 5-year transitional minimum applies.
- **B:** Invents a 10-year existence requirement; only the loss data length matters.
- **C:** Wrongly says data reliability is not required; the 5 years must be high quality, and age alone is irrelevant.

Q738

Schweser Module Quiz 65.1 Q1

REBUILT — Distractors changed to realistic errors (flat 12% or 15% on all BI).

A bank has a Business Indicator (BI) of EUR 13 billion. Using the standard marginal coefficients (12% up to EUR 1bn, 15% from EUR 1bn to EUR 30bn, 18% above EUR 30bn), the Business Indicator Component (BIC) is closest to:

A. EUR 1.56 billion
B. EUR 1.92 billion
C. EUR 1.95 billion
D. EUR 13.00 billion

Answer: B

Step by step

1. BI = 13 is between 1 and 30, so the bank is in bucket 2.
2. $\text{BIC} = 0.12 \times 1 + 0.15 \times (13 - 1) = 0.12 + 1.80 = 1.92$ bn.

Why B is correct: The first EUR 1bn is charged at 12% and the remaining EUR 12bn at 15%.

Why the others are wrong

- **A:** Applied 12% to the entire BI (13 x 12% = 1.56bn).
- **C:** Applied 15% to the entire BI (13 x 15% = 1.95bn).
- **D:** Used the BI itself as the BIC; no marginal coefficients applied.

Q739

Schweser Module Quiz 65.1 Q3

Which of the following components within the Business Indicator (BI) calculation takes into account a bank's trading and banking book P&L results?

A. Loss component.
B. Services component.
C. Financial component.
D. Interest, lease, dividend component.

Answer: C

Step by step

1. FC = abs(net P&L trading book) + abs(net P&L banking book), each averaged over three years.
2. The financial component is therefore the BI element capturing trading and banking book P&L.

Why C is correct: The financial component is defined as the absolute values of net trading book P&L and net banking book P&L.

Why the others are wrong

- **A:** The loss component is not part of the BI; it feeds the ILM and reflects internal operational losses.
- **B:** The services component covers other operating income/expense and fee income/expense.
- **D:** The ILDC covers net interest income (capped at 2.25% of interest-earning assets) and dividend income.

Q740

Schweser Module Quiz 65.1 Q2

Which of the following items from the profit and loss (P&L) statement should be included in the Business Indicator (BI) component calculation?

A. Administrative expenses.
B. Insurance premiums paid.
C. Depreciation related to capitalized equipment.
D. Provision reversals related to operational loss events.

Answer: D

Step by step

1. P&L items that do not contribute to the BI include administrative expenses (staff, outsourcing), premiums paid and reimbursements on insurance purchased, and depreciation/amortisation of tangible and intangible assets.
2. Provisions and reversals of provisions are also excluded, except those related to operational loss events.
3. Only D falls within that exception, so it is included in the BI.

Why D is correct: Provisions/reversals are excluded from BI unless they relate to operational loss events, in which case they are captured (in other operating expenses/income).

Why the others are wrong

- **A:** Administrative expenses, including staff and outsourcing costs, are explicitly excluded from the BI.
- **B:** Insurance premiums paid (and reimbursements received) are excluded from the BI.
- **C:** Depreciation/amortisation of tangible and intangible assets is excluded from the BI.

LO 65.b

Compare the Standardized Measurement Approach (SMA) to earlier methods of calculating operational risk capital, including the Advanced Measurement Approaches (AMA).

Q741

Schweser Module Quiz 65.1 Q4

Which of the following statements best describes a difference between the standardized approach and older operational risk capital approaches?

A. The advanced measurement approach (AMA) was introduced as part of the Basel III revisions.

B. The AMA was more flexible in its application than the standardized approach.

C. The standardized approach accounts for internal loss experiences that were not factored into the AMA.

D. The standardized approach uses a model-based methodology, while the AMA was more flexible and principles-based.

Answer: B

Step by step

1. The AMA (Basel II, 2006) let banks estimate capital using a wide range of internal modeling practices; it was a principles-based framework with significant flexibility.
2. The standardized approach is a single, non-model-based method combining financial statement information (the business indicator) with the bank's own loss experience (via the ILM).
3. Compare each option with these facts: only B correctly states that the AMA was the more flexible approach.

Why B is correct: The AMA was principles-based and allowed many internal models, so it was more flexible than the single, non-model-based standardized approach, which was created to improve comparability and reduce complexity.

Why the others are wrong

- **A:** Wrong: the AMA came from Basel II (2006), not Basel III; confuses the framework that introduced the AMA with the one that replaced it.
- **C:** Wrong: the AMA already used internal loss data; internal losses are not a new feature of the standardized approach.
- **D:** Wrong: reverses the descriptions; the standardized approach is non-model-based and the AMA was the model-based, flexible one (the second half is true but the first half is false).

Q742

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank that has used an internally developed advanced measurement approach (AMA) to set operational risk capital is preparing for the Basel III standardized approach. The head of operational risk asks an analyst to explain why the Basel Committee replaced the earlier approaches, including the AMA. Which of the following statements is most accurate?

A. The AMA permitted such a wide range of internal modeling practices that capital outcomes were difficult to compare across banks, and the standardized approach replaces it with a single non-model-based method.

B. The AMA ignored bank-specific loss experience, so the standardized approach was created to introduce internal loss data into the capital calculation for the first time.

C. The AMA was a rigid, rules-based framework, so the standardized approach was created to give banks more discretion over how they model operational risk.

D. The standardized approach is an optional alternative, and banks that have received supervisory approval may continue to use the AMA for regulatory capital.

Answer: A

Step by step

1. The AMA (Basel II, 2006) was principles-based and let banks estimate capital using a wide range of internal models.
2. Hoped-for convergence to best practice never happened, leaving comparability problems and overly complex calculations.
3. The standardized approach is a single, non-model-based method combining financial statement information (BI) with the bank's own loss data, and it replaces all existing Basel II approaches.
4. Therefore A is correct.

Why A is correct: The AMA's flexibility produced inconsistent, hard-to-compare and complex capital results; the standardized approach was designed for comparability and simplicity and replaces the earlier approaches.

Why the others are wrong

- **B:** The AMA did incorporate internal loss data; internal losses were not new to the standardized approach. Confuses AMA with the earlier basic indicator approach.
- **C:** Reverses the facts: the AMA was the flexible, principles-based framework, and the standardized approach reduces discretion.
- **D:** The standardized approach replaces all existing approaches in the Basel II framework, so continued AMA use is not permitted; this treats it as an optional add-on.

Q743 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Two internationally active banks each report a business indicator (BI) of EUR 10 billion. The business indicator component (BIC) of each is therefore EUR 1.47 billion. Bank X has a loss component well above its BIC, whereas Bank Y has a loss component well below its BIC. A risk consultant compares how the Basel III standardized approach and the earlier AMA would treat the two banks. Which of the following statements is correct?

A. Under the standardized approach, both banks would hold EUR 1.47 billion because bank-specific losses affect capital only under the AMA.

B. Under the standardized approach, Bank X would hold more capital than Bank Y because its internal loss multiplier exceeds one, whereas under the AMA capital differences also depended on each bank's choice of model.

C. Under the standardized approach, Bank Y would hold more capital than Bank X because a low loss component raises the internal loss multiplier above one.

D. Under the standardized approach, differences in the two banks' capital would depend on each bank's internal model of its loss distribution, as was the case under the AMA.

Answer: B

Step by step

1. BIC = 12% x 1 + 15% x 9 = 1.47 (EUR bn), identical for both banks.
2. ORC = BIC x ILM. ILM > 1 when LC > BIC (Bank X); ILM < 1 when LC < BIC (Bank Y).
3. So Bank X holds more than 1.47bn and Bank Y less than 1.47bn, through a prescribed formula.
4. Under the AMA, capital also reflected each bank's own modeling choices, harming comparability. B is correct.

Why B is correct: The standardized approach differentiates the banks only via the prescribed ILM driven by loss experience, while the AMA's results also depended on bank-specific modeling choices.

Why the others are wrong

- **A:** Losses do enter the standardized approach through the ILM for banks above bucket 1 (BI above EUR 1 billion); both are in bucket 2, so capital differs from BIC.
- **C:** Reverses the ILM logic: a loss component below the BIC gives ILM below one and lower capital.
- **D:** The standardized approach is non-model-based; the ILM follows a prescribed formula, so internal loss-distribution models play no role.

LO 65.c

Describe general and specific criteria recommended by the Basel Committee for the identification, collection, and treatment of operational loss data.

Q744 BT Mock Q43 · BT answers P2.T7.20.12.2 · BT notes P2.T7.20.12.2

Acme Financial's Business Indicator (BI) is well in excess of EUR 1.0 billion, so it must use loss data as a direct input into its operational risk capital calculation under the standardized approach. Regarding the general and specific criteria recommended by the Basel Committee for the identification, collection and treatment of operational loss data, which of the following is TRUE?

A. P&L provisions or reserves against potential loss impacts are included in the gross loss computation.
B. All internal loss events must be included in the gross loss computation regardless of the size of the loss.
C. Legal expenses are excluded from the gross loss computation because they represent the realization of legal risk.
D. Receivables should be counted as recoveries unless the particular receivable causes the current ratio to exceed an internal limit.

Answer: A

Step by step

1. Gross loss must include: direct P&L charges, impairments and settlements; external expenses directly linked to the event (e.g. legal and advisor fees) and repair/replacement costs; provisions or reserves booked in the P&L against the potential loss; and material pending losses in suspense accounts.
2. A matches the third item, so A is true.
3. B: a minimum collection threshold applies (EUR 20,000, which supervisors may raise to EUR 100,000 for buckets 2 and 3), so small events need not be included.
4. C: legal expenses directly related to the event are included in gross loss, not excluded.
5. D: recoveries reduce losses only after payment is actually received; receivables are never counted as recoveries.

Why A is correct: Provisions and reserves charged to the P&L for a potential operational loss are an explicit component of gross loss, which is recognised at the date of accounting.

Why the others are wrong

- **B:** Ignores the minimum threshold (EUR 20,000); events below it are not required in the loss data set.
- **C:** Reverses the rule: legal expenses with a direct link to the event are included in gross loss.
- **D:** Receivables do not count as recoveries; only payments actually received can reduce a loss. The 'current ratio' test is invented.

Q745 BT answers P2.T7.601.3 · BT notes P2.T7.601.3

In regard to a recent loss event, your firm's accounting staff has itemized the following amounts associated with the event:
- Gross loss due to the event: $5,000,000
- Forgone revenue (opportunity cost): $3,000,000
- Estimated, non-rapid recovery with 50% probability: $1,500,000
- Insurance premiums (paid by the firm while the event occurred): $12,500

Which is the operational loss amount that will be calculated in association with the event under the Basel framework?

A. $3,500,000
B. $4,250,000
C. $5,000,000
D. $6,487,500

Answer: C

Step by step

1. Start with the gross loss: $5,000,000.
2. Forgone revenue (opportunity cost) is not a direct P&L charge, expense or provision, so it is not part of the gross loss.
3. An estimated recovery that has not been received is not a recovery: recoveries reduce losses only once payment is received. No netting: 5,000,000 - 0 = 5,000,000.
4. Insurance premiums are explicitly excluded from gross loss.

5. Operational loss = $5,000,000.

Why C is correct: Only the realised gross loss counts; opportunity costs and insurance premiums are excluded and expected-but-unreceived recoveries cannot be netted.

Why the others are wrong

- **A:** 5,000,000 - 1,500,000 = 3,500,000: deducts the full expected recovery although no payment has been received.
- **B:** 5,000,000 - 0.5 x 1,500,000 = 4,250,000: deducts a probability-weighted recovery; recoveries count only when actually received.
- **D:** 5,000,000 + 1,500,000 - 12,500 = 6,487,500: wrongly adds the expected recovery and nets the insurance premium, which is simply excluded.

Q746 Schweser Module Quiz 65.1 Q5

In deriving the standardized approach loss dataset for an individual bank, each of the following items will most likely be included in the gross loss calculation EXCEPT:

A. legal fees of €900,000 associated with an unusual risk event.
B. a €2 million settlement tied to a recent operational risk event.
C. a €1.4 million reserve booked to the income statement to cover a potential operational loss.
D. €1.75 million spent on maintenance contracts tied to the bank's property, plant and equipment.

Answer: D

Step by step

1. Included in gross loss: external expenses directly tied to the event (legal and advisor fees), settlements and other direct P&L charges, and provisions/reserves booked to the P&L against the potential loss.
2. A (legal fees), B (settlement) and C (reserve) are all included.
3. Excluded from gross loss: general maintenance contract costs on property, plant and equipment, post-event improvement spending, and insurance premiums.
4. D is therefore the exception.

Why D is correct: Costs of general maintenance contracts on PP&E are expressly excluded from gross loss; the other three are listed inclusions.

Why the others are wrong

- **A:** Legal fees with a direct link to the event are included in gross loss, so this is not the exception.
- **B:** Settlements are direct charges to the P&L and are included.
- **C:** Provisions or reserves booked in the P&L against a potential operational loss are included.